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(54) **SYNERGISTIC BACTERIAL COMPOSITIONS AND METHODS OF PRODUCTION AND USE THEREOF**

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(58) **Field of Classification Search**

None

See application file for complete search history.

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(57) **ABSTRACT**

Provided are therapeutic compositions containing microbial populations for prevention, treatment and reduction of symptoms associated with a dysbiosis of a mammalian subject such as a human.

20 Claims, 7 Drawing Sheets

Specification includes a Sequence Listing.

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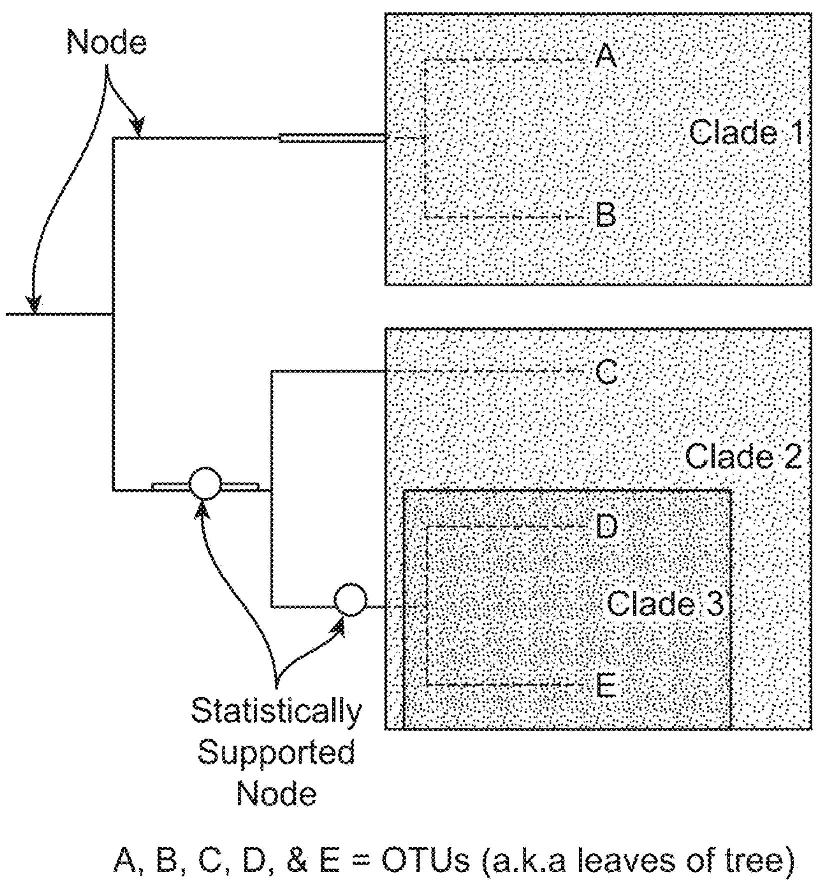

Node
A
Clade 1
B
C
Clade 2
D
Clade 3
E
Statistically Supported Node
A, B, C, D, & E = OTUs (a.k.a leaves of tree)

FIG. 1

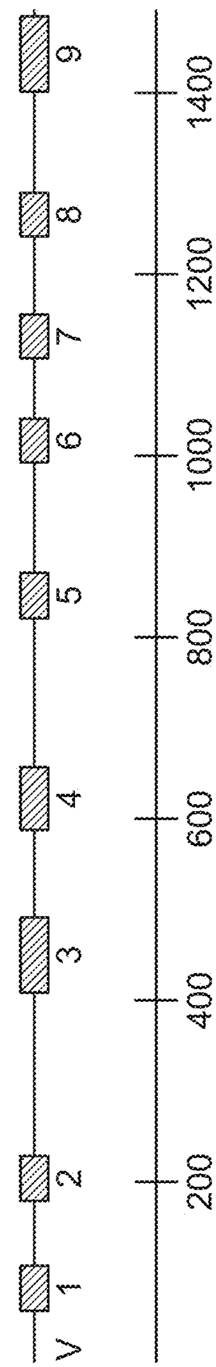
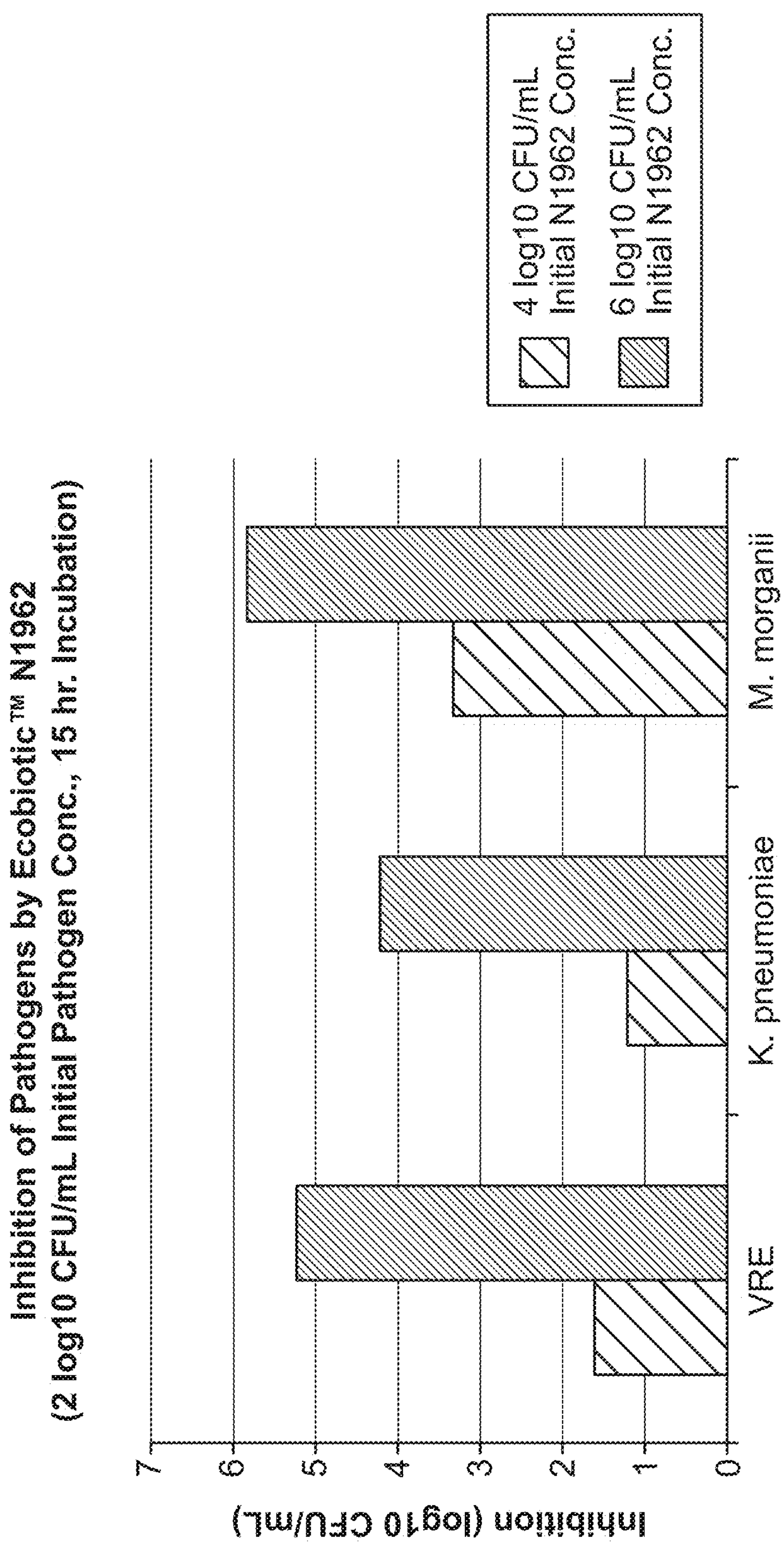


FIG. 2

1 AAATTGAAGAGTTTGATCATGGCTCAGATTGAACGCTGGCGGCAGGCCTA
51 ACACATGCAAGTCGAACG**GTAACAGGAAGAAGCTTGCTCTTTGCTGACG**A
101 GTGGCGGACGGGTGAGTAATGTCTGGGAAACTGCCT**GATGGAGGGGGATA**
151 **ACTACTGGAAACGGTAGCTAATACCGCATAACGTCGCAAGACCAAAGAGG**
201 **GGGACCTTCGGGCCTCTTGCCATCGGATGTGCCCAGATGGGA**TTAGCTAG
251 TAGGTGGGGTAACGGCTCACCTAGGCGACGATCCCTAGCTGGTCTGAGAG
301 GATGACCAGCCACACTGGAACTGAGACACGGTCCAGACTCCTACGGGAGG
351 CAGCAGTGGGGAATATTGCACAATGGGCGCAAGCCTGATGCAGCCATGCC
401 GCGTGTATGAAGAAGGCCTTCGGGTTGTAAAG**TACTTTCAGCGGGGAGGA**
451 **AGGGAGTAAAGTTAATACCTTTGCTCATTGACGTTACCCGCAGAAGA**AGC
501 ACCGGCTAACTCGTGCCCAGGCATGCGCAGGAATACGGAGGTGCAAGCGT
551 TAATCGGAATTACTGGGCGTAAAGC**GCACGCAGGCGGTTTGTTAAGTCAG**
601 **ATGTGAAATCCCCGGGCTCAACCTGGGAACTGCATCTGATACTGGCAAGC**
651 **TTGAGTCTCGTAGAGGGGGGTAGAATTCCAGG**TGTAGCGGTGAAATGCGT
701 AGAGATCTGGAGGAATACCGGTGGCGAAGGCGGCCCCCTGGACGAAGACT
751 CACGCTCAGGTGCGAAAGCGTGGGGAGCAAACAGGATTAGATACCCTGGT
801 AGTCCACGCCGTAAACGATGT**CGACTTGGAGGTTGTGCCCTTGAGGCGTG**
851 **GCTTCCGGAGCTAACGCGTTAAGTCGACC**GCCTGGGGAGTACGGCCGCAA
901 GGTTAAAACTCAAATGAATTGACGGGGGCCCGCACAAGCGGTGGAGCATG
951 TGGTTTAATTCGATGCAACGCGAAGAACCTTACCT**GGTCTTGACATCCAC**
1001 **GGAAGTTTTCAGAGATGAGAATGTGCCTTCGGGAACCGTGAGA**CAGGTGC
1051 TGCATGGCTGTCGTCAGCTCGTGTTGTGAAATGTTGGGTTAAGTCCCGCA
1101 ACGAGCGCAACCCTTA**TCCTTTGTTGCCAGCGGTCCGGCCGGGAACTCAA**
1151 **AGGAGACTGCCAGTGATAAACTG**GAGGAAGGTGGGGATGACGTCAAGTCA
1201 TCATGGCCCTTACGACCAGGGCTACACACGTGCTACAATGGC**GCATACAA**
1251 **AGAGAAGCGACCTCGCGAGAGCAAGCGGACCTCATAAAGTGCGT**CGTAGT
1301 CCGGATTGGAGTCTGCAACTCGACTCCATGAAGTCGGAATCGCTAGTAAT
1351 CGTGGATCAGAATGCCACGGTGAATACGTTCCCGGGCCTTGTACACACCG
1401 CCCGMCACACCATGGGAGTGGGTTGCAAAAGAAG**TAGGTAGCTTAACCTT**
1451 **CGGGAGGGCGCTTAC**CACTTTGTGATTCATGACTGGGGTGAAGTCGTAAC
1501 AAGGTAACCGTAGGGGAACCTGCGGTTGGATCACCTCCTTA

FIG. 3



Inhibition of Pathogens by Ecobiotic™ N1962
(2 log10 CFU/mL Initial Pathogen Conc., 15 hr. Incubation)
Inhibition (log10 CFU/mL)
0
1
2
3
4
5
6
7
VRE
K. pneumoniae
M. morganii
4 log10 CFU/mL Initial N1962 Conc.
6 log10 CFU/mL Initial N1962 Conc.

FIG. 4

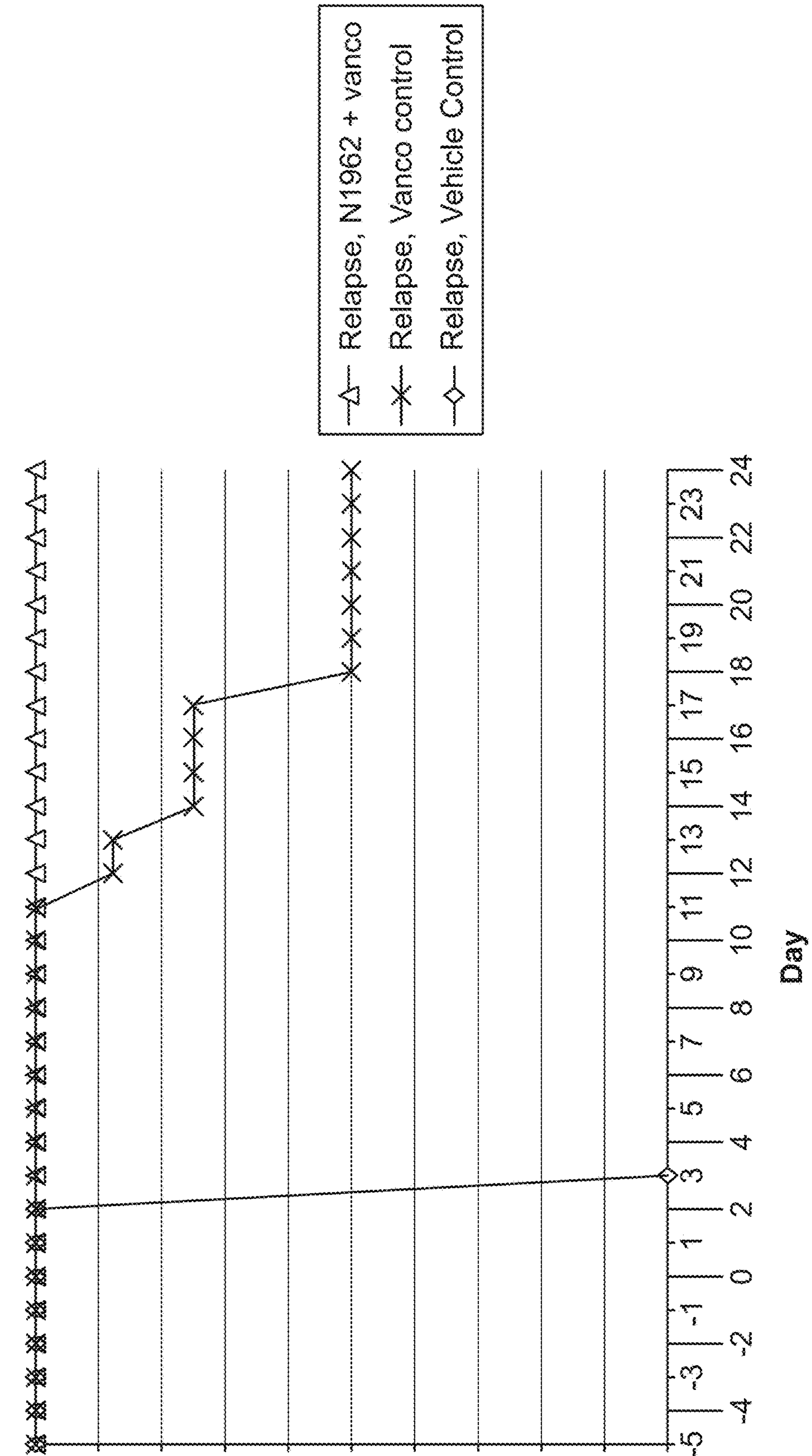

Relapse, N1962 + vanco
Relapse, Vanco control
Relapse, Vehicle Control
% Survival
100
90
80
70
60
50
40
30
20
10
0
Day
-5
-4
-3
-2
-1
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24

FIG. 5

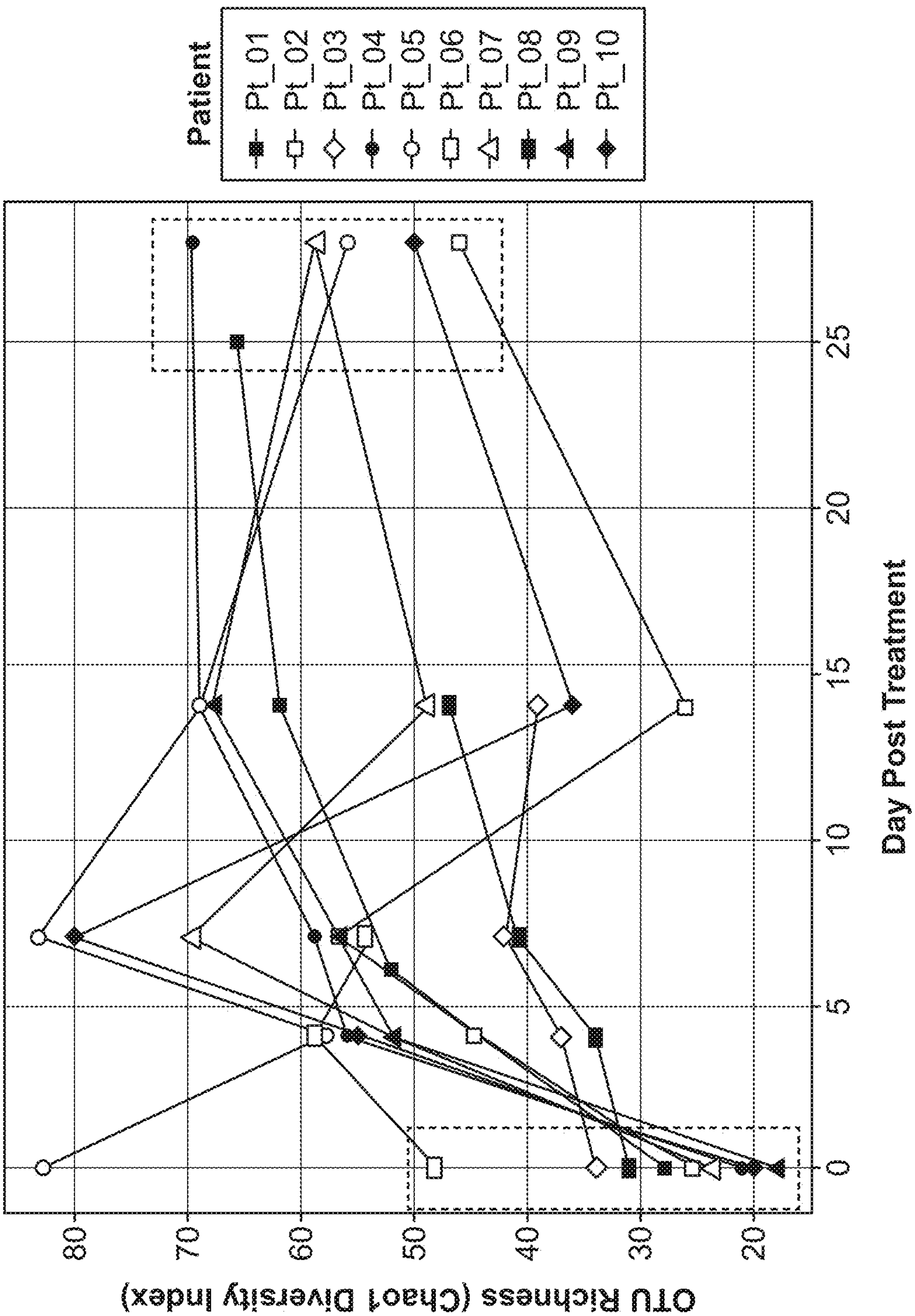

Patient
Pt_01
Pt_02
Pt_03
Pt_04
Pt_05
Pt_06
Pt_07
Pt_08
Pt_09
Pt_10
OTU Richness (Chao1 Diversity Index)
20
30
40
50
60
70
80
Day Post Treatment
0
5
10
15
20
25

FIG. 6

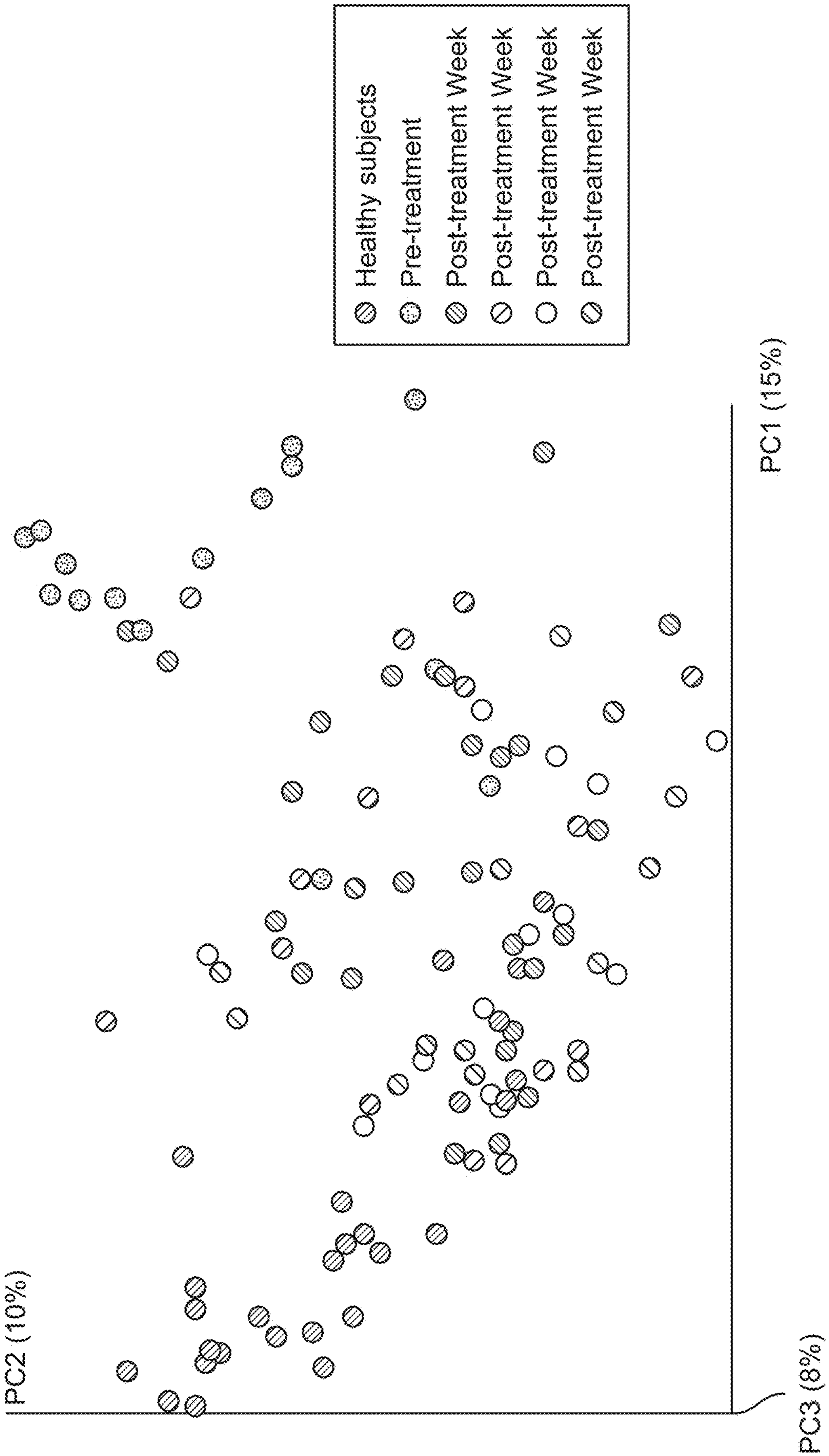

Healthy subjects
Pre-treatment
Post-treatment Week
Post-treatment Week
Post-treatment Week
Post-treatment Week
PC1 (15%)
PC2 (10%)
PC3 (8%)

FIG. 7

SYNERGISTIC BACTERIAL COMPOSITIONS AND METHODS OF PRODUCTION AND USE THEREOF

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a divisional application of U.S. patent application Ser. No. 17/588,122, filed Jan. 28, 2022 (now U.S. Pat. No. 11,918,612, issued on Mar. 5, 2024), which is a divisional application of U.S. patent application Ser. No. 16/230,807, filed Dec. 21, 2018, 2016 (now U.S. Pat. No. 11,266,699, issued on Mar. 8, 2022), which is a divisional application of U.S. patent application Ser. No. 15/039,007, filed May 24, 2016 (now U.S. Pat. No. 10,258,655, issued on Apr. 16, 2019), which is the National Stage of International Application No. PCT/US2014/067491, filed Nov. 25, 2014, which claims the benefit of U.S. Provisional Application No. 61/908,698, filed Nov. 25, 2013; U.S. Provisional Patent Application No. 61/908,702, filed Nov. 25, 2013; and U.S. Provisional Patent Application No. 62/004,187, filed May 28, 2014, the entire disclosures of which are hereby incorporated by reference in their entirety for all purposes.

REFERENCE TO A SEQUENCE LISTING SUBMITTED ELECTRONICALLY

The content of the electronically submitted sequence listing in (Name: 4268_0390007_SequenceListing_ST26.xml; Size: 5,546,501 bytes; and Date of Creation: May 23, 2024) filed with the application is herein incorporated by reference in its entirety.

BACKGROUND

Mammals are colonized by microbes in the gastrointestinal (GI) tract, on the skin, and in other epithelial and tissue niches such as the oral cavity, eye surface and vagina. The gastrointestinal tract harbors an abundant and diverse microbial community. It is a complex system, providing an environment or niche for a community of many different species or organisms, including diverse strains of bacteria. Hundreds of different species may form a commensal community in the GI tract in a healthy person, and this complement of organisms evolves from the time of birth to ultimately form a functionally mature microbial population by about 3 years of age. Interactions between microbial strains in these populations and between microbes and the host, e.g., the host immune system, shape the community structure, with availability of and competition for resources affecting the distribution of microbes. Such resources may be food, location and the availability of space to grow or a physical structure to which the microbe may attach. For example, host diet is involved in shaping the GI tract flora.

A healthy microbiota provides the host with multiple benefits, including colonization resistance to a broad spectrum of pathogens, essential nutrient biosynthesis and absorption, and immune stimulation that maintains a healthy gut epithelium and an appropriately controlled systemic immunity. In settings of 'dysbiosis' or disrupted symbiosis, microbiota functions can be lost or deranged, resulting in increased susceptibility to pathogens, altered metabolic profiles, or induction of proinflammatory signals that can result in local or systemic inflammation or autoimmunity. Thus, the intestinal microbiota plays a significant role in the pathogenesis of many diseases and disorders, including a variety of pathogenic infections of the gut. For example, subjects become more susceptible to pathogenic infections when the normal intestinal microbiota has been disturbed due to use of broad-spectrum antibiotics. Some of these diseases and disorders are chronic conditions that significantly decrease a subject's quality of life and ultimately some can be fatal.

Fecal transplantation has been shown to sometimes be an effective treatment for subjects suffering from severe or refractory GI infections and other disorders by repopulating the gut with a diverse array of microbes that control key pathogens by creating an ecological environment inimical to their proliferation and survival. Such approaches have demonstrated potential to decrease host susceptibility to infection. Fecal transplantation, however, is generally used only for recurrent cases because it has the potential to transmit infectious or allergenic agents between hosts, involves the transmission of potentially hundreds of unknown strains from donor to subject, and is difficult to perform on a mass scale. Additionally, fecal transplantation is inherently non-standardized and different desired and/or undesired material may be transmitted in any given donation. Thus, there is a need for defined compositions that can be used to decrease susceptibility to infection and/or that facilitate restoration of a healthy gut microbiota.

In addition, practitioners have a need for safe and reproducible treatments for disorders currently treated on an experimental basis using fecal transplantation.

SUMMARY OF THE INVENTION

To meet the need for safe, reproducible treatments for disorders that can be modulated by the induction of a healthy GI microbiome and to treat diseases associated with the GI microbiome, Applicants have designed bacterial compositions of isolated bacterial strains with a plurality of functional properties, in particular that are useful for treating dysbiosis (e.g., restoring a GI microbiome to a state of health), and for treating disorders associated with infection or imbalance of microbial species found in the gut that are based on Applicants discoveries related to those bacterial strains and analysis and insights into properties related to those strains and combinations of those strains, leading to the inventions disclosed herein.

In a first aspect, provided are compositions comprising an effective amount of a bacterial composition comprising at least a first type of isolated bacterium capable of forming a spore, a second type of isolated bacterium capable of forming a spore and optionally a third type of isolated bacterium capable of forming a spore, wherein the first type, the second type and the optional third type are not identical, and wherein at least two of the first type, the second type and the optional third type are capable of synergistically decreasing and/or inhibiting the growth and/or colonization of at least one type of pathogenic bacteria. In some embodiments, the bacterial composition comprises at least about 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 types of isolated bacteria capable of forming spores. In other embodiments, the bacterial composition comprises at least about 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 types of isolated bacteria not containing at least one sporulation-associated gene. In further embodiments, the bacterial composition comprises at least about 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 types of isolated bacteria in spore form. In further embodiments, the bacterial composition comprises at least about 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 types of isolated bacteria in vegetative form. In further embodiments, the bacterial composition comprises at least about 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 types of isolated bacteria in spore form, and wherein the bacterial composition further comprises at least about 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 types of isolated bacteria in vegetative form. In further embodiments, the bacterial composition comprises at least about 5 types of isolated bacteria and at least about 20% of the isolated bacteria are capable of forming spores or are in spore form. In further embodiments, the bacterial composition comprises at least about 5 types of isolated bacteria and at least 2 of the isolated bacteria are capable of forming spores or are in spore form. In further embodiments, the first type, second type and optional third type are present in the composition in approximately equal concentrations. In further embodiments, the first type and the third type are present in the composition in approximately equal concentrations. In further embodiments, the second type and the third type are present in the composition in approximately equal concentrations. In further embodiments, the first type is present in the composition in at least about 150% the concentration of the second type and/or the third type. In further embodiments, the first type, second type and optional third type are individually present in the composition in at least about 150% the concentration of the third type. In further embodiments, the composition consists essentially of between two and about twenty types of isolated bacteria, wherein at least two types of the isolated bacteria are independently capable of spore formation. In further embodiments, at least two types of the isolated bacteria are in spore form. In further embodiments, the first, second and third types are independently selected from Table 1. In further embodiments, the first, second and third types comprise an operational taxonomic unit (OTU) distinction. In further embodiments, the OTU distinction comprises 16S rRDNA sequence similarity below about 95% identity. In further embodiments, the first, second and third types independently comprise bacteria that comprise 16S rDNA sequence at least 95% identical to 16S rDNA sequence present in a bacterium selected from Table 1.

In another aspect, provided are compositions comprising an effective amount of a bacterial composition comprising a first type of isolated bacterium; a second type of isolated bacterium; and a third type of isolated bacterium, wherein at least one of the first, second and third types are capable of forming a spore, wherein the first, second and third types are not identical, and wherein a combination of at least two of the first, second and third types are inhibitory to at least one type of pathogenic bacteria. In some embodiments, a combination of the first, second and third types is capable of being inhibitory to the pathogenic bacterium. In other embodiments, a combination of the first, second and third types is capable of being cytotoxic or cytostatic to the pathogenic bacterium. In further embodiments, a combination of the first, second and third types is capable of being cytotoxic or cytostatic to the pathogenic bacterium. In further embodiments, a combination of the first, second and third types is capable of inhibiting proliferation of the pathogenic bacterial present at a concentration at least equal to the concentration of the combination of the first, second and third types. In further embodiments, the pathogenic bacterium is selected from the group consisting of *Yersinia, Vibrio, Treponema, Streptococcus, Staphylococcus, Shigella, Salmonella, Rickettsia, Pseudomonas, Neisseria, Mycoplasma, Mycobacterium, Listeria, Leptospira, Legionella, Helicobacter, Haemophilus, Francisella, Escherichia, Enterococcus, Klebsiella, Corynebacterium, Clostridium, Chlamydia, Chlamydophila, Campylobacter, Brucella, Borrelia*, and *Bordetella*. In further embodiments, the first, second and third types synergistically interact. In further embodiments, at least one of the first, second and third types are capable of independently forming a spore. In further embodiments, at least two of the first, second and third types are capable of independently forming a spore. In further embodiments, the first, second and third types are capable of independently forming a spore.

In further embodiments, wherein the first, second and third types are capable of functionally populating the gastrointestinal tract of a human subject to whom the composition is administered. In further embodiments, the functional populating of the gastrointestinal tract comprises preventing a dysbiosis of the gastrointestinal tract. In further embodiments, the functional populating of the gastrointestinal tract comprises treating a dysbiosis of the gastrointestinal tract. In further embodiments, the functional populating of the gastrointestinal tract comprises reducing the severity of a dysbiosis of the gastrointestinal tract. In further embodiments, the functional populating of the gastrointestinal tract comprises reducing one or more symptoms of a dysbiosis of the gastrointestinal tract. In further embodiments, the functional populating of the gastrointestinal tract comprises preventing colonization of the gastrointestinal tract by a pathogenic bacterium. In further embodiments, the functional populating of the gastrointestinal tract comprises reducing colonization of the gastrointestinal tract by a pathogenic bacterium. In further embodiments, the functional populating of the gastrointestinal tract comprises reducing the number of one or more types of pathogenic bacteria in the gastrointestinal tract. In further embodiments, the functional populating of the gastrointestinal tract comprises increasing the number of one or more non-pathogenic bacteria in the gastrointestinal tract. Also provided are single dose units comprising the bacterial compositions provided herein, for example, dose units comprising at least 1×10^7, 1×10^8, 1×10^9, 1×10^{10}, 1×10^{11}, or 1×10^{12} colony forming units (CFUs) of viable bacteria. Also provided are pharmaceutical formulations comprising an effective amount of the compositions provided herein, and further comprising an effective amount of an anti-bacterial agent, a pharmaceutical formulation comprising an effective amount of the bacterial composition, and further comprising an effective amount of an anti-fungal agent, a pharmaceutical formulation comprising an effective amount of the bacterial composition, and further comprising an effective amount of an anti-viral agent, and a pharmaceutical formulation comprising an effective amount of the bacterial composition, and further comprising an effective amount of an anti-parasitic agent.

In another aspect, provided are methods comprising administering to a human subject in need thereof an effective amount of the bacterial compositions, and further comprising administering to the human subject an effective amount of an anti-biotic agent. In some embodiments, the bacterial composition and the anti-biotic agent are administered simultaneously. In other embodiments, the bacterial composition is administered prior to administration of the anti-biotic agent. In further embodiments, provided are methods in which the number of pathogenic bacteria present in the gastrointestinal tract of the human subject is not detectably increased or is detectably decreased over a period of time. In other embodiments, the human subject is diagnosed as having a dysbiosis of the gastrointestinal tract. In other embodiments, the human subject is diagnosed as infected with a pathogenic bacterium selected from the group consisting of *Yersinia, Vibrio, Treponema, Streptococcus, Staphylococcus, Shigella, Salmonella, Rickettsia, Pseudomonas, Neisseria, Mycoplasma, Mycobacterium, Listeria, Leptospira, Legionella, Helicobacter, Haemophilus, Francisella, Escherichia, Enterococcus, Klebsiella, Corynebacterium, Clostridium, Chlamydia, Chlamydophila, Campylobacter, Brucella, Borrelia*, and *Bordetella*. In other embodiments, the anti-bacterial agent is administered to the human subject prior to administration of the bacteria composition. In other embodiments, the number of pathogenic bacteria present in or excreted from the gastrointestinal tract of the human subject is detectably reduced within two weeks of administration of the bacterial composition.

In another aspect, provided are methods of functionally populating the gastrointestinal tract of a human subject, comprising administering to the subject an effective amount of the bacterial composition of the present invention, under conditions such that the first, second and third types functionally populate the gastrointestinal tract of the human subject. In some embodiments, the bacterial composition is orally administered, rectally administered, or the combination of orally and rectally administered. In other embodiments, the bacterial composition is topically or nasally administered or inhaled.

Also provided are methods of preparing a comestible product, comprising combining with a comestible carrier the bacterial compositions of the present invention, wherein the comestible product is substantially free of non-comestible materials.

In one aspect, provided are compositions comprising an effective amount of a bacterial composition comprising at least a first type of isolated bacterium capable of forming a spore and a second type of isolated bacterium capable of forming a spore, wherein the first type, second type and optional third type are not identical, and wherein at least one of the first type, second type and optional third type are capable of decreasing and/or inhibiting the growth and/or colonization of at least one type of pathogenic bacteria. In an embodiment, the bacterial composition comprises at least about 3, 4, 5, 6, 7, 8, 9 or 10 types of isolated bacteria. In an embodiment, the bacterial composition comprises at least about 3, 4, 5, 6, 7, 8, 9, or 10 types of isolated bacteria and at least 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 of the isolated bacteria are capable of forming spores. In an embodiment, the bacterial composition comprise at least about 5 types of isolated bacteria and at least 2 of the isolated bacteria are capable of forming spores. In an embodiment, the bacterial composition comprises i) at least about 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30 or more types of isolated bacteria capable of forming spores, ii) at least about 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30 or more types of isolated bacteria not known to be capable of forming spores, or iii) any combination of i) and ii). In an embodiment, the first type, the second type and the optional third type are present in the composition in approximately equal concentrations or activity levels. In an embodiment, the first type, the second type and the optional third type are present in the composition in not substantially equal concentrations. In an embodiment, the first type is present in the composition in at least about 150% the concentration of the second type, or wherein the second type is present in the composition in at least about 150% the concentration of the first type. In an embodiment, the composition consists essentially of: i) between two and about twenty types of isolated bacteria, wherein at least two types of isolated bacteria are independently capable of spore formation; ii) between two and about twenty types of isolated bacteria, wherein at least two types of isolated bacteria not known to be capable of spore formation, or iii) any combination of i) and ii). In an embodiment, the first type of isolated bacterium and the second type of isolated bacterium are selected from Table 1. In an embodiment, the first type of isolated bacterium, the second type of isolated bacterium and the optional third type of isolated bacterium comprise an operational taxonomic unit (OTU) distinction. In an embodiment, the OTU distinction comprises 16S rDNA sequence similarity below about 95% identity. In an embodiment, the first type of isolated bacterium and the second type of isolated bacterium independently comprise bacteria that comprise 16S rDNA sequence at least 95% identical to 16S rDNA sequence present in a bacterium selected from Table 1. In an embodiment, a combination of the first type, second type and optional third type are: i) cytotoxic, ii) cytostatic, iii) capable of decreasing the growth of the pathogenic bacterium, iv) capable of inhibiting the growth of the pathogenic bacterium, v) capable of decreasing the colonization of the pathogenic bacterium, vi) capable of inhibiting the colonization of the pathogenic bacterium, or vii) any combination of i)-vi). In an embodiment, the combination is capable of inhibiting proliferation of the pathogenic bacteria present at a concentration at least equal to the concentration of the combination of the first type, the second type and the optional third type. In an embodiment, the combination is capable of inhibiting proliferation of the pathogenic bacterial present at a concentration at least about twice the concentration of the combination of the first type, the second type and the optional third type. In an embodiment, the combination is capable of inhibiting proliferation of the pathogenic bacterial present at a concentration at least about ten times the concentration of the combination of the first type, the second type and the optional third type. In an embodiment, the combination is capable of proliferating in the presence of the pathogenic bacteria. In an embodiment, the pathogenic bacterium is selected from the group consisting of *Yersinia, Vibrio, Treponema, Streptococcus, Staphylococcus, Shigella, Salmonella, Rickettsia, Orientia, Pseudomonas, Providencia, Proteus, Propionibacterium, Neisseria, Mycoplasma, Mycobacterium, Morganella, Listeria, Leptospira, Legionella, Klebsiella, Helicobacter, Haemophilus, Fusobacterium, Francisella, Escherichia, Ehrlichia, Enterococcus, Coxiella, Corynebacterium, Clostridium, Chlamydia, Chlamydophila, Campylobacter, Burkholderia, Brucella, Borrelia, Bordetella, Bifidobacterium, Bacillus*, multi-drug resistant bacteria, carbapenem-resistant Enterobacteriaceae (CRE), extended spectrum beta-lactam resistant Enterococci (ESBL), and vancomycin-resistant Enterococci (VRE). In an embodiment, the first type, the second type and the optional third type synergistically interact. In an embodiment, the first type, the second type and the optional third type synergistically interact to inhibit the pathogenic bacterium. In an embodiment, the composition comprises a combination of bacteria described in any row of Table 4a or Table 4b, or a combination of bacteria described in any row of Table 4a that has a ++++ or a +++ designation, or a combination of bacteria described in any row of Table 4a that has a 75^{th} percentile designation.

In another aspect, provided are compositions comprising an effective amount of a bacterial composition comprising at least a first type of isolated bacterium, a second type of isolated bacterium and an optional third type of isolated bacterium, wherein only one of the first type, the second type and the optional third type is capable of forming a spore, and wherein at least one of the first type, the second type and the optional third type is capable of decreasing the growth and/or colonization of at least one type of pathogenic bacteria.

In another aspect, provided are compositions comprising an effective amount of a bacterial composition comprising at least a first type of isolated bacterium, a second type of isolated bacterium and an optional third type of isolated bacterium, wherein the first type, the second type and the optional third type are not spores or known to be capable of forming a spore, and wherein at least one of the first type, the second type and the optional third type are capable of decreasing the growth and/or colonization of at least one type of pathogenic bacteria.

In an embodiment, at least one of the first type, second type and optional third type are capable of reducing the growth rate of at least one type of pathogenic bacteria. In an embodiment, at least one of the first type, second type and optional third type are cytotoxic to at least one type of pathogenic bacteria. In an embodiment, at least one of the first type, second type and optional third type are cytostatic to at least one type of pathogenic bacteria. In an embodiment, the first type, second type and optional third type are selected from Table 1. In an embodiment, the first type, second type and optional third type comprise different species. In an embodiment, the first type, second type and optional third type comprise different genera. In an embodiment, the first type, second type and optional third type comprise different families. In an embodiment, the first type, second type and optional third type comprise different orders. In an embodiment, the first type, second type and optional third type comprise a combination of bacteria described in any row of Table 4a or Table 4b, a combination of bacteria described in any row of Table 4a that has a ++++ or a +++ designation, or any or of Table 4a that has a 75^{th} percentile designation.

In another aspect, provided are compositions comprising an effective amount of a bacterial composition comprising at least a first type of isolated bacterium and a second type of isolated bacterium, wherein: i) the first type, second type and optional third type are independently capable of forming a spore; ii) only one of the first type, second type and optional third type is capable of forming a spore or iii) neither the first type nor the second type is capable of forming a spore, wherein the first type, second type and optional third type are not identical, wherein the first type, second type and optional third type are capable of functionally populating the gastrointestinal tract of a human subject to whom the composition is administered. In an embodiment, the first type, second type and optional third type comprise a combination of bacteria described in any row of Table 4a or Table 4b, a combination of bacteria described in any row of Table 4a that has a ++++ or a +++ designation, or any or of Table 4a that has a 75^{th} percentile designation. In an embodiment, the functional populating of the gastrointestinal tract comprises preventing a dysbiosis of the gastrointestinal tract. In an embodiment, the functional populating of the gastrointestinal tract comprises treating a dysbiosis of the gastrointestinal tract. In an embodiment, the functional populating of the gastrointestinal tract comprises reducing the severity of a dysbiosis of the gastrointestinal tract. In an embodiment, the functional populating of the gastrointestinal tract comprises reducing one or more symptoms of a dysbiosis of the gastrointestinal tract. In an embodiment, the functional populating of the gastrointestinal tract comprises preventing growth and/or colonization of the gastrointestinal tract by a pathogenic bacterium. In an embodiment, the functional populating of the gastrointestinal tract comprises reducing growth and/or colonization of the gastrointestinal tract by a pathogenic bacterium. In an embodiment, the functional populating of the gastrointestinal tract comprises reducing the number of one or more types of pathogenic bacteria in the gastrointestinal tract. In an embodiment, the functional populating of the gastrointestinal tract comprises increasing the number of one or more non-pathogenic bacteria in the gastrointestinal tract. In an embodiment, the bacterial composition comprises 0, 1, 2, 3 or greater than 3 types of isolated bacteria capable of forming spores. In an embodiment, the bacterial composition comprises at least about 5 types of isolated bacteria capable of forming spores. In an embodiment, the bacterial composition comprises at least about 7 types of isolated bacteria capable of forming spores. In an embodiment, the first type, second type and optional third type are present in the composition in not substantially equal concentrations. In an embodiment, the first type, second type and optional third type are present in the composition in approximately equal concentrations. In an embodiment, the first type is present in the composition in at least about 150% the concentration of the second type. In an embodiment, the second type is present in the composition in at least about 150% the concentration of the first type. In an embodiment, the composition consists essentially of between two and about ten types of isolated bacteria, wherein at least one type of isolated bacteria are independently capable of spore formation. In an embodiment, the first type of isolated bacterium and the second type of isolated bacterium are selected from Table 1. In an embodiment, the first type of isolated bacterium, the second type of isolated bacterium and the optional third type of isolated bacterium comprise an operational taxonomic unit (OTU) distinction. In an embodiment, the OTU distinction comprises 16S rDNA sequence similarity below about 95% identity. In an embodiment, the first type of isolated bacterium and the second type of isolated bacterium independently comprise bacteria that comprise 16S rDNA sequence at least 95% identical to 16S rDNA sequence present in a bacterium selected from Table 1. In an embodiment, a combination of the first type, second type and optional third type are cytotoxic or cytostatic to the pathogenic bacterium. In an embodiment, the combination is capable of inhibiting proliferation of the pathogenic bacteria present at a concentration at least equal to the concentration of the combination of the first type, second type and optional third type. In an embodiment, the combination is capable of inhibiting proliferation of the pathogenic bacterial present at a concentration at least about twice the concentration of the combination of the first type, second type and optional third type. In an embodiment, the combination is capable of inhibiting proliferation of the pathogenic bacteria present at a concentration at least about ten times the concentration of the combination of the first type, second type and optional third type. In an embodiment, the pathogenic bacterium is selected from the group consisting of *Yersinia, Vibrio, Treponema, Streptococcus, Staphylococcus, Shigella, Salmonella, Rickettsia, Orientia, Pseudomonas, Providencia, Proteus, Propionibacterium, Neisseria, Mycoplasma, Mycobacterium, Morganella, Listeria, Leptospira, Legionella, Klebsiella, Helicobacter, Haemophilus, Fusobacterium, Francisella, Escherichia, Ehrlichia, Enterococcus, Coxiella, Corynebacterium, Clostridium, Chlamydia, Chlamydophila, Campylobacter, Burkholderia, Brucella, Borrelia, Bordetella, Bifidobacterium, Bacillus*, multi-drug resistant bacteria, Carbapenem-resistant Enterobacteriaceae (CRE), extended spectrum beta-lactam resistant Enterococci (ESBL), and vancomycin-resistant Enterococci (VRE). In an embodiment, the first type, second type and optional third type synergistically interact to be cytotoxic to the pathogenic bacterium. In an embodiment, wherein the first type, second type and optional third type synergistically interact to be cytostatic to the pathogenic bacterium.

In another aspect, provided are single dose units comprising the compositions of the present invention. In an embodiment, the dose unit comprises at least 1×10^4, 1×10^5, 1×10^6, 1×10^7, 1×10^8, 1×10^9, 1×10^{10}, 1×10^{11} or greater than 1×10^{11} colony forming units (CFUs) of either spores or vegetative bacterial cells. In an embodiment, the dose unit comprises a pharmaceutically acceptable excipient, an enteric coating or a combination thereof. In an embodiment, the dose unit further comprises a drug selected from corticosteroids, mesalazine, mesalamine, sulfasalazine, sulfasalazine derivatives, immunosuppressive drugs, cyclosporin A, mercaptopurine, azathiopurine, prednisone, methotrexate, antihistamines, glucocorticoids, epinephrine, theophylline, cromolyn sodium, anti-leukotrienes, anti-cholinergic drugs for rhinitis, anti-cholinergic decongestants, mast-cell stabilizers, monoclonal anti-IgE antibodies, vaccines, and combinations thereof, wherein the drug is present in an amount effective to modulate the amount and/or activity of at least one pathogen. In an embodiment, the dose unit is formulated for oral administration, rectal administration, or the combination of oral and rectal administration, or is formulated for topical, nasal or inhalation administration. In an embodiment, the dose unit comprises a of bacteria described in any row of Table 4a or Table 4b, a combination of bacteria described in any row of Table 4a that has a ++++ or a +++ designation, or any or of Table 4a that has a 75^{th} percentile designation.

In another aspect, provided are kits comprising in one or more containers: a first purified population of a first type of bacterial spores substantially free of viable vegetal bacterial cells; a second purified population of a second type of bacterial spores substantially free of viable vegetal bacterial cells; and optionally a third purified population of a third type of bacterial spores substantially free of viable vegetal bacterial cells, wherein the first type, second type and optional third type of bacterial spores are not identical, and wherein the first type, second type and optional third type of bacterial spores, when co-localized in a target region of a gastrointestinal tract of a human subject in need thereof, are capable of functionally populating the gastrointestinal tract. In an embodiment, the first purified population and the second purified population are present in a single container. In an embodiment, the first purified population, the second purified population and the optional third purified population present in two or optionally three containers. In an embodiment, the first purified population and the second purified population are lyophilized or substantially dehydrated. In an embodiment, the kit further comprises in one or more containers an effective amount of an anti-bacterial agent, an effective amount of an anti-viral agent, an effective amount of an anti-fungal agent, an effective amount of an anti-parasitic agent, or a combination thereof in one or more containers. In an embodiment, the kit further comprises a pharmaceutically acceptable excipient or diluent. In an embodiment, the first purified population, the second purified population and the optional third purified population comprise a combination of bacteria described in any row of Table 4a or Table 4b, a combination of bacteria described in any row of Table 4a that has a ++++ or a +++ designation, or any or of Table 4a that has a 75^{th} percentile designation.

Also provided are pharmaceutical formulations comprising an effective amount of the compositions of the invention, and further comprising an effective amount of an anti-bacterial agent, an effective amount of an anti-fungal agent, an effective amount of an anti-viral agent, an effective amount of an anti-parasitic agent.

Also provided are comestible products comprising a first purified population of a first type of bacterial spores, a second purified population of a second type of bacterial spores and optionally a third purified population of a third type of bacterial spores, wherein the first type, second type and optional third type of bacterial spores are not identical, wherein the comestible product is substantially free of viable vegetal bacterial cells, and wherein the first type, second type and optional third type of bacterial spores, when administered to a human subject in need thereof, are capable of functionally populating the gastrointestinal tract of the human subject. In an embodiment, the comestible product comprises a food or food additive, a beverage or beverage additive, or a medical food. In an embodiment, the comestible product comprises at least 1×10^4, 1×10^5, 1×10^6, 1×10^7, 1×10^8, 1×10^9, 1×10^{10}, 1×10^{11} or greater than 1×10^{11} colony forming units (CFUs) of viable spores. In an embodiment, the comestible product comprises a first type of bacterial spores and a second type of bacterial spores selected from Table 1, or where the first type of bacterial spores and the second type of bacterial spores independently comprise bacterial spores that comprise 16S rDNA sequence at least 95% identical to 16S rDNA sequence present in a bacterium selected from Table 1. In an embodiment, the first purified population, the second purified population and the optional third purified population comprise a combination of bacteria described in any row of Table 4a or Table 4b, a combination of bacteria described in any row of Table 4a that has a ++++ or a +++ designation, or any row of Table 4a that has a 75^{th} percentile designation.

Also provided are methods comprising administering to a human subject in need thereof an effective amount of a bacterial composition comprising at least a first type of isolated bacterium, a second type of isolated bacterium and optionally a third type of isolated bacterium, wherein: the first type, second type and optional third type are independently capable of forming a spore; only one of the first type, second type and optional third type is capable of forming a spore; or none of the first type, the second typeand optional third type is capable of forming a spore, wherein the first type, second type and optional third type are not identical, and wherein at least one of the first type, second type and optional third type exert an inhibitory-effect on a pathogenic bacterium present in the gastrointestinal tract of the human subject, such that the number of pathogenic bacteria present in the gastrointestinal tract is not detectably increased or is detectably decreased over a period of time. In an embodiment, the composition comprise a combination of bacteria described in any row of Table 4a or Table 4b, a combination of bacteria described in any row of Table 4a that has a ++++ or a +++ designation, or any or of Table 4a that has a 75^{th} percentile designation. In an embodiment, the human subject is diagnosed as having a dysbiosis of the gastrointestinal tract. In an embodiment, the human subject is diagnosed as infected with a pathogenic bacterium selected from the group consisting of *Yersinia, Vibrio, Treponema, Streptococcus, Staphylococcus, Shigella, Salmonella, Rickettsia, Orientia, Pseudomonas, Providencia, Proteus, Propionibacterium, Neisseria, Mycoplasma, Mycobacterium, Morganella, Listeria, Leptospira, Legionella, Klebsiella, Helicobacter, Haemophilus, Fusobacterium, Francisella, Escherichia, Ehrlichia, Enterococcus, Coxiella, Corynebacterium, Clostridium, Chlamydia, Chlamydophila, Campylo-*

bacter, Burkholderia, Brucella, Borrelia, Bordetella, Bifidobacterium, Bacillus, multi-drug resistant bacteria, Carbapenem-resistant Enterobacteriaceae (CRE), extended spectrum beta-lactam resistant Enterococci (ESBL), and vancomycin-resistant Enterococci (VRE). In an embodiment, the bacterial composition is administered simultaneously with i) an antibiotic, ii) a prebiotic, or iii) a combination of i) and ii). In an embodiment, the bacterial composition is administered prior to administration of i) an antibiotic, ii) a prebiotic, or iii) a combination of i) and ii). In an embodiment, the bacterial composition is administered subsequent to administration of i) an antibiotic, ii) a prebiotic, or iii) a combination of i) and ii). In an embodiment, the number of pathogenic bacterium present in or excreted from the gastrointestinal tract of the human subject is detectably reduced within one month, within two weeks, or within one week of administration of the bacterial composition. In an embodiment, the number of pathogenic bacterium present in or excreted from the gastrointestinal tract of the human subject is detectably reduced within three days, two days or one day of administration of the bacterial composition. In an embodiment, the human subject is detectably free of the pathogenic bacterium within one month, two weeks, one week, three days or one day of administration of the bacterial composition. In an embodiment, the bacterial composition comprises at least about 3, 4, 5, 6, 7, 8, 9, or 10 types of isolated bacteria. In an embodiment, the bacterial composition comprises at least about 3, 4, 5, 6, 7, 8, 9, or 10 types of isolated bacteria and at least 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 of the isolated bacteria are capable of forming spores. In an embodiment, the bacterial composition comprises at least about 5 types of isolated bacteria and at least 2 of the isolated bacteria are capable of forming spores. In an embodiment, the bacterial composition comprises: i) at least about 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30 or more types of isolated bacteria capable of forming spores, ii) at least about 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30 or more types of isolated bacteria not known to be capable of forming spores, or iii) any combination of i) and ii). In an embodiment, the bacterial composition comprises at least about 5 types of isolated bacteria and at least 1 of the isolated bacteria are capable of forming spores. In an embodiment, the bacterial composition comprises at least about 5 types of isolated bacteria and at least 1 of the isolated bacteria is not capable of forming spores. In an embodiment, the bacterial composition comprises at least about 3, 4, 5, 6, 7, 8, 9 or 10 types of isolated bacteria, wherein i) at least 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 types of isolated bacteria are capable of forming spores, ii) at least 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 types of isolated bacteria are not capable of forming spores, or iii) any combination of i) and ii). In an embodiment, the first type, second type and optional third type are present in the composition in approximately equal concentrations. In an embodiment, the first type, second type and optional third type are present in the composition in not substantially equal concentrations. In an embodiment, the first type is present in the composition in at least about 150% the concentration of the second type, or wherein the second type is present in the composition in at least about 150% the concentration of the first type. In an embodiment, the composition consists essentially of between two and about ten types of isolated bacteria, wherein at least two types of isolated bacteria are independently capable of spore formation. In an embodiment, the composition consists essentially of between two and about ten types of isolated bacteria, wherein at least two types of isolated bacteria are not capable of spore formation. In an embodiment, the first type of isolated bacterium and the second type of isolated bacterium are selected from Table 1. In an embodiment, the first type of isolated bacterium, the second type of isolated bacterium and the optional third type of isolated bacterium comprise an operational taxonomic unit (OTU) distinction. In an embodiment, the OTU distinction comprises 16S rDNA sequence similarity below about 95% identity. In an embodiment, the first type of isolated bacterium and the second type of isolated bacterium independently comprise bacteria that comprise 16S rDNA sequence at least 95% identical to 16S rDNA sequence present in a bacterium selected from Table 1. In an embodiment, a combination of the first type, second type and optional third type are cytotoxic or cytostatic to the pathogenic bacterium. In an embodiment, the combination is capable of inhibiting proliferation of the pathogenic bacterial present at a concentration at least equal to the concentration of the combination of the first type, second type and optional third type. In an embodiment, the combination is capable of inhibiting proliferation of the pathogenic bacterial present at a concentration at least about twice the concentration of the combination of the first type, second type and optional third type. In an embodiment, the combination is capable of inhibiting proliferation of the pathogenic bacterial present at a concentration at least about ten times the concentration of the combination of the first type, second type and optional third type. In an embodiment, the pathogenic bacterium is selected from the group consisting of *Yersinia, Vibrio, Treponema, Streptococcus, Staphylococcus, Shigella, Salmonella, Rickettsia, Orientia, Pseudomonas, Providencia, Proteus, Propionibacterium, Neisseria, Mycoplasma, Mycobacterium, Morganella, Listeria, Leptospira, Legionella, Klebsiella, Helicobacter, Haemophilus, Fusobacterium, Francisella, Escherichia, Ehrlichia, Enterococcus, Coxiella, Corynebacterium, Clostridium, Chlamydia, Chlamydophila, Campylobacter, Burkholderia, Brucella, Borrelia, Bordetella, Bifidobacterium, Bacillus*, multi-drug resistant bacteria, Carbapenem-resistant Enterobacteriaceae (CRE), extended spectrum beta-lactam resistant Enterococci (ESBL), and vancomycin-resistant Enterococci (VRE). In an embodiment, the first type, second type and optional third type synergistically interact to be cytotoxic to the pathogenic bacterium. In an embodiment, the first type, second type and optional third type synergistically interact to be cytostatic to the pathogenic bacterium.

Also provided are methods of functionally populating the gastrointestinal tract of a human subject, comprising administering to the subject an effective amount of a bacterial composition comprising at least a first type of isolated bacterium, a second type of isolated bacterium, and optionally a third type of isolated bacterium wherein i) the first type, second type and optional third type are independently capable of forming a spore; ii) only one of the first type, second type and optional third type is capable of forming a spore or iii) none of the first type, the second type and the optional third type is capable of forming a spore, wherein the first type, second type and optional third type are not identical, under conditions such that the first type, second type and optional third type functionally populate the gastrointestinal tract of the human subject. In an embodiment, the composition comprises a combination of bacteria described in any row of Table 4a or Table 4b, a combination of bacteria described in any row of Table 4a that has a ++++ or a +++ designation, or any row of Table 4a that has a 75^{th} percentile designation. In an embodiment, the bacterial composition is orally administered, rectally administered, or the combination of orally and rectally administered. In an embodiment, the bacterial composition is topically or nasally administered or inhaled. In an embodiment, the first type of isolated bacteria and the second type of isolated bacteria are selected from Table 1. In an embodiment, the bacterial composition consists essentially of spores, wherein the spores comprise spores of the first type of isolated bacteria, spores of the second type of isolated bacteria and spores of the optional third type of isolated bacteria. In an embodiment, the first type of isolated bacteria and the second type of isolated bacteria independently comprise bacterial spores that comprise 16S rDNA sequence at least 95% identical to 16S rDNA sequence present in a bacterium selected from Table 1. In an embodiment, the functional populating of the gastrointestinal tract comprises preventing a dysbiosis of the gastrointestinal tract. In an embodiment, the functional populating of the gastrointestinal tract comprises treating a dysbiosis of the gastrointestinal tract. In an embodiment, the functional populating of the gastrointestinal tract comprises reducing the severity of a dysbiosis of the gastrointestinal tract. In an embodiment, the functional populating of the gastrointestinal tract comprises reducing one or more symptoms of a dysbiosis of the gastrointestinal tract. In an embodiment, the functional populating of the gastrointestinal tract comprises preventing colonization of the gastrointestinal tract by a pathogenic bacterium. In an embodiment, the functional populating of the gastrointestinal tract comprises reducing colonization of the gastrointestinal tract and/or growth by a pathogenic bacterium. In an embodiment, wherein the functional populating of the gastrointestinal tract comprises reducing the number of one or more types of pathogenic bacteria in the gastrointestinal tract. In an embodiment, the functional populating of the gastrointestinal tract comprises increasing the number of one or more non-pathogenic bacteria in the gastrointestinal tract. In an embodiment, the bacterial composition comprises at least about 3, 5, 7 or 9 types of isolated bacteria capable of forming spores. In an embodiment, the bacterial composition comprises at least about 5 types of isolated bacteria and at least 20% of the isolated bacteria are capable of forming spores. In an embodiment, the bacterial composition comprises at least about 5 types of isolated bacteria and at least 2 of the isolated bacteria are capable of forming spores. In an embodiment, the bacterial composition comprises at least about 3, 4, 5, 6, 7, 8, 9 or 10 types of isolated bacteria, wherein i) at least 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 types of isolated bacteria are capable of forming spores, ii) at least 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 types of isolated bacteria are not capable of forming spores, or iii) any combination of i) and ii). In an embodiment, the first type, second type and optional third type are present in the composition in approximately equal concentrations. In an embodiment, the first type, second type and optional third type are present in the composition in not substantially equal concentrations. In an embodiment, the first type is present in the composition in at least about 150% the concentration of the second type, or wherein the second type is present in the composition in at least about 150% the concentration of the first type. In an embodiment, the composition consists essentially of between two and about ten types of isolated bacteria, wherein i) at least one type of isolated bacteria is capable of spore formation, ii) at least one type of isolated bacteria is not capable of spore formation, or iii) a combination of i) and ii). In an embodiment, a combination of the first type, second type and optional third type are inhibitory to the pathogenic bacterium. In an embodiment, the combination reduces the growth rate of the pathogenic bacterium. In an embodiment, the combination is cytostatic or cytotoxic to the pathogenic bacterium. In an embodiment, the combination is capable of inhibiting growth of the pathogenic bacterial present at a concentration at least equal to the concentration of the combination of the first type, second type and optional third type. In an embodiment, the combination is capable of inhibiting growth of the pathogenic bacterial present at a concentration at least about twice the concentration of the combination of the first type, second type and optional third type. In an embodiment, the combination is capable of inhibiting proliferation of the pathogenic bacterial present at a concentration at least about ten times the concentration of the combination of the first type, second type and optional third type. In an embodiment, the pathogenic bacterium is selected from the group consisting of *Yersinia, Vibrio, Treponema, Streptococcus, Staphylococcus, Shigella, Salmonella, Rickettsia, Orientia, Pseudomonas, Providencia, Proteus, Propionibacterium, Neisseria, Mycoplasma, Mycobacterium, Morganella, Listeria, Leptospira, Legionella, Klebsiella, Helicobacter, Haemophilus, Fusobacterium, Francisella, Escherichia, Ehrlichia, Enterococcus, Coxiella, Corynebacterium, Clostridium, Chlamydia, Chlamydophila, Campylobacter, Burkholderia, Brucella, Borrelia, Bordetella, Bifidobacterium, Bacillus*, multi-drug resistant bacteria, Carbapenem-resistant Enterobacteriaceae (CRE), extended spectrum beta-lactam resistant Enterococci (ESBL), and vancomycin-resistant Enterococci (VRE). In an embodiment, the first type, second type and optional third type synergistically interact to reduce or inhibit the growth of the pathogenic bacterium. In an embodiment, the first type, second type and optional third type synergistically interact to reduce or inhibit the colonization of the pathogenic bacterium. In an embodiment, the method comprises administering to the human subject a single dose unit comprising at least 1×10^4, 1×10^5, 1×10^6, 1×10^7, 1×10^8, 1×10^9, 1×10^{10}, 1×10^{11} or greater than 1×10^{11} colony forming units (CFUs) of viable bacteria. In an embodiment, the dose unit comprises a bacterial population substantially in the form of spores. In an embodiment, the dose unit comprises a pharmaceutically acceptable excipient and/or an enteric coating. In an embodiment, the unit dose is formulated for oral administration, rectal administration, or the combination of oral and rectal administration. In an embodiment, the unit dose is formulated for topical or nasal administration or for inhalation.

In another aspect, provided are methods of reducing the number of pathogenic bacteria present in the gastrointestinal tract of a human subject, comprising administering to the subject an effective amount of a pharmaceutical formulation comprising an effective amount of the composition of the present disclosure, and further comprising an effective amount of an anti-microbial agent, under conditions such that the number of pathogenic bacteria present in the gastrointestinal tract of the human subject is reduced within about one month of administration of the pharmaceutical formulation. In an embodiment, the number of pathogenic bacteria present in the gastrointestinal tract of the human subject is reduced within about two weeks of administration of the pharmaceutical formulation. In an embodiment, the number of pathogenic bacteria present in the gastrointestinal tract of the human subject is reduced within about one week of administration of the pharmaceutical formulation. In an embodiment, the number of pathogenic bacteria present in the gastrointestinal tract of the human subject is reduced within about three days of administration of the pharmaceutical formulation. In an embodiment, the number of pathogenic bacteria present in the gastrointestinal tract of the human subject is reduced within about one day of administration of the pharmaceutical formulation. In an embodiment, the anti-microbial agent comprises anti-bacterial agent. In an embodiment, the anti-microbial agent comprises anti-fungal agent. In an embodiment, the anti-microbial agent comprises anti-viral agent. In an embodiment, the anti-microbial agent comprises anti-parasitic agent.

In another aspect, provided are methods of preparing a comestible product, comprising combining with a comestible carrier a first purified population comprising at least a first type of isolated bacterium, a second purified population comprising at least a second type of isolated bacterium and optionally a third purified population comprising at least a third type of isolated bacterium, wherein: i) the first type, second type and optional third type are independently capable of forming a spore; ii) only one of the first type, second type and optional third type is capable of forming a spore or iii) none of the first type, the second type and the optional third type is capable of forming a spore, wherein the first type, second type and optional third type of bacteria are not identical, wherein the comestible product is substantially free of non-comestible materials. In an embodiment, at least one of the first purified population, the second purified population and the optional third purified population consist essentially of viable spores. In an embodiment, the first purified population, the second purified population and the optional third purified population consist essentially of viable spores. In an embodiment, the comestible product is substantially free of viable vegetal bacterial cells. In an embodiment, the viable spores, when the comestible product is consumed by a human subject in need thereof, are capable of functionally populating the gastrointestinal tract of the human subject. In an embodiment, the comestible product comprises a food or food additive. In an embodiment, the comestible product comprises a beverage or beverage additive. In an embodiment, the comestible product comprises a medical food. In an embodiment, the comestible product comprises at least 1×10^4, 1×10^5, 1×10^6, 1×10^7, 1×10^8, 1×10^9, 1×10^{10}, 1×10^{11} or greater than 1×10^{11} colony forming units (CFUs) of viable spores. In an embodiment, the first purified population, the second purified population and the optional third purified population comprise a combination of bacteria described in any row of Table 4a or Table 4b, or any row of Table 4a that has a ++++ designation or a +++ designation, or any row of Table 4a that has a 75^{th} percentile designation. In an embodiment, spores are of a bacterium selected from Table 1. In an embodiment, the first purified population and the second purified population independently comprise bacterial spores that comprise 16S rDNA sequence at least 95% identical to 16S rDNA sequence present in a bacterium selected from Table 1.

Also provided are methods of reducing the abundance of a pathogen in the gastrointestinal tract of a subject comprising administering a composition of in a therapeutically effective amount and allowing the bacterial composition to compete with the pathogen in the gastrointestinal tract of a subject.

Further provided are methods of treating diarrhea comprising administering a bacterial composition in a therapeutically effective amount and allowing the bacterial composition to reduce the diarrheal effect of a pathogen in the gastrointestinal tract of a subject. In an embodiment, the pathogen is *Yersinia, Vibrio, Treponema, Streptococcus, Staphylococcus, Shigella, Salmonella, Rickettsia, Orientia, Pseudomonas, Providencia, Proteus, Propionibacterium, Neisseria, Mycoplasma, Mycobacterium, Morganella, Listeria, Leptospira, Legionella, Klebsiella, Helicobacter, Haemophilus, Fusobacterium, Francisella, Escherichia, Ehrlichia, Enterococcus, Coxiella, Corynebacterium, Clostridium, Chlamydia, Chlamydophila, Campylobacter, Burkholderia, Brucella, Borrelia, Bordetella, Bifidobacterium, Bacillus*, multi-drug resistant bacteria, Carbapenem-resistant Enterobacteriaceae (CRE), extended spectrum beta-lactam resistant Enterococci (ESBL), and vancomycin-resistant Enterococci (VRE). In an embodiment, the pathogen is *Clostridium difficile, Salmonella* spp., pathogenic *Escherichia coli*, or vancomycin-resistant *Enterococcus* spp. In an embodiment, the pathogen is *Clostridium difficile*. In an embodiment, the composition is administered orally. In an embodiment the composition comprises a combination of bacteria described in any row of Table 4a, or Table 4b, or any row of Table 4a that has a ++++ designation or a +++ designation, or any row of Table 4a that has a 75^{th} percentile designation.

In some aspects, the invention relates to a composition comprising a network ecology selected from Table 10. In some embodiments, the network ecology comprises network clades provided in Table 10. In other embodiments, the network ecology comprises network OTUs provided in Table 10. In some cases the composition comprises *Blautia producta, Clostridium disporicum, Clostridium innocuum, Clostridium mayombei, Clostridium orbiscindens, Clostridium symbiosum*, and *Lachnospiraceae bacterium* 5_1_57FAA. In some embodiments, the composition the composition is effective for treating at least one sign or symptom of a dysbiosis, for example, the is effective for reducing at least one sign or symptom of infection or dysbiosis associated with *C. difficile, Klebsiella pneumonii, Morganella morganii*, or vancomycin-resistant Enterococci (VRE).

In another aspect, the invention relates to a composition comprising a bacterial heterotrimer selected from a heterotrimer identified in Table 4a, Table 4b, or Table 12, such that the heterotrimer can e.g., inhibit growth of a pathobiont in a CivSim assay.

In some aspects, the invention relates to a composition comprising a bacterial heterotrimer selected from a heterotrimer identified in Table 14, Table 15, Table 16, Table 17, Table 17, Table 18, Table 19, Table 20, or Table 21, such that the organisms of the heterotrimer can augment and/or engraft in a human gastrointestinal tract. In some embodiments, the engraftment and/or augmentation can occur after administration of the composition to a human having a dysbiosis. In some embodiments, the dysbiosis is associated with the presence of *C. difficile* in the gastrointestinal tract of the human.

Additional objects and advantages will be set forth in part in the description which follows, and in part will be obvious from the description, or may be learned by practice of the embodiments. The objects and advantages will be realized and attained by means of the elements and combinations particularly pointed out in the appended claims.

It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the claims.

The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate several embodiments and together with the description, serve to further explain the embodiments.

BRIEF DESCRIPTION OF TABLES

Table 1 is a list of Operational Taxonomic Units (OTU) with taxonomic assignments made to genus, species, and phylogenetic clade. Clade membership of bacterial OTUs is based on 16S sequence data. Clades are defined based on the topology of a phylogenetic tree that is constructed from full-length 16S sequences using maximum likelihood methods familiar to individuals with ordinary skill in the art of phylogenetics. Clades are constructed to ensure that all OTUs in a given clade are: (i) within a specified number of bootstrap supported nodes from one another, and (ii) within 5% genetic similarity. OTUs that are within the same clade can be distinguished as genetically and phylogenetically distinct from OTUs in a different clade based on 16S-V4 sequence data, while OTUs falling within the same clade are closely related. OTUs falling within the same clade are evolutionarily closely related and may or may not be distinguishable from one another using 16S-V4 sequence data. Members of the same clade, due to their evolutionary relatedness, play similar functional roles in a microbial ecology such as that found in the human gut. Compositions substituting one species with another from the same clade are likely to have conserved ecological function and therefore are useful in the present invention. All OTUs are denoted as to their putative capacity to form spores and whether they are a pathogen or pathobiont (see Definitions for description of "Pathobiont"). NIAID (National Institute of Allergy and Infectious Disease) Priority Pathogens are denoted as 'Category-A', 'Category-B', or 'Category-C', and opportunistic pathogens are denoted as 'OP'. OTUs that are not pathogenic or for which their ability to exist as a pathogen is unknown are denoted as 'N'. The 'SEQ ID Number' denotes the identifier of the OTU in the Sequence Listing File and 'Public DB Accession' denotes the identifier of the OTU in a public sequence repository.

Table 2 provides phylogenetic clades and their members determined using 16S full-length and V4 sequencing.

Table 3 is a list of human diseases, disorders and conditions for which the provided bacterial compositions are useful.

Table 4a. Provides representative combinations of the present invention tested in vitro.

Table 4b. Provides representative combinations of the present invention tested in vitro

Table 5 provides data from testing of representative ternary OTU combinations of the present invention in a CivSim assay and in vivo.

Table 6 provides data on the ability of a 15 member bacterial composition to inhibit VRE in vitro.

Table 7 provides data on the ability of a 15 member bacterial composition to inhibit *K. pneumoniae* in vitro.

Table 8 provides data on the ability of a 15 member bacterial composition to inhibit *M. morganii* in vitro.

Table 9 provides data demonstrating the efficacy of combinations of the present invention against *C. difficile* infection in a preventive murine model.

Table 10. Provides exemplary combinations of the present invention that were tested against *C. difficile* infection in a preventive murine model.

Table 11. Provides bacterial OTUs associated with a bacterial composition used to treat patients with *C. difficile* associated diarrheal disease, and to OTUs comprising the OTUs undergo engraftment and ecological augmentation to establish a more diverse microbial ecology in patients post-treatment. OTUs that comprise an augmented ecology are not present in the patient prior to treatment and/or exist at extremely low frequencies such that they do not comprise a significant fraction of the total microbial carriage and are not detectable by genomic and/or microbiological assay methods. OTUs that are members of the engrafting and augmented ecologies were identified by characterizing the OTUs that increase in their relative abundance post treatment and that respectively are: (i) present in the ethanol-treated spore preparation and absent in the patient pretreatment, or (ii) absent in the ethanol-treated spore preparation, but increase in their relative abundance through time post treatment with the preparation due to the formation of favorable growth conditions by the treatment. Notably, augmenting OTUs can grow from low frequency reservoirs in the subject, or be introduced from exogenous sources such as diet. OTUs that comprise a "core" composition in the treatment bacterial composition are denoted.

Table 12 provides bacterial compositions that exhibited inhibition against *C. difficile* as measured by a mean log inhibition greater than the 99% confidence interval (C.I.) of the null hypothesis (see Example 6, ++++) and that are identified in at least one spore ecology treatment or in a human subject microbiome after treatment with a composition.

Table 13 provides exemplary of 4-mer to 10-mer bacterial compositions that were comprised in a bacterial therapy administered to subjects with *C. difficile*-associated diarrheal disease.

Table 14 provides exemplary ternary OTUs that either engrafted or augmented in at least one patient (of 29 that responded to treatment) after treatment with a spore ecology composition. Each ternary combination was either in all doses or the organisms of the ternary combination were present together in all subjects at some post-treatment time.

Table 15 provides exemplary OTUs that engrafted in at least one subject. The ternary combinations were found in 95% of the doses of administered spore ecology compositions.

Table 16 provides exemplary OTUs that augmented in at least one patient post treatment with a spore ecology composition. The ternary combinations were found together in at least 75% of the subjects at some post-treatment timepoint.

Table 17 provides exemplary OTU combinations that were present in at least 75% of the doses of administered spore ecology compositions. All administered doses containing the listed ternary combinations had the OTU *Clostridiales* sp. SM4/1 as either augmenting or engrafting in the subjects given doses containing the ternary composition.

Table 18 provides exemplary ternary OTU combinations that were present in at least 75% of the doses of administered spore ecology compositions. All administered doses containing the listed ternary combinations had the OTU *Clostridiales* sp. SSC/2 as either augmenting or engrafting in the subjects given a composition containing the ternary combination.

Table 19 provides exemplary ternary combinations of OTUs that were present in at least 75% of the doses of administered spore ecology compositions. All administered doses containing the listed ternary combinations had the OTU *Clostridium* sp. NML 04A032 as either augmenting or engrafting in the subjects given a composition containing the ternary combination.

Table 20 provides exemplary ternary combinations of OTUs that were present in at least 75% of the doses of administered spore ecology compositions. All administered doses containing the listed ternary combinations had the OTUs *Clostridium* sp. NML 04A032, *Ruminococcus lactaris*, and *Ruminococcus torques* as either augmenting or engrafting in the subjects given a composition containing the ternary combination.

Table 21 provides exemplary ternary combinations of OTUs that are present in at least 75% of the doses of administered spore ecology compositions. All administered doses containing the listed ternary combinations had the OTUs *Eubacterium* rectale, *Faecalibacterium prausnitzii, Oscillibacter* sp. G2, *Ruminococcus lactaris*, and *Ruminococcus torques* as either augmenting or engrafting in the subjects given a composition containing the ternary combination.

Table 22 provides alternate names of organisms found in OTUs of the embodiments of the present invention.

BRIEF DESCRIPTION OF FIGURES

FIG. **1** shows an exemplary phylogenetic tree and the relationship of OTUs and Clades. A, B, C, D, and E represent OTUs, also known as leaves in the tree. Clade 1 comprises OTUs A and B, Clade 2 comprises OTUs C, D and E, and Clade 3 is a subset of Clade 2 comprising OTUs D and E. Nodes in a tree that define clades in the tree can be either statistically supported or not statistically supported. OTUs within a clade are more similar to each other than to OTUs in another clade; the robustness the clade assignment is denoted by the degree of statistical support for a node upstream of the OTUs in the clade.

FIG. **2** provides a schematic of 16S rDNA gene and denotes the coordinates of hypervariable regions 1-9 (V1-V9), according to an embodiment of the invention. Coordinates of V1-V9 are 69-99, 137-242, 433-497, 576-682, 822-879, 986-1043, 1117-1173, 1243-1294, and 1435-1465 respectively, based on numbering using *E. coli* system of nomenclature defined by Brosius et al., Complete nucleotide sequence of a 16S ribosomal RNA gene (16S rRNA) from *Escherichia coli*, PNAS 75 (10): 4801-4805 (1978).

FIG. **3** highlights in bold the nucleotide sequences for each hypervariable region in the exemplary reference *E. coli* 16S sequence (SEQ ID NO: 2047) described by Brosius et al., supra.

FIG. **4** provides representative combinations of the present invention tested in vitro and their respective inhibition of pathogen growth.

FIG. **5** shows an in vivo hamster *Clostridium difficile* relapse prevention model to validate efficacy of network ecology bacterial composition, according to an embodiment of the invention.

FIG. **6** shows the increase in the total microbial diversity (measured using the Chao-1 diversity index) in the gut of human subjects with recurrent *Clostridium difficile* associated disease pretreatment and post-treatment with a microbial spore ecology.

FIG. **7** shows the compositional change in the microbiome (measured using the Bray-Curtis PCoA metric) in the gut of human subjects with recurrent *Clostridium difficile* associated disease pretreatment and post-treatment with a microbial spore ecology.

DEFINITIONS

As used herein, the term "antioxidant" refers to, without limitation, any one or more of various substances such as beta-carotene (a vitamin A precursor), vitamin C, vitamin E, and selenium that inhibit oxidation or reactions promoted by Reactive Oxygen Species ("ROS") and other radical and non-radical species. Additionally, antioxidants are molecules capable of slowing or preventing the oxidation of other molecules. Non-limiting examples of antioxidants include astaxanthin, carotenoids, coenzyme Q10 ("CoQ10"), flavonoids, glutathione, Goji (wolfberry), hesperidin, lactowolfberry, lignan, lutein, lycopene, polyphenols, selenium, vitamin A, vitamin C, vitamin E, zeaxanthin, or combinations thereof.

"Backbone Network Ecology" or simply "Backbone Network" or "Backbone" are compositions of microbes that form a foundational composition that can be built upon or subtracted from to optimize a Network Ecology or Functional Network Ecology to have specific biological characteristics or to comprise desired functional properties, respectively. Microbiome therapeutics can be comprised of these "Backbone Networks Ecologies" in their entirety, or the "Backbone Networks" can be modified by the addition or subtraction of "R-Groups" to give the network ecology desired characteristics and properties. "R-Groups" as used herein, can be defined in multiple terms including, but not limited to: individual OTUs, individual or multiple OTUs derived from a specific phylogenetic clade or a desired phenotype such as the ability to form spores, or functional bacterial compositions that comprise. "Backbone Networks" can comprise a computationally derived Network Ecology in its entirety or can be subsets of the computed network that represent key nodes in the network that contributed to efficacy such as but not limited to a composition of Keystone OTUs. The number of organisms in the human gastrointestinal tract, as well as the diversity between healthy individuals, is indicative of the functional redundancy of a healthy gut microbiome ecology. See The Human Microbiome Consortia. 2012. Structure, function and diversity of the healthy human microbiome. Nature 486:207-214. This redundancy makes it highly likely that non-obvious subsets of OTUs or functional pathways (i.e., "Backbone Networks") are critical to maintaining states of health and or catalyzing a shift from a dysbiotic state to one of health. One way of exploiting this redundancy is through the substitution of OTUs that share a given clade (see below) or of adding members of a clade not found in the Backbone Network.

"Bacterial Composition" refers to a consortium of microbes comprising two or more OTUs. Backbone Network Ecologies, Functional Network Ecologies, Network Classes, and Core Ecologies are all types of bacterial compositions. A "Bacterial Composition" can also refer to a composition of enzymes that are derived from a microbe or multiple microbes. As used herein, Bacterial Composition includes a therapeutic microbial composition, a prophylactic microbial composition, a Spore Population, a Purified Spore Population, or ethanol treated spore population.

"Clade" refers to the OTUs or members of a phylogenetic tree that are downstream of a statistically valid node in a phylogenetic tree (FIG. **1**). The clade comprises a set of terminal leaves in the phylogenetic tree (i.e., tips of the tree) that are a distinct monophyletic evolutionary unit and that share some extent of sequence similarity. Clades are hierarchical. In one embodiment, the node in a phylogenetic tree that is selected to define a clade is dependent on the level of resolution suitable for the underlying data used to compute the tree topology. Exemplary clades are delineated in Table 1 and Table 2. As used herein, clade membership of bacterial OTUs is based on 16S sequence data. Clades are defined based on the topology of a phylogenetic tree that is constructed from full-length 16S sequences using maximum likelihood methods familiar to individuals with ordinary skill in the art of phylogenetics. Clades are constructed to ensure that all OTUs in a given clade are (i) within a specified number of bootstrap supported nodes from one another, and (ii) within 5% genetic identity. OTUs that are within the same clade can be distinguished as genetically and phylogenetically distinct from OTUs in a different clade based on 16S-V4 sequence data, while OTUs falling within the same clade are closely related. OTUs falling within the same clade are evolutionarily closely related and may or may not be distinguishable from one another using 16S-V4 sequence data. Members of the same clade, due to their evolutionary relatedness, play or are predicted to play similar functional roles in a microbial ecology such as that found in the human gut. In some embodiments, one OTU from a clade can be substituted in a composition by a different OTU from the same clade.

The "Colonization" of a host organism includes the non-transitory residence of a bacterium or other microscopic organism. As used herein, "reducing colonization" of a host subject's gastrointestinal tract (or any other microbiotal niche) by a pathogenic or non-pathogenic bacterium includes a reduction in the residence time of the bacterium the gastrointestinal tract as well as a reduction in the number (or concentration) of the bacterium in the gastrointestinal tract or adhered to the luminal surface of the gastrointestinal tract. The reduction in colonization can be permanent or occur during a transient period of time. Reductions of adherent pathogens can be demonstrated directly, e.g., by determining pathogenic burden in a biopsy sample, or reductions may be measured indirectly, e.g., by measuring the pathogenic burden in the stool of a mammalian host.

A "Combination" of two or more bacteria includes the physical co-existence of the two bacteria, either in the same material or product or in physically connected products, as well as the temporal co-administration or co-localization of the two bacteria.

The term "consisting essentially of" as used herein conforms to the definition as provided in the Manual of Patent Examination and Procedure (MPEP; March 2014). The basic and novel characteristics of inventions claimed herein include the ability to catalyze changes in a microbiome ecology of a mammalian subject, e.g., a human, from dysbiotic to a more normative state, and to promote engraftment and augmentation of microbiome component as set out in the specification, e.g., see Tables 14-21. A more normative state can include, in a non-limiting example, a decrease in a sign or symptom of a disease or disorder associated with a dysbiosis.

"Cytotoxic" activity of bacterium includes the ability to kill a bacterial cell, such as a pathogenic bacterial cell. A "cytostatic" activity or bacterium includes the ability to inhibit, partially or fully, growth, metabolism, and/or proliferation of a bacterial cell, such as a pathogenic bacterial cell. Cytotoxic activity may also apply to other cell types such as but not limited to Eukaryotic cells.

"Dimer" refers to a combination of bacteria that is comprised of two OTUs. The descriptions "homodimer" and "heterodimer" refer to combinations where the two OTUs are the same or different, respectively.

"Dysbiosis" refers to a state of the microbiota or microbiome of the gut or other body area, including mucosal or skin surfaces in which the normal diversity and/or function of the ecological network is disrupted. Any disruption from a preferred (e.g., ideal) state of the microbiota can be considered a dysbiosis, even if such dysbiosis does not result in a detectable decrease in health. This state of dysbiosis may be unhealthy, it may be unhealthy under only certain conditions, or it may prevent a subject from becoming healthier. Dysbiosis may be due to a decrease in diversity, the overgrowth of one or more pathogens or pathobionts, symbiotic organisms able to cause disease only when certain genetic and/or environmental conditions are present in a subject, or the shift to an ecological network that no longer provides a beneficial function to the host and therefore no longer promotes health.

"Ecological Niche" or simply "Niche" refers to the ecological space in which an organism or group of organisms occupies. Niche describes how an organism or population or organisms responds to the distribution of resources, physical parameters (e.g., host tissue space) and competitors (e.g., by growing when resources are abundant, and when predators, parasites and pathogens are scarce) and how it in turn alters those same factors (e.g., limiting access to resources by other organisms, acting as a food source for predators and a consumer of prey).

"Germinant" is a material or composition or physical-chemical process capable of inducing vegetative growth of a bacterium that is in a dormant spore form, or group of bacteria in the spore form, either directly or indirectly in a host organism and/or in vitro.

"Inhibition" of a pathogen or non-pathogen encompasses the inhibition of any desired function or activity of the bacterial compositions of the present invention. Demonstrations of inhibition, such as decrease in the growth of a pathogenic bacterium or reduction in the level of colonization of a pathogenic bacterium are provided herein and otherwise recognized by one of ordinary skill in the art. Inhibition of a pathogenic or non-pathogenic bacterium's "growth" may include inhibiting the increase in size of the pathogenic or non-pathogenic bacterium and/or inhibiting the proliferation (or multiplication) of the pathogenic or non-pathogenic bacterium. Inhibition of colonization of a pathogenic or non-pathogenic bacterium may be demonstrated by measuring the amount or burden of a pathogen before and after a treatment. An "inhibition" or the act of "inhibiting" includes the total cessation and partial reduction of one or more activities of a pathogen, such as growth, proliferation, colonization, and function. Inhibition of function includes, for example, the inhibition of expression of pathogenic gene products such as a toxin or invasive pilus induced by the bacterial composition.

"Isolated" encompasses a bacterium or other entity or substance that has been (1) separated from at least some of the components with which it was associated when initially produced (whether in nature or in an experimental setting), and/or (2) produced, prepared, purified, and/or manufactured by the hand of man. Isolated bacteria include those bacteria that are cultured, even if such cultures are not monocultures. Isolated bacteria may be separated from at least about 10%, about 20%, about 30%, about 40%, about 50%, about 60%, about 70%, about 80%, about 90%, or more of the other components with which they were initially associated. In some embodiments, isolated bacteria are more than about 80%, about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, about 99%, or more than about 99% pure. In some embodiments, isolated bacteria are separated from 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of the other components with which they were initially associated. In some embodiments, isolated bacteria are more than 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more than 99% pure. As used herein, a substance is "pure" if it is substantially free of other components. The terms "purify," "purifying" and "purified" refer to a bacterium or other material that has been separated from at least some of the components with which it was associated either when initially produced or generated (e.g., whether in nature or in an experimental setting), or during any time after its initial production. A bacterium or a bacterial population may be considered purified if it is isolated at or after production, such as from a material or environment containing the bacterium or bacterial population, or by passage through culture, and a purified bacterium or bacterial population may contain other materials up to about 10%, about 20%, about 30%, about 40%, about 50%, about 60%, about 70%, about 80%, about 90%, or above about 90% and still be considered "isolated." In other embodiments, a purified bacterium or bacterial population may contain other materials up to 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or 90% and still be considered "isolated." In some embodiments, purified bacteria and bacterial populations are more than about 80%, about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, about 99%, or more than about 99% pure. In some embodiments, purified bacteria and bacterial populations are more than 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more than 99% pure. In the instance of bacterial compositions provided herein, the one or more bacterial types present in the composition can be independently purified from one or more other bacteria produced and/or present in the material or environment containing the bacterial type. Bacterial compositions and the bacterial components thereof are generally purified from residual habitat products.

"Keystone OTU" or "Keystone Function" refers to one or more OTUs or Functional Pathways (e.g., KEGG or COG pathways) that are common to many network ecologies or functional network ecologies and are members of networks that occur in many subjects (i.e., are pervasive). Due to the ubiquitous nature of Keystone OTUs and their associated Functions Pathways, they are central to the function of network ecologies in healthy subjects and are often missing or at reduced levels in subjects with disease. Keystone OTUs and their associated functions may exist in low, moderate, or high abundance in subjects. "Non-Keystone OTU" or "non-Keystone Function" refers to an OTU or Function that is observed in a Network Ecology or a Functional Network Ecology and is not a keystone OTU or Function.

"Microbiota" refers to the community of microorganisms that occur (sustainably or transiently) in and on an animal subject, typically a mammal such as a human, including eukaryotes, archaea, bacteria, and viruses (including bacterial viruses, i.e., phage).

"Microbiome" refers to the genetic content of the communities of microbes that live in and on the human body, both sustainably and transiently, including eukaryotes, archaea, bacteria, and viruses (including bacterial viruses (i.e., phage)), wherein "genetic content" includes genomic DNA, RNA such as ribosomal RNA, the epigenome, plasmids, and all other types of genetic information.

"Microbial Carriage" or simply "Carriage" refers to the population of microbes inhabiting a niche within or on humans. Carriage is often defined in terms of relative abundance. For example, OTU1 comprises 60% of the total microbial carriage, meaning that OTU1 has a relative abundance of 60% compared to the other OTUs in the sample from which the measurement was made. Carriage is most often based on genomic sequencing data where the relative abundance or carriage of a single OTU or group of OTUs is defined by the number of sequencing reads that are assigned to that OTU/s relative to the total number of sequencing reads for the sample. Alternatively, Carriage may be measured using microbiological assays.

"Microbial Augmentation" or simply "augmentation" refers to the establishment or significant increase of a population of microbes that are (i) absent or undetectable (as determined by the use of standard genomic and microbiological techniques) from the administered therapeutic microbial composition, (ii) absent, undetectable, or present at low frequencies in the host niche (for example: gastrointestinal tract, skin, anterior-nares, or vagina) before the delivery of the microbial composition, and (iii) are found after the administration of the microbial composition or significantly increased, for example, 2-fold, 5-fold, 1×10^2, 1×10^3, 1×10^4, 1×10^5, 1×10^6, 1×10^7, or greater than 1×10^8, in cases where they were present at low frequencies. The microbes that comprise an augmented ecology can be derived from exogenous sources such as food and the environment, or grow out from micro-niches within the host where they reside at low frequency. The administration of a bacterial microbial composition induces an environmental shift in the target niche that promotes favorable conditions for the growth of these commensal microbes. In the absence of treatment with a bacterial composition, the host can be constantly exposed to these microbes; however, sustained growth and the positive health effects associated with the stable population of increased levels of the microbes comprising the augmented ecology are not observed.

"Microbial Engraftment" or simply "engraftment" refers to the establishment of OTUs present in the bacterial composition in a target niche that are absent in the treated host prior to treatment. The microbes that comprise the engrafted ecology are found in the therapeutic microbial composition and establish as constituents of the host microbial ecology upon treatment. Engrafted OTUs can establish for a transient period of time, or demonstrate long-term stability in the microbial ecology that populates the host post treatment with a bacterial composition. The engrafted ecology can induce an environmental shift in the target niche that promotes favorable conditions for the growth of commensal microbes capable of catalyzing a shift from a dysbiotic ecology to one representative of a health state.

As used herein, the term "Minerals" is understood to include boron, calcium, chromium, copper, iodine, iron, magnesium, manganese, molybdenum, nickel, phosphorus, potassium, selenium, silicon, tin, vanadium, zinc, or combinations thereof.

"Network Ecology" refers to a consortium of clades or OTUs that co-occur in some number of subjects. As used herein, a "network" is defined mathematically by a graph delineating how specific nodes (i.e., clades or OTUs) and edges (connections between specific clades or OTUs) relate to one another to define the structural ecology of a consortium of clades or OTUs. Any given Network Ecology will possess inherent phylogenetic diversity and functional properties. A Network Ecology can also be defined in terms of its functional capabilities where for example the nodes would be comprised of elements such as, but not limited to, enzymes, clusters of orthologous groups (COGS; http://www.ncbi.nlm.nih.gov/books/NBK21090/), or KEGG Orthology Pathways (www.genome.jp/kegg/); these networks are referred to as a "Functional Network Ecology". Functional Network Ecologies can be reduced to practice by defining the group of OTUs that together comprise the functions defined by the Functional Network Ecology.

"Network Class" and "Network Class Ecology" refer to a group of network ecologies that in general are computationally determined to comprise ecologies with similar phylogenetic and/or functional characteristics. A Network Class therefore contains important biological features, defined either phylogenetically or functionally, of a group (i.e., a cluster) of related network ecologies. One representation of a Network Class Ecology is a designed consortium of microbes, typically non-pathogenic bacteria, that represents core features of a set of phylogenetically or functionally related network ecologies seen in many different subjects. In some occurrences, a Network Class, while designed as described herein, exists as a Network Ecology observed in one or more subjects. Network Class ecologies are useful for reversing or reducing a dysbiosis in subjects where the underlying, related Network Ecology has been disrupted.

To be free of "non-comestible products" means that a bacterial composition or other material provided herein does not have a substantial amount of a non-comestible product, e.g., a product or material that is inedible, harmful or otherwise undesired in a product suitable for administration, e.g., oral administration, to a human subject. Non-comestible products are often found in preparations of bacteria from the prior art.

"Operational taxonomic units" and "OTU" (or plural, "OTUs") refer to a terminal leaf in a phylogenetic tree and is defined by a nucleic acid sequence, e.g., the entire genome, or a specific genetic sequence, and all sequences that share sequence identity to this nucleic acid sequence at the level of species. In some embodiments the specific genetic sequence may be the 16S sequence or a portion of the 16S sequence. In other embodiments, the entire genomes of two entities are sequenced and compared. In another embodiment, select regions such as multilocus sequence tags (MLST), specific genes, or sets of genes may be genetically compared. In 16S embodiments, OTUs that share ≥97% average nucleotide identity across the entire 16S or some variable region of the 16S are considered the same OTU. See e.g., Claesson et al., 2010. Comparison of two next-generation sequencing technologies for resolving highly complex microbiota composition using tandem variable 16S rRNA gene regions. Nucleic Acids Res 38: e200. Konstantinidis et al., 2006. The bacterial species definition in the genomic era. Philos Trans R Soc Lond B Biol Sci 361:1929-1940. In embodiments involving the complete genome, MLSTs, specific genes, other than 16S, or sets of genes OTUs that share ≥95% average nucleotide identity are considered the same OTU. See e.g., Achtman and Wagner. 2008. Microbial diversity and the genetic nature of microbial species. Nat. Rev. Microbiol. 6:431-440; Konstantinidis et al., 2006, supra. The bacterial species definition in the genomic era. Philos Trans R Soc Lond B Biol Sci 361:1929-1940. OTUs can be defined by comparing sequences between organisms. Generally, sequences with less than 95% sequence identity are not considered to form part of the same OTU. OTUs may also be characterized by any combination of nucleotide markers or genes, in particular highly conserved genes (e.g., "house-keeping" genes), or a combination thereof. Such characterization employs, e.g., WGS data or a whole genome sequence. As used herein, a "type" of bacterium refers to an OTU that can be at the level of a strain, species, clade, or family.

Table 1 below shows a List of Operational Taxonomic Units (OTU) with taxonomic assignments made to genus, species, and phylogenetic clade. Clade membership of bacterial OTUs is based on 16S sequence data. Clades are defined based on the topology of a phylogenetic tree that is constructed from full-length 16S sequences using maximum likelihood methods familiar to individuals with ordinary skill in the art of phylogenetics. Clades are constructed to ensure that all OTUs in a given clade are: (i) within a specified number of bootstrap supported nodes from one another, and (ii) within 5% genetic similarity. OTUs that are within the same clade can be distinguished as genetically and phylogenetically distinct from OTUs in a different clade based on 16S-V4 sequence data, while OTUs falling within the same clade are closely related. OTUs falling within the same clade are evolutionarily closely related and may or may not be distinguishable from one another using 16S-V4 sequence data. Members of the same clade, due to their evolutionary relatedness, play similar functional roles in a microbial ecology such as that found in the human gut. Compositions substituting one species with another from the same clade are likely to have conserved ecological function and therefore are useful in the present invention. All OTUs are denoted as to their putative capacity to form spores and whether they are a Pathogen or Pathobiont (see Definitions for description of "pathobiont"). NIAID Priority Pathogens are denoted as 'Category-A', 'Category-B', or 'Category-C', and Opportunistic Pathogens are denoted as 'OP'. OTUs that are not pathogenic or for which their ability to exist as a pathogen is unknown are denoted as 'N'. The 'SEQ ID Number' denotes the identifier of the OTU in the Sequence Listing File and 'Public DB Accession' denotes the identifier of the OTU in a public sequence repository.

"Pathobionts" or "opportunistic pathogens" refers to specific bacterial species found in healthy hosts that may trigger immune-mediated pathology and/or disease in response to certain genetic or environmental factors (Chow et al., 2011. Curr Op Immunol. Pathobionts of the intestinal microbiota and inflammatory disease. 23:473-80). A pathobiont is an opportunistic microbe that is mechanistically distinct from an acquired infectious organism. The term "pathogen" as used herein includes both acquired infectious organisms and pathobionts.

"Pathogen," "pathobiont" and "pathogenic" in reference to a bacterium or any other organism or entity that includes any such organism or entity that is capable of causing or affecting a disease, disorder or condition of a host organism containing the organism or entity, including but not limited to pre-diabetes, type 1 diabetes or type 2 diabetes.

"Phenotype" refers to a set of observable characteristics of an individual entity. As example an individual subject may have a phenotype of "health" or "disease". Phenotypes describe the state of an entity and all entities within a phenotype share the same set of characteristics that describe the phenotype. The phenotype of an individual results in part, or in whole, from the interaction of the entity's genome and/or microbiome with the environment, especially including diet.

"Phylogenetic Diversity" is a biological characteristic that refers to the biodiversity present in a given Network Ecology or Network Class Ecology based on the OTUs that comprise the network. Phylogenetic diversity is a relative term, meaning that a Network Ecology or Network Class that is comparatively more phylogenetically diverse than another network contains a greater number of unique species, genera, and taxonomic families. Uniqueness of a species, genera, or taxonomic family is generally defined using a phylogenetic tree that represents the genetic diversity all species, genera, or taxonomic families relative to one another. In another embodiment phylogenetic diversity may be measured using the total branch length or average branch length of a phylogenetic tree. Phylogenetic Diversity may be optimized in a bacterial composition by including a wide range of biodiversity.

"Phylogenetic tree" refers to a graphical representation of the evolutionary relationships of one genetic sequence to another that is generated using a defined set of phylogenetic reconstruction algorithms (e.g., parsimony, maximum likelihood, or Bayesian). Nodes in the tree represent distinct ancestral sequences and the confidence of any node is provided by a bootstrap or Bayesian posterior probability, which measures branch uncertainty.

"Prediabetes" refers a condition in which blood glucose levels are higher than normal, but not high enough to be classified as diabetes. Individuals with pre-diabetes are at increased risk of developing type 2 diabetes within a decade. According to CDC, prediabetes can be diagnosed by fasting glucose levels between 100-125 mg/dL, 2 hour post-glucose load plasma glucose in oral glucose tolerance test (OGTT) between 140 and 199 mg/dL, or hemoglobin A1c test between 5.7%-6.4%.

"rDNA," "rRNA," "16S-rDNA," "16S-IRNA," "16S," "16S sequencing," "16S-NGS," "18S," "18S-rRNA," "18S-rDNA," "18S sequencing," and "18S-NGS" refer to the nucleic acids that encode for the RNA subunits of the ribosome. rDNA refers to the gene that encodes the rRNA that comprises the RNA subunits. There are two RNA subunits in the ribosome termed the small subunit (SSU) and large subunit (LSU); the RNA genetic sequences (rRNA) of these subunits is related to the gene that encodes them (rDNA) by the genetic code. rDNA genes and their complementary RNA sequences are widely used for determination of the evolutionary relationships amount organisms as they are variable, yet sufficiently conserved to allow cross organism molecular comparisons. Typically 16S rDNA sequence (approximately 1542 nucleotides in length) of the 30S SSU is used for molecular-based taxonomic assignments of prokaryotes and the 18S rDNA sequence (approximately 1869 nucleotides in length) of 40S SSU is used for eukaryotes. 16S sequences are used for phylogenetic reconstruction as they are in general highly conserved, but contain specific hypervariable regions that harbor sufficient nucleotide diversity to differentiate genera and species of most bacteria.

"Residual habitat products" refers to material derived from the habitat for microbiota within or on a human or animal. For example, microbiota live in stool in the gastrointestinal tract, on the skin itself, in saliva, mucus of the respiratory tract, or secretions of the genitourinary tract (i.e., biological matter associated with the microbial community). Substantially free of residual habitat products means that the bacterial composition no longer contains the biological matter associated with the microbial environment on or in the human or animal subject and is 100% free, 99% free, 98% free, 97% free, 96% free, or 95% free of any contaminating biological matter associated with the microbial community. Residual habitat products can include abiotic materials (including undigested food) or it can include unwanted microorganisms. Substantially free of residual habitat products may also mean that the bacterial composition contains no detectable cells from a human or animal and that only microbial cells are detectable. In one embodiment, substantially free of residual habitat products may also mean that the bacterial composition contains no detectable viral (including bacterial viruses, i.e., phage), fungal, mycoplasmal contaminants. In another embodiment, it means that fewer than 1×10−2%, 1×10−3%, 1×10−4%, 1×10−5%, 1×10−6%, 1×10−7%, 1×10−8 of the viable cells in the bacterial composition are human or animal, as compared to microbial cells. There are multiple ways to accomplish this degree of purity, none of which are limiting. Thus, contamination may be reduced by isolating desired constituents through multiple steps of streaking to single colonies on solid media until replicate (such as, but not limited to, two) streaks from serial single colonies have shown only a single colony morphology. Alternatively, reduction of contamination can be accomplished by multiple rounds of serial dilutions to single desired cells (e.g., a dilution of 10-8 or 10-9), such as through multiple 10-fold serial dilutions. This can further be confirmed by showing that multiple isolated colonies have similar cell shapes and Gram staining behavior. Other methods for confirming adequate purity include genetic analysis (e.g., PCR, DNA sequencing), serology and antigen analysis, enzymatic and metabolic analysis, and methods using instrumentation such as flow cytometry with reagents that distinguish desired constituents from contaminants.

"Synergy" refers to an effect produced by a combination, e.g., of two microbes (for example, microbes or two different species or two different clades) that is greater than the expected additive effectives of the combination components. In certain embodiments, "synergy" between two or more microbes can result in the inhibition of a pathogens ability to grow. For example, ternary combinations synergistically inhibit *C. difficile* if their mean log inhibition is greater than the sum of the log inhibition of homotrimers of each constituent bacterium divided by three to account for the three-fold higher dose of each strain or for binary combinations, the log inhibition of a homodimer of each constituent bacterium divided by two. In another example, synergy can be calculated by defining the OTU compositions that demonstrate greater inhibition than that represented by the sum of the log inhibition of each bacterium tested separately. In other embodiments, synergy can be defined as a property of compositions that exhibit inhibition greater than the maximum log inhibition among those of each constituent bacterium's homodimer or homotrimer measured independently. As used herein, "synergy" or "synergistic interactions" refers to the interaction or cooperation of two or more microbes to produce a combined effect greater than the sum of their separate effects.

"Spore" or a population of "spores" includes bacteria (or other single-celled organisms) that are generally viable, more resistant to environmental influences such as heat and bacteriocidal agents than vegetative forms of the same bacteria, and typically capable of germination and outgrowth. Spores are characterized by the absence of active metabolism until they respond to specific environmental signals, causing them to germinate. "Spore-formers" or bacteria "capable of forming spores" are those bacteria containing the genes and other necessary abilities to produce spores under suitable environmental conditions.

"Spore population" refers to a plurality of spores present in a composition. Synonymous terms used herein include spore composition, spore preparation, ethanol-treated spore fraction and spore ecology. A spore population may be purified from a fecal donation, e.g., via ethanol or heat treatment, or a density gradient separation or any combination of methods described herein to increase the purity, potency and/or concentration of spores in a sample. Alternatively, a spore population may be derived through culture methods starting from isolated spore former species or spore former OTUs or from a mixture of such species, either in vegetative or spore form.

In one embodiment, the spore preparation comprises spore forming species wherein residual non-spore forming species have been inactivated by chemical or physical treatments including ethanol, detergent, heat, sonication, and the like; or wherein the non-spore forming species have been removed from the spore preparation by various separations steps including density gradients, centrifugation, filtration and/or chromatography; or wherein inactivation and separation methods are combined to make the spore preparation. In yet another embodiment, the spore preparation comprises spore forming species that are enriched over viable non-spore formers or vegetative forms of spore formers. In this embodiment, spores are enriched by 2-fold, 5-fold, 10-fold, 50-fold, 100-fold, 1000-fold, 10,000-fold or greater than 10,000-fold compared to all vegetative forms of bacteria. In yet another embodiment, the spores in the spore preparation undergo partial germination during processing and formulation such that the final composition comprises spores and vegetative bacteria derived from spore forming species.

"Sporulation induction agent" is a material or physical-chemical process that is capable of inducing sporulation in a bacterium, either directly or indirectly, in a host organism and/or in vitro.

To "increase production of bacterial spores" includes an activity or a sporulation induction agent. "Production" includes conversion of vegetative bacterial cells into spores and augmentation of the rate of such conversion, as well as decreasing the germination of bacteria in spore form, decreasing the rate of spore decay in vivo, or ex vivo, or to increasing the total output of spores (e.g., via an increase in volumetric output of fecal material).

"Subject" refers to any animal subject including humans, laboratory animals (e.g., non-human primates, rats, mice), livestock (e.g., cows, sheep, goats, pigs, turkeys, and chickens), and household pets (e.g., dogs, cats, and rodents). The subject may be suffering from a dysbiosis, that contributes to or causes a condition classified as diabetes or pre-diabetes, including but not limited to mechanisms such as metabolic endotoxemia, altered metabolism of primary bile acids, immune system activation, or an imbalance or reduced production of short chain fatty acids including butyrate, propionate, acetate, and branched chain fatty acids.

"Trimer" refers to a combination of bacteria that is comprised of three OTUs. The descriptions "homotrimer" and "heterotrimer" refer to combinations where all three OTUs are the same or different, respectively. A "semi-heterotrimer" refers to combinations where two constituents are the same with a third that is different

As used herein the term "vitamin" is understood to include any of various fat-soluble or water-soluble organic substances (non-limiting examples include vitamin A, vitamin B1 (thiamine), vitamin B2 (riboflavin), vitamin B3 (niacin or niacinamide), vitamin B5 (pantothenic acid), vitamin B6 (pyridoxine, pyridoxal, or pyridoxamine, or pyridoxine hydrochloride), vitamin B7 (biotin), vitamin B9 (folic acid), and Vitamin B12 (various cobalamins; commonly cyanocobalamin in vitamin supplements), vitamin C, vitamin D, vitamin E, vitamin K, K1 and K2 (i.e., MK-4, MK-7), folic acid and biotin) essential in minute amounts for normal growth and activity of the body and obtained naturally from plant and animal foods or synthetically made, pro-vitamins, derivatives, analogs.

"V1-V9 regions" or "16S V1-V9 regions" refers to the first through ninth hypervariable regions of the 16S rDNA gene that are used for genetic typing of bacterial samples. These regions in bacteria are defined by nucleotides 69-99, 137-242, 433-497, 576-682, 822-879, 986-1043, 1117-1173, 1243-1294 and 1435-1465 respectively using numbering based on the *E. coli* system of nomenclature (Brosius et al., 1978. Complete nucleotide sequence of a 16S ribosomal RNA gene from *Escherichia coli*, PNAS USA 75 (10): 4801-4805). In some embodiments, at least one of the V1, V2, V3, V4, V5, V6, V7, V8, and V9 regions are used to characterize an OTU. In one embodiment, the V1, V2, and V3 regions are used to characterize an OTU. In another embodiment, the V3, V4, and V5 regions are used to characterize an OTU. In another embodiment, the V4 region is used to characterize an OTU. A person of ordinary skill in the art can identify the specific hypervariable regions of a candidate 16S rDNA by comparing the candidate sequence in question to a reference sequence and identifying the hypervariable regions based on similarity to the reference hypervariable regions, or alternatively, one can employ Whole Genome Shotgun (WGS) sequence characterization of microbes or a microbial community.

DETAILED DESCRIPTION

Applicants have discovered combinations of bacteria that, when present, are associated with improvement in the microbiome of a subject, e.g., a subject having a dysbiosis such as a dysbiosis associated with *C. difficile*. In addition, combinations of microorganisms have been identified that are associated with the engraftment and/or augmentation of organisms associated with a healthy microbiome. Applicants have also identified combinations of microorganisms that can be useful for treating antibiotic-resistant or other pathogenic bacterial conditions. In some embodiments the microbial content of such a composition comprises the organisms, in other embodiments, the microbial content of the composition consists essentially of the organisms, and in other embodiments, the microbial content of the composition consists of the organisms. In all cases, the composition may include non-microbial components. In some cases, the composition comprises at least two organisms (e.g., three organisms) or more, as are described herein.

Emergence of Antibiotic Resistance in Bacteria

Antibiotic resistance is an emerging public health issue (Carlet et al., 2011. Society's failure to protect a precious resource: antibiotics. Lancet 378:369-371). Numerous genera of bacteria harbor species that are developing resistance to antibiotics. These include but are not limited to vancomycin resistant *Enterococcus* (VRE) and carbapenem resistant *Klebsiella* (CRKp). *Klebsiella pneumoniae* and *Escherichia coli* strains are becoming resistant to carbapenems and require the use of old antibiotics characterized by high toxicity, such as colistin (Cantón et al. 2012. Rapid evolution and spread of carbapenemases among Enterobacteriaceae in Europe. Clin Microbiol Infect 18:413-431). Additional multiply drug resistant strains of multiple species, including *Pseudomonas aeruginosa, Enterobacter* spp., and *Acinetobacter* spp. are observed clinically including isolates that are highly resistant to ceftazidime, carbapenems, and quinolones (European Centre for Disease Prevention and Control: EARSS net database. http://ecdc.europa.eu.). The Centers for Disease Control and Prevention in 2013 released a Threat Report (http://www.cdc.gov/drugresistance/threat-report-2013/) citing numerous bacterial infection threats that included *Clostridium difficile*, carbapenem-resistant Enterobacteriaceae (CRE), multidrug-resistant *Acinetobacter*, drug-resistant *Campylobacter*, extended spectrum ß-lactamase producing Enterobacteriaceae (ESBLs), vancomycin-resistant *Enterococcus* (VRE), multidrug-resistant *Pseudomonas aeruginosa*, drug-resistant non-typhoidal *Salmonella*, drug-resistant *Salmonella typhi*, drug-resistant *Shigella*, methicillin-resistant *Staphylococcus aureus* (MRSA), drug-resistant *Streptococcus pneumoniae*, vancomycin-resistant *Staphylococcus aureus* (VRSA), erythromycin-resistant Group A *Streptococcus*, and clindamycin-resistant Group B *Streptococcus*. The increasing failure of antibiotics due the rise of resistant microbes demands new therapeutics to treat bacterial infections. Administration of a microbiome therapeutic bacterial composition offers potential for such therapies.

Applicants have discovered that subjects suffering from recurrent *C. difficile* associated diarrhea (CDAD) often harbor antibiotic resistant Gram-negative bacteria, in particular Enterobacteriaceae and that treatment with a bacterial composition effectively treats CDAD and reduces the antibiotic resistant Gram-negative bacteria. The gastrointestinal tract is implicated as a reservoir for many of these organisms including VRE, MRSA, *Pseudomonas aeruginosa, Acinetobacter* and the yeast *Candida* (Donskey, Clinical Infectious Diseases 2004 39:214, The Role of the Intestinal Tract as a Reservoir and Source for Transmission of Nosocomial Pathogens), and thus as a source of nosocomial infections. Antibiotic treatment and other decontamination procedures are among the tools in use to reduce colonization of these organisms in susceptible subjects including those who are immunosuppressed. Bacterial-based therapeutics would provide a new tool for decolonization, with a key benefit of not promoting antibiotic resistance as antibiotic therapies do.

Bacterial Compositions

Provided are bacteria and combinations of bacteria of the human gut microbiota with the capacity to meaningfully provide functions of a healthy microbiota or catalyze an augmentation to the resident microbiome when administered to mammalian hosts. In particular, provided are synergistic combinations that treat, prevent, delay or reduce the symptoms of diseases, disorders and conditions associated with a dysbiosis. Representative diseases, disorders and conditions potentially associated with a dysbiosis, which are suitable for treatment with the compositions and methods as described herein, are provided in Table 3. Without being limited to a specific mechanism, it is thought that such compositions inhibit the growth, proliferation, and/or colonization of one or a plurality of pathogenic bacteria in the dysbiotic microbiotal niche, so that a healthy, diverse and protective microbiota colonizes and populates the intestinal lumen to establish or reestablish ecological control over pathogens or potential pathogens (e.g., some bacteria are pathogenic bacteria only when present in a dysbiotic environment). Inhibition of pathogens includes those pathogens such as *C. difficile, Salmonella* spp., enteropathogenic *E. coli*, multi-drug resistant bacteria such as *Klebsiella*, and *E. coli*, carbapenem-resistant Enterobacteriaceae (CRE), extended spectrum beta-lactam resistant Enterococci (ESBL), and vancomycin-resistant Enterococci (VRE).

The bacterial compositions provided herein are produced and the efficacy thereof in inhibiting pathogenic bacteria is demonstrated as provided in further detail herein.

In particular, in order to characterize those antagonistic relationships between gut commensals that are relevant to the dynamics of the mammalian gut habitat, provided is an in vitro microplate-based screening system that demonstrates the efficacy of those bacterial compositions, including the ability to inhibit (or antagonize) the growth of a bacterial pathogen or pathobiont, typically a gastrointestinal microorganism. These methods provide novel combinations of gut microbiota species and OTUs that are able to restore or enhance ecological control over important gut pathogens or pathobionts in vivo.

Bacterial compositions may comprise two types of bacteria (termed "binary combinations" or "binary pairs") or greater than two types of bacteria. Bacterial compositions that comprise three types of bacteria are termed "ternary combinations". For instance, a bacterial composition may comprise at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, at least 20, or at least 21, 22, 23, 24, 25, 26, 27, 28, 29 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, or at least 40, at least 50 or greater than 50 types of bacteria, as defined by species or operational taxonomic unit (OTU), or otherwise as provided herein. In one embodiment, the composition comprises at least two types of bacteria chosen from Table 1.

In another embodiment, the number of types of bacteria present in a bacterial composition is at or below a known value. For example, in such embodiments the bacterial composition comprises 50 or fewer types of bacteria, such as 49, 48, 47, 46, 45, 44, 43, 42, 41, 40, 39, 38, 37, 36, 35, 34, 33, 32, 31, 30, 29, 28, 27, 26, 25, 24, 23, 22, 21, 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, or 10 or fewer, or 9 or fewer types of bacteria, 8 or fewer types of bacteria, 7 or fewer types of bacteria, 6 or fewer types of bacteria, 5 or fewer types of bacteria, 4 or fewer types of bacteria, or 3 or fewer types of bacteria. In another embodiment, a bacterial composition comprises from 2 to no more than 40, from 2 to no more than 30, from 2 to no more than 20, from 2 to no more than 15, from 2 to no more than 10, or from 2 to no more than 5 types of bacteria.

In some embodiments, bacterial compositions are provided with the ability to exclude pathogenic bacteria. Exemplary bacterial compositions are demonstrated to reduce the growth rate of one pathogen, *C. difficile*, as provided in the Examples, wherein the ability of the bacterial compositions is demonstrated by assessing the antagonism activity of a combination of OTUs or strains towards a given pathogen using in vitro assays.

In some embodiments, bacterial compositions with the capacity to durably exclude *C. difficile*, are developed using a methodology for estimating an Ecological Control Factor (ECF) for constituents within the human microbiota. The ECF is determined by assessing the antagonistic activity of a given commensal strain or combination of strains towards a given pathogen using an in vitro assay, resulting in observed levels of ecological control at various concentrations of the added commensal strains. The ECF for a commensal strain or combination of strains is somewhat analogous to the longstanding minimal inhibitory concentration (MIC) assessment that is employed in the assessment of antibiotics. The ECF allows for the assessment and ranking of relative potencies of commensal strains and combinations of strains for their ability to antagonize gastrointestinal pathogens. The ECF of a commensal strain or combination of strains may be calculated by assessing the concentration of that composition that is able to mediate a given percentage of inhibition (e.g., at least 10%, 20%, 50%, 70%, 75%, 80%, 85%, 90%, 95%, or 100%) of a target pathogen in the in vitro assay. Provided herein are combinations of strains or OTUs within the human microbiota that are able to significantly reduce the rate of gastrointestinal pathogen replication within the in vitro assay. These compositions are capable of providing a safe and effective means by which to affect the growth, replication, and disease severity of such bacterial pathogens.

Bacterial compositions may be prepared comprising at least two types of isolated bacteria, wherein a first type and a second type are independently chosen from the species or OTUs listed in Table 1. Certain embodiments of bacterial compositions with at least two types of isolated bacteria containing binary pairs are reflected in Table 4a. Additionally, a bacterial composition may be prepared comprising at least two types of isolated bacteria, wherein a first OTU and a second OTU are independently characterized by, i.e., at least 95%, 96%, 97%, 98%, 99% or including 100% sequence identity to, sequences listed in Table 1. Generally, the first bacteria and the second bacteria are not the same OTU. The sequences provided in the sequencing listing file for OTUs in Table 1 are full 16S sequences. Therefore, in one embodiment, the first and/or second OTUs may be characterized by the full 16S sequences of OTUs listed in Table 1. In another embodiment, the first and/or second OTUs may be characterized by one or more of the variable regions of the 16S sequence (V1-V9). These regions in bacteria are defined by nucleotides 69-99, 137-242, 433-497, 576-682, 822-879, 986-1043, 1117-1173, 1243-1294 and 1435-1465 respectively using numbering based on the *E. coli* system of nomenclature. (See, e.g., Brosius et al., Complete nucleotide sequence of a 16S ribosomal RNA gene from *Escherichia coli*, PNAS 75 (10): 4801-4805 (1978)). In some embodiments, at least one of the V1, V2, V3, V4, V5, V6, V7, V8, and V9 regions are used to characterize an OTU. In one embodiment, the V1, V2, and V3 regions are used to characterize an OTU. In another embodiment, the V3, V4, and V5 regions are used to characterize an OTU. In another embodiment, the V4 region is used to characterize an OTU.

Bacterial Compositions Described by Species

In some embodiments, compositions are defined by species included in the composition. Methods of identifying species are known in the art.

Bacterial Compositions Described by Operational Taxonomic Units (OTUs)

OTUs may be defined either by full 16S sequencing of the rDNA gene, by sequencing of a specific hypervariable region of this gene (i.e., V1, V2, V3, V4, V5, V6, V7, V8, or V9), or by sequencing of any combination of hypervariable regions from this gene (e.g., V1-3 or V3-5). The bacterial 16S rDNA is approximately 1500 nucleotides in length and is used in reconstructing the evolutionary relationships and sequence similarity of one bacterial isolate to another using phylogenetic approaches. 16S sequences are used for phylogenetic reconstruction as they are in general highly conserved, but contain specific hypervariable regions that harbor sufficient nucleotide diversity to differentiate genera and species of most microbes. Using well known techniques, in order to determine the full 16S sequence or the sequence of any hypervariable region of the 16S sequence, genomic DNA is extracted from a bacterial sample, the 16S rDNA (full region or specific hypervariable regions) amplified using polymerase chain reaction (PCR), the PCR products cleaned, and nucleotide sequences delineated to determine the genetic composition of 16S gene or subdomain of the gene. If full 16S sequencing is performed, the sequencing method used may be, but is not limited to, Sanger sequencing. If one or more hypervariable regions are used, such as the V4 region, the sequencing may be, but is not limited to being, performed using the Sanger method or using a next-generation sequencing method, such as an Illumina (sequencing by synthesis) method using barcoded primers allowing for multiplex reactions.

Bacterial Compositions Exclusive of Certain Bacterial Species or Strains

In one embodiment, the bacterial composition does not comprise at least one of *Enterococcus faecalis* (previously known as *Streptococcus faecalis*), *Clostridium innocuum, Clostridium ramosum, Bacteroides ovatus, Bacteroides vulgatus, Bacteroides thetaoiotaomicron, Escherichia coli* (1109 and 1108-1), *Clostridium bifermentans*, and *Blautia producta* (previously known as *Peptostreptococcus productus*).

In another embodiment, the bacterial composition does not comprise at least one of *Acidaminococcus intestinalis, Bacteroides ovatus*, two species of *Bifidobacterium adolescentis*, two species of *Bifidobacterium longum, Collinsella aerofaciens*, two species of *Dorea longicatena, Escherichia coli, Eubacterium eligens, Eubacterium limosum*, four species of *Eubacterium rectale, Eubacterium ventriosumi, Faecalibacterium prausnitzii, Lactobacillus casei, Lactobacillus paracasei, Paracateroides distasonis, Raoultella* sp., one species of *Roseburia* (chosen from *Roseburia faecalis* or *Roseburia faecis*), *Roseburia intestinalis*, two species of *Ruminococcus torques*, and *Streptococcus mitis.*

In another embodiment, the bacterial composition does not comprise at least one of Barnesiella intestinihominis; *Lactobacillus reuteri*; a species characterized as one of *Enterococcus hirae, Enterococus faecium*, or *Enterococcus durans*; a species characterized as one of *Angerostipes caccae* or *Clostridium indolis*; a species characterized as one of *Staphylococcus warneri* or *Staphylococcus pasteuri*; and *Adlercreutzia equolifaciens.*

In another embodiment, the bacterial composition does not comprise at least one of *Clostridium absonum, Clostridium argentinense, Clostridium baratii, Clostridium bifermentans, Clostridium botulinum, Clostridium butyricum, Clostridium cadaveris, Clostridium camis, Clostridium celatum, Clostridium chauvoei, Clostridium clostridioforme, Clostridium cochlearium, Clostridium difficile, Clostridium fallax, Clostridium felsineum, Clostridium ghonii, Clostridium glycolicum, Clostridium haemolyticum, Clostridium hastiforme, Clostridium histolyticum, Clostridium indolis, Clostridium innocuum, Clostridium irregulare, Clostridium limosum, Clostridium malenominatum, Clostridium novyi, Clostridium oroticum, Clostridium paraputrificum, Clostridium perfringens, Clostridium piliforme, Clostridium putrefaciens, Clostridium putrificum, Clostridium ramosum, Clostridium sardiniense, Clostridium sartagoforme, Clostridium scindens, Clostridium septicum, Clostridium sordellii, Clostridium sphenoides, Clostridium spiroforme, Clostridium sporogenes, Clostridium subterminale, Clostridium symbiosum, Clostridium tertium, Clostridium tetani, Clostridium welchii*, and *Clostridium villosum.*

In another embodiment, the bacterial composition does not comprise at least one of *Clostridium innocuum, Clostridum bifermentans, Clostridium butyricum, Bacteroides fragilis, Bacteroides thetaiotaomicron, Bacteroides uniformis*, three strains of *Escherichia coli*, and *Lactobacillus* sp.

In another embodiment, the bacterial composition does not comprise at least one of *Clostridium bifermentans, Clostridium innocuum, Clostridium butyricum*, three strains of *Escherichia coli*, three strains of *Bacteroides*, and *Blautia producta* (previously known as *Peptostreptococcus productus*).

In another embodiment, the bacterial composition does not comprise at least one of *Bacteroides* sp., *Escherichia coli*, and non-pathogenic *Clostridia*, including *Clostridium innocuum, Clostridium bifermentans* and *Clostridium ramosum.*

In another embodiment, the bacterial composition does not comprise at least one of more than one *Bacteroides* species, *Escherichia coli* and non-pathogenic *Clostridia*, such as *Clostridium butyricum, Clostridium bifermentans* and *Clostridium innocuum.*

In another embodiment, the bacterial composition does not comprise at least one of *Bacteroides caccae, Bacteroides capillosus, Bacteroides coagulans, Bacteroides distasonis, Bacteroides eggerthii, Bacteroides forsythus, Bacteroides fragilis, Bacteroides fragilis-ryhm, Bacteroides gracilis, Bacteroides levii, Bacteroides macacae, Bacteroides mer-*

dae, *Bacteroides ovatus, Bacteroides pneumosintes, Bacteroides putredinis, Bacteroides pyogenes, Bacteroides splanchnicus, Bacteroides stercoris, Bacteroides tectum, Bacteroides thetaiotaomicron, Bacteroides uniformis, Bacteroides ureolyticus*, and *Bacteroides vulgatus.*

In another embodiment, the bacterial composition does not comprise at least one of *Bacteroides, Eubacteria, Fusobacteria, Propionibacteria, Lactobacilli*, anaerobic cocci, *Ruminococcus, Escherichia coli, Gemmiger, Desulfomonas*, and *Peptostreptococcus.*

In another embodiment, the bacterial composition does not comprise at least one of *Bacteroides fragilis* ss. *Vulgatus, Eubacterium aerofaciens, Bacteroides fragilis* ss. *Thetaiotaomicron, Blautia producta* (previously known as *Peptostreptococcus productus* II), *Bacteroides fragilis* ss. *Distasonis, Fusobacterium prausnitzii, Coprococcus eutactus, Eubacterium aerofaciens* III, *Blautia producta* (previously known as *Peptostreptococcus productus* I), *Ruminococcus bromii, Bifidobacterium adolescentis, Gemmiger formicilis, Bifidobacterium longum, Eubacterium siraeum, Ruminococcus torques, Eubacterium rectale* III-H, *Eubacterium rectale* IV. *Eubacterium eligens, Bacteroides eggerthii, Clostridium leptum, Bacteroides fragilis* ss. A, *Eubacterium biforme, Bifidobacterium infantis, Eubacterium rectale* III-F, *Coprococcus comes, Bacteroides capillosus, Ruminococcus albus, Eubacterium formicigenerans, Eubacterium hallii, Eubacterium ventriosum* I, *Fusobacterium russii, Ruminococcus obeum, Eubacterium rectale* II, *Clostridium ramosum* I, *Lactobacillus leichmanii, Ruminococcus cailidus, Butyrivibrio crossotus, Acidaminococcus fermentans, Eubacterium ventriosum, Bacteroides fragilis* ss. *fragilis, Bacteroides* AR, *Coprococcus catus, Eubacterium hadrum, Eubacterium cylindroides, Eubacterium ruminantium, Eubacterium* CH-1, *Staphylococcus epidermidis, Peptostreptococcus* BL, *Eubacterium limosum, Bacteroides praeacutus, Bacteroides* L. *Fusobacterium mortiferum* I, *Fusobacterium naviforme, Clostridium innocuum, Clostridium ramosum, Propionibacterium acnes, Ruminococcus flavefaciens, Ruminococcus* AT, *Peptococcus* AU-1, *Eubacterium* AG, -AK, -AL, -AL-1, -AN; *Bacteroides fragilis* ss. *ovatus*, -ss. d, -ss. f; *Bacteroides* L-1, L-5; *Fusobacterium micleatum, Fusobacterium mortiferum, Escherichia coli, Streptococcus morbiliorum, Peptococcus magmis, Peptococcus* G, AU-2; *Streptococcus intermedius, Ruminococcus lactaris, Ruminococcus* CO *Gemmiger* X, *Coprococcus* BH, -CC; *Eubacterium tenue, Eubacterium ramulus, Eubacterium* AE, -AG-H, -AG-M, -AJ, -BN-1; *Bacteroides clostridiiformis* ss. *clostridliformis, Bacteroides coagulans, Bacteroides orails, Bacteroides ruminicola* ss. *brevis*, -ss. *ruminicola, Bacteroides splanchnicus, Desuifomonas pigra, Bacteroides* L-4, -N-i; *Fusobacterium* H, *Lactobacillus* G, and *Succinivibrio* A.

In another embodiment, the bacterial composition does not comprise at least one of *Bifidobacterium bifidum* W23, *Bifidobacterium lactis* W18, *Bifidobacterium longum* W51, *Enterococcus faecium* W54, *Lactobacillus plantarum* W62, *Lactobacillus rhamnosus* W71, *Lactobacillus acidophilus* W37, *Lactobacillus acidophilus* W55, *Lactobacillus paracasei* W20, and *Lactobacillus salivarius* W24.

In another embodiment, the bacterial composition does not comprise at least one of *Anaerotruncus colihominis* DSM 17241, *Blautia producta* JCM 1471, *Clostridiales bacterium* 1 7 47FAA, *Clostridium asparagiforme* DSM 15981, *Clostridium bacterium* JC13, *Clostridium bolteae* ATCC BAA 613, *Clostridium hathewayi* DSM 13479. *Clostridium indolis* CM971, *Clostridium ramosum* DSM 1402, *Clostridium saccharogumia* SDG Mt85 3 Db, *Clostridium scindens* VP 12708, *Clostridium* sp 7 3 54FAA, *Eubacterium contortum* DSM 3982, *Lachnospiraceae bacterium* 3 1 57FAA CT1, *Lachnospiraceae bacterium* 7 1 58FAA, and *Ruminococcus* sp ID8.

In another embodiment, the bacterial composition does not comprise at least one of *Anaerotruncus colihominis* DSM 17241, *Blautia producta* JCM 1471, *Clostridiales bacterium* 1 7 47FAA, *Clostridium asparagiforme* DSM 15981, *Clostridium bacterium* JC13, *Clostridium bolteae* ATCC BAA 613, *Clostridium hathewayi* DSM 13479, *Clostridium indolis* CM971, *Clostridium ramosum* DSM 1402, *Clostridium saccharogumia* SDG M185 3 Db, *Clostridium scindens* VP 12708, *Clostridium* sp 7 3 54FAA, *Eubacterium contortum* DSM 3982, *Lachnospiraceae bacterium* 3 1 57FAA CT1, *Lachnospiraceae bacterium* 7 1 58FAA, *Oscillospiraceae bacterium* NML 061048, and *Ruminococcus* sp ID8.

In another embodiment, the bacterial composition does not comprise at least one of *Clostridium ramosum* DSM 1402, *Clostridium saccharogumia* SDG Mt85 3 Db, and *Lachnospiraceae bacterium* 7 1 58FAA.

In another embodiment, the bacterial composition does not comprise at least one of *Clostridium hathewayi* DSM 13479, *Clostridium saccharogumia* SDG Mt85 3 Db, *Clostridium* sp 7 3 54FAA, and *Lachnospiraceae bacterium* 3 1 57FAA CT1.

In another embodiment, the bacterial composition does not comprise at least one of *Anaerotruncus colihominis* DSM 17241, *Blautia producta* JCM 1471, *Clostridium bacterium* JC13, *Clostridium scindens* VP 12708, and *Ruminococcus* sp ID8.

In another embodiment, the bacterial composition does not comprise at least one of *Clostridiales bacterium* 1 7 47FAA, *Clostridium asparagiforme* DSM 15981, *Clostridium bolteae* ATCC BAA 613, *Clostridium indolis* CM971, and *Lachnospiraceae bacterium* 7 1 58FAA.

Inhibition of Bacterial Pathogens

The bacterial compositions offer a protective or therapeutic effect against infection by one or more GI pathogens of interest, some of which are listed in Table 3.

In some embodiments, the pathogenic bacterium is selected from the group consisting of *Yersinia, Vibrio, Treponema, Streptococcus, Staphylococcus, Shigella, Salmonella, Rickettsia, Orientia, Pseudomonas, Neisseria, Mycoplasma, Mycobacterium, Listeria, Leptospira, Legionella, Klebsiella, Helicobacter, Haemophilus, Francisella, Escherichia, Ehrlichia, Enterococcus, Coxiella, Corynebacterium, Clostridium, Chlamydia, Chlamydophila, Campylobacter, Burkholderia, Brucella, Borrelia, Bordetella, Bifidobacterium, Bacillus*, multi-drug resistant bacteria, extended spectrum beta-lactam resistant Enterococci (ESBL), carbapenem-resistant Enterobacteriaceae (CRE), and vancomycin-resistant Enterococci (VRE).

In some embodiments, these pathogens include, but are not limited to, *Aeromonas hydrophila, Campylobacter fetus, Plesiomonas shigelloides, Bacillus cereus, Campylobacter jejuni, Clostridium botulinum, Clostridium difficile, Clostridium perfringens*, enteroaggregative *Escherichia coli*, enterohemorrhagic *Escherichia coli*, enteroinvasive *Escherichia coli*, enterotoxigenic *Escherichia coli* (such as, but not limited to, LT and/or ST), *Escherichia coli* 0157: H7, *Fusarium* spp., *Helicobacter pylori, Klebsiella pneumonia, Klebsiella oxytoca, Lysteria monocytogenes, Morganella* spp., *Plesiomonas shigelloides, Proteus* spp., *Providencia* spp., *Salmonella* spp., *Salmonella typhi, Salmonella paratyphi, Shigella* spp., *Staphylococcus* spp., *Staphylococcus* *aureus*, vancomycin-resistant *enterococcus* spp., *Vibrio* spp., *Vibrio cholerae, Vibrio parahaemolyticus, Vibrio vulnificus*, and *Yersinia enterocolitica*.

In one embodiment, the pathogen of interest is at least one pathogen chosen from *Clostridium difficile, Salmonella* spp., pathogenic *Escherichia coli*, vancomycin-resistant *Enterococcus* spp., and extended spectrum beta-lactam resistant Enterococci (ESBL).

In Vitro Assays Substantiating Protective Effect of Bacterial Compositions

In one embodiment, provided is an in vitro assay utilizing competition between the bacterial compositions or subsets thereof and *C. difficile*. Exemplary embodiments of the assay are provided herein and in the Examples.

In another embodiment, provided is an in vitro assay utilizing 10% (wt/vol) Sterile-Filtered Stool (SFS). Provided is an in vitro assay to test for the protective effect of the bacterial compositions and to screen in vitro for combinations of microbes that inhibit the growth of a pathogen. The assay can operate in automated high-throughput or manual modes. Under either system, human or animal stool may be re-suspended in an anaerobic buffer solution, such as pre-reduced PBS or other suitable buffer, the particulate removed by centrifugation, and filter sterilized. This 10% sterile-filtered stool material serves as the base media for the in vitro assay. To test a bacterial composition, an investigator may add it to the sterile-filtered stool material for a first incubation period and then may inoculate the incubated microbial solution with the pathogen of interest for a second incubation period. The resulting titer of the pathogen may be quantified by any number of methods such as those described below, and the change in the amount of pathogen is compared to standard controls including the pathogen cultivated in the absence of the bacterial composition. The assay is conducted using at least one control. Stool from a healthy subject may be used as a positive control. As a negative control, antibiotic-treated stool or heat-treated stool may be used. Various bacterial compositions may be tested in this material and the bacterial compositions optionally compared to the positive and/or negative controls. The ability to inhibit the growth of the pathogen may be measured by plating the incubated material on *C. difficile* selective media and counting colonies. After competition between the bacterial composition and *C. difficile*, each well of the in vitro assay plate is serially diluted ten-fold six times, and plated on selective media, such as but not limited to cycloserine cefoxitin mannitol agar (CCMA) or cycloserine cefoxitin fructose agar (CCFA), and incubated. Colonies of *C. difficile* are then counted to calculate the concentration of viable cells in each well at the end of the competition. Colonies of *C. difficile* are confirmed by their characteristic diffuse colony edge morphology as well as fluorescence under UV light.

In another embodiment, the in vitro assay utilizes Antibiotic-Treated Stool. In an alternative embodiment, and instead of using 10% sterile-filtered stool, human or animal stool may be resuspended in an anaerobic buffer solution, such as pre-reduced PBS or other suitable buffer. The resuspended stool is treated with an antibiotic, such as clindamycin, or a cocktail of several antibiotics in order to reduce the ability of stool from a healthy subject to inhibit the growth of *C. difficile*; this material is termed the antibiotic-treated matrix. While not being bound by any mechanism, it is believed that beneficial bacteria in healthy subjects protects them from infection by competing out *C. difficile*. Treating stool with antibiotics kills or reduces the population of those beneficial bacteria, allowing *C. difficile* to grow in this assay matrix. Antibiotics in addition to clindamycin that inhibit the normal flora include ceftriaxone and piperacillin-tazobactam and may be substituted for the clindamycin. The antibiotic-treated matrix is centrifuged, the supernatant removed, and the pelleted material resuspended in filter-sterilized, diluted stool in order to remove any residual antibiotic. This washed antibiotic-treated matrix may be used in the in vitro assay described above in lieu of the 10% sterile-filtered stool.

Also provided is an In Vitro Assay utilizing competition between the bacterial compositions or subsets thereof and Vancomycin-resistant *Enterococcus faecium*. Exemplary embodiments of this Assay are provided herein and in the Examples.

Also provided is an in vitro assay utilizing competition between the bacterial compositions or subsets thereof and *Morganella morganii*. Exemplary embodiments of this Assay are provided herein and in the Examples.

Also provided is an in vitro assay utilizing competition between the bacterial compositions or subsets thereof and *Klebsiella pneumoniae*. Exemplary embodiments of this Assay are provided herein and in the Examples.

Alternatively, the ability to inhibit the growth of the pathogen may be measured by quantitative PCR (qPCR). Standard techniques may be followed to generate a standard curve for the pathogen of interest. Genomic DNA may be extracted from samples using commercially available kits, such as the Mo Bio Powersoil®-htp 96 Well Soil DNA Isolation Kit (Mo Bio Laboratories, Carlsbad, CA), the Mo Bio Powersoil® DNA Isolation Kit (Mo Bio Laboratories, Carlsbad, CA), or the QIAamp DNA Stool Mini Kit (QIAGEN, Valencia, CA) according to the manufacturer's instructions. The qPCR may be conducted using HotMasterMix (5PRIME, Gaithersburg, MD) and primers specific for the pathogen of interest, and may be conducted on a MicroAmp® Fast Optical 96-well Reaction Plate with Barcode (0.1 mL) (Life Technologies, Grand Island, NY) and performed on a BioRad C1000™ Thermal Cycler equipped with a CFX96™ Real-Time System (BioRad, Hercules, CA), with fluorescent readings of the FAM and ROX channels. The Cq value for each well on the FAM channel is determined by the CFX Manager™ software version 2.1. The $\log_{10}$ (cfu/ml) of each experimental sample is calculated by inputting a given sample's Cq value into linear regression model generated from the standard curve comparing the Cq values of the standard curve wells to the known $\log_{10}$ (cfu/ml) of those samples. The skilled artisan may employ alternative qPCR modes.

In Vivo Assay Establishing Protective Effect of Bacterial Compositions.

Provided is an in vivo mouse model to test for the protective effect of the bacterial compositions against *C. difficile*. In this model (based on Chen, et al., A mouse model of *Clostridium difficile* associated disease, Gastroenterology 135 (6): 1984-1992 (2008)), mice are made susceptible to *C. difficile* by a 7 day treatment (days −12 to −5 of experiment) with 5 to 7 antibiotics (including kanamycin, colistin, gentamycin, metronidazole and vancomycin and optionally including ampicillin and ciprofloxacin) delivered via their drinking water, followed by a single dose with Clindamycin on day −3, then challenged three days later on day 0 with 10^4 spores of *C. difficile* via oral gavage (i.e., oro-gastric lavage). Bacterial compositions may be given either before (prophylactic treatment) or after (therapeutic treatment) *C. difficile* gavage. Further, bacterial compositions may be given after (optional) vancomycin treatment (see below) to assess their ability to prevent recurrence and thus suppress the pathogen in vivo. The outcomes assessed each day from day −1 to day 6 (or beyond, for prevention of recurrence) are weight, clinical signs, mortality and shedding of *C. difficile* in the stool. Weight loss, clinical signs of disease, and *C. difficile* shedding are typically observed without treatment. Vancomycin provided by oral gavage on days −1 to 4 protects against these outcomes and serves as a positive control. Clinical signs are subjective, and scored each day by the same experienced observer. Animals that lose greater than or equal to 25% of their body weight are euthanized and counted as infection-related mortalities. Stool are gathered from mouse cages (5 mice per cage) each day, and the shedding of *C. difficile* spores is detected in the stool using a selective plating assay as described for the in vitro assay above, or via qPCR for the toxin gene as described herein. The effects of test materials including 10% suspension of human stool (as a positive control), bacterial compositions, or PBS (as a negative vehicle control), are determined by introducing the test article in a 0.2 mL volume into the mice via oral gavage on day −1, one day prior to *C. difficile* challenge, on day 1, 2 and 3 as treatment or post-vancomycin treatment on days 5, 6, 7 and 8. Vancomycin, as discussed above, is given on days 1 to 4 as another positive control. Alternative dosing schedules and routes of administration (e.g., rectal) may be employed, including multiple doses of test article, and 10^3 to 10^{10} of a given organism or composition may be delivered.

Methods for Preparing a Bacterial Composition for Administration to a Subject

Methods for producing bacterial compositions may include three main processing steps, combined with one or more mixing steps. The steps are: organism banking, organism production, and preservation.

For banking, the strains included in the bacterial composition may be (1) isolated directly from a specimen or taken from a banked stock, (2) optionally cultured on a nutrient agar or broth that supports growth to generate viable biomass, and (3) the biomass optionally preserved in multiple aliquots in long-term storage.

In embodiments using a culturing step, the agar or broth may contain nutrients that provide essential elements and specific factors that enable growth. An example would be a medium composed of 20 g/L glucose, 10 g/L yeast extract, 10 g/L soy peptone, 2 g/L citric acid, 1.5 g/L sodium phosphate monobasic, 100 mg/L ferric ammonium citrate, 80 mg/L magnesium sulfate, 10 mg/L hemin chloride, 2 mg/L calcium chloride, 1 mg/L menadione. A variety of microbiological media and variations are well known in the art (e.g., R. M. Atlas, *Handbook of Microbiological Media* (2010) CRC Press). Medium can be added to the culture at the start, may be added during the culture, or may be intermittently/continuously flowed through the culture. The strains in the bacterial composition may be cultivated alone, as a subset of the bacterial composition, or as an entire collection comprising the bacterial composition. As an example, a first strain may be cultivated together with a second strain in a mixed continuous culture, at a dilution rate lower than the maximum growth rate of either cell to prevent the culture from washing out of the cultivation.

The inoculated culture is incubated under favorable conditions for a time sufficient to build biomass. For bacterial compositions for human use this is often at 37° C. temperature, pH, and other parameter with values similar to the normal human niche. The environment may be actively controlled, passively controlled (e.g., via buffers), or allowed to drift. For example, for anaerobic bacterial compositions (e.g., gut microbiota), an anoxic/reducing environment may be employed. This can be accomplished by addition of reducing agents such as cysteine to the broth, and/or stripping it of oxygen. As an example, a culture of a bacterial composition may be grown at 37° C., pH 7, in the medium above, pre-reduced with 1 g/L cysteine HCl.

When the culture has generated sufficient biomass, it may be preserved for banking. The organisms may be placed into a chemical milieu that protects from freezing (adding 'cryoprotectants'), drying ('lyoprotectants'), and/or osmotic shock ('osmoprotectants'), dispensing into multiple (optionally identical) containers to create a uniform bank, and then treating the culture for preservation. Containers are generally impermeable and have closures that assure isolation from the environment. Cryopreservation treatment is accomplished by freezing a liquid at ultra-low temperatures (e.g., at or below −80° C.). Dried preservation removes water from the culture by evaporation (in the case of spray drying or 'cool drying') or by sublimation (e.g., for freeze drying, spray freeze drying). Removal of water improves long-term bacterial composition storage stability at temperatures elevated above cryogenic. If the bacterial composition comprises spore forming species and results in the production of spores, the final composition may be purified by additional means such as density gradient centrifugation preserved using the techniques described above. Bacterial composition banking may be done by culturing and preserving the strains individually, or by mixing the strains together to create a combined bank. As an example of cryopreservation, a bacterial composition culture may be harvested by centrifugation to pellet the cells from the culture medium, the supernatant decanted and replaced with fresh culture broth containing 15% glycerol. The culture can then be aliquoted into 1 mL cryotubes, sealed, and placed at −80° C. for long-term viability retention. This procedure achieves acceptable viability upon recovery from frozen storage.

Organism production may be conducted using similar culture steps to banking, including medium composition and culture conditions. It may be conducted at larger scales of operation, especially for clinical development or commercial production. At larger scales, there may be several subcultivations of the bacterial composition prior to the final cultivation. At the end of cultivation, the culture is harvested to enable further formulation into a dosage form for administration. This can involve concentration, removal of undesirable medium components, and/or introduction into a chemical milieu that preserves the bacterial composition and renders it acceptable for administration via the chosen route. For example, a bacterial composition may be cultivated to a concentration of 10^{10} CFU/mL, then concentrated 20-fold by tangential flow microfiltration; the spent medium may be exchanged by diafiltering with a preservative medium consisting of 2% gelatin, 100 mM trehalose, and 10 mM sodium phosphate buffer. The suspension can then be freeze-dried to a powder and titrated.

After drying, the powder may be blended to an appropriate potency, and mixed with other cultures and/or a filler such as microcrystalline cellulose for consistency and ease of handling, and the bacterial composition formulated as provided herein.

Formulations

Provided are formulations for administration to humans and other subjects in need thereof. Generally the bacterial compositions are combined with additional active and/or inactive materials to produce a final product, which may be in single dosage unit or in a multi-dose format.

In some embodiments the composition comprises at least one carbohydrate. A "carbohydrate" refers to a sugar or polymer of sugars. The terms "saccharide," "polysaccharide," "carbohydrate," and "oligosaccharide" may be used interchangeably. Most carbohydrates are aldehydes or ketones with many hydroxyl groups, usually one on each carbon atom of the molecule. Carbohydrates generally have the molecular formula $C_nH_{2n}O_n$. A carbohydrate may be a monosaccharide, a disaccharide, trisaccharide, oligosaccharide, or polysaccharide. The most basic carbohydrate is a monosaccharide, such as glucose, sucrose, galactose, mannose, ribose, arabinose, xylose, and fructose. Disaccharides are two joined monosaccharides. Exemplary disaccharides include sucrose, maltose, cellobiose, and lactose. Typically, an oligosaccharide includes between three and six monosaccharide units (e.g., raffinose, stachyose), and polysaccharides include six or more monosaccharide units. Exemplary polysaccharides include starch, glycogen, and cellulose. Carbohydrates may contain modified saccharide units such as 2'-deoxyribose wherein a hydroxyl group is removed, 2'-fluororibose wherein a hydroxyl group is replace with a fluorine, or N-acetylglucosamine, a nitrogen-containing form of glucose (e.g., 2'-fluororibose, deoxyribose, and hexose). Carbohydrates may exist in many different forms, for example, conformers, cyclic forms, acyclic forms, stereoisomers, tautomers, anomers, and isomers.

In some embodiments the composition comprises at least one lipid. As used herein a "lipid" includes fats, oils, triglycerides, cholesterol, phospholipids, fatty acids in any form including free fatty acids. Fats, oils and fatty acids can be saturated, unsaturated (cis or trans) or partially unsaturated (cis or trans). In some embodiments the lipid comprises at least one fatty acid selected from lauric acid (12:0), myristic acid (14:0), palmitic acid (16:0), palmitoleic acid (16:1), margaric acid (17:0), heptadecenoic acid (17:1), stearic acid (18:0), oleic acid (18:1), linoleic acid (18:2), linolenic acid (18:3), octadecatetraenoic acid (18:4), arachidic acid (20:0), eicosenoic acid (20:1), eicosadienoic acid (20:2), eicosatetraenoic acid (20:4), eicosapentaenoic acid (20:5) (EPA), docosanoic acid (22:0), docosenoic acid (22:1), docosapentaenoic acid (22:5), docosahexaenoic acid (22:6) (DHA), and tetracosanoic acid (24:0). In some embodiments the composition comprises at least one modified lipid, for example a lipid that has been modified by cooking.

In some embodiments the composition comprises at least one supplemental mineral or mineral source. Examples of minerals include, without limitation: chloride, sodium, calcium, iron, chromium, copper, iodine, zinc, magnesium, manganese, molybdenum, phosphorus, potassium, and selenium. Suitable forms of any of the foregoing minerals include soluble mineral salts, slightly soluble mineral salts, insoluble mineral salts, chelated minerals, mineral complexes, non-reactive minerals such as carbonyl minerals, and reduced minerals, and combinations thereof.

In some embodiments the composition comprises at least one supplemental vitamin. The at least one vitamin can be fat-soluble or water soluble vitamins. Suitable vitamins include but are not limited to vitamin C, vitamin A, vitamin E, vitamin B12, vitamin K, riboflavin, niacin, vitamin D, vitamin B6, folic acid, pyridoxine, thiamine, pantothenic acid, and biotin. Suitable forms of any of the foregoing are salts of the vitamin, derivatives of the vitamin, compounds having the same or similar activity of the vitamin, and metabolites of the vitamin.

In some embodiments the composition comprises an excipient. Non-limiting examples of suitable excipients include a buffering agent, a preservative, a stabilizer, a binder, a compaction agent, a lubricant, a dispersion enhancer, a disintegration agent, a flavoring agent, a sweetener, and a coloring agent.

In some embodiments the excipient is a buffering agent. Non-limiting examples of suitable buffering agents include sodium citrate, magnesium carbonate, magnesium bicarbonate, calcium carbonate, and calcium bicarbonate.

In some embodiments the excipient comprises a preservative. Non-limiting examples of suitable preservatives include antioxidants, such as alpha-tocopherol and ascorbate, and antimicrobials, such as parabens, chlorobutanol, and phenol.

In some embodiments the composition comprises a binder as an excipient. Non-limiting examples of suitable binders include starches, pregelatinized starches, gelatin, polyvinylpyrolidone, cellulose, methylcellulose, sodium carboxymethylcellulose, ethylcellulose, polyacrylamides, polyvinyloxoazolidone, polyvinylalcohols, C_{12}-C_{18} fatty acid alcohol, polyethylene glycol, polyols, saccharides, oligosaccharides, and combinations thereof.

In some embodiments the composition comprises a lubricant as an excipient. Non-limiting examples of suitable lubricants include magnesium stearate, calcium stearate, zinc stearate, hydrogenated vegetable oils, sterotex, polyoxyethylene monostearate, talc, polyethyleneglycol, sodium benzoate, sodium lauryl sulfate, magnesium lauryl sulfate, and light mineral oil.

In some embodiments the composition comprises a dispersion enhancer as an excipient. Non-limiting examples of suitable dispersants include starch, alginic acid, polyvinylpyrrolidones, guar gum, kaolin, bentonite, purified wood cellulose, sodium starch glycolate, isoamorphous silicate, and microcrystalline cellulose as high HLB emulsifier surfactants.

In some embodiments the composition comprises a disintegrant as an excipient. In some embodiments the disintegrant is a non-effervescent disintegrant. Non-limiting examples of suitable non-effervescent disintegrants include starches such as corn starch, potato starch, pregelatinized and modified starches thereof, sweeteners, clays, such as bentonite, micro-crystalline cellulose, alginates, sodium starch glycolate, gums such as agar, guar, locust bean, karaya, pecitin, and tragacanth. In some embodiments the disintegrant is an effervescent disintegrant. Non-limiting examples of suitable effervescent disintegrants include sodium bicarbonate in combination with citric acid, and sodium bicarbonate in combination with tartaric acid.

In some embodiments the excipient comprises a flavoring agent. Flavoring agents can be chosen from synthetic flavor oils and flavoring aromatics; natural oils; extracts from plants, leaves, flowers, and fruits; and combinations thereof. In some embodiments the flavoring agent is selected from cinnamon oils; oil of wintergreen; peppermint oils; clover oil; hay oil; anise oil; eucalyptus; vanilla; citrus oil such as lemon oil, orange oil, grape and grapefruit oil; and fruit essences including apple, peach, pear, strawberry, raspberry, cherry, plum, pineapple, and apricot.

In some embodiments the excipient comprises a sweetener. Non-limiting examples of suitable sweeteners include glucose (corn syrup), dextrose, invert sugar, fructose, and mixtures thereof (when not used as a carrier); saccharin and its various salts such as the sodium salt; dipeptide sweeteners such as aspartame; dihydrochalcone compounds, glycyrrhizin; *Stevia rebaudiana* (Stevioside); chloro derivatives of sucrose such as sucralose; and sugar alcohols such as sorbitol, mannitol, sylitol, and the like. Also contemplated are hydrogenated starch hydrolysates and the synthetic sweetener 3,6-dihydro-6-methyl-1,2,3-oxathiazin-4-one-2,2-dioxide, particularly the potassium salt (acesulfame-K), and sodium and calcium salts thereof.

In some embodiments the composition comprises a coloring agent. Non-limiting examples of suitable color agents include food, drug and cosmetic colors (FD&C), drug and cosmetic colors (D&C), and external drug and cosmetic colors (Ext. D&C). The coloring agents can be used as dyes or their corresponding lakes.

The weight fraction of the excipient or combination of excipients in the formulation is usually about 99% or less, such as about 95% or less, about 90% or less, about 85% or less, about 80% or less, about 75% or less, about 70% or less, about 65% or less, about 60% or less, about 55% or less, 50% or less, about 45% or less, about 40% or less, about 35% or less, about 30% or less, about 25% or less, about 20% or less, about 15% or less, about 10% or less, about 5% or less, about 2% or less, or about 1% or less of the total weight of the composition.

The bacterial compositions disclosed herein can be formulated into a variety of forms and administered by a number of different means. The compositions can be administered orally, rectally, or parenterally, in formulations containing conventionally acceptable carriers, adjuvants, and vehicles as desired. The term "parenteral" as used herein includes subcutaneous, intravenous, intramuscular, or intrasternal injection and infusion techniques. In an exemplary embodiment, the bacterial composition is administered orally.

Solid dosage forms for oral administration include capsules, tablets, caplets, pills, troches, lozenges, powders, and granules. A capsule typically comprises a core material comprising a bacterial composition and a shell wall that encapsulates the core material. In some embodiments the core material comprises at least one of a solid, a liquid, and an emulsion. In some embodiments the shell wall material comprises at least one of a soft gelatin, a hard gelatin, and a polymer. Suitable polymers include, but are not limited to: cellulosic polymers such as hydroxypropyl cellulose, hydroxyethyl cellulose, hydroxypropyl methyl cellulose (HPMC), methyl cellulose, ethyl cellulose, cellulose acetate, cellulose acetate phthalate, cellulose acetate trimellitate, hydroxypropylmethyl cellulose phthalate, hydroxypropylmethyl cellulose succinate and carboxymethylcellulose sodium; acrylic acid polymers and copolymers, such as those formed from acrylic acid, methacrylic acid, methyl acrylate, ammonio methylacrylate, ethyl acrylate, methyl methacrylate and/or ethyl methacrylate (e.g., those copolymers sold under the trade name "Eudragit"); vinyl polymers and copolymers such as polyvinyl pyrrolidone, polyvinyl acetate, polyvinylacetate phthalate, vinylacetate crotonic acid copolymer, and ethylene-vinyl acetate copolymers; and shellac (purified lac). In some embodiments at least one polymer functions as taste-masking agents.

Tablets, pills, and the like can be compressed, multiply compressed, multiply layered, and/or coated. The coating can be single or multiple. In one embodiment, the coating material comprises at least one of a saccharide, a polysaccharide, and glycoproteins extracted from at least one of a plant, a fungus, and a microbe. Non-limiting examples include corn starch, wheat starch, potato starch, tapioca starch, cellulose, hemicellulose, dextrans, maltodextrin, cyclodextrins, inulins, pectin, mannans, gum arabic, locust bean gum, mesquite gum, guar gum, gum karaya, gum ghatti, tragacanth gum, funori, carrageenans, agar, alginates, chitosans, or gellan gum. In some embodiments the coating material comprises a protein. In some embodiments the coating material comprises at least one of a fat and an oil. In some embodiments the at least one of a fat and an oil is high temperature melting. In some embodiments the at least one of a fat and an oil is hydrogenated or partially hydrogenated. In some embodiments the at least one of a fat and an oil is derived from a plant. In some embodiments the at least one of a fat and an oil comprises at least one of glycerides, free fatty acids, and fatty acid esters. In some embodiments the coating material comprises at least one edible wax. The edible wax can be derived from animals, insects, or plants. Non-limiting examples include beeswax, lanolin, bayberry wax, carnauba wax, and rice bran wax. Tablets and pills can additionally be prepared with enteric coatings.

Alternatively, powders or granules embodying the bacterial compositions disclosed herein can be incorporated into a food product. In some embodiments the food product is a drink for oral administration. Non-limiting examples of a suitable drink include fruit juice, a fruit drink, an artificially flavored drink, an artificially sweetened drink, a carbonated beverage, a sports drink, a liquid diary product, a shake, an alcoholic beverage, a caffeinated beverage, infant formula and so forth. Other suitable means for oral administration include aqueous and nonaqueous solutions, emulsions, suspensions and solutions and/or suspensions reconstituted from non-effervescent granules, containing at least one of suitable solvents, preservatives, emulsifying agents, suspending agents, diluents, sweeteners, coloring agents, and flavoring agents.

In some embodiments the food product is a solid foodstuff. Suitable examples of a solid foodstuff include without limitation a food bar, a snack bar, a cookie, a brownie, a muffin, a cracker, an ice cream bar, a frozen yogurt bar, and the like.

In some embodiments, the compositions disclosed herein are incorporated into a therapeutic food. In some embodiments, the therapeutic food is a ready-to-use food that optionally contains some or all essential macronutrients and micronutrients. In some embodiments, the compositions disclosed herein are incorporated into a supplementary food that is designed to be blended into an existing meal. In some embodiments, the supplemental food contains some or all essential macronutrients and micronutrients. In some embodiments, the bacterial compositions disclosed herein are blended with or added to an existing food to fortify the food's protein nutrition. Examples include food staples (grain, salt, sugar, cooking oil, margarine), beverages (coffee, tea, soda, beer, liquor, sports drinks), snacks, sweets and other foods.

In one embodiment, the formulations are filled into gelatin capsules for oral administration. An example of an appropriate capsule is a 250 mg gelatin capsule containing from 10 (up to 100 mg) of lyophilized powder (10^8 to 10^{11} bacteria), 160 mg microcrystalline cellulose, 77.5 mg gelatin, and 2.5 mg magnesium stearate. In an alternative embodiment, from 10^5 to 10^{12} bacteria may be used, 10^5 to 10^7, 10^6 to 10^7, or 10^8 to 10^{10}, with attendant adjustments of the excipients if necessary. In an alternative embodiment an enteric-coated capsule or tablet or with a buffering or protective composition may be used.

In one embodiment, the number of bacteria of each type may be present in the same amount or in different amounts. For example, in a bacterial composition with two types of bacteria, the bacteria may be present in from a 1:10,000 ratio to a 1:1 ratio, from a 1:10,000 ratio to a 1:1,000 ratio, from a 1:1,000 ratio to a 1:100 ratio, from a 1:100 ratio to a 1:50 ratio, from a 1:50 ratio to a 1:20 ratio, from a 1:20 ratio to a 1:10 ratio, from a 1:10 ratio to a 1:1 ratio. For bacterial compositions comprising at least three types of bacteria, the ratio of type of bacteria may be chosen pairwise from ratios for bacterial compositions with two types of bacteria. For example, in a bacterial composition comprising bacteria A, B, and C, at least one of the ratio between bacteria A and B, the ratio between bacteria B and C, and the ratio between bacteria A and C may be chosen, independently, from the pairwise combinations above.

Methods of Treating a Subject

In some embodiments the proteins and compositions disclosed herein are administered to a subject or a user (sometimes collectively referred to as a "subject"). As used herein "administer" and "administration" encompasses embodiments in which one person directs another to consume a bacterial composition in a certain manner and/or for a certain purpose, and also situations in which a user uses a bacteria composition in a certain manner and/or for a certain purpose independently of or in variance to any instructions received from a second person. Non-limiting examples of embodiments in which one person directs another to consume a bacterial composition in a certain manner and/or for a certain purpose include when a physician prescribes a course of conduct and/or treatment to a subject, when a parent commands a minor user (such as a child) to consume a bacterial composition, when a trainer advises a user (such as an athlete) to follow a particular course of conduct and/or treatment, and when a manufacturer, distributer, or marketer recommends conditions of use to an end user, for example through advertisements or labeling on packaging or on other materials provided in association with the sale or marketing of a product.

The bacterial compositions offer a protective and/or therapeutic effect against infection by one or more GI pathogens of interest and thus may be administered after an acute case of infection has been resolved in order to prevent relapse, during an acute case of infection as a complement to antibiotic therapy if the bacterial composition is not sensitive to the same antibiotics as the GI pathogen, or to prevent infection or reduce transmission from disease carriers. These pathogens include, but are not limited to, *Aeromonas hydrophila, Campylobacter fetus, Plesiomonas shigelloides, Bacillus cereus, Campylobacter jejuni, Clostridium botulinum, Clostridium difficile, Clostridium perfringens*, enteroaggregative *Escherichia coli*, enterohemorrhagic *Escherichia coli*, enteroinvasive *Escherichia coli*, enterotoxigenic *Escherichia coli* (LT and/or ST), *Escherichia coli* 0157: H7, *Helicobacter pylori, Klebsiella pneumonia, Lysteria monocytogenes, Plesiomonas shigelloides, Salmonella* spp., *Salmonella typhi, Shigella* spp., *Staphylococcus, Staphylococcus aureus*, vancomycin-resistant *Enterococcus* spp., *Vibrio* spp., *Vibrio cholerae, Vibrio parahaemolyticus, Vibrio vulnificus*, and *Yersinia enterocolitica.*

In one embodiment, the pathogen may be *Clostridium difficile, Salmonella* spp., pathogenic *Escherichia coli*, carbapenem-resistant Enterobacteriaceae (CRE), extended spectrum beta-lactam resistant Enterococci (ESBL) and vancomycin-resistant Enterococci (VRE). In yet another embodiment, the pathogen may be *Clostridium difficile.*

The present bacterial compositions may be useful in a variety of clinical situations. For example, the bacterial compositions may be administered alone, as a complementary treatment to antibiotics (e.g., when a subject is suffering from an acute infection, to reduce the risk of recurrence after an acute infection has subsided or, or when a subject will be in close proximity to others with or at risk of serious gastrointestinal infections (physicians, nurses, hospital workers, family members of those who are ill or hospitalized).

The present bacterial compositions may be administered to animals, including humans, laboratory animals (e.g., primates, rats, mice), livestock (e.g., cows, sheep, goats, pigs, turkeys, chickens), and household pets (e.g., dogs, cats, rodents).

In the present method, the bacterial composition is administered enterically, in other words by a route of access to the gastrointestinal tract. This includes oral administration, rectal administration (including enema, suppository, or colonoscopy), by an oral or nasal tube (nasogastric, nasojejunal, oral gastric, or oral jejunal), as detailed more fully herein.

It has been reported that a GI dysbiosis is associated with diabetes (Qin et al., 2012. Nature 490:55). In some embodiments, a composition provided herein can be used to alter the microbiota of a subject having or susceptible diabetes. Typically, such a composition provides at least one, two, or three OTUs identified in the art as associated with an improvement in insulin sensitivity or other sign or symptom associated with diabetes, e.g., Type 2 or Type 1 diabetes. In some embodiments, the composition is associated with an increase in engraftment and/or augmentation of at least one, two, or three OTUs associated with an improvement in at least one sign or symptom of diabetes.

Pretreatment Protocols

Prior to administration of the bacterial composition, the subject may optionally have a pretreatment protocol to prepare the gastrointestinal tract to receive the bacterial composition. In certain embodiments, the pretreatment protocol is advisable, such as when a subject has an acute infection with a highly resilient pathogen. In other embodiments, the pretreatment protocol is entirely optional, such as when the pathogen causing the infection is not resilient, or the subject has had an acute infection that has been successfully treated but where the physician is concerned that the infection may recur. In these instances, the pretreatment protocol may enhance the ability of the bacterial composition to affect the subject's microbiome.

As one way of preparing the subject for administration of the microbial ecosystem, at least one antibiotic may be administered to alter the bacteria in the subject. As another way of preparing the subject for administration of the microbial ecosystem, a standard colon-cleansing preparation may be administered to the subject to substantially empty the contents of the colon, such as used to prepare a subject for a colonscopy. By "substantially emptying the contents of the colon," this application means removing at least 75%, at least 80%, at least 90%, at least 95%, or about 100% of the contents of the ordinary volume of colon contents. Antibiotic treatment may precede the colon-cleansing protocol.

If a subject has received an antibiotic for treatment of an infection, or if a subject has received an antibiotic as part of a specific pretreatment protocol, in one embodiment the antibiotic should be stopped in sufficient time to allow the antibiotic to be substantially reduced in concentration in the gut before the bacterial composition is administered. In one embodiment, the antibiotic may be discontinued 1, 2, or 3 days before the administration of the bacterial composition. In one embodiment, the antibiotic may be discontinued 3, 4, 5, 6, or 7 antibiotic half-lives before administration of the bacterial composition. In another embodiment, the antibiotic may be chosen so the constituents in the bacterial composition have an MIC50 that is higher than the concentration of the antibiotic in the gut.

MIC50 of a bacterial composition or the elements in the composition may be determined by methods well known in the art. Reller et al., Antimicrobial Susceptibility Testing: A Review of General Principles and Contemporary Practices, Clinical Infectious Diseases 49 (11): 1749-1755 (2009). In such an embodiment, the additional time between antibiotic administration and administration of the bacterial composition is not necessary. If the pretreatment protocol is part of treatment of an acute infection, the antibiotic may be chosen so that the infection is sensitive to the antibiotic, but the constituents in the bacterial composition are not sensitive to the antibiotic.

Routes of Administration

The bacterial compositions of the invention are suitable for administration to mammals and non-mammalian animals in need thereof. In certain embodiments, the mammalian subject is a human subject who has one or more symptoms of a dysbiosis.

When the mammalian subject is suffering from a disease, disorder or condition characterized by an aberrant microbiota, the bacterial compositions described herein are suitable for treatment thereof. In some embodiments, the mammalian subject has not received antibiotics in advance of treatment with the bacterial compositions. For example, the mammalian subject has not been administered at least two doses of vancomycin, metronidazole and/or or similar antibiotic compound within one week prior to administration of the therapeutic composition. In other embodiments, the mammalian subject has not previously received an antibiotic compound in the one month prior to administration of the therapeutic composition. In other embodiments, the mammalian subject has received one or more treatments with one or more different antibiotic compounds and such treatment (s) resulted in no improvement or a worsening of symptoms.

In some embodiments, the gastrointestinal disease, disorder or condition is diarrhea caused by *C. difficile* including recurrent *C. difficile* infection, ulcerative colitis, colitis, Crohn's disease, or irritable bowel disease. Beneficially, the therapeutic composition is administered only once prior to improvement of the disease, disorder or condition. In some embodiments the therapeutic composition is administered at intervals greater than two days, such as once every three, four, five or six days, or every week or less frequently than every week. Or the preparation may be administered intermittently according to a set schedule, e.g., once a day, once weekly, or once monthly, or when the subject relapses from the primary illness. In another embodiment, the preparation may be administered on a long-term basis to subjects who are at risk for infection with or who may be carriers of these pathogens, including subjects who will have an invasive medical procedure (such as surgery), who will be hospitalized, who live in a long-term care or rehabilitation facility, who are exposed to pathogens by virtue of their profession (livestock and animal processing workers), or who could be carriers of pathogens (including hospital workers such as physicians, nurses, and other health care professionals).

In embodiments, the bacterial composition is administered enterically. This preferentially includes oral administration, or by an oral or nasal tube (including nasogastric, nasojejunal, oral gastric, or oral jejunal). In other embodiments, administration includes rectal administration (including enema, suppository, or colonoscopy). The bacterial composition may be administered to at least one region of the gastrointestinal tract, including the mouth, esophagus, stomach, small intestine, large intestine, and rectum. In some embodiments it is administered to all regions of the gastrointestinal tract. The bacterial compositions may be administered orally in the form of medicaments such as powders, capsules, tablets, gels or liquids. The bacterial compositions may also be administered in gel or liquid form by the oral route or through a nasogastric tube, or by the rectal route in a gel or liquid form, by enema or instillation through a colonoscope or by a suppository.

If the composition is administered colonoscopically and, optionally, if the bacterial composition is administered by other rectal routes (such as an enema or suppository) or even if the subject has an oral administration, the subject may have a colon-cleansing preparation. The colon-cleansing preparation can facilitate proper use of the colonoscope or other administration devices, but even when it does not serve a mechanical purpose it can also maximize the proportion of the bacterial composition relative to the other organisms previously residing in the gastrointestinal tract of the subject. Any ordinarily acceptable colon-cleansing preparation may be used such as those typically provided when a subject undergoes a colonoscopy.

Dosages and Schedule for Administration

In some embodiments the bacteria and bacterial compositions are provided in a dosage form. In some embodiments the dosage form is designed for administration of at least one OTU or combination thereof disclosed herein, wherein the total amount of bacterial composition administered is selected from 0.1 ng to 10 g, 10 ng to 1 g, 100 ng to 0.1 g, 0.1 mg to 500 mg, 1 mg to 100 mg, or from 10-15 mg. In some embodiments the bacterial composition is consumed at a rate of from 0.1 ng to 10 g a day, 10 ng to 1 g a day, 100 ng to 0.1 g a day, 0.1 mg to 500 mg a day, 1 mg to 100 mg a day, or from 10-15 mg a day, or more.

In some embodiments the treatment period is at least 1 day, at least 2 days, at least 3 days, at least 4 days, at least 5 days, at least 6 days, at least 1 week, at least 2 weeks, at least 3 weeks, at least 4 weeks, at least 1 month, at least 2 months, at least 3 months, at least 4 months, at least 5 months, at least 6 months, or at least 1 year. In some embodiments the treatment period is from 1 day to 1 week, from 1 week to 4 weeks, from 1 month, to 3 months, from 3 months to 6 months, from 6 months to 1 year, or for over a year.

In one embodiment, from 10^5 and 10^{12} microorganisms total may be administered to the subject in a given dosage form. In one mode, an effective amount may be provided in from 1 to 500 ml or from 1 to 500 grams of the bacterial composition having from 10^7 to 10^{11} bacteria per ml or per gram, or a capsule, tablet or suppository having from 1 mg to 1000 mg lyophilized powder having from 10^7 to 10^{11} bacteria. Those receiving acute treatment may receive higher doses than those who are receiving chronic administration (such as hospital workers or those admitted into long-term care facilities).

Any of the preparations described herein may be administered once on a single occasion or on multiple occasions, such as once a day for several days or more than once a day on the day of administration (including twice daily, three times daily, or up to five times daily). Or the preparation may be administered intermittently according to a set schedule, e.g., once weekly, once monthly, or when the subject relapses from the primary illness. In another embodiment, the preparation may be administered on a long-term basis to individuals who are at risk for infection with or who may be carriers of these pathogens, including individuals who will have an invasive medical procedure (such as surgery), who will be hospitalized, who live in a long-term care or rehabilitation facility, who are exposed to pathogens by virtue of their profession (livestock and animal processing workers), or who could be carriers of pathogens (including hospital workers such as physicians, nurses, and other health care professionals).

Subject Selection

Particular bacterial compositions may be selected for individual subjects or for subjects with particular profiles. For example, 16S sequencing may be performed for a given subject to identify the bacteria present in his or her microbiota. The sequencing may either profile the subject's entire microbiome using 16S sequencing (to the family, genera, or species level), a portion of the subject's microbiome using 16S sequencing, or it may be used to detect the presence or absence of specific candidate bacteria that are biomarkers for health or a particular disease state, such as markers of multi-drug resistant organisms or specific genera of concern such as *Escherichia*. Based on the biomarker data, a particular composition may be selected for administration to a subject to supplement or complement a subject's microbiota in order to restore health or treat or prevent disease. In another embodiment, subjects may be screened to determine the composition of their microbiota to determine the likelihood of successful treatment.

Combination Therapy

The bacterial compositions may be administered with other agents in a combination therapy mode, including anti-microbial agents and prebiotics. Administration may be sequential, over a period of hours or days, or simultaneous.

In one embodiment, the bacterial compositions are included in combination therapy with one or more anti-microbial agents, which include anti-bacterial agents, anti-fungal agents, anti-viral agents and anti-parasitic agents.

Anti-bacterial agents include cephalosporin antibiotics (cephalexin, cefuroxime, cefadroxil, cefazolin, cephalothin, cefaclor, cefamandole, cefoxitin, cefprozil, and ceftobiprole); fluoroquinolone antibiotics (cipro, Levaquin, floxin, tequin, avelox, and norflox); tetracycline antibiotics (tetracycline, minocycline, oxytetracycline, and doxycycline); penicillin antibiotics (amoxicillin, ampicillin, penicillin V, dicloxacillin, carbenicillin, vancomycin, and methicillin); and carbapenem antibiotics (ertapenem, doripenem, imipenem/cilastatin, and meropenem).

Anti-viral agents include Abacavir, Acyclovir, Adefovir, Amprenavir, Atazanavir, Cidofovir, Darunavir, Delavirdine, Didanosine, Docosanol, Efavirenz, Elvitegravir, Emtricitabine, Enfuvirtide, Etravirine, Famciclovir, Foscarnet, Fomivirsen, Ganciclovir, Indinavir, Idoxuridine, Lamivudine, Lopinavir Maraviroc, MK-2048, Nelfinavir, Nevirapine, Penciclovir, Raltegravir, Rilpivirine, Ritonavir, Saquinavir, Stavudine, Tenofovir Trifluridine, Valaciclovir, Valganciclovir, Vidarabine, Ibacitabine, Amantadine, Oseltamivir, Rimantidine, Tipranavir, Zalcitabine, Zanamivir and Zidovudine.

Examples of antifungal compounds include, but are not limited to polyene antifungals such as natamycin, rimocidin, filipin, nystatin, amphotericin B, candicin, and hamycin; imidazole antifungals such as miconazole, ketoconazole, clotrimazole, econazole, omoconazole, bifonazole, butoconazole, fenticonazole, isoconazole, oxiconazole, sertaconazole, sulconazole, and tioconazole; triazole antifungals such as fluconazole, itraconazole, isavuconazole, ravuconazole, posaconazole, voriconazole, terconazole, and albaconazole; thiazole antifungals such as abafungin; allylamine antifungals such as terbinafine, naftifine, and butenafine; and echinocandin antifungals such as anidulafungin, caspofungin, and micafungin. Other compounds that have antifungal properties include, but are not limited to polygodial, benzoic acid, ciclopirox, tolnaftate, undecylenic acid, flucytosine or 5-fluorocytosine, griseofulvin, and haloprogin.

In one embodiment, the bacterial compositions are included in combination therapy with one or more corticosteroids, mesalazine, mesalamine, sulfasalazine, sulfasalazine derivatives, immunosuppressive drugs, cyclosporin A, mercaptopurine, azathiopurine, prednisone, methotrexate, antihistamines, glucocorticoids, epinephrine, theophylline, cromolyn sodium, anti-leukotrienes, anti-cholinergic drugs for rhinitis, anti-cholinergic decongestants, mast-cell stabilizers, monoclonal anti-IgE antibodies, vaccines, and combinations thereof.

A prebiotic is a selectively fermented ingredient that allows specific changes, both in the composition and/or activity in the gastrointestinal microbiota that confers benefits upon host well being and health. Prebiotics may include complex carbohydrates, amino acids, peptides, or other essential nutritional components for the survival of the bacterial composition. Prebiotics include, but are not limited to, amino acids, biotin, fructooligosaccharide, galactooligosaccharides, inulin, lactulose, mannan oligosaccharides, oligofructose-enriched inulin, oligofructose, oligodextrose, tagatose, trans-galactooligosaccharide, and xylooligosaccharides.

Methods for Characterization of Bacterial Compositions

In certain embodiments, provided are methods for testing certain characteristics of bacterial compositions. For example, the sensitivity of bacterial compositions to certain environmental variables is determined, e.g., in order to select for particular desirable characteristics in a given composition, formulation and/or use. For example, the constituents in the bacterial composition may be tested for pH resistance, bile acid resistance, and/or antibiotic sensitivity, either individually on a constituent-by-constituent basis or collectively as a bacterial composition comprised of multiple bacterial constituents (collectively referred to in this section as bacterial composition).

pH Sensitivity Testing. If a bacterial composition will be administered other than to the colon or rectum (i.e., through, for example, but not limited to, an oral route), optionally testing for pH resistance enhances the selection of bacterial compositions that will survive at the highest yield possible through the varying pH environments of the distinct regions of the GI tract. Understanding how the bacterial compositions react to the pH of the GI tract also assists in formulation, so that the number of bacteria in a dosage form can be increased if beneficial and/or so that the composition may be administered in an enteric-coated capsule or tablet or with a buffering or protective composition. As the pH of the stomach can drop to a pH of 1 to 2 after a high-protein meal for a short time before physiological mechanisms adjust it to a pH of 3 to 4 and often resides at a resting pH of 4 to 5, and as the pH of the small intestine can range from a pH of 6 to 7.4, bacterial compositions can be prepared that survive these varying pH ranges (specifically wherein at least 1%, 5%, 10%, 15%, 20%, 25%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or as much as 100% of the bacteria can survive gut transit times through various pH ranges). This may be tested by exposing the bacterial composition to varying pH ranges for the expected gut transit times through those pH ranges. Therefore, as a nonlimiting example only, 18-hour cultures of bacterial compositions may be grown in standard media, such as gut microbiota medium ("GMM", see Goodman et al., Extensive personal human gut microbiota culture collections characterized and manipulated in gnotobiotic mice, PNAS 108 (15): 6252-6257 (2011)) or another animal-products-free medium, with the addition of pH adjusting agents for a pH of 1 to 2 for 30 minutes, a pH of 3 to 4 for 1 hour, a pH of 4 to 5 for 1 to 2 hours, and a pH of 6 to 7.4 for 2.5 to 3 hours. An alternative method for testing stability to acid is described in U.S. Pat. No. 4,839,281. Survival of bacteria may be determined by culturing the bacteria and counting colonies on appropriate selective or non-selective media.

Bile Acid Sensitivity Testing. Additionally, in some embodiments, testing for bile-acid resistance enhances the selection of bacterial compositions that will survive exposures to bile acid during transit through the GI tract. Bile acids are secreted into the small intestine and can, like pH, affect the survival of bacterial compositions. This may be tested by exposing the bacterial compositions to bile acids for the expected gut exposure time to bile acids. For example, bile acid solutions may be prepared at desired concentrations using 0.05 mM Tris at pH 9 as the solvent. After the bile acid is dissolved, the pH of the solution may be adjusted to 7.2 with 10% HCl. Bacterial compositions may be cultured in 2.2 ml of a bile acid composition mimicking the concentration and type of bile acids in the subject, 1.0 ml of 10% sterile-filtered stool media and 0.1 ml of an 18-hour culture of the given strain of bacteria. Incubations may be conducted for from 2.5 to 3 hours or longer. An alternative method for testing stability to bile acid is described in U.S. Pat. No. 4,839,281. Survival of bacteria may be determined by culturing the bacteria and counting colonies on appropriate selective or non-selective media.

Antibiotic Sensitivity Testing. As a further optional sensitivity test, bacterial compositions may be tested for sensitivity to antibiotics. In one embodiment, bacterial compositions may be chosen so that the bacterial constituents are sensitive to antibiotics such that if necessary they can be eliminated or substantially reduced from the subject's gastrointestinal tract by at least one antibiotic targeting the bacterial composition.

Adherence to Gastrointestinal Cells. The bacterial compositions may optionally be tested for the ability to adhere to gastrointestinal cells. A method for testing adherence to gastrointestinal cells is described in U.S. Pat. No. 4,839,281.

The specification is most thoroughly understood in light of the teachings of the references cited within the specification. The embodiments within the specification provide an illustration of embodiments and should not be construed to limit the scope. The skilled artisan readily recognizes that many other embodiments are encompassed. All publications and patents cited in this disclosure are incorporated by reference in their entirety. To the extent the material incorporated by reference contradicts or is inconsistent with this specification, the specification will supersede any such material. The citation of any references herein is not an admission that such references are prior art.

Unless otherwise indicated, all numbers expressing quantities of ingredients, reaction conditions, and so forth used in the specification, including claims, are to be understood as being modified in all instances by the term "about." Accordingly, unless otherwise indicated to the contrary, the numerical parameters are approximations and may vary depending upon the desired properties sought to be obtained. At the very least, and not as an attempt to limit the application of the doctrine of equivalents to the scope of the claims, each numerical parameter should be construed in light of the number of significant digits and ordinary rounding approaches.

Unless otherwise indicated, the term "at least" preceding a series of elements is to be understood to refer to every element in the series.

EXAMPLES

Below are examples of specific embodiments for carrying out the present invention. The examples are offered for illustrative purposes only, and are not intended to limit the scope of the present invention in any way. Efforts have been made to ensure accuracy with respect to numbers used (e.g., amounts, temperatures, etc.), but some experimental error and deviation should, of course, be allowed for. Examples of the techniques and protocols described herein with regard to therapeutic compositions can be found in, e.g., Remington's Pharmaceutical Sciences, 16th edition, Osol, A. (ed), 1980.

Example 1. Construction of Binary Pairs in a High-Throughput 96-Well Format/Plate Preparation

Pairs of bacteria were used to identify binary pairs useful for inhibition of *C. difficile*. To prepare plates for the high-throughput screening assay of binary pairs, vials of −80° C. glycerol stock bacterial banks were thawed and diluted to 1e8 CFU/mL. Each bacterial strain was then diluted 10× (to a final concentration of 1e7 CFU/mL of each strain) into 200 uL of PBS+15% glycerol in the wells of a 96-well plate. Plates were then frozen at −80° C. When needed, plates were removed from −80° C. and thawed at room temperature under anaerobic conditions for testing in a CivSim assay with *C. difficile.*

Example 2. Construction of Ternary Combinations in a High-Throughput 96-Well Format

Triplet combinations of bacteria were used to identify ternary combinations useful for inhibition of *C. difficile*. To prepare plates for high-throughput screening of ternary combinations, vials of −80° C. glycerol bacterial stock banks were thawed and diluted to 1e8 CFU/mL. Each bacterial strain was then diluted 10× (to a final concentration of 1e7 CFU/mL of each strain) into 200 uL of PBS+15% glycerol in the wells of a 96-well plate. Plates were then frozen at −80° C. When needed for the assay, plates were removed from −80° C. and thawed at room temperature under anaerobic conditions when testing in a CivSim assay with *Clostridium difficile.*

Example 3. Construction of a CivSim Assay to Screen for Bacterial Compositions Inhibitory to the Growth of *Clostridium difficile*

A competition assay (CivSim assay) was used to identify compositions that can inhibit the growth of *C. difficile*. Briefly, an overnight culture of *C. difficile* was grown under anaerobic conditions in SweetB-FosIn for the growth of *C. difficile*. In some cases, other suitable media can be used. SweetB-FosIn is a version of BHI (Remel R452472) supplemented with several components as follows: Components per liter: 37 g BHI powder (Remel R452472), supplemented with 5 g yeast extract UF (Difco 210929), 1 g cysteine-HCl (Spectrum C1473), 1 g cellobiose (Sigma C7252), 1 g maltose (Spectrum MA155), 1.5 ml hemin solution, 1 g soluble starch (Sigma-Aldrich S9765), 1 g fructooligosaccharides/inulin (Jarrow Formulas 103025) and 50 mL 1 M MOPS/KOH PH 7. To prepare the hemin solution, hemin (Sigma 51280) was dissolved in 0.1 M NaOH to make a 10 mg/mL stock.

After 24 hours of growth the culture was diluted 100,000 fold into a complex medium SweetB-FosIn. In some embodiments a medium is selected for use in which all desired organisms can grow, i.e., which is suitable for the growth of a wide variety of anaerobic, and, in some cases facultative anaerobic bacterial species. The diluted *C. difficile* mixture was then aliquoted to wells of a 96-well plate (180 uL to each well). 20 uL of a bacterial composition was then added to each well at a final concentration of 1e6 CFU/mL of each or two or three species. Alternatively the assay can be tested with binary pairs at different initial concentrations (1e9 CFU/mL, 1e8 CFU/mL, 1e7 CFU/mL, 1e5 CFU/mL, 1e4 CFU/mL, 1e3 CFU/mL, 1e2 CFU/mL). Control wells only inoculated with *C. difficile* were included for a comparison to the growth of *C. difficile* without inhibition. Additional wells were used for controls that either inhibit or do not inhibit the growth of *C. difficile*. One example of a positive control that inhibits growth was a combination of *Blautia producta, Clostridium bifermentans* and *Escherichia coli*. One example of a control that shows reduced inhibition of *C. difficile* growth as a combination of *Bacteroides thetaiotaomicron, Bacteroides ovatus* and *Bacteroides vulgatus*. Plates were wrapped with parafilm and incubated for 24 hours at 37° C. under anaerobic conditions. After 24 hours, the wells containing *C. difficile* alone were serially diluted and plated to determine titer. The 96-well plate was then frozen at −80 C before quantifying *C. difficile* by qPCR assay (see Example 6). Experimental combinations that inhibit *C. difficile* in this assay are useful in compositions for prevention or treatment of *C. difficile* infection.

Example 4. Construction of a CivSim Assay to Screen for Bacterial Compositions that Produce Diffusible Products Inhibitory to the Growth of *Clostridium difficile* Using a Filter Insert

To identify bacterial compositions that can produce diffusible products that inhibit *C. difficile* a modified CivSim assay was designed. In this experiment, the CivSim assay described above was modified by using a 0.22 uM filter insert (Millipore™ MultiScreen™ 96-Well Assay Plates—Item MAGVS2210) in 96-well format to physically separate *C. difficile* from the bacterial compositions. The *C. difficile* was aliquoted into the 96-well plate while the bacterial compositions were aliquoted into media on the filter overlay. The nutrient/growth medium is in contact on both sides of the 0.22 uM filter, allowing exchange of nutrients, small molecules and many macromolecules (e.g., bacteriocins, cell-surface proteins, or polysaccharides) by diffusion. In this embodiment, after a 24 hour incubation, the filter insert containing the bacterial compositions was removed. The plate containing *C. difficile* was then transferred to a 96-well plate reader suitable for measuring optical density (OD) at 600 nm. The growth of *C. difficile* in the presence of different bacterial compositions was compared based on the OD measurement. The results of these experiments demonstrated that compositions that can inhibit *C. difficile* when grown in shared medium under conditions that do not permit contact between the bacteria in the composition and *C. difficile* can be identified. Such compositions are candidates for producing diffusible products that are effective for treating *C. difficile* infection and can serve as part of a process for isolating such diffusible products, e.g., for use in treating infection.

Example 5. Construction of a CivSim Assay to Screen for Bacterial Compositions Inhibitory to the Growth of *Clostridium difficile* Using *Clostridium difficile* Selective Media for Quantification

The CivSim assay described above can be modified to determine final *C. difficile* titer by serially diluting and plating to *C. difficile* selective media (Bloedt et al. 2009) such as CCFA (cycloserine cefoxitin fructose agar, Anaerobe Systems), CDSA (*Clostridium difficile* selective agar, which is cycloserine cefoxitin mannitol agar, Becton Dickinson).

Example 6. Quantification of *C. difficile* Using Quantitative PCR (qPCR)

A. Standard Curve Preparation

To quantitate *C. difficile*, a standard curve was generated from a well on each assay plate in, e.g., a CivSim assay, containing only pathogenic *C. difficile* grown in SweetB+FosIn media as provided herein and quantified by selective spot plating. Serial dilutions of the culture were performed in sterile phosphate-buffered saline. Genomic DNA was extracted from the standard curve samples along with the other wells.

B. Genomic DNA Extraction

Genomic DNA was extracted from 5 μl of each sample using a dilution, freeze/thaw, and heat lysis protocol. 5 μL of thawed samples were added to 45 μL of UltraPure water (Life Technologies, Carlsbad, CA) and mixed by pipetting. The plates with diluted samples were frozen at −20° C. until use for qPCR which included a heated lysis step prior to amplification. Alternatively the genomic DNA could be isolated using the Mo Bio Powersoil®-htp 96 Well Soil DNA Isolation Kit (Mo Bio Laboratories, Carlsbad, CA), Mo Bio Powersoil® DNA Isolation Kit (Mo Bio Laboratories, Carlsbad, CA), or the QIAamp DNA Stool Mini Kit (QIAGEN, Valencia, CA) according to the manufacturer's instructions.

C. qPCR Composition and Conditions

The qPCR reaction mixture contained 1×SsoAdvanced Universal Probes Supermix, 900 nM of Wr-tcdB-F primer (AGCAGTTGAATATAGTGGTTTAGTTAGAGTTG (SEQ ID NO: 2033), IDT, Coralville, IA), 900 nM of Wr-tcdB-R primer (CATGCTTTTTTAGTTTCTGGATTGAA (SEQ ID NO: 2034), IDT, Coralville, IA), 250 nM of Wr-tcdB-P probe (6FAM-CATCCAGTCTCAATTGTATATGTTTCTCCA (SEQ ID NO: 2035)—MGB, Life Technologies, Grand Island, NY), and Molecular Biology Grade Water (Mo Bio Laboratories, Carlsbad, CA) to 18 μl (Primers adapted from: Wroblewski, D. et al., Rapid Molecular Characterization of *Clostridium difficile* and Assessment of Populations of *C. difficile* in Stool Specimens, Journal of Clinical Microbiology 47:2142-2148 (2009)). This reaction mixture was aliquoted to wells of a Hard-shell Low-Profile Thin Wall 96-well Skirted PCR Plate (BioRad, Hercules, CA). To this reaction mixture, 2 μl of diluted, frozen, and thawed samples were added and the plate sealed with a Microseal 'B' Adhesive Seal (BioRad, Hercules, CA). The qPCR was performed on a BioRad C1000™ Thermal Cycler equipped with a CFX96™ Real-Time System (BioRad, Hercules, CA). The thermocycling conditions were 95° C. for 15 minutes followed by 45 cycles of 95° C. for 5 seconds, 60° C. for 30 seconds, and fluorescent readings of the FAM channel. Alternatively, the qPCR could be performed with other standard methods known to those skilled in the art.

D. Data Analysis

The Cq value for each well on the FAM channel was determined by the CFX Manager™ 3.0 software. The $\log_{10}$ (cfu/mL) of *C. difficile* each experimental sample was calculated by inputting a given sample's Cq value into a linear regression model generated from the standard curve comparing the Cq values of the standard curve wells to the known $\log_{10}$ (cfu/mL) of those samples. The log inhibition was calculated for each sample by subtracting the $\log_{10}$ (cfu/mL) of *C. difficile* in the sample from the $\log_{10}$ (cfu/mL) of *C. difficile* in the sample on each assay plate used for the generation of the standard curve that has no additional bacteria added. The mean log inhibition was calculated for all replicates for each composition.

A histogram of the range and standard deviation of each composition was plotted. Ranges or standard deviations of the log inhibitions that were distinct from the overall distribution were examined as possible outliers. If the removal of a single log inhibition datum from one of the binary pairs that were identified in the histograms would bring the range or standard deviation in line with those from the majority of the samples, that datum was removed as an outlier, and the mean log inhibition was recalculated.

The pooled variance of all samples evaluated in the assay was estimated as the average of the sample variances weighted by the sample's degrees of freedom. The pooled standard error was then calculated as the square root of the pooled variance divided by the square root of the number of samples. Confidence intervals for the null hypothesis were determined by multiplying the pooled standard error to the z score corresponding to a given percentage threshold. Mean log inhibitions outside the confidence interval were considered to be inhibitory if positive or stimulatory if negative with the percent confidence corresponding to the interval used. Samples with mean log inhibition greater than the 99% confidence interval (C.I) of the null hypothesis are reported as ++++, those with a 95%<C.I.<99% as +++, those with a 90%<C.I.<95% as ++, those with a 80%<C.I.<90% as + while samples with mean log inhibition less than the 99% confidence interval (C.I) of the null hypothesis are reported as −−−, those with a 95%<C.I.<99% as −−−, those with a 90%<C.I.<95% as −−, those with a 80%<C.I.<90% as −.

Example 7. Inhibition of *C. difficile* Growth by Bacterial Compositions

Using methods described herein, binary pairs were identified that can inhibit *C. difficile* (see Table 4). 622 of 989 combinations showed inhibition with a confidence interval >80%; 545 of 989 with a C.I. >90%; 507 of 989 with a C.I. >95%; 430 of 989 with a C.I. of >99%. Non-limiting but exemplary binary pairs include those with mean log reduction greater than 0.366, e.g., *Allistipes shahii* paired with *Blautia producta, Clostridium hathaweyi*, or *Collinsella aerofaciens*, or *Clostridium mayombei* paired with *C. innocuum, C. tertium, Collinsella aerofaciens*, or any of the other 424 combinations shown in Table 4. Equally important, the CivSim assay describes binary pairs that do not effectively inhibit *C. difficile*. 188 of 989 combinations promote growth with >80% confidence; 52 of 989 show a lack of inhibition with >90% confidence; 22 of 989 show a lack of inhibition with >95% confidence; 3 of 989, including *B. producta* combined with *Coprococcus catus, Alistipes shahii* combined with *Dorea formicigenerans*, and *Eubacterium rectale* combined with *Roseburia intestinalis*, show a lack of inhibition with >99% confidence. 249 of 989 combinations are neutral in the assay, meaning they neither promote nor inhibit *C. difficile* growth to the limit of measurement.

Ternary combinations with mean log inhibition greater than the 99% confidence interval (C.I) of the null hypothesis are reported as ++++, those with a 95%<C.I.<99% as +++, those with a 90%<C.I.<95% as ++, those with a 80%<C.I.<90% as + while samples with mean log inhibition less than the 99% confidence interval (C.I) of the null hypothesis are reported as −−−, those with a 95%<C.I.<99% as −−−, those with a 90%<C.I.<95% as −−, those with a 80%<C.I.<90% as −.

The CivSim assay results demonstrate that many ternary combinations can inhibit *C. difficile* (Table 4). 516 of 632 ternary combinations show inhibition with a confidence interval >80%; 507 of 632 with a C.I. >90%; 496 of 632 with a C.I. >95%; 469 of 632 with a C.I. of >99%. Non-limiting but exemplary ternary combinations include those with a score of ++++, such as *Colinsella aerofaciens, Coprococcus comes*, and *Blautia producta*. The CivSim assay also describes ternary combinations that do not effectively inhibit *C. difficile*. 76 of 632 combinations promote growth with >80% confidence; 67 of 632 promote growth with >90% confidence; 61 of 632, promote growth with >95% confidence; and 49 of 632 combinations such as, but not limited to, *Clostridium orbiscendens, Coprococcus comes*, and *Faecalibacterium prausnitzii* promote growth with >99% confidence. 40 of 632 combinations are neutral in the assay, meaning they neither promote nor inhibit *C. difficile* growth to the limit of confidence.

Of the ternary combinations that inhibit *C. difficile* with >99% confidence, those that strongly inhibit *C. difficile* can be identified by comparing their mean log inhibition to the distribution of all results for all ternary combinations tested. Those above the 75th percentile can be considered to strongly inhibit *C. difficile*. Alternatively, those above the 50th, 60th, 70th, 80th, 90th, 95th, or 99th percentile can be considered to strongly inhibit *C. difficile*. Non-limiting but exemplary ternary combinations above the 75^{th} percentile include *Blautia producta, Clostridium tertium*, and *Ruminococcus gnavus* and *Eubacterium rectale, Clostridium mayombei*, and *Ruminococcus bromii*.

In addition to the demonstration that many binary and ternary combinations inhibit *C. difficile*, the CivSim demonstrates that many of these combinations synergistically inhibit *C. difficile*. Exemplary ternary combinations that demonstrate synergy in the inhibition of *C. difficile* growth include, but are not limited to, *Blautia producta, Clostridium innocuum, Clostridium orbiscendens* and *Colinsella aerofaciens, Blautia producta*, and *Eubacterium rectale*. Additional useful combinations are provided throughout, e.g., in Tables 4a, 4b, and 14-21.

Two higher-order bacterial compositions were tested in the CivSim assay for inhibition of *C. difficile*. N1962 (a.k.a. S030 and N1952), a 15 member composition, inhibited *C. difficile* by an average of 2.73 log 10 CFU/mL with a standard deviation of 0.58 log 10 CFU/mL while N1984 (a.k.a. S075), a 9 member composition, inhibited *C. difficile* by an average of 1.42 log 10 CFU/mL with a standard deviation of 0.45 log 10 CFU/mL.

These data collectively demonstrate that the CivSim assay can be used to identify compositions containing multiple species that are effective at inhibiting growth, that promote growth, or do not have an effect on growth of an organism, e.g., a pathogenic organism such as *C. difficile*.

Example 8. In Vivo Validation of Ternary Combinations' Efficacy in a Murine Model of *Clostridium difficile* Infection

To test the therapeutic potential of a bacterial composition such as but not limited to a spore population, a prophylactic mouse model of *C. difficile* infection was used (model based on Chen et al. 2008. A mouse model of *Clostridium difficile*-associated disease. Gastroenterology 135:1984-1992). Briefly, two cages of five mice each were tested for each arm of the experiment. All mice received an antibiotic cocktail consisting of 10% glucose, kanamycin (0.5 mg/ml), gentamicin (0.044 mg/ml), colistin (1062.5 U/ml), metronidazole (0.269 mg/ml), ciprofloxacin (0.156 mg/ml), ampicillin (0.1 mg/ml) and vancomycin (0.056 mg/ml) in their drinking water on days −14 through −5 and a dose of 10 mg/kg clindamycin by oral gavage on day −3. On day −1, test compositions were spun for 5 minutes at 12,100 rcf, their supernatants' removed, and the remaining pellets were resuspended in sterile PBS, prereduced if bacterial composition was not in spore form, and delivered via oral gavage. On day 0 the mice were challenged by administration of approximately 4.5 log 10 cfu of *C. difficile* (ATCC 43255) or sterile PBS (for the naive arm) via oral gavage. Mortality, weight and clinical scoring of *C. difficile* symptoms based upon a 0-4 scale by combining scores for appearance (0-2 points based on normal, hunched, piloerection, or lethargic), and clinical signs (0-2 points based on normal, wet tail, cold-to-the-touch, or isolation from other animals), with a score of 4 in the case of death, were assessed every day from day −2 through day 6. Mean minimum weight relative to day −1 and mean maximum clinical score as well as average cumulative mortality were calculated. Reduced mortality, increased mean minimum weight relative to day −1, and reduced mean maximum clinical score with death assigned to a score of 4 relative to the vehicle control were used to assess the ability of the test composition to inhibit infection by *C. difficile*.

Ternary combinations were tested in the murine model described above at 1e9 CFU/mL per strain. The results are shown in Table 5. The data demonstrate that the CivSim assay results are highly predictive of the ability of a combination to inhibit weight loss in *C. difficile* infection. Weight loss in this model is generally considered to be indicative of disease.

In one embodiment, compositions to screen for efficacy in vivo can be selected by ranking the compositions based on a functional metric such as but not limited to in vitro growth inhibition scores; compositions that are ranked ≥ the 75th percentile can be considered to strongly inhibit growth and be selected for in vivo validation of the functional phenotype. In other embodiments, compositions above the 50th, 60th, 70th, 80th, 90th, 95th, or 99th percentile can be considered to be the optimal candidates. In another embodiment, combinations with mean log inhibition greater than the 99% confidence interval (C.I) of the null hypothesis are selected. In other embodiments, compositions greater than the 95%, 90%, 85%, or 80% confidence interval (C.I.) are selected. In another embodiment, compositions demonstrated to have synergistic inhibition are selected (see Example 7) for testing in an in vivo model such as that described above.

Compositions selected to screen for efficacy in in vivo models can also be selected using a combination of growth inhibition metrics. In a non-limiting example: (i) compositions are selected based on their log inhibition being greater than the 99% confidence interval (C.I.) of the null hypothesis, (ii) the selected subset of compositions is further selected to represent those that are ranked ≥ the 75th percentile in the distribution of all inhibition scores, (iii) the subset of (ii) is then further selected based on compositions that demonstrate synergistic inhibition. In some embodiments, different confidence intervals (C.I.) and percentiles are used to create the composition subsets, e.g., see Table 4b.

Of the twelve exemplary ternary combinations selected, all were demonstrated to inhibit *C. difficile* in the CivSim assay (see Example 6) with >99% confidence. Ten of the twelve compositions demonstrated a protective effect when compared to a vehicle control with respect to the Mean Minimum Relative Weight. All twelve compositions outperformed vehicle with respect to Mean Maximum Clinical Score while eleven of twelve compositions surpassed the vehicle control by Cumulative Mortality. A non-limiting, but exemplary ternary combination, *Collinsella aerofaciens, Clostridium buytricum*, and *Ruminococcus gnavus*, was protective against symptoms of *C. difficile* infection, producing a Mean Minimum Relative Weight of 0.96, a Mean Maximum Clinical Score of 0.2, and Cumulative Mortality of 0% compared to the vehicle control of 0.82, 2.6, and 30%, respectively. These results demonstrate that the in vitro CivSim assay can be used to identify compositions that are protective in an in vivo murine model. This is surprising given the inherent dynamic nature of in vivo biological systems and the inherent simplification of the in vitro assays; it would not be expected that there is a direct correlation of in vitro in in vivo measures of inhibition and efficacy. This is in part because of the complexity of the in vivo system into which a composition is administered for treatment in which it might have been expected that confounding factors would obscure or affect the ability of a composition deemed effective in vitro to be effective in vivo.

Example 9. Construction of a CivSim Assay to Screen for Bacterial Compositions Inhibitory to the Growth of Vancomycin-Resistant *Enterococcus* (VRE) Using Vancomycin-Resistant *Enterococcus* Selective Medium for Quantification and Composition Screening

To determine the ability of a composition to compete with a pathogenic bacterium, e.g., vancomycin-resistant *Enterococcus*, a competition assay was developed. In these experiments, an overnight culture of a vancomycin-resistant strain of *Enterococcus faecium* was grown anaerobically in SweetB-FosIn for 24 hours. A glycerol stock of a bacterial composition was thawed from −80° C. and diluted to 1e6 CFU/mL per strain in SweetB-FosIn in the appropriate wells of a 96-well plate. The plate was incubated anaerobically at 37° C. for 1 hour to allow the previously frozen bacteria to revive. After the 1 hour initial incubation, VRE was inoculated into appropriate wells at target concentrations of 1e2 or 1e3 CFU/mL. Wells were also inoculated with VRE alone, without a bacterial composition. The plate was incubated anaerobically at 37° C. for 24 hours. Aliquots were removed at 15 hours and 24 hours and the VRE titers determined. At each time-point, well contents were serially diluted and plated to agar plates selective for VRE (Enterococcosel Agar+8 ug/mL vancomycin hydrochloride) (Enterococcosel Agar from BBL 212205, vancomycin hydrochloride from Sigma 94747). The selective plates were incubated aerobically at 37° C. for 24 hours before counting colonies to determine final titer of VRE in each well of the CivSim plate. Log Inhibition of VRE was determined by subtracting the final titer of a competition well from the final titer of a well containing VRE alone. Multiple ratios of the starting concentrations of VRE and bacterial compositions were tested to optimize for conditions resulting in the greatest signal. A competition time of 15 hours, a starting concentration of VRE at 1e2 CFU/mL and a starting concentration of N1962 (a.k.a. S030 and N1952) at 1e6 CFU/mL showed the greatest inhibition of growth compared to control.

Using the conditions described above, one 15-member and 44 heterotrimeric bacterial compositions were tested in the assay, the results of which are provided in Tables 4 and

6. Of the 44 heterotrimeric compositions tested, 43 inhibited VRE with >80% confidence, 41 inhibited VRE with >95% confidence, and 39 inhibited VRE with >99% confidence. One ternary composition tested did not demonstrate inhibition or induction with >80% confidence.

Of the ternary combinations that inhibit VRE with >99% confidence, those that strongly inhibit VRE can be identified by comparing their mean log inhibition to the distribution of all results for all ternary combinations tested. Those above the 75th percentile can be considered to strongly inhibit VRE. Alternatively, those above the 50th, 60th, 70th, 80th, 90th, 95th, or 99th percentile can be considered to strongly inhibit VRE. Non-limiting but exemplary ternary combinations that inhibit VRE with >99% confidence and above the 75th percentile include *Blautia producta, Clostridium innocuum*, and *Ruminococcus gnavus* and *Blautia producta, Clostridium butyricum*, and *Clostridium hylemonde.*

The 15-member composition, N1962 (a.k.a. S030 and N1952), inhibited VRE by at least 0.7 log 10 CFU/mL across all of the conditions tested and demonstrating inhibition of 5.7 log 10 CFU/mL in the optimal conditions.

These data demonstrate methods of identifying compositions useful for prophylaxis and treatment of VRE infection.

Example 10. Construction of a CivSim Assay to Screen for Bacterial Compositions Inhibitory to the Growth of *Klebsiella pneumoniae* Using *Klebsiella* Selective Medium for Quantification

To determine the ability of a composition to compete with a pathogenic bacterium, e.g., *Klebsiella pneumoniae*, a competition assay was developed. In these experiments, an overnight culture of a vancomycin-resistant strain of *Klebsiella pneumoniae* was grown anaerobically in SweetB-FosIn for 24 hours. A glycerol stock of a bacterial composition (N1962) was thawed from −80° C. and diluted to 1e6 CFU/mL per strain in SweetB-FosIn in the appropriate wells of a 96-well plate. The plate was incubated anaerobically at 37° C. for 1 hour to allow the previously frozen bacteria to revive. After the 1 hour initial incubation, *K. pneumoniae* was inoculated into appropriate wells at target concentrations of 1e2 or 1e3 CFU/mL. Wells were also inoculated with *K. pneumoniae* alone, without a bacterial composition. The plate was incubated anaerobically at 37° C. for 24 hours. Aliquots were removed at 15 hours and 24 hours to titer for the final concentration of *K. pneumoniae* at the end of competition. At each time-point, wells were serially diluted and plated to agar plates selective for *K. pneumoniae* (MacConkey Lactose Agar, Teknova M0149). The selective plates were incubated aerobically at 37° C. for 24 hours before counting colonies to determine final titer of *K. pneumoniae* in each well of the CivSim plate. Log Inhibition of *K. pneumoniae* was determined by subtracting the final titer of a competition well from the final titer of a well containing *K. pneumoniae* alone. Multiple ratios of the starting concentrations of *K. pneumoniae* and bacterial compositions were tested to optimize for conditions giving the greatest signal. The results of the assay are provided in Table 7. A competition time of 15 hours, a starting concentration of *K. pneumoniae* at 1e2 CFU/mL and a starting concentration of N1962 (a.k.a. S030 and N1952) at 1e6 CFU/mL showed the greatest inhibition of growth compared to control. N1962 (a.k.a. S030 and N1952) inhibited *K. pneumoniae* by 0.1-4.2 log 10 CFU/mL across the conditions tested.

Example 11. Construction of a CivSim Assay to Screen for Bacterial Compositions Inhibitory to the Growth of *Morganella morganii* Using *Morganella* Selective Media for Quantification

To determine the ability of a composition to compete with a pathogenic bacterium, e.g., *Morganella morganii*, a competition assay was developed. In this experiment, an overnight culture of a vancomycin-resistant strain of *Morganella morganii* was grown anaerobically in SweetB-FosIn for 24 hours. A glycerol stock of a bacterial composition, N1962, was thawed from −80° C. and diluted to 1e6 CFU/mL per strain in SweetB-FosIn in the appropriate wells of a 96-well plate. The plate was incubated anaerobically at 37° C. for 1 hour to allow the previously frozen bacteria to revive. After the 1 hour initial incubation, *M. morganii* was inoculated into appropriate wells at target concentrations of 1e2 or 1e3 CFU/mL. Wells were also inoculated with *M. morganii* alone, without a bacterial composition. The plate was incubated anaerobically at 37° C. for 24 hours. Aliquots were removed at 15 hours and 24 hours to titer for the final concentration of *M. morganii* at the end of competition. At each time-point, wells were serially diluted and plated to agar plates selective for *M. morganii* (MacConkey Lactose Agar, Teknova M0149). The selective plates were incubated aerobically at 37° C. for 24 hours before counting colonies to determine final titer of *M. morganii* in each well of the CivSim plate. Log Inhibition of *M. morganii* was determined by subtracting the final titer of a competition well from the final titer of a well containing *M. morganii* alone. Multiple ratios of the starting concentrations of *M. morganii* and bacterial compositions were tested to optimize for conditions providing the greatest signal. A competition time of 15 hours, a starting concentration of *M. morganii* at 1e2 CFU/mL and a starting concentration of N1962 (a.k.a. S030 and N1952) at 1e6 CFU/mL showed the greatest inhibition of growth compared to control.

A 15-member bacterial composition, N1962 (a.k.a. S030 and N1952), was tested in the assay, the results of which are provided in Table 8. N1962 (a.k.a. S030 and N1952) inhibited *M. morganii* by 1.4 to 5.8 log 10 CFU/mL across the conditions tested.

Example 12. Sequence-Based Genomic Characterization of Operational Taxonomic Units (OTU) and Functional Genes

Method for Determining 16S rDNA Gene Sequence

As described above, OTUs are defined either by full 16S sequencing of the rDNA gene, by sequencing of a specific hypervariable region of this gene (i.e., V1, V2, V3, V4, V5, V6, V7, V8, or V9), or by sequencing of any combination of hypervariable regions from this gene (e.g., V1-3 or V3-5). The bacterial 16S rDNA gene is approximately 1500 nucleotides in length and is used in reconstructing the evolutionary relationships and sequence similarity of one bacterial isolate to another using phylogenetic approaches. 16S sequences are used for phylogenetic reconstruction as they are in general highly conserved, but contain specific hypervariable regions that harbor sufficient nucleotide diversity to differentiate genera and species of most microbes. rDNA gene sequencing methods are applicable to both the analysis of non-enriched samples, but also for identification of microbes after enrichment steps that either enrich the microbes of interest from a microbial composition or a microbial sample and/or the nucleic acids that harbor the appropriate rDNA gene sequences as described below. For example, enrichment treatments prior to 16S rDNA gene characterization will increase the sensitivity of 16S as well as other molecular-based characterization nucleic acid purified from the microbes.

Using techniques known in the art, to determine the full 16S sequence or the sequence of any hypervariable region of the 16S rDNA sequence, genomic DNA is extracted from a bacterial sample, the 16S rDNA (full region or specific hypervariable regions) amplified using polymerase chain reaction (PCR), the PCR products cleaned, and nucleotide sequences delineated to determine the genetic composition of 16S gene or subdomain of the gene. If full 16S sequencing is performed, the sequencing method used may be, but is not limited to, Sanger sequencing. If one or more hypervariable regions are used, such as the V4 region, the sequencing may be, but is not limited to being, performed using the Sanger method or using a next-generation sequencing method, such as an Illumina (sequencing by synthesis) method using barcoded primers allowing for multiplex reactions.

Method for Determining 18S rDNA and ITS Gene Sequence

Methods to assign and identify fungal OTUs by genetic means can be accomplished by analyzing 18S sequences and the internal transcribed spacer (ITS). The rRNA of fungi that forms the core of the ribosome is transcribed as a single gene and consists of the 8S, 5.8S and 28S regions with ITS4 and 5 between the 8S and 5.8S and 5.8S and 28S regions, respectively. These two intercistronic segments between the 18S and 5.8S and 5.8S and 28S regions are removed by splicing and contain significant variation between species for barcoding purposes as previously described (Schoch et al. Nuclear ribosomal internal transcribed spacer (ITS) region as a universal DNA barcode marker for Fungi. PNAS USA 109:6241-6246. 2012). 18S rDNA is typically used for phylogenetic reconstruction however the ITS can serve this function as it is generally highly conserved but contains hypervariable regions that harbor sufficient nucleotide diversity to differentiate genera and species of most fungus.

Using techniques known in the art, to determine the full 18S and ITS sequences or a smaller hypervariable section of these sequences, genomic DNA is extracted from a microbial sample, the rDNA amplified using polymerase chain reaction (PCR), the PCR products cleaned, and nucleotide sequences delineated to determine the genetic composition rDNA gene or subdomain of the gene. The sequencing method used may be, but is not limited to, Sanger sequencing or using a next-generation sequencing method, such as an Illumina (sequencing by synthesis) method using barcoded primers allowing for multiplex reactions.

Method for Determining Other Marker Gene Sequences

In addition to the 16S and 18S rDNA gene, an OTU can be defined by sequencing a selected set of genes or portions of genes that are known marker genes for a given species or taxonomic group of OTUs. These genes may alternatively be assayed using a PCR-based screening strategy. For example, various strains of pathogenic *Escherichia coli* can be distinguished using DNAs from the genes that encode heat-labile (LTI, LTIIa, and LTIIb) and heat-stable (STI and STII) toxins, verotoxin types 1, 2, and 2e (VT1, VT2, and VT2e, respectively), cytotoxic necrotizing factors (CNF1 and CNF2), attaching and effacing mechanisms (eaeA), enteroaggregative mechanisms (Eagg), and enteroinvasive mechanisms (Einv). The optimal genes to utilize for taxonomic assignment of OTUs by use of marker genes will be familiar to one with ordinary skill in the art of sequence based taxonomic identification.

Genomic DNA Extraction

Genomic DNA can be extracted from pure or enriched microbial cultures using a hot alkaline lysis method. For example, 1 μl of microbial culture is added to 9 μl of Lysis Buffer (25 mM NaOH, 0.2 mM EDTA) and the mixture is incubated at 95° C. for 30 minutes. Subsequently, the samples are cooled to 4° C. and neutralized by the addition of 10 μl of Neutralization Buffer (40 mM Tris-HCl) and then diluted 10-fold in Elution Buffer (10 mM Tris-HCl). Alternatively, genomic DNA is extracted from pure or enriched microbial cultures using commercially available kits such as the Mo Bio Ultraclean® Microbial DNA Isolation Kit (Mo Bio Laboratories, Carlsbad, CA) or by methods known to those skilled in the art. For fungal samples, DNA extraction can be performed by methods described previously (e.g., see US20120135127) for producing lysates from fungal fruiting bodies by mechanical grinding methods.

Amplification of 16S Sequences for Downstream Sanger Sequencing

To amplify bacterial 16S rDNA (e.g., in FIG. **2** and FIG. **3**), 2 μl of extracted gDNA is added to a 20 μl final volume PCR reaction. For full-length 16 sequencing the PCR reaction also contains 1× HotMaster®Mix (5PRIME, Gaithersburg, MD), 250 nM of 27f (AGRGTTTGATCMTGGCTCAG (SEQ ID NO: 2036), IDT, Coralville, IA), and 250 nM of 1492r (TACGGYTACCTTGTTAYGACTT (SEQ ID NO: 2037), IDT, Coralville, IA), with PCR Water (Mo Bio Laboratories, Carlsbad, CA) for the balance of the volume.

FIG. **2** shows the hypervariable regions mapped onto a 16s sequence and the sequence regions corresponding to these sequences on a sequence map. A schematic is shown of a 16S rDNA gene and the figure denotes the coordinates of hypervariable regions 1-9 (V1-V9), according to an embodiment of the invention. Coordinates of V1-V9 are 69-99, 137-242, 433-497, 576-682, 822-879, 986-1043, 1117-1173, 1243-1294, and 1435-1465 respectively, based on numbering using *E. coli* system of nomenclature defined by Brosius et al. (Complete nucleotide sequence of a 16S ribosomal RNA gene (16S rRNA) from *Escherichia coli*, PNAS USA 75 (10): 4801-4805. 1978).

Alternatively, other universal bacterial primers or thermostable polymerases known to those skilled in the art are used. For example, primers are available to those skilled in the art for the sequencing of the "V1-V9 regions" of the 16S rDNA (e.g., FIG. **2**). These regions refer to the first through ninth hypervariable regions of the 16S rDNA gene that are used for genetic typing of bacterial samples. These regions in bacteria are defined by nucleotides 69-99, 137-242, 433-497, 576-682, 822-879, 986-1043, 1117-1173, 1243-1294 and 1435-1465 respectively using numbering based on the *E. coli* system of nomenclature. See Brosius et al., 1978, supra. In some embodiments, at least one of the V1, V2, V3, V4, V5, V6, V7, V8, and V9 regions are used to characterize an OTU. In one embodiment, the V1, V2, and V3 regions are used to characterize an OTU. In another embodiment, the V3, V4, and V5 regions are used to characterize an OTU. In another embodiment, the V4 region is used to characterize an OTU. A person of ordinary skill in the art can identify the specific hypervariable regions of a candidate 16S rDNA (e.g., FIG. **2**) by comparing the candidate sequence in question to the reference sequence (as in FIG. **3**) and identifying the hypervariable regions based on similarity to the reference hypervariable regions. FIG. **3** highlights in bold the nucleotide sequences for each hypervariable region in the exemplary reference *E. coli* 16S sequence described by Brosius et al., supra.

The PCR is typically performed on commercially available thermocyclers such as a BioRad MyCycler™ Thermal Cycler (BioRad, Hercules, CA). The reactions are run at 94° C. for 2 minutes followed by 30 cycles of 94° C. for 30 seconds, 51° C. for 30 seconds, and 68° C. for 1 minute 30 seconds, followed by a 7 minute extension at 72° C. and an indefinite hold at 4° C. Following PCR, gel electrophoresis of a portion of the reaction products is used to confirm successful amplification of a ~1.5 kb product.

To remove nucleotides and oligonucleotides from the PCR products, 2 μl of HT ExoSap-IT® (Affymetrix, Santa Clara, CA) is added to 5 μl of PCR product followed by a 15 minute incubation at 37° C. and then a 15 minute inactivation at 80° C.

Amplification of 16S Sequences for Downstream Characterization By Massively Parallel Sequencing Technologies

Amplification performed for downstream sequencing by short read technologies such as Illumina require amplification using primers known to those skilled in the art that additionally include a sequence-based barcoded tag. For example, to amplify the 16s hypervariable region V4 region of bacterial 16S rDNA, 2 μl of extracted gDNA is added to a 20 μl final volume PCR reaction. The PCR reaction also contains 1× HotMasterMix (5PRIME, Gaithersburg, MD), 200 nM of V4_515f_adapt (AATGATACGGCGACCACCGAGATCTACACTATGGTAATTGTGTGCCAGCMGCCGCGGT AA (SEQ ID NO: 2038), IDT, Coralville, IA), and 200 nM of barcoded 806rbc (CAAGCAGAAGACGGCATACGAGAT_12bpGolayBarcode_AGTCAGTCAGCCGGACTACHV GGGTWTCTAAT (SEQ ID NO: 2039), IDT, Coralville, IA), with PCR Water (Mo Bio Laboratories, Carlsbad, CA) for the balance of the volume. In the preceding primer sequences non-ACTG nucleotide designations refer to conventional degenerate codes as are used in the art. These primers incorporate barcoded adapters for Illumina sequencing by synthesis. Optionally, identical replicate, triplicate, or quadruplicate reactions may be performed. Alternatively other universal bacterial primers or thermostable polymerases known to those skilled in the art are used to obtain different amplification and sequencing error rates as well as results on alternative sequencing technologies.

The PCR amplification is performed on commercially available thermocyclers such as a BioRad MyCycler™ Thermal Cycler (BioRad, Hercules, CA). The reactions are run at 94° C. for 3 minutes followed by 25 cycles of 94° C. for 45 seconds, 50° C. for 1 minute, and 72° C. for 1 minute 30 seconds, followed by a 10 minute extension at 72° C. and a indefinite hold at 4° C. Following PCR, gel electrophoresis of a portion of the reaction products is used to confirm successful amplification of a ~1.5 kb product. PCR cleanup is performed as described above.

Sanger Sequencing of Target Amplicons from Pure Homogeneous Samples

To detect nucleic acids for each sample, two sequencing reactions are performed to generate a forward and reverse sequencing read. For full-length 16s sequencing primers 27f and 1492r are used. 40 ng of ExoSap-IT-cleaned PCR products are mixed with 25 pmol of sequencing primer and Mo Bio Molecular Biology Grade Water (Mo Bio Laboratories, Carlsbad, CA) to 15 μl total volume. This reaction is submitted to a commercial sequencing organization such as Genewiz (South Plainfield, NJ) for Sanger sequencing.

Amplification of 18S and ITS Regions for Downstream Sequencing

To amplify the 18S or ITS regions, 2 μL fungal DNA were amplified in a final volume of 30 μL with 15 μL AmpliTaq Gold 360 Mastermix, PCR primers, and water. The forward and reverse primers for PCR of the ITS region are 5'-TCCTCCGCTTATTGATATGC-3' (SEQ ID NO: 2040) and 5'-GGAAGTAAAAGTCGTAACAAGG-3' (SEQ ID NO: 2041) and are added at 0.2 uM concentration each. The forward and reverse primers for the 18s region are 5'-GTAGTCATATGCTTGTCTC-3' (SEQ ID NO: 2042) and 5'-CTTCCGTCAATTCCTTTAAG-3' (SEQ ID NO: 2043) and are added at 0.4 uM concentration each. PCR is performed with the following protocol: 95° C. for 10 minutes, 35 cycles of 95° C. for 15 seconds, 52° C. for 30 seconds, 72° C. for 1.5 seconds; and finally 72° C. for 7 minutes followed by storage at 4° C. All forward primers contained the M13F-20 sequencing primer, and reverse primers included the M13R-27 sequencing primer. PCR products (3 μL) were enzymatically cleaned before cycle sequencing with 1 μL ExoSap-IT and 1 μL Tris EDTA and incubated at 37° C. for 20 minutes followed by 80° C. for 15 minutes. Cycle sequencing reactions contained 5 μL cleaned PCR product, 2 μL BigDye® Terminator v3.1 Ready Reaction Mix, 1 μL 5× Sequencing Buffer, 1.6 pmol of appropriate sequencing primers designed by one skilled in the art, and water in a final volume of 10 μL. The standard cycle sequencing protocol is 27 cycles of 10 seconds at 96° C., 5 seconds at 50° C., 4 minutes at 60° C., and hold at 4° C. Sequencing cleaning is performed with the BigDye XTerminator Purification Kit as recommended by the manufacturer for 10 μL volumes. The genetic sequence of the resulting 18S and ITS sequences is performed using methods familiar to one with ordinary skill in the art using either Sanger sequencing technology or next-generation sequencing technologies such as but not limited to Illumina.

Preparation of Extracted Nucleic Acids for Metagenomic Characterization by Massively Parallel Sequencing Technologies

Extracted nucleic acids (DNA or RNA) are purified and prepared by downstream sequencing using standard methods familiar to one with ordinary skill in the art and as described by the sequencing technology's manufactures instructions for library preparation. In short, RNA or DNA are purified using standard purification kits such as but not limited to Qiagen's RNeasy® Kit or Promega's Genomic DNA purification kit. For RNA, the RNA is converted to cDNA prior to sequence library construction. Following purification of nucleic acids, RNA is converted to cDNA using reverse transcription technology such as but not limited to Nugen Ovation® RNA-Seq System or Illumina Truseq as per the manufacturer's instructions. Extracted DNA or transcribed cDNA are sheared using physical (e.g., Hydroshear), acoustic (e.g., Covaris), or molecular (e.g., Nextera) technologies and then size selected as per the sequencing technologies manufacturer's recommendations. Following size selection, nucleic acids are prepared for sequencing as per the manufacturer's instructions for sample indexing and sequencing adapter ligation using methods familiar to one with ordinary skill in the art of genomic sequencing.

Massively Parallel Sequencing of Target Amplicons from Heterogeneous Samples

DNA Quantification & Library Construction

The cleaned PCR amplification products are quantified using the Quant-iT™ PicoGreen® dsDNA Assay Kit (Life Technologies, Grand Island, NY) according to the manufacturer's instructions. Following quantification, the barcoded cleaned PCR products are combined such that each distinct PCR product is at an equimolar ratio to create a prepared Illumina library.

Nucleic Acid Detection

The prepared library is sequenced on Illumina HiSeq or MiSeq sequencers (Illumina, San Diego, CA) with cluster generation, template hybridization, isothermal amplification, linearization, blocking and denaturation and hybridization of the sequencing primers performed according to the manufacturer's instructions. 16SV4SeqFw (TATGGTAATTGTGTGCCAGCMGCCGCGGTAA (SEQ ID NO: 2044)), 16SV4SeqRev (AGTCAGTCAGCCGGACTACHVGGGTWTCTAAT (SEQ ID NO: 2045)), and 16SV4Index (ATTAGAWACCCBDGTAGTCCGGCTGACTGACT (SEQ ID NO: 2046)) (IDT, Coralville, IA) are used for sequencing. Other sequencing technologies can be used such as but not limited to 454, Pacific Biosciences, Helicos, Ion Torrent, and Nanopore using protocols that are standard to someone skilled in the art of genomic sequencing.

Example 13. Sequence Read Annotation

Primary Read Annotation

Nucleic acid sequences are analyzed and annotated to define taxonomic assignments using sequence similarity and phylogenetic placement methods or a combination of the two strategies. A similar approach can be used to annotate protein names, protein function, transcription factor names, and any other classification schema for nucleic acid sequences. Sequence similarity based methods include those familiar to individuals skilled in the art including, but not limited to BLAST, BLASTx, BLASTn, tBLASTx, RDP-classifier, DNAclust, and various implementations of these algorithms such as Qiime or Mothur. These methods rely on mapping a sequence read to a reference database and selecting the match with the best score and e-value. Common databases include, but are not limited to the Human Microbiome Project, NCBI non-redundant database, Greengenes, RDP, and Silva for taxonomic assignments. For functional assignments reads are mapped to various functional databases such as but not limited to COG, KEGG, BioCyc, and MetaCyc. Further functional annotations can be derived from 16S taxonomic annotations using programs such as PICRUST (M. Langille, et al. 2013. Nature Biotechnology 31, 814-821). Phylogenetic methods can be used in combination with sequence similarity methods to improve the calling accuracy of an annotation or taxonomic assignment. Tree topologies and nodal structure are used to refine the resolution of the analysis. In this approach we analyze nucleic acid sequences using one of numerous sequence similarity approaches and leverage phylogenetic methods that are known to those skilled in the art, including but not limited to maximum likelihood phylogenetic reconstruction (see e.g., Liu et al., 2011. RAxML and FastTree: Comparing Two Methods for Large-Scale Maximum Likelihood Phylogeny Estimation. PLOS ONE 6: e27731; McGuire et al., 2001. Models of sequence evolution for DNA sequences containing gaps. Mol. Biol. Evol 18:481-490; Wróbel B. 2008. Statistical measures of uncertainty for branches in phylogenetic trees inferred from molecular sequences by using model-based methods. J. Appl. Genet. 49:49-67). Sequence reads (e.g., 16S, 18S, or ITS) are placed into a reference phylogeny comprised of appropriate reference sequences. Annotations are made based on the placement of the read in the phylogenetic tree. The certainty or significance of the OTU annotation is defined based on the OTU's sequence similarity to a reference nucleic acid sequence and the proximity of the OTU sequence relative to one or more reference sequences in the phylogeny. As an example, the specificity of a taxonomic assignment is defined with confidence at the level of Family, Genus, Species, or Strain with the confidence determined based on the position of bootstrap supported branches in the reference phylogenetic tree relative to the placement of the OTU sequence being interrogated. Nucleic acid sequences can be assigned functional annotations using the methods described above.

Clade Assignments

Clade assignments were generally made using full-length sequences of 16S rDNA and of V4. The ability of 16S-V4 OTU identification to assign an OTU as a specific species depends in part on the resolving power of the 16S-V4 region of the 16S gene for a particular species or group of species. Both the density of available reference 16S sequences for different regions of the tree as well as the inherent variability in the 16S gene between different species will determine the definitiveness of a taxonomic annotation. Given the topological nature of a phylogenetic tree and the fact that tree represents hierarchical relationships of OTUs to one another based on their sequence similarity and an underlying evolutionary model, taxonomic annotations of a read can be rolled up to a higher level using a clade-based assignment procedure. Using this approach, clades are defined based on the topology of a phylogenetic tree that is constructed from full-length 16S sequences using maximum likelihood or other phylogenetic models familiar to individuals with ordinary skill in the art of phylogenetics. Clades are constructed to ensure that all OTUs in a given clade are: (i) within a specified number of bootstrap supported nodes from one another (generally, 1-5 bootstraps), and (ii) share a defined percent similarity (for 16S molecular data typically set to 95%-97% sequence similarity). OTUs that are within the same clade can be distinguished as genetically and phylogenetically distinct from OTUs in a different clade based on 16S-V4 sequence data. OTUs falling within the same clade are evolutionarily closely related and may or may not be distinguishable from one another using 16S-V4 sequence data. The power of clade based analysis is that members of the same clade, due to their evolutionary relatedness, are likely to play similar functional roles in a microbial ecology such as that found in the human gut. Compositions substituting one species with another from the same clade are likely to have conserved ecological function and therefore are useful in the present invention. Notably in addition to 16S-V4 sequences, clade-based analysis can be used to analyze 18S, ITS, and other genetic sequences.

Notably, 16S sequences of isolates of a given OTU are phylogenetically placed within their respective clades, sometimes in conflict with the microbiological-based assignment of species and genus that may have preceded 16S-based assignment. Discrepancies between taxonomic assignments based on microbiological characteristics versus genetic sequencing are known to exist from the literature.

For a given network ecology or functional network ecology one can define a set of OTUs from the network's representative clades. As example, if a network was comprised of clade_100 and clade_102 it can be said to be comprised of at least one OTU from the group consisting of *Corynebacterium coyleae, Corynebacterium mucifaciens*, and *Corynebacterium ureicelerivorans*, and at least one OTU from the group consisting of *Corynebacterium appendicis, Corynebacterium genitalium, Corynebacterium glaucum, Corynebacterium imitans, Corynebacterium riegelii, Corynebacterium* sp. L_2012475, *Corynebacterium* sp. NML 93_0481, *Corynebacterium sundsvallense*, and *Corynebacterium tuscaniae* (see Table 1). Conversely as example, if a network was said to consist of *Corynebacte-*

rium coyleae and/or *Corynebacterium mucifaciens* and/or *Corynebacterium ureicelerivorans*, and also consisted of *Corynebacterium appendicis* and/or *Corynebacterium genitalium* and/or *Corynebacterium glaucum* and/or *Corynebacterium imitans* and/or *Corynebacterium riegelii* and/or *Corynebacterium* sp. L_2012475 and/or *Corynebacterium* sp. NML 93_0481 and/or *Corynebacterium sundsvallense* and/or *Corynebacterium tuscaniae* it can be said to be comprised of clade_100 and clade_102.

The applicants made clade assignments to all OTUs disclosed herein using the above described method and these assignments are reported in Table 1. Results of the network analysis provides, in some embodiments, e.g., of compositions, substitution of clade_172 by clade_172i. In another embodiment, the network analysis provides substitution of clade_198 by clade_198i. In another embodiment, the network analysis permits substitution of clade_260 by clade_260c, clade_260g or clade_260h. In another embodiment, the network analysis permits substitution of clade_262 by clade_262i. In another embodiment, the network analysis permits substitution of clade_309 by clade_309c, clade_309e, clade_309g, clade_309h or clade 309i. In another embodiment, the network analysis permits substitution of clade 313 by clade 313f. In another embodiment, the network analysis permits substitution of clade_325 by clade 325f. In another embodiment, the network analysis permits substitution of clade_335 by clade 335i. In another embodiment, the network analysis permits substitution of clade_351 by clade_351e. In another embodiment, the network analysis permits substitution of clade_354 by clade_354e. In another embodiment, the network analysis permits substitution of clade_360 by clade_360c, clade_360g, clade_360h, or clade_360i. In another embodiment, the network analysis permits substitution of clade_378 by clade_378e. In another embodiment, the network analysis permits substitution of clade_38 by clade_38e or clade_38i. In another embodiment, the network analysis permits substitution of clade_408 by clade_408b, clade_408d, clade_408f, clade_408g or clade_408h. In another embodiment, the network analysis permits substitution of clade_420 by clade_420f. In another embodiment, the network analysis permits substitution of clade_444 by clade_444i. In another embodiment, the network analysis permits substitution of clade_478 by clade_478i. In another embodiment, the network analysis permits substitution of clade_479 by clade_479c, by clade_479g or by clade_479h. In another embodiment, the network analysis permits substitution of clade_481 by clade_481a, clade_481b, clade_481e, clade_481g, clade_481h or by clade_481i. In another embodiment, the network analysis substitution of clade_497 by clade_497e or by clade_497f. In another embodiment, the network analysis permits substitution of clade_512 by clade_512i. In another embodiment, the network analysis permits the network analysis permits substitutions of clade_516 by clade_516c, by clade_516g or by clade_516h. In another embodiment, the network analysis permits the network analysis permits substitutions of clade_522 by clade_522i. In another embodiment, the network analysis permits the network analysis permits substitutions of clade_553 by clade_553i. In another embodiment, the network analysis permits the network analysis permits substitutions of clade_566 by clade_566f. In another embodiment, the network analysis permits the network analysis permits substitutions of clade_572 by clade_572i. In another embodiment, the network analysis permits the network analysis permits substitutions of clade_65 by clade_65e. In another embodiment, the network analysis permits the network analysis permits substitutions of clade_92 by clade_92e or by clade_92i. In another embodiment, the network analysis permits the network analysis permits substitutions of clade_96 by clade_96g or by clade_96h. In another embodiment, the network analysis permits the network analysis permits substitutions of clade_98 by clade_98i. These permitted clade substitutions are described in Table 2.

Metagenomic Read Annotation

Metagenomic or whole genome shotgun sequence data is annotated as described above, with the additional step that sequences are either clustered or assembled prior to annotation. Following sequence characterization as described above, sequence reads are demultiplexed using the indexing (i.e. barcodes). Following demultiplexing sequence reads are either: (i) clustered using a rapid clustering algorithm such as but not limited to UCLUST (http://drive5.com/usearch/manual/uclust_algo.html) or hash methods such VICUNA (Xiao Yang, Patrick Charlebois, Sante Gnerre, Matthew G Coole, Niall J. Lennon, Joshua Z. Levin, James Qu, Elizabeth M. Ryan, Michael C. Zody, and Matthew R. Henn. 2012. De novo assembly of highly diverse viral populations. BMC Genomics 13:475). Following clustering a representative read for each cluster is identified based and analyzed as described above in "Primary Read Annotation". The result of the primary annotation is then applied to all reads in a given cluster. (ii) A second strategy for metagenomic sequence analysis is genome assembly followed by annotation of genomic assemblies using a platform such as but not limited to MetAMOS (Treangen et al. 2013 Genome Biology 14: R2), HUMAaN (Abubucker et al. 2012. Metabolic Reconstruction for Metagenomic Data and Its Application to the Human Microbiome ed. J. A. Eisen. PLOS Computational Biology 8: e1002358) and other methods familiar to one of skill in the art.

Example 14. OTU Identification Using Microbial Culturing Techniques

The identity of the bacterial species that grow up from a complex fraction can be determined in multiple ways. For example, individual colonies can be picked into liquid media in a 96 well format, grown up and saved as 15% glycerol stocks at −80° C. Aliquots of the cultures can be placed into cell lysis buffer and colony PCR methods can be used to amplify and sequence the 16S rDNA gene (Example 1). Alternatively, colonies may be streaked to purity in several passages on solid media. Well-separated colonies are streaked onto the fresh plates of the same kind and incubated for 48-72 hours at 37° C. The process is repeated multiple times to ensure purity. Pure cultures can be analyzed by phenotypic- or sequence-based methods, including 16S rDNA amplification and sequencing as described in Example 1. Sequence characterization of pure isolates or mixed communities e.g., plate scrapes and spore fractions can also include whole genome shotgun sequencing. The latter is valuable to determine the presence of genes associated with sporulation, antibiotic resistance, pathogenicity, and virulence. Colonies can also be scraped from plates en masse and sequenced using a massively parallel sequencing method as described in Example 1 such that individual 16S signatures can be identified in a complex mixture. Optionally, the sample can be sequenced prior to germination (if appropriate DNA isolation procedures are used to lyse and release the DNA from spores) in order to compare the diversity of germinable species with the total number of species in a spore sample. As an alternative or complementary approach to 16S analysis, MALDI-TOF-mass spec can also be used for species identification (Barreau et al., 2013. Improving the identification of anaerobes in the clinical microbiology laboratory through MALDI-TOF mass spectrometry. Anaerobe 22:123-125).

Example 15. Microbiological Strain Identification Approaches

Pure bacterial isolates can be identified using microbiological methods as described in Wadsworth-KTL Anaerobic Microbiology Manual (Jouseimies-Somer et al., 2002. Wadsworth-KTL Anaerobic Bacteriology Manual), and The Manual of Clinical Microbiology (ASM Press, 10th Edition). These methods rely on phenotypes of strains and include Gram-staining to confirm Gram positive or negative staining behavior of the cell envelope, observance of colony morphologies on solid media, motility, cell morphology observed microscopically at 60× or 100× magnification including the presence of bacterial endospores and flagella. Biochemical tests that discriminate between genera and species are performed using appropriate selective and differential agars and/or commercially available kits for identification of Gram-negative and Gram-positive bacteria and yeast, for example, RapID tests (Remel) or API tests (bioMerieux). Similar identification tests can also be performed using instrumentation such as the Vitek 2 system (bioMerieux). Phenotypic tests that discriminate between genera and species and strains (for example the ability to use various carbon and nitrogen sources) can also be performed using growth and metabolic activity detection methods, for example the Biolog Microbial identification microplates. The profile of short chain fatty acid production during fermentation of particular carbon sources can also be used as a way to discriminate between species (Wadsworth-KTL Anaerobic Microbiology Manual, Jousimies-Somer, et al 2002). MALDI-TOF-mass spectrometry can also be used for species identification (as reviewed in Anaerobe 22:123).

Example 16. Construction of an In Vitro Assay to Screen for Combinations of Microbes Inhibitory to the Growth of Pathogenic *E. coli*

A modification of the in vitro assay described herein is used to screen for combinations of bacteria inhibitory to the growth of *E. coli*. In general, the assay is modified by using a medium suitable for growth of the pathogen inoculum. For example, suitable media include Reinforced Clostridial Media (RCM), Brain Heart Infusion Broth (BHI) or Luria Bertani Broth (LB) (also known as Lysogeny Broth). *E. coli* is quantified by using alternative selective media specific for *E. coli* or using qPCR probes specific for the pathogen. For example, aerobic growth on MacConkey lactose medium selects for enteric Gram-negative bacteria, including *E. coli*. qPCR is conducted using probes specific for the shiga toxin of pathogenic *E. coli*.

In general, the method can be used to test compositions in vitro for their ability to inhibit growth of any pathogen that can be cultured.

Example 17. Construction of an In Vitro Assay to Screen for Combinations of Microbes Inhibitory to the Growth of Vancomycin-Resistant *Enterococcus* (VRE)

The in vitro assay can be used to screen for combinations of bacteria inhibitory to the growth of vancomycin-resistant *Enterococcus* spp. (VRE) by modifying the media used for growth of the pathogen inoculum. Several choices of media can be used for growth of the pathogen such as Reinforced Clostridial Media (RCM), Brain Heart Infusion Broth (BHI) or Luria Bertani Broth (LB). VRE is quantified by using alternative selective media specific for VRE or using qPCR probes specific for the pathogen. For example, m-*Enterococcus* agar containing sodium azide is selective for *Enterococcus* spp. and a small number of other species. Probes known in the art that are specific to the van genes conferring vancomycin resistance are used in the qPCR or such probes can be designed using methods known in the art.

Example 18. In Vitro Assay Screening Bacterial Compositions for Inhibition of *Salmonella*

The in vitro assay described herein is used to screen for combinations of bacteria inhibitory to the growth of *Salmonella* spp. by modifying the media used for growth of the pathogen inoculum. Several choices of media are used for growth of the pathogen such as Reinforced Clostridial Media (RCM), Brain Heart Infusion Broth (BHI) or Luria Bertani Broth (LB). *Salmonella* spp. are quantified by using alternative selective media specific for *Salmonella* spp. or using qPCR probes specific for the pathogen. For example, MacConkey agar is used to select for *Salmonella* spp. and the invA gene is targeted with qPCR probes; this gene encodes an invasion protein carried by many pathogenic *Salmonella* spp. and is used in invading eukaryotic cells.

Example 19. In Vivo Validation of the Efficacy of Network Ecology Bacterial Compositions for Prevention of *Clostridium difficile* Infection in a Murine Model

To test the therapeutic potential of the bacterial composition, a prophylactic mouse model of *C. difficile* infection was used (model based on Chen et al., 2008. A mouse model of *Clostridium difficile*-associated disease. Gastroenterology 135:1984-1992). Two cages of five mice each were tested for each arm of the experiment. All mice received an antibiotic cocktail consisting of 10% glucose, kanamycin (0.5 mg/ml), gentamicin (0.044 mg/ml), colistin (1062.5 U/ml), metronidazole (0.269 mg/ml), ciprofloxacin (0.156 mg/ml), ampicillin (0.1 mg/ml) and vancomycin (0.056 mg/ml) in their drinking water on days −14 through −5 and a dose of 10 mg/kg clindamycin by oral gavage on day −3. On day −1, test articles were spun for 5 minutes at 12,100 rcf, their supernatants' removed, and the remaining pellets were resuspended in sterile PBS, prereduced if bacterial composition was not in spore form, and delivered via oral gavage. On day 0 they were challenged by administration of approximately 4.5 log 10 cfu of *C. difficile* (ATCC 43255) or sterile PBS (for the naive arm) via oral gavage. Optionally a positive control group received vancomycin from day −1 through day 3 in addition to the antibiotic protocol and *C. difficile* challenge specified above. Stool were collected from the cages for analysis of bacterial carriage. Mortality, weight and clinical scoring of *C. difficile* symptoms based upon a 0-4 scale by combining scores for appearance (0-2 points based on normal, hunched, piloerection, or lethargic), and clinical signs (0-2 points based on normal, wet tail, cold-to-the-touch, or isolation from other animals) are assessed every day from day −2 through day 6. Mean minimum weight relative to day −1 and mean maximum clinical score where a death was assigned a clinical score of 4 as well as average cumulative mortality are calculated. Reduced mortality, increased mean minimum weight relative to day −1, and reduced mean maximum clinical score with death assigned to a score of 4 relative to the vehicle control are used to assess the success of the test article.

Table 9 and Table 10 report results for 14 experiments in the prophylactic mouse model of *C. difficile* infection where treatment was with a bacterial composition. In the 14 experiments, 157 of the arms tested network ecologies, with 86 distinct networks ecologies tested (Table 10). Indicia of efficacy of a composition (test article) in these experiments is a low cumulative mortality for the test composition relative to the vehicle control, a mean minimum relative weight of at least 0.85 (e.g., at least 0.90, at least 0.95, or at least 0.97), and a mean maximum clinical score less than 1, e.g., 0.9, 0.8, 0.7, 0.5, 0.2, or 0. Of the 157 arms of the experiment, 136 of the arms and 73 of the networks performed better than the respective experiment's vehicle control arm by at least one of the following metrics: cumulative mortality, mean minimum relative weight, and mean maximum clinical score. Examples of efficacious networks include but are not limited to networks N1979 as tested in SP-361 which had 0% cumulative mortality, 0.97 mean minimum relative weight, and 0 mean maximum clinical score or N2007 which had 10% cumulative mortality, 0.91 mean minimum relative weight, and 0.9 mean maximum clinical score with both networks compared to the vehicle control in SP-361 which had 30% cumulative mortality, 0.88 mean minimum relative weight, and 2.4 mean maximum clinical score. In SP-376, N1962 had no cumulative mortality, mean maximum clinical scores of 0 at both target doses tested with mean minimum relative weights of 0.98 and 0.95 for target doses of 1e8 and 1e7 CFU/OTU/mouse respectively. These results confirm that bacterial compositions comprised of binary and ternary and combinations thereof are efficacious as demonstrated using the mouse model.

Example 20. In Vivo Validation of Network Ecology Bacterial Composition Efficacy in Prophylactic and Relapse Prevention Hamster Model

Previous studies with hamsters using toxigenic and nontoxigenic strains of *C. difficile* demonstrated the utility of the hamster model in examining relapse post antibiotic treatment and the effects of prophylaxis treatments with cecal flora in *C. difficile* infection (Wilson et al., 1981. Infect Immun 34:626-628), Wilson et al., 1983. J Infect Dis 147:733, Borriello et al., 1985. J Med Microbiol 19:339-350) and more broadly in gastrointestinal infectious disease. Accordingly, to demonstrate prophylactic use of bacterial compositions comprising specific operational taxonomic units to ameliorate *C. difficile* infection, the following hamster model was used. Clindamycin (10 mg/kg s.c.) was administered to animals on day −5, the test composition or control was administered on day −3, and *C. difficile* challenge occurred on day 0. In the positive control arm, vancomycin was then administered on days 1-5 (and vehicle control was delivered on day −3). Stool were collected on days −5, −4, −1, 1, 3, 5, 7, 9 and fecal samples were assessed for pathogen carriage and reduction by microbiological methods. 16S sequencing approaches or other methods could also be utilized by one skilled in the art. Mortality was assessed multiple times per day through 21 days post *C. difficile* challenge. The percentage survival curves showed that a bacterial composition (N1962) comprised of OTUs that were shown to be inhibitory against *C. difficile* in an in vitro inhibition assay (see above examples) better protected the hamsters compared to the vancomycin control, and vehicle control (FIG. **5**).

These data demonstrate the efficacy of a composition in vivo, as well as the utility of using an in vitro inhibition method as described herein to predict compositions that have activity in vivo.

Example 21. Method of Preparing a Bacterial Composition for Administration to a Subject

Two or more strains that comprise the bacterial composition are independently cultured and mixed together before administration. Both strains are independently be grown at 37° C., pH 7, in a GMM or other animal-products-free medium, pre-reduced with 1 g/L cysteine HCl. After each strain reaches a sufficient biomass, it is preserved for banking by adding 15% glycerol and then frozen at −80° C. in 1 ml cryotubes.

Each strain is then be cultivated to a concentration of 10^{10} CFU/mL, then concentrated 20-fold by tangential flow microfiltration; the spent medium is exchanged by diafiltering with a preservative medium consisting of 2% gelatin, 100 mM trehalose, and 10 mM sodium phosphate buffer, or other suitable preservative medium. The suspension is freeze-dried to a powder and titrated.

After drying, the powder is blended with microcrystalline cellulose and magnesium stearate and formulated into a 250 mg gelatin capsule containing 10 mg of lyophilized powder (10^8 to 10^{11} bacteria), 160 mg microcrystalline cellulose, 77.5 mg gelatin, and 2.5 mg magnesium stearate.

A bacterial composition can be derived by selectively fractionating the desired bacterial OTUs from a raw material such as but not limited to stool. As an example, a 10% w/v suspension of human stool material in PBS was prepared that was filtered, centrifuged at low speed, and then the supernatant containing spores was mixed with absolute ethanol in a 1:1 ratio and vortexed to mix. The suspension was incubated at room temperature for 1 hour. After incubation the suspension was centrifuged at high speed to concentrate spores into a pellet containing a purified spore-containing preparation. The supernatant was discarded and the pellet resuspended in an equal mass of glycerol, and the purified spore preparation was placed into capsules and stored at −80° C.; this preparation is referred to as an ethanol-treated spore population.

Example 22. Method of Treating a Subject with Recurrent *C. difficile* Infection with a Bacterial Composition

In one example, a subject has suffered from recurrent bouts of *C. difficile*. In the most recent acute phase of the illness, the subject is treated with an antibiotic sufficient to ameliorate the symptoms of the illness. To prevent another relapse of *C. difficile* infection, a bacterial composition described herein is administered to the subject. For example, the subject is administered one of the present bacterial compositions at a dose in the range of $1e10^7$ to $1e10^{12}$ in, e.g., a lyophilized form, in one or more gelatin capsules (e.g., 2, 3, 4, 5, 10, 15 or more capsules) containing 10 mg of lyophilized bacteria and stabilizing components. The capsule is administered by mouth and the subject resumes a normal diet after 4, 8, 12, or 24 hours. In another embodiment, the subject may take the capsule by mouth before, during, or immediately after a meal. In a further embodiment, the subject takes the dose daily for a specified period of time.

Stool is collected from the subject before and after treatment. In one embodiment stool is collected at 1 day, 3 days, 1 week, and 1 month after administration. The presence of *C. difficile* is found in the stool before administration of the bacterial composition, but stool collections after administration show a reduction in the level of *C. difficile* in the stool (for example, at least 50% less, 60%, 70%, 80%, 90%, or 95%) to no detectable levels of *C. difficile*, as measured by qPCR and if appropriate, compared to a healthy reference subject microbiome, as described above. Typically, the quantitation is performed using material extracted from the same amounts of starting material, e.g., stool. ELISA for toxin protein or traditional microbiological identification techniques may also be used. Effective treatment is defined as a reduction in the amount of *C. difficile* present after treatment.

In some cases, effective treatment, i.e., a positive response to treatment with a composition disclosed herein is defined as absence of diarrhea, which itself is defined as 3 or more loose or watery stools per day for at least 2 consecutive days or 8 or more loose or watery stools in 48 hours, or persisting diarrhea (due to other causes) with repeating (three times) negative stool tests for toxins of *C. difficile.*

Treatment failure is defined as persisting diarrhea with a positive *C. difficile* toxin stool test or no reduction in levels of *C. difficile*, as measured by qPCR sequencing. ELISA or traditional microbiological identification techniques may also be used.

In some cases, effective treatment is determined by the lack of recurrence of signs or symptoms of *C. difficile* infection within, e.g., 2 weeks, 3 weeks, 4 weeks, 5 weeks, 10 weeks, 12 weeks, 16 weeks, 20 weeks, or 24 weeks after the treatment.

Example 23. Treatment of Subjects with *Clostridium difficile* Associated Diarrheal Disease with a Bacterial Composition

Microbial Population Engraftment, Augmentation, and Reduction of Pathogen Carriage in Patients Treated with Spore Compositions

Complementary genomic and microbiological methods were used to characterize the composition of the microbiota of 15 subjects with recurrent *C. difficile* associated disease (CDAD) that were treated with a bacterial composition. The microbiome of these subjects was characterized pretreatment and initially up to 4 weeks post-treatment and further to 24 weeks. An additional 15 subjects were treated and data for those subjects was collected to at least 8 weeks post-treatment and up to 24 weeks post-treatment. The bacterial compositions used for treatment were comprised of spore forming bacteria and constitute a microbial spore ecology derived from healthy human stool. Methods for preparing such compositions can be found in PCT/US2014/014715.

Non-limiting exemplary OTUs and clades of the spore forming microbes identified in the initial compositions are provided in Table 11. OTUs and clades in the spore ecology treatment were observed in 1 to 15 of the initial 15 subjects treated (Table 11) and in subsequently treated subjects. Treatment of the subjects with the microbial spore ecology resolved *C. difficile* associated disease (CDAD) in all subjects treated. In addition, treatment with the microbial spore composition led to the reduction or removal of Gram (−) and Gram (+) pathobionts including but not limited to pathobionts with multi-drug resistance such as but not limited to vancomycin-resistant Enterococci (VRE) and carbapenem- or imipenem resistant bacteria. Additionally, treatment led to an increase in the total microbial diversity of the subjects gut microbiome (FIG. **6**) and the resulting microbial community that established as the result of treatment with the microbial spore ecology was different from the microbiome pretreatment and more closely represented that of a healthy individual than that of an individual with CDAD (FIG. **7**).

Using novel computational approaches, applicants delineated bacterial OTUs associated with engraftment and ecological augmentation and establishment of a more diverse microbial ecology in patients treated with an ethanol-treated spore preparation (Table 11). OTUs that comprise an augmented ecology are those below the limit of detection in the patient prior to treatment and/or exist at extremely low frequencies such that they do not comprise a significant fraction of the total microbial carriage and are not detectable by genomic and/or microbiological assay methods in the bacterial composition. OTUs that are members of the engrafting and augmented ecologies were identified by characterizing the OTUs that increase in their relative abundance post treatment and that respectively are: (i) present in the ethanol-treated spore preparation and not detectable in the patient pretreatment (engrafting OTUs), or (ii) absent in the ethanol-treated spore preparation, but increase in their relative abundance in the patient through time post treatment with the preparation due to the formation of favorable growth conditions by the treatment (augmenting OTUs). Augmenting OTUs can grow from low frequency reservoirs in the patient, or can be introduced from exogenous sources such as diet.

Notably, 16S sequences of isolates of a given OTU are phylogenetically placed within their respective clades despite that the actual taxonomic assignment of species and genus may suggest they are taxonomically distinct from other members of the clades in which they fall. Discrepancies between taxonomic names given to an OTU is based on microbiological characteristics versus genetic sequencing are known to exist from the literature. The OTUs footnoted in this table are known to be discrepant between the different methods for assigning a taxonomic name.

Rational Design of Therapeutic Compositions from Core Ecologies

To define the Core Ecology underlying the remarkable clinical efficacy of the microbial spore bacterial the following analysis was carried out. The OTU composition of the microbial spore ecology was determined by 16S-V4 rDNA sequencing and computational assignment of OTUs per Example 13. A requirement to detect at least ten sequence reads in the microbial spore ecology was set as a conservative threshold to define only OTUs that were highly unlikely to arise from errors during amplification or sequencing. Methods routinely employed by those familiar to the art of genomic-based microbiome characterization use a read relative abundance threshold of 0.005% (see e.g., Bokulich et al. 2013. Quality-filtering vastly improves diversity estimates from Illumina amplicon sequencing. Nature Methods 10:57-59), which would equate to ≥2 reads given the sequencing depth obtained for the samples analyzed in this example, as cut-off which is substantially lower than the ≥10 reads used in this analysis. All taxonomic and clade assignments were made for each OTU as described in Example 13. The resulting list of OTUs, clade assignments, and frequency of detection in the spore preparations are shown in Table 11.

In one embodiment, OTUs that comprise a "core" bacterial composition of a microbial spore ecology, augmented ecology or engrafted ecology can be defined by the percentage of total subjects in which they are observed; the greater this percentage the more likely they are to be part of a core ecology responsible for catalyzing a shift away from a dysbiotic ecology. In one embodiment, therapeutic bacterial compositions are rationally designed by identifying the OTUs that occur in the greatest number of subjects evaluated. In one embodiment OTUs that occur in 100% of subjects define a therapeutic bacterial composition. In other embodiments, OTUs that are defined to occur in ≥90%, ≥80%, ≥70%, ≥60%, or ≥50% of the subjects evaluated comprise the therapeutic bacterial composition. In a further embodiment, OTUs that are in either 100%, ≥90%, ≥80%, ≥70%, ≥60%, or ≥50% are further refined to rationally design a therapeutic bacterial composition using phylogenetic parameters or other features such as but not limited to their capacity to metabolize secondary bile acids, illicit TH17 immune signaling, or produce short-chain fatty acids.

In an additional embodiment, the dominant OTUs in an ecology can be identified using several methods including but not limited to defining the OTUs that have the greatest relative abundance in either the augmented or engrafted ecologies and defining a total relative abundance threshold. As example, the dominant OTUs in the augmented ecology of Patient-1 were identified by defining the OTUs with the greatest relative abundance, which together comprise 60% of the microbial carriage in this patient's augmented ecology by day 25 post-treatment.

In a further embodiment, an OTU is assigned to be a member of the Core Ecology of the bacterial composition, that OTU must be shown to engraft in a patient. Engraftment is important for at least two reasons. First, engraftment is believed to be a sine qua non of the mechanism to reshape the microbiome and eliminate *C. difficile* colonization. OTUs that engraft with higher frequency are highly likely to be a component of the Core Ecology of the spore preparation or broadly speaking a set bacterial composition. Second, OTUs detected by sequencing a bacterial composition may include non-viable cells or other contaminant DNA molecules not associated with the composition. The requirement that an OTU must be shown to engraft in the patient eliminates OTUs that represent non-viable cells or contaminating sequences. OTUs that are present in a large percentage of the bacterial composition, e.g., ethanol spore preparations analyzed and that engraft in a large number of patients represent a subset of the Core Ecology that are highly likely to catalyze the shift from a dysbiotic disease ecology to a healthy microbiome. OTUs from which to define such therapeutic bacterial compositions derived of OTUs that engraft are denoted in Table 11.

A third lens was applied to further refine discoveries into the Core Ecology of the bacterial composition (e.g., microbial spore ecology). Computational-based, network analysis has enabled the description of microbial ecologies that are present in the microbiota of a broad population of healthy individuals. These network ecologies are comprised of multiple OTUs, some of which are defined as Keystone OTUs. Keystone OTUs are computationally defined OTUs that occur in a large percentage of computed networks and meet the networks in which they occur are highly prevalent in the population of subjects evaluated. Keystone OTUs form a foundation to the microbially ecologies in that they are found and as such are central to the function of network ecologies in healthy subjects. Keystone OTUs associated with microbial ecologies associated with healthy subjects are often are missing or exist at reduced levels in subjects with disease. Keystone OTUs may exist in low, moderate, or high abundance in subjects.

There are several important findings from these data. A relatively small number of species, 11 in total, are detected in all of the spore preparations from 6 donors and 10 donations. This is surprising because the HMP database (www.hmpdacc.org) describes the enormous variability of commensal species across healthy individuals. The presence of a small number of consistent OTUs lends support to the concept of a Core Ecology and Backbone Networks. The engraftment data further supports this conclusion.

In another embodiment, three factors—prevalence in the bacterial composition such as but not limited to a spore preparation, frequency of engraftment, and designation as a Keystone OTUs—enabled the creation of a "Core Ecology Score" (CES) to rank individual OTUs. CES was defined as follows:

- 40% weighting for presence of OTU in spore preparation
  - multiplier of 1 for presence in 1-3 spore preparations
  - multiplier of 2.5 for presence in 4-8 spore preparations
  - multiplier of 5 for presences in ≥9 spore preparations
- 40% weighting for engraftment in a patient
  - multiplier of 1 for engraftment in 1-4 patients
  - multiplier of 2.5 for engraftment in 5-6 patients
  - multiplier of 5 for engraftment in ≥7 patients
- 20% weighting to Keystone OTUs
  - multiplier of 1 for a Keystone OTU
  - multiplier of 0 for a non-Keystone OTU

Using this guide, the CES has a maximum possible score of 5 and a minimum possible score of 0.8. As an example, an OTU found in 8 of the 10 bacterial composition such as but not limited to a spore preparations that engrafted in 3 patients and was a Keystone OTU would be assigned the follow CES:

$$CES=(0.4\times2.5)+(0.4\times1)+(0.2\times1)=1.6$$

Table 11 provides a rank of OTUs by CES. Bacterial compositions rationally designed using a CES score are highly likely to catalyze the shift from a dysbiotic disease ecology to a healthy microbiome. In additional embodiments, the CES score can be combined with other factors to refine the rational design of a therapeutic bacterial composition. Such factors include but are not limited to: using phylogenetic parameters or other features such as but not limited to their capacity to metabolize secondary bile acids, illicit TH17 immune signaling, or produce short-chain fatty acids. In an additional embodiment, refinement can be done by identifying the OTUs that have the greatest relative abundance in either the augmented or engrafted ecologies and defining a total relative abundance threshold.

The number of organisms in the human gastrointestinal tract, as well as the diversity between healthy individuals, is indicative of the functional redundancy of a healthy gut microbiome ecology (see The Human Microbiome Consortia. 2012. Structure, function and diversity of the healthy human microbiome. Nature 486:207-214). This redundancy makes it highly likely that subsets of the Core Ecology describe therapeutically beneficial components of the bacterial composition such as but not limited to an ethanol-treated spore preparation and that such subsets may themselves be useful compositions for populating the GI tract and for the treatment of *C. difficile* infection given the ecologies functional characteristics. Using the CES, as well as other key metrics as defined above, individual OTUs can be prioritized for evaluation as an efficacious subset of the Core Ecology.

Another aspect of functional redundancy is that evolutionarily related organisms (i.e., those close to one another on the phylogenetic tree, e.g., those grouped into a single clade) will also be effective substitutes in the Core Ecology or a subset thereof for treating *C. difficile.*

To one skilled in the art, the selection of appropriate OTU subsets for testing in vitro or in vivo is straightforward. Subsets may be selected by picking any 2, 3, 4, 5, 6, 7, 8, 9, 10, or more than 10 OTUs from Table 11, typically selecting those with higher CES. In addition, using the clade relationships defined in Example 13 above and Table 11, related OTUs can be selected as substitutes for OTUs with acceptable CES values. These organisms can be cultured anaerobically in vitro using the appropriate media, and then combined in a desired ratio. A typical experiment in the mouse *C. difficile* model utilizes at least 104 and preferably at least 105, 106, 107, 108, 109 or more than 109 colony forming units of a each microbe in the composition. In some compositions, organisms are combined in unequal ratios, for example, due to variations in culture yields, e.g., 1:10, 1:100, 1:1,000, 1:10,000, 1:100,000, or greater than 1:100,000. What is important in these compositions is that each strain be provided in a minimum amount so that the strain's contribution to the efficacy of the Core Ecology subset can be therapeutically effective, and in some cases, measured. Using the principles and instructions described here, one of skill in the art can make clade-based substitutions to test the efficacy of subsets of the Core Ecology. Table 11 and Table 2 describe the clades for each OTU from which such substitutions can be derived.

Rational Design of Therapeutic Compositions by Integration of In Vitro and Clinical Microbiome Data

In one embodiment, efficacious subsets of the treatment microbial spore ecology as well as subsets of the microbial ecology of the subject post-treatment are defined by rationally interrogating and the composition of these ecologies with respect to compositions comprising 2, 3, 4, 5, 6, 7, 8, 9, 10, or some larger number of OTUs. In one embodiment, the bacterial compositions that have demonstrated efficacy in an in vitro pathogen inhibition assay and that are additionally identified as constituents of the ecology of the treatment itself and/or the microbial ecology of 100%, ≥90%, >80%, ≥70%, ≥60%, or ≥50% of the subject's can by an individual with ordinary skill in the art be prioritize for functional screening. Functional screens can include but are not limited to in vivo screens using various pathogen or non-pathogen models (as example, murine models, hamster models, primate models, or human). Table 12 provides bacterial compositions that exhibited inhibition against *C. difficile* as measured by a mean log inhibition greater than the 99% confidence interval (C.I) of the null hypothesis (see Example 6, ++++) and that are identified in at least one spore ecology treatment or subject post-treatment. In another embodiment compositions found in the 95%, 90%, or 80% confidence intervals (C.I.) and occurring in the treatment and post-treatment ecologies are selected. In other embodiments, bacterial compositions are selected for screening for therapeutic potential by selecting OTUs that occur in the treatment or post-treatment ecologies and the measured growth inhibition of the composition is ranked ≥ the 75^{th} percentile of all growth inhibition scores. In other embodiments, compositions ranked ≥ the 50th, 60th, 70th, 80th, 90th, 95th, or 99th percentiles are selected. In another embodiment, compositions demonstrated to have synergistic inhibition are selected (see Example 7). In yet a further embodiment, compositions selected to screen for efficacy in in vivo models are selected using a combination of growth inhibition metrics. As non-limiting example: (i) compositions are first selected based on their log inhibition being greater than the 99% confidence interval (C.I.) of the null hypothesis, (ii) then this subset of compositions further selected to represent those that are ranked ≥ the 75^{th} percentile in the distribution of all inhibition scores, (iii) this subset is then further selected based on compositions that demonstrate synergistic inhibition. In some embodiments, different confidence intervals (C.I.) and percentiles are used to subset and rationally select the compositions. In yet another embodiment, bacterial compositions are further rationally defined for their therapeutic potential using phylogenetic criteria, such as but not limited to, the presence of particular phylogenetic clade, or other features such as but not limited to their capacity to metabolize secondary bile acids, illicit TH17 immune signaling, or produce short-chain fatty acids.

In a related embodiment, all unique bacterial compositions that can be delineated in silico using the OTUs that occur in 100% of the dose spore ecologies are defined; exemplary bacterial compositions are denoted in Table 13. In other embodiments, compositions are derived form OTUs that occur in ≥90%, ≥80%, ≥70%, ≥60%, or ≥50% of the dose spore ecology or the subject's post-treatment ecologies. One with ordinary skill in the art can interrogate the resulting bacterial compositions and using various metrics including, but not limited to the percentage of spore formers, the presence of keystone OTUs, phylogenetic composition, or the OTUs' ability to metabolize secondary bile acids or the ability to produce short-chain fatty acids to rationally define bacterial compositions with suspected efficacy and suitability for further screening.

Example 24. Computational Analysis of Administered Spore Ecology Dose Compositions, and Augmentation and Engraftment Following Administration of Spore Ecology Doses

The clinical trial described in Example 23 enrolled 15 additional subjects. Further analyses were carried out on information combining data from all subjects responding to treatment in the trial (29 of 30 subjects). The treatment was with a complex formulation of microbes derived from human stool. Analyses of these results are provided in Tables 14-21. Table 22 is provided for convenience, and lists alternative names for certain organisms. Typically, the presence of an OTU is made using a method known in the art, for example, using qPCR under conditions known in the art and described herein.

The set of doses used in the trial is the collection of doses that was provided to at least one patient. Thus, a dose is implicitly a member of the set of doses. Consequently, the set of all OTUs in doses is defined as the unique set of OTUs such that each OTU is present in at least one dose.

As described herein, an engrafting OTU is an OTU that is not detectable in a patient, e.g., in their stool, pre-treatment, but is present in the composition delivered to the subject and is detected in the subject, (e.g., in the subject's stool) in at least one post-treatment sample from the subject. The set of all engrafting OTUs is defined as the unique set of engrafting OTUs found in at least one subject. An augmenting OTU is an OTU detected in a subject that is not engrafting and has an abundance ten times greater than the pre-treatment abundance at some post-treatment time point. The set of all augmenting OTUs is the unique set of augmenting OTUs found in at least one subject. The set of all augmenting and engrafting OTUs is defined as the unique set of OTUs that either augment or engraft in at least one subject.

The set of all unique ternary combinations can be generated from the experimentally derived set of OTUs by considering the all combinations of OTUs such that 1) each OTU of the ternary is different and 2) the three OTUs were not used together previously. A computer program can be used to generate such combinations.

Table 14 is generated from the set of all augmenting and engrafting OTUs and provides the OTUs that either were found to engraft or augment in at least one subject after they were treated with the composition. Each listed ternary combination is either in all doses provided to subjects or were detected together in all patients for at least one post-treatment time point. Typically, a useful composition includes at least one of the ternary compositions. In some embodiments, all three members of the ternary composition either engraft or augment in at least, e.g., 68%, 70%, 71%, 75%, 79%, 86%, 89%, 93%, or 100% of subjects. Because all subjects analyzed responded to treatment, the ternaries listed in the Table are useful in compositions for treatment of a dysbiosis.

Table 15 provides the list of unique ternary combinations of OTUs that were present in at least 95% of doses (rounding to the nearest integer) and that engrafted in at least one subject. Note that ternary combinations that were present in 100% of doses are listed in Table 14. Compositions that include a ternary combination are useful in compositions for treating a dysbiosis.

Table 16 provides the set of all unique ternary combinations of augmenting OTUs such that each ternary combination was detected in at least 75% of the subjects at a post-treatment time point.

Table 17 provides the set of all unique ternary combinations that were present in at least 75% of doses and for which the subject receiving the dose containing the ternary combination had *Clostridiales* sp. SM4/1 present as either an engrafting or augmenting OTU. Accordingly, in some embodiments, a composition consisting of, consisting essentially of, or comprising a ternary combination selected from Table 17 is useful for increasing *Clostridiales* sp. SM4/1 in a subject.

Table 18 provides the set of all unique ternary combinations generated from the set of all OTUs in doses such that each ternary is present at least 75% of the doses and for which the subject receiving the dose containing the ternary combination had *Clostridiales* sp. SSC/2 present as either an engrafting or augmenting OTU after treatment. Accordingly, in some embodiments, a composition consisting of, consisting essentially of, or comprising a ternary combination selected from Table 18 is useful for increasing *Clostridiales* sp. SSC/2 in a subject.

Table 19 provides the set of all unique ternary combinations generated from the set of all OTUs present in doses such that each ternary is present at least 75% of the doses and for which the subject to whom the doses containing the ternary combination was administered had *Clostridium* sp. NML 04A032 present as either an engrafting or augmenting OTU after treatment. Accordingly, in some embodiments, a composition consisting of, consisting essentially of, or comprising a ternary combination selected from Table 19 is useful for increasing *Clostridium* sp. NML 04A032 in a subject.

Table 20 provides the set of all unique ternary combinations generated from the set of all OTUs in doses such that the ternary is present at least 75% of the doses and for which the subject to whom the dose containing the ternary was administered had *Clostridium* sp. NML 04A032, *Ruminococcus lactaris*, and *Ruminococcus torques* present as either an engrafting or augmenting OTUs. Accordingly, in some embodiments, a composition consisting of, consisting essentially of, or comprising a ternary combination selected from Table 20 is useful for increasing *Clostridium* sp. NML 04A032, *Ruminococcus lactaris*, and *Ruminococcus torques* in a subject.

Table 21 shows the set of all unique ternary combinations generated from the set of all OTUs in doses such that each ternary is present at least 75% of the doses and for which the subject to whom the dose containing the ternary combination was administered has *Eubacterium* rectale, *Faecalibacterium prausnitzii, Oscillibacter* sp. G2, *Ruminococcus lactaris*, and *Ruminococcus torques* present as either an engrafting or augmenting OTU. Accordingly, in some embodiments, a composition consisting of, consisting essentially of, or comprising a ternary combination selected from Table 21 is useful for increasing *Eubacterium* rectale, *Faecalibacterium prausnitzii, Oscillibacter* sp. G2, *Ruminococcus lactaris*, and *Ruminococcus torques* in a subject.

Other embodiments will be apparent to those skilled in the art from consideration of the specification and practice of the embodiments. Consider the specification and examples as exemplary only, with a true scope and spirit being indicated by the following claims.

Tables

TABLE 1

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Corynebacterium coyleae	697	X96497	clade_100	N	N
Corynebacterium mucifaciens	711	NR_026396	clade_100	N	N
Corynebacterium ureicelerivorans	733	AM397636	clade_100	N	N
Corynebacterium appendicis	684	NR_028951	clade_102	N	N
Corynebacterium genitalium	698	ACLJ01000031	clade_102	N	N
Corynebacterium glaucum	699	NR_028971	clade_102	N	N
Corynebacterium imitans	703	AF537597	clade_102	N	N
Corynebacterium riegelii	719	EU848548	clade_102	N	N
Corynebacterium sp. L_2012475	723	HE575405	clade_102	N	N
Corynebacterium sp. NML 93_0481	724	GU238409	clade_102	N	N
Corynebacterium sundsvallense	728	Y09655	clade_102	N	N
Corynebacterium tuscaniae	730	AY677186	clade_102	N	N
Prevotella maculosa	1504	AGEK01000035	clade_104	N	N
Prevotella oris	1513	ADDV01000091	clade_104	N	N
Prevotella salivae	1517	AB108826	clade_104	N	N
Prevotella sp. ICM55	1521	HQ616399	clade_104	N	N
Prevotella sp. oral clone AA020	1528	AY005057	clade_104	N	N

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Prevotella sp. oral clone GI032	1538	AY349396	clade_104	N	N
Prevotella sp. oral taxon G70	1558	GU432179	clade_104	N	N
Prevotella corporis	1491	L16465	clade_105	N	N
Bacteroides sp. 4_1_36	312	ACTC01000133	clade_110	N	N
Bacteroides sp. AR20	315	AF139524	clade_110	N	N
Bacteroides sp. D20	319	ACPT01000052	clade_110	N	N
Bacteroides sp. F_4	322	AB470322	clade_110	N	N
Bacteroides uniformis	329	AB050110	clade_110	N	N
Prevotella nanceiensis	1510	JN867228	clade_127	N	N
Prevotella sp. oral taxon 299	1548	ACWZ01000026	clade_127	N	N
Prevotella bergensis	1485	ACKS01000100	clade_128	N	N
Prevotella buccalis	1489	JN867261	clade_129	N	N
Prevotella timonensis	1564	ADEF01000012	clade_129	N	N
Prevotella oralis	1512	AEPE01000021	clade_130	N	N
Prevotella sp. SEQ072	1525	JN867238	clade_130	N	N
Leuconostoc carnosum	1177	NR_040811	clade_135	N	N
Leuconostoc gasicomitatum	1179	FN822744	clade_135	N	N
Leuconostoc inhae	1180	NR_025204	clade_135	N	N
Leuconostoc kimchii	1181	NR_075014	clade_135	N	N
Edwardsiella tarda	777	CP002154	clade_139	N	N
Photorhabdus asymbiotica	1466	Z76752	clade_139	N	N
Psychrobacter arcticus	1607	CP000082	clade_141	N	N
Psychrobacter cibarius	1608	HQ698586	clade_141	N	N
Psychrobacter cryohalolentis	1609	CP000323	clade_141	N	N
Psychrobacter faecalis	1610	HQ698566	clade_141	N	N
Psychrobacter nivimaris	1611	HQ698587	clade_141	N	N
Psychrobacter pulmonis	1612	HQ698582	clade_141	N	N
Pseudomonas aeruginosa	1592	AABQ07000001	clade_154	N	N
Pseudomonas sp. 2_1_26	1600	ACWU01000257	clade_154	N	N
Corynebacterium confusum	691	Y15886	clade_158	N	N
Corynebacterium propinquum	712	NR_037038	clade_158	N	N
Corynebacterium pseudodiphtheriticum	713	X84258	clade_158	N	N
Bartonella bacilliformis	338	NC_008783	clade_159	N	N
Bartonella grahamii	339	CP001562	clade_159	N	N
Bartonella henselae	340	NC_005956	clade_159	N	N
Bartonella quintana	341	BX897700	clade_159	N	N
Bartonella tamiae	342	EF672728	clade_159	N	N
Bartonella washoensis	343	FJ719017	clade_159	N	N
Brucella abortus	430	ACBJ01000075	clade_159	N	Category-B
Brucella canis	431	NR_044652	clade_159	N	Category-B
Brucella ceti	432	ACJD01000006	clade_159	N	Category-B
Brucella melitensis	433	AE009462	clade_159	N	Category-B
Brucella microti	434	NR_042549	clade_159	N	Category-B
Brucella ovis	435	NC_009504	clade_159	N	Category-B
Brucella sp. 83_13	436	ACBQ01000040	clade_159	N	Category-B
Brucella sp. BO1	437	EU053207	clade_159	N	Category-B
Brucella suis	438	ACBK01000034	clade_159	N	Category-B
Ochrobactrum anthropi	1360	NC_009667	clade_159	N	N
Ochrobactrum intermedium	1361	ACQA01000001	clade_159	N	N
Ochrobactrum pseudintermedium	1362	DQ365921	clade_159	N	N
Prevotella genomosp. C2	1496	AY278625	clade_164	N	N
Prevotella multisaccharivorax	1509	AFJE01000016	clade_164	N	N
Prevotella sp. oral clone IDR_CEC_0055	1543	AY550997	clade_164	N	N
Prevotella sp. oral taxon 292	1547	GQ422735	clade_164	N	N
Prevotella sp. oral taxon 300	1549	GU409549	clade_164	N	N
Prevotella marshii	1505	AEEI01000070	clade_166	N	N
Prevotella sp. oral clone IK053	1544	AY349401	clade_166	N	N
Prevotella sp. oral taxon 781	1554	GQ422744	clade_166	N	N
Prevotella stercorea	1562	AB244774	clade_166	N	N
Prevotella brevis	1487	NR_041954	clade_167	N	N
Prevotella ruminicola	1516	CP002006	clade_167	N	N
Prevotella sp. sp24	1560	AB003384	clade_167	N	N
Prevotella sp. sp34	1561	AB003385	clade_167	N	N
Prevotella albensis	1483	NR_025300	clade_168	N	N
Prevotella copri	1490	ACBX02000014	clade_168	N	N
Prevotella oulorum	1514	L16472	clade_168	N	N
Prevotella sp. BI_42	1518	AJ581354	clade_168	N	N
Prevotella sp. oral clone P4PB_83 P2	1546	AY207050	clade_168	N	N
Prevotella sp. oral taxon G60	1557	GU432133	clade_168	N	N
Prevotella amnii	1484	AB547670	clade_169	N	N
Bacteroides caccae	268	EU136686	clade_170	N	N
Bacteroides finegoldii	277	AB222699	clade_170	N	N
Bacteroides intestinalis	283	ABJL02000006	clade_171	N	N
Bacteroides sp. XB44A	326	AM230649	clade_171	N	N
Bifidobacteriaceae genomosp. C1	345	AY278612	clade_172	N	N
Bifidobacterium adolescentis	346	AAXD02000018	clade_172	N	N
Bifidobacterium angulatum	347	ABYS02000004	clade_172	N	N

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Bifidobacterium animalis	348	CP001606	clade_172	N	N
Bifidobacterium breve	350	CP002743	clade_172	N	N
Bifidobacterium catenulatum	351	ABXY01000019	clade_172	N	N
Bifidobacterium dentium	352	CP001750	clade_172	N	OP
Bifidobacterium gallicum	353	ABXB03000004	clade_172	N	N
Bifidobacterium infantis	354	AY151398	clade_172	N	N
Bifidobacterium kashiwanohense	355	AB491757	clade_172	N	N
Bifidobacterium longum	356	ABQQ01000041	clade_172	N	N
Bifidobacterium pseudocatenulatum	357	ABXX02000002	clade_172	N	N
Bifidobacterium pseudolongum	358	NR_043442	clade_172	N	N
Bifidobacterium scardovii	359	AJ307005	clade_172	N	N
Bifidobacterium sp. HM2	360	AB425276	clade_172	N	N
Bifidobacterium sp. HMLN12	361	JF519685	clade_172	N	N
Bifidobacterium sp. M45	362	HM626176	clade_172	N	N
Bifidobacterium sp. MSX5B	363	HQ616382	clade_172	N	N
Bifidobacterium sp. TM_7	364	AB218972	clade_172	N	N
Bifidobacterium thermophilum	365	DQ340557	clade_172	N	N
Leuconostoc citreum	1178	AM157444	clade_175	N	N
Leuconostoc lactis	1182	NR_040823	clade_175	N	N
Eubacterium saburreum	858	AB525414	clade_178	Y	N
Eubacterium sp. oral clone IR009	866	AY349376	clade_178	Y	N
Lachnospiraceae bacterium ICM62	1061	HQ616401	clade_178	Y	N
Lachnospiraceae bacterium MSX33	1062	HQ616384	clade_178	Y	N
Lachnospiraceae bacterium oral taxon 107	1063	ADDS01000069	clade_178	Y	N
Alicyclobacillus acidocaldarius	122	NR_074721	clade_179	Y	N
Alicyclobacillus acidoterrestris	123	NR_040844	clade_179	N	N
Alicyclobacillus cycloheptanicus	125	NR_024754	clade_179	N	N
Acinetobacter baumannii	27	ACYQ01000014	clade_181	N	N
Acinetobacter calcoaceticus	28	AM157426	clade_181	N	N
Acinetobacter genomosp. C1	29	AY278636	clade_181	N	N
Acinetobacter haemolyticus	30	ADMT01000017	clade_181	N	N
Acinetobacter johnsonii	31	ACPL01000162	clade_181	N	N
Acinetobacter junii	32	ACPM01000135	clade_181	N	N
Acinetobacter lwoffii	33	ACPN01000204	clade_181	N	N
Acinetobacter parvus	34	AIEB01000124	clade_181	N	N
Acinetobacter schindleri	36	NR_025412	clade_181	N	N
Acinetobacter sp. 56A1	37	GQ178049	clade_181	N	N
Acinetobacter sp. CIP 101934	38	JQ638573	clade_181	N	N
Acinetobacter sp. CIP 102143	39	JQ638578	clade_181	N	N
Acinetobacter sp. M16_22	41	HM366447	clade_181	N	N
Acinetobacter sp. RUH2624	42	ACQF01000094	clade_181	N	N
Acinetobacter sp. SH024	43	ADCH01000068	clade_181	N	N
Lactobacillus jensenii	1092	ACQD01000066	clade_182	N	N
Alcaligenes faecalis	119	AB680368	clade_183	N	N
Alcaligenes sp. CO14	120	DQ643040	clade_183	N	N
Alcaligenes sp. S3	121	HQ262549	clade_183	N	N
Oligella ureolytica	1366	NR_041998	clade_183	N	N
Oligella urethralis	1367	NR_041753	clade_183	N	N
Eikenella corrodens	784	ACEA01000028	clade_185	N	N
Kingella denitrificans	1019	AEWV01000047	clade_185	N	N
Kingella genomosp. P1 oral cone MB2_C20	1020	DQ003616	clade_185	N	N
Kingella kingae	1021	AFHS01000073	clade_185	N	N
Kingella oralis	1022	ACJW02000005	clade_185	N	N
Kingella sp. oral clone ID059	1023	AY349381	clade_185	N	N
Neisseria elongate	1330	ADBF01000003	clade_185	N	N
Neisseria genomosp. P2 oral clone MB5_P15	1332	DQ003630	clade_185	N	N
Neisseria sp. oral clone JC012	1345	AY349388	clade_185	N	N
Neisseria sp. SMC_A9199	1342	FJ763637	clade_185	N	N
Simonsiella muelleri	1731	ADCY01000105	clade_185	N	N
Corynebacterium glucuronolyticum	700	ABYP01000081	clade_193	N	N
Corynebacterium pyruviciproducens	716	FJ185225	clade_193	N	N
Rothia aeria	1649	DQ673320	clade_194	N	N
Rothia dentocariosa	1650	ADDW01000024	clade_194	N	N
Rothia sp. oral taxon 188	1653	GU470892	clade_194	N	N
Corynebacterium accolens	681	ACGD01000048	clade_195	N	N
Corynebacterium macginleyi	707	AB359393	clade_195	N	N
Corynebacterium pseudogenitalium	714	ABYQ01000237	clade_195	N	N
Corynebacterium tuberculostearicum	729	ACVP01000009	clade_195	N	N
Lactobacillus casei	1074	CP000423	clade_198	N	N
Lactobacillus paracasei	1106	ABQV01000067	clade_198	N	N
Lactobacillus zeae	1143	NR_037122	clade_198	N	N
Prevotella dentalis	1492	AB547678	clade_205	N	N
Prevotella sp. oral clone ASCG10	1529	AY923148	clade_206	N	N
Prevotella sp. oral clone HF050	1541	AY349399	clade_206	N	N
Prevotella sp. oral clone ID019	1542	AY349400	clade_206	N	N

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Prevotella sp. oral clone IK062	1545	AY349402	clade_206	N	N
Prevotella genomosp. P9 oral clone MB7_G16	1499	DQ003633	clade_207	N	N
Prevotella sp. oral clone AU069	1531	AY005062	clade_207	N	N
Prevotella sp. oral clone CY006	1532	AY005063	clade_207	N	N
Prevotella sp. oral clone FL019	1534	AY349392	clade_207	N	N
Actinomyces genomosp. C1	56	AY278610	clade_212	N	N
Actinomyces genomosp. C2	57	AY278611	clade_212	N	N
Actinomyces genomosp. P1 oral clone MB6_C03	58	DQ003632	clade_212	N	N
Actinomyces georgiae	59	GU561319	clade_212	N	N
Actinomyces israelii	60	AF479270	clade_212	N	N
Actinomyces massiliensis	61	AB545934	clade_212	N	N
Actinomyces meyeri	62	GU561321	clade_212	N	N
Actinomyces odontolyticus	66	ACYT01000123	clade_212	N	N
Actinomyces orihominis	68	AJ575186	clade_212	N	N
Actinomyces sp. CCUG 37290	71	AJ234058	clade_212	N	N
Actinomyces sp. ICM34	75	HQ616391	clade_212	N	N
Actinomyces sp. ICM41	76	HQ616392	clade_212	N	N
Actinomyces sp. ICM47	77	HQ616395	clade_212	N	N
Actinomyces sp. ICM54	78	HQ616398	clade_212	N	N
Actinomyces sp. oral clone IP081	87	AY349366	clade_212	N	N
Actinomyces sp. oral taxon 178	91	AEUH01000060	clade_212	N	N
Actinomyces sp. oral taxon 180	92	AEPP01000041	clade_212	N	N
Actinomyces sp. TeJ5	80	GU561315	clade_212	N	N
Haematobacter sp. BC14248	968	GU396991	clade_213	N	N
Paracoccus denitrificans	1424	CP000490	clade_213	N	N
Paracoccus marcusii	1425	NR_044922	clade_213	N	N
Grimontia hollisae	967	ADAQ01000013	clade_216	N	N
Shewanella putrefaciens	1723	CP002457	clade_216	N	N
Afipia genomosp. 4	111	EU117385	clade_217	N	N
Rhodopseudomonas palustris	1626	CP000301	clade_217	N	N
Methylobacterium extorquens	1223	NC_010172	clade_218	N	N
Methylobacterium podarium	1224	AY468363	clade_218	N	N
Methylobacterium radiotolerans	1225	GU294320	clade_218	N	N
Methylobacterium sp. 1sub	1226	AY468371	clade_218	N	N
Methylobacterium sp. MM4	1227	AY468370	clade_218	N	N
Clostridium baratii	555	NR_029229	clade_223	Y	N
Clostridium colicanis	576	FJ957863	clade_223	Y	N
Clostridium paraputrificum	611	AB536771	clade_223	Y	N
Clostridium sardiniense	621	NR_041006	clade_223	Y	N
Eubacterium budayi	837	NR_024682	clade_223	Y	N
Eubacterium moniliforme	851	HF558373	clade_223	Y	N
Eubacterium multiforme	852	NR_024683	clade_223	Y	N
Eubacterium nitritogenes	853	NR_024684	clade_223	Y	N
Achromobacter denitrificans	18	NR_042021	clade_224	N	N
Achromobacter piechaudii	19	ADMS01000149	clade_224	N	N
Achromobacter xylosoxidans	20	ACRC01000072	clade_224	N	N
Bordetella bronchiseptica	384	NR_025949	clade_224	N	OP
Bordetella holmesii	385	AB683187	clade_224	N	OP
Bordetella parapertussis	386	NR_025950	clade_224	N	OP
Bordetella pertussis	387	BX640418	clade_224	N	OP
Microbacterium chocolatum	1230	NR_037045	clade_225	N	N
Microbacterium flavescens	1231	EU714363	clade_225	N	N
Microbacterium lacticum	1233	EU714351	clade_225	N	N
Microbacterium oleivorans	1234	EU714381	clade_225	N	N
Microbacterium oxydans	1235	EU714348	clade_225	N	N
Microbacterium paraoxydans	1236	AJ491806	clade_225	N	N
Microbacterium phyllosphaerae	1237	EU714359	clade_225	N	N
Microbacterium schleiferi	1238	NR_044936	clade_225	N	N
Microbacterium sp. 768	1239	EU714378	clade_225	N	N
Microbacterium sp. oral strain C24KA	1240	AF287752	clade_225	N	N
Microbacterium testaceum	1241	EU714365	clade_225	N	N
Corynebacterium atypicum	686	NR_025540	clade_229	N	N
Corynebacterium mastitidis	708	AB359395	clade_229	N	N
Corynebacterium sp. NML 97_0186	725	GU238411	clade_229	N	N
Mycobacterium elephantis	1275	AF385898	clade_237	N	OP
Mycobacterium paraterrae	1288	EU919229	clade_237	N	OP
Mycobacterium phlei	1289	GU142920	clade_237	N	OP
Mycobacterium sp. 1776	1293	EU703152	clade_237	N	N
Mycobacterium sp. 1781	1294	EU703147	clade_237	N	N
Mycobacterium sp. AQ1GA4	1297	HM210417	clade_237	N	N
Mycobacterium sp. GN_10546	1299	FJ497243	clade_237	N	N
Mycobacterium sp. GN_10827	1300	FJ497247	clade_237	N	N
Mycobacterium sp. GN_11124	1301	FJ652846	clade_237	N	N
Mycobacterium sp. GN_9188	1302	FJ497240	clade_237	N	N
Mycobacterium sp. GR_2007_210	1303	FJ555538	clade_237	N	N
Anoxybacillus contaminans	172	NR_029006	clade_238	N	N
Anoxybacillus flavithermus	173	NR_074667	clade_238	Y	N

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Bacillus aeolius	195	NR_025557	clade_238	N	N
Bacillus aerophilus	196	NR_042339	clade_238	Y	N
Bacillus aestuarii	197	GQ980243	clade_238	Y	N
Bacillus amyloliquefaciens	199	NR_075005	clade_238	Y	N
Bacillus anthracis	200	AAEN01000020	clade_238	Y	Category-A
Bacillus atrophaeus	201	NR_075016	clade_238	Y	OP
Bacillus badius	202	NR_036893	clade_238	Y	OP
Bacillus cereus	203	ABDJ01000015	clade_238	Y	OP
Bacillus circulans	204	AB271747	clade_238	Y	OP
Bacillus firmus	207	NR_025842	clade_238	Y	OP
Bacillus flexus	208	NR_024691	clade_238	Y	OP
Bacillus fordii	209	NR_025786	clade_238	Y	OP
Bacillus halmapalus	211	NR_026144	clade_238	Y	OP
Bacillus herbersteinensis	213	NR_042286	clade_238	Y	OP
Bacillus idriensis	215	NR_043268	clade_238	Y	OP
Bacillus lentus	216	NR_040792	clade_238	Y	OP
Bacillus licheniformis	217	NC_006270	clade_238	Y	OP
Bacillus megaterium	218	GU252124	clade_238	Y	OP
Bacillus nealsonii	219	NR_044546	clade_238	Y	OP
Bacillus niabensis	220	NR_043334	clade_238	Y	OP
Bacillus niacini	221	NR_024695	clade_238	Y	OP
Bacillus pocheonensis	222	NR_041377	clade_238	Y	OP
Bacillus pumilus	223	NR_074977	clade_238	Y	OP
Bacillus safensis	224	JQ624766	clade_238	Y	OP
Bacillus simplex	225	NR_042136	clade_238	Y	OP
Bacillus sonorensis	226	NR_025130	clade_238	Y	OP
Bacillus sp. 10403023 MM10403188	227	CAET01000089	clade_238	Y	OP
Bacillus sp. 2_A_57_CT2	230	ACWD01000095	clade_238	Y	OP
Bacillus sp. 2008724126	228	GU252108	clade_238	Y	OP
Bacillus sp. 2008724139	229	GU252111	clade_238	Y	OP
Bacillus sp. 7_16AIA	231	FN397518	clade_238	Y	OP
Bacillus sp. AP8	233	JX101689	clade_238	Y	OP
Bacillus sp. B27(2008)	234	EU362173	clade_238	Y	OP
Bacillus sp. BT1B_CT2	235	ACWC01000034	clade_238	Y	OP
Bacillus sp. GB1.1	236	FJ897765	clade_238	Y	OP
Bacillus sp. GB9	237	FJ897766	clade_238	Y	OP
Bacillus sp. HU19.1	238	FJ897769	clade_238	Y	OP
Bacillus sp. HU29	239	FJ897771	clade_238	Y	OP
Bacillus sp. HU33.1	240	FJ897772	clade_238	Y	OP
Bacillus sp. JC6	241	JF824800	clade_238	Y	OP
Bacillus sp. oral taxon F79	248	HM099654	clade_238	Y	OP
Bacillus sp. SRC_DSF1	243	GU797283	clade_238	Y	OP
Bacillus sp. SRC_DSF10	242	GU797292	clade_238	Y	OP
Bacillus sp. SRC_DSF2	244	GU797284	clade_238	Y	OP
Bacillus sp. SRC_DSF6	245	GU797288	clade_238	Y	OP
Bacillus sp. tc09	249	HQ844242	clade_238	Y	OP
Bacillus sp. zh168	250	FJ851424	clade_238	Y	OP
Bacillus sphaericus	251	DQ286318	clade_238	Y	OP
Bacillus sporothermodurans	252	NR_026010	clade_238	Y	OP
Bacillus subtilis	253	EU627588	clade_238	Y	OP
Bacillus thermoamylovorans	254	NR_029151	clade_238	Y	OP
Bacillus thuringiensis	255	NC_008600	clade_238	Y	OP
Bacillus weihenstephanensis	256	NR_074926	clade_238	Y	OP
Brevibacterium frigoritolerans	422	NR_042639	clade_238	N	N
Geobacillus kaustophilus	933	NR_074989	clade_238	Y	N
Geobacillus sp. E263	934	DQ647387	clade_238	N	N
Geobacillus sp. WCH70	935	CP001638	clade_238	N	N
Geobacillus stearothermophilus	936	NR_040794	clade_238	Y	N
Geobacillus thermocatenulatus	937	NR_043020	clade_238	N	N
Geobacillus thermodenitrificans	938	NR_074976	clade_238	Y	N
Geobacillus thermoglucosidasius	939	NR_043022	clade_238	Y	N
Geobacillus thermoleovorans	940	NR_074931	clade_238	N	N
Lysinibacillus fusiformis	1192	FN397522	clade_238	N	N
Lysinibacillus sphaericus	1193	NR_074883	clade_238	Y	N
Planomicrobium koreense	1468	NR_025011	clade_238	N	N
Sporosarcina newyorkensis	1754	AFPZ01000142	clade_238	N	N
Sporosarcina sp. 2681	1755	GU994081	clade_238	N	N
Ureibacillus composti	1968	NR_043746	clade_238	N	N
Ureibacillus suwonensis	1969	NR_043232	clade_238	N	N
Ureibacillus terrenus	1970	NR_025394	clade_238	N	N
Ureibacillus thermophilus	1971	NR_043747	clade_238	N	N
Ureibacillus thermosphaericus	1972	NR_040961	clade_238	N	N
Prevotella micans	1507	AGWK01000061	clade_239	N	N
Prevotella sp. oral clone DA058	1533	AY005065	clade_239	N	N
Prevotella sp. SEQ053	1523	JN867222	clade_239	N	N
Treponema socranskii	1937	NR_024868	clade_240	N	OP
Treponema sp. 6:H:D15A_4	1938	AY005083	clade_240	N	N

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Treponema sp. oral taxon 265	1953	GU408850	clade_240	N	N
Treponema sp. oral taxon G85	1958	GU432215	clade_240	N	N
Porphyromonas endodontalis	1472	ACNN01000021	clade_241	N	N
Porphyromonas sp. oral clone BB134	1478	AY005068	clade_241	N	N
Porphyromonas sp. oral clone F016	1479	AY005069	clade_241	N	N
Porphyromonas sp. oral clone P2PB_52 P1	1480	AY207054	clade_241	N	N
Porphyromonas sp. oral clone P4GB_100 P2	1481	AY207057	clade_241	N	N
Acidovorax sp. 98_63833	26	AY258065	clade_245	N	N
Comamonadaceae bacterium NML000135	663	JN585335	clade_245	N	N
Comamonadaceae bacterium NML790751	664	JN585331	clade_245	N	N
Comamonadaceae bacterium NML910035	665	JN585332	clade_245	N	N
Comamonadaceae bacterium NML910036	666	JN585333	clade_245	N	N
Comamonas sp. NSP5	668	AB076850	clade_245	N	N
Delftia acidovorans	748	CP000884	clade_245	N	N
Xenophilus aerolatus	2018	JN585329	clade_245	N	N
Clostridiales sp. SS3/4	543	AY305316	clade_246	Y	N
Oribacfcerium sp. oral taxon 078	1380	ACIQ02000009	clade_246	N	N
Oribacterium sp. oral taxon 102	1381	GQ422713	clade_246	N	N
Weissella cibaria	2007	NR_036924	clade_247	N	N
Weissella confusa	2008	NR_040816	clade_247	N	N
Weissella hellenica	2009	AB680902	clade_247	N	N
Weissella kandleri	2010	NR_044659	clade_247	N	N
Weissella koreensis	2011	NR_075058	clade_247	N	N
Weissella paramesenteroides	2012	ACKU01000017	clade_247	N	N
Weissella sp. KLDS 7.0701	2013	EU600924	clade_247	N	N
Mobiluncus curtisii	1251	AEPZ01000013	clade_249	N	N
Clostridium beijerinckii	557	NR_074434	clade_252	Y	N
Clostridium botulinum	560	NC_010723	clade_252	Y	Category-A
Clostridium butyricum	561	ABDT01000017	clade_252	Y	N
Clostridium chauvoei	568	EU106372	clade_252	Y	N
Clostridium favososporum	582	X76749	clade_252	Y	N
Clostridium histolyticum	592	HF558362	clade_252	Y	N
Clostridium isatidis	597	NR_026347	clade_252	Y	N
Clostridium limosum	602	FR870444	clade_252	Y	N
Clostridium sartagoforme	622	NR_026490	clade_252	Y	N
Clostridium septicum	624	NR_026020	clade_252	Y	N
Clostridium sp. 7_2_43FAA	626	ACDK01000101	clade_252	Y	N
Clostridium sporogenes	645	ABKW02000003	clade_252	Y	N
Clostridium tertium	653	Y18174	clade_252	Y	N
Clostridium carnis	564	NR_044716	clade_253	Y	N
Clostridium celatum	565	X77844	clade_253	Y	N
Clostridium disporicum	579	NR_026491	clade_253	Y	N
Clostridium gasigenes	585	NR_024945	clade_253	Y	N
Clostridium quinii	616	NR_026149	clade_253	Y	N
Enhydrobacter aerosaccus	785	ACYI01000081	clade_256	N	N
Moraxella osloensis	1262	JN175341	clade_256	N	N
Moraxella sp. GM2	1264	JF837191	clade_256	N	N
Brevibacterium casei	420	JF951998	clade_257	N	N
Brevibacterium epidermidis	421	NR_029262	clade_257	N	N
Brevibacterium sanguinis	426	NR_028016	clade_257	N	N
Brevibacterium sp. H15	427	AB177640	clade_257	N	N
Clostridium hylemonae	593	AB023973	clade_260	Y	N
Clostridium scindens	623	AF262238	clade_260	Y	N
Lachnospiraceae bacterium 5_1_57FAA	1054	ACTR01000020	clade_260	Y	N
Acinetobacter radioresistens	35	ACVR01000010	clade_261	N	N
Clostridium glycyrrhizinilyticum	588	AB233029	clade_262	Y	N
Clostridium nexile	607	X73443	clade_262	Y	N
Coprococcus comes	674	ABVR01000038	clade_262	Y	N
Lachnospiraceae bacterium 1_1_57FAA	1048	ACTM01000065	clade_262	Y	N
Lachnospiraceae bacterium 1_4_56FAA	1049	ACTN01000028	clade_262	Y	N
Lachnospiraceae bacterium 8_1_57FAA	1057	ACWQ01000079	clade_262	Y	N
Ruminococcus lactaris	1663	ABOU02000049	clade_262	Y	N
Ruminococcus torques	1670	AAVP02000002	clade_262	Y	N
Lactobacillus alimentarius	1068	NR_044701	clade_263	N	N
Lactobacillus farciminis	1082	NR_044707	clade_263	N	N
Lactobacillus kimchii	1097	NR_025045	clade_263	N	N
Lactobacillus nodensis	1101	NR_041629	clade_263	N	N
Lactobacillus tucceti	1138	NR_042194	clade_263	N	N
Pseudomonas mendocina	1595	AAUL01000021	clade_265	N	N
Pseudomonas pseudoalcaligenes	1598	NR_037000	clade_265	N	N
Pseudomonas sp. NP522b	1602	EU723211	clade_265	N	N
Pseudomonas stutzeri	1603	AM905854	clade_265	N	N
Paenibacillus barcinonensis	1390	NR_042272	clade_270	N	N
Paenibacillus barengoltzii	1391	NR_042756	clade_270	N	N
Paenibacillus chibensis	1392	NR_040885	clade_270	N	N
Paenibacillus cookii	1393	NR_025372	clade_270	N	N
Paenibacillus durus	1394	NR_037017	clade_270	N	N

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Paenibacillus glucanolyticus	1395	D78470	clade_270	N	N
Paenibacillus lactis	1396	NR_025739	clade_270	N	N
Paenibacillus lautus	1397	NR_040882	clade_270	Y	N
Paenibacillus pabuli	1398	NR_040853	clade_270	N	N
Paenibacillus polymyxa	1399	NR_037006	clade_270	Y	N
Paenibacillus popilliae	1400	NR_040888	clade_270	N	N
Paenibacillus sp. CIP 101062	1401	HM212646	clade_270	N	N
Paenibacillus sp. HGF5	1402	AEXS01000095	clade_270	Y	N
Paenibacillus sp. HGF7	1403	AFDH01000147	clade_270	Y	N
Paenibacillus sp. JC66	1404	JF824808	clade_270	N	N
Paenibacillus sp. R_27413	1405	HE586333	clade_270	N	N
Paenibacillus sp. R_27422	1406	HE586338	clade_270	N	N
Paenibacillus timonensis	1408	NR_042844	clade_270	N	N
Rothia mucilaginosa	1651	ACVO01000020	clade_271	N	N
Rothia nasimurium	1652	NR_025310	clade_271	N	N
Prevotella sp. oral taxon 302	1550	ACZK01000043	clade_280	N	N
Prevotella sp. oral taxon F68	1556	HM099652	clade_280	N	N
Prevotella tannerae	1563	ACIJ02000018	clade_280	N	N
Prevotellaceae bacterium P4P_62 P1	1566	AY207061	clade_280	N	N
Porphyromonas asaccharolytica	1471	AENO01000048	clade_281	N	N
Porphyromonas gingivails	1473	AE015924	clade_281	N	N
Porphyromonas macacae	1475	NR_025908	clade_281	N	N
Porphyromonas sp. UQD 301	1477	EU012301	clade_281	N	N
Porphyromonas uenonis	1482	ACLR01000152	clade_281	N	N
Leptotrichia buccalis	1165	CP001685	clade_282	N	N
Leptotrichia hofstadii	1168	ACVB02000032	clade_282	N	N
Leptotrichia sp. oral clone HE012	1173	AY349386	clade_282	N	N
Leptotrichia sp. oral taxon 223	1176	GU408547	clade_282	N	N
Bacteroides fluxus	278	AFBN01000029	clade_285	N	N
Bacteroides helcogenes	281	CP002352	clade_285	N	N
Parabacteroides johnsonii	1419	ABYH01000014	clade_286	N	N
Parabacteroides merdae	1420	EU136685	clade_286	N	N
Treponema denticola	1926	ADEC01000002	clade_288	N	OP
Treponema genomosp. P5 oral clone MB3_P23	1929	DQ003624	clade_288	N	N
Treponema putidum	1935	AJ543428	clade_288	N	OP
Treponema sp. oral clone P2PB_53 P3	1942	AY207055	clade_288	N	N
Treponema sp. oral taxon 247	1949	GU408748	clade_288	N	N
Treponema sp. oral taxon 250	1950	GU408776	clade_288	N	N
Treponema sp. oral taxon 251	1951	GU408781	clade_288	N	N
Anaerococcus hydrogenalis	144	ABXA01000039	clade_289	N	N
Anaerococcus sp. 8404299	148	HM587318	clade_289	N	N
Anaerococcus sp. gpac215	156	AM176540	clade_289	N	N
Anaerococcus vaginalis	158	ACXU01000016	clade_289	N	N
Propionibacterium acidipropionici	1569	NC_019395	clade_290	N	N
Propionibacterium avidum	1571	AJ003055	clade_290	N	N
Propionibacterium granulosum	1573	FJ785716	clade_290	N	N
Propionibacterium jensenii	1574	NR_042269	clade_290	N	N
Propionibacterium propionicum	1575	NR_025277	clade_290	N	N
Propionibacterium sp. H456	1577	AB177643	clade_290	N	N
Propionibacterium thoenii	1581	NR_042270	clade_290	N	N
Bifidobacterium bifidum	349	ABQP01000027	clade_293	N	N
Leuconostoc mesenteroides	1183	ACKV01000113	clade_295	N	N
Leuconostoc pseudomesenteroides	1184	NR_040814	clade_295	N	N
Eubacterium sp. oral clone JI012	868	AY349379	clade_298	Y	N
Johnsonella ignava	1016	X87152	clade_298	N	N
Propionibacterium acnes	1570	ADJM01000010	clade_299	N	N
Propionibacterium sp. 434_HC2	1576	AFIL01000035	clade_299	N	N
Propionibacterium sp. LG	1578	AY354921	clade_299	N	N
Propionibacterium sp. S555a	1579	AB264622	clade_299	N	N
Alicyclobacillus contaminans	124	NR_041475	clade_301	Y	N
Alicyclobacillus herbarius	126	NR_024753	clade_301	Y	N
Alicyclobacillus pomorum	127	NR_024801	clade_301	Y	N
Alicyclobacillus sp. CCUG 53762	128	HE613268	clade_301	N	N
Actinomyces cardiffensis	53	GU470888	clade_303	N	N
Actinomyces funkei	55	HQ906497	clade_303	N	N
Actinomyces sp. HKU31	74	HQ335393	clade_303	N	N
Actinomyces sp. oral taxon C55	94	HM099646	clade_303	N	N
Kerstersia gyiorum	1018	NR_025669	clade_307	N	N
Pigmentiphaga daeguensis	1467	JN585327	clade_307	N	N
Aeromonas allosaccharophila	104	S39232	clade_308	N	N
Aeromonas enteropelogenes	105	X71121	clade_308	N	N
Aeromonas hydrophila	106	NC_008570	clade_308	N	N
Aeromonas jandaei	107	X60413	clade_308	N	N
Aeromonas salmonicida	108	NC_009348	clade_308	N	N
Aeromonas trota	109	X60415	clade_308	N	N
Aeromonas veronii	110	NR_044845	clade_308	N	N
Blautia coccoides	373	AB571656	clade_309	Y	N

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Blautia glucerasea	374	AB588023	clade_309	Y	N
Blautia glucerasei	375	AB439724	clade_309	Y	N
Blautia hansenii	376	ABYU02000037	clade_309	Y	N
Blautia luti	378	AB691576	clade_309	Y	N
Blautia producta	379	AB600998	clade_309	Y	N
Blautia schinkii	380	NR_026312	clade_309	Y	N
Blautia sp. M25	381	HM626178	clade_309	Y	N
Blautia stercoris	382	HM626177	clade_309	Y	N
Blautia wexlerae	383	EF036467	clade_309	Y	N
Bryantella formatexigens	439	ACCL02000018	clade_309	Y	N
Clostridium coccoides	573	EF025906	clade_309	Y	N
Eubacterium cellulosolvens	839	AY178842	clade_309	Y	N
Lachnospiraceae bacterium 6_1_63FAA	1056	ACTV01000014	clade_309	Y	N
Marvinbryantia formatexigens	1196	AJ505973	clade_309	N	N
Ruminococcus hansenii	1662	M59114	clade_309	Y	N
Ruminococcus obeum	1664	AY169419	clade_309	Y	N
Ruminococcus sp. 5_1_39BFAA	1666	ACII01000172	clade_309	Y	N
Ruminococcus sp. K_1	1669	AB222208	clade_309	Y	N
Syntrophococcus sucromutans	1911	NR_036869	clade_309	Y	N
Rhodobacter sp. oral taxon C30	1620	HM099648	clade_310	N	N
Rhodobacter sphaeroides	1621	CP000144	clade_310	N	N
Lactobacillus antri	1071	ACLL01000037	clade_313	N	N
Lactobacillus coleohominis	1076	ACOH01000030	clade_313	N	N
Lactobacillus fermentum	1083	CP002033	clade_313	N	N
Lactobacillus gastricus	1085	AICN01000060	clade_313	N	N
Lactobacillus mucosae	1099	FR693800	clade_313	N	N
Lactobacillus oris	1103	AEKL01000077	clade_313	N	N
Lactobacillus pontis	1111	HM218420	clade_313	N	N
Lactobacillus reuteri	1112	ACGW02000012	clade_313	N	N
Lactobacillus sp. KLDS 1.0707	1127	EU600911	clade_313	N	N
Lactobacillus sp. KLDS 1.0709	1128	EU600913	clade_313	N	N
Lactobacillus sp. KLDS 1.0711	1129	EU600915	clade_313	N	N
Lactobacillus sp. KLDS 1.0713	1131	EU600917	clade_313	N	N
Lactobacillus sp. KLDS 1.0716	1132	EU600921	clade_313	N	N
Lactobacillus sp. KLDS 1.0718	1133	EU600922	clade_313	N	N
Lactobacillus sp. oral taxon 052	1137	GQ422710	clade_313	N	N
Lactobacillus vaginalis	1140	ACGV01000168	clade_313	N	N
Brevibacterium aurantiacum	419	NR_044854	clade_314	N	N
Brevibacterium linens	423	AJ315491	clade_314	N	N
Lactobacillus pentosus	1108	JN813103	clade_315	N	N
Lactobacillus plantarum	1110	ACGZ02000033	clade_315	N	N
Lactobacillus sp. KLDS 1.0702	1123	EU600906	clade_315	N	N
Lactobacillus sp. KLDS 1.0703	1124	EU600907	clade_315	N	N
Lactobacillus sp. KLDS 1.0704	1125	EU600908	clade_315	N	N
Lactobacillus sp. KLDS 1.0705	1126	EU600909	clade_315	N	N
Agrobacterium radiobacter	115	CP000628	clade_316	N	N
Agrobacterium tumefaciens	116	AJ389893	clade_316	N	N
Corynebacterium argentoratense	685	EF463055	clade_317	N	N
Corynebacterium diphtheriae	693	NC_002935	clade_317	N	OP
Corynebacterium pseudotuberculosis	715	NR_037070	clade_317	N	N
Corynebacterium renale	717	NR_037069	clade_317	N	N
Corynebacterium ulcerans	731	NR_074467	clade_317	N	N
Aurantimonas coralicida	191	AY065627	clade_318	N	N
Aureimonas altamirensis	192	FN658986	clade_318	N	N
Lactobacillus acidipiscis	1066	NR_024718	clade_320	N	N
Lactobacillus salivarius	1117	AEBA01000145	clade_320	N	N
Lactobacillus sp. KLDS 1.0719	1134	EU600923	clade_320	N	N
Lactobacillus buchneri	1073	ACGH01000101	clade_321	N	N
Lactobacillus genomosp. C1	1086	AY278619	clade_321	N	N
Lactobacillus genomosp. C2	1087	AY278620	clade_321	N	N
Lactobacillus hilgardii	1089	ACGP01000200	clade_321	N	N
Lactobacillus kefiri	1096	NR_042230	clade_321	N	N
Lactobacillus parabuchneri	1105	NR_041294	clade_321	N	N
Lactobacillus parakefiri	1107	NR_029039	clade_321	N	N
Lactobacillus curvatus	1079	NR_042437	clade_322	N	N
Lactobacillus sakei	1116	DQ989236	clade_322	N	N
Aneurinibacillus aneurinilyticus	167	AB101592	clade_323	N	N
Aneurinibacillus danicus	168	NR_028657	clade_323	N	N
Aneurinibacillus migulanus	169	NR_036799	clade_323	N	N
Aneurinibacillus terranovensis	170	NR_042271	clade_323	N	N
Staphylococcus aureus	1757	CP002643	clade_325	N	Category-B
Staphylococcus auricularis	1758	JQ624774	clade_325	N	N
Staphylococcus capitis	1759	ACFR01000029	clade_325	N	N
Staphylococcus caprae	1760	ACRH01000033	clade_325	N	N
Staphylococcus carnosus	1761	NR_075003	clade_325	N	N
Staphylococcus cohnii	1762	JN175375	clade_325	N	N
Staphylococcus condimenti	1763	NR_029345	clade_325	N	N

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Staphylococcus epidermidis	1764	ACHE01000056	clade_325	N	N
Staphylococcus equorum	1765	NR_027520	clade_325	N	N
Staphylococcus haemolyticus	1767	NC_007168	clade_325	N	N
Staphylococcus hominis	1768	AM157418	clade_325	N	N
Staphylococcus lugdunensis	1769	AEQA01000024	clade_325	N	N
Staphylococcus pasteuri	1770	FJ189773	clade_325	N	N
Staphylococcus pseudintermedius	1771	CP002439	clade_325	N	N
Staphylococcus saccharolyticus	1772	NR_029158	clade_325	N	N
Staphylococcus saprophyticus	1773	NC_007350	clade_325	N	N
Staphylococcus sp. clone bottae7	1777	AF467424	clade_325	N	N
Staphylococcus sp. H292	1775	AB177642	clade_325	N	N
Staphylococcus sp. H780	1776	AB177644	clade_325	N	N
Staphylococcus succinus	1778	NR_028667	clade_325	N	N
Staphylococcus warneri	1780	ACPZ01000009	clade_325	N	N
Staphylococcus xylosus	1781	AY395016	clade_325	N	N
Cardiobacterium hominis	490	ACKY01000036	clade_326	N	N
Cardiobacterium valvarum	491	NR_028847	clade_326	N	N
Pseudomonas fluorescens	1593	AY622220	clade_326	N	N
Pseudomonas gessardii	1594	FJ943496	clade_326	N	N
Pseudomonas monteilii	1596	NR_024910	clade_326	N	N
Pseudomonas poae	1597	GU188951	clade_326	N	N
Pseudomonas putida	1599	AF094741	clade_326	N	N
Pseudomonas sp. G1229	1601	DQ910482	clade_326	N	N
Pseudomonas tolaasii	1604	AF320988	clade_326	N	N
Pseudomonas viridiflava	1605	NR_042764	clade_326	N	N
Bacillus alcalophilus	198	X76436	clade_327	Y	N
Bacillus clausii	205	FN397477	clade_327	Y	OP
Bacillus gelatini	210	NR_025595	clade_327	Y	OP
Bacillus halodurans	212	AY144582	clade_327	Y	OP
Bacillus sp. oral taxon F26	246	HM099642	clade_327	Y	OP
Listeria grayi	1185	ACCR02000003	clade_328	N	OP
Listeria innocua	1186	JF967625	clade_328	N	N
Listeria ivanovii	1187	X56151	clade_328	N	N
Listeria monocytogenes	1188	CP002003	clade_328	N	Category-B
Listeria welshimeri	1189	AM263198	clade_328	N	OP
Capnocytophaga sp. oral clone ASCH05	484	AY923149	clade_333	N	N
Capnocytophaga sputigena	489	ABZV01000054	clade_333	N	N
Leptotrichia genomosp. C1	1166	AY278621	clade_334	N	N
Leptotrichia shahii	1169	AY029806	clade_334	N	N
Leptotrichia sp. neutropenicPatient	1170	AF189244	clade_334	N	N
Leptotrichia sp. oral clone GT018	1171	AY349384	clade_334	N	N
Leptotrichia sp. oral clone GT020	1172	AY349385	clade_334	N	N
Bacteroides sp. 20_3	296	ACRQ01000064	clade_335	N	N
Bacteroides sp. 3_1_19	307	ADCJ01000062	clade_335	N	N
Bacteroides sp. 3_2_5	311	ACIB01000079	clade_335	N	N
Parabacteroides distasonis	1416	CP000140	clade_335	N	N
Parabacteroides goldsteinii	1417	AY974070	clade_335	N	N
Parabacteroides gordonii	1418	AB470344	clade_335	N	N
Parabacteroides sp. D13	1421	ACPW01000017	clade_335	N	N
Capnocytophaga genomosp. C1	477	AY278613	clade_336	N	N
Capnocytophaga ochracea	480	AEOH01000054	clade_336	N	N
Capnocytophaga sp. GEJ8	481	GU561335	clade_336	N	N
Capnocytophaga sp. oral strain A47ROY	486	AY005077	clade_336	N	N
Capnocytophaga sp. S1b	482	U42009	clade_336	N	N
Paraprevotella clara	1426	AFFY01000068	clade_336	N	N
Bacteroides heparinolyticus	282	JN867284	clade_338	N	N
Prevotella heparinolytica	1500	GQ422742	clade_338	N	N
Treponema genomosp. P4 oral clone MB2_G19	1928	DQ003618	clade_339	N	N
Treponema genomosp. P6 oral clone MB4_G11	1930	DQ003625	clade_339	N	N
Treponema sp. oral taxon 254	1952	GU408803	clade_339	N	N
Treponema sp. oral taxon 508	1956	GU413616	clade_339	N	N
Treponema sp. oral taxon 518	1957	GU413640	clade_339	N	N
Chlamydia muridarum	502	AE002160	clade_341	N	OP
Chlamydia trachomatis	504	U68443	clade_341	N	OP
Chlamydia psittaci	503	NR_036864	clade_342	N	Category-B
Chiamydophila pneumoniae	509	NC_002179	clade_342	N	OP
Chlamydophila psittaci	510	D85712	clade_342	N	OP
Anaerococcus octavius	146	NR_026360	clade_343	N	N
Anaerococcus sp. 8405254	149	HM587319	clade_343	N	N
Anaerococcus sp. 9401487	150	HM587322	clade_343	N	N
Anaerococcus sp. 9403502	151	HM587325	clade_343	N	N
Gardnerella vaginalis	923	CP001849	clade_344	N	N
Campylobacter lari	466	CP000932	clade_346	N	OP
Anaerobiospirillum succiniciproducens	142	NR_026075	clade_347	N	N
Anaerobiospirillum thomasii	143	AJ420985	clade_347	N	N
Ruminobacter amylophilus	1654	NR_026450	clade_347	N	N
Succinatimonas hippei	1897	AEVO01000027	clade_347	N	N

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OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Actinomyces europaeus	54	NR_026363	clade_348	N	N
Actinomyces sp. oral clone GU009	82	AY349361	clade_348	N	N
Moraxella catarrhalis	1260	CP002005	clade_349	N	N
Moraxella lincolnii	1261	FR822735	clade_349	N	N
Moraxella sp. 16285	1263	JF682466	clade_349	N	N
Psychrobacter sp. 13983	1613	HM212668	clade_349	N	N
Actinobaculum massiliae	49	AF487679	clade_350	N	N
Actinobaculum schaalii	50	AY957507	clade_350	N	N
Actinobaculum sp. BM#101342	51	AY282578	clade_350	N	N
Actinobaculum sp. P2P_19 P1	52	AY207066	clade_350	N	N
Actinomyces sp. oral clone IO076	84	AY349363	clade_350	N	N
Actinomyces sp. oral taxon 848	93	ACUY01000072	clade_350	N	N
Clostridium innocuum	595	M23732	clade_351	Y	N
Clostridium sp. HGF2	628	AENW01000022	clade_351	Y	N
Actinomyces neuii	65	X71862	clade_352	N	N
Mobiluncus mulieris	1252	ACKW01000035	clade_352	N	N
Clostridium perfringens	612	ABDW01000023	clade_353	Y	Category-B
Sarcina ventriculi	1687	NR_026146	clade_353	Y	N
Clostridium bartlettii	556	ABEZ02000012	clade_354	Y	N
Clostridium bifermentans	558	X73437	clade_354	Y	N
Clostridium ghonii	586	AB542933	clade_354	Y	N
Clostridium glycolicum	587	FJ384385	clade_354	Y	N
Clostridium mayombei	605	FR733682	clade_354	Y	N
Clostridium sordellii	625	AB448946	clade_354	Y	N
Clostridium sp. MT4 E	635	FJ159523	clade_354	Y	N
Eubacterium tenue	872	M59118	clade_354	Y	N
Clostridium argentinense	553	NR_029232	clade_355	Y	N
Clostridium sp. JC122	630	CAEV01000127	clade_355	Y	N
Clostridium sp. NMBHI_1	636	JN093130	clade_355	Y	N
Clostridium subterminale	650	NR_041795	clade_355	Y	N
Clostridium sulfidigenes	651	NR_044161	clade_355	Y	N
Blastomonas natatoria	372	NR_040824	clade_356	N	N
Novospbingobium aromaticivorans	1357	AAAV03000008	clade_356	N	N
Sphingomonas sp. oral clone FI012	1745	AY349411	clade_356	N	N
Sphingopyxis alaskensis	1749	CP000356	clade_356	N	N
Oxalobacter formigenes	1389	ACDQ01000020	clade_357	N	N
Veillonella atypica	1974	AEDS01000059	clade_358	N	N
Veillonella dispar	1975	ACIK02000021	clade_358	N	N
Veillonella genomosp. P1 oral clone MB5_P17	1976	DQ003631	clade_358	N	N
Veillonella parvula	1978	ADFU01000009	clade_358	N	N
Veillonella sp. 3_1_44	1979	ADCV01000019	clade_358	N	N
Veillonella sp. 6_1_27	1980	ADCW01000016	clade_358	N	N
Veillonella sp. ACP1	1981	HQ616359	clade_358	N	N
Veillonella sp. AS16	1982	HQ616365	clade_358	N	N
Veillonella sp. BS32b	1983	HQ616368	clade_358	N	N
Veillonella sp. ICM51a	1984	HQ616396	clade_358	N	N
Veillonella sp. MSA12	1985	HQ616381	clade_358	N	N
Veillonella sp. NVG 100cf	1986	EF108443	clade_358	N	N
Veillonella sp. OK11	1987	JN695650	clade_358	N	N
Veillonella sp. oral clone ASCG01	1990	AY923144	clade_358	N	N
Veillonella sp. oral clone ASCG02	1991	AY953257	clade_358	N	N
Veillonella sp. oral clone OH1A	1992	AY947495	clade_358	N	N
Veillonella sp. oral taxon 158	1993	AENU01000007	clade_358	N	N
Dorea formicigenerans	773	AAXA02000006	clade_360	Y	N
Dorea longicatena	774	AJ132842	clade_360	Y	N
Lachnospiraceae bacterium 2_1_46FAA	1050	ADLB01000035	clade_360	Y	N
Lachnospiraceae bacterium 2_1_58FAA	1051	ACTO01000052	clade_360	Y	N
Lachnospiraceae bacterium 4_1_37FAA	1053	ADCR01000030	clade_360	Y	N
Lachnospiraceae bacterium 9_1_43BFAA	1058	ACTX01000023	clade_360	Y	N
Ruminococcus gnavus	1661	X94967	clade_360	Y	N
Ruminococcus sp. ID8	1668	AY960564	clade_360	Y	N
Kocuria marina	1040	GQ260086	clade_365	N	N
Kocuria rhizophila	1042	AY030315	clade_365	N	N
Kocuria rosea	1043	X87756	clade_365	N	N
Kocuria varians	1044	AF542074	clade_365	N	N
Blautia hydrogenotrophica	377	ACBZ01000217	clade_368	Y	N
Clostridiaceae bacterium END_2	531	EF451053	clade_368	N	N
Lactonifactor longoviformis	1147	DQ100449	clade_368	Y	N
Robinsoniella peoriensis	1633	AF445258	clade_368	Y	N
Micrococcus antarcticus	1242	NR_025285	clade_371	N	N
Micrococcus luteus	1243	NR_075062	clade_371	N	N
Micrococcus lylae	1244	NR_026200	clade_371	N	N
Micrococcus sp. 185	1245	EU714334	clade_371	N	N
Lactobacillus brevis	1072	EU194349	clade_372	N	N
Lactobacillus parabrevis	1104	NR_042456	clade_372	N	N
Pediococcus acidilactici	1436	ACXB01000026	clade_372	N	N
Pediococcus pentosaceus	1437	NR_075052	clade_372	N	N

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OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Lactobacillus dextrinicus	1081	NR_036861	clade_373	N	N
Lactobacillus perolens	1109	NR_029360	clade_373	N	N
Lactobacillus rhamnosus	1113	ABWJ01000068	clade_373	N	N
Lactobacillus saniviri	1118	AB602569	clade_373	N	N
Lactobacillus sp. BT6	1121	HQ616370	clade_373	N	N
Mycobacterium mageritense	1282	FR798914	clade_374	N	OP
Mycobacterium neoaurum	1286	AF268445	clade_374	N	OP
Mycobacterium smegmatis	1291	CP000480	clade_374	N	OP
Mycobacterium sp. HE5	1304	AJ012738	clade_374	N	N
Dysgonomonas gadei	775	ADLV01000001	clade_377	N	N
Dysgonomonas mossii	776	ADLW01000023	clade_377	N	N
Porphyromonas levii	1474	NR_025907	clade_377	N	N
Porphyromonas somerae	1476	AB547667	clade_377	N	N
Bacteroides barnesiae	267	NR_041446	clade_378	N	N
Bacteroides coprocola	272	ABIY02000050	clade_378	N	N
Bacteroides coprophilus	273	ACBW01000012	clade_378	N	N
Bacteroides dorei	274	ABWZ01000093	clade_378	N	N
Bacteroides massiliensis	284	AB200226	clade_378	N	N
Bacteroides plebeius	289	AB200218	clade_378	N	N
Bacteroides sp. 3_1_33FAA	309	ACPS01000085	clade_378	N	N
Bacteroides sp. 3_1_40A	310	ACRT01000136	clade_378	N	N
Bacteroides sp. 4_3_47FAA	313	ACDR02000029	clade_378	N	N
Bacteroides sp. 9_1_42FAA	314	ACAA01000096	clade_378	N	N
Bacteroides sp. NB_8	323	AB117565	clade_378	N	N
Bacteroides vulgatus	331	CP000139	clade_378	N	N
Bacteroides ovatus	287	ACWH01000036	clade_38	N	N
Bacteroides sp. 1_1_30	294	ADCL01000128	clade_38	N	N
Bacteroides sp. 2_1_22	297	ACPQ01000117	clade_38	N	N
Bacteroides sp. 2_2_4	299	ABZZ01000168	clade_38	N	N
Bacteroides sp. 3_1_23	308	ACRS01000081	clade_38	N	N
Bacteroides sp. D1	318	ACAB02000030	clade_38	N	N
Bacteroides sp. D2	321	ACGA01000077	clade_38	N	N
Bacteroides sp. D22	320	ADCK01000151	clade_38	N	N
Bacteroides xylanisolvens	332	ADKP01000087	clade_38	N	N
Treponema lecithinolyticum	1931	NR_026247	clade_380	N	OP
Treponema parvum	1933	AF302937	clade_380	N	OP
Treponema sp. oral clone JU025	1940	AY349417	clade_380	N	N
Treponema sp. oral taxon 270	1954	GQ422733	clade_380	N	N
Parascardovia denticolens	1428	ADEB01000020	clade_381	N	N
Scardovia inopinata	1688	AB029087	clade_381	N	N
Scardovia wiggsiae	1689	AY278626	clade_381	N	N
Clostridiales bacterium 9400853	533	HM587320	clade_384	N	N
Eubacterium infirmum	849	U13039	clade_384	Y	N
Eubacterium sp. WAL 14571	864	FJ687606	clade_384	Y	N
Mogibacterium diversum	1254	NR_027191	clade_384	N	N
Mogibacterium neglectum	1255	NR_027203	clade_384	N	N
Mogibacterium pumilum	1256	NR_028608	clade_384	N	N
Mogibacterium timidum	1257	Z36296	clade_384	N	N
Erysipeiotrichaceae bacterium 5_2_54FAA	823	ACZW01000054	clade_385	Y	N
Eubacterium biforme	835	ABYT01000002	clade_385	Y	N
Eubacterium cylindroides	842	FP929041	clade_385	Y	N
Eubacterium dolichum	844	L34682	clade_385	Y	N
Eubacterium sp. 3_1_31	861	ACTL01000045	clade_385	Y	N
Eubacterium tortuosum	873	NR_044648	clade_385	Y	N
Borrelia burgdorferi	389	ABGI01000001	clade_386	N	OP
Borrelia garinii	392	ABJV01000001	clade_386	N	OP
Borrelia sp. NE49	397	AJ224142	clade_386	N	OP
Caldimonas manganoxidans	457	NR_040787	clade_387	N	N
Comamonadaceae bacterium oral taxon F47	667	HM099651	clade_387	N	N
Lautropia mirabilis	1149	AEQP01000026	clade_387	N	N
Lautropia sp. oral clone AP009	1150	AY005030	clade_387	N	N
Bulleidia extructa	441	ADFR01000011	clade_388	Y	N
Solobacterium moorei	1739	AECQ01000039	clade_388	Y	N
Peptoniphilus asaccharolyticus	1441	D14145	clade_389	N	N
Peptoniphilus duerdenii	1442	EU526290	clade_389	N	N
Peptoniphilus harei	1443	NR_026358	clade_389	N	N
Peptoniphilus indolicus	1444	AY153431	clade_389	N	N
Peptoniphilus lacrimalis	1446	ADDO01000050	clade_389	N	N
Peptoniphilus sp. gpac077	1450	AM176527	clade_389	N	N
Peptoniphilus sp. JC140	1447	JF824803	clade_389	N	N
Peptoniphilus sp. oral taxon 386	1452	ADCS01000031	clade_389	N	N
Peptoniphilus sp. oral taxon 836	1453	AEAA01000090	clade_389	N	N
Peptostreptococcaceae bacterium ph1	1454	JN837495	clade_389	N	N
Dialister pneumosintes	765	HM596297	clade_390	N	N
Dialister sp. oral taxon 502	767	GQ422739	clade_390	N	N
Cupriavidus metallidurans	741	GU230889	clade_391	N	N
Herbaspirillum seropedicae	1001	CP002039	clade_391	N	N

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Herbaspirillum sp. JC206	1002	JN657219	clade_391	N	N
Janthinobacterium sp. SY12	1015	EF455530	clade_391	N	N
Massilia sp. CCUG 43427A	1197	FR773700	clade_391	N	N
Ralstonia pickettii	1615	NC_010682	clade_391	N	N
Ralstonia sp. 5_7_47FAA	1616	ACUF01000076	clade_391	N	N
Francisella novicida	889	ABSS01000002	clade_392	N	N
Francisella philomiragia	890	AY928394	clade_392	N	N
Francisella tularensis	891	ABAZ01000082	clade_392	N	Category-A
Ignatzschineria indica	1009	HQ823562	clade_392	N	N
Ignatzschineria sp. NML 95_0260	1010	HQ823559	clade_392	N	N
Coprococcus catus	673	EU266552	clade_393	Y	N
Lachnospiraceae bacterium oral taxon F15	1064	HM099641	clade_393	Y	N
Streptococcus mutans	1814	AP010655	clade_394	N	N
Clostridium cochlearium	574	NR_044717	clade_395	Y	N
Clostridium malenominatum	604	FR749893	clade_395	Y	N
Clostridium tetani	654	NC_004557	clade_395	Y	N
Acetivibrio ethanolgignens	6	FR749897	clade_396	Y	N
Anaerosporobacter mobilis	161	NR_042953	clade_396	Y	N
Bacteroides pectinophilus	288	ABVQ01000036	clade_396	Y	N
Clostridium aminovalericum	551	NR_029245	clade_396	Y	N
Clostridium phytofermentans	613	NR_074652	clade_396	Y	N
Eubacterium hallii	848	L34621	clade_396	Y	N
Eubacterium xylanophilum	875	L34628	clade_396	Y	N
Lactobacillus gasseri	1084	ACOZ01000018	clade_398	N	N
Lactobacillus hominis	1090	FR681902	clade_398	N	N
Lactobacillus iners	1091	AEKJ01000002	clade_398	N	N
Lactobacillus johnsonii	1093	AE017198	clade_398	N	N
Lactobacillus senioris	1119	AB602570	clade_398	N	N
Lactobacillus sp. oral clone HT002	1135	AY349382	clade_398	N	N
Weissella beninensis	2006	EU439435	clade_398	N	N
Sphingomonas echinoides	1744	NR_024700	clade_399	N	N
Sphingomonas sp. oral taxon A09	1747	HM099639	clade_399	N	N
Sphingomonas sp. oral taxon F71	1748	HM099645	clade_399	N	N
Zymomonas mobilis	2032	NR_074274	clade_399	N	N
Arcanobacterium haemolyticum	174	NR_025347	clade_400	N	N
Arcanobacterium pyogenes	175	GU585578	clade_400	N	N
Trueperella pyogenes	1962	NR_044858	clade_400	N	N
Lactococcus garvieae	1144	AF061005	clade_401	N	N
Lactococcus lactis	1145	CP002365	clade_401	N	N
Brevibacterium mcbrellneri	424	ADNU01000076	clade_402	N	N
Brevibacterium paucivorans	425	EU086796	clade_402	N	N
Brevibacterium sp. JC43	428	JF824806	clade_402	N	N
Selenomonas artemidis	1692	HM596274	clade_403	N	N
Selenomonas sp. FOBRC9	1704	HQ616378	clade_403	N	N
Selenomonas sp. oral taxon 137	1715	AENV01000007	clade_403	N	N
Desmospora activa	751	AM940019	clade_404	N	N
Desmospora sp. 8437	752	AFHT01000143	clade_404	N	N
Paenibacillus sp. oral taxon F45	1407	HM099647	clade_404	N	N
Corynebacterium ammoniagenes	682	ADNS01000011	clade_405	N	N
Corynebacterium aurimucosum	687	ACLH01000041	clade_405	N	N
Corynebacterium bovis	688	AF537590	clade_405	N	N
Corynebacterium canis	689	GQ871934	clade_405	N	N
Corynebacterium casei	690	NR_025101	clade_405	N	N
Corynebacterium durum	694	Z97069	clade_405	N	N
Corynebacterium efficiens	695	ACLI01000121	clade_405	N	N
Corynebacterium falsenii	696	Y13024	clade_405	N	N
Corynebacterium flavescens	697	NR_037040	clade_405	N	N
Corynebacterium glutamicum	701	BA000036	clade_405	N	N
Corynebacterium jeikeium	704	ACYW01000001	clade_405	N	OP
Corynebacterium kroppenstedtii	705	NR_026380	clade_405	N	N
Corynebacterium lipophiloflavum	706	ACHJ01000075	clade_405	N	N
Corynebacterium matruchotii	709	ACSH02000003	clade_405	N	N
Corynebacterium minutissimum	710	X82064	clade_405	N	N
Corynebacterium resistens	718	ADGN01000058	clade_405	N	N
Corynebacterium simulans	720	AF537604	clade_405	N	N
Corynebacterium singulare	721	NR_026394	clade_405	N	N
Corynebacterium sp. 1 ex sheep	722	Y13427	clade_405	N	N
Corynebacterium sp. NML 99_0018	726	GU238413	clade_405	N	N
Corynebacterium striatum	727	ACGE01000001	clade_405	N	OP
Corynebacterium urealyticum	732	X81913	clade_405	N	OP
Corynebacterium variabile	734	NR_025314	clade_405	N	N
Ruminococcus callidus	1658	NR_029160	clade_406	Y	N
Ruminococcus champanellensis	1659	FP929052	clade_406	Y	N
Ruminococcus sp. 18P13	1665	AJ515913	clade_406	Y	N
Ruminococcus sp. 9SE51	1667	FM954974	clade_406	Y	N
Aerococcus sanguinicola	98	AY837833	clade_407	N	N
Aerococcus urinae	99	CP002512	clade_407	N	N

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Aerococcus urinaeequi	100	NR_043443	clade_407	N	N
Aerococcus viridans	101	ADNT01000041	clade_407	N	N
Anaerostipes caccae	162	ABAX03000023	clade_408	Y	N
Anaerostipes sp. 3_2_56FAA	163	ACWB01000002	clade_408	Y	N
Clostridiales bacterium 1_7_47FAA	541	ABQR01000074	clade_408	Y	N
Clostridiales sp. SM4_1	542	FP929060	clade_408	Y	N
Clostridiales sp. SSC_2	544	FP929061	clade_408	Y	N
Clostridium aerotolerans	546	X76163	clade_408	Y	N
Clostridium aldenense	547	NR_043680	clade_408	Y	N
Clostridium algidixylanolyticum	550	NR_028726	clade_408	Y	N
Clostridium amygdalinum	552	AY353957	clade_408	Y	N
Clostridium asparagiforme	554	ACCJ01000522	clade_408	Y	N
Clostridium bolteae	559	ABCC02000039	clade_408	Y	N
Clostridium celerecrescens	566	JQ246092	clade_408	Y	N
Clostridium citroniae	569	ADLJ01000059	clade_408	Y	N
Clostridium clostridiiformes	571	M59089	clade_408	Y	N
Clostridium clostridioforme	572	NR_044715	clade_408	Y	N
Clostridium hathewayi	590	AY552788	clade_408	Y	N
Clostridium indolis	594	AF028351	clade_408	Y	N
Clostridium lavalense	600	EF564277	clade_408	Y	N
Clostridium saccharolyticum	620	CP002109	clade_408	Y	N
Clostridium sp. M62_1	633	ACFX02000046	clade_408	Y	N
Clostridium sp. SS2_1	638	ABGC03000041	clade_408	Y	N
Clostridium sphenoides	643	X73449	clade_408	Y	N
Clostridium symbiosum	652	ADLQ01000114	clade_408	Y	N
Clostridium xylanolyticum	658	NR_037068	clade_408	Y	N
Eubacterium hadrum	847	FR749933	clade_408	Y	N
Fusobacterium naviforme	898	HQ223106	clade_408	N	N
Lachnospiraceae bacterium 3_1_57FAA	1052	ACTP01000124	clade_408	Y	N
Lachnospiraceae bacterium 5_1_63FAA	1055	ACTS01000081	clade_408	Y	N
Lachnospiraceae bacterium A4	1059	DQ789118	clade_408	Y	N
Lachnospiraceae bacterium DJF VP30	1060	EU728771	clade_408	Y	N
Lachnospiraceae genomosp. C1	1065	AY278618	clade_408	Y	N
Moryella indoligenes	1268	AF527773	clade_408	N	N
Clostridium difficile	578	NC_013315	clade_409	Y	OP
Selenomonas genomosp. P5	1697	AY341820	clade_410	N	N
Selenomonas sp. oral clone IQ048	1710	AY349408	clade_410	N	N
Selenomonas sputigena	1717	ACKP02000033	clade_410	N	N
Hyphomicrobium sulfonivorans	1007	AY468372	clade_411	N	N
Methylocella silvestris	1228	NR_074237	clade_411	N	N
Legionella pneumophila	1153	NC_002942	clade_412	N	OP
Lactobacillus coryniformis	1077	NR_044705	clade_413	N	N
Arthrobacter agilis	178	NR_026198	clade_414	N	N
Arthrobacter arilaitensis	179	NR_074608	clade_414	N	N
Arthrobacter bergerei	180	NR_025612	clade_414	N	N
Arthrobacter globiformis	181	NR_026187	clade_414	N	N
Arthrobacter nicotianae	182	NR_026190	clade_414	N	N
Mycobacterium abscessus	1269	AGQU01000002	clade_418	N	OP
Mycobacterium chelonae	1273	AB548610	clade_418	N	OP
Bacteroides salanitronis	291	CP002530	clade_419	N	N
Paraprevotella xylaniphila	1427	AFBR01000011	clade_419	N	N
Barnesiella intestinihominis	336	AB370251	clade_420	N	N
Barnesiella viscericola	337	NR_041508	clade_420	N	N
Parabacteroides sp. NS31_3	1422	JN029805	clade_420	N	N
Porphyromonadaceae bacterium NML 060648	1470	EF184292	clade_420	N	N
Tannerella forsythia	1913	CP003191	clade_420	N	N
Tannerella sp. 6_1_58FAA_CT1	1914	ACWX01000068	clade_420	N	N
Mycoplasma amphoriforme	1311	AY531656	clade_421	N	N
Mycoplasma genitalium	1317	L43967	clade_421	N	N
Mycoplasma pneumoniae	1322	NC_000912	clade_421	N	N
Mycoplasma penetrans	1321	NC_004432	clade_422	N	N
Ureaplasma parvum	1966	AE002127	clade_422	N	N
Ureaplasma urealyticum	1967	AAYN01000002	clade_422	N	N
Treponema genomosp. P1	1927	AY341822	clade_425	N	N
Treponema sp. oral taxon 228	1943	GU408580	clade_425	N	N
Treponema sp. oral taxon 230	1944	GU408603	clade_425	N	N
Treponema sp. oral taxon 231	1945	GU408631	clade_425	N	N
Treponema sp. oral taxon 232	1946	GU408646	clade_425	N	N
Treponema sp. oral taxon 235	1947	GU408673	clade_425	N	N
Treponema sp. ovine footrot	1959	AJ010951	clade_425	N	N
Treponema vincentii	1960	ACYH01000036	clade_425	N	OP
Eubacterium sp. AS15b	862	HQ616364	clade_428	Y	N
Eubacterium sp. OBRC9	863	HQ616354	clade_428	Y	N
Eubacterium sp. oral clone OH3A	871	AY947497	clade_428	Y	N
Eubacterium yurii	876	AEES01000073	clade_428	Y	N
Clostridium acetobutylicum	545	NR_074511	clade_430	Y	N
Clostridium algidicarnis	549	NR_041746	clade_430	Y	N

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Clostridium cadaveris	562	AB542932	clade_430	Y	N
Clostridium carboxidivorans	563	FR733710	clade_430	Y	N
Clostridium estertheticum	580	NR_042153	clade_430	Y	N
Clostridium fallax	581	NR_044714	clade_430	Y	N
Clostridium felsineum	583	AF270502	clade_430	Y	N
Clostridium frigidicarnis	584	NR_024919	clade_430	Y	N
Clostridium kluyveri	598	NR_074165	clade_430	Y	N
Clostridium magnum	603	X77835	clade_430	Y	N
Clostridium putrefaciens	615	NR_024995	clade_430	Y	N
Clostridium sp. HPB_46	629	AY862516	clade_430	Y	N
Clostridium tyrobutyricum	656	NR_044718	clade_430	Y	N
Burkholderiales bacterium 1_1_47	452	ADCQ01000066	clade_432	N	OP
Parasutterella excrementihominis	1429	AFBP01000029	clade_432	N	N
Parasutterella secunda	1430	AB491209	clade_432	N	N
Sutterella morbirenis	1898	AJ832129	clade_432	N	N
Sutterella parvirubra	1899	AB300989	clade_432	Y	N
Sutterella sanguinus	1900	AJ748647	clade_432	N	N
Sutterella sp. YIT 12072	1901	AB491210	clade_432	N	N
Sutterella stercoricanis	1902	NR_025600	clade_432	N	N
Sutterella wadsworthensis	1903	ADMF01000048	clade_432	N	N
Propionibacterium freudenreichii	1572	NR_036972	clade_433	N	N
Propionibacterium sp. oral taxon 192	1580	GQ422728	clade_433	N	N
Tessaracoccus sp. oral taxon F04	1917	HM099640	clade_433	N	N
Peptoniphilus ivorii	1445	Y07840	clade_434	N	N
Peptoniphilus sp. gpac007	1448	AM176517	clade_434	N	N
Peptoniphilus sp. gpac018A	1449	AM176519	clade_434	N	N
Peptoniphilus sp. gpac148	1451	AM176535	clade_434	N	N
Flexispira rappini	887	AY126479	clade_436	N	N
Helicobacter bilis	993	ACDN01000023	clade_436	N	N
Helicobacter cinaedi	995	ABQT01000054	clade_436	N	N
Helicobacter sp. None	998	U44756	clade_436	N	N
Brevundimonas subvibrioides	429	CP002102	clade_438	N	N
Hyphomonas neptunium	1008	NR_074092	clade_438	N	N
Phenylobacterium zucineum	1465	AY628697	clade_438	N	N
Acetanaerobaeterium elongatum	4	NR_042930	clade_439	Y	N
Clostridium cellulosi	567	NR_044624	clade_439	Y	N
Ethanoligenens harbinense	832	AY675965	clade_439	Y	N
Streptococcus downei	1793	AEKN01000002	clade_441	N	N
Streptococcus sp. SHV515	1848	Y07601	clade_441	N	N
Acinetobacter sp. CIP 53.82	40	JQ638584	clade_443	N	N
Halomonas elongata	990	NR_074782	clade_443	N	N
Halomonas johnsoniae	991	FR775979	clade_443	N	N
Butyrivibrio fibrisolvens	456	U41172	clade_444	N	N
Eubacterium rectale	856	FP929042	clade_444	Y	N
Eubacterium sp. oral clone GI038	865	AY349374	clade_444	Y	N
Lachnobacterium bovis	1045	GU324407	clade_444	Y	N
Roseburia cecicola	1634	GU233441	clade_444	Y	N
Roseburia faecalis	1635	AY804149	clade_444	Y	N
Roseburia faecis	1636	AY305310	clade_444	Y	N
Roseburia hominis	1637	AJ270482	clade_444	Y	N
Roseburia intestinalis	1638	FP929050	clade_444	Y	N
Roseburia inulinivorans	1639	AJ270473	clade_444	Y	N
Roseburia sp. 11SE37	1640	FM954975	clade_444	N	N
Roseburia sp. 11SE38	1641	FM954976	clade_444	N	N
Shuttleworthia satelles	1728	ACIP02000004	clade_444	N	N
Shuttleworthia sp. MSX8B	1729	HQ616383	clade_444	N	N
Shuttleworthia sp. oral taxon G69	1730	GU432167	clade_444	N	N
Bdellovibrio sp. MPA	344	AY294215	clade_445	N	N
Desulfobulbus sp. oral clone CH031	755	AY005036	clade_445	N	N
Desulfovibrio desulfuricans	757	DQ092636	clade_445	N	N
Desulfovibrio fairfieldensis	758	U42221	clade_445	N	N
Desulfovibrio piger	759	AF192152	clade_445	N	N
Desulfovibrio sp. 3_1_syn3	760	ADDR01000239	clade_445	N	N
Geobacter bemidjiensis	941	CP001124	clade_445	N	N
Brachybacterium alimentarium	401	NR_026269	clade_446	N	N
Brachybacterium conglomeratum	402	AB537169	clade_446	N	N
Brachybacterium tyrofermentans	403	NR_026272	clade_446	N	N
Dermabacter hominis	749	FJ263375	clade_446	N	N
Aneurinibacillus thermoaerophilus	171	NR_029303	clade_448	N	N
Brevibacillus agri	409	NR_040983	clade_448	N	N
Brevibacillus brevis	410	NR_041524	clade_448	Y	N
Brevibacillus centrosporus	411	NR_043414	clade_448	N	N
Brevibacillus choshinensis	412	NR_040980	clade_448	N	N
Brevibacillus invocatus	413	NR_041836	clade_448	N	N
Brevibacillus laterosporus	414	NR_037005	clade_448	Y	N
Brevibacillus parabrevis	415	NR_040981	clade_448	N	N
Brevibacillus reuszeri	416	NR_040982	clade_448	N	N

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Brevibacillus sp. phR	417	JN837488	clade_448	N	N
Brevibacillus thermoruber	418	NR_026514	clade_448	N	N
Lactobacillus murinus	1100	NR_042231	clade_449	N	N
Lactobacillus oeni	1102	NR_043095	clade_449	N	N
Lactobacillus ruminis	1115	ACGS02000043	clade_449	N	N
Lactobacillus vini	1141	NR_042196	clade_449	N	N
Gemella haemolysans	924	ACDZ02000012	clade_450	N	N
Gemella morbillorum	925	NR_025904	clade_450	N	N
Gemella morbillorum	926	ACRX01000010	clade_450	N	N
Gemella sanguinis	927	ACRY01000057	clade_450	N	N
Gemella sp. oral clone ASCE02	929	AY923133	clade_450	N	N
Gemella sp. oral clone ASCF04	930	AY923139	clade_450	N	N
Gemella sp. oral clone ASCF12	931	AY923143	clade_450	N	N
Gemella sp. WAL 1945J	928	EU427463	clade_450	N	N
Bacillus coagulans	206	DQ297928	clade_451	Y	OP
Sporolactobacillus inulinus	1752	NR_040962	clade_451	Y	N
Sporolactobacillus nakayamae	1753	NR_042247	clade_451	N	N
Gluconacetobacter entanii	945	NR_028909	clade_452	N	N
Gluconacetobacter europaeus	946	NR_026513	clade_452	N	N
Gluconacetobacter hansenii	947	NR_026133	clade_452	N	N
Gluconacetobacter oboediens	949	NR_041295	clade_452	N	N
Gluconacetobacter xylinus	950	NR_074338	clade_452	N	N
Auritibacter ignavus	193	FN554542	clade_453	N	N
Dermacoccus sp. Ellin185	750	AEIQ01000090	clade_453	N	N
Janibacter limosus	1013	NR_026362	clade_453	N	N
Janibacter melonis	1014	EF063716	clade_453	N	N
Kocuria palustris	1041	EU333884	clade_453	Y	N
Acetobacter aceti	7	NR_026121	clade_454	N	N
Acetobacter fabarum	8	NR_042678	clade_454	N	N
Acetobacter lovaniensis	9	NR_040832	clade_454	N	N
Acetobacter malorum	10	NR_025513	clade_454	N	N
Acetobacter orientalis	11	NR_028625	clade_454	N	N
Acetobacter pasteurianus	12	NR_026107	clade_454	N	N
Acetobacter pomorum	13	NR_042112	clade_454	N	N
Acetobacter syzygii	14	NR_040868	clade_454	N	N
Acetobacter tropicalis	15	NR_036881	clade_454	N	N
Gluconacetobacter azotocaptans	943	NR_028767	clade_454	N	N
Gluconacetobacter diazotrophicus	944	NR_074292	clade_454	N	N
Gluconacetobacter johannae	948	NR_024959	clade_454	N	N
Nocardia brasiliensis	1351	AIHV01000038	clade_455	N	N
Nocardia cyriacigeorgica	1352	HQ009486	clade_455	N	N
Nocardia farcinica	1353	NC_006361	clade_455	Y	N
Nocardia puris	1354	NR_028994	clade_455	N	N
Nocardia sp. 01_Je_025	1355	GU574059	clade_455	N	N
Rhodococcus equi	1623	ADNW01000058	clade_455	N	N
Bacillus sp. oral taxon F28	247	HM099650	clade_456	Y	OP
Oceanobacillus caeni	1358	NR_041533	clade_456	N	N
Oceanobacillus sr. Ndiop	1359	CAER01000083	clade_456	N	N
Ornithinibacillus bavariensis	1384	NR_044923	clade_456	N	N
Ornithinibacillus sp. 7_10AIA	1385	FN397526	clade_456	N	N
Virgibacillus proomii	2005	NR_025308	clade_456	N	N
Corynebacterium amycolatum	683	ABZU01000033	clade_457	N	OP
Corynebacterium hansenii	702	AM946639	clade_457	N	N
Corynebacterium xerosis	735	FN179330	clade_457	N	OP
Staphylococcaceae bacterium NML 92 0017	1756	AY841362	clade_458	N	N
Staphylococcus fleurettii	1766	NR_041326	clade_458	N	N
Staphylococcus sciuri	1774	NR_025520	clade_458	N	N
Staphylococcus vitulinus	1779	NR_024670	clade_458	N	N
Stenotrophomonas maltophilia	1782	AAVZ01000005	clade_459	N	N
Stenotrophomonas sp. FG_6	1783	EF017810	clade_459	N	N
Mycobacterium africanum	1270	AF480605	clade_46	N	OP
Mycobacterium alsiensis	1271	AJ938169	clade_46	N	OP
Mycobacterium avium	1272	CP000479	clade_46	N	OP
Mycobacterium colombiense	1274	AM062764	clade_46	N	OP
Mycobacterium gordonae	1276	GU142930	clade_46	N	OP
Mycobacterium intracellulare	1277	GQ153276	clade_46	N	OP
Mycobacterium kansasii	1278	AF480601	clade_46	N	OP
Mycobacterium lacus	1279	NR_025175	clade_46	N	OP
Mycobacterium leprae	1280	FM211192	clade_46	N	OP
Mycobacterium lepromatosis	1281	EU203590	clade_46	N	OP
Mycobacterium mantenii	1283	FJ042897	clade_46	N	OP
Mycobacterium marinum	1284	NC_010612	clade_46	N	OP
Mycobacterium microti	1285	NR_025234	clade_46	N	OP
Mycobacterium parascrofulaceum	1287	ADNV01000350	clade_46	N	OP
Mycobacterium seoulense	1290	DQ536403	clade_46	N	OP
Mycobacterium sp. 1761	1292	EU703150	clade_46	N	N
Mycobacterium sp. 1791	1295	EU703148	clade_46	N	N

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OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Mycobacterium sp. 1797	1296	EU703149	clade_46	N	N
Mycobacterium sp. B10_07.09.0206	1298	HQ174245	clade_46	N	N
Mycobacterium sp. NLA001000736	1305	HM627011	clade_46	N	N
Mycobacterium sp. W	1306	DQ437715	clade_46	N	N
Mycobacterium tuberculosis	1307	CP001658	clade_46	N	Category-C
Mycobacterium ulcerans	1308	AB548725	clade_46	N	OP
Mycobacterium vulneris	1309	EU834055	clade_46	N	OP
Xanthomonas campestris	2016	EF101975	clade_461	N	N
Xanthomonas sp. kmd_489	2017	EU723184	clade_461	N	N
Dietzia natronolimnaea	769	GQ870426	clade_462	N	N
Dietzia sp. BBDP51	770	DQ337512	clade_462	N	N
Dietzia sp. CA149	771	GQ870422	clade_462	N	N
Dietzia timorensis	772	GQ870424	clade_462	N	N
Gordonia bronchialis	951	NR_027594	clade_463	N	N
Gordonia polyisoprenivorans	952	DQ385609	clade_463	N	N
Gordonia sp. KTR9	953	DQ068383	clade_463	N	N
Gordonia sputi	954	FJ536304	clade_463	N	N
Gordonia terrae	955	GQ848239	clade_463	N	N
Leptotrichia goodfellowii	1167	ADAD01000110	clade_465	N	N
Leptotrichia sp. oral clone IK040	1174	AY349387	clade_465	N	N
Leptotrichia sp. oral clone P2PB_51 P1	1175	AY207053	clade_465	N	N
Bacteroidales genomosp. P7 oral clone MB3_P19	264	DQ003623	clade_466	N	N
Butyricimonas virosa	454	AB443949	clade_466	N	N
Odoribacter laneus	1363	AB490805	clade_466	N	N
Odoribacter splanchnicus	1364	CP002544	clade_466	N	N
Capnocytophaga gingivalis	478	ACLQ01000011	clade_467	N	N
Capnocytophaga granulosa	479	X97248	clade_467	N	N
Capnocytophaga sp. oral clone AH015	483	AY005074	clade_467	N	N
Copnocytophoga sp. oral strain S3	487	AY005073	clade_467	N	N
Copnocytophaga sp. oral taxon 338	488	AEXX01000050	clade_467	N	N
Capnocytophaga canimorsus	476	CP002113	clade_468	N	N
Copnocytophoga sp. oral clone ID062	485	AY349368	clade_468	N	N
Catenibacterium mitsuokai	495	AB030224	clade_469	Y	N
Clostridium sp. TM_40	640	AB249652	clade_469	Y	N
Coprobacillus cateniformis	670	AB030218	clade_469	Y	N
Coprobacillus sp. 29_1	671	ADKX01000057	clade_469	Y	N
Lactobacillus catenaformis	1075	M23729	clade_469	N	N
Lactobacillus vitulinus	1142	NR_041305	clade_469	N	N
Cetobacterium somerae	501	AJ438155	clade_470	N	N
Clostridium rectum	618	NR_029271	clade_470	Y	N
Fusobacterium gonidiaformans	896	ACET01000043	clade_470	N	N
Fusobacterium mortiferum	897	ACDB02000034	clade_470	N	N
Fusobacterium necrogenes	899	X55408	clade_470	N	N
Fusobacterium necrophorum	900	AM905356	clade_470	N	N
Fusobacterium sp. 12_1B	905	AGWJ01000070	clade_470	N	N
Fusobacterium sp. 3_1_5R	911	ACDD01000078	clade_470	N	N
Fusobacterium sp. D12	918	ACDG02000036	clade_470	N	N
Fusobacterium ulcerans	921	ACDH01000090	clade_470	N	N
Fusobacterium varium	922	ACIE01000009	clade_470	N	N
Mycoplasma arthritidis	1312	NC_011025	clade_473	N	N
Mycoplasma faucium	1314	NR_024983	clade_473	N	N
Mycoplasma hominis	1318	AF443616	clade_473	N	N
Mycoplasma orale	1319	AY796060	clade_473	N	N
Mycoplasma salivarium	1324	M24661	clade_473	N	N
Mitsuokella jalaludinii	1247	NR_028840	clade_474	N	N
Mitsuokella multacida	1248	ABWK02000005	clade_474	N	N
Mitsuokella sp. oral taxon 521	1249	GU413658	clade_474	N	N
Mitsuokella sp. oral taxon G68	1250	GU432166	clade_474	N	N
Selenomonas genomosp. C1	1695	AY278627	clade_474	N	N
Selenomonas genomosp. P8 oral clone MB5_P06	1700	DQ003628	clade_474	N	N
Selenomonas ruminantium	1703	NR_075026	clade_474	N	N
Veillonellaceae bacterium oral taxon 131	1994	GU402916	clade_474	N	N
Alloscardoria omnicolens	139	NR_042583	clade_475	N	N
Alloscardovia sp. OB7196	140	AB425070	clade_475	N	N
Bifidobacterium urinalis	366	AJ278695	clade_475	N	N
Eubacterium nodatum	854	U13041	clade_476	Y	N
Eubacterium saphenum	859	NR_026031	clade_476	Y	N
Eubacterium sp. oral clone JH012	867	AY349373	clade_476	Y	N
Eubacterium sp. oral clone JS001	870	AY349378	clade_476	Y	N
Faecalibacterium prausnitzii	880	ACOP02000011	clade_478	Y	N
Gemmiger formicilis	932	GU562446	clade_478	Y	N
Subdoligranulum variabile	1896	AJ518869	clade_478	Y	N
Clostridiaceae bacterium JC13	532	JF824807	clade_479	Y	N
Clostridium sp. MLG055	634	AF304435	clade_479	Y	N
Erysipelotrichaceae bacterium 3_1_53	822	ACTJ01000113	clade_479	Y	N
Prevotella loescheii	1503	JN867231	clade_48	N	N
Prevotella sp. oral clone ASCG12	1530	DQ272511	clade_48	N	N

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OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Prevotella sp. oral clone GU027	1540	AY349398	clade_48	N	N
Prevotella sp. oral taxon 472	1553	ACZS01000106	clade_48	N	N
Selenomonas dianae	1693	GQ422719	clade_480	N	N
Selenomonas flueggei	1694	AF287803	clade_480	N	N
Selenomonas genomosp. C2	1696	AY278628	clade_480	N	N
Selenomonas genomosp. P6 oral clone MB3_C41	1698	DQ003636	clade_480	N	N
Selenomonas genomosp. P7 oral clone MB5_C08	1699	DQ003627	clade_480	N	N
Selenomonas infelix	1701	AF287802	clade_480	N	N
Selenomonas noxia	1702	GU470909	clade_480	N	N
Selenomonas sp. oral clone FT050	1705	AY349403	clade_480	N	N
Selenomonas sp. oral clone GI064	1706	AY349404	clade_480	N	N
Selenomonas sp. oral clone GT010	1707	AY349405	clade_480	N	N
Selenomonas sp. oral clone HU051	1708	AY349406	clade_480	N	N
Selenomonas sp. oral clone IK004	1709	AY349407	clade_480	N	N
Selenomonas sp. oral clone JI021	1711	AY349409	clade_480	N	N
Selenomonas sp. oral clone JS031	1712	AY349410	clade_480	N	N
Selenomonas sp. oral clone OH4A	1713	AY947498	clade_480	N	N
Selenomonas sp. oral clone P2PA_80 P4	1714	AY207052	clade_480	N	N
Selenomonas sp. oral taxon 149	1716	AEEJ01000007	clade_480	N	N
Veillonellaceae bacterium oral taxon 155	1995	GU470897	clade_480	N	N
Clostridium cocleatum	575	NR_026495	clade_481	Y	N
Clostridium ramosum	617	M23731	clade_481	Y	N
Clostridium saccharogumia	619	DQ100445	clade_481	Y	N
Clostridium spiroforme	644	X73441	clade_481	Y	N
Coprobacillus sp. D7	672	ACDT01000199	clade_481	Y	N
Clostridiales bacterium SY8519	535	AB477431	clade_482	Y	N
Clostridium sp. SY8519	639	AP012212	clade_482	Y	N
Eubacterium ramulus	855	AJ011522	clade_482	Y	N
Agrococcus jenensis	117	NR_026275	clade_484	Y	N
Microbacterium gubbeenense	1232	NR_025098	clade_484	N	N
Pseudoclavibacter sp. Timone	1590	FJ375951	clade_484	N	N
Tropheryma whipplei	1961	BX251412	clade_484	N	N
Zimmermannella bifida	2031	AB012592	clade_484	N	N
Erysipelothrix inopinata	819	NR_025594	clade_485	Y	N
Erysipelothrix rhusiopathiae	820	ACLK01000021	clade_485	Y	N
Erysipelothrix tonsillarum	821	NR_040871	clade_485	Y	N
Holdemania filiformis	1004	Y11466	clade_485	Y	N
Mollicutes bacterium pACH93	1258	AY297808	clade_485	Y	N
Coxiella burnetii	736	CP000890	clade_486	Y	Category-B
Legionella hackeliae	1151	M36028	clade_486	N	OP
Legionella longbeachae	1152	M36029	clade_486	N	OP
Legionella sp. D3923	1154	JN380999	clade_486	N	OP
Legionella sp. D4088	1155	JN381012	clade_486	N	OP
Legionella sp. H63	1156	JF831047	clade_486	N	OP
Legionella sp. NML 93L054	1157	GU062706	clade_486	N	OP
Legionella steelei	1158	HQ398202	clade_486	N	OP
Tatlockia micdadei	1915	M36032	clade_486	N	N
Clostridium hiranonis	591	AB023970	clade_487	Y	N
Clostridium irregulare	596	NR_029249	clade_487	Y	N
Helicobacter pullorum	996	ABQU01000097	clade_489	N	N
Acetobacteraceae bacterium AT_5844	16	AGEZ01000040	clade_490	N	N
Roseomonas cervicalis	1643	ADVL01000363	clade_490	N	N
Roseomonas mucosa	1644	NR_028857	clade_490	N	N
Roseomonas sp. NML94_0193	1645	AF533357	clade_490	N	N
Roseomonas sp. NML97_0121	1646	AF533359	clade_490	N	N
Roseornonas sp. NML98_0009	1647	AF533358	clade_490	N	N
Roseomonas sp. NML98_0157	1648	AF533360	clade_490	N	N
Rickettsia akari	1627	CP000847	clade_492	N	OP
Rickettsia conorii	1628	AE008647	clade_492	N	OP
Rickettsia prowazekii	1629	M21789	clade_492	N	Category-B
Rickettsia rickettsii	1630	NC_010263	clade_492	N	OP
Rickettsia slovaca	1631	L36224	clade_492	N	OP
Rickettsia typhi	1632	AE017197	clade_492	N	OP
Anaeroglobus geminatus	160	AGCJ01000054	clade_493	N	N
Megasphaera genomosp. C1	1201	AY278622	clade_493	N	N
Megasphaera micronuciformis	1203	AECS01000020	clade_493	N	N
Clostridium orbiscindens	609	Y18187	clade_494	Y	N
Clostridium sp. NML 04A032	637	EU815224	clade_494	Y	N
Flavonifractor plautii	886	AY724678	clade_494	Y	N
Pseudoflavonifractor capillosus	1591	AY136666	clade_494	Y	N
Ruminococcaceae bacterium D16	1655	ADDX01000083	clade_494	Y	N
Acetivibrio cellulolyticus	5	NR_025917	clade_495	Y	N
Clostridiales genomosp. BVAB3	540	CP001850	clade_495	N	N
Clostridium aldrichii	548	NR_026099	clade_495	Y	N
Clostridium clariflavum	570	NR_041235	clade_495	Y	N
Clostridium stercorarium	647	NR_025100	clade_495	Y	N
Clostridium straminisolvens	649	NR_024829	clade_495	Y	N

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Clostridium thermocellum	655	NR_074629	clade_495	Y	N
Tsukamurella paurometabola	1963	X80628	clade_496	N	N
Tsukamurella tyrosinosolvens	1964	AB478958	clade_496	N	N
Abiotrophia para_adiacens	2	AB022027	clade_497	N	N
Carnobacterium divergens	492	NR_044706	clade_497	N	N
Carnobacterium maltaromaticum	493	NC_019425	clade_497	N	N
Enterococcus avium	800	AF133535	clade_497	N	N
Enterococcus caccae	801	AY943820	clade_497	N	N
Enterococcus casseliflavus	802	AEWT01000047	clade_497	N	N
Enterococcus durans	803	AJ276354	clade_497	N	N
Enterococcus faecalis	804	AE016830	clade_497	N	N
Enterococcus faecium	805	AM157434	clade_497	N	N
Enterococcus gallinarum	806	AB269767	clade_497	N	N
Enterococcus gilvus	807	AY033814	clade_497	N	N
Enterococcus hawaiiensis	808	AY321377	clade_497	N	N
Enterococcus hirae	809	AF061011	clade_497	N	N
Enterococcus italicus	810	AEPV01000109	clade_497	N	N
Enterococcus mundtii	811	NR_024906	clade_497	N	N
Enterococcus raffinosus	812	FN600541	clade_497	N	N
Enterococcus sp. BV2CASA2	813	JN809766	clade_497	N	N
Enterococcus sp. CCRI 16620	814	GU457263	clade_497	N	N
Enterococcus sp. F95	815	FJ463817	clade_497	N	N
Enterococcus sp. RfL6	816	AJ133478	clade_497	N	N
Enterococcus thailandicus	817	AY321376	clade_497	N	N
Fusobacterium canifelinum	893	AY162222	clade_497	N	N
Fusobacterium genomosp. C1	894	AY278616	clade_497	N	N
Fusobacterium genomosp. C2	895	AY278617	clade_497	N	N
Fusobacterium nucleatum	901	ADVK01000034	clade_497	Y	N
Fusobacterium periodonticum	902	ACJY01000002	clade_497	N	N
Fusobacterium sp. 1_1_41FAA	906	ADGG01000053	clade_497	N	N
Fusobacterium sp. 11_3_2	904	ACUO01000052	clade_497	N	N
Fusobacterium sp. 2_1_31	907	ACDC02000018	clade_497	N	N
Fusobacterium sp. 3_1_27	908	ADGF01000045	clade_497	N	N
Fusobacterium sp. 3_1_33	909	ACQE01000178	clade_497	N	N
Fusobacterium sp. 3_1_36A2	910	ACPU01000044	clade_497	N	N
Fusobacterium sp. AC18	912	HQ616357	clade_497	N	N
Fusobacterium sp. ACB2	913	HQ616358	clade_497	N	N
Fusobacterium sp. AS2	914	HQ616361	clade_497	N	N
Fusobacterium sp. CM1	915	HQ616371	clade_497	N	N
Fusobacterium sp. CM21	916	HQ616375	clade_497	N	N
Fusobacterium sp. CM22	917	HQ616376	clade_497	N	N
Fusobacterium sp. oral clone ASCF06	919	AY923141	clade_497	N	N
Fusobacterium sp. oral clone ASCF11	920	AY953256	clade_497	N	N
Granulicatella adiacens	959	ACKZ01000002	clade_497	N	N
Granulicatella elegans	960	AB252689	clade_497	N	N
Granulicatella paradiacens	961	AY879298	clade_497	N	N
Granulicatella sp. oral clone ASC02	963	AY923126	clade_497	N	N
Granulicatella sp. oral clone ASCA05	964	DQ341469	clade_497	N	N
Granulicatella sp. oral clone ASCB09	965	AY953251	clade_497	N	N
Granulicatella sp. oral. clone ASCG05	966	AY923146	clade_497	N	N
Tetragenococcus halophilus	1918	NR_075020	clade_497	N	N
Tetragenococcus koreensis	1919	NR_043113	clade_497	N	N
Vagococcus fluvialis	1973	NR_026489	clade_497	N	N
Chryseobacterium anthropi	514	AM982793	clade_498	N	N
Chryseobacterium gleum	515	ACKQ02000003	clade_498	N	N
Chryseobacterium hominis	516	NR_042517	clade_498	N	N
Treponema refringens	1936	AF426101	clade_499	N	OP
Treponema sp. oral clone JU031	1941	AY349416	clade_499	N	N
Treponema sp. oral taxon 239	1948	GU408738	clade_499	N	N
Treponema sp. oral taxon 271	1955	GU408871	clade_499	N	N
Alistipes finegoldii	129	NR_043064	clade_500	N	N
Alistipes onderdonkii	131	NR_043318	clade_500	N	N
Alistipes putredinis	132	ABFK02000017	clade_500	N	N
Alistipes shahii	133	FP929032	clade_500	N	N
Alistipes sp. HGB5	134	AENZ01000082	clade_500	N	N
Alistipes sp. JC50	135	JF824804	clade_500	N	N
Alistipes sp. RMA 9912	136	GQ140629	clade_500	N	N
Mycoplasma agalactiae	1310	AF010477	clade_501	N	N
Mycoplasma bovoculi	1313	NR_025987	clade_501	N	N
Mycoplasma fermentans	1315	CP002458	clade_501	N	N
Mycoplasma flocculare	1316	X62699	clade_501	N	N
Mycoplasma ovipneumoniae	1320	NR_025989	clade_501	N	N
Arcobacter butzleri	176	AEPT01000071	clade_502	N	N
Arcobacter cryaerophilus	177	NR_025905	clade_502	N	N
Campylobacter curvus	461	NC_009715	clade_502	N	OP
Campylobacter rectus	467	ACFU01000050	clade_502	N	OP
Campylobacter showae	468	ACVQ01000030	clade_502	N	OP

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Campylobacter sp. FOBRC14	469	HQ616379	clade_502	N	OP
Campylobacter sp. FOBRC15	470	HQ616380	clade_502	N	OP
Campylobacter sp. oral clone BB120	471	AY005038	clade_502	N	OP
Campylobacter sputorum	472	NR_044839	clade_502	N	OP
Bacteroides ureolyticus	330	GQ167666	clade_504	N	N
Campylobacter gracilis	463	ACYG01000026	clade_504	N	OP
Campylobacter hominis	464	NC_009714	clade_504	N	OP
Dialister invisus	762	ACIM02000001	clade_506	N	N
Dialister micraerophilus	763	AFBB01000028	clade_506	N	N
Dialister microaerophilus	764	AENT01000008	clade_506	N	N
Dialister propionicifaciens	766	NR_043231	clade_506	N	N
Dialister succinatiphilus	768	AB370249	clade_506	N	N
Megasphaera elsdenii	1200	AY038996	clade_506	N	N
Megasphaera genomosp. type_1	1202	ADGP01000010	clade_506	N	N
Megasphaera sp. BLPYG_07	1204	HM990964	clade_506	N	N
Megasphaera sp. UPII 199_6	1205	AFIJ01000040	clade_506	N	N
Chromobacterium violaceum	513	NC_005085	clade_507	N	N
Laribacter hongkongensis	1148	CP001154	clade_507	N	N
Methylophilus sp. ECd5	1279	AY436794	clade_507	N	N
Finegoldia magna	883	ACHM02000001	clade_509	N	N
Parvimonas micra	1431	AB729072	clade_509	N	N
Parvimonas sp. oral taxon 110	1432	AFII01000002	clade_509	N	N
Peptostreptococcus micros	1456	AM176538	clade_509	N	N
Peptostreptococcus sp. oral clone FJ023	1460	AY349390	clade_509	N	N
Peptostreptococcus sp. P4P_31 P3	1458	AY207059	clade_509	N	N
Helicobacter pylori	997	CP00001.2	clade_510	N	OP
Anaplasma marginale	165	ABOR01000019	clade_511	N	N
Anaplasma phagocytophilum	166	NC_007797	clade_511	N	N
Ehrlichia chaffeensis	783	AAIF01000035	clade_511	N	OP
Neorickettsia risticii	1349	CP001431	clade_511	N	N
Neorickettsia sennetsu	1350	NC_007798	clade_511	N	N
Eubacterium barkeri	834	NR_044661	clade_512	Y	N
Eubacterium callanderi	838	NR_026310	clade_512	N	N
Eubacterium limosum	850	CP002273	clade_512	Y	N
Pseudoramibacter alactolyticus	1606	AB036759	clade_512	N	N
Veillonella montpellierensis	1977	AF473836	clade_513	N	N
Veillonella sp. oral clone ASCA08	1988	AY923118	clade_513	N	N
Veillonella sp. oral clone ASCB03	1989	AY923122	clade_513	N	N
Inquilinus limosus	1012	NR_029046	clade_514	N	N
Sphingomonas sp. oral clone FZ016	1746	AY349412	clade_514	N	N
Anaerococcus lactolyticus	145	ABYO01000217	clade_515	N	N
Anaerococcus prevotii	147	CP001708	clade_515	N	N
Anaerococcus sp. gpac104	152	AM176528	clade_515	N	N
Anaerococcus sp. gpac126	153	AM176530	clade_515	N	N
Anaerococcus sp. gpac155	154	AM176536	clade_515	N	N
Anaerococcus sp. gpac199	155	AM176539	clade_515	N	N
Anaerococcus tetradius	157	ACGC01000107	clade_515	N	N
Bacteroides coagulans	271	AB547639	clade_515	N	N
Clostridiales bacterium 9403326	534	HM587324	clade_515	N	N
Clostridiales bacterium ph2	539	JN837487	clade_515	N	N
Peptostreptococcus sp. 9succ1	1457	X90471	clade_515	N	N
Peptostreptococcus sp. oral clone AP24	1459	AB175072	clade_515	N	N
Tissierella praeacuta	1924	NR_044860	clade_515	N	N
Anaerotruncus colihominis	164	ABGD02000021	clade_516	Y	N
Clostridium methylpentosum	606	ACEC01000059	clade_516	Y	N
Clostridium sp. YIT 12070	642	AB491208	clade_516	Y	N
Hydrogenoanaerobacterium saccharovorans	1005	NR_044425	clade_516	Y	N
Ruminococcus albus	1656	AY445600	clade_516	Y	N
Ruminococcus flavefaciens	1660	NR_025931	clade_516	Y	N
Clostridium haemolyticum	589	NR_024749	clade_517	Y	N
Clostridium novyi	608	NR_074343	clade_517	Y	N
Clostridium sp. LMG 16094	632	X95274	clade_517	Y	N
Helicobacter canadensis	994	ABQS01000108	clade_518	N	N
Eubacterium ventriosum	874	L34421	clade_519	Y	N
Peptostreptococcus anaerobius	1455	AY326462	clade_520	N	N
Peptostreptococcus stomatis	1461	ADGQ01000048	clade_520	N	N
Bilophila wadsworthia	367	ADCP01000166	clade_521	N	N
Desulfovibrio vulgaris	761	NR_074897	clade_521	N	N
Bacteroides galacturonicus	280	DQ497994	clade_522	Y	N
Eubacterium eligens	845	CP001104	clade_522	Y	N
Lachnospira multipara	1046	FR733699	clade_522	Y	N
Lachnospira pectinoschiza	1047	L14675	clade_522	Y	N
Lactobacillus rogosae	1114	GU269544	clade_522	Y	N
Actinomyces nasicola	64	AJ508455	clade_523	N	N
Cellulosimicrobium funkei	500	AY501364	clade_523	N	N
Lactococcus raffinolactis	1146	NR_044359	clade_524	N	N
Bacillus horti	214	NR_036860	clade_527	Y	OP

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Bacillus sp. 9_3AIA	232	FN397519	clade_527	Y	OP
Bacteroidales genomosp. P1	258	AY341819	clade_529	N	N
Bacteroidales genomosp. P2 oral clone MB1_G13	259	DQ003613	clade_529	N	N
Bacteroidales genomosp. P3 oral clone MB1_G34	260	DQ003615	clade_529	N	N
Bacteroidales genomosp. P4 oral clone MB2_G17	261	DQ003617	clade_529	N	N
Bacteroidales genomosp. P5 oral clone MB2_P04	262	DQ003619	clade_529	N	N
Bacteroidales genomosp. P6 oral clone MB3_C19	263	DQ003634	clade_529	N	N
Bacteroidales genomosp. P8 oral clone MB4_G15	265	DQ003626	clade_529	N	N
Bacteroidetes bacterium oral taxon D27	333	HM099638	clade_530	N	N
Bacteroidetes bacterium oral taxon F31	334	HM099643	clade_530	N	N
Bacteroidetes bacterium oral taxon F44	335	HM099649	clade_530	N	N
Flavobacterium sp. NF2_1	885	FJ195988	clade_530	N	N
Myroides odoratimimus	1326	NR_042354	clade_530	N	N
Myroides sp. MY15	1327	GU253339	clade_530	N	N
Chlamydiales bacterium NS16	507	JN606076	clade_531	N	N
Chlamydophila pecorum	508	D88317	clade_531	N	OP
Parachlamydia sp. UWE25	1423	BX908798	clade_531	N	N
Fusobacterium russii	903	NR_044687	clade_532	N	N
Streptobacillus moniliformis	1784	NR_027615	clade_532	N	N
Eubacteriaceae bacterium P4P_50 P4	833	AY207060	clade_533	N	N
Eubacterium brachy	836	U13038	clade_533	Y	N
Filifactor alocis	881	CP002390	clade_533	Y	N
Filifactor villosus	882	NR_041928	clade_533	Y	N
Abiotrophia defectiva	1	ACIN02000016	clade_534	N	N
Abiotrophia sp. oral clone P4PA_155 P1	3	AY207063	clade_534	N	N
Catonella genomosp. P1 oral clone MB5_P12	496	DQ003629	clade_534	N	N
Catonella morbi	497	ACIL02000016	clade_534	N	N
Catonella sp. oral clone FL037	498	AY349369	clade_534	N	N
Eremococcus coleocola	818	AENN01000008	clade_534	N	N
Facklamia hominis	879	Y10772	clade_534	N	N
Granulicatella sp. M658_99_3	962	AJ271861	clade_534	N	N
Campylobacter coli	459	AAFL01000004	clade_535	N	OP
Campylobacter concisus	460	CP000792	clade_535	N	OP
Campylobacter fetus	462	ACLG01001177	clade_535	N	OP
Campylobacter jejuni	465	AL139074	clade_535	N	Category-B
Campylobacter upsaliensis	473	AEPU01000040	clade_535	N	OP
Clostridium leptum	601	AJ305238	clade_537	Y	N
Clostridium sp. YIT 12069	641	AB491207	clade_537	Y	N
Clostridium sporosphaeroides	646	NR_044835	clade_537	Y	N
Eubacterium coprostanoligenes	841	HM037995	clade_537	Y	N
Ruminococcus bromii	1657	EU266549	clade_537	Y	N
Eubacterium siraeum	860	ABCA03000054	clade_538	Y	N
Atopobium minutum	183	HM007583	clade_539	N	N
Atopobium parvulum	184	CP001721	clade_539	N	N
Atopobium rimae	185	ACFE01000007	clade_539	N	N
Atopobium sp. BS2	186	HQ616367	clade_539	N	N
Atopobium sp. F0209	187	EU592966	clade_539	N	N
Atopobium sp. ICM42b10	188	HQ616393	clade_539	N	N
Atopobium sp. ICM57	189	HQ616400	clade_539	N	N
Atopobium vaginae	190	AEDQ01000024	clade_539	N	N
Coriobacteriaceae bacterium BV3Ac1	677	JN809768	clade_539	N	N
Actinomyces naeslundii	63	X81062	clade_54	N	N
Actinomyces oricola	67	NR_025559	clade_54	N	N
Actinomyces oris	69	BABV01000070	clade_54	N	N
Actinomyces sp. 7400942	70	EU484334	clade_54	N	N
Actinomyces sp. ChDC B197	72	AF543275	clade_54	N	N
Actinomyces sp. GEJ15	73	GU561313	clade_54	N	N
Actinomyces sp. M2231_94_1	79	AJ234063	clade_54	N	N
Actinomyces sp. oral clone GU067	83	AY349362	clade_54	N	N
Actinomyces sp. oral clone IO077	85	AY349364	clade_54	N	N
Actinomyces sp. oral clone IP073	86	AY349365	clade_54	N	N
Actinomyces sp. oral clone JA063	88	AY349367	clade_54	N	N
Actinomyces sp. oral taxon 170	89	AFBL01000010	clade_54	N	N
Actinomyces sp. oral taxon 171	90	AECW01000034	clade_54	N	N
Actinomyces urogenitalis	95	ACFH01000038	clade_54	N	N
Actinomyces viscosus	96	ACRE01000096	clade_54	N	N
Clostridium viride	657	NR_026204	clade_540	Y	N
Oscillibacter sp. G2	1386	HM626173	clade_540	Y	N
Oscillibacter valericigenes	1387	NR_074793	clade_540	Y	N
Oscillospira guilliermondii	1388	AB040495	clade_540	Y	N
Orientia tsutsugamushi	1383	AP008981	clade_541	N	OP
Megamonas funiformis	1198	AB300988	clade_542	N	N
Megamonas hypermegale	1199	AJ420107	clade_542	N	N
Butyrivibrio crossotus	455	ABWN01000012	clade_543	Y	N
Clostridium sp. L2_50	631	AAYW02000018	clade_543	Y	N
Coprococcus eutactus	675	EF031543	clade_543	Y	N
Coprococcus sp. ART55_1	676	AY350746	clade_543	Y	N

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Eubacterium ruminantium	857	NR_024661	clade_543	Y	N
Aeromicrobium marinum	102	NR_025681	clade_544	N	N
Aeromicrobium sp. JC14	103	JF824798	clade_544	N	N
Luteococcus sanguinis	1190	NR_025507	clade_544	N	N
Propionibacteriaceae bacterium NML 02_0265	1568	EF599122	clade_544	N	N
Rhodococcus corynebacterioides	1622	X80615	clade_546	N	N
Rhodococcus erythropolis	1624	ACNO01000030	clade_546	N	N
Rhodococcus fascians	1625	NR_037021	clade_546	N	N
Segniliparus rotundus	1690	CP001958	clade_546	N	N
Segniliparus rugosus	1691	ACZI01000025	clade_546	N	N
Exiguobacterium acetylicum	878	FJ970034	clade_547	N	N
Micrococcus caseolyticus	1194	NR_074941	clade_547	N	N
Streptomyces sp. 1 AIP_2009	1890	FJ176782	clade_548	N	N
Streptomyces sp. SD 524	1892	EU544234	clade_548	N	N
Streptomyces sp. SD 528	1893	EU544233	clade_548	N	N
Streptomyces thermoviolaceus	1895	NR_027616	clade_548	N	N
Borrelia afzelii	388	ABCU01000001	clade_549	N	OP
Borrelia crocidurae	390	DQ057990	clade_549	N	OP
Borrelia duttonii	391	NC_011229	clade_549	N	OP
Borrelia hermsii	393	AY597657	clade_549	N	OP
Borrelia hispanica	394	DQ057988	clade_549	N	OP
Borrelia persica	395	HM161645	clade_549	N	OP
Borrelia recurrentis	396	AF107367	clade_549	N	OP
Borrelia spielmanii	398	ABKB01000002	clade_549	N	OP
Borrelia turicatae	399	NC_008710	clade_549	N	OP
Borrelia valaisiana	400	ABCY01000002	clade_549	N	OP
Providencia alcalifaciens	1586	ABXW01000071	clade_55	N	N
Providencia rettgeri	1587	AM040492	clade_55	N	N
Providencia rustigianii	1588	AM040489	clade_55	N	N
Providencia stuartii	1589	AF008581	clade_55	N	N
Treponema pallidum	1932	CP001752	clade_550	N	OP
Treponema phagedenis	1934	AEFH01000172	clade_550	N	N
Treponema sp. clone DDKL_4	1939	Y08894	clade_550	N	N
Acholeplasma laidlawii	17	NR_074448	clade_551	N	N
Mycoplasma putrefaciens	1323	U26055	clade_551	N	N
Mycoplasmataceae genomosp P1 oral clone	1325	DQ003614	clade_551	N	N
Spiroplasma insolitum	1750	NR_025705	clade_551	N	N
Collinsella aerofaciens	659	AAVN02000007	clade_553	Y	N
Collinsella intestinalis	660	ABXH02000037	clade_553	N	N
Collinsella stercoris	661	ABXJ01000150	clade_553	N	N
Collinsella tanakaei	662	AB490807	clade_553	N	N
Alkaliphilus metalliredigenes	137	AY137848	clade_554	Y	N
Alkaliphilus oremlandii	138	NR_043674	clade_554	Y	N
Caminicella sporogenes	458	NR_025485	clade_554	N	N
Clostridium sticklandii	648	L04167	clade_554	Y	N
Turicibacter sanguinis	1965	AF349724	clade_555	Y	N
Acidaminococcus fermentans	21	CP001859	clade_556	N	N
Acidaminococcus intestini	22	CP003058	clade_556	N	N
Acidaminococcus sp. D21	23	ACGB01000071	clade_556	N	N
Phascolarctobacterium faecium	1462	NR_026111	clade_556	N	N
Phascolarctobacterium sp. YIT 12068	1463	AB490812	clade_556	N	N
Phascolarctobacterium succinatutens	1464	AB490811	clade_556	N	N
Acidithiobacillus ferrivorans	25	NR_074660	clade_557	N	N
Fulvimonas sp. NML 060897	892	EF589680	clade_557	Y	N
Xanthomonadaceae bacterium NML 03_0222	2015	EU313791	clade_557	N	N
Catabacter hongkongensis	494	AB671763	clade_558	N	N
Christensenella minuta	512	AB490809	clade_558	N	N
Clostridiales bacterium oral clone P4PA	536	AY207065	clade_558	N	N
Clostridiales bacterium oral taxon 093	537	GQ422712	clade_558	N	N
Desulfitobacterium frappieri	753	AJ276701	clade_560	Y	N
Desulfitobacterium hafniense	754	NR_074996	clade_560	Y	N
Desulfotomaculum nigrificans	756	NR_044832	clade_560	Y	N
Heliobacterium modesticaldum	1000	NR_074517	clade_560	N	N
Alistipes indistinctus	130	AB490804	clade_561	N	N
Bacteroidales bacterium ph8	257	JN837494	clade_561	N	N
Candidatus Sulcia muelleri	475	CP002163	clade_561	N	N
Cytophaga xylanolytica	742	FR733683	clade_561	N	N
Flavobacteriaceae genomosp. C1	884	AY278614	clade_561	N	N
Gramella forsetii	958	NR_074707	clade_561	N	N
Sphingobacterium faecium	1740	NR_025537	clade_562	N	N
Sphingobacterium mizutaii	1741	JF708889	clade_562	N	N
Sphingobacterium multivorum	1742	NR_040953	clade_562	N	N
Sphingobacterium spiritivorum	1743	ACHA02000013	clade_562	N	N
Jonquetella anthropi	1017	ACOO02000004	clade_563	N	N
Pyramidobacter piscolens	1614	AY207056	clade_563	N	N
Synergistes genomosp. C1	1904	AY278615	clade_563	N	N
Synergistes sp. RMA 14551	1905	DQ412722	clade_563	N	N

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Synergistetes bacterium ADV897	1906	GQ258968	clade_563	N	N
Candidatus Arthromitus sp. SFB_mouse_Yit	474	NR_074460	clade_564	N	N
Gracilibacter thermotolerans	957	NR_043559	clade_564	N	N
Lutispora thermophila	1191	NR_041236	clade_564	Y	N
Brachyspira aalborgi	404	FM178386	clade_565	N	N
Brachyspira pilosicoli	405	NR_075069	clade_565	Y	N
Brachyspira sp. HIS3	406	FM178387	clade_565	N	N
Brachyspira sp. HIS4	407	FM178388	clade_565	N	N
Brachyspira sp. HIS5	408	FM178389	clade_565	N	N
Adlercreutzia equolifaciens	97	AB306661	clade_566	N	N
Coriobacteriaceae bacterium JC110	678	CAEM01000062	clade_566	N	N
Coriobacteriaceae bacterium phI	679	JN837493	clade_566	N	N
Cryptobacterium curtum	740	GQ422741	clade_566	N	N
Eggerthella lenta	778	AF292375	clade_566	Y	N
Eggerthella sinensis	779	AY321958	clade_566	N	N
Eggerthella sp. 1_3_56FAA	780	ACWN01000099	clade_566	N	N
Eggerthella sp. HGA1	781	AEXR01000021	clade_566	N	N
Eggerthella sp. YY7918	782	AP012211	clade_566	N	N
Gordonibacter pamelaeae	680	AM886059	clade_566	N	N
Gordonibacter pamelaeae	956	FP929047	clade_566	N	N
Slackia equolifaciens	1732	EU377663	clade_566	N	N
Slackia exigua	1733	ACUX01000029	clade_566	N	N
Slackia faecicanis	1734	NR_042220	clade_566	N	N
Slackia heliotrinireducens	1735	NR_074439	clade_566	N	N
Slackia isoflavoniconvertens	1736	AB566418	clade_566	N	N
Slackia piriformis	1737	AB490806	clade_566	N	N
Slackia sp. NATTS	1738	AB505075	clade_566	N	N
Streptomyces albus	1888	AJ697941	clade_566	Y	N
Chlamydiales bacterium NS11	505	JN606074	clade_567	Y	N
Chlamydiales bacterium NS13	506	JN606075	clade_567	N	N
Victivallaceae bacterium NML 080035	2003	FJ394915	clade_567	N	N
Victivallis vadensis	2004	ABDE02000010	clade_567	N	N
Anaerofustis stercorihominis	159	ABIL02000005	clade_570	Y	N
Butyricicoccus pullicaecorum	453	HH793440	clade_572	Y	N
Eubacterium desmolans	843	NR_044644	clade_572	Y	N
Papillibacter cinnamivorans	1415	NR_025025	clade_572	Y	N
Sporobacter termitidis	1751	NR_044972	clade_572	Y	N
Streptomyces griseus	1889	NR_074787	clade_573	N	N
Streptomyces sp. SD 511	1891	EU544231	clade_573	N	N
Streptomyces sp. SD 534	1894	EU544232	clade_573	N	N
Cloacibacillus evryensis	530	GQ258966	clade_575	N	N
Deferribacteres sp. oral clone JV001	743	AY349370	clade_575	N	N
Deferribacteres sp. oral clone JV006	744	AY349371	clade_575	Y	N
Deferribacteres sp. oral clone JV023	745	AY349372	clade_575	N	N
Synergistetes bacterium LBVCM1157	1907	GQ258969	clade_575	N	N
Synergistetes bacterium oral taxon 362	1909	GU410752	clade_575	N	N
Synergistetes bacterium oral taxon D48	1910	GU430992	clade_575	N	N
Clostridium colinum	577	NR_026151	clade_576	Y	N
Clostridium lactatifermentans	599	NR_025651	clade_576	Y	N
Clostridium piliforme	614	D14639	clade_576	Y	N
Peptococcus sp. oral clone JM048	1439	AY349389	clade_576	N	N
Helicobacter winghamensis	999	ACDO01000013	clade_577	N	N
Wolinella succinogenes	2014	BX571657	clade_577	N	N
Olsenella genomosp. C1	1368	AY278623	clade_578	N	N
Olsenella profusa	1369	FN178466	clade_578	N	N
Olsenella sp. F0004	1370	EU592964	clade_578	N	N
Olsenella sp. oral taxon 809	1371	ACVE01000002	clade_578	N	N
Olsenella uli	1372	CP002106	clade_578	N	N
Nocardiopsis dassonvillei	1356	CP002041	clade_579	N	N
Saccharomonospora viridis	1671	X54286	clade_579	Y	N
Thermobifida fusca	1921	NC_007333	clade_579	Y	N
Peptococcus niger	1438	NR_029221	clade_580	N	N
Peptococcus sp. oral taxon 167	1440	GQ422727	clade_580	N	N
Akkermansia muciniphila	118	CP001071	clade_583	N	N
Opitutus terrae	1373	NR_074978	clade_583	N	N
Clostridiales bacterium oral taxon F32	538	HM099644	clade_584	N	N
Leptospira borgpetersenii	1161	NC_008508	clade_585	N	OP
Leptospira broomii	1162	NR_043200	clade_585	N	OP
Leptospira interrogans	3163	NC_005823	clade_585	N	OP
Leptospira licerasiae	1164	EF612284	clade_585	Y	OP
Methanobrevibacter gottschalkii	1213	NR_044789	clade_587	N	N
Methanobrevibacter millerae	1214	NR_042785	clade_587	N	N
Methanobrevibacter oralis	1216	HE654003	clade_587	N	N
Methanobrevibacter thaueri	1219	NR_044787	clade_587	N	N
Methanobrevibacter smithii	1218	ABYV02000002	clade_588	N	N
Deinococcus radiodurans	746	AE000513	clade_589	N	N
Deinococcus sp. R_43890	747	FR682752	clade_589	N	N

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Thermus aquaticus	1923	NR_025900	clade_589	N	N
Actinomyces sp. c109	81	AB167239	clade_590	N	N
Moorella thermoacetica	1259	NR_075001	clade_590	Y	N
Syntrophomonadaceae genomosp. P1	1912	AY341821	clade_590	N	N
Thermoanaerobacter pseudethanolicus	1920	CP000924	clade_590	Y	N
Anaerobaculum hydrogeniformans	141	ACJX02000009	clade_591	N	N
Flexistipes sinusarabici	888	NR_074881	clade_591	Y	N
Microcystis aeruginosa	1246	NC_010296	clade_592	N	N
Prochlorococcus marinus	1567	CP000551	clade_592	N	N
Methanobrevibacter acididurans	1208	NR_028779	clade_593	N	N
Methanobrevibacter arboriphilus	1209	NR_042783	clade_593	N	N
Methanobrevibacter curvatus	1210	NR_044796	clade_593	N	N
Methanobrevibacter cuticularis	1211	NR_044776	clade_593	N	N
Methanobrevibacter filiformis	1212	NR_044801	clade_593	N	N
Methanobrevibacter woesei	1220	NR_044788	clade_593	N	N
Roseiflexus castenholzii	1642	CP000804	clade_594	N	N
Methanobrevibacter olleyae	1215	NR_043024	clade_595	N	N
Methanobrevibacter ruminantium	1217	NR_042784	clade_595	N	N
Methanobrevibacter wolinii	1221	NR_044790	clade_595	N	N
Methanosphaera stadtmanae	1222	AY196684	clade_595	N	N
Chloroflexi genomosp. P1	511	AY331414	clade_596	N	N
Gloeobacter violaceus	942	NR_074282	clade_596	Y	N
Halorubrum lipolyticum	992	AB477978	clade_597	N	N
Methanobacterium formicicum	1207	NR_025028	clade_597	N	N
Acidilobus saccharovorans	24	AY350586	clade_598	N	N
Hyperthermus butylicus	1006	CP000493	clade_598	N	N
Ignicoccus islandicus	1011	X99562	clade_598	N	N
Metallosphaera sedula	1206	D26491	clade_598	N	N
Thermofilum pendens	1922	X14835	clade_598	N	N
Prevotella melaninogenica	1506	CP002122	clade_6	N	N
Prevotella sp. ICM1	1520	HQ616385	clade_6	N	N
Prevotella sp. oral clone FU048	1535	AY349393	clade_6	N	N
Prevotella sp. oral done GI030	1537	AY349395	clade_6	N	N
Prevotella sp. SEQ116	1526	JN867246	clade_6	N	N
Streptococcus anginosus	1787	AECT01000011	clade_60	N	N
Streptococcus milleri	1812	X81023	clade_60	N	N
Streptococcus sp. 16362	1829	JN590019	clade_60	N	N
Streptococcus sp. 69130	1832	X78825	clade_60	N	N
Streptococcus sp. AC15	1833	HQ616356	clade_60	N	N
Streptococcus sp. CM7	1839	HQ616373	clade_60	N	N
Streptococcus sp. OBRC6	1847	HQ616352	clade_60	N	N
Burkholderia ambifaria	442	AAUZ01000009	clade_61	N	OP
Burkholderia cenocepacia	443	AAHI001000060	clade_61	N	OP
Burkholderia cepacia	444	NR_041719	clade_61	N	OP
Burkholderia mallei	445	CP000547	clade_61	N	Category-B
Burkholderia multivorans	446	NC_010086	clade_61	N	OP
Burkholderia oklahomensis	447	DQ108388	clade_61	N	OP
Burkholderia pseudomallei	448	CP001408	clade_61	N	Category-B
Burkholderia rhizoxinica	449	HQ005410	clade_61	N	OP
Burkholderia sp. 383	450	CP000151	clade_61	N	OP
Burkholderia xenovorans	451	U86373	clade_61	N	OP
Prevotella buccae	1488	ACRB01000001	clade_62	N	N
Prevotella genomosp. P8 oral clone MB3_P13	1498	DQ003622	clade_62	N	N
Prevotella sp. oral clone FW035	1536	AY349394	clade_62	N	N
Prevotella bivia	1486	ADFO01000096	clade_63	N	N
Prevotella disiens	1494	AEDO01000026	clade_64	N	N
Bacteroides faecis	276	GQ496624	clade_65	N	N
Bacteroides fragilis	279	AP006841	clade_65	N	N
Bacteroides nordii	285	NR_043017	clade_65	N	N
Bacteroides salyersiae	292	EU136690	clade_65	N	N
Bacteroides sp. 1_1_14	293	ACRP01000155	clade_65	N	N
Bacteroides sp. 1_1_6	295	ACIC01000215	clade_65	N	N
Bacteroides sp. 2_1_56FAA	298	ACWI01000065	clade_65	N	N
Bacteroides sp. AR29	316	AF139525	clade_65	N	N
Bacteroides sp. B2	317	EU722733	clade_65	N	N
Bacteroides thetaiotaomicron	328	NR_074277	clade_65	N	N
Actinobacillus minor	45	ACFT01000025	clade_69	N	N
Haemophilus parasuis	978	GU226366	clade_69	N	N
Vibrio cholerae	1996	AAUR01000095	clade_71	N	Category-B
Vibrio fluvialis	1997	X76335	clade_71	N	Category-B
Vibrio furnissii	1998	CP002377	clade_71	N	Category-B
Vibrio mimicus	1999	ADAF01000001	clade_71	N	Category-B
Vibrio parahaemolyticus	2000	AAWQ01000116	clade_71	N	Category-B
Vibrio sp. RC341	2001	ACZT01000024	clade_71	N	Category-B
Vibrio vulnificus	2002	AE016796	clade_71	N	Category-B
Lactobacillus acidophilus	1067	CP000033	clade_72	N	N
Lactobacillus amylolyticus	1069	ADNY01000006	clade_72	N	N

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Lactobacillus amylovorus	1070	CP002338	clade_72	N	N
Lactobacillus crispatus	1078	ACOG01000151	clade_72	N	N
Lactobacillus delbrueckii	1080	CP002341	clade_72	N	N
Lactobacillus helveticus	1088	ACLM01000202	clade_72	N	N
Lactobacillus kalixensis	1094	NR_029083	clade_72	N	N
Lactobacillus kefiranofaciens	1095	NR_042440	clade_72	N	N
Lactobacillus leichmannii	1098	JX986966	clade_72	N	N
Lactobacillus sp. 66c	1120	FR681900	clade_72	N	N
Lactobacillus sp. KLDS 1.0701	1122	EU600905	clade_72	N	N
Lactobacillus sp. KLDS 1.0712	1130	EU600916	clade_72	N	N
Lactobacillus sp. oral clone HT070	1136	AY349383	clade_72	N	N
Lactobacillus ultunensis	1139	ACGU01000081	clade_72	N	N
Prevotella intermedia	1502	AF414829	clade_81	N	N
Prevotella nigrescens	1511	AFPX01000069	clade_81	N	N
Prevotella pallens	1515	AFPY01000135	clade_81	N	N
Prevotella sp. oral taxon 310	1551	GQ422737	clade_81	N	N
Prevotella genomosp. C1	1495	AY278624	clade_82	N	N
Prevotella sp. CM38	1519	HQ610181	clade_82	N	N
Prevotella sp. oral taxon 317	1552	ACQH01000158	clade_82	N	N
Prevotella sp. SG12	1527	GU561343	clade_82	N	N
Prevotella denticola	1493	CP002589	clade_83	N	N
Prevotella genomosp. P7 oral clone MB2_P31	1497	DQ003620	clade_83	N	N
Prevotella histicola	1501	JN867315	clade_83	N	N
Prevotella multiformis	1508	AEWX01000054	clade_83	N	N
Prevotella sp. JCM 6330	1522	AB547699	clade_83	N	N
Prevotella sp. oral clone GI059	1539	AY349397	clade_83	N	N
Prevotella sp. oral taxon 782	1555	GQ422745	clade_83	N	N
Prevotella sp. oral taxon G71	1559	GU432180	clade_83	N	N
Prevotella sp. SEQ065	1524	JN867234	clade_83	N	N
Prevotella veroralis	1565	ACVA01000027	clade_83	N	N
Bacteroides acidifaciens	266	NR_028607	clade_85	N	N
Bacteroides cellulosilyticus	269	ACCH01000108	clade_85	N	N
Bacteroides clarus	270	AFBM01000011	clade_85	N	N
Bacteroides eggerthii	275	ACWG01000065	clade_85	N	N
Bacteroides oleiciplenus	286	AB547644	clade_85	N	N
Bacteroides pyogenes	290	NR_041280	clade_85	N	N
Bacteroides sp. 315_5	300	FJ848547	clade_85	N	N
Bacteroides sp. 31SF15	301	AJ583248	clade_85	N	N
Bacteroides sp. 31SF18	302	AJ583249	clade_85	N	N
Bacteroides sp. 35AE31	303	AJ583244	clade_85	N	N
Bacteroides sp. 35AE37	304	AJ583245	clade_85	N	N
Bacteroides sp. 35BE34	305	AJ583246	clade_85	N	N
Bacteroides sp. 35BE35	306	AJ583247	clade_85	N	N
Bacteroides sp. WH2	324	AY895180	clade_85	N	N
Bacteroides sp. XB12B	325	AM230648	clade_85	N	N
Bacteroides stercoris	327	ABFZ02000022	clade_85	N	N
Actinobacillus pleuropneumoniae	46	NR_074857	clade_88	N	N
Actinobacillus ureae	48	AEVG01000167	clade_88	N	N
Haemophilus aegyptius	969	AFBC01000053	clade_88	N	N
Haemophilus ducreyi	970	AE017143	clade_88	N	OP
Haemophilus haemolyticus	973	JN175335	clade_88	N	N
Haemophilus influenzae	974	AADP01000001	clade_88	N	OP
Haemophilus parahaemolyticus	975	GU561425	clade_88	N	N
Haemophilus parainfluenzae	976	AEWU01000024	clade_88	N	N
Haemophilus paraphrophaemolyticus	977	M75076	clade_88	N	N
Haemophilus somnus	979	NC_008309	clade_88	N	N
Haemophilus sp. 70334	980	HQ680854	clade_88	N	N
Haemophilus sp. HK445	981	FJ685624	clade_88	N	N
Haemophilus sp. oral clone ASCA07	982	AY923117	clade_88	N	N
Haemophilus sp. oral clone ASCG06	983	AY923147	clade_88	N	N
Haemophilus sp. oral clone BJ021	984	AY005034	clade_88	N	N
Haemophilus sp. oral clone BJ095	985	AY005033	clade_88	N	N
Haemophilus sp. oral taxon 851	987	AGRK01000004	clade_88	N	N
Haemophilus sputorum	988	AFNK01000005	clade_88	N	N
Histophilus somni	1003	AF549387	clade_88	N	N
Mannheimia haemolytica	1195	ACZX01000102	clade_88	N	N
Pasteurella bettyae	1433	L06088	clade_88	N	N
Moellerella wisconsensis	1253	JN175344	clade_89	N	N
Morganella morganii	1265	AJ301681	clade_89	N	N
Morganella sp. JB_T16	1266	AJ781005	clade_89	N	N
Proteus mirabilis	1582	ACLE01000013	clade_89	N	N
Proteus penneri	1583	ABVP01000020	clade_89	N	N
Proteus sp. HS7514	1584	DQ512963	clade_89	N	N
Proteus vulgaris	1585	AJ233425	clade_89	N	N
Eubacterium sp. oral clone JN088	869	AY349377	clade_90	Y	N
Oribacterium sinus	1374	ACKX01000142	clade_90	N	N
Oribacterium sp. ACB1	1375	HM120210	clade_90	N	N

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Oribacterium sp. ACB7	1376	HM120211	clade_90	N	N
Oribacterium sp. CM12	1377	HQ616374	clade_90	N	N
Oribacterium sp. ICM51	1378	HQ616397	clade_90	N	N
Oribacterium sp. OBRC12	1379	HQ616355	clade_90	N	N
Oribacterium sp. oral taxon 108	1382	AFIH01000001	clade_90	N	N
Actinobacillus actinomycetemcomitans	44	AY362885	clade_92	N	N
Actinobacillus succinogenes	47	CP000746	clade_92	N	N
Aggregatibacter actinomycetemcomitans	112	CP001733	clade_92	N	N
Aggregatibacter aphrophilus	113	CP001607	clade_92	N	N
Aggregatibacter segnis	114	AEPS01000017	clade_92	N	N
Averyella dalhousiensis	194	DQ481464	clade_92	N	N
Bisgaard Taxon	368	AY683487	clade_92	N	N
Bisgaard Taxon	369	AY683489	clade_92	N	N
Bisgaard Taxon	370	AY683491	clade_92	N	N
Bisgaard Taxon	371	AY683492	clade_92	N	N
Buchnera aphidicola	440	NR_074609	clade_92	N	N
Cedecea davisae	499	AF493976	clade_92	N	N
Citrobacter amalonaticus	517	FR870441	clade_92	N	N
Citrobacter braakii	518	NR_028687	clade_92	N	N
Citrobacter farmeri	519	AF025371	clade_92	N	N
Citrobacter freundii	520	NR_028894	clade_92	N	N
Citrobacter gillenii	521	AF025367	clade_92	N	N
Citrobacter koseri	522	NC_009792	clade_92	N	N
Citrobacter murliniae	523	AF025369	clade_92	N	N
Citrobacter rodentium	524	NR_074903	clade_92	N	N
Citrobacter sedlakii	525	AF025364	clade_92	N	N
Citrobacter sp. 30_2	526	ACDJ01000053	clade_92	N	N
Citrobacter sp. KMSI_3	527	GQ468398	clade_92	N	N
Citrobacter werkmanii	528	AF025373	clade_92	N	N
Citrobacter youngae	529	ABWL02000011	clade_92	N	N
Cronobacter malonaticus	737	GU122174	clade_92	N	N
Cronobacter sakazakii	738	NC_009778	clade_92	N	N
Cronobacter turicensis	739	FN543093	clade_92	N	N
Enterobacter aerogenes	786	AJ251468	clade_92	N	N
Enterobacter asburiae	787	NR_024640	clade_92	N	N
Enterobacter cancerogenus	788	Z96078	clade_92	N	N
Enterobacter cloacae	789	FP929040	clade_92	N	N
Enterobacter cowanii	790	NR_025566	clade_92	N	N
Enterobacter hormaechei	791	AFHR01000079	clade_92	N	N
Enterobacter sp. 247BMC	792	HQ122932	clade_92	N	N
Enterobacter sp. 638	793	NR_074777	clade_92	N	N
Enterobacter sp. JC163	794	JN657217	clade_92	N	N
Enterobacter sp. SCSS	795	HM007811	clade_92	N	N
Enterobacter sp. TSE38	796	HM156134	clade_92	N	N
Enterobacteriaceae bacterium 9_2_54FAA	797	ADCU01000033	clade_92	N	N
Enterobacteriaceae bacterium CF01Ent_1	798	AJ489826	clade_92	N	N
Enterobacteriaceae bacterium Smarlab 3302238	799	AY538694	clade_92	N	N
Escherichia albertii	824	ABKX01000012	clade_92	N	N
Escherichia coli	825	NC_008563	clade_92	N	Category-B
Escherichia fergusonii	826	CU928158	clade_92	N	N
Escherichia hermannii	827	HQ407266	clade_92	N	N
Escherichia sp. 1_1_43	828	ACID01000033	clade_92	N	N
Escherichia sp. 4_1_40B	829	ACDM02000056	clade_92	N	N
Escherichia sp. B4	830	EU722735	clade_92	N	N
Escherichia vulneris	831	NR_041927	clade_92	N	N
Ewingella americana	877	JN175329	clade_92	N	N
Haemophilus genomosp. P2 oral clone MB3_C24	971	DQ003621	clade_92	N	N
Haemophilus genomosp. P3 oral clone MB3_C38	972	DQ003635	clade_92	N	N
Haemophilus sp. oral clone JM053	986	AY349380	clade_92	N	N
Hafnia alvei	989	DQ412565	clade_92	N	N
Klebsiella oxytoca	1024	AY292871	clade_92	N	OP
Klebsiella pneumoniae	1025	CP000647	clade_92	N	OP
Klebsiella sp. AS10	1026	HQ616362	clade_92	N	N
Klebsiella sp. Co9935	1027	DQ068764	clade_92	N	N
Klebsiella sp. enrichment culture clone *SRC* _DSD25	1036	HM195210	clade_92	N	N
Klebsiella sp. OBRC7	1028	HQ616353	clade_92	N	N
Klebsiella sp. SP_BA	1029	FJ999767	clade_92	N	N
Klebsiella sp. SRC_DSD1	1033	GU797254	clade_92	N	N
Klebsiella sp. SRC_DSD11	1030	GU797263	clade_92	N	N
Klebsiella sp. SRC_DSD12	1031	GU797264	clade_92	N	N
Klebsiella sp. SRC_DSD15	1032	GU797267	clade_92	N	N
Klebsiella sp. SRC_DSD2	1034	GU797253	clade_92	N	N
Klebsiella sp. SRC_DSD6	1035	GU797258	clade_92	N	N
Klebsiella variicola	1037	CP001891	clade_92	N	N
Kluyvera ascorbata	1038	NR_028677	clade_92	N	N
Kluyvera cryocrescens	1039	NR_028803	clade_92	N	N
Leminorella grimontii	1159	AJ233421	clade_92	N	N

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Leminorella richardii	1160	HF558368	clade_92	N	N
Pantoea agglomerans	1409	AY335552	clade_92	N	N
Pantoea ananatis	1410	CP001875	clade_92	N	N
Pantoea brenneri	1411	EU216735	clade_92	N	N
Pantoea citrea	1412	EF688008	clade_92	N	N
Pantoea conspicua	1413	EU216737	clade_92	N	N
Pantoea septica	1414	EU216734	clade_92	N	N
Pasteurella dagmatis	1434	ACZR01000003	clade_92	N	N
Pasteurella multocida	1435	NC_002663	clade_92	N	N
Plesiomonas shigelloides	1469	X60418	clade_92	N	N
Raoultella ornithinolytica	1617	AB364958	clade_92	N	N
Raoultella planticola	1618	AF129443	clade_92	N	N
Raoultella terrigena	1619	NR_037085	clade_92	N	N
Salmonella bongori	1683	NR_041699	clade_92	N	Category-B
Salmonella enterica	1672	NC_011149	clade_92	N	Category-B
Salmonella enterica	1673	NC_011205	clade_92	N	Category-B
Salmonella enterica	1674	DQ344532	clade_92	N	Category-B
Salmonella enterica	1675	ABEH02000004	clade_92	N	Category-B
Salmonella enterica	1676	ABAK02000001	clade_92	N	Category-B
Salmonella enterica	1677	NC_011080	clade_92	N	Category-B
Salmonella enterica	1678	EU118094	clade_92	N	Category-B
Salmonella enterica	1679	NC_011094	clade_92	N	Category-B
Salmonella enterica	1680	AE014613	clade_92	N	Category-B
Salmonella enterica	1682	ABFH02000001	clade_92	N	Category-B
Salmonella enterica	1684	ABEM01000001	clade_92	N	Category-B
Salmonella enterica	1685	ABAM02000001	clade_92	N	Category-B
Salmonella typhimurium	1681	DQ344533	clade_92	N	Category-B
Salmonella typhimurium	1686	AF170176	clade_92	N	Category-B
Serratia fonticola	1718	NR_025339	clade_92	N	N
Serratia liquefaciens	1719	NR_042062	clade_92	N	N
Serratia marcescens	1720	GU826157	clade_92	N	N
Serratia odorifera	1721	ADBY01000001	clade_92	N	N
Serratia proteamaculans	1722	AAUN01000015	clade_92	N	N
Shigella boydii	1724	AAKA01000007	clade_92	N	Category-B
Shigella dysenteriae	1725	NC_007606	clade_92	N	Category-B
Shigella flexneri	1726	AE005674	clade_92	N	Category-B
Shigella sonnei	1727	NC_007384	clade_92	N	Category-B
Tatumella ptyseos	1916	NR_025342	clade_92	N	N
Trabulsiella guamensis	1925	AY373830	clade_92	N	N
Yersinia aldovae	2019	AJ871363	clade_92	N	OP
Yersinia aleksiciae	2020	AJ627597	clade_92	N	OP
Yersinia bercovieri	2021	AF366377	clade_92	N	OP
Yersinia enterocolitica	2022	FR729477	clade_92	N	Category-B
Yersinia frederiksenii	2023	AF366379	clade_92	N	OP
Yersinia intermedia	2024	AF366380	clade_92	N	OP
Yersinia kristensenii	2025	ACCA01000078	clade_92	N	OP
Yersinia mollaretii	2026	NR_027546	clade_92	N	OP
Yersinia pestis	2027	AE013632	clade_92	N	Category-A
Yersinia pseudotuberculosis	2028	NC_009708	clade_92	N	OP
Yersinia rohdei	2029	ACCD01000071	clade_92	N	OP
Yokenella regensburgei	2030	AB273739	clade_92	N	N
Conchiformibius kuhniae	669	NR_041821	clade_94	N	N
Morococcus cerebrosus	1267	JN175352	clade_94	N	N
Neisseria bacilliformis	1328	AFAY01000058	clade_94	N	N
Neisseria cinerea	1329	ACDY01000037	clade_94	N	N
Neisseria flavescens	1331	ACQV01000025	clade_94	N	N
Neisseria gonorrhoeae	1333	CP002440	clade_94	N	OP
Neisseria lactamica	1334	ACEQ01000095	clade_94	N	N
Neisseria macacae	1335	AFQE01000146	clade_94	N	N
Neisseria meningitidis	1336	NC_003112	clade_94	N	OP
Neisseria mucosa	1337	ACDX01000110	clade_94	N	N
Neisseria pharyngis	1338	AJ239281	clade_94	N	N
Neisseria polysaccharea	1339	ADBE01000137	clade_94	N	N
Neisseria sicca	1340	ACKO02000016	clade_94	N	N
Neisseria sp. KEM232	1341	GQ203291	clade_94	N	N
Neisseria sp. oral clone AP132	1344	AY005027	clade_94	N	N
Neisseria sp. oral strain B33KA	1346	AY005028	clade_94	N	N
Neisseria sp. oral taxon 014	1347	ADEA01000039	clade_94	N	N
Neisseria sp. TM10_1	1343	DQ279352	clade_94	N	N
Neisseria subflava	1348	ACEO01000067	clade_94	N	N
Clostridium oroticum	610	FR749922	clade_96	Y	N
Clostridium sp. D5	627	ADBG01000142	clade_96	Y	N
Eubacterium contortum	840	FR749946	clade_96	Y	N
Eubacterium fissicatena	846	FR749935	clade_96	Y	N
Okadaella gastrococcus	1365	HQ699465	clade_98	N	N
Streptococcus agalactiae	1785	AAJO01000130	clade_98	N	N
Streptococcus alactolyticus	1786	NR_041781	clade_98	N	N

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Streptococcus australis	1788	AEQR01000024	clade_98	N	N
Streptococcus bovis	1789	AEEL01000030	clade_98	N	N
Streptococcus canis	1790	AJ413203	clade_98	N	N
Streptococcus constellatus	1791	AY277942	clade_98	N	N
Streptococcus cristatus	1792	AEVC01000028	clade_98	N	N
Streptococcus dysgalactiae	1794	AP010935	clade_98	N	N
Streptococcus equi	1795	CP001129	clade_98	N	N
Streptococcus equinus	1796	AEVB01000043	clade_98	N	N
Streptococcus gallolyticus	1797	FR824043	clade_98	N	N
Streptococcus genomosp. C1	1798	AY278629	clade_98	N	N
Streptococcus genomosp. C2	1799	AY278630	clade_98	N	N
Streptococcus genomosp. C3	1800	AY278631	clade_98	N	N
Streptococcus genomosp. C4	1801	AY278632	clade_98	N	N
Streptococcus genomosp. C5	1802	AY278633	clade_98	N	N
Streptococcus genomosp. C6	1803	AY278634	clade_98	N	N
Streptococcus genomosp. C7	1804	AY278635	clade_98	N	N
Streptococcus genomosp. C8	1805	AY278609	clade_98	N	N
Streptococcus gordonii	1806	NC_009785	clade_98	N	N
Streptococcus infantarius	1807	ABJK02000017	clade_98	N	N
Streptococcus infantis	1808	AFNN01000024	clade_98	N	N
Streptococcus intermedius	1809	NR_028736	clade_98	N	N
Streptococcus lutetiensis	1810	NR_037096	clade_98	N	N
Streptococcus massiliensis	1811	AY769997	clade_98	N	N
Streptococcus mitis	1813	AM157420	clade_98	N	N
Streptococcus oligofermentans	1815	AY099095	clade_98	N	N
Streptococcus oralis	1816	ADMV01000001	clade_98	N	N
Streptococcus parasanguinis	1817	AEKM01000012	clade_98	N	N
Streptococcus pasteurianus	1818	AP012054	clade_98	N	N
Streptococcus peroris	1819	AEVF01000016	clade_98	N	N
Streptococcus pneumoniae	1820	AE008537	clade_98	N	N
Streptococcus porcinus	1821	EF121439	clade_98	N	N
Streptococcus pseudopneumoniae	1822	FJ827123	clade_98	N	N
Streptococcus pseudoporcinus	1823	AENS01000003	clade_98	N	N
Streptococcus pyogenes	1824	AE006496	clade_98	N	OP
Streptococcus ratti	1825	X58304	clade_98	N	N
Streptococcus salivarius	1826	AGBV01000001	clade_98	N	N
Streptococcus sanguinis	1827	NR_074974	clade_98	N	N
Streptococcus sinensis	1828	AF432857	clade_98	N	N
Streptococcus sp. 2_1_36FAA	1831	ACOI01000028	clade_98	N	N
Streptococcus sp. 2285_97	1830	AJ131965	clade_98	N	N
Streptococcus sp. ACS2	1834	HQ616360	clade_98	N	N
Streptococcus sp. AS20	1835	HQ616366	clade_98	N	N
Streptococcus sp. BS35a	1836	HQ616369	clade_98	N	N
Streptococcus sp. C150	1837	ACRI01000045	clade_98	N	N
Streptococcus sp. CM6	1838	HQ616372	clade_98	N	N
Streptococcus sp. ICM10	1840	HQ616389	clade_98	N	N
Streptococcus sp. ICM12	1841	HQ616390	clade_98	N	N
Streptococcus sp. ICM2	1842	HQ616386	clade_98	N	N
Streptococcus sp. ICM4	1844	HQ616387	clade_98	N	N
Streptococcus sp. ICM45	1843	HQ616394	clade_98	N	N
Streptococcus sp. M143	1845	ACRK01000025	clade_98	N	N
Streptococcus sp. M334	1846	ACRL01000052	clade_98	N	N
Streptococcus sp. oral clone ASB02	1849	AY923121	clade_98	N	N
Streptococcus sp. oral clone ASCA03	1850	DQ272504	clade_98	N	N
Streptococcus sp. oral clone ASCA04	1851	AY923116	clade_98	N	N
Streptococcus sp. oral clone ASCA09	1852	AY923119	clade_98	N	N
Streptococcus sp. oral clone ASCB04	1853	AY923123	clade_98	N	N
Streptococcus sp. oral clone ASCB06	1854	AY923124	clade_98	N	N
Streptococcus sp. oral clone ASCC04	1855	AY923127	clade_98	N	N
Streptococcus sp. oral clone ASCC05	1856	AY923128	clade_98	N	N
Streptococcus sp. oral clone ASCC12	1857	DQ272507	clade_98	N	N
Streptococcus sp. oral clone ASCD01	1858	AY923129	clade_98	N	N
Streptococcus sp. oral clone ASCD09	1859	AY923130	clade_98	N	N
Streptococcus sp. oral clone ASCD10	1860	DQ272509	clade_98	N	N
Streptococcus sp. oral clone ASCE03	1861	AY923134	clade_98	N	N
Streptococcus sp. oral clone ASCE04	1862	AY953253	clade_98	N	N
Streptococcus sp. oral clone ASCE05	1863	DQ272510	clade_98	N	N
Streptococcus sp. oral clone ASCE06	1864	AY923135	clade_98	N	N
Streptococcus sp. oral clone ASCE09	1865	AY923136	clade_98	N	N
Streptococcus sp. oral clone ASCE10	1866	AY923137	clade_98	N	N
Streptococcus sp. oral clone ASCE12	1867	AY923138	clade_98	N	N
Streptococcus sp. oral clone ASCF05	1868	AY923140	clade_98	N	N
Streptococcus sp. oral clone ASCF07	1869	AY953255	clade_98	N	N
Streptococcus sp. oral clone ASCF09	1870	AY923142	clade_98	N	N
Streptococcus sp. oral clone ASCG04	1871	AY923145	clade_98	N	N
Streptococcus sp. oral clone BW009	1872	AY005042	clade_98	N	N
Streptococcus sp. oral clone CH016	1873	AY005044	clade_98	N	N

TABLE 1-continued

OTU	SEQ ID Number	Public DB Accession	Phylogenetic Clade	Spore Former	Pathogen Status
Streptococcus sp. oral clone GK051	1874	AY349413	clade_98	N	N
Streptococcus sp. oral clone GM006	1875	AY349414	clade_98	N	N
Streptococcus sp. oral clone P2PA_41 P2	1876	AY207051	clade_98	N	N
Streptococcus sp. oral clone P4PA_30 P4	1877	AY207064	clade_98	N	N
Streptococcus sp. oral taxon 071	1878	AEEP01000019	clade_98	N	N
Streptococcus sp. oral taxon G59	1879	GU432132	clade_98	N	N
Streptococcus sp. oral taxon G62	1880	GU432146	clade_98	N	N
Streptococcus sp. oral taxon G63	1881	GU432150	clade_98	N	N
Streptococcus suis	1882	FM252032	clade_98	N	N
Streptococcus thermophilus	1883	CP000419	clade_98	N	N
Streptococcus uberis	1884	HQ391900	clade_98	N	N
Streptococcus urinalis	1885	DQ303194	clade_98	N	N
Streptococcus vestibularis	1886	AEKO01000008	clade_98	N	N
Streptococcus viridans	1887	AF076036	clade_98	N	N
Synergistetes bacterium oral clone 03 5 D05	1908	GU227192	clade_98	N	N

TABLE 2

Phylogenetic Clade	OTUs in clade
clade_172	Bifidobacteriaceae genomosp. C1, *Bifidobacterium adolescentis*, *Bifidobacterium angulatum*, *Bifidobacterium animalis*, *Bifidobacterium breve*, *Bifidobacterium catenulatum*, *Bifidobacterium dentium*, *Bifidobacterium gallicuin*, *Bifidobacterium infantis*, *Bifidobacterium kashiwanohense*, *Bifidobacterium longum*, *Bifidobacterium pseudocatenulatum*, *Bifidobacterium pseudolongum*, *Bifidobacterium scardovii*, *Bifidobacterium* sp. HM2, *Bifidobacterium* sp. HMLN12, *Bifidobacterium* sp. M45, *Bifidobacterium* sp. MSX5B, *Bifidobacterium* sp. TM_7, *Bifidobacterium thermophilum*
clade_172i	Bifidobacteriaceae genomosp. C1, *Bifidobacterium angulatum*, *Bifidobacterium animalis*, *Bifidobacterium breve*, *Bifidobacterium catenulatum*, *Bifidobacterium dentium*, *Bifidobacterium gallicum*, *Bifidobacterium infantis*, *Bifidobacterium kashiwanohense*, *Bifidobacterium pseudocatenulatum*, *Bifidobacterium pseudolongum*, *Bifidobacterium scardovii*, *Bifidobacterium* sp. HM2, *Bifidobacterium* sp. HMLN12, *Bifidobacterium* sp. M45, *Bifidobacterium* sp. MSX5B, *Bifidobacterium* sp. TM 7, *Bifidobacterium thermophilum*
clade_198	*Lactobacillus casei*, *Lactobacillus paracasei*, *Lactobacillus zeae*
clade_198i	*Lactobacillus zeae*
clade_260	*Clostridium hylemonae*, *Clostridium scindens*, Lachnospiraceae bacterium 5_1_57FAA
clade_260c	*Clostridium hylemonae*, Lachnospiraceae bacterium 5_1_57FAA
clade_260g	*Clostridium hylemonae*, Lachnospiraceae bacterium 5_1_57FAA
clade_260h	*Clostridium hylemonae*, Lachnospiraceae bacterium 5_1_57FAA
clade_262	*Clostridium glycyrrhizinilyticum*, *Clostridium nexile*, *Coprococcus comes*, Lachnospiraceae bacterium 1_1_57FAA, Lachnospiraceae bacterium 1_4_56FAA, Lachnospiraceae bacterium 8_1_57FAA, *Ruminococcus lactaris*, *Rummococcus torques*
clade_262i	*Clostridium glycyrrhizinilyticum*, *Clostridium nexile*, *Coprococcus comes*, Lachnospiraceae bacterium 1_1_57FAA, Lachnospiraceae bacterium 1_4_56FAA, Lachnospiraceae bacterium 8_1_57FAA, *Ruminococcus lactaris*
clade_309	*Blautia coccoides*, *Blautia glucerasea*, *Blautia glucerasei*, *Blautia hansenii*, *Blautia luti*, *Blautia producta*, *Blautia schinkii*, *Blautia* sp. M25, *Blautia stercoris*, *Blautia wexlerae*, *Bryantella formatexigens*, *Clostridium coccoides*, *Eubacterium cellulosolveris*, Lachnospiraceae bacterium 6_1_63FAA, *Marvinbryantia formatexigens*, *Ruminococcus hansenii*, *Ruminococcus obeum*, *Ruminococcus* sp. 5_1_39BFAA, *Ruminococcus* sp. K_1, *Syntrophococcus sucromutans*
clade_309c	*Blautia coccoides*, *Blautia glucerasea*, *Blautia glucerasei*, *Blautia hansenii*, *Blautia luti*, *Blautia schinkii*, *Blautia* sp. M25, *Blautia stercoris*, *Blautia wexlerae*, *Bryantella formatexigens*, *Clostridium coccoides*, *Eubacterium cellulosolvens*, Lachnospiraceae bacterium 6_1_63FAA, *Marvinbryantia formatexigens*, *Ruminococcus hansenii*, *Ruminococcus obeum*, *Ruminococcus* sp. 5_1_39BFAA, *Ruminococcus* sp. K_1, *Syntrophococcus sucromutans*
clade_309e	*Blautia coccoides*, *Blautia glucerasea*, *Blautia glucerasei*, *Blautia hansenii*, *Blautia luti*, *Blautia schinkii*, *Blautia* sp. M25, *Blautia stercoris*, *Blautia wexlerae*, *Bryantella formatexigens*, *Clostridium coccoides*, *Eubacterium cellulosolvens*, Lachnospiraceae bacterium 6_1_63FAA, *Marvinbryantia formatexigens*, *Ruminococcus hansenii*, *Ruminococcus obeum*, *Ruminococcus* sp. 5_1_39BFAA, *Ruminococcus* sp. K_1, *Syntrophococcus sucromutans*
clade_309g	*Blautia coccoides*, *Blautia glucerasea*, *Blautia glucerasei*, *Blautia hansenii*, *Blautia luti*, *Blautia schinkii*, *Blautia* sp. M25, *Blautia stercoris*, *Blautia wexlerae*, *Bryantella formatexigens*, *Clostridium coccoides*, *Eubacterium cellulosolvens*, Lachnospiraceae bacterium 6_1_63FAA, *Marvinbryantia formatexigens*, *Ruminococcus hansenii*, *Ruminococcus obeum*, *Ruminococcus* sp. 5_1_39BFAA, *Ruminococcus* sp. K_1, *Syntrophococcus sucromutans*
clade_309h	*Blautia coccoides*, *Blautia glucerasea*, *Blautia glucerasei*, *Blautia hansenii*, *Blautia luti*, *Blautia schinkii*, *Blautia* sp. M25, *Blautia stercoris*, *Blautia wexlerae*, *Bryantella formatexigens*, *Clostridium coccoides*, *Eubacterium cellulosolvens*, Lachnospiraceae bacterium 6_1_63FAA, *Marvinbryantia formatexigens*, *Ruminococcus hansenii*, *Ruminococcus obeum*, *Ruminococcus* sp. 5_1_39BFAA, *Ruminococcus* sp. K_1, *Syntrophococcus sucromutans*

TABLE 2-continued

Phylogenetic Clade	OTUs in clade
clade_309i	*Blautia coccoides*, *Blautia glucerasea*, *Blautia glucerasei*, *Blautia hansenii*, *Blautia luti*, *Blautia schinkii*, *Blautia* sp. M25, *Blautia stercoris*, *Blautia wexlerae*, *Bryantella formatexigens*, *Clostridium coccoides*, *Eubacterium cellulosolvens*, Lachnospiraceae bacterium 6_1_63FAA, *Marvinbryantia formatexigens*, *Ruminococcus hansenii*, *Ruminococcus* sp. 5_1_39BFAA, *Ruminococcus* sp. K_1, *Syntrophococcus sucromutans*
clade_313	*Lactobacillus antri*, *Lactobacillus coleohominis*, *Lactobacillus fermentum*, *Lactobacillus gastricus*, *Lactobacillus mucosae*, *Lactobacillus oris*, *Lactobacillus pontis*, *Lactobacillus reuteri*, *Lactobacillus* sp. KLDS 1.0707, *Lactobacillus* sp. KLDS 1.0709, *Lactobacillus* sp. KLDS 1.0711, *Lactobacillus* sp. KLDS 1.0713, *Lactobacillus* sp. KLDS 1.0716, *Lactobacillus* sp. KLDS 1.0718, *Lactobacillus* sp. oral taxon 052, *Lactobacillus vaginalis*
clade_313f	*Lactobacillus antri*, *Lactobacillus coleohominis*, *Lactobacillus fermentum*, *Lactobacillus gastricus*, *Lactobacillus mucosae*, *Lactobacillus oris*, *Lactobacillus pontis*, *Lactobacillus* sp. KLDS 1.0707, *Lactobacillus* sp. KLDS 1.0709, *Lactobacillus* sp. KLDS 1.0711, *Lactobacillus* sp. KLDS 1.0713, *Lactobacillus* sp. KLDS 1.0716, *Lactobacillus* sp. KLDS 1.0718, *Lactobacillus* sp. oral taxon 052, *Lactobacillus vaginalis*
clade_325	*Staphylococcus aureus*, *Staphylococcus auricularis*, *Staphylococcus capitis*, *Staphylococcus caprae*, *Staphylococcus carnosus*, *Staphylococcus cohnii*, *Staphylococcus condimenti*, *Staphylococcus epidermidis*, *Staphylococcus equorum*, *Staphylococcus haemolyticus*, *Staphylococcus hominis*, *Staphylococcus lugdunensis*, *Staphylococcus pasteuri*, *Staphylococcus pseudintermedius*, *Staphylococcus saccharolyticus*, *Staphylococcus saprophyticus*, *Staphylococcus* sp. H292, *Staphylococcus* sp. H780, *Staphylococcus* sp. clone bottae7, *Staphylococcus succinus*, *Staphylococcus warneri*, *Staphylococcus xylosus*
clade_325f	*Staphylococcus aureus*, *Staphylococcus auricularis*, *Staphylococcus capitis*, *Staphylococcus caprae*, *Staphylococcus carnosus*, *Staphylococcus cohnii*, *Staphylococcus condimenti*, *Staphylococcus epidermidis*, *Staphylococcus equorum*, *Staphylococcus haemolyticus*, *Staphylococcus hominis*, *Staphylococcus lugdunensis*, *Staphylococcus pseudintermedius*, *Staphylococcus saccharolyticus*, *Staphylococcus saprophyticus*, *Staphylococcus* sp. H292, *Staphylococcus* sp. H780, *Staphylococcus* sp. clone bottae7, *Staphylococcus succinus*, *Staphylococcus xylosus*
clade_335	*Bacteroides* sp. 20_3, *Bacteroides* sp. 3_1_19, *Bacteroides* sp. 3_2_5, *Parabacteroides distasonis*, *Parabacteroides goldsteinii*, *Parabacteroides gordonii*, *Parabacteroides* sp. D13
clade_335i	*Bacteroides* sp. 20_3, *Bacteroides* sp. 3_1_19, *Bacteroides* sp. 3_2_5, *Parabacteroides goldsteinii*, *Parabacteroides gordonii*, *Parabacteroides* sp. D13
clade_351	*Clostridium innocuum*, *Clostridium* sp. HGF2
clade_351e	*Clostridium* sp. HGF2
clade_354	*Clostridium bartlettii*, *Clostridium bifermentans*, *Clostridium ghonii*, *Clostridium glycolicum*, *Clostridium mayombei*, *Clostridium sordellii*, *Clostridium* sp. MT4 E, *Eubacterium tenue*
clade_354e	*Clostridium bartlettii*, *Clostridium ghonii*, *Clostridium glycolicum*, *Clostridium mayombei*, *Clostridium sordellii*, *Clostridium* sp. MT4 E, *Eubacterium tenue*
clade_360	*Dorea formicigenerans*, *Dorea longicatena*, Lachnospiraceae bacterium 2_1_46FAA, Lachnospiraceae bacterium 2_1_58FAA, Lachnospiraceae bacterium 4_1_37FAA, Lachnospiraceae bacterium 9_1_43BFAA, *Ruminococcus gnavus*, *Ruminococcus* sp. ID8
clade_360c	*Dorea formicigenerans*, *Dorea longicatena*, Lachnospiraceae bacterium 2_1_46FAA, Lachnospiraceae bacterium 2_1_58FAA, Lachnospiraceae bacterium 4_1_37FAA, Lachnospiraceae bacterium 9_1_43BFAA, *Ruminococcus gnavus*
clade_360g	*Dorea formicigenerans*, *Dorea longicatena*, Lachnospiraceae bacterium 2_1_46FAA, Lachnospiraceae bacterium 2_1_58FAA, Lachnospiraceae bacterium 4_1_37FAA, Lachnospiraceae bacterium 9_1_43BFAA, *Ruminococcus gnavus*
clade_360h	*Dorea formicigenerans*, *Dorea longicatena*, Lachnospiraceae bacterium 2_1_46FAA, Lachnospiraceae bacterium 2_1_58FAA, Lachnospiraceae bacterium 4_1_37FAA, Lachnospiraceae bacterium 9_1_43BFAA, *Ruminococcus gnavus*
clade_360i	*Dorea formicigenerans*, Lachnospiraceae bacterium 2_1_46FAA, Lachnospiraceae bacterium 2_1_58FAA, Lachnospiraceae bacterium 4_1_37FAA, Lachnospiraceae bacterium 9_1_43BFAA, *Ruminococcus gnavus*, *Ruminococcus* sp. ID8
clade_378	*Bacteroides barnesiae*, *Bacteroides coprocola*, *Bacteroides coprophilus*, *Bacteroides dorei*, *Bacteroides massiliensis*, *Bacteroides plebeius*, *Bacteroides* sp. 3_1_33FAA, *Bacteroides* sp. 3_1_40A, *Bacteroides* sp. 4_3_47FAA, *Bacteroides* sp. 9_1_42FAA, *Bacteroides* sp. NB_8, *Bacteroides vulgatus*
clade_378e	*Bacteroides barnesiae*, *Bacteroides coprocola*, *Bacteroides coprophilus*, *Bacteroides dorei*, *Bacteroides massiliensis*, *Bacteroides plebeius*, *Bacteroides* sp. 3_1_33FAA, *Bacteroides* sp. 3_1_40A, *Bacteroides* sp. 4_3_47FAA, *Bacteroides* sp. 9_1_42FAA, *Bacteroides* sp. NB_8
clade_38	*Bacteroides ovatus*, *Bacteroides* sp. 1_1_30, *Bacteroides* sp. 2_1_22, *Bacteroides* sp. 2_2_4, *Bacteroides* sp. 3_1_23, *Bacteroides* sp. D1, *Bacteroides* sp. D2, *Bacteroides* sp. D22, *Bacteroides xylanisolvens*
clade_38e	*Bacteroides* sp. 1_1_30, *Bacteroides* sp. 2_1_22, *Bacteroides* sp. 2_2_4, *Bacteroides* sp. 3_1_23, *Bacteroides* sp. D1, *Bacteroides* sp. D2, *Bacteroides* sp. D22, *Bacteroides xylanisolvens*
clade_38i	*Bacteroides* sp. 1_1_30, *Bacteroides* sp. 2_1_22, *Bacteroides* sp. 2_2_4, *Bacteroides* sp. 3_1_23, *Bacteroides* sp. D1, *Bacteroides* sp. D2, *Bacteroides* sp. D22, *Bacteroides xylanisolvens*
clade_408	*Anaerostipes caccae*, *Anaerostipes* sp. 3_2_56FAA, Clostridiales bacterium 1_7_47FAA, Clostridiales sp. SM4_1, Clostridiales sp. SSC_2, *Clostridium aerotolerans*, *Clostridium aldenense*, *Clostridium algidixylanolyticum*, *Clostridium amygdalinum*, *Clostridium asparagiforme*, *Clostridium bolteae*, *Clostridium celerecrescens*, *Clostridium citroniae*, *Clostridium clostridiiformes*, *Clostridium clostridioforme*, *Clostridium hathewayi*, *Clostridium indolis*, *Clostridium lavalense*, *Clostridium saccharolyticum*, *Clostridium* sp. M62_1, *Clostridium* sp. SS2_1, *Clostridium sphenoides*, *Clostridium symbiosum*, *Clostridium xylanolyticum*, *Eubacterium hadrum*, *Fusobacterium naviforme*, Lachnospiraceae bacterium 3_1_57FAA, Lachnospiraceae bacterium 5_1_63FAA, Lachnospiraceae bacterium A4, Lachnospiraceae bacterium DJF VP30, Lachnospiraceae genomosp. C1, *Moryella indoligenes*

TABLE 2-continued

Phylogenetic Clade	OTUs in clade
clade_408b	*Anaerostipes caccae*, *Anaerostipes* sp. 3_2_56FAA, Clostridiales bacterium 1_7_47FAA, Clostridiales sp. SM4_1, Clostridiales sp. SSC_2, *Clostridium aerotolerans*, *Clostridium aldenense*, *Clostridium algidixylanolyticum*, *Clostridium amygdalinum*, *Clostridium asparagiforme*, *Clostridium bolteae*, *Clostridium celerecrescens*, *Clostridium citroniae*, *Clostridium clostridiiformes*, *Clostridium clostridioforme*, *Clostridium indolis*, *Clostridium lavalense*, *Clostridium saccharolyticum*, *Clostridium* sp. M62_1, *Clostridium* sp. SS2_1, *Clostridium sphenoides*, *Clostridium symbiosum*, *Clostridium xylanolyticum*, *Eubacterium hadrum*, *Fusobacterium naviforme*, Lachnospiraceae bacterium 5_1_63FAA, Lachnospiraceae bacterium A4, Lachnospiraceae bacterium DJF VP30, Lachnospiraceae genomosp. C1, *Moryella indoligenes*
clade_408d	*Anaerostipes caccae*, *Anaerostipes* sp. 3_2_56FAA, Clostridiales sp. SM4_1, Clostridiales sp. SSC_2, *Clostridium aerotolerans*, *Clostridium aldenense*, *Clostridium algidixylanolyticum*, *Clostridium amygdalinum*, *Clostridium celerecrescens*, *Clostridium citroniae*, *Clostridium clostridiiformes*, *Clostridium clostridioforme*, *Clostridium hathewayi*, *Clostridium lavalense*, *Clostridium saccharolyticum*, *Clostridium* sp. M62_1, *Clostridium* sp. SS2_1, *Clostridium sphenoides*, *Clostridium symbiosum*, *Clostridium xylanolyticum*, *Eubacterium hadrum*, *Fusobacterium naviforme*, Lachnospiraceae bacterium 3_1_57FAA, Lachnospiraceae bacterium 5_1_63FAA, Lachnospiraceae bacterium A4, Lachnospiraceae bacterium DJF VP30, Lachnospiraceae genomosp. C1, *Moryella indoligenes*
clade_408f	*Anaerostipes* sp. 3_2_56FAA, Clostridiales bacterium 1_7_47FAA, Clostridiales sp. SM4_1, Clostridiales sp. SSC_2, *Clostridium aerotolerans*, *Clostridium aldenense*, *Clostridium algidixylanolyticum*, *Clostridium amygdalinum*, *Clostridium asparagiforme*, *Clostridium bolteae*, *Clostridium celerecrescens*, *Clostridium citroniae*, *Clostridium clostridiiformes*, *Clostridium clostridioforme*, *Clostridium hathewayi*, *Clostridium lavalense*, *Clostridium saccharolyticum*, *Clostridium* sp. M62_1, *Clostridium* sp. SS2_1, *Clostridium sphenoides*, *Clostridium symbiosum*, *Clostridium xylanolyticum*, *Eubacterium hadrum*, *Fusobacterium naviforme*, Lachnospiraceae bacterium 3_1_57FAA, Lachnospiraceae bacterium 5_1_63FAA, Lachnospiraceae bacterium A4, Lachnospiraceae bacterium DJF VP30, Lachnospiraceae genomosp. C1, *Moryella indoligenes*
clade_408g	*Anaerostipes caccae*, *Anaerostipes* sp. 3_2_56FAA, Clostridiales sp. SM4_1, Clostridiales sp. SSC_2, *Clostridium aerotolerans*, *Clostridium aldenense*, *Clostridium algidixylanolyticum*, *Clostridium amygdalinum*, *Clostridium celerecrescens*, *Clostridium citroniae*, *Clostridium clostridiiformes*, *Clostridium clostridioforme*, *Clostridium lavalense*, *Clostridium saccharolyticum*, *Clostridium* sp. M62_1, *Clostridium* sp. SS2_1, *Clostridium sphenoides*, *Clostridium symbiosum*, *Clostridium xylanolyticum*, *Eubacterium hadrum*, *Fusobacterium naviforme*, Lachnospiraceae bacterium 5_1_63FAA, Lachnospiraceae bacterium A4, Lachnospiraceae bacterium DJF VP30, Lachnospiraceae genomosp. C1, *Moryella indoligenes*
clade_408h	*Anaerostipes caccae*, *Anaerostipes* sp. 3_2_56FAA, Clostridiales sp. SM4_1, Clostridiales sp. SSC_2, *Clostridium aerotolerans*, *Clostridium aldenense*, *Clostridium algidixylanolyticum*, *Clostridium amygdalinum*, *Clostridium celerecrescens*, *Clostridium citroniae*, *Clostridium clostridiiformes*, *Clostridium clostridioforme*, *Clostridium lavalense*, *Clostridium saccharolyticum*, *Clostridium* sp. M62_1, *Clostridium* sp. SS2_1, *Clostridium sphenoides*, *Clostridium symbiosum*, *Clostridium xylanolyticum*, *Eubacterium hadrum*, *Fusobacterium naviforme*, Lachnospiraceae bacterium 5_1_63FAA, Lachnospiraceae bacterium A4, Lachnospiraceae bacterium DJF VP30, Lachnospiraceae genomosp. C1, *Moryella indoligenes*
clade_420	*Barnesiella intestinihominis*, *Barnesiella viscericola*, *Parabacteroides* sp. NS31_3, Porphyromonadaceae bacterium NML 060648, *Tannerella forsythia*, *Tannerella* sp. 6_1_58FAA_CT1
clade_420f	*Barnesiella viscericola*, *Parabacteroides* sp. NS31_3. Porphyromonadaceae bacterium NML 060648, *Tannerella forsythia*, *Tannerella* sp. 6_1_58FAA_CT1
clade_444	*Butyrivibrio fibrisolvens*, *Eubacterium rectale*, *Eubacterium* sp. oral clone GI038, *Lachnobacterium bovis*, *Roseburia cecicola*, *Roseburia faecalis*, *Roseburia faecis*, *Roseburia hominis*, *Roseburia intestinalis*, *Roseburia inulinivorans*, *Roseburia* sp. 11SE37, *Roseburia* sp. 11SE38, *Shuttleworthia satelles*, *Shuttleworthia* sp. MSX8B, *Shuttleworthia* sp. oral taxon G69
clade_444i	*Butyrivibrio fibrisolvens*, *Eubacterium* sp. oral clone GI038, *Lachnobacterium bovis*, *Roseburia cecicola*, *Roseburia faecis*, *Roseburia hominis*, *Roseburia inulinivorans*, *Roseburia* sp. 11SE37, *Roseburia* sp. 11SE38, *Shuttleworthia satelles*, *Shuttleworthia* sp. MSX8B, *Shuttleworthia* sp. oral taxon G69
clade_478	*Faecalibacterium prausnitzii*, *Gemmiger formicilis*, *Subdoligranulum variabile*
clade_478i	*Gemmiger formicilis*, *Subdoligranulum variabile*
clade_479	Clostridiaceae bacterium JC13, *Clostridium* sp. MLG055, Erysipelotrichaceae bacterium 3_1_53
clade_479c	*Clostridium* sp. MLG055, Erysipelotrichaceae bacterium 3_1_53
clade_479g	*Clostridium* sp. MLG055, Erysipelotrichaceae bacterium 3_1_53
clade_479h	*Clostridium* sp. MLG055, Erysipelotrichaceae bacterium 3_1_53
clade_481	*Clostridium cocleatum*, *Clostridium ramosum*, *Clostridium saccharogumia*, *Clostridium spiroforme*, *Coprobacillus* sp. D7
clade_481a	*Clostridium cocleatum*, *Clostridium spiroforme*, *Coprobacillus* sp. D7
clade_481b	*Clostridium cocleatum*, *Clostridium ramosum*, *Clostridium spiroforme*, *Coprobacillus* sp. D7
clade_481e	*Clostridium cocleatum*, *Clostridium saccharogumia*, *Clostridium spiroforme*, *Coprobacillus* sp. D7
clade_481g	*Clostridium cocleatum*, *Clostridium spiroforme*, *Coprobacillus* sp. D7
clade_481h	*Clostridium cocleatum*, *Clostridium spiroforme*, *Coprobacillus* sp. D7
clade_481i	*Clostridium ramosum*, *Clostridium saccharogumia*, *Clostridium spiroforme*, *Coprobacillus* sp. D7
clade_497	*Abiotrophia para_adiacens*, *Carnobacterium divergens*, *Carnobacterium maltaromaticum*, *Enterococcus avium*, *Enterococcus caccae*, *Enterococcus casseliflavus*, *Enterococcus durans*, *Enterococcus faecalis*, *Enterococcus faecium*, *Enterococcus gallinarum*, *Enterococcus gilvus*, *Enterococcus hawaiiensis*, *Enterococcus hirae*, *Enterococcus italicus*, *Enterococcus mundtii*, *Enterococcus raffinosus*, *Enterococcus* sp. BV2CASA2, *Enterococcus* sp. CCRI 16620, *Enterococcus* sp. F95, *Enterococcus* sp. RfL6, *Enterococcus thailandieus*, *Fusobacterium canifelinum*, *Fusobacterium* genomosp. C1, *Fusobacterium* genomosp. C2, *Fusobacterium nucleatum*, *Fusobacterium periodonticum*, *Fusobacterium* sp. 11_3_2, *Fusobacterium* sp. 1_1_41FAA, *Fusobacterium* sp. 2_1_31, *Fusobacterium* sp. 3_1_27, *Fusobacterium*

TABLE 2-continued

Phylogenetic Clade	OTUs in clade
	sp. 3_1_33, *Fusobacterium* sp. 3_1_36A2, *Fusobacterium* sp. AC18, *Fusobacterium* sp. ACB2, *Fusobacterium* sp. AS2, *Fusobacterium* sp. CM1, *Fusobacterium* sp. CM21, *Fusobacterium* sp. CM22, *Fusobacterium* sp. oral clone ASCF06, *Fusobacterium* sp. oral clone ASCF11, *Granulicatella adiacens*, *Granulicatella elegans*, *Granulicatella paradiacens*, *Granulicatella* sp. oral clone ASC02, *Granulicatella* sp. oral clone ASCA05, *Granulicatella* sp. oral clone ASCB09, *Granulicatella* sp. oral clone ASCG05, *Tetragenococcus halophilus*, *Tetragenococcus koreensis*, *Vagococcus fluvialis*
clade_497e	*Abiotrophia para_adiacens*, *Carnobacterium divergens*, *Carnobacterium maltaromaticum*, *Enterococcus avium*, *Enterococcus caccae*, *Enterococcus casseliflavus*, *Enterococcus durans*, *Enterococcus faecium*, *Enterococcus gallinarum*, *Enterococcus gilvus*, *Enterococcus hawaiiensis*, *Enterococcus hirae*, *Enterococcus italicus*, *Enterococcus mundtii*, *Enterococcus raffinosus*, *Enterococcus* sp. BV2CASA2, *Enterococcus* sp. CCRI 16620, *Enterococcus* sp. F95, *Enterococcus* sp. RfL6, *Enterococcus thailandicus*, *Fusobacterium canifelinum*, *Fusobacterium* genomosp. C1, *Fusobacterium* genomosp. C2, *Fusobacterium nucleatum*, *Fusobacterium periodonticum*, *Fusobacterium* sp. 11_3_2, *Fusobacterium* sp. 1_1_41FAA, *Fusobacterium* sp. 2_1_31, *Fusobacterium* sp. 3_1_27, *Fusobacterium* sp. 3_1_33, *Fusobacterium* sp. 3_1_36A2, *Fusobacterium* sp. AC18, *Fusobacterium* sp. ACB2, *Fusobacterium* sp. AS2, *Fusobacterium* sp. CM1, *Fusobacterium* sp. CM21, *Fusobacterium* sp. CM22, *Fusobacterium* sp. oral clone ASCF06, *Fusobacterium* sp. oral clone ASCF11, *Granulicatella adiacens*, *Granulicatella elegans*, *Granulicatella paradiacens*, *Granulicatella* sp. oral clone ASC02, *Granulicatella* sp. oral clone ASCA05, *Granulicatella* sp. oral clone ASCB09, *Granulicatella* sp. oral clone ASCG05, *Tetragenococcus halophilus*, *Tetragenococcus koreensis*, *Vagococcus fluvialis*
clade_497f	*Abiotrophia para_adiacens*, *Carnobacterium divergens*, *Carnobacterium maltaromaticum*, *Enterococcus avium*, *Enterococcus caccae*, *Enterococcus casseliflavus*, *Enterococcus faecalis*, *Enterococcus gallinarum*, *Enterococcus gilvus*, *Enterococcus hawaiiensis*, *Enterococcus italicus*, *Enterococcus mundtii*, *Enterococcus raffinosus*, *Enterococcus* sp. BV2CASA2, *Enterococcus* sp. CCRI 16620, *Enterococcus* sp. F95, *Enterococcus* sp. RfL6, *Enterococcus thailandicus*, *Fusobacterium canifelinum*, *Fusobacterium* genomosp. C1, *Fusobacterium* genomosp. C2, *Fusobacterium nucleatum*, *Fusobacterium periodonticum*, *Fusobacterium* sp. 11_3_2, *Fusobacterium* sp. 1_1_41FAA, *Fusobacterium* sp. 2_1_31, *Fusobacterium* sp. 3_1_27, *Fusobacterium* sp. 3_1_33, *Fusobacterium* sp. 3_1_36A2, *Fusobacterium* sp. AC18, *Fusobacterium* sp. ACB2, *Fusobacterium* sp. AS2, *Fusobacterium* sp. CM1, *Fusobacterium* sp. CM21, *Fusobacterium* sp. CM22, *Fusobacterium* sp. oral clone ASCF06, *Fusobacterium* sp. oral clone ASCF11, *Granulicatella adiacens*, *Granulicatella elegans*, *Granulicatella paradiacens*, *Granulicatella* sp. oral clone ASC02, *Granulicatella* sp. oral clone ASCA05, *Granulicatella* sp. oral clone ASCB09, *Granulicatella* sp. oral clone ASCG05, *Tetragenococcus halophilus*, *Tetragenococcus koreensis*, *Vagococcus fluvialis*
clade_512	*Eubacterium barkeri*, *Eubacterium callanderi*, *Eubacterium limosum*, *Pseudoramibacter alactolyticus*
clade_512i	*Eubacterium barkeri*, *Eubacterium callanderi*, *Pseudoramibacter alactolyticus*
clade_516	*Anaerotruncus colihominis*, *Clostridium methylpentosum*, *Clostridium* sp. YIT 12070, *Hydrogenoanaerobacterium saccharovorans*, *Ruminococcus albus*, *Ruminococcus flavefaciens*
clade_516c	*Clostridium methylpentosum*, *Clostridium* sp. YIT 12070, *Hydrogenoanaerobacterium saccharovorans*, *Ruminococcus albus*, *Ruminococcus flavefaciens*
clade_516g	*Clostridium methylpentosum*, *Clostridium* sp. YIT 12070, *Hydrogenoanaerobacterium saccharovorans*, *Ruminococcus albus*, *Ruminococcus flavefaciens*
clade_516h	*Clostridium methylpentosum*, *Clostridium* sp. YIT 12070, *Hydrogenoanaerobacterium saccharovorans*, *Ruminococcus albus*, *Ruminococcus flavefaciens*
clade_519	*Eubacterium ventriosum*
clade_522	*Bacteroides galacturonicus*, *Eubacterium eligens*, *Lachnospira multipara*, *Lachnospira pectinoschiza*, *Lactobacillus rogosae*
clade_522i	*Bacteroides galacturonicus*, *Lachnospira multipara*, *Lachnospira pectinoschiza*, *Lactobacillus rogosae*
clade_553	*Collinsella aerofaciens*, *Collinsella intestinalis*, *Collinsella stercoris*, *Collinsella tanakaei*
clade_553i	*Collinsella intestinalis*, *Collinsella stercoris*, *Collinsella tanakaei*
clade_566	*Adlercreutzia equolifaciens*, Coriobacteriaceae bacterium JC110, Coriobacteriaceae bacterium phI, *Cryptobacterium curtum*, *Eggerthella lenta*, *Eggerthella sinensis*, *Eggerthella* sp. 1_3_56FAA, *Eggerthella* sp. HGA1, *Eggerthella* sp. YY7918, *Gordonibacter pamelaeae*, *Slackia equolifaciens*, *Slackia exigua*, *Slackia faecicanis*, *Slackia heliotrinireducens*, *Slackia isoflavoniconvertens*, *Slackia piriformis*, *Slackia* sp. NATTS, *Streptomyces albus*
clade_566f	Coriobacteriaceae bacterium JC110, Coriobacteriaceae bacterium phI, *Cryptobacterium curtum*, *Eggerthella lenta*, *Eggerthella sinensis*, *Eggerthella* sp. 1_3_56FAA, *Eggerthella* sp. HGA1, *Eggerthella* sp. YY7918, *Gordonibacter pamelaeae*, *Slackia equolifaciens*, *Slackia exigua*, *Slackia faecicanis*, *Slackia heliotrinireducens*, *Slackia isoflavoniconvertens*, *Slackia piriformis*, *Slackia* sp. NATTS, *Streptomyces albus*
clade_572	*Butyricicoccus pullicaecorum*, *Eubacterium desmolans*, *Papillibacter cinnamivorans*, *Sporobacter termitidis*
clade_572i	*Butyricicoccus pullicaecorum*, *Papillibacter cinnamivorans*, *Sporobacter termitidis*
clade_65	*Bacteroides faecis*, *Bacteroides fragilis*, *Bacteroides nordii*, *Bacteroides salyersiae*, *Bacteroides* sp. 1_1_14, *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 2_1_56FAA, *Bacteroides* sp. AR29, *Bacteroides* sp. B2, *Bacteroides thetaiotaomicron*
clade_65e	*Bacteroides faecis*, *Bacteroides fragilis*, *Bacteroides nordii*, *Bacteroides salyersiae*, *Bacteroides* sp. 1_1_14, *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 2_1_56FAA, *Bacteroides* sp. AR29, *Bacteroides* sp. B2
clade_92	*Actinobacillus actinomycetemcomitans*, *Actinobacillus succinogenes*, *Aggregatibacter actinomycetemcomitans*, *Aggregatibacter aphrophilus*, *Aggregatibacter segnis*, *Averyella dalhousiensis*, Bisgaard Taxon, *Buchnera aphidicola*, *Cedecea davisae*, *Citrobacter amalonaticus*, *Citrobacter braakii*, *Citrobacter farmeri*, *Citrobacter freundii*, *Citrobacter gillenii*, *Citrobacter koseri*, *Citrobacter murliniae*, *Citrobacter rodentium*, *Citrobacter sedlakii*, *Citrobacter* sp. 30_2, *Citrobacter* sp. KMSI_3, *Citrobacter werkmanii*, *Citrobacter youngae*, *Cronobacter malonaticus*, *Cronobacter sakazakii*, *Cronobacter turicensis*, *Enterobacter aerogenes*, *Enterobacter asburiae*, *Enterobacter cancerogenus*, *Enterobacter cloacae*, *Enterobacter cowanii*, *Enterobacter hormaechei*, *Enterobacter* sp. 247BMC, *Enterobacter* sp. 638, *Enterobacter* sp. JC163, *Enterobacter* sp. SCSS, *Enterobacter* sp. TSE38, Enterobacteriaceae

TABLE 2-continued

Phylogenetic Clade	OTUs in clade
	bacterium 9_2_54FAA, Enterobacteriaceae bacterium CF01Ent_1, Enterobacteriaceae bacterium Smarlab 3302238, *Escherichia albertii*, *Escherichia coli*, *Escherichia fergusonii*, *Escherichia hermannii*, *Escherichia* sp. 1_1_43, *Escherichia* sp. 4_1_40B, *Escherichia* sp. B4, *Escherichia vulneris*, *Ewingella americana*, *Haemophilus* genomosp. P2 oral clone MB3_C24, *Haemophilus* genomosp. P3 oral clone MB3_C38, *Haemophilus* sp. oral clone JM053, *Hafnia alvei*, *Klebsiella oxytoca*, *Klebsiella pneumoniae*, *Klebsiella* sp. AS10, *Klebsiella* sp. Co9935, *Klebsiella* sp. OBRC7, *Klebsiella* sp. SP_BA, *Klebsiella* sp. SRC_DSD1, *Klebsiella* sp. SRC_DSD11, *Klebsiella* sp. SRC_DSD12, *Klebsiella* sp. SRC_DSD15, *Klebsiella* sp. SRC_DSD2, *Klebsiella* sp. SRC_DSD6, *Klebsiella* sp. enrichment culture clone SRC_DSD25, *Klebsiella variicola*, *Kluyvera ascorbata*, *Kluyvera cryocrescens*, *Leminorella grimontii*, *Leminorella richardii*, *Pantoea agglomerans*, *Pantoea ananatis*, *Pantoea brenneri*, *Pantoea citrea*, *Pantoea conspicua*, *Pantoea septica*, *Pasteurella dagmatis*, *Pasteurella multocida*, *Plesiomonas shigelloides*, *Raoultella ornithinolytica*, *Raoultella planticola*, *Raoultella terrigena*, *Salmonella bongori*, *Salmonella enterica*, *Salmonella typhimurium*, *Serratia fonticola*, *Serratia liquefaciens*, *Serratia marcescens*, *Serratia odorifera*, *Serratia proteamaculans*, *Shigella boydii*, *Shigella dysenteriae*, *Shigella flexneri*, *Shigella sonnei*, *Tatumella ptyseos*, *Trabulsiella guamensis*, *Yersinia aldovae*, *Yersinia aleksiciae*, *Yersinia bercovieri*, *Yersinia enterocolitica*, *Yersinia frederiksenii*, *Yersinia intermedia*, *Yersinia kristensenii*, *Yersinia mollaretii*, *Yersinia pestis*, *Yersinia pseudotuberculosis*, *Yersinia rohdei*, *Yokenella regensburgei*
clade_92e	*Actinobacillus actinomycetemcomitans*, *Actinobacillus succinogenes*, *Aggregatibacter actinomycetemcomitans*, *Aggregatibacter aphrophilus*, *Aggregatibacter segnis*, *Averyella dalhousiensis*, Bisgaard Taxon, *Buchnera aphidicola*, *Cedecea davisae*, *Citrobacter amalonaticus*, *Citrobacter braakii*, *Citrobacter farmeri*, *Citrobacter freundii*, *Citrobacter gillenii*, *Citrobacter koseri*, *Citrobacter murliniae*, *Citrobacter rodentium*, *Citrobacter sedlakii*, *Citrobacter* sp. 30_2, *Citrobacter* sp. KMSI_3, *Citrobacter werkmanii*, *Citrobacter youngae*, *Cronobacter malonaticus*, *Cronobacter sakazakii*, *Cronobacter turicensis*, *Enterobacter aerogenes*, *Enterobacter asburiae*, *Enterobacter cancerogenus*, *Enterobacter cloacae*, *Enterobacter cowanii*, *Enterobacter hormaechei*, *Enterobacter* sp. 247BMC, *Enterobacter* sp. 638, *Enterobacter* sp. JC163, *Enterobacter* sp. SCSS, *Enterobacter* sp. TSE38, Enterobacteriaceae bacterium 9_2_54FAA, Enterobacteriaceae bacterium CF01Ent_1, Enterobacteriaceae bacterium Smarlab 3302238, *Escherichia albertii*, *Escherichia fergusonii*, *Escherichia hermannii*, *Escherichia* sp. 1_1_43, *Escherichia* sp. 4_1_40B, *Escherichia* sp. B4, *Escherichia vulneris*, *Ewingella americana*, *Haemophilus* genomosp. P2 oral clone MB3_C24, *Haemophilus* genomosp. P3 oral clone MB3_C38, *Haemophilus* sp. oral clone JM053, *Hafnia alvei*, *Klebsiella oxytoca*, *Klebsiella pneumoniae*, *Klebsiella* sp. AS10, *Klebsiella* sp. Co9935, *Klebsiella* sp. OBRC7, *Klebsiella* sp. SP_BA, *Klebsiella* sp. SRC_DSD1, *Klebsiella* sp. SRC_DSD11, *Klebsiella* sp. SRC_DSD12, *Klebsiella* sp. SRC_DSD15, *Klebsiella* sp. SRC_DSD2, *Klebsiella* sp. SRC_DSD6, *Klebsiella* sp. enrichment culture clone SRC_DSD25, *Klebsiella variicola*, *Kluyvera ascorbata*, *Kluyvera cryocrescens*, *Leminorella grimontii*, *Leminorella richardii*, *Pantoea agglomerans*, *Pantoea ananatis*, *Pantoea brenneri*, *Pantoea citrea*, *Pantoea conspicua*, *Pantoea septica*, *Pasteurella dagmatis*, *Pasteurella multocida*, *Plesiomonas shigelloides*, *Raoultella ornithinolytica*, *Raoultella planticola*, *Raoultella terrigena*, *Salmonella bongori*, *Salmonella enterica*, *Salmonella typhimurium*, *Serratia fonticola*, *Serratia liquefaciens*, *Serratia marcescens*, *Serratia odorifera*, *Serratia proteamaculans*, *Shigella boydii*, *Shigella dysenteriae*, *Shigella flexneri*, *Shigella sonnei*, *Tatumella ptyseos*, *Trabulsiella guamensis*, *Yersinia aldovae*, *Yersinia aleksiciae*, *Yersinia bercovieri*, *Yersinia enterocolitica*, *Yersinia frederiksenii*, *Yersinia intermedia*, *Yersinia kristensenii*, *Yersinia mollaretii*, *Yersinia pestis*, *Yersinia pseudotuberculosis*, *Yersinia rohdei*, *Yokenella regensburgei*
clade_92i	*Actinobacillus actinomycetemcomitans*, *Actinobacillus succinogenes*, *Aggregatibacter actinomycetemcomitans*, *Aggregatibacter aphrophilus*, *Aggregatibacter segnis*, *Averyella dalhousiensis*, Bisgaard Taxon, *Buchnera aphidicola*, *Cedecea davisae*, *Citrobacter amalonaticus*, *Citrobacter braakii*, *Citrobacter farmeri*, *Citrobacter freundii*, *Citrobacter gillenii*, *Citrobacter koseri*, *Citrobacter murliniae*, *Citrobacter rodentium*, *Citrobacter sedlakii*, *Citrobacter* sp. 30_2, *Citrobacter* sp. KMSI_3, *Citrobacter werkmanii*, *Citrobacter youngae*, *Cronobacter malonaticus*, *Cronobacter sakazakii*, *Cronobacter turicensis*, *Enterobacter aerogenes*, *Enterobacter asburiae*, *Enterobacter cancerogenus*, *Enterobacter cloacae*, *Enterobacter cowanii*, *Enterobacter hormaechei*, *Enterobacter* sp. 247BMC, *Enterobacter* sp. 638, *Enterobacter* sp. JC163, *Enterobacter* sp. SCSS, *Enterobacter* sp. TSE38, Enterobacteriaceae bacterium 9_2_54FAA, Enterobacteriaceae bacterium CF01Ent_1, Enterobacteriaceae bacterium Smarlab 3302238, *Escherichia albertii*, *Escherichia fergusonii*, *Escherichia hermannii*, *Escherichia* sp. 1_1_43, *Escherichia* sp. 4_1_40B, *Escherichia* sp. B4, *Escherichia vulneris*, *Ewingella americana*, *Haemophilus* genomosp. P2 oral clone MB3_C24, *Haemophilus* genomosp. P3 oral clone MB3_C38, *Haemophilus* sp. oral clone JM053, *Hafnia alvei*, *Klebsiella oxytoca*, *Klebsiella pneumoniae*, *Klebsiella* sp. AS10, *Klebsiella* sp. Co9935, *Klebsiella* sp. OBRC7, *Klebsiella* sp. SP_BA, *Klebsiella* sp. SRC_DSD1, *Klebsiella* sp. SRC_DSD11, *Klebsiella* sp. SRC_DSD12, *Klebsiella* sp. SRC_DSD15, *Klebsiella* sp. SRC_DSD2, *Klebsiella* sp. SRC_DSD6, *Klebsiella* sp. enrichment culture clone SRC_DSD25, *Klebsiella variicola*, *Kluyvera ascorbata*, *Kluyvera cryocrescens*, *Leminorella grimontii*, *Leminorella richardii*, *Pantoea agglomerans*, *Pantoea ananatis*, *Pantoea brenneri*, *Pantoea citrea*, *Pantoea conspicua*, *Pantoea septica*, *Pasteurella dagmatis*, *Pasteurella multocida*, *Plesiomonas shigelloides*, *Raoultella ornithinolytica*, *Raoultella planticola*, *Raoultella terrigena*, *Salmonella bongori*, *Salmonella enterica*, *Salmonella typhimurium*, *Serratia fonticola*, *Serratia liquefaciens*, *Serratia marcescens*, *Serratia odorifera*, *Serratia proteamaculans*, *Shigella boydii*, *Shigella dysenteriae*, *Shigella flexneri*, *Shigella sonnei*, *Tatumella ptyseos*, *Trabulsiella guamensis*, *Yersinia aldovae*, *Yersinia aleksiciae*, *Yersinia bercovieri*, *Yersinia enterocolitica*, *Yersinia frederiksenii*, *Yersinia intermedia*, *Yersinia kristensenii*, *Yersinia mollaretii*, *Yersinia pestis*, *Yersinia pseudotuberculosis*, *Yersinia rohdei*, *Yokenella regensburgei*
clade_96	*Clostridium oroticum*, *Clostridium* sp. D5, *Eubacterium contortum*, *Eubacterium fissicatena*
clade_96g	*Clostridium oroticum*, *Clostridium* sp. D5, *Eubacterium fissicatena*
clade_96h	*Clostridium oroticum*, *Clostridium* sp. D5, *Eubacterium fissicatena*
clade_98	*Okadaella gastrococcus*, *Streptococcus agalactiae*, *Streptococcus alactolyticus*, *Streptococcus australis*, *Streptococcus bovis*, *Streptococcus canis*, *Streptococcus constellatus*, *Streptococcus cristatus*,

TABLE 2-continued

Phylogenetic Clade	OTUs in clade
	Streptococcus dysgalactiae, *Streptococcus equi*, *Streptococcus equinus*, *Streptococcus gallolyticus*, *Streptococcus* genomosp. C1, *Streptococcus* genomosp. C2, *Streptococcus* genomosp. C3, *Streptococcus* genomosp. C4, *Streptococcus* genomosp. C5, *Streptococcus* genomosp. C6, *Streptococcus* genomosp. C7, *Streptococcus* genomosp. C8, *Streptococcus gordonii*, *Streptococcus infantarius*, *Streptococcus infantis*, *Streptococcus intermedius*, *Streptococcus lutetiensis*, *Streptococcus massiliensis*, *Streptococcus mitis*, *Streptococcus oligofermentans*, *Streptococcus oralis*, *Streptococcus parasanguinis*, *Streptococcus pasteurianus*, *Streptococcus peroris*, *Streptococcus pneumoniae*, *Streptococcus porcinus*, *Streptococcus pseudopneumoniae*, *Streptococcus pseudoporcinus*, *Streptococcus pyogenes*, *Streptococcus ratti*, *Streptococcus salivarius*, *Streptococcus sanguinis*, *Streptococcus sinensis*, *Streptococcus* sp. 2285_97, *Streptococcus* sp. 2_1_36FAA, *Streptococcus* sp. ACS2, *Streptococcus* sp. AS20, *Streptococcus* sp. BS35a, *Streptococcus* sp. C150, *Streptococcus* sp. CM6, *Streptococcus* sp. ICM10, *Streptococcus* sp. ICM12, *Streptococcus* sp. ICM2, *Streptococcus* sp. ICM4, *Streptococcus* sp. ICM45, *Streptococcus* sp. M143, *Streptococcus* sp. M334, *Streptococcus* sp. oral clone ASB02, *Streptococcus* sp. oral clone ASCA03, *Streptococcus* sp. oral clone ASCA04, *Streptococcus* sp. oral clone ASCA09, *Streptococcus* sp. oral clone ASCB04, *Streptococcus* sp. oral clone ASCB06, *Streptococcus* sp. oral clone ASCC04, *Streptococcus* sp. oral clone ASCC05, *Streptococcus* sp. oral clone ASCC12, *Streptococcus* sp. oral clone ASCD01, *Streptococcus* sp. oral clone ASCD09, *Streptococcus* sp. oral clone ASCD10, *Streptococcus* sp. oral clone ASCE03, *Streptococcus* sp. oral clone ASCE04, *Streptococcus* sp. oral clone ASCE05, *Streptococcus* sp. oral clone ASCE06, *Streptococcus* sp. oral clone ASCE09, *Streptococcus* sp. oral clone ASCE10, *Streptococcus* sp. oral clone ASCE12, *Streptococcus* sp. oral clone ASCF05, *Streptococcus* sp. oral clone ASCF07, *Streptococcus* sp. oral clone ASCF09, *Streptococcus* sp. oral clone ASCG04, *Streptococcus* sp. oral clone RW009, *Streptococcus* sp. oral clone CH016, *Streptococcus* sp. oral clone GK051, *Streptococcus* sp. oral clone GM006, *Streptococcus* sp. oral clone P2PA_41 P2, *Streptococcus* sp. oral clone P4PA_30 P4, *Streptococcus* sp. oral taxon 071, *Streptococcus* sp. oral taxon G59, *Streptococcus* sp. oral taxon G62, *Streptococcus* sp. oral taxon G63, *Streptococcus suis*, *Streptococcus thermophilus*, *Streptococcus uberis*, *Streptococcus urinalis*, *Streptococcus vestibularis*, *Streptococcus viridans*, Synergistetes bacterium oral clone 03 5 D05
clade_98i	*Okadaella gastrococcus*, *Streptococcus agalactiae*, *Streptococcus alactolyticus*, *Streptococcus australis*, *Streptococcus bovis*, *Streptococcus canis*, *Streptococcus constellatus*, *Streptococcus cristatus*, *Streptococcus dysgalactiae*, *Streptococcus equi*, *Streptococcus equinus*, *Streptococcus gallolyticus*, *Streptococcus* genomosp. C1, *Streptococcus* genomosp. C2, *Streptococcus* genomosp. C3, *Streptococcus* genomosp. C4, *Streptococcus* genomosp. C5, *Streptococcus* genomosp. C6, *Streptococcus* genomosp. C7, *Streptococcus* genomosp. C8, *Streptococcus gordonii*, *Streptococcus infantarius*, *Streptococcus infantis*, *Streptococcus intermedius*, *Streptococcus lutetiensis*, *Streptococcus massiliensis*, *Streptococcus oligofermentans*, *Streptococcus oralis*, *Streptococcus parasanguinis*, *Streptococcus pasteurianus*, *Streptococcus peroris*, *Streptococcus pneumoniae*, *Streptococcus porcinus*, *Streptococcus pseudopneumoniae*, *Streptococcus pseudoporcinus*, *Streptococcus pyogenes*, *Streptococcus ratti*, *Streptococcus salivarius*, *Streptococcus sanguinis*, *Streptococcus sinensis*, *Streptococcus* sp. 2285_97, *Streptococcus* sp. 2_1_36FAA, *Streptococcus* sp. ACS2, *Streptococcus* sp. AS20, *Streptococcus* sp. BS35a, *Streptococcus* sp. C150, *Streptococcus* sp. CM6, *Streptococcus* sp. ICM10, *Streptococcus* sp. ICM12, *Streptococcus* sp. ICM2, *Streptococcus* sp. ICM4, *Streptococcus* sp. ICM45, *Streptococcus* sp. M143, *Streptococcus* sp. M334, *Streptococcus* sp. oral clone ASB02, *Streptococcus* sp. oral clone ASCA03, *Streptococcus* sp. oral clone ASCA04, *Streptococcus* sp. oral clone ASCA09, *Streptococcus* sp. oral clone ASCB04, *Streptococcus* sp. oral clone ASCB06, *Streptococcus* sp. oral clone ASCC04, *Streptococcus* sp. oral clone ASCC05, *Streptococcus* sp. oral clone ASCC12, *Streptococcus* sp. oral clone ASCD01, *Streptococcus* sp. oral clone ASCD09, *Streptococcus* sp. oral clone ASCD10, *Streptococcus* sp. oral clone ASCE03, *Streptococcus* sp. oral clone ASCE04, *Streptococcus* sp. oral clone ASCE05, *Streptococcus* sp. oral clone ASCE06, *Streptococcus* sp. oral clone ASCE09, *Streptococcus* sp. oral clone ASCE10, *Streptococcus* sp. oral clone ASCE12, *Streptococcus* sp. oral clone ASCF05, *Streptococcus* sp. oral clone ASCF07, *Streptococcus* sp. oral clone ASCF09, *Streptococcus* sp. oral clone ASCG04, *Streptococcus* sp. oral clone BW009, *Streptococcus* sp. oral clone CH016, *Streptococcus* sp. oral clone GK051, *Streptococcus* sp. oral clone GM006, *Streptococcus* sp. oral clone P2PA_41 P2, *Streptococcus* sp. oral clone P4PA_30 P4, *Streptococcus* sp. oral taxon 071, *Streptococcus* sp. oral taxon G59, *Streptococcus* sp. oral taxon G62, *Streptococcus* sp. oral taxon G63, *Streptococcus suis*, *Streptococcus thermophilus*, *Streptococcus uberis*, *Streptococcus urinalis*, *Streptococcus vestibularis*, *Streptococcus viridans*, Synergistetes bacterium oral clone 03 5 D05

TABLE 3

Disease Indication
Abdominal cavity inflammation
Absidia infection
Acinetobacter baumanii infection
Acinetobacter infection
Acinetobacter lwoffii infection
Acne
Actinomyces israelii infection
Adenovirus infection
Adult varicella zoster virus infection
Aging
Alcoholism (and effects)
Allergic conjunctivitis
Allergic rhinitis

TABLE 3-continued

Disease Indication
Allergy
ALS
Alzheimers disease
Amoeba infection
Anal cancer
Antibiotic treatment
Antitbiotic associated diarrhea
Arteriosclerosis
Arthritis
Aspergillus fumigatus infection
Aspergillus infection
Asthma
Atherosclerosis

TABLE 3-continued

Disease Indication
Atopic dermatitis
Atopy/Allergic Sensitivity
Autism
Autoimmune disease
Bacillus anthracis infection
Bacillus infection
Bacterial endocarditis
Bacterial eye infection
Bacterial infection
Bacterial meningitis
Bacterial pneumonia
Bacterial respiratory tract infection
Bacterial skin infection
Bacterial susceptibility
Bacterial urinary tract infection
Bacterial vaginosis
Bacteroides caccae infection
Bacteroides fragilis infection
Bacteroides infection
Bacteroides thetaiotaomicron infection
Bacteroides uniformis infection
Bacteroides vulgatus infection
Bartonella bacilliformis infection
Bartonella infection
Bifidobacterium infection
Biliary cancer
Biliary cirrhosis
Biliary tract disease
Biliary tract infection
Biliary tumor
BK virus infection
Blastomyces infection
Bone and joint infection
Bone infection
Bordetella pertussis infection
Borrelia burgdorferi infection
Borrelia recurrentis infection
Brucella infection
Burkholderia infection
Cachexia
Campylobacter fetus infection
Campylobacter infection
Campylobacter jejuni infection
Cancer
Candida albicans infection
Candida infection
Candida krusei infection
Celiac Disease
Cervix infection
Chemotherapy-induced diarrhea
Chlamydia infection
Chlamydia pneumoniae infection
Chlamydia trachomatis infection
Chlamydiae infection
Chronic fatigue syndrome
Chronic infection
Chronic inflammatory demyelinating polyneuropathy
Chronic Polio Shedders
Circadian rhythm sleep disorder
Cirrhosis
Citrobacter infection
Cladophialophora infection
Clostridiaceae infection
Clostridium botulinum infection
Clostridium difficile infection
Clostridium infection
Clostridium tetani infection
Coccidioides infection
Colitis
Colon cancer
Colorectal cancer
Common cold
Compensated liver cirrhosis
Complicated skin and skin structure infection
Complicated urinary tract infection
Constipation
Constipation predominant irritable bowel syndrome
Corynebacterium diphtheriae infection

TABLE 3-continued

Disease Indication
Corynebacterium infection
Coxiella infection
carbapenem-resistant Enterobacteriaceae (CRE) infection
Crohns disease
Cryptococcus infection
Cryptococcus neoformans infection
Cryptosporidium infection
Cutaneous lupus erythematosus
Cystic fibrosis
Cystitis
Cytomegalovirus infection
Dementia
Dengue virus infection
Depression
Dermatitis
Diabetes mellitus
Diabetic complication
Diabetic foot ulcer
Diarrhea
Diarrhea predominant irritable bowel syndrome
Discoid lupus erythematosus
Diverticulitis
DNA virus infection
Duodenal ulcer
Ebola virus infection
Entamoeba histolytica infection
Enterobacter aerogenes infection
Enterobacter cloacae infection
Enterobacter infection
Enterobacteriaceae infection
Enterococcus faecalis infection
Enterococcus faecium infection
Enterococcus infection
Enterocolitis
Enterovirus 71 infection
Epidermophyton infection
Epstein Barr virus infection
ESBL (Extended Spectrum Beta Lactamase) Producing Bacterial Infection
Escherichia coli infection
Esophageal cancer
Exophiala infection
Familial cold autoinflammatory syndrome
Fasciola hepatica infection
Female genital tract infection
Female genital tract tumor
Female infertility
Fibrosis
Flavivirus infection
Food Allergy
Francisella tularensis infection
Functional bowel disorder
Fungal infection
Fungal respiratory tract infection
Fungal urinary tract infection
Fusarium infection
Fusobacterium infection
Gastric ulcers
Gastrointestinal infection
Gastrointestinal pain
Gastrointestinal ulcer
Genital tract infection
Genitourinary disease
Genitourinary tract rumor
Gestational diabetes
Giardia lamblia infection
Gingivitis
Gram negative bacterium infection
Gram positive bacterium infection
Haemophilus aegyptus infection
Haemophilus ducreyi infection
Haemophilus infection
Haemophilus influenzae infection
Haemophilus parainfluenzae infection
Hantavirus infection
Helicobacter pylori infection
Helminth infection
Hepatitis A virus infection

TABLE 3-continued

Disease Indication
Hepatitis B virus infection
Hepatitis C virus infection
Hepatitis D virus infection
Hepatitis E virus infection
Hepatitis virus infection
Herpes simplex virus infection
Herpesvirus infection
Histoplasma infection
HIV infection
HIV-1 infection
HIV-2 infection
HSV-1 infection
HSV-2 infection
Human T cell leukemia virus 1 infection
Hypercholesterolemia
Hyperoxaluria
Hypertension
Infectious arthritis
Infectious disease
Infectious endocarditis
Infertility
Inflammatory bowel disease
Inflammatory disease
Influenza virus A infection
Influenza virus B infection
Influenza virus infection
Insomnia
Insulin dependent diabetes
Intestine infection
Irritable bowel syndrome
Japanese encephalitis virus infection
Joint infection
Juvenile rheumatoid arthritis
Klebsiella granulomatis infection
Klebsiella infection
Klebsiella pneumoniae infection
Legionella infection
Legionella pneumophila infection
Leishmania braziliensis infection
Leishmania donovani infection
Leishmania infection
Leishmania tropica infection
Leptospiraceae infection
Listeria monocytogenes infection
Listerosis
Liver cirrhosis
Liver fibrosis
Lower respiratory tract infection
Lung infection
Lung inflammation
Lupus erythematosus panniculitis
Lupus nephritis
Lyme disease
Male infertility
Marburg virus infection
Measles virus infection
Metabolic disorder
Metabolic Syndrome
Metastatic colon cancer
Metastatic colorectal cancer
Metastatic esophageal cancer
Metastatic gastrointestinal cancer
Metastatic stomach cancer
Micrococcaceae infection
Micrococcus infection
Microsporidial infection
Microsporum infection
Molluscum contagiosum infection
Monkeypox virus infection
Moraxella catarrhalis infection
Moraxella infection
Morganella infection
Morganella morganii infection
MRSA infection
Mucor infection
Multidrug resistant infection
Multiple sclerosis
Mumps virus infection

TABLE 3-continued

Disease Indication
Musculoskeletal system inflammation
Mycobacterium infection
Mycobacterium leprae infection
Mycobacterium tuberculosis infection
Mycoplasma infection
Mycoplasma pneumoniae infection
Necrotizing enterocolitis
Necrotizing Pancreatitis
Neisseria gonorrhoeae infection
Neisseria infection
Neisseria meningitidis infection
Nematode infection
Non alcoholic fatty liver disease
Non-alcoholic steatohepatitis
Non-insulin dependent diabetes
Obesity
Ocular infection
Ocular inflammation
Orbital inflammatory disease
Osteoarthritis
Otorhinolaryngological infection
Pain
Papillomavirus infection
Parasitic infection
Parkinsons disease
Pediatric varicella zoster virus infection
Pelvic inflammatory disease
Peptostreptococcus infection
Perennial allergic rhinitis
Periarthritis
Pink eye infection
Plasmodium falciparum infection
Plasmodium infection
Plasmodium malariae infection
Plasmodium vivax infection
Pneumocystis carinii infection
Poliovirus infection
Polyomavirus infection
Post-surgical bacterial leakage
Pouchitis
Prevotella infection
Primary biliary cirrhosis
Primary sclerosing cholangitis
Propionibacterium acnes infection
Propionibacterium infection
Prostate cancer
Proteus infection
Proteus mirabilis infection
Protozoal infection
Providencia infection
Pseudomonas aeruginosa infection
Pseudomonas infection
Psoriasis
Psoriatic arthritis
Pulmonary fibrosis
Rabies virus infection
Rectal cancer
Respiratory syncytial virus infection
Respiratory tract infection
Respiratory tract inflammation
Rheumatoid arthritis
Rhinitis
Rhizomucor infection
Rhizopus infection
Rickettsia infection
Ross River virus infection
Rotavirus infection
Rubella virus infection
Salmonella infection
Salmonella typhi infection
Sarcopenia
SARS coronavirus infection
Scabies infection
Scedosporium infection
Scleroderma
Seasonal allergic rhinitis
Serratia infection
Serratia marcescens infection

TABLE 3-continued

Disease Indication
Shigella boydii infection
Shigella dysenteriae infection
Shigella flexneri infection
Shigella infection
Shigella sonnei infection
Short bowel syndrome
Skin allergy
Skin cancer
Skin infection
Skin Inflammatory disease
Sleep disorder
Spondylarthritis
Staphylococcus aureus infection
Staphylococcus epidermidis infection
Staphylococcus infection
Staphylococcus saprophyticus infection
Stenotrophomonas maltophilia infection
Stomach cancer
Stomach infection
Stomach ulcer
Streptococcus agalactiae infection
Streptococcus constellatus infection
Streptococcus infection
Streptococcus intermedius infection
Streptococcus mitis infection
Streptococcus oralis infection
Streptococcus pneumoniae infection
Streptococcus pyogenes infection
Systemic lupus erythematosus
Traveler's diarrhea
Trench mouth
Treponema infection
Treponema pallidum infection
Trichomonas infection
Trichophyton infection
Trypanosoma brucei infection
Trypanosoma cruzi infection
Type 1 Diabetes
Type 2 Diabetes
Ulcerative colitis
Upper respiratory tract infection
Ureaplasma urealyticum infection
Urinary tract disease
Urinary tract infection
Urinary tract tumor
Urogenital tract infection
Uterus infection
Vaccinia virus infection
Vaginal infection
Varicella zoster virus infection
Variola virus infection
Vibrio cholerae infection
Viral eye infection
Viral infection
Viral respiratory tract infection
Viridans group *Streptococcus* infection
Vancomycin-Resistant *Enterococcus* infection
Wasting Syndrome
Weight loss
West Nile virus infection
Whipple's disease
Xenobiotic metabolism
Yellow fever virus infection
Yersinia pestis infection
Flatulence
Gastrointestinal Disorder
General Inflammation

TABLE 4a

OTU1 of Composition	OTU2 of Composition	OTU3 of Composition	Strain ID OTU1
Escherichia_coli	*Escherichia_coli*		SPC00001
Escherichia_coli	*Bacteroides_vulgatus*		SPC00001
Escherichia_coli	*Bacteroides*_sp_1_1_6		SPC00001
Escherichia_coli	*Bacteroides*_sp_3_1_23		SPC00001
Escherichia_coli	*Enterococcus_faecalis*		SPC00001
Escherichia_coli	*Coprobacillus*_sp_D7		SPC00001
Escherichia_coli	*Streptococcus_thermophilus*		SPC00001
Escherichia_coli	*Dorea_formicigenerans*		SPC00001
Escherichia_coli	*Blautia_producta*		SPC00001
Escherichia_coli	*Eubacterium_eligens*		SPC00001
Escherichia_coli	*Clostridium_nexile*		SPC00001
Escherichia_coli	*Clostridium*_sp_HGF2		SPC00001
Escherichia_coli	*Faecalibacterium_prausnitzii*		SPC00001
Escherichia_coli	*Odoribacter_splanchnicus*		SPC00001
Escherichia_coli	*Dorea_longicatena*		SPC00001
Escherichia_coli	*Roseburia_intestinalis*		SPC00001
Escherichia_coli	*Coprococcus_catus*		SPC00001
Escherichia_coli	Erysipelotrichaceae_*bacterium*_3_1_53		SPC00001
Escherichia_coli	*Bacteroides*_sp_D20		SPC00001
Escherichia_coli	*Bacteroides_ovatus*		SPC00001
Escherichia_coli	*Parabacteroides_merdae*		SPC00001
Escherichia_coli	*Bacteroides_vulgatus*		SPC00001
Escherichia_coli	*Collinsella_aerofaciens*		SPC00001
Escherichia_coli	*Escherichia_coli*		SPC00001
Escherichia_coli	*Ruminococcus_obeum*		SPC00001
Escherichia_coli	*Bacteroides_caccae*		SPC00001
Escherichia_coli	*Bacteroides_eggerthii*		SPC00001
Escherichia_coli	*Ruminococcus_torques*		SPC00001
Escherichia_coli	*Clostridium_hathewayi*		SPC00001
Escherichia_coli	*Bifidobacterium_pseudocatenulatum*		SPC00001
Escherichia_coli	*Bifidobacterium_adolescentis*		SPC00001
Escherichia_coli	*Coprococcus_comes*		SPC00001
Escherichia_coli	*Clostridium_symbiosum*		SPC00001
Escherichia_coli	*Eubacterium_rectale*		SPC00001
Escherichia_coli	*Faecalibacterium_prausnitzii*		SPC00001
Escherichia_coli	*Odoribacter_splanchnicus*		SPC00001

TABLE 4a-continued

Escherichia_coli	Lachnospiraceae_bacterium_5_1_57FAA	SPC00001
Escherichia_coli	Blautia_schinkii	SPC00001
Escherichia_coli	Alistipes_shahii	SPC00001
Escherichia_coli	Blautia_producta	SPC00001
Bacteroides_vulgatus	Bacteroides_vulgatus	SPC00005
Bacteroides_vulgatus	Bacteroides_sp_1_1_6	SPC00005
Bacteroides_vulgatus	Bacteroides_sp_3_1_23	SPC00005
Bacteroides_vulgatus	Enterococcus_faecalis	SPC00005
Bacteroides_vulgatus	Coprobacillus_sp_D7	SPC00005
Bacteroides_vulgatus	Streptococcus_thermophilus	SPC00005
Bacteroides_vulgatus	Dorea_formicigenerans	SPC00005
Bacteroides_vulgatus	Blautia_producta	SPC00005
Bacteroides_vulgatus	Eubacterium_eligens	SPC00005
Bacteroides_vulgatus	Clostridium_nexile	SPC00005
Bacteroides_vulgatus	Clostridium_sp_HGF2	SPC00005
Bacteroides_vulgatus	Faecalibacterium_prausnitzii	SPC00005
Bacteroides_vulgatus	Odoribacter_splanchnicus	SPC00005
Bacteroides_vulgatus	Dorea_longicatena	SPC00005
Bacteroides_vulgatus	Roseburia_intestinalis	SPC00005
Bacteroides_vulgatus	Coprococcus_catus	SPC00005
Bacteroides_vulgatus	Erysipelotrichaceae_bacterium_3_1_53	SPC00005
Bacteroides_vulgatus	Bacteroides_sp_D20	SPC00005
Bacteroides_vulgatus	Bacteroides_ovatus	SPC00005
Bacteroides_vulgatus	Parabacteroides_merdae	SPC00005
Bacteroides_vulgatus	Bacteroides_vulgatus	SPC00005
Bacteroides_vulgatus	Collinsella_aerofaciens	SPC00005
Bacteroides_vulgatus	Escherichia_coli	SPC00005
Bacteroides_vulgatus	Ruminococcus_obeum	SPC00005
Bacteroides_vulgatus	Bacteroides_caccae	SPC00005
Bacteroides_vulgatus	Bacteroides_eggerthii	SPC00005
Bacteroides_vulgatus	Ruminococcus_torques	SPC00005
Bacteroides_vulgatus	Clostridium_hathewayi	SPC00005
Bacteroides_vulgatus	Bifidobacterium_pseudocatenulatum	SPC00005
Bacteroides_vulgatus	Bifidobacterium_adolescentis	SPC00005
Bacteroides_vulgatus	Coprococcus_comes	SPC00005
Bacteroides_vulgatus	Clostridium_symbiosum	SPC00005
Bacteroides_vulgatus	Eubacterium_rectale	SPC00005
Bacteroides_vulgatus	Faecalibacterium_prausnitzii	SPC00005
Bacteroides_vulgatus	Odoribacter_splanchnicus	SPC00005
Bacteroides_vulgatus	Lachnospiraceae_bacterium_5_1_57FAA	SPC00005
Bacteroides_vulgatus	Blautia_schinkii	SPC00005
Bacteroides_vulgatus	Alistipes_shahii	SPC00005
Bacteroides_vulgatus	Blautia_producta	SPC00005
Bacteroides_sp_1_1_6	Bacteroides_sp_1_1_6	SPC00006
Bacteroides_sp_1_1_6	Bacteroides_sp_3_1_23	SPC00006
Bacteroides_sp_1_1_6	Enterococcus_faecalis	SPC00006
Bacteroides_sp_1_1_6	Coprobacillus_sp_D7	SPC00006
Bacteroides_sp_1_1_6	Streptococcus_thermophilus	SPC00006
Bacteroides_sp_1_1_6	Dorea_formicigenerans	SPC00006
Bacteroides_sp_1_1_6	Blautia_producta	SPC00006
Bacteroides_sp_1_1_6	Eubacterium_eligens	SPC00006
Bacteroides_sp_1_1_6	Clostridium_nexile	SPC00006
Bacteroides_sp_1_1_6	Clostridium_sp_HGF2	SPC00006
Bacteroides_sp_1_1_6	Faecalibacterium_prausnitzii	SPC00006
Bacteroides_sp_1_1_6	Odoribacter_splanchnicus	SPC00006
Bacteroides_sp_1_1_6	Dorea_longicatena	SPC00006
Bacteroides_sp_1_1_6	Roseburia_intestinalis	SPC00006
Bacteroides_sp_1_1_6	Coprococcus_catus	SPC00006
Bacteroides_sp_1_1_6	Erysipelotrichaceae_bacterium_3_1_53	SPC00006
Bacteroides_sp_1_1_6	Bacteroides_sp_D20	SPC00006
Bacteroides_sp_1_1_6	Bacteroides_ovatus	SPC00006
Bacteroides_sp_1_1_6	Parabacteroides_merdae	SPC00006
Bacteroides_sp_1_1_6	Bacteroides_vulgatus	SPC00006
Bacteroides_sp_1_1_6	Collinsella_aerofaciens	SPC00006
Bacteroides_sp_1_1_6	Escherichia_coli	SPC00006
Bacteroides_sp_1_1_6	Ruminococcus_obeum	SPC00006
Bacteroides_sp_1_1_6	Bacteroides_caccae	SPC00006
Bacteroides_sp_1_1_6	Bacteroides_eggerthii	SPC00006
Bacteroides_sp_1_1_6	Ruminococcus_torques	SPC00006
Bacteroides_sp_1_1_6	Clostridium_hathewayi	SPC00006
Bacteroides_sp_1_1_6	Bifidobacterium_pseudocatenulatum	SPC00006
Bacteroides_sp_1_1_6	Bifidobacterium_adolescentis	SPC00006
Bacteroides_sp_1_1_6	Coprococcus_comes	SPC00006
Bacteroides_sp_1_1_6	Clostridium_symbiosum	SPC00006
Bacteroides_sp_1_1_6	Eubacterium_rectale	SPC00006
Bacteroides_sp_1_1_6	Faecalibacterium_prausnitzii	SPC00006
Bacteroides_sp_1_1_6	Odoribacter_splanchnicus	SPC00006
Bacteroides_sp_1_1_6	Lachnospiraceae_bacterium_5_1_57FAA	SPC00006
Bacteroides_sp_1_1_6	Blautia_schinkii	SPC00006
Bacteroides_sp_1_1_6	Alistipes_shahii	SPC00006

TABLE 4a-continued

Bacteroides_sp_1_1_6	Blautia_producta	SPC00006
Bacteroides_sp_3_1_23	Bacteroides_sp_3_1_23	SPC00007
Bacteroides_sp_3_1_23	Enterococcus_faecalis	SPC00007
Bacteroides_sp_3_1_23	Coprobacillus_sp_D7	SPC00007
Bacteroides_sp_3_1_23	Streptococcus_thermophilus	SPC00007
Bacteroides_sp_3_1_23	Dorea_formicigenerans	SPC00007
Bacteroides_sp_3_1_23	Blautia_producta	SPC00007
Bacteroides_sp_3_1_23	Eubacterium_eligens	SPC00007
Bacteroides_sp_3_1_23	Clostridium_nexile	SPC00007
Bacteroides_sp_3_1_23	Clostridium_sp_HGF2	SPC00007
Bacteroides_sp_3_1_23	Faecalibacterium_prausnitzii	SPC00007
Bacteroides_sp_3_1_23	Odoribacter_splanchnicus	SPC00007
Bacteroides_sp_3_1_23	Dorea_longicatena	SPC00007
Bacteroides_sp_3_1_23	Roseburia_intestinalis	SPC00007
Bacteroides_sp_3_1_23	Coprococcus_catus	SPC00007
Bacteroides_sp_3_1_23	Erysipelotrichaceae_bacterium_3_1_53	SPC00007
Bacteroides_sp_3_1_23	Bacteroides_sp_D20	SPC00007
Bacteroides_sp_3_1_23	Bacteroides_ovatus	SPC00007
Bacteroides_sp_3_1_23	Parabacteroides_merdae	SPC00007
Bacteroides_sp_3_1_23	Bacteroides_vulgatus	SPC00007
Bacteroides_sp_3_1_23	Collinsella_aerofaciens	SPC00007
Bacteroides_sp_3_1_23	Escherichia_coli	SPC00007
Bacteroides_sp_3_1_23	Ruminococcus_obeum	SPC00007
Bacteroides_sp_3_1_23	Bacteroides_caccae	SPC00007
Bacteroides_sp_3_1_23	Bacteroides_eggerthii	SPC00007
Bacteroides_sp_3_1_23	Ruminococcus_torques	SPC00007
Bacteroides_sp_3_1_23	Clostridium_hathewayi	SPC00007
Bacteroides_sp_3_1_23	Bifidobacterium_pseudocatenulatum	SPC00007
Bacteroides_sp_3_1_23	Bifidobacterium_adolescentis	SPC00007
Bacteroides_sp_3_1_23	Coprococcus_comes	SPC00007
Bacteroides_sp_3_1_23	Clostridium_symbiosum	SPC00007
Bacteroides_sp_3_1_23	Eubacterium_rectale	SPC00007
Bacteroides_sp_3_1_23	Faecalibacterium_prausnitzii	SPC00007
Bacteroides_sp_3_1_23	Odoribacter_splanchnicus	SPC00007
Bacteroides_sp_3_1_23	Lachnospiraceae_bacterium_5_1_57FAA	SPC00007
Bacteroides_sp_3_1_23	Blautia_schinkii	SPC00007
Bacteroides_sp_3_1_23	Alistipes_shahii	SPC00007
Bacteroides_sp_3_1_23	Blautia_producta	SPC00007
Enterococcus_faecalis	Enterococcus_faecalis	SPC00008
Enterococcus_faecalis	Coprobacillus_sp_D7	SPC00008
Enterococcus_faecalis	Streptococcus_thermophilus	SPC00008
Enterococcus_faecalis	Dorea_formicigenerans	SPC00008
Enterococcus_faecalis	Blautia_producta	SPC00008
Enterococcus_faecalis	Eubacterium_eligens	SPC00008
Enterococcus_faecalis	Clostridium_nexile	SPC00008
Enterococcus_faecalis	Clostridium_sp_HGF2	SPC00008
Enterococcus_faecalis	Faecalibacterium_prausnitzii	SPC00008
Enterococcus_faecalis	Odoribacter_splanchnicus	SPC00008
Enterococcus_faecalis	Dorea_longicatena	SPC00008
Enterococcus_faecalis	Roseburia_intestinalis	SPC00008
Enterococcus_faecalis	Coprococcus_catus	SPC00008
Enterococcus_faecalis	Erysipelotrichaceae_bacterium_3_1_53	SPC00008
Enterococcus_faecalis	Bacteroides_sp_D20	SPC00008
Enterococcus_faecalis	Bacteroides_ovatus	SPC00008
Enterococcus_faecalis	Parabacteroides_merdae	SPC00008
Enterococcus_faecalis	Bacteroides_vulgatus	SPC00008
Enterococcus_faecalis	Collinsella_aerofaciens	SPC00008
Enterococcus_faecalis	Escherichia_coli	SPC00008
Enterococcus_faecalis	Ruminococcus_obeum	SPC00008
Enterococcus_faecalis	Bacteroides_caccae	SPC00008
Enterococcus_faecalis	Bacteroides_eggerthii	SPC00008
Enterococcus_faecalis	Ruminococcus_torques	SPC00008
Enterococcus_faecalis	Clostridium_hathewayi	SPC00008
Enterococcus_faecalis	Bifidobacterium_pseudocatenulatum	SPC00008
Enterococcus_faecalis	Bifidobacterium_adolescentis	SPC00008
Enterococcus_faecalis	Coprococcus_comes	SPC00008
Enterococcus_faecalis	Clostridium_symbiosum	SPC00008
Enterococcus_faecalis	Eubacterium_rectale	SPC00008
Enterococcus_faecalis	Faecalibacterium_prausnitzii	SPC00008
Enterococcus_faecalis	Odoribacter_splanchnicus	SPC00008
Enterococcus_faecalis	Lachnospiraceae_bacterium_5_1_57FAA	SPC00008
Enterococcus_faecalis	Blautia_schinkii	SPC00008
Enterococcus_faecalis	Alistipes_shahii	SPC00008
Enterococcus_faecalis	Blautia_producta	SPC00008
Coprobacillus_sp_D7	Coprobacillus_sp_D7	SPC00009
Coprobacillus_sp_D7	Streptococcus_thermophilus	SPC00009
Coprobacillus_sp_D7	Dorea_formicigenerans	SPC00009
Coprobacillus_sp_D7	Blautia_producta	SPC00009
Coprobacillus_sp_D7	Eubacterium_eligens	SPC00009
Coprobacillus_sp_D7	Clostridium_nexile	SPC00009

TABLE 4a-continued

Coprobacillus_sp_D7	Clostridium_sp_HGF2	SPC00009
Coprobacillus_sp_D7	Faecalibacterium_prausnitzii	SPC00009
Coprobacillus_sp_D7	Odoribacter_splanchnicus	SPC00009
Coprobacillus_sp_D7	Dorea_longicatena	SPC00009
Coprobacillus_sp_D7	Roseburia_intestinalis	SPC00009
Coprobacillus_sp_D7	Coprococcus_catus	SPC00009
Coprobacillus_sp_D7	Erysipelotrichaceae_bacterium_3_1_53	SPC00009
Coprobacillus_sp_D7	Bacteroides_sp_D20	SPC00009
Coprobacillus_sp_D7	Bacteroides_ovatus	SPC00009
Coprobacillus_sp_D7	Parabacteroides_merdae	SPC00009
Coprobacillus_sp_D7	Bacteroides_vulgatus	SPC00009
Coprobacillus_sp_D7	Collinsella_aerofaciens	SPC00009
Coprobacillus_sp_D7	Escherichia_coli	SPC00009
Coprobacillus_sp_D7	Ruminococcus_obeum	SPC00009
Coprobacillus_sp_D7	Bacteroides_caccae	SPC00009
Coprobacillus_sp_D7	Bacteroides_eggerthii	SPC00009
Coprobacillus_sp_D7	Ruminococcus_torques	SPC00009
Coprobacillus_sp_D7	Clostridium_hathewayi	SPC00009
Coprobacillus_sp_D7	Bifidobacterium_pseudocatenulatum	SPC00009
Coprobacillus_sp_D7	Bifidobacterium_adolescentis	SPC00009
Coprobacillus_sp_D7	Coprococcus_comes	SPC00009
Coprobacillus_sp_D7	Clostridium_symbiosum	SPC00009
Coprobacillus_sp_D7	Eubacterium_rectale	SPC00009
Coprobacillus_sp_D7	Faecalibacterium_prausnitzii	SPC00009
Coprobacillus_sp_D7	Odoribacter_splanchnicus	SPC00009
Coprobacillus_sp_D7	Lachnospiraceae_bacterium_5_1_57FAA	SPC00009
Coprobacillus_sp_D7	Blautia_schinkii	SPC00009
Coprobacillus_sp_D7	Alistipes_shahii	SPC00009
Coprobacillus_sp_D7	Blautia_producta	SPC00009
Streptococcus_thermophilus	Streptococcus_thermophilus	SPC00015
Streptococcus_thermophilus	Dorea_formicigenerans	SPC00015
Streptococcus_thermophilus	Blautia_producta	SPC00015
Streptococcus_thermophilus	Eubacterium_eligens	SPC00015
Streptococcus_thermophilus	Clostridium_nexile	SPC00015
Streptococcus_thermophilus	Clostridium_sp_HGF2	SPC00015
Streptococcus_thermophilus	Faecalibacterium_prausnitzii	SPC00015
Streptococcus_thermophilus	Odoribacter_splanchnicus	SPC00015
Streptococcus_thermophilus	Dorea_longicatena	SPC00015
Streptococcus_thermophilus	Roseburia_intestinalis	SPC00015
Streptococcus_thermophilus	Coprococcus_catus	SPC00015
Streptococcus_thermophilus	Erysipelotrichaceae_bacterium_3_1_53	SPC00015
Streptococcus_thermophilus	Bacteroides_sp_D20	SPC00015
Streptococcus_thermophilus	Bacteroides_ovatus	SPC00015
Streptococcus_thermophilus	Parabacteroides_merdae	SPC00015
Streptococcus_thermophilus	Bacteroides_vulgatus	SPC00015
Streptococcus_thermophilus	Collinsella_aerofaciens	SPC00015
Streptococcus_thermophilus	Escherichia_coli	SPC00015
Streptococcus_thermophilus	Ruminococcus_obeum	SPC00015
Streptococcus_thermophilus	Bacteroides_caccae	SPC00015
Streptococcus_thermophilus	Bacteroides_eggerthii	SPC00015
Streptococcus_thermophilus	Ruminococcus_torques	SPC00015
Streptococcus_thermophilus	Clostridium_hathewayi	SPC00015
Streptococcus_thermophilus	Bifidobacterium_pseudocatenulatum	SPC00015
Streptococcus_thermophilus	Bifidobacterium_adolescentis	SPC00015
Streptococcus_thermophilus	Coprococcus_comes	SPC00015
Streptococcus_thermophilus	Clostridium_symbiosum	SPC00015
Streptococcus_thermophilus	Eubacterium_rectale	SPC00015
Streptococcus_thermophilus	Faecalibacterium_prausnitzii	SPC00015
Streptococcus_thermophilus	Odoribacter_splanchnicus	SPC00015
Streptococcus_thermophilus	Lachnospiraceae_bacterium_5_1_57FAA	SPC00015
Streptococcus_thermophilus	Blautia_schinkii	SPC00015
Streptococcus_thermophilus	Alistipes_shahii	SPC00015
Streptococcus_thermophilus	Blautia_producta	SPC00015
Dorea_formicigenerans	Dorea_formicigenerans	SPC00018
Dorea_formicigenerans	Blautia_producta	SPC00018
Dorea_formicigenerans	Eubacterium_eligens	SPC00018
Dorea_formicigenerans	Clostridium_nexile	SPC00018
Dorea_formicigenerans	Clostridium_sp_HGF2	SPC00018
Dorea_formicigenerans	Faecalibacterium_prausnitzii	SPC00018
Dorea_formicigenerans	Odoribacter_splanchnicus	SPC00018
Dorea_formicigenerans	Dorea_longicatena	SPC00018
Dorea_formicigenerans	Roseburia_intestinalis	SPC00018
Dorea_formicigenerans	Coprococcus_catus	SPC00018
Dorea_formicigenerans	Erysipelotrichaceae_bacterium_3_1_53	SPC00018
Dorea_formicigenerans	Bacteroides_sp_D20	SPC00018
Dorea_formicigenerans	Bacteroides_ovatus	SPC00018
Dorea_formicigenerans	Parabacteroides_merdae	SPC00018
Dorea_formicigenerans	Bacteroides_vulgatus	SPC00018
Dorea_formicigenerans	Collinsella_aerofaciens	SPC00018
Dorea_formicigenerans	Escherichia_coli	SPC00018

TABLE 4a-continued

Dorea_formicigenerans	*Ruminococcus_obeum*	SPC00018
Dorea_formicigenerans	*Bacteroides_caccae*	SPC00018
Dorea_formicigenerans	*Bacteroides_eggerthii*	SPC00018
Dorea_formicigenerans	*Ruminococcus_torques*	SPC00018
Dorea_formicigenerans	*Clostridium_hathewayi*	SPC00018
Dorea_formicigenerans	*Bifidobacterium_pseudocatenulatum*	SPC00018
Dorea_formicigenerans	*Bifidobacterium_adolescentis*	SPC00018
Dorea_formicigenerans	*Coprococcus_comes*	SPC00018
Dorea_formicigenerans	*Clostridium_symbiosum*	SPC00018
Dorea_formicigenerans	*Eubacterium_rectale*	SPC00018
Dorea_formicigenerans	*Faecalibacterium_prausnitzii*	SPC00018
Dorea_formicigenerans	*Odoribacter_splanchnicus*	SPC00018
Dorea_formicigenerans	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC00018
Dorea_formicigenerans	*Blautia_schinkii*	SPC00018
Dorea_formicigenerans	*Alistipes_shahii*	SPC00018
Dorea_formicigenerans	*Blautia_producta*	SPC00018
Blautia_producta	*Blautia_producta*	SPC00021
Blautia_producta	*Eubacterium_eligens*	SPC00021
Blautia_producta	*Clostridium_nexile*	SPC00021
Blautia_producta	*Clostridium*_sp_HGF2	SPC00021
Blautia_producta	*Faecalibacterium_prausnitzii*	SPC00021
Blautia_producta	*Odoribacter_splanchnicus*	SPC00021
Blautia_producta	*Dorea_longicatena*	SPC00021
Blautia_producta	*Roseburia_intestinalis*	SPC00021
Blautia_producta	*Coprococcus_catus*	SPC00021
Blautia_producta	Erysipelotrichaceae_*bacterium*_3_1_53	SPC00021
Blautia_producta	*Bacteroides*_sp_D20	SPC00021
Blautia_producta	*Bacteroides_ovatus*	SPC00021
Blautia_producta	*Parabacteroides_merdae*	SPC00021
Blautia_producta	*Bacteroides_vulgatus*	SPC00021
Blautia_producta	*Collinsella_aerofaciens*	SPC00021
Blautia_producta	*Escherichia_coli*	SPC00021
Blautia_producta	*Ruminococcus_obeum*	SPC00021
Blautia_producta	*Bacteroides_caccae*	SPC00021
Blautia_producta	*Bacteroides_eggerthii*	SPC00021
Blautia_producta	*Ruminococcus_torques*	SPC00021
Blautia_producta	*Clostridium_hathewayi*	SPC00021
Blautia_producta	*Bifidobacterium_pseudocatenulatum*	SPC00021
Blautia_producta	*Bifidobacterium_adolescentis*	SPC00021
Blautia_producta	*Coprococcus_comes*	SPC00021
Blautia_producta	*Clostridium_symbiosum*	SPC00021
Blautia_producta	*Eubacterium_rectale*	SPC00021
Blautia_producta	*Faecalibacterium_prausnitzii*	SPC00021
Blautia_producta	*Odoribacter_splanchnicus*	SPC00021
Blautia_producta	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC00021
Blautia_producta	*Blautia_schinkii*	SPC00021
Blautia_*producta*	*Alistipes_shahii*	SPC00021
Blautia_producta	*Blautia_producta*	SPC00021
Eubacterium_eligens	*Eubacterium_eligens*	SPC00022
Eubacterium_eligens	*Clostridium_nexile*	SPC00022
Eubacterium_eligens	*Clostridium*_sp_HGF2	SPC00022
Eubacterium_eligens	*Faecalibacterium_prausnitzii*	SPC00022
Eubacterium_eligens	*Odoribacter_splanchnicus*	SPC00022
Eubacterium_eligens	*Dorea_longicatena*	SPC00022
Eubacterium_eligens	*Roseburia_intestinalis*	SPC00022
Eubacterium_eligens	*Coprococcus_catus*	SPC00022
Eubacterium_eligens	Erysipelotrichaceae_*bacterium*_3_1_53	SPC00022
Eubacterium_eligens	*Bacteroides*_sp_D20	SPC00022
Eubacterium_eligens	*Bacteroides_ovatus*	SPC00022
Eubacterium_eligens	*Parabacteroides_merdae*	SPC00022
Eubacterium_eligens	*Bacteroides_vulgatus*	SPC00022
Eubacterium_eligens	*Collinsella_aerofaciens*	SPC00022
Eubacterium_eligens	*Escherichia_coli*	SPC00022
Eubacterium_eligens	*Ruminococcus_obeum*	SPC00022
Eubacterium_eligens	*Bacteroides_caccae*	SPC00022
Eubacterium_eligens	*Bacteroides_eggerthii*	SPC00022
Eubacterium_eligens	*Ruminococcus_torques*	SPC00022
Eubacterium_eligens	*Clostridium_hathewayi*	SPC00022
Eubacterium_eligens	*Bifidobacterium_pseudocatenulatum*	SPC00022
Eubacterium_eligens	*Bifidobacterium_adolescentis*	SPC00022
Eubacterium_eligens	*Coprococcus_comes*	SPC00022
Eubacterium_eligens	*Clostridium_symbiosum*	SPC00022
Eubacterium_eligens	*Eubacterium_rectale*	SPC00022
Eubacterium_eligens	*Faecalibacterium_prausnitzii*	SPC00022
Eubacterium_eligens	*Odoribacter_splanchnicus*	SPC00022
Eubacterium_eligens	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC00022
Eubacterium_eligens	*Blautia_schinkii*	SPC00022
Eubacterium_eligens	*Alistipes_shahii*	SPC00022
Eubacterium_eligens	*Blautia_producta*	SPC00022
Clostridium_nexile	*Clostridium_nexile*	SPC00026

TABLE 4a-continued

Clostridium_nexile	*Clostridium*_sp_HGF2	SPC00026
Clostridium_nexile	*Faecalibacterium_prausnitzii*	SPC00026
Clostridium_nexile	*Odoribacter_splanchnicus*	SPC00026
Clostridium_nexile	*Dorea_longicatena*	SPC00026
Clostridium_nexile	*Roseburia_intestinalis*	SPC00026
Clostridium_nexile	*Coprococcus_catus*	SPC00026
Clostridium_nexile	Erysipelotrichaceae_*bacterium*_3_1_53	SPC00026
Clostridium_nexile	*Bacteroides*_sp_D20	SPC00026
Clostridium_nexile	*Bacteroides_ovatus*	SPC00026
Clostridium_nexile	*Parabacteroides_merdae*	SPC00026
Clostridium_nexile	*Bacteroides_vulgatus*	SPC00026
Clostridium_nexile	*Collinsella_aerofaciens*	SPC00026
Clostridium_nexile	*Escherichia_coli*	SPC00026
Clostridium_nexile	*Ruminococcus_obeum*	SPC00026
Clostridium_nexile	*Bacteroides_caccae*	SPC00026
Clostridium_nexile	*Bacteroides_eggerthii*	SPC00026
Clostridium_nexile	*Ruminococcus_torques*	SPC00026
Clostridium_nexile	*Clostridium_hathewayi*	SPC00026
Clostridium_nexile	*Bifidobacterium_pseudocatenulatum*	SPC00026
Clostridium_nexile	*Bifidobacterium_adolescentis*	SPC00026
Clostridium_nexile	*Coprococcus_comes*	SPC00026
Clostridium_nexile	*Clostridium_symbiosum*	SPC00026
Clostridium_nexile	*Eubacterium_rectale*	SPC00026
Clostridium_nexile	*Faecalibacterium_prausnitzii*	SPC00026
Clostridium_nexile	*Odoribacter_splanchnicus*	SPC00026
Clostridium_nexile	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC00026
Clostridium_nexile	*Blautia_schinkii*	SPC00026
Clostridium_nexile	*Alistipes_shahii*	SPC00026
Clostridium_nexile	*Blautia_producta*	SPC00026
*Clostridium*_sp_HGF2	*Clostridium*_sp_HGF2	SPC00027
*Clostridium*_sp_HGF2	*Faecalibacterium_prausnitzii*	SPC00027
Clostridium_sp_HGF2	*Odoribacter_splanchnicus*	SPC00027
*Clostridium*_sp_HGF2	*Dorea_longicatena*	SPC00027
*Clostridium*_sp_HGF2	*Roseburia_intestinalis*	SPC00027
*Clostridium*_sp_HGF2	*Coprococcus_catus*	SPC00027
*Clostridium*_sp_HGF2	Erysipelotrichaceae_*bacterium*_3_1_53	SPC00027
*Clostridium*_sp_HGF2	*Bacteroides*_sp_D20	SPC00027
*Clostridium*_sp_HGF2	*Bacteroides_ovatus*	SPC00027
*Clostridium*_sp_HGF2	*Parabacteroides_merdae*	SPC00027
*Clostridium*_sp_HGF2	*Bacteroides_vulgatus*	SPC00027
*Clostridium*_sp_HGF2	*Collinsella_aerofaciens*	SPC00027
*Clostridium*_sp_HGF2	*Escherichia_coli*	SPC00027
*Clostridium*_sp_HGF2	*Ruminococcus_obeum*	SPC00027
*Clostridium*_sp_HGF2	*Bacteroides_caccae*	SPC00027
*Clostridium*_sp_HGF2	*Bacteroides_eggerthii*	SPC00027
*Clostridium*_sp_HGF2	*Ruminococcus_torques*	SPC00027
*Clostridium*_sp_HGF2	*Clostridium_hathewayi*	SPC00027
*Clostridium*_sp_HGF2	*Bifidobacterium_pseudocatenulatum*	SPC00027
*Clostridium*_sp_HGF2	*Bifidobacterium_adolescentis*	SPC00027
*Clostridium*_sp_HGF2	*Coprococcus_comes*	SPC00027
*Clostridium*_sp_HGF2	*Clostridium_symbiosum*	SPC00027
*Clostridium*_sp_HGF2	*Eubacterium_rectale*	SPC00027
*Clostridium*_sp_HGF2	*Faecalibacterium_prausnitzii*	SPC00027
*Clostridium*_sp_HGF2	*Odoribacter_splanchnicus*	SPC00027
*Clostridium*_sp_HGF2	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC00027
*Clostridium*_sp_HGF2	*Blautia_schinkii*	SPC00027
*Clostridium*_sp_HGF2	*Alistipes_shahii*	SPC00027
*Clostridium*_sp_HGF2	*Blautia_producta*	SPC00027
Faecalibacterium_prausnitzii	*Faecalibacterium_prausnitzii*	SPC00054
Faecalibacterium_prausnitzii	*Odoribacter_splanchnicus*	SPC00054
Faecalibacterium_prausnitzii	*Dorea_longicatena*	SPC00054
Faecalibacterium_prausnitzii	*Roseburia_intestinalis*	SPC00054
Faecalibacterium_prausnitzii	*Coprococcus_catus*	SPC00054
Faecalibacterium_prausnitzii	Erysipelotrichaceae_*bacterium*_3_1_53	SPC00054
Faecalibacterium_prausnitzii	*Bacteroides*_sp_D20	SPC00054
Faecalibacterium_prausnitzii	*Bacteroides_ovatus*	SPC00054
Faecalibacterium_prausnitzii	*Parabacteroides_merdae*	SPC00054
Faecalibacterium_prausnitzii	*Bacteroides_vulgatus*	SPC00054
Faecalibacterium_prausnitzii	*Collinsella_aerofaciens*	SPC00054
Faecalibacterium_prausnitzii	*Escherichia_coli*	SPC00054
Faecalibacterium_prausnitzii	*Ruminococcus_obeum*	SPC00054
Faecalibacterium_prausnitzii	*Bacteroides_caccae*	SPC00054
Faecalibacterium_prausnitzii	*Bacteroides_eggerthii*	SPC00054
Faecalibacterium_prausnitzii	*Ruminococcus_torques*	SPC00054
Faecalibacterium_prausnitzii	*Clostridium_hathewayi*	SPC00054
Faecalibacterium_prausnitzii	*Bifidobacterium_pseudocatenulatum*	SPC00054
Faecalibacterium_prausnitzii	*Bifidobacterium_adolescentis*	SPC00054
Faecalibacterium_prausnitzii	*Coprococcus_comes*	SPC00054
Faecalibacterium_prausnitzii	*Clostridium_symbiosum*	SPC00054
Faecalibacterium_prausnitzii	*Eubacterium_rectale*	SPC00054

TABLE 4a-continued

Faecalibacterium_prausnitzii	Faecalibacterium_prausnitzii	SPC00054
Faecalibacterium_prausnitzii	Odoribacter_splanchnicus	SPC00054
Faecalibacterium_prausnitzii	Lachnospiraceae_bacterium_5_1_57FAA	SPC00054
Faecalibacterium_prausnitzii	Blautia_schinkii	SPC00054
Faecalibacterium_prausnitzii	Alistipes_shahii	SPC00054
Faecalibacterium_prausnitzii	Blautia_producta	SPC00054
Odoribacter_splanchnicus	Odoribacter_splanchnicus	SPC00056
Odoribacter_splanchnicus	Dorea_longicatena	SPC00056
Odoribacter_splanchnicus	Roseburia_intestinalis	SPC00056
Odoribacter_splanchnicus	Coprococcus_catus	SPC00056
Odoribacter_splanchnicus	Erysipelotrichaceae_bacterium_3_1_53	SPC00056
Odoribacter_splanchnicus	Bacteroides_sp_D20	SPC00056
Odoribacter_splanchnicus	Bacteroides_ovatus	SPC00056
Odoribacter_splanchnicus	Parabacteroides_merdae	SPC00056
Odoribacter_splanchnicus	Bacteroides_vulgatus	SPC00056
Odoribacter_splanchnicus	Collinsella_aerofaciens	SPC00056
Odoribacter_splanchnicus	Escherichia_coli	SPC00056
Odoribacter_splanchnicus	Ruminococcus_obeum	SPC00056
Odoribacter_splanchnicus	Bacteroides_caccae	SPC00056
Odoribacter_splanchnicus	Bacteroides_eggerthii	SPC00056
Odoribacter_splanchnicus	Ruminococcus_torques	SPC00056
Odoribacter_splanchnicus	Clostridium_hathewayi	SPC00056
Odoribacter_splanchnicus	Bifidobacterium_pseudocatenulatum	SPC00056
Odoribacter_splanchnicus	Bifidobacterium_adolescentis	SPC00056
Odoribacter_splanchnicus	Coprococcus_comes	SPC00056
Odoribacter_splanchnicus	Clostridium_symbiosum	SPC00056
Odoribacter_splanchnicus	Eubacterium_rectale	SPC00056
Odoribacter_splanchnicus	Faecalibacterium_prausnitzii	SPC00056
Odoribacter_splanchnicus	Odoribacter_splanchnicus	SPC00056
Odoribacter_splanchnicus	Lachnospiraceae_bacterium_5_1_57FAA	SPC00056
Odoribacter_splanchnicus	Blautia_schinkii	SPC00056
Odoribacter_splanchnicus	Alistipes_shahii	SPC00056
Odoribacter_splanchnicus	Blautia_producta	SPC00056
Dorea_longicatena	Dorea_longicatena	SPC00057
Dorea_longicatena	Roseburia_intestinalis	SPC00057
Dorea_longicatena	Coprococcus_catus	SPC00057
Dorea_longicatena	Erysipelotrichaceae_bacterium_3_1_53	SPC00057
Dorea_longicatena	Bacteroides_sp_D20	SPC00057
Dorea_longicatena	Bacteroides_ovatus	SPC00057
Dorea_longicatena	Parabacteroides_merdae	SPC00057
Dorea_longicatena	Bacteroides_vulgatus	SPC00057
Dorea_longicatena	Collinsella_aerofaciens	SPC00057
Dorea_longicatena	Escherichia_coli	SPC00057
Dorea_longicatena	Ruminococcus_obeum	SPC00057
Dorea_longicatena	Bacteroides_caccae	SPC00057
Dorea_longicatena	Bacteroides_eggerthii	SPC00057
Dorea_longicatena	Ruminococcus_torques	SPC00057
Dorea_longicatena	Clostridium_hathewayi	SPC00057
Dorea_longicatena	Bifidobacterium_pseudocatenulatum	SPC00057
Dorea_longicatena	Bifidobacterium_adolescentis	SPC00057
Dorea_longicatena	Coprococcus_comes	SPC00057
Dorea_longicatena	Clostridium_symbiosum	SPC00057
Dorea_longicatena	Eubacterium_rectale	SPC00057
Dorea_longicatena	Faecalibacterium_prausnitzii	SPC00057
Dorea_longicatena	Odoribacter_splanchnicus	SPC00057
Dorea_longicatena	Lachnospiraceae_bacterium_5_1_57FAA	SPC00057
Dorea_longicatena	Blautia_schinkii	SPC00057
Dorea_longicatena	Alistipes_shahii	SPC00057
Dorea_longicatena	Blautia_producta	SPC00057
Roseburia_intestinalis	Roseburia_intestinalis	SPC00061
Roseburia_intestinalis	Coprococcus_catus	SPC00061
Roseburia_intestinalis	Erysipelotrichaceae_bacterium_3_1_53	SPC00061
Roseburia_intestinalis	Bacteroides_sp_D20	SPC00061
Roseburia_intestinalis	Bacteroides_ovatus	SPC00061
Roseburia_intestinalis	Parabacteroides_merdae	SPC00061
Roseburia_intestinalis	Bacteroides_vulgatus	SPC00061
Roseburia_intestinalis	Collinsella_aerofaciens	SPC00061
Roseburia_intestinalis	Escherichia_coli	SPC00061
Roseburia_intestinalis	Ruminococcus_obeum	SPC00061
Roseburia_intestinalis	Bacteroides_caccae	SPC00061
Roseburia_intestinalis	Bacteroides_eggerthii	SPC00061
Roseburia_intestinalis	Ruminococcus_torques	SPC00061
Roseburia_intestinalis	Clostridium_hathewayi	SPC00061
Roseburia_intestinalis	Bifidobacterium_pseudocatenulatum	SPC00061
Roseburia_intestinalis	Bifidobacterium_adolescentis	SPC00061
Roseburia_intestinalis	Coprococcus_comes	SPC00061
Roseburia_intestinalis	Clostridium_symbiosum	SPC00061
Roseburia_intestinalis	Eubacterium_rectale	SPC00061
Roseburia_intestinalis	Faecalibacterium_prausnitzii	SPC00061
Roseburia_intestinalis	Odoribacter_splanchnicus	SPC00061

TABLE 4a-continued

Roseburia_intestinalis	Lachnospiraceae_bacterium_5_1_57FAA	SPC00061
Roseburia_intestinalis	Blautia_schinkii	SPC00061
Roseburia_intestinalis	Alistipes_shahii	SPC00061
Roseburia_intestinalis	Blautia_producta	SPC00061
Coprococcus_catus	Coprococcus_catus	SPC00080
Coprococcus_catus	Erysipelotricliaceae_bacterium_3_1_53	SPC00080
Coprococcus_catus	Bacteroides_sp_D20	SPC00080
Coprococcus_catus	Bacteroides_ovatus	SPC00080
Coprococcus_catus	Parabacteroides_merdae	SPC00080
Coprococcus_catus	Bacteroides_vulgatus	SPC00080
Coprococcus_catus	Collinsella_aerofaciens	SPC00080
Coprococcus_catus	Escherichia_coli	SPC00080
Coprococcus_catus	Ruminococcus_obeum	SPC00080
Coprococcus_catus	Bacteroides_caccae	SPC00080
Coprococcus_catus	Bacteroides_eggerthii	SPC00080
Coprococcus_catus	Ruminococcus_torques	SPC00080
Coprococcus_catus	Clostridium_hathewayi	SPC00080
Coprococcus_catus	Bifidobacterium_pseudocatenulatum	SPC00080
Coprococcus_catus	Bifidobacterium_adolescentis	SPC00080
Coprococcus_catus	Coprococcus_comes	SPC00080
Coprococcus_catus	Clostridium_symbiosum	SPC00080
Coprococcus_catus	Eubacterium_rectale	SPC00080
Coprococcus_catus	Faecalibacterium_prausnitzii	SPC00080
Coprococcus_catus	Odoribacter_splanchnicus	SPC00080
Coprococcus_catus	Lachnospiraceae_bacterium_5_1_57FAA	SPC00080
Coprococcus_catus	Blautia_schinkii	SPC00080
Coprococcus_catus	Alistipes_shahii	SPC00080
Coprococcus_catus	Blautia_producta	SPC00080
Erysipelotrichaceae_bacterium_3_1_53	Erysipelotrichaceae_bacterium_3_1_53	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Bacteroides_sp_D20	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Bacteroides_ovatus	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Parabacteroides_merdae	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Bacteroides_vulgatus	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Collinsella_aerofaciens	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Escherichia_coli	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Ruminococcus_obeum	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Bacteroides_caccae	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Bacteroides_eggerthii	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Ruminococcus_torques	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Clostridium_hathewayi	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Bifidobacterium_pseudocatenulatum	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Bifidobacterium_adolescentis	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Coprococcus_comes	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Clostridium_symbiosum	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Eubacterium_rectale	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Faecalibacterium_prausnitzii	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Odoribacter_splanchnicus	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Lachnospiraceae_bacterium_5_1_57FAA	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Blautia_schinkii	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Alistipes_shahii	SPC10001
Erysipelotrichaceae_bacterium_3_1_53	Blautia_producta	SPC10001
Bacteroides_sp_D20	Bacteroides_sp_D20	SPC10019
Bacteroides_sp_D20	Bacteroides_ovatus	SPC10019
Bacteroides_sp_D20	Parabacteroides_merdae	SPC10019
Bacteroides_sp_D20	Bacteroides_vulgatus	SPC10019
Bacteroides_sp_D20	Collinsella_aerofaciens	SPC10019
Bacteroides_sp_D20	Escherichia_coli	SPC10019
Bacteroides_sp_D20	Ruminococcus_obeum	SPC10019
Bacteroides_sp_D20	Bacteroides_caccae	SPC10019
Bacteroides_sp_D20	Bacteroides_eggerthii	SPC10019
Bacteroides_sp_D20	Ruminococcus_torques	SPC10019
Bacteroides_sp_D20	Clostridium_hathewayi	SPC10019
Bacteroides_sp_D20	Bifidobacterium_pseudocatenulatum	SPC10019
Bacteroides_sp_D20	Bifidobacterium_adolescentis	SPC10019
Bacteroides_sp_D20	Coprococcus_comes	SPC10019
Bacteroides_sp_D20	Clostridium_symbiosum	SPC10019
Bacteroides_sp_D20	Eubacterium_rectale	SPC10019
Bacteroides_sp_D20	Faecalibacterium_prausnitzii	SPC10019
Bacteroides_sp_D20	Odoribacter_splanchnicus	SPC10019
Bacteroides_sp_D20	Lachnospiraceae_bacterium_5_1_57FAA	SPC10019
Bacteroides_sp_D20	Blautia_schinkii	SPC10019
Bacteroides_sp_D20	Alistipes_shahii	SPC10019
Bacteroides_sp_D20	Blautia_producta	SPC10019
Bacteroides_ovatus	Bacteroides_ovatus	SPC10030
Bacteroides_ovatus	Parabacteroides_merdae	SPC10030
Bacteroides_ovatus	Bacteroides_vulgatus	SPC10030
Bacteroides_ovatus	Collinsella_aerofaciens	SPC10030
Bacteroides_ovatus	Escherichia_coli	SPC10030
Bacteroides_ovatus	Ruminococcus_obeum	SPC10030
Bacteroides_ovatus	Bacteroides_caccae	SPC10030

TABLE 4a-continued

Bacteroides_ovatus	*Bacteroides_eggerthii*	SPC10030
Bacteroides_ovatus	*Ruminococcus_torques*	SPC10030
Bacteroides_ovatus	*Clostridium_hathewayi*	SPC10030
Bacteroides_ovatus	*Bifidobacterium_pseudocatenulatum*	SPC10030
Bacteroides_ovatus	*Bifidobacterium_adolescentis*	SPC10030
Bacteroides_ovatus	*Coprococcus_comes*	SPC10030
Bacteroides_ovatus	*Clostridium_symbiosum*	SPC10030
Bacteroides_ovatus	*Eubacterium_rectale*	SPC10030
Bacteroides_ovatus	*Faecalibacterium_prausnitzii*	SPC10030
Bacteroides_ovatus	*Odoribacter_splanchnicus*	SPC10030
Bacteroides_ovatus	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10030
Bacteroides_ovatus	*Blautia_schinkii*	SPC10030
Bacteroides_ovatus	*Alistipes_shahii*	SPC10030
Bacteroides_ovatus	*Blautia_producta*	SPC10030
Parabacteroides_merdae	*Parabacteroides_merdae*	SPC10048
Parabacteroides_merdae	*Bacteroides_vulgatus*	SPC10048
Parabacteroides_merdae	*Collinsella_aerofaciens*	SPC10048
Parabacteroides_merdae	*Escherichia_coli*	SPC10048
Parabacteroides_merdae	*Ruminococcus_obeum*	SPC10048
Parabacteroides_merdae	*Bacteroides_caccae*	SPC10048
Parabacteroides_merdae	*Bacteroides_eggerthii*	SPC10048
Parabacteroides_merdae	*Ruminococcus_torques*	SPC10048
Parabacteroides_merdae	*Clostridium_hathewayi*	SPC10048
Parabacteroides_merdae	*Bifidobacterium_pseudocatenulatum*	SPC10048
Parabacteroides_merdae	*Bifidobacterium_adolescentis*	SPC10048
Parabacteroides_merdae	*Coprococcus_comes*	SPC10048
Parabacteroides_merdae	*Clostridium_symbiosum*	SPC10048
Parabacteroides_merdae	*Eubacterium_rectale*	SPC10048
Parabacteroides_merdae	*Faecalibacterium_prausnitzii*	SPC10048
Parabacteroides_merdae	*Odoribacter_splanchnicus*	SPC10048
Parabacteroides_merdae	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10048
Parabacteroides_merdae	*Blautia_schinkii*	SPC10048
Parabacteroides_merdae	*Alistipes_shahii*	SPC10048
Parabacteroides_merdae	*Blautia_producta*	SPC10048
Bacteroides_vulgatus	*Bacteroides_vulgatus*	SPC10081
Bacteroides_vulgatus	*Collinsella_aerofaciens*	SPC10081
Bacteroides_vulgatus	*Escherichia_coli*	SPC10081
Bacteroides_vulgatus	*Ruminococcus_obeum*	SPC10081
Bacteroides_vulgatus	*Bacteroides_caccae*	SPC10081
Bacteroides_vulgatus	*Bacteroides_eggerthii*	SPC10081
Bacteroides_vulgatus	*Ruminococcus_torques*	SPC10081
Bacteroides_vulgatus	*Clostridium_hathewayi*	SPC10081
Bacteroides_vulgatus	*Bifidobacterium_pseudocatenulatum*	SPC10081
Bacteroides_vulgatus	*Bifidobacterium_adolescentis*	SPC10081
Bacteroides_vulgatus	*Coprococcus_comes*	SPC10081
Bacteroides_vulgatus	*Clostridium_symbiosum*	SPC10081
Bacteroides_vulgatus	*Eubacterium_rectale*	SPC10081
Bacteroides_vulgatus	*Faecalibacterium_prausnitzii*	SPC10081
Bacteroides_vulgatus	*Odoribacter_splanchnicus*	SPC10081
Bacteroides_vulgatus	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10081
Bacteroides_vulgatus	*Blautia_schinkii*	SPC10081
Bacteroides_*vulgatus*	*Alistipes_shahii*	SPC10081
Bacteroides_vulgatus	*Blautia_producta*	SPC10081
Collinsella_aerofaciens	*Collinsella_aerofaciens*	SPC10097
Collinsella_aerofaciens	*Escherichia_coli*	SPC10097
Collinsella_aerofaciens	*Ruminococcus_obeum*	SPC10097
Collinsella_aerofaciens	*Bacteroides_caccae*	SPC10097
Collinsella_aerofaciens	*Bacteroides_eggerthii*	SPC10097
Collinsella_aerofaciens	*Ruminococcus_torques*	SPC10097
Collinsella_aerofaciens	*Clostridium_hathewayi*	SPC10097
Collinsella_aerofaciens	*Bifidobacterium_pseudocatenulatum*	SPC10097
Collinsella_aerofaciens	*Bifidobacterium_adolescentis*	SPC10097
Collinsella_aerofaciens	*Coprococcus_comes*	SPC10097
Collinsella_aerofaciens	*Clostridium_symbiosum*	SPC10097
Collinsella_aerofaciens	*Eubacterium_rectale*	SPC10097
Collinsella_aerofaciens	*Faecalibacterium_prausnitzii*	SPC10097
Collinsella_aerofaciens	*Odoribacter_splanchnicus*	SPC10097
Collinsella_aerofaciens	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10097
Collinsella_aerofaciens	*Blautia_schinkii*	SPC10097
Collinsella_aerofaciens	*Alistipes_shahii*	SPC10097
Collinsella_aerofaciens	*Blautia_producta*	SPC10097
Escherichia_coli	*Escherichia_coli*	SPC10110
Escherichia_coli	*Ruminococcus_obeum*	SPC10110
Escherichia_coli	*Bacteroides_caccae*	SPC10110
Escherichia_coli	*Bacteroides_eggerthii*	SPC10110
Escherichia_coli	*Ruminococcus_torques*	SPC10110
Escherichia_coli	*Clostridium_hathewayi*	SPC10110
Escherichia_coli	*Bifidobacterium_pseudocatenulatum*	SPC10110
Escherichia_coli	*Bifidobacterium_adolescentis*	SPC10110
Escherichia_coli	*Coprococcus_comes*	SPC10110

TABLE 4a-continued

Escherichia_coli	*Clostridium_symbiosum*	SPC10110
Escherichia_coli	*Eubacterium_rectale*	SPC10110
Escherichia_coli	*Faecalibacterium_prausnitzii*	SPC10110
Escherichia_coli	*Odoribacter_splanchnicus*	SPC10110
Escherichia_coli	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10110
Escherichia_coli	*Blautia_schinkii*	SPC10110
Escherichia_coli	*Alistipes_shahii*	SPC10110
Escherichia_coli	*Blautia_producta*	SPC10110
Ruminococcus_obeum	*Ruminococcus_obeum*	SPC10197
Ruminococcus_obeum	*Bacteroides_caccae*	SPC10197
Ruminococcus_obeum	*Bacteroides_eggerthii*	SPC10197
Ruminococcus_obeum	*Ruminococcus_torques*	SPC10197
Ruminococcus_obeum	*Clostridium_hathewayi*	SPC10197
Ruminococcus_obeum	*Bifidobacterium_pseudocatenulatum*	SPC10197
Ruminococcus_obeum	*Bifidobacterium_adolescentis*	SPC10197
Ruminococcus_obeum	*Coprococcus_comes*	SPC10197
Ruminococcus_obeum	*Clostridium_symbiosum*	SPC10197
Ruminococcus_obeum	*Eubacterium_rectale*	SPC10197
Ruminococcus_obeum	*Faecalibacterium_prausnitzii*	SPC10197
Ruminococcus_obeum	*Odoribacter_splanchnicus*	SPC10197
Ruminococcus_obeum	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10197
Ruminococcus_obeum	*Blautia_schinkii*	SPC10197
Ruminococcus_obeum	*Alistipes_shahii*	SPC10197
Ruminococcus_obeum	*Blautia_producta*	SPC10197
Bacteroides_caccae	*Bacteroides_caccae*	SPC10211
Bacteroides_caccae	*Bacteroides_eggerthii*	SPC10211
Bacteroides_caccae	*Ruminococcus_torques*	SPC10211
Bacteroides_caccae	*Clostridium_hathewayi*	SPC10211
Bacteroides_caccae	*Bifidobacterium_pseudocatenulatum*	SPC10211
Bacteroides_caccae	*Bifidobacterium_adolescentis*	SPC10211
Bacteroides_caccae	*Coprococcus_comes*	SPC10211
Bacteroides_caccae	*Clostridium_symbiosum*	SPC10211
Bacteroides_caccae	*Eubacterium_rectale*	SPC10211
Bacteroides_caccae	*Faecalibacterium_prausnitzii*	SPC10211
Bacteroides_caccae	*Odoribacter_splanchnicus*	SPC10211
Bacteroides_caccae	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10211
Bacteroides_*caccae*	*Blautia_schinkii*	SPC10211
Bacteroides_caccae	*Alistipes_shahii*	SPC10211
Bacteroides_caccae	*Blautia_producta*	SPC10211
Bacteroides_eggerthii	*Bacteroides_eggerthii*	SPC10213
Bacteroides_eggerthii	*Ruminococcus_torques*	SPC10213
Bacteroides_eggerthii	*Clostridium_hathewayi*	SPC10213
Bacteroides_eggerthii	*Bifidobacterium_pseudocatenulatum*	SPC10213
Bacteroides_eggerthii	*Bifidobacterium_adolescentis*	SPC10213
Bacteroides_eggerthii	*Coprococcus_comes*	SPC10213
Bacteroides_eggerthii	*Clostridium_symbiosum*	SPC10213
Bacteroides_eggerthii	*Eubacterium_rectale*	SPC10213
Bacteroides_eggerthii	*Faecalibacterium_prausnitzii*	SPC10213
Bacteroides_eggerthii	*Odoribacter_splanchnicus*	SPC10213
Bacteroides_eggerthii	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10213
Bacteroides_eggerthii	*Blautia_schinkii*	SPC10213
Bacteroides_eggerthii	*Alistipes_shahii*	SPC10213
Bacteroides_eggerthii	*Blautia_producta*	SPC10213
Ruminococcus_torques	*Ruminococcus_torques*	SPC10233
Ruminococcus_torques	*Clostridium_hathewayi*	SPC10233
Ruminococcus_torques	*Bifidobacterium_pseudocatenulatum*	SPC10233
Ruminococcus_torques	*Bifidobacterium_adolescentis*	SPC10233
Ruminococcus_torques	*Coprococcus_comes*	SPC10233
Ruminococcus_torques	*Clostridium_symbiosum*	SPC10233
Ruminococcus_torques	*Eubacterium_rectale*	SPC10233
Ruminococcus_torques	*Faecalibacterium_prausnitzii*	SPC10233
Ruminococcus_torques	*Odoribacter_splanchnicus*	SPC10233
Ruminococcus_torques	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10233
Ruminococcus_torques	*Blautia_schinkii*	SPC10233
Ruminococcus_torques	*Alistipes_shahii*	SPC10233
Ruminococcus_torques	*Blautia_producta*	SPC10233
Clostridium_hathewayi	*Clostridium_hathewayi*	SPC10243
Clostridium_hathewayi	*Bifidobacterium_pseudocatenulatum*	SPC10243
Clostridium_hathewayi	*Bifidobacterium_adolescentis*	SPC10243
Clostridium_hathewayi	*Coprococcus_comes*	SPC10243
Clostridium_hathewayi	*Clostridium_symbiosum*	SPC10243
Clostridium_hathewayi	*Eubacterium_rectale*	SPC10243
Clostridium_hathewayi	*Faecalibacterium_prausnitzii*	SPC10243
Clostridium_hathewayi	*Odoribacter_splanchnicus*	SPC10243
Clostridium_hathewayi	Lachospiraceae_*bacterium*_5_1_57FAA	SPC10243
Clostridium_hathewayi	*Blautia_schinkii*	SPC10243
Clostridium_hathewayi	*Alistipes_shahii*	SPC10243
Clostridium_hathewayi	*Blautia_producta*	SPC10243
Bifidobacterium_pseudocatenulatum	*Bifidobacterium_pseudocatenulatum*	SPC10298
Bifidobacterium_pseudocatenulatum	*Bifidobacterium_adolescentis*	SPC10298

TABLE 4a-continued

Bifidobacterium_pseudocatenulatum	*Coprococcus_comes*	SPC10298
Bifidobacterium_pseudocatenulatum	*Clostridium_symbiosum*	SPC10298
Bifidobacterium_pseudocatenulatum	*Eubacterium_rectale*	SPC10298
Bifidobacterium_pseudocatenulatum	*Faecalibacterium_prausnitzii*	SPC10298
Bifidobacterium_pseudocatenulatum	*Odoribacter_splanchnicus*	SPC10298
Bifidobacterium_pseudocatenulatum	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10298
Bifidobacterium_pseudocatenulatum	*Blautia_schinkii*	SPC10298
Bifidobacterium_pseudocatenulatum	*Alistipes_shahii*	SPC10298
Bifidobacterium_pseudocatenulatum	*Blautia_producta*	SPC10298
Bifidobacterium_adolescentis	*Bifidobacterium_adolescentis*	SPC10301
Bifidobacterium_adolescentis	*Coprococcus_comes*	SPC10301
Bifidobacterium_adolescentis	*Clostridium_symbiosum*	SPC10301
Bifidobacterium_adolescentis	*Eubacterium_rectale*	SPC10301
Bifidobacterium_adolescentis	*Faecalibacterium_prausnitzii*	SPC10301
Bifidobacterium_adolescentis	*Odoribacter_splanchnicus*	SPC10301
Bifidobacterium_adolescentis	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10301
Bifidobacterium_adolescentis	*Blautia_schinkii*	SPC10301
Bifidobacterium_adolescentis	*Alistipes_shahii*	SPC10301
Bifidobacterium_adolescentis	*Blautia_producta*	SPC10301
Coprococcus_comes	*Coprococcus_comes*	SPC10304
Coprococcus_comes	*Clostridium_symbiosum*	SPC10304
Coprococcus_*comes*	*Eubacterium_rectale*	SPC10304
Coprococcus_comes	*Faecalibacterium_prausnitzii*	SPC10304
Coprococcus_comes	*Odoribacter_splanchnicus*	SPC10304
Coprococcus_comes	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10304
Coprococcus_comes	*Blautia_schinkii*	SPC10304
Coprococcus_comes	*Alistipes_shahii*	SPC10304
Coprococcus_comes	*Blautia_producta*	SPC10304
Clostridium_symbiosum	*Clostridium_symbiosum*	SPC10355
Clostridium_symbiosum	*Eubacterium_rectale*	SPC10355
Clostridium_symbiosum	*Faecalibacterium_prausnitzii*	SPC10355
Clostridium_symbiosum	*Odoribacter_splanchnicus*	SPC10355
Clostridium_symbiosum	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10355
Clostridium_symbiosum	*Blautia_schinkii*	SPC10355
Clostridium_symbiosum	*Alistipes_shahii*	SPC10355
Clostridium_symbiosum	*Blautia_producta*	SPC10355
Eubacterium_rectale	*Eubacterium_rectale*	SPC10363
Eubacterium_rectale	*Faecalibacterium_prausnitzii*	SPC10363
Eubacterium_rectale	*Odoribacter_splanchnicus*	SPC10363
Eubacterium_rectale	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10363
Eubacterium_rectale	*Blautia_schinkii*	SPC10363
Eubacterium_rectale	*Alistipes_shahii*	SPC10363
Eubacterium_rectale	*Blautia_producta*	SPC10363
Faecalibacterium_prausnitzii	*Faecalibacterium_prausnitzii*	SPC10386
Faecalibacterium_prausnitzii	*Odoribacter_splanchnicus*	SPC10386
Faecalibacterium_prausnitzii	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10386
Faecalibacterium_prausnitzii	*Blautia_schinkii*	SPC10386
Faecalibacterium_prausnitzii	*Alistipes_shahii*	SPC10386
Faecalibacterium_prausnitzii	*Blautia_producta*	SPC10386
Odoribacter_splanchnicus	*Odoribacter_splanchnicus*	SPC10388
Odoribacter_splanchnicus	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10388
Odoribacter_splanchnicus	*Blautia_schinkii*	SPC10388
Odoribacter_splanchnicus	*Alistipes_shahii*	SPC10388
Odoribacter_splanchnicus	*Blautia_producta*	SPC10388
Lachnospiraceae_*bacterium*_5_1_57FAA	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Blautia_schinkii*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Alistipes_shahii*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Blautia_producta*	SPC10390
Blautia_schinkii	*Blautia_schinkii*	SPC10403
Blautia_schinkii	*Alistipes_shahii*	SPC10403
Blautia_schinkii	*Blautia_producta*	SPC10403
Alistipes_shahii	*Alistipes_shahii*	SPC10414
Alistipes_shahii	*Blautia_producta*	SPC10414
Blautia_producta	*Blautia_producta*	SPC10415
Collinsella_aerofaciens	*Clostridium_tertium*	SPC10097
Collinsella_aerofaciens	*Clostridium_disporicum*	SPC10097
Collinsella_aerofaciens	*Clostridium_innocuum*	SPC10097
Collinsella_aerofaciens	*Clostridium_mayombei*	SPC10097
Collinsella_aerofaciens	*Clostridium_butyricum*	SPC10097
Collinsella_aerofaciens	*Clostridium_hylemonae*	SPC10097
Collinsella_aerofaciens	*Clostridium_bolteae*	SPC10097
Collinsella_aerofaciens	*Clostridium_orbiscindens*	SPC10097
Collinsella_aerofaciens	*Ruminococcus_gnavus*	SPC10097
Collinsella_aerofaciens	*Ruminococcus_bromii*	SPC10097
Collinsella_aerofaciens	*Eubacterium_rectale*	SPC10097
Clostridium_tertium	*Collinsella_aerofaciens*	SPC10155
Clostridium_tertium	*Clostridium_tertium*	SPC10155
Clostridium_tertium	*Clostridium_disporicum*	SPC10155
Clostridium_tertium	*Clostridium_innocuum*	SPC10155
Clostridium_tertium	*Clostridium_mayombei*	SPC10155

TABLE 4a-continued

Clostridium_tertium	*Clostridium_butyricum*	SPC10155
Clostridium_tertium	*Coprococcus_comes*	SPC10155
Clostridium_tertium	*Clostridium_hylemonae*	SPC10155
Clostridium_tertium	*Clostridium_bolteae*	SPC10155
Clostridium_tertium	*Clostridium_symbiosum*	SPC10155
Clostridium_tertium	*Clostridium_orbiscindens*	SPC10155
Clostridium_*tertium*	*Faecalibacterium_prausnitzii*	SPC10155
Clostridium_tertium	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10155
Clostridium_tertium	*Blautia_producta*	SPC10155
Clostridium_tertium	*Ruminococcus_gnavus*	SPC10155
Clostridium_tertium	*Ruminococcus_bromii*	SPC10155
Clostridium_tertium	*Eubacterium_rectale*	SPC10155
Clostridium_disporicum	*Clostridium_tertium*	SPC10167
Clostridium_disporicum	*Clostridium_disporicum*	SPC10167
Clostridium_disporicum	*Clostridium_innocuum*	SPC10167
Clostridium_disporicum	*Clostridium_mayombei*	SPC10167
Clostridium_disporicum	*Clostridium_butyricum*	SPC10167
Clostridium_disporicum	*Coprococcus_comes*	SPC10167
Clostridium_disporicum	*Clostridium_hylemonae*	SPC10167
Clostridium_disporicum	*Clostridium_bolteae*	SPC10167
Clostridium_disporicum	*Clostridium_symbiosum*	SPC10167
Clostridium_disporicum	*Clostridium_orbiscindens*	SPC10167
Clostridium_disporicum	*Faecalibacterium_prausnitzii*	SPC10167
Clostridium_disporicum	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10167
Clostridium_disporicum	*Blautia_producta*	SPC10167
Clostridium_disporicum	*Ruminococcus_gnavus*	SPC10167
Clostridium_disporicum	*Ruminococcus_bromii*	SPC10167
Clostridium_disporicum	*Eubacterium_rectale*	SPC10167
Clostridium_innocuum	*Clostridium_disporicum*	SPC10202
Clostridium_innocuum	*Clostridium_innocuum*	SPC10202
Clostridium_innocuum	*Clostridium_mayombei*	SPC10202
Clostridium_innocuum	*Clostridium_butyricum*	SPC10202
Clostridium_innocuum	*Coprococcus_comes*	SPC10202
Clostridium_innocuum	*Clostridium_hylemonae*	SPC10202
Clostridium_innocuum	*Clostridium_bolteae*	SPC10202
Clostridium_innocuum	*Clostridium_symbiosum*	SPC10202
Clostridium_innocuum	*Clostridium_orbiscindens*	SPC10202
Clostridium_innocuum	*Faecalibacterium_prausnitzii*	SPC10202
Clostridium_innocuum	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10202
Clostridium_innocuum	*Blautia_producta*	SPC10202
Clostridium_innocuum	*Ruminococcus_gnavus*	SPC10202
Clostridium_innocuum	*Ruminococcus_bromii*	SPC10202
Clostridium_innocuum	*Eubacterium_rectale*	SPC10202
Clostridium_mayombei	*Clostridium_innocuum*	SPC10238
Clostridium_mayombei	*Clostridium_mayombei*	SPC10238
Clostridium_mayombei	*Clostridium_butyricum*	SPC10238
Clostridium_mayombei	*Coprococcus_comes*	SPC10238
Clostridium_mayombei	*Clostridium_hylemonae*	SPC10238
Clostridium_mayombei	*Clostridium_bolteae*	SPC10238
Clostridium_mayombei	*Clostridium_symbiosum*	SPC10238
Clostridium_mayombei	*Clostridium_orbiscindens*	SPC10238
Clostridium_mayombei	*Faecalibacterium_prausnitzii*	SPC10238
Clostridium_mayombei	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10238
Clostridium_mayombei	*Blautia_producta*	SPC10238
Clostridium_mayombei	*Ruminococcus_gnavus*	SPC10238
Clostridium_mayombei	*Ruminococcus_bromii*	SPC10238
Clostridium_mayombei	*Eubacterium_rectale*	SPC10238
Clostridium_butyricum	*Clostridium_mayombei*	SPC10256
Clostridium_butyricum	*Clostridium_butyricum*	SPC10256
Clostridium_butyricum	*Coprococcus_comes*	SPC10256
Clostridium_butyricum	*Clostridium_hylemonae*	SPC10256
Clostridium_butyricum	*Clostridium_bolteae*	SPC10256
Clostridium_butyricum	*Clostridium_symbiosum*	SPC10256
Clostridium_butyricum	*Clostridium_orbiscindens*	SPC10256
Clostridium_butyricum	*Faecalibacterium_prausnitzii*	SPC10256
Clostridium_butyricum	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10256
Clostridium_butyricum	*Blautia_producta*	SPC10256
Clostridium_butyricum	*Ruminococcus_gnavus*	SPC10256
Clostridium_butyricum	*Ruminococcus_bromii*	SPC10256
Clostridium_butyricum	*Eubacterium_rectale*	SPC10256
Coprococcus_comes	*Clostridium_butyricum*	SPC10304
Coprococcus_*comes*	*Clostridium_hylemonae*	SPC10304
Coprococcus_comes	*Clostridium_bolteae*	SPC10304
Coprococcus_comes	*Clostridium_orbiscindens*	SPC10304
Coprococcus_comes	*Ruminococcus_gnavus*	SPC10304
Coprococcus_comes	*Ruminococcus_bromii*	SPC10304
Coprococcus_comes	*Eubacterium_rectale*	SPC10304
Clostridium_hylemonae	*Coprococcus_comes*	SPC10313
Clostridium_hylemonae	*Clostridium_hylemonae*	SPC10313
Clostridium_hylemonae	*Clostridium_bolteae*	SPC10313

TABLE 4a-continued

Clostridium_hylemonae	*Clostridium_symbiosum*		SPC10313
Clostridium_hylemonae	*Clostridium_orbiscindens*		SPC10313
Clostridium_hylemonae	*Faecalibacterium_prausnitzii*		SPC10313
Clostridium_hylemonae	Lachnospiraceae_*bacterium*_5_1_57FAA		SPC10313
Clostridium_hylemonae	*Blautia_producta*		SPC10313
Clostridium_hylemonae	*Ruminococcus_gnavus*		SPC10313
Clostridium_hylemonae	*Ruminococcus_bromii*		SPC10313
Clostridium_hylemonae	*Eubacterium_rectale*		SPC10313
Clostridium_bolteae	*Clostridium_hylemonae*		SPC10325
Clostridium_bolteae	*Clostridium_bolteae*		SPC10325
Clostridium_bolteae	*Clostridium_symbiosum*		SPC10325
Clostridium_bolteae	*Clostridium_orbiscindens*		SPC10325
Clostridium_bolteae	*Faecalibacterium_prausnitzii*		SPC10325
Clostridium_bolteae	Lachnospiraceae_*bacterium*_5_1_57FAA		SPC10325
Clostridium_bolteae	*Blautia_producta*		SPC10325
Clostridium_bolteae	*Ruminococcus_gnavus*		SPC10325
Clostridium_bolteae	*Ruminococcus_bromii*		SPC10325
Clostridium_bolteae	*Eubacterium_rectale*		SPC10325
Clostridium_symbiosum	*Clostridium_bolteae*		SPC10355
Clostridium_symbiosum	*Clostridium_orbiscindens*		SPC10355
Clostridium_symbiosum	*Ruminococcus_gnavus*		SPC10355
Clostridium_symbiosum	*Ruminococcus_bromii*		SPC10355
Clostridium_symbiosum	*Eubacterium_rectale*		SPC10355
Clostridium_orbiscindens	*Clostridium_symbiosum*		SPC10358
Clostridium_orbiscindens	*Clostridium_orbiscindens*		SPC10358
Clostridium_orbiscindens	*Faecalibacterium_prausnitzii*		SPC10358
Clostridium_orbiscindens	Lachnospiraceae_*bacterium*_5_1_57FAA		SPC10358
Clostridium_orbiscindens	*Blautia_producta*		SPC10358
Clostridium_orbiscindens	*Ruminococcus_gnavus*		SPC10358
Clostridium_orbiscindens	*Ruminococcus_bromii*		SPC10358
Clostridium_orbiscindens	*Eubacterium_rectale*		SPC10358
Faecalibacterium_prausnitzii	*Clostridium_orbiscindens*		SPC10386
Faecalibacterium_prausnitzii	*Ruminococcus_gnavus*		SPC10386
Faecalibacterium_prausnitzii	*Ruminococcus_bromii*		SPC10386
Faecalibacterium_prausnitzii	*Eubacterium_rectale*		SPC10386
Lachnospiraceae_*bacterium*_5_1_57FAA	*Ruminococcus_gnavus*		SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Ruminococcus_bromii*		SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Eubacterium_rectale*		SPC10390
Blautia_producta	*Ruminococcus_gnavus*		SPC10415
Blautia_producta	*Ruminococcus_bromii*		SPC10415
Blautia_producta	*Eubacterium_rectale*		SPC10415
Ruminococcus_gnavus	*Blautia_producta*		SPC10468
Ruminococcus_gnavus	*Ruminococcus_gnavus*		SPC10468
Ruminococcus_gnavus	*Ruminococcus_bromii*		SPC10468
Ruminococcus_gnavus	*Eubacterium_rectale*		SPC10468
Ruminococcus_bromii	*Ruminococcus_gnavus*		SPC10470
Ruminococcus_bromii	*Ruminococcus_bromii*		SPC10470
Ruminococcus_bromii	*Eubacterium_rectale*		SPC10470
Eubacterium_rectale	*Ruminococcus_bromii*		SPC10567
Eubacterium_rectale	*Eubacterium_rectale*		SPC10567
Collinsella_aerofaciens	*Collinsella_aerofaciens*	*Collinsella_aerofaciens*	SPC10097
Collinsella_aerofaciens	*Collinsella_aerofaciens*	*Coprococcus_comes*	SPC10097
Collinsella_aerofaciens	*Collinsella_aerofaciens*	*Clostridium_bolteae*	SPC10097
Collinsella_aerofaciens	*Collinsella_aerofaciens*	*Clostridium_symbiosum*	SPC10097
Collinsella_aerofaciens	*Collinsella_aerofaciens*	*Faecalibacterium_prausnitzii*	SPC10097
Collinsella_aerofaciens	*Collinsella_aerofaciens*	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10097
Collinsella_*aerofaciens*	*Collinsella_aerofaciens*	*Blautia_producta*	SPC10097
Collinsella_aerofaciens	*Collinsella_aerofaciens*	*Eubacterium_rectale*	SPC10097
Collinsella_aerofaciens	*Coprococcus_comes*	*Coprococcus_comes*	SPC10097
Collinsella_aerofaciens	*Coprococcus_comes*	*Clostridium_bolteae*	SPC10097
Collinsella_aerofaciens	*Coprococcus_comes*	*Clostridium_symbiosum*	SPC10097
Collinsella_aerofaciens	*Coprococcus_comes*	*Faecalibacterium_prausnitzii*	SPC10097
Collinsella_aerofaciens	*Coprococcus_comes*	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10097
Collinsella_aerofaciens	*Coprococcus_comes*	*Blautia_producta*	SPC10097
Collinsella_aerofaciens	*Coprococcus_comes*	*Eubacterium_rectale*	SPC10097
Collinsella_aerofaciens	*Clostridium_bolteae*	*Clostridium_bolteae*	SPC10097
Collinsella_aerofaciens	*Clostridium_bolteae*	*Clostridium_symbiosum*	SPC10097
Collinsella_aerofaciens	*Clostridium_bolteae*	*Faecalibacterium_prausnitzii*	SPC10097
Collinsella_aerofaciens	*Clostridium_bolteae*	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10097
Collinsella_aerofaciens	*Clostridium_bolteae*	*Blautia_producta*	SPC10097
Collinsella_aerofaciens	*Clostridium_bolteae*	*Eubacterium_rectale*	SPC10097
Collinsella_aerofaciens	*Clostridium_symbiosum*	*Clostridium_symbiosum*	SPC10097
Collinsella_aerofaciens	*Clostridium_symbiosum*	*Faecalibacterium_prausnitzii*	SPC10097
Collinsella_aerofaciens	*Clostridium_symbiosum*	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10097
Collinsella_aerofaciens	*Clostridium_symbiosum*	*Blautia_producta*	SPC10097
Collinsella_aerofaciens	*Clostridium_symbiosum*	*Eubacterium_rectale*	SPC10097
Collinsella_aerofaciens	*Faecalibacterium_prausnitzii*	*Faecalibacterium_prausnitzii*	SPC10097
Collinsella_aerofaciens	*Faecalibacterium_prausnitzii*	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10097
Collinsella_aerofaciens	*Faecalibacterium_prausnitzii*	*Blautia_producta*	SPC10097
Collinsella_aerofaciens	*Faecalibacterium_prausnitzii*	*Eubacterium_rectale*	SPC10097

TABLE 4a-continued

Collinsella_aerofaciens	Lachnospiraceae_bacterium_5_1_57FAA	Lachnospiraceae_bacterium_5_1_57FAA	SPC10097
Collinsella_aerofaciens	Lachnospiraceae_bacterium_5_1_57FAA	Blautia_producta	SPC10097
Collinsella_aerofaciens	Lachnospiraceae_bacterium_5_1_57FAA	Eubacterium_rectale	SPC10097
Collinsella_aerofaciens	Blautia_producta	Blautia_producta	SPC10097
Collinsella_aerofaciens	Blautia_producta	Eubacterium_rectale	SPC10097
Collinsella_aerofaciens	Eubacterium_rectale	Eubacterium_rectale	SPC10097
Coprococcus_comes	Coprococcus_comes	Coprococcus_comes	SPC10304
Coprococcus_comes	Coprococcus_comes	Clostridium_bolteae	SPC10304
Coprococcus_comes	Coprococcus_comes	Clostridium_symbiosum	SPC10304
Coprococcus_comes	Coprococcus_comes	Faecalibacterium_prausnitzii	SPC10304
Coprococcus_comes	Coprococcus_comes	Lachnospiraceae_bacterium_5_1_57FAA	SPC10304
Coprococcus_comes	Coprococcus_comes	Blautia_producta	SPC10304
Coprococcus_comes	Coprococcus_comes	Eubacterium_rectale	SPC10304
Coprococcus_comes	Clostridium_bolteae	Clostridium_bolteae	SPC10304
Coprococcus_comes	Clostridium_bolteae	Clostridium_symbiosum	SPC10304
Coprococcus_comes	Clostridium_bolteae	Faecalibacterium_prausnitzii	SPC10304
Coprococcus_comes	Clostridium_bolteae	Lachnospiraceae_bacterium_5_1_57FAA	SPC10304
Coprococcus_comes	Clostridium_bolteae	Blautia_producta	SPC10304
Coprococcus_comes	Clostridium_bolteae	Eubacterium_rectale	SPC10304
Coprococcus_comes	Clostridium_symbiosum	Clostridium_symbiosum	SPC10304
Coprococcus_comes	Clostridium_symbiosum	Faecalibacterium_prausnitzii	SPC10304
Coprococcus_comes	Clostridium_symbiosum	Lachnospiraceae_bacterium_5_1_57FAA	SPC10304
Coprococcus_comes	Clostridium_symbiosum	Blautia_producta	SPC10304
Coprococcus_comes	Clostridium_symbiosum	Eubacterium_rectale	SPC10304
Coprococcus_comes	Faecalibacterium_prausnitzii	Faecalibacterium_prausnitzii	SPC10304
Coprococcus_comes	Faecalibacterium_prausnitzii	Lachnospiraceae_bacterium_5_1_57FAA	SPC10304
Coprococcus_comes	Faecalibacterium_prausnitzii	Blautia_producta	SPC10304
Coprococcus_comes	Faecalibacterium_prausnitzii	Eubacterium_rectale	SPC10304
Coprococcus_comes	Lachnospiraceae_bacterium_5_1_57FAA	Lachnospiraceae_bacterium_5_1_57FAA	SPC10304
Coprococcus_comes	Lachnospiraceae_bacterium_5_1_57FAA	Blautia_producta	SPC10304
Coprococcus_comes	Lachnospiraceae_bacterium_5_1_57FAA	Eubacterium_rectale	SPC10304
Coprococcus_comes	Blautia_producta	Blautia_producta	SPC10304
Coprococcus_comes	Blautia_producta	Eubacterium_rectale	SPC10304
Coprococcus_comes	Eubacterium_rectale	Eubacterium_rectale	SPC10304
Clostridium_bolteae	Clostridium_bolteae	Clostridium_bolteae	SPC10325
Clostridium_bolteae	Clostridium_bolteae	Clostridium_symbiosum	SPC10325
Clostridium_bolteae	Clostridium_bolteae	Faecalibacterium_prausnitzii	SPC10325
Clostridium_bolteae	Clostridium_bolteae	Lachnospiraceae_bacterium_5_1_57FAA	SPC10325
Clostridium_bolteae	Clostridium_bolteae	Blautia_producta	SPC10325
Clostridium_bolteae	Clostridium_bolteae	Eubacterium_rectale	SPC10325
Clostridium_bolteae	Clostridium_symbiosum	Clostridium_symbiosum	SPC10325
Clostridium_bolteae	Clostridium_symbiosum	Faecalibacterium_prausnitzii	SPC10325
Clostridium_bolteae	Clostridium_symbiosum	Lachnospiraceae_bacterium_5_1_57FAA	SPC10325
Clostridium_bolteae	Clostridium_symbiosum	Blautia_producta	SPC10325
Clostridium_bolteae	Clostridium_symbiosum	Eubacterium_rectale	SPC10325
Clostridium_bolteae	Faecalibacterium_prausnitzii	Faecalibacterium_prausnitzii	SPC10325
Clostridium_bolteae	Faecalibacterium_prausnitzii	Lachnospiraceae_bacterium_5_1_57FAA	SPC10325
Clostridium_bolteae	Faecalibacterium_prausnitzii	Blautia_producta	SPC10325
Clostridium_bolteae	Faecalibacterium_prausnitzii	Eubacterium_rectale	SPC10325
Clostridium_bolteae	Lachnospiraceae_bacterium_5_1_57FAA	Lachnospiraceae_bacterium_5_1_57FAA	SPC10325
Clostridium_bolteae	Lachnospiraceae_bacterium_5_1_57FAA	Blautia_producta	SPC10325
Clostridium_bolteae	Lachnospiraceae_bacterium_5_1_57FAA	Eubacterium_rectale	SPC10325
Clostridium_bolteae	Blautia_producta	Blautia_producta	SPC10325
Clostridium_bolteae	Blautia_producta	Eubacterium_rectale	SPC10325
Clostridium_bolteae	Eubacterium_rectale	Eubacterium_rectale	SPC10325
Clostridium_symbiosum	Clostridium_symbiosum	Clostridium_symbiosum	SPC10355
Clostridium_symbiosum	Clostridium_symbiosum	Faecalibacterium_prausnitzii	SPC10355
Clostridium_symbiosum	Clostridium_symbiosum	Lachnospiraceae_bacterium_5_1_57FAA	SPC10355
Clostridium_symbiosum	Clostridium_symbiosum	Blautia_producta	SPC10355
Clostridium_symbiosum	Clostridium_symbiosum	Eubacterium_rectale	SPC10355
Clostridium_symbiosum	Faecalibacterium_prausnitzii	Faecalibacterium_prausnitzii	SPC10355
Clostridium_symbiosum	Faecalibacterium_prausnitzii	Lachnospiraceae_bacterium_5_1_57FAA	SPC10355
Clostridium_symbiosum	Faecalibacterium_prausnitzii	Blautia_producta	SPC10355
Clostridium_symbiosum	Faecalibacterium_prausnitzii	Eubacterium_rectale	SPC10355
Clostridium_symbiosum	Lachnospiraceae_bacterium_5_1_57FAA	Lachnospiraceae_bacterium_5_1_57FAA	SPC10355
Clostridium_symbiosum	Lachnospiraceae_bacterium_5_1_57FAA	Blautia_producta	SPC10355
Clostridium_symbiosum	Lachnospiraceae_bacterium_5_1_57FAA	Eubacterium_rectale	SPC10355
Clostridium_symbiosum	Blautia_producta	Blautia_producta	SPC10355
Clostridium_symbiosum	Blautia_producta	Eubacterium_rectale	SPC10355
Clostridium_symbiosum	Eubacterium_rectale	Eubacterium_rectale	SPC10355
Faecalibacterium_prausnitzii	Faecalibacterium_prausnitzii	Faecalibacterium_prausnitzii	SPC10386
Faecalibacterium_prausnitzii	Faecalibacterium_prausnitzii	Lachnospiraceae_bacterium_5_1_57FAA	SPC10386
Faecalibacterium_prausnitzii	Faecalibacterium_prausnitzii	Blautia_producta	SPC10386
Faecalibacterium_prausnitzii	Faecalibacterium_prausnitzii	Eubacterium_rectale	SPC10386
Faecalibacterium_prausnitzii	Lachnospiraceae_bacterium_5_1_57FAA	Lachnospiraceae_bacterium_5_1_57FAA	SPC10386
Faecalibacterium_prausnitzii	Lachnospiraceae_bacterium_5_1_57FAA	Blautia_producta	SPC10386
Faecalibacterium_prausnitzii	Lachnospiraceae_bacterium_5_1_57FAA	Eubacterium_rectale	SPC10386
Faecalibacterium_prausnitzii	Blautia_producta	Blautia_producta	SPC10386
Faecalibacterium_prausnitzii	Blautia_producta	Eubacterium_rectale	SPC10386
Faecalibacterium_prausnitzii	Eubacterium_rectale	Eubacterium_rectale	SPC10386

TABLE 4a-continued

Lachnospiraceae_bacterium_5_1_57FAA	Lachnospiraceae_bacterium_5_1_57FAA	Lachnospiraceae_bacterium_5_1_57FAA	SPC10390
Lachnospiraceae_bacterium_5_1_57FAA	Lachnospiraceae_bacterium_5_1_57FAA	Blautia_producta	SPC10390
Lachnospiraceae_bacterium_5_1_57FAA	Lachnospiraceae_bacterium_5_1_57FAA	Eubacterium_rectale	SPC10390
Lachnospiraceae_bacterium_5_1_57FAA	Blautia_producta	Blautia_producta	SPC10390
Lachnospiraceae_bacterium_5_1_57FAA	Blautia_producta	Eubacterium_rectale	SPC10390
Lachnospiraceae_bacterium_5_1_57FAA	Eubacterium_rectale	Eubacterium_rectale	SPC10390
Blautia_producta	Blautia_producta	Blautia_producta	SPC10415
Blautia_producta	Blautia_producta	Eubacterium_rectale	SPC10415
Blautia_producta	Eubacterium_rectale	Eubacterium_rectale	SPC10415
Eubacterium_rectale	Eubacterium_rectale	Eubacterium_rectale	SPC10567
Collinsella_aerofaciens	Clostridium_tertium	Clostridium_tertium	SPC10097
Collinsella_aerofaciens	Clostridium_tertium	Clostridium_disporicum	SPC10097
Collinsella_aerofaciens	Clostridium_tertium	Clostridium_innocuum	SPC10097
Collinsella_aerofaciens	Clostridium_tertium	Clostridium_mayombei	SPC10097
Collinsella_aerofaciens	Clostridium_tertium	Clostridium_butyricum	SPC10097
Collinsella_aerofaciens	Clostridium_tertium	Clostridium_hylemonae	SPC10097
Collinsella_aerofaciens	Clostridium_tertium	Clostridium_orbiscindens	SPC10097
Collinsella_aerofaciens	Clostridium_tertium	Ruminococcus_gnavus	SPC10097
Collinsella_aerofaciens	Clostridium_tertium	Ruminococcus_bromii	SPC10097
Collinsella_aerofaciens	Clostridium_tertium	Blautia_sp_M25	SPC10097
Collinsella_aerofaciens	Clostridium_disporicum	Clostridium_disporicum	SPC10097
Collinsella_aerofaciens	Clostridium_disporicum	Clostridium_innocuum	SPC10097
Collinsella_aerofaciens	Clostridium_disporicum	Clostridium_mayombei	SPC10097
Collinsella_aerofaciens	Clostridium_disporicum	Clostridium_butyricum	SPC10097
Collinsella_aerofaciens	Clostridium_disporicum	Clostridium_hylemonae	SPC10097
Collinsella_aerofaciens	Clostridium_disporicum	Clostridium_orbiscindens	SPC10097
Collinsella_aerofaciens	Clostridium_disporicum	Ruminococcus_gnavus	SPC10097
Collinsella_aerofaciens	Clostridium_disporicum	Ruminococcus_bromii	SPC10097
Collinsella_aerofaciens	Clostridium_disporicum	Blautia_sp_M25	SPC10097
Collinsella_aerofaciens	Clostridium_innocuum	Clostridium_innocuum	SPC10097
Collinsella_aerofaciens	Clostridium_innocuum	Clostridium_mayombei	SPC10097
Collinsella_aerofaciens	Clostridium_innocuum	Clostridium_butyricum	SPC10097
Collinsella_aerofaciens	Clostridium_innocuum	Clostridium_hylemonae	SPC10097
Collinsella_aerofaciens	Clostridium_innocuum	Clostridium_orbiscindens	SPC10097
Collinsella_aerofaciens	Clostridium_innocuum	Ruminococcus_gnavus	SPC10097
Collinsella_aerofaciens	Clostridium_innocuum	Ruminococcus_bromii	SPC10097
Collinsella_aerofaciens	Clostridium_innocuum	Blautia_sp_M25	SPC10097
Collinsella_aerofaciens	Clostridium_mayombei	Clostridium_mayombei	SPC10097
Collinsella_aerofaciens	Clostridium_mayombei	Clostridium_butyricum	SPC10097
Collinsella_aerofaciens	Clostridium_mayombei	Clostridium_hylemonae	SPC10097
Collinsella_aerofaciens	Clostridium_mayombei	Clostridium_orbiscindens	SPC10097
Collinsella_aerofaciens	Clostridium_mayombei	Ruminococcus_gnavus	SPC10097
Collinsella_aerofaciens	Clostridium_mayombei	Ruminococcus_bromii	SPC10097
Collinsella_aerofaciens	Clostridium_mayombei	Blautia_sp_M25	SPC10097
Collinsella_aerofaciens	Clostridium_butyricum	Clostridium_butyricum	SPC10097
Collinsella_aerofaciens	Clostridium_butyricum	Clostridium_hylemonae	SPC10097
Collinsella_aerofaciens	Clostridium_butyricum	Clostridium_orbiscindens	SPC10097
Collinsella_aerofaciens	Clostridium_butyricum	Ruminococcus_gnavus	SPC10097
Collinsella_aerofaciens	Clostridium_butyricum	Ruminococcus_bromii	SPC10097
Collinsella_aerofaciens	Clostridium_butyricum	Blautia_sp_M25	SPC10097
Collinsella_aerofaciens	Clostridium_hylemonae	Clostridium_hylemonae	SPC10097
Collinsella_aerofaciens	Clostridium_hylemonae	Clostridium_orbiscindens	SPC10097
Collinsella_aerofaciens	Clostridium_hylemonae	Ruminococcus_gnavus	SPC10097
Collinsella_aerofaciens	Clostridium_hylemonae	Ruminococcus_bromii	SPC10097
Collinsella_aerofaciens	Clostridium_hylemonae	Blautia_sp_M25	SPC10097
Collinsella_aerofaciens	Clostridium_orbiscindens	Clostridium_orbiscindens	SPC10097
Collinsella_aerofaciens	Clostridium_orbiscindens	Ruminococcus_gnavus	SPC10097
Collinsella_aerofaciens	Clostridium_orbiscindens	Ruminococcus_bromii	SPC10097
Collinsella_aerofaciens	Clostridium_orbiscindens	Blautia_sp_M25	SPC10097
Collinsella_aerofaciens	Ruminococcus_gnavus	Ruminococcus_gnavus	SPC10097
Collinsella_aerofaciens	Ruminococcus_gnavus	Ruminococcus_bromii	SPC10097
Collinsella_aerofaciens	Ruminococcus_gnavus	Blautia_sp_M25	SPC10097
Coprococcus_comes	Clostridium_tertium	Clostridium_tertium	SPC10304
Coprococcus_comes	Clostridium_tertium	Clostridium_disporicum	SPC10304
Coprococcus_comes	Clostridium_tertium	Clostridium_innocuum	SPC10304
Coprococcus_comes	Clostridium_tertium	Clostridium_mayombei	SPC10304
Coprococcus_comes	Clostridium_tertium	Clostridium_butyricum	SPC10304
Coprococcus_comes	Clostridium_tertium	Clostridium_hylemonae	SPC10304
Coprococcus_comes	Clostridium_tertium	Clostridium_orbiscindens	SPC10304
Coprococcus_comes	Clostridium_tertium	Ruminococcus_gnavus	SPC10304
Coprococcus_comes	Clostridium_tertium	Ruminococcus_bromii	SPC10304
Coprococcus_comes	Clostridium_tertium	Blautia_sp_M25	SPC10304
Coprococcus_comes	Clostridium_disporicum	Clostridium_disporicum	SPC10304
Coprococcus_comes	Clostridium_disporicum	Clostridium_innocuum	SPC10304
Coprococcus_comes	Clostridium_disporicum	Clostridium_mayombei	SPC10304
Coprococcus_comes	Clostridium_disporicum	Clostridium_butyricum	SPC10304
Coprococcus_comes	Clostridium_disporicum	Clostridium_hylemonae	SPC10304
Coprococcus_comes	Clostridium_disporicum	Clostridium_orbiscindens	SPC10304
Coprococcus_comes	Clostridium_disporicum	Ruminococcus_gnavus	SPC10304
Coprococcus_comes	Clostridium_disporicum	Ruminococcus_bromii	SPC10304

TABLE 4a-continued

Coprococcus_comes	Clostridium_disporicum	Blautia_sp_M25	SPC10304
Coprococcus_comes	Clostridium_innocuum	Clostridium_innocuum	SPC10304
Coprococcus_comes	Clostridium_innocuum	Clostridium_mayombei	SPC10304
Coprococcus_comes	Clostridium_innocuum	Clostridium_butyricum	SPC10304
Coprococcus_comes	Clostridium_innocuum	Clostridium_hylemonae	SPC10304
Coprococcus_comes	Clostridium_innocuum	Clostridium_orbiscindens	SPC10304
Coprococcus_comes	Clostridium_innocuum	Ruminococcus_gnavus	SPC10304
Coprococcus_comes	Clostridium_innocuum	Ruminococcus_bromii	SPC10304
Coprococcus_comes	Clostridium_innocuum	Blautia_sp_M25	SPC10304
Coprococcus_comes	Clostridium_mayombei	Clostridium_mayombei	SPC10304
Coprococcus_comes	Clostridium_mayombei	Clostridium_butyricum	SPC10304
Coprococcus_comes	Clostridium_mayombei	Clostridium_hylemonae	SPC10304
Coprococcus_comes	Clostridium_mayombei	Clostridium_orbiscindens	SPC10304
Coprococcus_comes	Clostridium_mayombei	Ruminococcus_gnavus	SPC10304
Coprococcus_comes	Clostridium_mayombei	Ruminococcus_bromii	SPC10304
Coprococcus_comes	Clostridium_mayombei	Blautia_sp_M25	SPC10304
Coprococcus_comes	Clostridium_butyricum	Clostridium_butyricum	SPC10304
Coprococcus_comes	Clostridium_butyricum	Clostridium_hylemonae	SPC10304
Coprococcus_comes	Clostridium_butyricum	Clostridium_orbiscindens	SPC10304
Coprococcus_comes	Clostridium_butyricum	Ruminococcus_gnavus	SPC10304
Coprococcus_comes	Clostridium_butyricum	Ruminococcus_bromii	SPC10304
Coprococcus_comes	Clostridium_butyricum	Blautia_sp_M25	SPC10304
Coprococcus_comes	Clostridium_hylemonae	Clostridium_hylemonae	SPC10304
Coprococcus_comes	Clostridium_hylemonae	Clostridium_orbiscindens	SPC10304
Coprococcus_comes	Clostridium_hylemonae	Ruminococcus_gnavus	SPC10304
Coprococcus_comes	Clostridium_hylemonae	Ruminococcus_bromii	SPC10304
Coprococcus_comes	Clostridium_hylemonae	Blautia_sp_M25	SPC10304
Coprococcus_comes	Clostridium_orbiscindens	Clostridium_orbiscindens	SPC10304
Coprococcus_comes	Clostridium_orbiscindens	Ruminococcus_gnavus	SPC10304
Coprococcus_comes	Clostridium_orbiscindens	Ruminococcus_bromii	SPC10304
Coprococcus_comes	Clostridium_orbiscindens	Blautia_sp_M25	SPC10304
Coprococcus_comes	Ruminococcus_gnavus	Ruminococcus_gnavus	SPC10304
Coprococcus_comes	Ruminococcus_gnavus	Ruminococcus_bromii	SPC10304
Coprococcus_comes	Ruminococcus_gnavus	Blautia_sp_M25	SPC10304
Clostridium_bolteae	Clostridium_tertium	Clostridium_tertium	SPC10325
Clostridium_bolteae	Clostridium_tertium	Clostridium_disporicum	SPC10325
Clostridium_bolteae	Clostridium_tertium	Clostridium_innocuum	SPC10325
Clostridium_bolteae	Clostridium_tertium	Clostridium_mayombei	SPC10325
Clostridium_bolteae	Clostridium_tertium	Clostridium_butyricum	SPC10325
Clostridium_bolteae	Clostridium_tertium	Clostridium_hylemonae	SPC10325
Clostridium_bolteae	Clostridium_tertium	Clostridium_orbiscindens	SPC10325
Clostridium_bolteae	Clostridium_tertium	Ruminococcus_gnavus	SPC10325
Clostridium_bolteae	Clostridium_tertium	Ruminococcus_bromii	SPC10325
Clostridium_bolteae	Clostridium_tertium	Blautia_sp_M25	SPC10325
Clostridium_bolteae	Clostridium_disporicum	Clostridium_disporicum	SPC10325
Clostridium_bolteae	Clostridium_disporicum	Clostridium_innocuum	SPC10325
Clostridium_bolteae	Clostridium_disporicum	Clostridium_mayombei	SPC10325
Clostridium_bolteae	Clostridium_disporicum	Clostridium_butyricum	SPC10325
Clostridium_bolteae	Clostridium_disporicum	Clostridium_hylemonae	SPC10325
Clostridium_bolteae	Clostridium_disporicum	Clostridium_orbiscindens	SPC10325
Clostridium_bolteae	Clostridium_disporicum	Ruminococcus_gnavus	SPC10325
Clostridium_bolteae	Clostridium_disporicum	Ruminococcus_bromii	SPC10325
Clostridium_bolteae	Clostridium_disporicum	Blautia_sp_M25	SPC10325
Clostridium_bolteae	Clostridium_innocuum	Clostridium_innocuum	SPC10325
Clostridium_bolteae	Clostridium_innocuum	Clostridium_mayombei	SPC10325
Clostridium_bolteae	Clostridium_innocuum	Clostridium_butyricum	SPC10325
Clostridium_bolteae	Clostridium_innocuum	Clostridium_hylemonae	SPC10325
Clostridium_bolteae	Clostridium_innocuum	Clostridium_orbiscindens	SPC10325
Clostridium_bolteae	Clostridium_innocuum	Ruminococcus_gnavus	SPC10325
Clostridium_bolteae	Clostridium_innocuum	Ruminococcus_bromii	SPC10325
Clostridium_bolteae	Clostridium_innocuum	Blautia_sp_M25	SPC10325
Clostridium_bolteae	Clostridium_mayombei	Clostridium_mayombei	SPC10325
Clostridium_bolteae	Clostridium_mayombei	Clostridium_butyricum	SPC10325
Clostridium_bolteae	Clostridium_mayombei	Clostridium_hylemonae	SPC10325
Clostridium_bolteae	Clostridium_mayombei	Clostridium_orbiscindens	SPC10325
Clostridium_bolteae	Clostridium_mayombei	Ruminococcus_gnavus	SPC10325
Clostridium_bolteae	Clostridium_mayombei	Ruminococcus_bromii	SPC10325
Clostridium_bolteae	Clostridium_mayombei	Blautia_sp_M25	SPC10325
Clostridium_bolteae	Clostridium_butyricum	Clostridium_butyricum	SPC10325
Clostridium_bolteae	Clostridium_butyricum	Clostridium_hylemonae	SPC10325
Clostridium_bolteae	Clostridium_butyricum	Clostridium_orbiscindens	SPC10325
Clostridium_bolteae	Clostridium_butyricum	Ruminococcus_gnavus	SPC10325
Clostridium_bolteae	Clostridium_butyricum	Ruminococcus_bromii	SPC10325
Clostridium_bolteae	Clostridium_butyricum	Blautia_sp_M25	SPC10325
Clostridium_bolteae	Clostridium_hylemonae	Clostridium_hylemonae	SPC10325
Clostridium_bolteae	Clostridium_hylemonae	Clostridium_orbiscindens	SPC10325
Clostridium_bolteae	Clostridium_hylemonae	Ruminococcus_gnavus	SPC10325
Clostridium_bolteae	Clostridium_hylemonae	Ruminococcus_bromii	SPC10325
Clostridium_bolteae	Clostridium_hylemonae	Blautia_sp_M25	SPC10325
Clostridium_bolteae	Clostridium_orbiscindens	Clostridium_orbiscindens	SPC10325

TABLE 4a-continued

Clostridium_bolteae	Clostridium_orbiscindens	Ruminococcus_gnavus	SPC10325
Clostridium_bolteae	Clostridium_orbiscindens	Ruminococcus_bromii	SPC10325
Clostridium_bolteae	Clostridium_orbiscindens	Blautia_sp_M25	SPC10325
Clostridium_bolteae	Ruminococcus_gnavus	Ruminococcus_gnavus	SPC10325
Clostridium_bolteae	Ruminococcus_gnavus	Ruminococcus_bromii	SPC10325
Clostridium_bolteae	Ruminococcus_gnavus	Blautia_sp_M25	SPC10325
Clostridium_symbiosum	Clostridium_tertium	Clostridium_tertium	SPC10355
Clostridium_symbiosum	Clostridium_tertium	Clostridium_disporicum	SPC10355
Clostridium_symbiosum	Clostridium_tertium	Clostridium_innocuum	SPC10355
Clostridium_symbiosum	Clostridium_tertium	Clostridium_mayombei	SPC10355
Clostridium_symbiosum	Clostridium_tertium	Clostridium_butyricum	SPC10355
Clostridium_symbiosum	Clostridium_tertium	Clostridium_hylemonae	SPC10355
Clostridium_symbiosum	Clostridium_tertium	Clostridium_orbiscindens	SPC10355
Clostridium_symbiosum	Clostridium_tertium	Ruminococcus_gnavus	SPC10355
Clostridium_symbiosum	Clostridium_tertium	Ruminococcus_bromii	SPC10355
Clostridium_symbiosum	Clostridium_tertium	Blautia_sp_M25	SPC10355
Clostridium_symbiosum	Clostridium_disporicum	Clostridium_disporicum	SPC10355
Clostridium_symbiosum	Clostridium_disporicum	Clostridium_innocuum	SPC10355
Clostridium_symbiosum	Clostridium_disporicum	Clostridium_mayombei	SPC10355
Clostridium_symbiosum	Clostridium_disporicum	Clostridium_butyricum	SPC10355
Clostridium_symbiosum	Clostridium_disporicum	Clostridium_hylemonae	SPC10355
Clostridium_symbiosum	Clostridium_disporicum	Clostridium_orbiscindens	SPC10355
Clostridium_symbiosum	Clostridium_disporicum	Ruminococcus_gnavus	SPC10355
Clostridium_symbiosum	Clostridium_disporicum	Ruminococcus_bromii	SPC10355
Clostridium_symbiosum	Clostridium_disporicum	Blautia_sp_M25	SPC10355
Clostridium_symbiosum	Clostridium_innocuum	Clostridium_innocuum	SPC10355
Clostridium_symbiosum	Clostridium_innocuum	Clostridium_mayombei	SPC10355
Clostridium_symbiosum	Clostridium_innocuum	Clostridium_butyricum	SPC10355
Clostridium_symbiosum	Clostridium_innocuum	Clostridium_hylemonae	SPC10355
Clostridium_symbiosum	Clostridium_innocuum	Clostridium_orbiscindens	SPC10355
Clostridium_symbiosum	Clostridium_innocuum	Ruminococcus_gnavus	SPC10355
Clostridium_symbiosum	Clostridium_innocuum	Ruminococcus_bromii	SPC10355
Clostridium_symbiosum	Clostridium_innocuum	Blautia_sp_M25	SPC10355
Clostridium_symbiosum	Clostridium_mayombei	Clostridium_mayombei	SPC10355
Clostridium_symbiosum	Clostridium_mayombei	Clostridium_butyricum	SPC10355
Clostridium_symbiosum	Clostridium_mayombei	Clostridium_hylemonae	SPC10355
Clostridium_symbiosum	Clostridium_mayombei	Clostridium_orbiscindens	SPC10355
Clostridium_symbiosum	Clostridium_mayombei	Ruminococcus_gnavus	SPC10355
Clostridium_symbiosum	Clostridium_mayombei	Ruminococcus_bromii	SPC10355
Clostridium_symbiosum	Clostridium_mayombei	Blautia_sp_M25	SPC10355
Clostridium_symbiosum	Clostridium_butyricum	Clostridium_butyricum	SPC10355
Clostridium_symbiosum	Clostridium_butyricum	Clostridium_hylemonae	SPC10355
Clostridium_symbiosum	Clostridium_butyricum	Clostridium_orbiscindens	SPC10355
Clostridium_symbiosum	Clostridium_butyricum	Ruminococcus_gnavus	SPC10355
Clostridium_symbiosum	Clostridium_butyricum	Ruminococcus_bromii	SPC10355
Clostridium_symbiosum	Clostridium_butyricum	Blautia_sp_M25	SPC10355
Clostridium_symbiosum	Clostridium_hylemonae	Clostridium_hylemonae	SPC10355
Clostridium_symbiosum	Clostridium_hylemonae	Clostridium_orbiscindens	SPC10355
Clostridium_symbiosum	Clostridium_hylemonae	Ruminococcus_gnavus	SPC10355
Clostridium_symbiosum	Clostridium_hylemonae	Ruminococcus_bromii	SPC10355
Clostridium_symbiosum	Clostridium_hylemonae	Blautia_sp_M25	SPC10355
Clostridium_symbiosum	Clostridium_orbiscindens	Clostridium_orbiscindens	SPC10355
Clostridium_symbiosum	Clostridium_orbiscindens	Ruminococcus_gnavus	SPC10355
Clostridium_symbiosum	Clostridium_orbiscindens	Ruminococcus_bromii	SPC10355
Clostridium_symbiosum	Clostridium_orbiscindens	Blautia_sp_M25	SPC10355
Clostridium_symbiosum	Ruminococcus_gnavus	Ruminococcus_gnavus	SPC10355
Clostridium_symbiosum	Ruminococcus_gnavus	Ruminococcus_bromii	SPC10355
Clostridium_symbiosum	Ruminococcus_gnavus	Blautia_sp_M25	SPC10355
Faecalibacterium_prausnitzii	Clostridium_tertium	Clostridium_tertium	SPC10386
Faecalibacterium_prausnitzii	Clostridium_tertium	Clostridium_disporicum	SPC10386
Faecalibacterium_prausnitzii	Clostridium_tertium	Clostridium_innocuum	SPC10386
Faecalibacterium_prausnitzii	Clostridium_tertium	Clostridium_mayombei	SPC10386
Faecalibacterium_prausnitzii	Clostridium_tertium	Clostridium_butyricum	SPC10386
Faecalibacterium_prausnitzii	Clostridium_tertium	Clostridium_hylemonae	SPC10386
Faecalibacterium_prausnitzii	Clostridium_tertium	Clostridium_orbiscindens	SPC10386
Faecalibacterium_prausnitzii	Clostridium_tertium	Ruminococcus_gnavus	SPC10386
Faecalibacterium_prausnitzii	Clostridium_tertium	Ruminococcus_bromii	SPC10386
Faecalibacterium_prausnitzii	Clostridium_tertium	Blautia_sp_M25	SPC10386
Faecalibacterium_prausnitzii	Clostridium_disporicum	Clostridium_disporicum	SPC10386
Faecalibacterium_prausnitzii	Clostridium_disporicum	Clostridium_innocuum	SPC10386
Faecalibacterium_prausnitzii	Clostridium_disporicum	Clostridium_mayombei	SPC10386
Faecalibacterium_prausnitzii	Clostridium_disporicum	Clostridium_butyricum	SPC10386
Faecalibacterium_prausnitzii	Clostridium_disporicum	Clostridium_hylemonae	SPC10386
Faecalibacterium_prausnitzii	Clostridium_disporicum	Clostridium_orbiscindens	SPC10386
Faecalibacterium_prausnitzii	Clostridium_disporicum	Ruminococcus_gnavus	SPC10386
Faecalibacterium_prausnitzii	Clostridium_disporicum	Ruminococcus_bromii	SPC10386
Faecalibacterium_prausnitzii	Clostridium_disporicum	Blautia_sp_M25	SPC10386
Faecalibacterium_prausnitzii	Clostridium_innocuum	Clostridium_innocuum	SPC10386
Faecalibacterium_prausnitzii	Clostridium_innocuum	Clostridium_mayombei	SPC10386
Faecalibacterium_prausnitzii	Clostridium_innocuum	Clostridium_butyricum	SPC10386

TABLE 4a-continued

Faecalibacterium_prausnitzii	*Clostridium_innocuum*	*Clostridium_hylemonae*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_innocuum*	*Clostridium_orbiscindens*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_innocuum*	*Ruminococcus_gnavus*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_innocuum*	*Ruminococcus_bromii*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_innocuum*	*Blautia*_sp_M25	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_mayombei*	*Clostridium_mayombei*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_mayombei*	*Clostridium_butyricum*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_mayombei*	*Clostridium_hylemonae*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_mayombei*	*Clostridium_orbiscindens*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_mayombei*	*Ruminococcus_gnavus*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_mayombei*	*Ruminococcus_bromii*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_mayombei*	*Blautia*_sp_M25	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_butyricum*	*Clostridium_butyricum*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_butyricum*	*Clostridium_hylemonae*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_butyricum*	*Clostridium_orbiscindens*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_butyricum*	*Ruminococcus_gnavus*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_butyricum*	*Ruminococcus_bromii*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_butyricum*	*Blautia*_sp_M25	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_hylemonae*	*Clostridium_hylemonae*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_hylemonae*	*Clostridium_orbiscindens*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_hylemonae*	*Ruminococcus_gnavus*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_hylemonae*	*Ruminococcus_bromii*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_hylemonae*	*Blautia*_sp_M25	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_orbiscindens*	*Clostridium_orbiscindens*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_orbiscindens*	*Ruminococcus_gnavus*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_orbiscindens*	*Ruminococcus_bromii*	SPC10386
Faecalibacterium_prausnitzii	*Clostridium_orbiscindens*	*Blautia*_sp_M25	SPC10386
Faecalibacterium_prausnitzii	*Ruminococcus_gnavus*	*Ruminococcus_gnavus*	SPC10386
Faecalibacterium_prausnitzii	*Ruminococcus_gnavus*	*Ruminococcus_bromii*	SPC10386
Faecalibacterium_prausnitzii	*Ruminococcus_gnavus*	*Blautia*_sp_M25	SPC10386
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_tertium*	*Clostridium_tertium*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_tertium*	*Clostridium_disporicum*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_tertium*	*Clostridium_innocuum*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_tertium*	*Clostridium_mayombei*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_tertium*	*Clostridium_butyricum*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_tertium*	*Clostridium_hylemonae*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_tertium*	*Clostridium_orbiscindens*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_tertium*	*Ruminococcus_gnavus*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_tertium*	*Ruminococcus_bromii*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_tertium*	*Blautia*_sp_M25	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_disporicum*	*Clostridium_disporicum*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_disporicum*	*Clostridium_innocuum*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_disporicum*	*Clostridium_mayombei*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_disporicum*	*Clostridium_butyricum*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_disporicum*	*Clostridium_hylemonae*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_disporicum*	*Clostridium_orbiscindens*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_disporicum*	*Ruminococcus_gnavus*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_disporicum*	*Ruminococcus_bromii*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_disporicum*	*Blautia*_sp_M25	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_innocuum*	*Clostridium_innocuum*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_innocuum*	*Clostridium_mayombei*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_innocuum*	*Clostridium_butyricum*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_innocuum*	*Clostridium_hylemonae*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_innocuum*	*Clostridium_orbiscindens*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_innocuum*	*Ruminococcus_gnavus*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_innocuum*	*Ruminococcus_bromii*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_innocuum*	*Blautia*_sp_M25	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_mayombei*	*Clostridium_mayombei*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_mayombei*	*Clostridium_butyricum*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_mayombei*	*Clostridium_hylemonae*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_mayombei*	*Clostridium_orbiscindens*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_mayombei*	*Ruminococcus_gnavus*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_mayombei*	*Ruminococcus_bromii*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_mayombei*	*Blautia*_sp_M25	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_butyricum*	*Clostridium_butyricum*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_butyricum*	*Clostridium_hylemonae*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_butyricum*	*Clostridium_orbiscindens*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_butyricum*	*Ruminococcus_gnavus*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_butyricum*	*Ruminococcus_bromii*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_butyricum*	*Blautia*_sp_M25	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_hylemonae*	*Clostridium_hylemonae*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_hylemonae*	*Clostridium_orbiscindens*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_hylemonae*	*Ruminococcus_gnavus*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_hylemonae*	*Ruminococcus_bromii*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_hylemonae*	*Blautia*_sp_M25	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_orbiscindens*	*Clostridium_orbiscindens*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_orbiscindens*	*Ruminococcus_gnavus*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_orbiscindens*	*Ruminococcus_bromii*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Clostridium_orbiscindens*	*Blautia*_sp_M25	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Ruminococcus_gnavus*	*Ruminococcus_gnavus*	SPC10390

TABLE 4a-continued

Lachnospiraceae_*bacterium*_5_1_57FAA	*Ruminococcus_gnavus*	*Ruminococcus_bromii*	SPC10390
Lachnospiraceae_*bacterium*_5_1_57FAA	*Ruminococcus_gnavus*	*Blautia*_sp_M25	SPC10390
Blautia_producta	*Clostridium_tertium*	*Clostridium_tertium*	SPC10415
Blautia_producta	*Clostridium_tertium*	*Clostridium_disporicum*	SPC10415
Blautia_producta	*Clostridium_tertium*	*Clostridium_innocuum*	SPC10415
Blautia_producta	*Clostridium_tertium*	*Clostridium_mayombei*	SPC10415
Blautia_producta	*Clostridium_tertium*	*Clostridium_butyricum*	SPC10415
Blautia_producta	*Clostridium_tertium*	*Clostridium_hylemonae*	SPC10415
Blautia_producta	*Clostridium_tertium*	*Clostridium_orbiscindens*	SPC10415
Blautia_producta	*Clostridium_tertium*	*Ruminococcus_gnavus*	SPC10415
Blautia_producta	*Clostridium_tertium*	*Ruminococcus_bromii*	SPC10415
Blautia_producta	*Clostridium_tertium*	*Blautia*_sp_M25	SPC10415
Blautia_producta	*Clostridium_disporicum*	*Clostridium_disporicum*	SPC10415
Blautia_producta	*Clostridium_disporicum*	*Clostridium_innocuum*	SPC10415
Blautia_producta	*Clostridium_disporicum*	*Clostridium_mayombei*	SPC10415
Blautia_producta	*Clostridium_disporicum*	*Clostridium_butyricum*	SPC10415
Blautia_producta	*Clostridium_disporicum*	*Clostridium_hylemonae*	SPC10415
Blautia_producta	*Clostridium_disporicum*	*Clostridium_orbiscindens*	SPC10415
Blautia_producta	*Clostridium_disporicum*	*Ruminococcus_gnavus*	SPC10415
Blautia_producta	*Clostridium_disporicum*	*Ruminococcus_bromii*	SPC10415
Blautia_producta	*Clostridium_disporicum*	*Blautia*_sp_M25	SPC10415
Blautia_producta	*Clostridium_innocuum*	*Clostridium_innocuum*	SPC10415
Blautia_producta	*Clostridium_innocuum*	*Clostridium_mayombei*	SPC10415
Blautia_producta	*Clostridium_innocuum*	*Clostridium_butyricum*	SPC10415
Blautia_producta	*Clostridium_innocuum*	*Clostridium_hylemonae*	SPC10415
Blautia_producta	*Clostridium_innocuum*	*Clostridium_orbiscindens*	SPC10415
Blautia_producta	*Clostridium_innocuum*	*Ruminococcus_gnavus*	SPC10415
Blautia_producta	*Clostridium_innocuum*	*Ruminococcus_bromii*	SPC10415
Blautia_producta	*Clostridium_innocuum*	*Blautia*_sp_M25	SPC10415
Blautia_producta	*Clostridium_mayombei*	*Clostridium_mayombei*	SPC10415
Blautia_producta	*Clostridium_mayombei*	*Clostridium_butyricum*	SPC10415
Blautia_*producta*	*Clostridium_mayombei*	*Clostridium_hylemonae*	SPC10415
Blautia_producta	*Clostridium_mayombei*	*Clostridium_orbiscindens*	SPC10415
Blautia_producta	*Clostridium_mayombei*	*Ruminococcus_gnavus*	SPC10415
Blautia_producta	*Clostridium_mayombei*	*Ruminococcus_bromii*	SPC10415
Blautia_producta	*Clostridium_mayombei*	*Blautia*_sp_M25	SPC10415
Blautia_producta	*Clostridium_butyricum*	*Clostridium_butyricum*	SPC10415
Blautia_producta	*Clostridium_butyricum*	*Clostridium_hylemonae*	SPC10415
Blautia_producta	*Clostridium_butyricum*	*Clostridium_orbiscindens*	SPC10415
Blautia_producta	*Clostridium_butyricum*	*Ruminococcus_gnavus*	SPC10415
Blautia_producta	*Clostridium_butyricum*	*Ruminococcus_bromii*	SPC10415
Blautia_producta	*Clostridium_butyricum*	*Blautia*_sp_M25	SPC10415
Blautia_producta	*Clostridium_hylemonae*	*Clostridium_hylemonae*	SPC10415
Blautia_producta	*Clostridium_hylemonae*	*Clostridium_orbiscindens*	SPC10415
Blautia_producta	*Clostridium_hylemonae*	*Ruminococcus_gnavus*	SPC10415
Blautia_producta	*Clostridium_hylemonae*	*Ruminococcus_bromii*	SPC10415
Blautia_producta	*Clostridium_hylemonae*	*Blautia*_sp_M25	SPC10415
Blautia_producta	*Clostridium_orbiscindens*	*Clostridium_orbiscindens*	SPC10415
Blautia_producta	*Clostridium_orbiscindens*	*Ruminococcus_gnavus*	SPC10415
Blautia_producta	*Clostridium_orbiscindens*	*Ruminococcus_bromii*	SPC10415
Blautia_producta	*Clostridium_orbiscindens*	*Blautia*_sp_M25	SPC10415
Blautia_producta	*Ruminococcus_gnavus*	*Ruminococcus_gnavus*	SPC10415
Blautia_producta	*Ruminococcus_gnavus*	*Ruminococcus_bromii*	SPC10415
Blautia_producta	*Ruminococcus_gnavus*	*Blautia*_sp_M25	SPC10415
Eubacterium_rectale	*Clostridium_tertium*	*Clostridium_tertium*	SPC10567
Eubacterium_rectale	*Clostridium_tertium*	*Clostridium_disporicum*	SPC10567
Eubacterium_rectale	*Clostridium_tertium*	*Clostridium_innocuum*	SPC10567
Eubacterium_rectale	*Clostridium_tertium*	*Clostridium_mayombei*	SPC10567
Eubacterium_rectale	*Clostridium_tertium*	*Clostridium_butyricum*	SPC10567
Eubacterium_rectale	*Clostridium_tertium*	*Clostridium_hylemonae*	SPC10567
Eubacterium_rectale	*Clostridium_tertium*	*Clostridium_orbiscindens*	SPC10567
Eubacterium_rectale	*Clostridium_tertium*	*Ruminococcus_gnavus*	SPC10567
Eubacterium_rectale	*Clostridium_tertium*	*Ruminococcus_bromii*	SPC10567
Eubacterium_rectale	*Clostridium_tertium*	*Blautia*_sp_M25	SPC10567
Eubacterium_rectale	*Clostridium_disporicum*	*Clostridium_disporicum*	SPC10567
Eubacterium_rectale	*Clostridium_disporicum*	*Clostridium_innocuum*	SPC10567
Eubacterium_rectale	*Clostridium_disporicum*	*Clostridium_mayombei*	SPC10567
Eubacterium_rectale	*Clostridium_disporicum*	*Clostridium_butyricum*	SPC10567
Eubacterium_rectale	*Clostridium_disporicum*	*Clostridium_hylemonae*	SPC10567
Eubacterium_rectale	*Clostridium_disporicum*	*Clostridium_orbiscindens*	SPC10567
Eubacterium_rectale	*Clostridium_disporicum*	*Ruminococcus_gnavus*	SPC10567
Eubacterium_rectale	*Clostridium_disporicum*	*Ruminococcus_bromii*	SPC10567
Eubacterium_rectale	*Clostridium_disporicum*	*Blautia*_sp_M25	SPC10567
Eubacterium_rectale	*Clostridium_innocuum*	*Clostridium_innocuum*	SPC10567
Eubacterium_rectale	*Clostridium_innocuum*	*Clostridium_mayombei*	SPC10567
Eubacterium_rectale	*Clostridium_innocuum*	*Clostridium_butyricum*	SPC10567
Eubacterium_rectale	*Clostridium_innocuum*	*Clostridium_hylemonae*	SPC10567
Eubacterium_rectale	*Clostridium_innocuum*	*Clostridium_orbiscindens*	SPC10567
Eubacterium_rectale	*Clostridium_innocuum*	*Ruminococcus_gnavus*	SPC10567
Eubacterium_rectale	*Clostridium_innocuum*	*Ruminococcus_bromii*	SPC10567

TABLE 4a-continued

Eubacterium_rectale	Clostridium_innocuum	Blautia_sp_M25	SPC10567
Eubacterium_rectale	Clostridium_mayombei	Clostridium_mayombei	SPC10567
Eubacterium_rectale	Clostridium_mayombei	Clostridium_butyricum	SPC10567
Eubacterium_rectale	Clostridium_mayombei	Clostridium_hylemonae	SPC10567
Eubacterium_rectale	Clostridium_mayombei	Clostridium_orbiscindens	SPC10567
Eubacterium_rectale	Clostridium_mayombei	Ruminococcus_gnavus	SPC10567
Eubacterium_rectale	Clostridium_mayombei	Ruminococcus_bromii	SPC10567
Eubacterium_rectale	Clostridium_mayombei	Blautia_sp_M25	SPC10567
Eubacterium_rectale	Clostridium_butyricum	Clostridium_butyricum	SPC10567
Eubacterium_rectale	Clostridium_butyricum	Clostridium_hylemonae	SPC10567
Eubacterium_rectale	Clostridium_butyricum	Clostridium_orbiscindens	SPC10567
Eubacterium_rectale	Clostridium_butyricum	Ruminococcus_gnavus	SPC10567
Eubacterium_rectale	Clostridium_butyricum	Ruminococcus_bromii	SPC10567
Eubacterium_rectale	Clostridium_butyricum	Blautia_sp_M25	SPC10567
Eubacterium_rectale	Clostridium_hylemonae	Clostridium_hylemonae	SPC10567
Eubacterium_rectale	Clostridium_hylemonae	Clostridium_orbiscindens	SPC10567
Eubacterium_rectale	Clostridium_hylemonae	Ruminococcus_gnavus	SPC10567
Eubacterium_rectale	Clostridium_hylemonae	Ruminococcus_bromii	SPC10567
Eubacterium_rectale	Clostridium_hylemonae	Blautia_sp_M25	SPC10567
Eubacterium_rectale	Clostridium_orbiscindens	Clostridium_orbiscindens	SPC10567
Eubacterium_rectale	Clostridium_orbiscindens	Ruminococcus_gnavus	SPC10567
Eubacterium_rectale	Clostridium_orbiscindens	Ruminococcus_bromii	SPC10567
Eubacterium_rectale	Clostridium_orbiscindens	Blautia_sp_M25	SPC10567
Eubacterium_rectale	Ruminococcus_gnavus	Ruminococcus_gnavus	SPC10567
Eubacterium_rectale	Ruminococcus_gnavus	Ruminococcus_bromii	SPC10567
Eubacterium_rectale	Ruminococcus_gnavus	Blautia_sp_M25	SPC10567
Clostridium_tertium	Collinsella_aerofaciens	Collinsella_aerofaciens	SPC10155
Clostridium_tertium	Collinsella_aerofaciens	Coprococcus_comes	SPC10155
Clostridium_tertium	Collinsella_aerofaciens	Clostridium_bolteae	SPC10155
Clostridium_tertium	Collinsella_aerofaciens	Clostridium_symbiosum	SPC10155
Clostridium_tertium	Collinsella_aerofaciens	Faecalibacterium_prausnitzii	SPC10155
Clostridium_tertium	Collinsella_aerofaciens	Lachnospiraceae_bacterium_5_1_57FAA	SPC10155
Clostridium_tertium	Collinsella_aerofaciens	Blautia_producta	SPC10155
Clostridium_tertium	Collinsella_aerofaciens	Eubacterium_rectale	SPC10155
Clostridium_tertium	Coprococcus_comes	Coprococcus_comes	SPC10155
Clostridium_tertium	Coprococcus_comes	Clostridium_bolteae	SPC10155
Clostridium_tertium	Coprococcus_comes	Clostridium_symbiosum	SPC10155
Clostridium_tertium	Coprococcus_comes	Faecalibacterium_prausnitzii	SPC10155
Clostridium_tertium	Coprococcus_comes	Lachnospiraceae_bacterium_5_1_57FAA	SPC10155
Clostridium_tertium	Coprococcus_comes	Blautia_producta	SPC10155
Clostridium_tertium	Coprococcus_comes	Eubacterium_rectale	SPC10155
Clostridium_tertium	Clostridium_bolteae	Clostridium_bolteae	SPC10155
Clostridium_tertium	Clostridium_bolteae	Clostridium_symbiosum	SPC10155
Clostridium_tertium	Clostridium_bolteae	Faecalibacterium_prausnitzii	SPC10155
Clostridium_tertium	Clostridium_bolteae	Lachnospiraceae_bacterium_5_1_57FAA	SPC10155
Clostridium_tertium	Clostridium_bolteae	Blautia_producta	SPC10155
Clostridium_tertium	Clostridium_bolteae	Eubacterium_rectale	SPC10155
Clostridium_tertium	Clostridium_symbiosum	Clostridium_symbiosum	SPC10155
Clostridium_tertium	Clostridium_symbiosum	Faecalibacterium_prausnitzii	SPC10155
Clostridium_tertium	Clostridium_symbiosum	Lachnospiraceae_bacterium_5_1_57FAA	SPC10155
Clostridium_tertium	Clostridium_symbiosum	Blautia_producta	SPC10155
Clostridium_tertium	Clostridium_symbiosum	Eubacterium_rectale	SPC10155
Clostridium_tertium	Faecalibacterium_prausnitzii	Faecalibacterium_prausnitzii	SPC10155
Clostridium_tertium	Faecalibacterium_prausnitzii	Lachnospiraceae_bacterium_5_1_57FAA	SPC10155
Clostridium_tertium	Faecalibacterium_prausnitzii	Blautia_producta	SPC10155
Clostridium_tertium	Faecalibacterium_prausnitzii	Eubacterium_rectale	SPC10155
Clostridium_tertium	Lachnospiraceae_bacterium_5_1_57FAA	Lachnospiraceae_bacterium_5_1_57FAA	SPC10155
Clostridium_tertium	Lachnospiraceae_bacterium_5_1_57FAA	Blautia_producta	SPC10155
Clostridium_tertium	Lachnospiraceae_bacterium_5_1_57FAA	Eubacterium_rectale	SPC10155
Clostridium_tertium	Blautia_producta	Blautia_producta	SPC10155
Clostridium_tertium	Blautia_producta	Eubacterium_rectale	SPC10155
Clostridium_tertium	Eubacterium_rectale	Eubacterium_rectale	SPC10155
Clostridium_disporicum	Collinsella_aerofaciens	Collinsella_aerofaciens	SPC10167
Clostridium_disporicum	Collinsella_aerofaciens	Coprococcus_comes	SPC10167
Clostridium_disporicum	Collinsella_aerofaciens	Clostridium_bolteae	SPC10167
Clostridium_disporicum	Collinsella_aerofaciens	Clostridium_symbiosum	SPC10167
Clostridium_disporicum	Collinsella_aerofaciens	Faecalibacterium_prausnitzii	SPC10167
Clostridium_disporicum	Collinsella_aerofaciens	Lachnospiraceae_bacterium_5_1_57FAA	SPC10167
Clostridium_disporicum	Collinsella_aerofaciens	Blautia_producta	SPC10167
Clostridium_disporicum	Collinsella_aerofaciens	Eubacterium_rectale	SPC10167
Clostridium_disporicum	Coprococcus_comes	Coprococcus_comes	SPC10167
Clostridium_disporicum	Coprococcus_comes	Clostridium_bolteae	SPC10167
Clostridium_disporicum	Coprococcus_comes	Clostridium_symbiosum	SPC10167
Clostridium_disporicum	Coprococcus_comes	Faecalibacterium_prausnitzii	SPC10167
Clostridium_disporicum	Coprococcus_comes	Lachnospiraceae_bacterium_5_1_57FAA	SPC10167
Clostridium_disporicum	Coprococcus_comes	Blautia_producta	SPC10167
Clostridium_disporicum	Coprococcus_comes	Eubacterium_rectale	SPC10167
Clostridium_disporicum	Clostridium_bolteae	Clostridium_bolteae	SPC10167
Clostridium_disporicum	Clostridium_bolteae	Clostridium_symbiosum	SPC10167
Clostridium_disporicum	Clostridium_bolteae	Faecalibacterium_prausnitzii	SPC10167

TABLE 4a-continued

Clostridium_disporicum	Clostridium_bolteae	Lachnospiraceae_bacterium_5_1_57FAA	SPC10167
Clostridium_disporicum	Clostridium_bolteae	Blautia_producta	SPC10167
Clostridium_disporicum	Clostridium_bolteae	Eubacterium_rectale	SPC10167
Clostridium_disporicum	Clostridium_symbiosum	Clostridium_symbiosum	SPC10167
Clostridium_disporicum	Clostridium_symbiosum	Faecalibacterium_prausnitzii	SPC10167
Clostridium_disporicum	Clostridium_symbiosum	Lachnospiraceae_bacterium_5_1_57FAA	SPC10167
Clostridium_disporicum	Clostridium_symbiosum	Blautia_producta	SPC10167
Clostridium_disporicum	Clostridium_symbiosum	Eubacterium_rectale	SPC10167
Clostridium_disporicum	Faecalibacterium_prausnitzii	Faecalibacterium_prausnitzii	SPC10167
Clostridium_disporicum	Faecalibacterium_prausnitzii	Lachnospiraceae_bacterium_5_1_57FAA	SPC10167
Clostridium_disporicum	Faecalibacterium_prausnitzii	Blautia_producta	SPC10167
Clostridium_disporicum	Faecalibacterium_prausnitzii	Eubacterium_rectale	SPC10167
Clostridium_disporicum	Lachnospiraceae_bacterium_5_1_57FAA	Lachnospiraceae_bacterium_5_1_57FAA	SPC10167
Clostridium_disporicum	Lachnospiraceae_bacterium_5_1_57FAA	Blautia_producta	SPC10167
Clostridium_disporicum	Lachnospiraceae_bacterium_5_1_57FAA	Eubacterium_rectale	SPC10167
Clostridium_disporicum	Blautia_producta	Blautia_producta	SPC10167
Clostridium_disporicum	Blautia_producta	Eubacterium_rectale	SPC10167
Clostridium_disporicum	Eubacterium_rectale	Eubacterium_rectale	SPC10167
Clostridium_innocuum	Collinsella_aerofaciens	Collinsella_aerofaciens	SPC10202
Clostridium_innocuum	Collinsella_aerofaciens	Coprococcus_comes	SPC10202
Clostridium_innocuum	Collinsella_aerofaciens	Clostridium_bolteae	SPC10202
Clostridium_innocuum	Collinsella_aerofaciens	Clostridium_symbiosum	SPC10202
Clostridium_innocuum	Collinsella_aerofaciens	Faecalibacterium_prausnitzii	SPC10202
Clostridium_innocuum	Collinsella_aerofaciens	Lachnospiraceae_bacterium_5_1_57FAA	SPC10202
Clostridium_innocuum	Collinsella_aerofaciens	Blautia_producta	SPC10202
Clostridium_innocuum	Collinsella_aerofaciens	Eubacterium_rectale	SPC10202
Clostridium_innocuum	Coprococcus_comes	Coprococcus_comes	SPC10202
Clostridium_innocuum	Coprococcus_comes	Clostridium_bolteae	SPC10202
Clostridium_innocuum	Coprococcus_comes	Clostridium_symbiosum	SPC10202
Clostridium_innocuum	Coprococcus_comes	Faecalibacterium_prausnitzii	SPC10202
Clostridium_innocuum	Coprococcus_comes	Lachnospiraceae_bacterium_5_1_57FAA	SPC10202
Clostridium_innocuum	Coprococcus_comes	Blautia_producta	SPC10202
Clostridium_innocuum	Coprococcus_comes	Eubacterium_rectale	SPC10202
Clostridium_innocuum	Clostridium_bolteae	Clostridium_bolteae	SPC10202
Clostridium_innocuum	Clostridium_bolteae	Clostridium_symbiosum	SPC10202
Clostridium_innocuum	Clostridium_bolteae	Faecalibacterium_prausnitzii	SPC10202
Clostridium_innocuum	Clostridium_bolteae	Lachnospiraceae_bacterium_5_1_57FAA	SPC10202
Clostridium_innocuum	Clostridium_bolteae	Blautia_producta	SPC10202
Clostridium_innocuum	Clostridium_bolteae	Eubacterium_rectale	SPC10202
Clostridium_innocuum	Clostridium_symbiosum	Clostridium_symbiosum	SPC10202
Clostridium_innocuum	Clostridium_symbiosum	Faecalibacterium_prausnitzii	SPC10202
Clostridium_innocuum	Clostridium_symbiosum	Lachnospiraceae_bacterium_5_1_57FAA	SPC10202
Clostridium_innocuum	Clostridium_symbiosum	Blautia_producta	SPC10202
Clostridium_innocuum	Clostridium_symbiosum	Eubacterium_rectale	SPC10202
Clostridium_innocuum	Faecalibacterium_prausnitzii	Faecalibacterium_prausnitzii	SPC10202
Clostridium_innocuum	Faecalibacterium_prausnitzii	Lachnospiraceae_bacterium_5_1_57FAA	SPC10202
Clostridium_innocuum	Faecalibacterium_prausnitzii	Blautia_producta	SPC10202
Clostridium_innocuum	Faecalibacterium_prausnitzii	Eubacterium_rectale	SPC10202
Clostridium_innocuum	Lachnospiraceae_bacterium_5_1_57FAA	Lachnospiraceae_bacterium_5_1_57FAA	SPC10202
Clostridium_innocuum	Lachnospiraceae_bacterium_5_1_57FAA	Blautia_producta	SPC10202
Clostridium_innocuum	Lachnospiraceae_bacterium_5_1_57FAA	Eubacterium_rectale	SPC10202
Clostridium_innocuum	Blautia_producta	Blautia_producta	SPC10202
Clostridium_innocuum	Blautia_producta	Eubacterium_rectale	SPC10202
Clostridium_innocuum	Eubacterium_rectale	Eubacterium_rectale	SPC10202
Clostridium_mayombei	Collinsella_aerofaciens	Collinsella_aerofaciens	SPC10238
Clostridium_mayombei	Collinsella_aerofaciens	Coprococcus_comes	SPC10238
Clostridium_mayombei	Collinsella_aerofaciens	Clostridium_bolteae	SPC10238
Clostridium_mayombei	Collinsella_aerofaciens	Clostridium_symbiosum	SPC10238
Clostridium_mayombei	Collinsella_aerofaciens	Faecalibacterium_prausnitzii	SPC10238
Clostridium_mayombei	Collinsella_aerofaciens	Lachnospiraceae_bacterium_5_1_57FAA	SPC10238
Clostridium_mayombei	Collinsella_aerofaciens	Blautia_producta	SPC10238
Clostridium_mayombei	Collinsella_aerofaciens	Eubacterium_rectale	SPC10238
Clostridium_mayombei	Coprococcus_comes	Coprococcus_comes	SPC10238
Clostridium_mayombei	Coprococcus_comes	Clostridium_bolteae	SPC10238
Clostridium_mayombei	Coprococcus_comes	Clostridium_symbiosum	SPC10238
Clostridium_mayombei	Coprococcus_comes	Faecalibacterium_prausnitzii	SPC10238
Clostridium_mayombei	Coprococcus_comes	Lachnospiraceae_bacterium_5_1_57FAA	SPC10238
Clostridium_mayombei	Coprococcus_comes	Blautia_producta	SPC10238
Clostridium_mayombei	Coprococcus_comes	Eubacterium_rectale	SPC10238
Clostridium_mayombei	Clostridium_bolteae	Clostridium_bolteae	SPC10238
Clostridium_mayombei	Clostridium_bolteae	Clostridium_symbiosum	SPC10238
Clostridium_mayombei	Clostridium_bolteae	Faecalibacterium_prausnitzii	SPC10238
Clostridium_mayombei	Clostridium_bolteae	Lachnospiraceae_bacterium_5_1_57FAA	SPC10238
Clostridium_mayombei	Clostridium_bolteae	Blautia_producta	SPC10238
Clostridium_mayombei	Clostridium_bolteae	Eubacterium_rectale	SPC10238
Clostridium_mayombei	Clostridium_symbiosum	Clostridium_symbiosum	SPC10238
Clostridium_mayombei	Clostridium_symbiosum	Faecalibacterium_prausnitzii	SPC10238
Clostridium_mayombei	Clostridium_symbiosum	Lachnospiraceae_bacterium_5_1_57FAA	SPC10238
Clostridium_mayombei	Clostridium_symbiosum	Blautia_producta	SPC10238
Clostridium_mayombei	Clostridium_symbiosum	Eubacterium_rectale	SPC10238

TABLE 4a-continued

Clostridium_mayombei	Faecalibacterium_prausnitzii	Faecalibacterium_prausnitzii	SPC10238
Clostridium_mayombei	Faecalibacterium_prausnitzii	Lachnospiraceae_bacterium_5_1_57FAA	SPC10238
Clostridium_mayombei	Faecalibacterium_prausnitzii	Blautia_producta	SPC10238
Clostridium_mayombei	Faecalibacterium_prausnitzii	Eubacterium_rectale	SPC10238
Clostridium_mayombei	Lachnospiraceae_bacterium_5_1_57FAA	Lachnospiraceae_bacterium_5_1_57FAA	SPC10238
Clostridium_mayombei	Lachnospiraceae_bacterium_5_1_57FAA	Blautia_producta	SPC10238
Clostridium_mayombei	Lachnospiraceae_bacterium_5_1_57FAA	Eubacterium_rectale	SPC10238
Clostridium_mayombei	Blautia_producta	Blautia_producta	SPC10238
Clostridium_mayombei	Blautia_producta	Eubacterium_rectale	SPC10238
Clostridium_mayombei	Eubacterium_rectale	Eubacterium_rectale	SPC10238
Clostridium_butyricum	Collinsella_aerofaciens	Collinsella_aerofaciens	SPC10256
Clostridium_butyricum	Collinsella_aerofaciens	Coprococcus_comes	SPC10256
Clostridium_butyricum	Collinsella_aerofaciens	Clostridium_bolteae	SPC10256
Clostridium_butyricum	Collinsella_aerofaciens	Clostridium_symbiosum	SPC10256
Clostridium_butyricum	Collinsella_aerofaciens	Faecalibacterium_prausnitzii	SPC10256
Clostridium_butyricum	Collinsella_aerofaciens	Lachnospiraceae_bacterium_5_1_57FAA	SPC10256
Clostridium_butyricum	Collinsella_aerofaciens	Blautia_producta	SPC10256
Clostridium_butyricum	Collinsella_aerofaciens	Eubacterium_rectale	SPC10256
Clostridium_butyricum	Coprococcus_comes	Coprococcus_comes	SPC10256
Clostridium_butyricum	Coprococcus_comes	Clostridium_bolteae	SPC10256
Clostridium_butyricum	Coprococcus_comes	Clostridium_symbiosum	SPC10256
Clostridium_butyricum	Coprococcus_comes	Faecalibacterium_prausnitzii	SPC10256
Clostridium_butyricum	Coprococcus_comes	Lachnospiraceae_bacterium_5_1_57FAA	SPC10256
Clostridium_butyricum	Coprococcus_comes	Blautia_producta	SPC10256
Clostridium_butyricum	Coprococcus_comes	Eubacterium_rectale	SPC10256
Clostridium_butyricum	Clostridium_bolteae	Clostridium_bolteae	SPC10256
Clostridium_butyricum	Clostridium_bolteae	Clostridium_symbiosum	SPC10256
Clostridium_butyricum	Clostridium_bolteae	Faecalibacterium_prausnitzii	SPC10256
Clostridium_butyricum	Clostridium_bolteae	Lachnospiraceae_bacterium_5_1_57FAA	SPC10256
Clostridium_butyricum	Clostridium_bolteae	Blautia_producta	SPC10256
Clostridium_butyricum	Clostridium_bolteae	Eubacterium_rectale	SPC10256
Clostridium_butyricum	Clostridium_symbiosum	Clostridium_symbiosum	SPC10256
Clostridium_butyricum	Clostridium_symbiosum	Faecalibacterium_prausnitzii	SPC10256
Clostridium_butyricum	Clostridium_symbiosum	Lachnospiraceae_bacterium_5_1_57FAA	SPC10256
Clostridium_butyricum	Clostridium_symbiosum	Blautia_producta	SPC10256
Clostridium_butyricum	Clostridium_symbiosum	Eubacterium_rectale	SPC10256
Clostridium_butyricum	Faecalibacterium_prausnitzii	Faecalibacterium_prausnitzii	SPC10256
Clostridium_butyricum	Faecalibacterium_prausnitzii	Lachnospiraceae_bacterium_5_1_57FAA	SPC10256
Clostridium_butyricum	Faecalibacterium_prausnitzii	Blautia_producta	SPC10256
Clostridium_butyricum	Faecalibacterium_prausnitzii	Eubacterium_rectale	SPC10256
Clostridium_butyricum	Lachnospiraceae_bacterium_5_1_57FAA	Lachnospiraceae_bacterium_5_1_57FAA	SPC10256
Clostridium_butyricum	Lachnospiraceae_bacterium_5_1_57FAA	Blautia_producta	SPC10256
Clostridium_butyricum	Lachnospiraceae_bacterium_5_1_57FAA	Eubacterium_rectale	SPC10256
Clostridium_butyricum	Blautia_producta	Blautia_producta	SPC10256
Clostridium_butyricum	Blautia_producta	Eubacterium_rectale	SPC10256
Clostridium_butyricum	Eubacterium_rectale	Eubacterium_rectale	SPC10256
Clostridium_hylemonae	Collinsella_aerofaciens	Collinsella_aerofaciens	SPC10313
Clostridium_hylemonae	Collinsella_aerofaciens	Coprococcus_comes	SPC10313
Clostridium_hylemonae	Collinsella_aerofaciens	Clostridium_bolteae	SPC10313
Clostridium_hylemonae	Collinsella_aerofaciens	Clostridium_symbiosum	SPC10313
Clostridium_hylemonae	Collinsella_aerofaciens	Faecalibacterium_prausnitzii	SPC10313
Clostridium_hylemonae	Collinsella_aerofaciens	Lachnospiraceae_bacterium_5_1_57FAA	SPC10313
Clostridium_hylemonae	Collinsella_aerofaciens	Blautia_producta	SPC10313
Clostridium_hylemonae	Collinsella_aerofaciens	Eubacterium_rectale	SPC10313
Clostridium_hylemonae	Coprococcus_comes	Coprococcus_comes	SPC10313
Clostridium_hylemonae	Coprococcus_comes	Clostridium_bolteae	SPC10313
Clostridium_hylemonae	Coprococcus_comes	Clostridium_symbiosum	SPC10313
Clostridium_hylemonae	Coprococcus_comes	Faecalibacterium_prausnitzii	SPC10313
Clostridium_hylemonae	Coprococcus_comes	Lachnospiraceae_bacterium_5_1_57FAA	SPC10313
Clostridium_hylemonae	Coprococcus_comes	Blautia_producta	SPC10313
Clostridium_hylemonae	Coprococcus_comes	Eubacterium_rectale	SPC10313
Clostridium_hylemonae	Clostridium_bolteae	Clostridium_bolteae	SPC10313
Clostridium_hylemonae	Clostridium_bolteae	Clostridium_symbiosum	SPC10313
Clostridium_hylemonae	Clostridium_bolteae	Faecalibacterium_prausnitzii	SPC10313
Clostridium_hylemonae	Clostridium_bolteae	Lachnospiraceae_bacterium_5_1_57FAA	SPC10313
Clostridium_hylemonae	Clostridium_bolteae	Blautia_producta	SPC10313
Clostridium_hylemonae	Clostridium_bolteae	Eubacterium_rectale	SPC10313
Clostridium_hylemonae	Clostridium_symbiosum	Clostridium_symbiosum	SPC10313
Clostridium_hylemonae	Clostridium_symbiosum	Faecalibacterium_prausnitzii	SPC10313
Clostridium_hylemonae	Clostridium_symbiosum	Lachnospiraceae_bacterium_5_1_57FAA	SPC10313
Clostridium_hylemonae	Clostridium_symbiosum	Blautia_producta	SPC10313
Clostridium_hylemonae	Clostridium_symbiosum	Eubacterium_rectale	SPC10313
Clostridium_hylemonae	Faecalibacterium_prausnitzii	Faecalibacterium_prausnitzii	SPC10313
Clostridium_hylemonae	Faecalibacterium_prausnitzii	Lachnospiraceae_bacterium_5_1_57FAA	SPC10313
Clostridium_hylemonae	Faecalibacterium_prausnitzii	Blautia_producta	SPC10313
Clostridium_hylemonae	Faecalibacterium_prausnitzii	Eubacterium_rectale	SPC10313
Clostridium_hylemonae	Lachnospiraceae_bacterium_5_1_57FAA	Lachnospiraceae_bacterium_5_1_57FAA	SPC10313
Clostridium_hylemonae	Lachnospiraceae_bacterium_5_1_57FAA	Blautia_producta	SPC10313
Clostridium_hylemonae	Lachnospiraceae_bacterium_5_1_57FAA	Eubacterium_rectale	SPC10313
Clostridium_hylemonae	Blautia_producta	Blautia_producta	SPC10313

TABLE 4a-continued

Clostridium_hylemonae	*Blautia_producta*	*Eubacterium_rectale*	SPC10313
Clostridium_hylemonae	*Eubacterium_rectale*	*Eubacterium_rectale*	SPC10313
Clostridium_orbiscindens	*Collinsella_aerofaciens*	*Collinsella_aerofaciens*	SPC10358
Clostridium_orbiscindens	*Collinsella_aerofaciens*	*Coprococcus_comes*	SPC10358
Clostridium_orbiscindens	*Collinsella_aerofaciens*	*Clostridium_bolteae*	SPC10358
Clostridium_orbiscindens	*Collinsella_aerofaciens*	*Clostridium_symbiosum*	SPC10358
Clostridium_orbiscindens	*Collinsella_aerofaciens*	*Faecalibacterium_prausnitzii*	SPC10358
Clostridium_orbiscindens	*Collinsella_aerofaciens*	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10358
Clostridium_orbiscindens	*Collinsella_aerofaciens*	*Blautia_producta*	SPC10358
Clostridium_orbiscindens	*Collinsella_aerofaciens*	*Eubacterium_rectale*	SPC10358
Clostridium_orbiscindens	*Coprococcus_comes*	*Coprococcus_comes*	SPC10358
Clostridium_orbiscindens	*Coprococcus_comes*	*Clostridium_bolteae*	SPC10358
Clostridium_orbiscindens	*Coprococcus_comes*	*Clostridium_symbiosum*	SPC10358
Clostridium_orbiscindens	*Coprococcus_comes*	*Faecalibacterium_prausnitzii*	SPC10358
Clostridium_orbiscindens	*Coprococcus_comes*	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10358
Clostridium_orbiscindens	*Coprococcus_comes*	*Blautia_producta*	SPC10358
Clostridium_orbiscindens	*Coprococcus_comes*	*Eubacterium_rectale*	SPC10358
Clostridium_orbiscindens	*Clostridium_bolteae*	*Clostridium_bolteae*	SPC10358
Clostridium_orbiscindens	*Clostridium_bolteae*	*Clostridium_symbiosum*	SPC10358
Clostridium_orbiscindens	*Clostridium_bolteae*	*Faecalibacterium_prausnitzii*	SPC10358
Clostridium_orbiscindens	*Clostridium_bolteae*	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10358
Clostridium_orbiscindens	*Clostridium_bolteae*	*Blautia_producta*	SPC10358
Clostridium_orbiscindens	*Clostridium_bolteae*	*Eubacterium_rectale*	SPC10358
Clostridium_orbiscindens	*Clostridium_symbiosum*	*Clostridium_symbiosum*	SPC10358
Clostridium_orbiscindens	*Clostridium_symbiosum*	*Faecalibacterium_prausnitzii*	SPC10358
Clostridium_orbiscindens	*Clostridium_symbiosum*	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10358
Clostridium_orbiscindens	*Clostridium_symbiosum*	*Blautia_producta*	SPC10358
Clostridium_orbiscindens	*Clostridium_symbiosum*	*Eubacterium_rectale*	SPC10358
Clostridium_orbiscindens	*Faecalibacterium_prausnitzii*	*Faecalibacterium_prausnitzii*	SPC10358
Clostridium_orbiscindens	*Faecalibacterium_prausnitzii*	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10358
Clostridium_orbiscindens	*Faecalibacterium_prausnitzii*	*Blautia_producta*	SPC10358
Clostridium_orbiscindens	*Faecalibacterium_prausnitzii*	*Eubacterium_rectale*	SPC10358
Clostridium_orbiscindens	Lachnospiraceae_*bacterium*_5_1_57FAA	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10358
Clostridium_orbiscindens	Lachnospiraceae_*bacterium*_5_1_57FAA	*Blautia_producta*	SPC10358
Clostridium_orbiscindens	Lachnospiraceae_*bacterium*_5_1_57FAA	*Eubacterium_rectale*	SPC10358
Clostridium_orbiscindens	*Blautia_producta*	*Blautia_producta*	SPC10358
Clostridium_*orbiscindens*	*Blautia_producta*	*Eubacterium_rectale*	SPC10358
Clostridium_orbiscindens	*Eubacterium_rectale*	*Eubacterium_rectale*	SPC10358
Ruminococcus_gnavus	*Collinsella_aerofaciens*	*Collinsella_aerofaciens*	SPC10468
Ruminococcus_gnavus	*Collinsella_aerofaciens*	*Coprococcus_comes*	SPC10468
Ruminococcus_gnavus	*Collinsella_aerofaciens*	*Clostridium_bolteae*	SPC10468
Ruminococcus_gnavus	*Collinsella_aerofaciens*	*Clostridium_symbiosum*	SPC10468
Ruminococcus_gnavus	*Collinsella_aerofaciens*	*Faecalibacterium_prausnitzii*	SPC10468
Ruminococcus_gnavus	*Collinsella_aerofaciens*	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10468
Ruminococcus_gnavus	*Collinsella_aerofaciens*	*Blautia_producta*	SPC10468
Ruminococcus_gnavus	*Collinsella_aerofaciens*	*Eubacterium_rectale*	SPC10468
Ruminococcus_gnavus	*Coprococcus_comes*	*Coprococcus_comes*	SPC10468
Ruminococcus_gnavus	*Coprococcus_comes*	*Clostridium_bolteae*	SPC10468
Ruminococcus_gnavus	*Coprococcus_comes*	*Clostridium_symbiosum*	SPC10468
Ruminococcus_gnavus	*Coprococcus_comes*	*Faecalibacterium_prausnitzii*	SPC10468
Ruminococcus_gnavus	*Coprococcus_comes*	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10468
Ruminococcus_gnavus	*Coprococcus_comes*	*Blautia_producta*	SPC10468
Ruminococcus_gnavus	*Coprococcus_comes*	*Eubacterium_rectale*	SPC10468
Ruminococcus_gnavus	*Clostridium_bolteae*	*Clostridium_bolteae*	SPC10468
Ruminococcus_gnavus	*Clostridium_bolteae*	*Clostridium_symbiosum*	SPC10468
Ruminococcus_gnavus	*Clostridium_bolteae*	*Faecalibacterium_prausnitzii*	SPC10468
Ruminococcus_gnavus	*Clostridium_bolteae*	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10468
Ruminococcus_gnavus	*Clostridium_bolteae*	*Blautia_producta*	SPC10468
Ruminococcus_gnavus	*Clostridium_bolteae*	*Eubacterium_rectale*	SPC10468
Ruminococcus_gnavus	*Clostridium_symbiosum*	*Clostridium_symbiosum*	SPC10468
Ruminococcus_gnavus	*Clostridium_symbiosum*	*Faecalibacterium_prausnitzii*	SPC10468
Ruminococcus_gnavus	*Clostridium_symbiosum*	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10468
Ruminococcus_gnavus	*Clostridium_symbiosum*	*Blautia_producta*	SPC10468
Ruminococcus_gnavus	*Clostridium_symbiosum*	*Eubacterium_rectale*	SPC10468
Ruminococcus_gnavus	*Faecalibacterium_prausnitzii*	*Faecalibacterium_prausnitzii*	SPC10468
Ruminococcus_gnavus	*Faecalibacterium_prausnitzii*	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10468
Ruminococcus_gnavus	*Faecalibacterium_prausnitzii*	*Blautia_producta*	SPC10468
Ruminococcus_gnavus	*Faecalibacterium_prausnitzii*	*Eubacterium_rectale*	SPC10468
Ruminococcus_gnavus	Lachnospiraceae_*bacterium*_5_1_57FAA	Lachnospiraceae_*bacterium*_5_1_57FAA	SPC10468
Ruminococcus_gnavus	Lachnospiraceae_*bacterium*_5_1_57FAA	*Blautia_producta*	SPC10468
Ruminococcus_gnavus	Lachnospiraceae_*bacterium*_5_1_57FAA	*Eubacterium_rectale*	SPC10468
Ruminococcus_gnavus	*Blautia_producta*	*Blautia_producta*	SPC10468
Ruminococcus_gnavus	*Blautia_producta*	*Eubacterium_rectale*	SPC10468
Ruminococcus_gnavus	*Eubacterium_rectale*	*Eubacterium_rectale*	SPC10468

TABLE 4a-continued

Strain ID OTU2	Strain ID OTU3 (if applicable)	Clade of OTU1	Clade of OTU2	Clade of OTU3 (if applicable)	Hetero/ Semi/ Homo	*C. diff* Inhibition Score	75th Percentile of *C. diff* Inhibition Score	*C. diff* Inhibition Synergy	VRE Inhibition Score	75th Percentile of VRE Inhibition Score
SPC00001		clade_92	clade_92		homo	+++	FALSE			
SPC00005		clade_92	clade_378		hetero	++++	FALSE	TRUE		
SPC00006		clade_92	clade_65		hetero	++++	TRUE	TRUE		
SPC00007		clade_92	clade_38		hetero	++++	TRUE	TRUE		
SPC00008		clade_92	clade_497		hetero	++++	TRUE	TRUE		
SPC00009		clade_92	clade_481		hetero	+	FALSE	TRUE		
SPC00015		clade_92	clade_98		hetero	+	FALSE	FALSE		
SPC00018		clade_92	clade_360		hetero	+	FALSE	TRUE		
SPC00021		clade_92	clade_309		hetero	++++	FALSE	FALSE		
SPC00022		clade_92	clade_522		hetero	+	FALSE	TRUE		
SPC00026		clade_92	clade_262		hetero	++++	TRUE	TRUE		
SPC00027		clade_92	clade_351		hetero	++++	FALSE	TRUE		
SPC00054		clade_92	clade_478		hetero	++++	TRUE	TRUE		
SPC00056		clade_92	clade_466		hetero	++++	TRUE	TRUE		
SPC00057		clade_92	clade_360		hetero	++++	TRUE	TRUE		
SPC00061		clade_92	clade_444		hetero	++++	FALSE	TRUE		
SPC00080		clade_92	clade_393		hetero	++++	FALSE	TRUE		
SPC10001		clade_92	clade_479		hetero	+++	FALSE	TRUE		
SPC10019		clade_92	clade_110		hetero	++++	TRUE	TRUE		
SPC10030		clade_92	clade_38		hetero	++++	TRUE	TRUE		
SPC10048		clade_92	clade_286		hetero	+++	FALSE	FALSE		
SPC10081		clade_92	clade_378		hetero	++++	TRUE	TRUE		
SPC10097		clade_92	clade_553		hetero	++++	FALSE	TRUE		
SPC10110		clade_92	clade_92		hetero	+++	FALSE	FALSE		
SPC10197		clade_92	clade_309		hetero	+++	FALSE	FALSE		
SPC10211		clade_92	clade_170		hetero	++++	TRUE	TRUE		
SPC10213		clade_92	clade_85		hetero	++++	TRUE	TRUE		
SPC10233		clade_92	clade_262		hetero	++++	FALSE	FALSE		
SPC10243		clade_92	clade_408		hetero	++++	FALSE	FALSE		
SPC10298		clade_92	clade_172		hetero	++++	TRUE	TRUE		
SPC10301		clade_92	clade_172		hetero	++++	TRUE	TRUE		
SPC10304		clade_92	clade_262		hetero	++++	FALSE	TRUE		
SPC10355		clade_92	clade_408		hetero	++++	TRUE	TRUE		
SPC10363		clade_92	clade_444		hetero	++++	TRUE	TRUE		
SPC10386		clade_92	clade_478		hetero	++++	FALSE	TRUE		
SPC10388		clade_92	clade_466		hetero	+++	FALSE	TRUE		
SPC10390		clade_92	clade_260		hetero	++++	TRUE	TRUE		
SPC10403		clade_92	clade_309		hetero	++++	TRUE	TRUE		
SPC10414		clade_92	clade_500		hetero	++++	TRUE	TRUE		
SPC10415		clade_92	clade_309		hetero	++++	FALSE	FALSE		
SPC00005		clade_378	clade_378		homo		FALSE			
SPC00006		clade_378	clade_65		hetero	+++	FALSE	TRUE		
SPC00007		clade_378	clade_38		hetero	+++	FALSE	TRUE		
SPC00008		clade_378	clade_497		hetero	++++	TRUE	TRUE		
SPC00009		clade_378	clade_481		hetero	+	FALSE	TRUE		
SPC00015		clade_378	clade_98		hetero	+	FALSE	TRUE		
SPC00018		clade_378	clade_360		hetero	+	FALSE	TRUE		
SPC00021		clade_378	clade_309		hetero	++++	TRUE	TRUE		
SPC00022		clade_378	clade_522		hetero	++	FALSE	TRUE		
SPC00026		clade_378	clade_262		hetero	+++	FALSE	TRUE		
SPC00027		clade_378	clade_351		hetero	+	FALSE	FALSE		
SPC00054		clade_378	clade_478		hetero	+++	FALSE	TRUE		
SPC00056		clade_378	clade_466		hetero	++	FALSE	TRUE		
SPC00057		clade_378	clade_360		hetero	++++	FALSE	TRUE		
SPC00061		clade_378	clade_444		hetero	+	FALSE	TRUE		
SPC00080		clade_378	clade_393		hetero		FALSE	TRUE		
SPC10001		clade_378	clade_479		hetero		FALSE	TRUE		
SPC10019		clade_378	clade_110		hetero		FALSE	TRUE		
SPC10030		clade_378	clade_38		hetero		FALSE	FALSE		
SPC10048		clade_378	clade_286		hetero	++++	FALSE	TRUE		
SPC10081		clade_378	clade_378		hetero	+	FALSE	TRUE		
SPC10097		clade_378	clade_553		hetero	++++	TRUE	TRUE		
SPC10110		clade_378	clade_92		hetero	+++	FALSE	TRUE		
SPC10197		clade_378	clade_309		hetero	++++	FALSE	TRUE		
SPC10211		clade_378	clade_170		hetero	++++	TRUE	TRUE		
SPC10213		clade_378	clade_85		hetero	++++	FALSE	TRUE		
SPC10233		clade_378	clade_262		hetero	++++	TRUE	TRUE		
SPC10243		clade_378	clade_408		hetero	++++	TRUE	TRUE		
SPC10298		clade_378	clade_172		hetero	++++	TRUE	TRUE		
SPC10301		clade_378	clade_172		hetero	++++	TRUE	TRUE		
SPC10304		clade_378	clade_262		hetero	++++	TRUE	TRUE		
SPC10355		clade_378	clade_408		hetero	++	FALSE	TRUE		
SPC10363		clade_378	clade_444		hetero	++++	FALSE	TRUE		
SPC10386		clade_378	clade_478		hetero	++	FALSE	TRUE		

TABLE 4a-continued

SPC10388	clade_378	clade_466	hetero	++	FALSE	TRUE
SPC10390	clade_378	clade_260	hetero	++	FALSE	TRUE
SPC10403	clade_378	clade_309	hetero	+	FALSE	TRUE
SPC10414	clade_378	clade_500	hetero	++	FALSE	TRUE
SPC10415	clade_378	clade_309	hetero	++++	FALSE	TRUE
SPC00006	clade_65	clade_65	homo	++++	TRUE	
SPC00007	clade_65	clade_38	hetero	+++	FALSE	FALSE
SPC00008	clade_65	clade_497	hetero	++++	TRUE	TRUE
SPC00009	clade_65	clade_481	hetero	++	FALSE	TRUE
SPC00015	clade_65	clade_98	hetero	++	FALSE	FALSE
SPC00018	clade_65	clade_360	hetero	++	FALSE	TRUE
SPC00021	clade_65	clade_309	hetero	++++	TRUE	TRUE
SPC00022	clade_65	clade_522	hetero	+++	FALSE	TRUE
SPC00026	clade_65	clade_262	hetero	+++	FALSE	FALSE
SPC00027	clade_65	clade_351	hetero	++++	TRUE	TRUE
SPC00054	clade_65	clade_478	hetero	++++	TRUE	TRUE
SPC00056	clade_65	clade_466	hetero	++++	FALSE	TRUE
SPC00057	clade_65	clade_360	hetero	++++	TRUE	TRUE
SPC00061	clade_65	clade_444	hetero	++++	FALSE	TRUE
SPC00080	clade_65	clade_393	hetero	+++	FALSE	TRUE
SPC10001	clade_65	clade_479	hetero	++++	FALSE	TRUE
SPC10019	clade_65	clade_110	hetero	++++	FALSE	TRUE
SPC10030	clade_65	clade_38	hetero	+++	FALSE	FALSE
SPC10048	clade_65	clade_286	hetero	+++	FALSE	FALSE
SPC10081	clade_65	clade_378	hetero		FALSE	FALSE
SPC10097	clade_65	clade_553	hetero	++++	TRUE	TRUE
SPC10110	clade_65	clade_92	hetero	+++	FALSE	FALSE
SPC10197	clade_65	clade_309	hetero	++	FALSE	FALSE
SPC10211	clade_65	clade_170	hetero	+++	FALSE	FALSE
SPC10213	clade_65	clade_85	hetero	+++	FALSE	FALSE
SPC10233	clade_65	clade_262	hetero	++++	TRUE	FALSE
SPC10243	clade_65	clade_408	hetero	++++	FALSE	FALSE
SPC10298	clade_65	clade_172	hetero	++++	TRUE	TRUE
SPC10301	clade_65	clade_172	hetero	++++	TRUE	TRUE
SPC10304	clade_65	clade_262	hetero	++	FALSE	FALSE
SPC10355	clade_65	clade_408	hetero	++++	TRUE	TRUE
SPC10363	clade_65	clade_444	hetero	++++	FALSE	TRUE
SPC10386	clade_65	clade_478	hetero	+++	FALSE	FALSE
SPC10388	clade_65	clade_466	hetero	+	FALSE	FALSE
SPC10390	clade_65	clade_260	hetero	++	FALSE	FALSE
SPC10403	clade_65	clade_309	hetero	++	FALSE	FALSE
SPC10414	clade_65	clade_500	hetero	+++	FALSE	TRUE
SPC10415	clade_65	clade_309	hetero	+++	FALSE	FALSE
SPC00007	clade_38	clade_38	homo		FALSE	
SPC00008	clade_38	clade_497	hetero	++++	TRUE	TRUE
SPC00009	clade_38	clade_481	hetero		FALSE	TRUE
SPC00015	clade_38	clade_98	hetero	++	FALSE	TRUE
SPC00018	clade_38	clade_360	hetero		FALSE	TRUE
SPC00021	clade_38	clade_309	hetero	++++	TRUE	TRUE
SPC00022	clade_38	clade_522	hetero		FALSE	TRUE
SPC00026	clade_38	clade_262	hetero	++++	TRUE	TRUE
SPC00027	clade_38	clade_351	hetero	+++	FALSE	TRUE
SPC00054	clade_38	clade_478	hetero	+	FALSE	TRUE
SPC00056	clade_38	clade_466	hetero		FALSE	TRUE
SPC00057	clade_38	clade_360	hetero	++	FALSE	TRUE
SPC00061	clade_38	clade_444	hetero		FALSE	TRUE
SPC00080	clade_38	clade_393	hetero		FALSE	FALSE
SPC10001	clade_38	clade_479	hetero		FALSE	FALSE
SPC10019	clade_38	clade_110	hetero	+++	FALSE	TRUE
SPC10030	clade_38	clade_38	hetero		FALSE	FALSE
SPC10048	clade_38	clade_286	hetero	++	FALSE	TRUE
SPC10081	clade_38	clade_378	hetero		FALSE	FALSE
SPC10097	clade_38	clade_553	hetero	++++	TRUE	TRUE
SPC10110	clade_38	clade_92	hetero	++++	TRUE	TRUE
SPC10197	clade_38	clade_309	hetero	++	FALSE	FALSE
SPC10211	clade_38	clade_170	hetero	++	FALSE	FALSE
SPC10213	clade_38	clade_85	hetero	+	FALSE	FALSE
SPC10233	clade_38	clade_262	hetero	+++	FALSE	FALSE
SPC10243	clade_38	clade_408	hetero	+++	FALSE	FALSE
SPC10298	clade_38	clade_172	hetero	++++	TRUE	TRUE
SPC10301	clade_38	clade_172	hetero	++++	TRUE	TRUE
SPC10304	clade_38	clade_262	hetero	+++	FALSE	TRUE
SPC10355	clade_38	clade_408	hetero	++++	TRUE	TRUE
SPC10363	clade_38	clade_444	hetero	++	FALSE	TRUE
SPC10386	clade_38	clade_478	hetero		FALSE	FALSE
SPC10388	clade_38	clade_466	hetero		FALSE	FALSE
SPC10390	clade_38	clade_260	hetero		FALSE	FALSE
SPC10403	clade_38	clade_309	hetero		FALSE	TRUE
SPC10414	clade_38	clade_500	hetero		FALSE	TRUE
SPC10415	clade_38	clade_309	hetero	+	FALSE	FALSE

TABLE 4a-continued

SPC00008	clade_497	clade_497	homo	++++	TRUE	
SPC00009	clade_497	clade_481	hetero	++++	TRUE	TRUE
SPC00015	clade_497	clade_98	hetero	++++	TRUE	TRUE
SPC00018	clade_497	clade_360	hetero	++++	TRUE	TRUE
SPC00021	clade_497	clade_309	hetero	++++	TRUE	TRUE
SPC00022	clade_497	clade_522	hetero	++++	TRUE	TRUE
SPC00026	clade_497	clade_262	hetero	++++	TRUE	TRUE
SPC00027	clade_497	clade_351	hetero	++++	TRUE	TRUE
SPC00054	clade_497	clade_478	hetero	++++	TRUE	TRUE
SPC00056	clade_497	clade_466	hetero	++++	TRUE	TRUE
SPC00057	clade_497	clade_360	hetero	++++	TRUE	TRUE
SPC00061	clade_497	clade_444	hetero	++++	TRUE	TRUE
SPC00080	clade_497	clade_393	hetero	++++	TRUE	TRUE
SPC10001	clade_497	clade_479	hetero	++++	TRUE	TRUE
SPC10019	clade_497	clade_110	hetero	++++	TRUE	TRUE
SPC10030	clade_497	clade_38	hetero	++++	TRUE	TRUE
SPC10048	clade_497	clade_286	hetero	++++	TRUE	TRUE
SPC10081	clade_497	clade_378	hetero	++++	TRUE	TRUE
SPC10097	clade_497	clade_553	hetero	++++	TRUE	TRUE
SPC10110	clade_497	clade_92	hetero	+++	FALSE	FALSE
SPC10197	clade_497	clade_309	hetero	++++	TRUE	TRUE
SPC10211	clade_497	clade_170	hetero	++++	TRUE	TRUE
SPC10213	clade_497	clade_85	hetero	++++	TRUE	TRUE
SPC10233	clade_497	clade_262	hetero	++++	TRUE	TRUE
SPC10243	clade_497	clade_408	hetero	++++	TRUE	TRUE
SPC10298	clade_497	clade_172	hetero	++++	TRUE	TRUE
SPC10301	clade_497	clade_172	hetero	++++	TRUE	TRUE
SPC10304	clade_497	clade_262	hetero	++++	TRUE	TRUE
SPC10355	clade_497	clade_408	hetero	++++	TRUE	TRUE
SPC10363	clade_497	clade_444	hetero	++++	TRUE	TRUE
SPC10386	clade_497	clade_478	hetero	++++	TRUE	TRUE
SPC10388	clade_497	clade_466	hetero	++++	TRUE	TRUE
SPC10390	clade_497	clade_260	hetero	++++	TRUE	TRUE
SPC10403	clade_497	clade_309	hetero	++++	TRUE	TRUE
SPC10414	clade_497	clade_500	hetero	++++	TRUE	TRUE
SPC10415	clade_497	clade_309	hetero	++++	TRUE	TRUE
SPC00009	clade_481	clade_481	homo		FALSE	
SPC00015	clade_481	clade_98	hetero		FALSE	TRUE
SPC00018	clade_481	clade_360	hetero		FALSE	TRUE
SPC00021	clade_481	clade_309	hetero	++++	TRUE	TRUE
SPC00022	clade_481	clade_522	hetero		FALSE	TRUE
SPC00026	clade_481	clade_262	hetero		FALSE	FALSE
SPC00027	clade_481	clade_351	hetero		FALSE	FALSE
SPC00054	clade_481	clade_478	hetero	−−	FALSE	FALSE
SPC00056	clade_481	clade_466	hetero		FALSE	FALSE
SPC00057	clade_481	clade_360	hetero		FALSE	TRUE
SPC00061	clade_481	clade_444	hetero		FALSE	FALSE
SPC00080	clade_481	clade_393	hetero	−−	FALSE	FALSE
SPC10001	clade_481	clade_479	hetero		FALSE	TRUE
SPC10019	clade_481	clade_110	hetero		FALSE	TRUE
SPC10030	clade_481	clade_38	hetero		FALSE	FALSE
SPC10048	clade_481	clade_286	hetero	−	FALSE	FALSE
SPC10081	clade_481	clade_378	hetero		FALSE	FALSE
SPC10097	clade_481	clade_553	hetero	++	FALSE	TRUE
SPC10110	clade_481	clade_92	hetero	+	FALSE	TRUE
SPC10197	clade_481	clade_309	hetero	+	FALSE	FALSE
SPC10211	clade_481	clade_170	hetero	++	FALSE	FALSE
SPC10213	clade_481	clade_85	hetero		FALSE	FALSE
SPC10233	clade_481	clade_262	hetero	+++	FALSE	FALSE
SPC10243	clade_481	clade_408	hetero	+++	FALSE	TRUE
SPC10298	clade_481	clade_172	hetero	+++	FALSE	TRUE
SPC10301	clade_481	clade_172	hetero	++++	TRUE	TRUE
SPC10304	clade_481	clade_262	hetero	++	FALSE	TRUE
SPC10355	clade_481	clade_408	hetero		FALSE	TRUE
SPC10363	clade_481	clade_444	hetero	++	FALSE	TRUE
SPC10386	clade_481	clade_478	hetero		FALSE	TRUE
SPC10388	clade_481	clade_466	hetero	++	FALSE	TRUE
SPC10390	clade_481	clade_260	hetero	+++	FALSE	TRUE
SPC10403	clade_481	clade_309	hetero	++++	FALSE	TRUE
SPC10414	clade_481	clade_500	hetero	+++	FALSE	TRUE
SPC10415	clade_481	clade_309	hetero	++++	TRUE	TRUE
SPC00015	clade_98	clade_98	homo		FALSE	
SPC00018	clade_98	clade_360	hetero		FALSE	TRUE
SPC00021	clade_98	clade_309	hetero	++++	TRUE	TRUE
SPC00022	clade_98	clade_522	hetero		FALSE	TRUE
SPC00026	clade_98	clade_262	hetero	+	FALSE	TRUE
SPC00027	clade_98	clade_351	hetero		FALSE	FALSE
SPC00054	clade_98	clade_478	hetero		FALSE	FALSE
SPC00056	clade_98	clade_466	hetero		FALSE	FALSE
SPC00057	clade_98	clade_360	hetero		FALSE	TRUE

TABLE 4a-continued

SPC00061	clade_98	clade_444	hetero		FALSE	FALSE
SPC00080	clade_98	clade_393	hetero		FALSE	FALSE
SPC10001	clade_98	clade_479	hetero		FALSE	FALSE
SPC10019	clade_98	clade_110	hetero	+	FALSE	TRUE
SPC10030	clade_98	clade_38	hetero	+	FALSE	TRUE
SPC10048	clade_98	clade_286	hetero		FALSE	FALSE
SPC10081	clade_98	clade_378	hetero	−	FALSE	FALSE
SPC10097	clade_98	clade_553	hetero		FALSE	TRUE
SPC10110	clade_98	clade_92	hetero	++	FALSE	FALSE
SPC10197	clade_98	clade_309	hetero	++	FALSE	FALSE
SPC10211	clade_98	clade_170	hetero	+	FALSE	FALSE
SPC10213	clade_98	clade_85	hetero		FALSE	FALSE
SPC10233	clade_98	clade_262	hetero		FALSE	FALSE
SPC10243	clade_98	clade_408	hetero	+	FALSE	FALSE
SPC10298	clade_98	clade_172	hetero	++++	TRUE	TRUE
SPC10301	clade_98	clade_172	hetero	++++	FALSE	FALSE
SPC10304	clade_98	clade_262	hetero	+	FALSE	TRUE
SPC10355	clade_98	clade_408	hetero		FALSE	FALSE
SPC10363	clade_98	clade_444	hetero		FALSE	FALSE
SPC10386	clade_98	clade_478	hetero		FALSE	TRUE
SPC10388	clade_98	clade_466	hetero		FALSE	TRUE
SPC10390	clade_98	clade_260	hetero		FALSE	FALSE
SPC10403	clade_98	clade_309	hetero		FALSE	TRUE
SPC10414	clade_98	clade_500	hetero		FALSE	FALSE
SPC10415	clade_98	clade_309	hetero		FALSE	FALSE
SPC00018	clade_360	clade_360	homo		FALSE	
SPC00021	clade_360	clade_309	hetero	++++	FALSE	TRUE
SPC00022	clade_360	clade_522	hetero	−	FALSE	FALSE
SPC00026	clade_360	clade_262	hetero		FALSE	TRUE
SPC00027	clade_360	clade_351	hetero		FALSE	FALSE
SPC00054	clade_360	clade_478	hetero		FALSE	TRUE
SPC00056	clade_360	clade_466	hetero		FALSE	TRUE
SPC00057	clade_360	clade_360	hetero	+	FALSE	TRUE
SPC00061	clade_360	clade_444	hetero		FALSE	TRUE
SPC00080	clade_360	clade_393	hetero		FALSE	TRUE
SPC10001	clade_360	clade_479	hetero	−	FALSE	FALSE
SPC10019	clade_360	clade_110	hetero		FALSE	TRUE
SPC10030	clade_360	clade_38	hetero		FALSE	FALSE
SPC10048	clade_360	clade_286	hetero		FALSE	TRUE
SPC10081	clade_360	clade_378	hetero		FALSE	TRUE
SPC10097	clade_360	clade_553	hetero	+	FALSE	TRUE
SPC10110	clade_360	clade_92	hetero	+++	FALSE	TRUE
SPC10197	clade_360	clade_309	hetero	+++	FALSE	TRUE
SPC10211	clade_360	clade_170	hetero	++	FALSE	TRUE
SPC10213	clade_360	clade_85	hetero		FALSE	FALSE
SPC10233	clade_360	clade_262	hetero	+++	FALSE	FALSE
SPC10243	clade_360	clade_408	hetero	+++	FALSE	TRUE
SPC10298	clade_360	clade_172	hetero	++	FALSE	TRUE
SPC10301	clade_360	clade_172	hetero	++++	TRUE	TRUE
SPC10304	clade_360	clade_262	hetero		FALSE	TRUE
SPC10355	clade_360	clade_408	hetero		FALSE	FALSE
SPC10363	clade_360	clade_444	hetero		FALSE	FALSE
SPC10386	clade_360	clade_478	hetero	−−	FALSE	FALSE
SPC10388	clade_360	clade_466	hetero		FALSE	FALSE
SPC10390	clade_360	clade_260	hetero	−	FALSE	FALSE
SPC10403	clade_360	clade_309	hetero		FALSE	FALSE
SPC10414	clade_360	clade_500	hetero	−−−	FALSE	FALSE
SPC10415	clade_360	clade_309	hetero	−	FALSE	FALSE
SPC00021	clade_309	clade_309	homo	++++	TRUE	
SPC00022	clade_309	clade_522	hetero	++++	TRUE	TRUE
SPC00026	clade_309	clade_262	hetero	++++	TRUE	TRUE
SPC00027	clade_309	clade_351	hetero	++++	TRUE	TRUE
SPC00054	clade_309	clade_478	hetero	++++	TRUE	TRUE
SPC00056	clade_309	clade_466	hetero	++++	TRUE	TRUE
SPC00057	clade_309	clade_360	hetero	++++	TRUE	TRUE
SPC00061	clade_309	clade_444	hetero	++++	TRUE	TRUE
SPC00080	clade_309	clade_393	hetero	++++	FALSE	TRUE
SPC10001	clade_309	clade_479	hetero	++++	FALSE	TRUE
SPC10019	clade_309	clade_110	hetero	++++	FALSE	TRUE
SPC10030	clade_309	clade_38	hetero	+++	FALSE	FALSE
SPC10048	clade_309	clade_286	hetero	++++	TRUE	TRUE
SPC10081	clade_309	clade_378	hetero	++++	TRUE	TRUE
SPC10097	clade_309	clade_553	hetero	++++	TRUE	TRUE
SPC10110	clade_309	clade_92	hetero	++++	TRUE	TRUE
SPC10197	clade_309	clade_309	hetero	++++	TRUE	FALSE
SPC10211	clade_309	clade_170	hetero	++++	TRUE	TRUE
SPC10213	clade_309	clade_85	hetero	++++	FALSE	FALSE
SPC10233	clade_309	clade_262	hetero	++++	TRUE	FALSE
SPC10243	clade_309	clade_408	hetero	++++	TRUE	FALSE
SPC10298	clade_309	clade_172	hetero		FALSE	FALSE

TABLE 4a-continued

SPC10301	clade_309	clade_172	hetero	++++	TRUE	TRUE
SPC10304	clade_309	clade_262	hetero	++++	TRUE	TRUE
SPC10355	clade_309	clade_408	hetero	++++	TRUE	TRUE
SPC10363	clade_309	clade_444	hetero	++++	TRUE	TRUE
SPC10386	clade_309	clade_478	hetero	++++	TRUE	TRUE
SPC10388	clade_309	clade_466	hetero	++++	TRUE	TRUE
SPC10390	clade_309	clade_260	hetero	++++	TRUE	TRUE
SPC10403	clade_309	clade_309	hetero	++++	TRUE	TRUE
SPC10414	clade_309	clade_500	hetero	++++	TRUE	TRUE
SPC10415	clade_309	clade_309	hetero	++++	TRUE	TRUE
SPC00022	clade_522	clade_522	homo		FALSE	
SPC00026	clade_522	clade_262	hetero		FALSE	TRUE
SPC00027	clade_522	clade_351	hetero		FALSE	FALSE
SPC00054	clade_522	clade_478	hetero		FALSE	FALSE
SPC00056	clade_522	clade_466	hetero		FALSE	FALSE
SPC00057	clade_522	clade_360	hetero	+	FALSE	TRUE
SPC00061	clade_522	clade_444	hetero		FALSE	TRUE
SPC00080	clade_522	clade_393	hetero		FALSE	FALSE
SPC10001	clade_522	clade_479	hetero	–	FALSE	FALSE
SPC10019	clade_522	clade_110	hetero	–	FALSE	FALSE
SPC10030	clade_522	clade_38	hetero		FALSE	FALSE
SPC10048	clade_522	clade_286	hetero		FALSE	FALSE
SPC10081	clade_522	clade_378	hetero		FALSE	FALSE
SPC10097	clade_522	clade_553	hetero	++	FALSE	TRUE
SPC10110	clade_522	clade_92	hetero	++++	FALSE	TRUE
SPC10197	clade_522	clade_309	hetero		FALSE	FALSE
SPC10211	clade_522	clade_170	hetero	+	FALSE	FALSE
SPC10213	clade_522	clade_85	hetero		FALSE	FALSE
SPC10233	clade_522	clade_262	hetero	+	FALSE	FALSE
SPC10243	clade_522	clade_408	hetero	+	FALSE	FALSE
SPC10298	clade_522	clade_172	hetero	++++	FALSE	TRUE
SPC10301	clade_522	clade_172	hetero	++++	FALSE	TRUE
SPC10304	clade_522	clade_262	hetero	+	FALSE	TRUE
SPC10355	clade_522	clade_408	hetero		FALSE	TRUE
SPC10363	clade_522	clade_444	hetero		FALSE	FALSE
SPC10386	clade_522	clade_478	hetero		FALSE	TRUE
SPC10388	clade_522	clade_466	hetero		FALSE	FALSE
SPC10390	clade_522	clade_260	hetero		FALSE	FALSE
SPC10403	clade_522	clade_309	hetero		FALSE	FALSE
SPC10414	clade_522	clade_500	hetero	–	FALSE	FALSE
SPC10415	clade_522	clade_309	hetero	––	FALSE	FALSE
SPC00026	clade_262	clade_262	homo	+	FALSE	
SPC00027	clade_262	clade_351	hetero	+++	FALSE	TRUE
SPC00054	clade_262	clade_478	hetero		FALSE	FALSE
SPC00056	clade_262	clade_466	hetero		FALSE	FALSE
SPC00057	clade_262	clade_360	hetero		FALSE	TRUE
SPC00061	clade_262	clade_444	hetero		FALSE	FALSE
SPC00080	clade_262	clade_393	hetero		FALSE	FALSE
SPC10001	clade_262	clade_479	hetero		FALSE	FALSE
SPC10019	clade_262	clade_110	hetero		FALSE	FALSE
SPC10030	clade_262	clade_38	hetero		FALSE	FALSE
SPC10048	clade_262	clade_286	hetero	+	FALSE	FALSE
SPC10081	clade_262	clade_378	hetero		FALSE	FALSE
SPC10097	clade_262	clade_553	hetero		FALSE	FALSE
SPC10110	clade_262	clade_92	hetero	+++	FALSE	FALSE
SPC10197	clade_262	clade_309	hetero		FALSE	FALSE
SPC10211	clade_262	clade_170	hetero		FALSE	FALSE
SPC10213	clade_262	clade_85	hetero		FALSE	FALSE
SPC10233	clade_262	clade_262	hetero		FALSE	FALSE
SPC10243	clade_262	clade_408	hetero		FALSE	FALSE
SPC10298	clade_262	clade_172	hetero	++++	TRUE	TRUE
SPC10301	clade_262	clade_172	hetero	++++	TRUE	FALSE
SPC10304	clade_262	clade_262	hetero		FALSE	FALSE
SPC10355	clade_262	clade_408	hetero	+	FALSE	FALSE
SPC10363	clade_262	clade_444	hetero	–	FALSE	FALSE
SPC10386	clade_262	clade_478	hetero	–	FALSE	FALSE
SPC10388	clade_262	clade_466	hetero	––	FALSE	FALSE
SPC10390	clade_262	clade_260	hetero	–	FALSE	FALSE
SPC10403	clade_262	clade_309	hetero	–	FALSE	FALSE
SPC10414	clade_262	clade_500	hetero		FALSE	FALSE
SPC10415	clade_262	clade_309	hetero		FALSE	FALSE
SPC00027	clade_351	clade_351	homo	++	FALSE	
SPC00054	clade_351	clade_478	hetero	+	FALSE	FALSE
SPC00056	clade_351	clade_466	hetero		FALSE	FALSE
SPC00057	clade_351	clade_360	hetero	–	FALSE	FALSE
SPC00061	clade_351	clade_444	hetero	––	FALSE	FALSE
SPC00080	clade_351	clade_393	hetero	––	FALSE	FALSE
SPC10001	clade_351	clade_479	hetero	–	FALSE	FALSE
SPC10019	clade_351	clade_110	hetero		FALSE	FALSE
SPC10030	clade_351	clade_38	hetero		FALSE	FALSE

TABLE 4a-continued

SPC10048	clade_351	clade_286	hetero	++	FALSE	FALSE
SPC10081	clade_351	clade_378	hetero	++	FALSE	TRUE
SPC10097	clade_351	clade_553	hetero	+++	FALSE	TRUE
SPC10110	clade_351	clade_92	hetero	+++	FALSE	FALSE
SPC10197	clade_351	clade_309	hetero	–	FALSE	FALSE
SPC10211	clade_351	clade_170	hetero		FALSE	FALSE
SPC10213	clade_351	clade_85	hetero		FALSE	FALSE
SPC10233	clade_351	clade_262	hetero		FALSE	FALSE
SPC10243	clade_351	clade_408	hetero		FALSE	FALSE
SPC10298	clade_351	clade_172	hetero	++	FALSE	FALSE
SPC10301	clade_351	clade_172	hetero	++++	FALSE	FALSE
SPC10304	clade_351	clade_262	hetero		FALSE	FALSE
SPC10355	clade_351	clade_408	hetero		FALSE	FALSE
SPC10363	clade_351	clade_444	hetero	–	FALSE	FALSE
SPC10386	clade_351	clade_478	hetero	–	FALSE	FALSE
SPC10388	clade_351	clade_466	hetero	–––	FALSE	FALSE
SPC10390	clade_351	clade_260	hetero		FALSE	FALSE
SPC10403	clade_351	clade_309	hetero		FALSE	FALSE
SPC10414	clade_351	clade_500	hetero		FALSE	FALSE
SPC10415	clade_351	clade_309	hetero		FALSE	FALSE
SPC00054	clade_478	clade_478	homo		FALSE	
SPC00056	clade_478	clade_466	hetero		FALSE	FALSE
SPC00057	clade_478	clade_360	hetero	–	FALSE	FALSE
SPC00061	clade_478	clade_444	hetero		FALSE	FALSE
SPC00080	clade_478	clade_393	hetero		FALSE	FALSE
SPC10001	clade_478	clade_479	hetero	–	FALSE	FALSE
SPC10019	clade_478	clade_110	hetero		FALSE	FALSE
SPC10030	clade_478	clade_38	hetero		FALSE	FALSE
SPC10048	clade_478	clade_286	hetero		FALSE	FALSE
SPC10081	clade_478	clade_378	hetero		FALSE	FALSE
SPC10097	clade_478	clade_553	hetero	++	FALSE	TRUE
SPC10110	clade_478	clade_92	hetero	+++	FALSE	FALSE
SPC10197	clade_478	clade_309	hetero		FALSE	FALSE
SPC10211	clade_478	clade_170	hetero		FALSE	FALSE
SPC10213	clade_478	clade_85	hetero		FALSE	FALSE
SPC10233	clade_478	clade_262	hetero		FALSE	FALSE
SPC10243	clade_478	clade_408	hetero		FALSE	FALSE
SPC10298	clade_478	clade_172	hetero	+	FALSE	FALSE
SPC10301	clade_478	clade_172	hetero		FALSE	FALSE
SPC10304	clade_478	clade_262	hetero		FALSE	FALSE
SPC10355	clade_478	clade_408	hetero		FALSE	FALSE
SPC10363	clade_478	clade_444	hetero	–	FALSE	FALSE
SPC10386	clade_478	clade_478	hetero		FALSE	FALSE
SPC10388	clade_478	clade_466	hetero		FALSE	FALSE
SPC10390	clade_478	clade_260	hetero		FALSE	FALSE
SPC10403	clade_478	clade_309	hetero		FALSE	FALSE
SPC10414	clade_478	clade_500	hetero		FALSE	FALSE
SPC10415	clade_478	clade_309	hetero		FALSE	FALSE
SPC00056	clade_466	clade_466	homo		FALSE	
SPC00057	clade_466	clade_360	hetero		FALSE	TRUE
SPC00061	clade_466	clade_444	hetero		FALSE	FALSE
SPC00080	clade_466	clade_393	hetero		FALSE	FALSE
SPC10001	clade_466	clade_479	hetero		FALSE	FALSE
SPC10019	clade_466	clade_110	hetero		FALSE	FALSE
SPC10030	clade_466	clade_38	hetero		FALSE	FALSE
SPC10048	clade_466	clade_286	hetero		FALSE	FALSE
SPC10081	clade_466	clade_378	hetero		FALSE	FALSE
SPC10097	clade_466	clade_553	hetero	++	FALSE	TRUE
SPC10110	clade_466	clade_92	hetero	++	FALSE	TRUE
SPC10197	clade_466	clade_309	hetero	–	FALSE	FALSE
SPC10211	clade_466	clade_170	hetero		FALSE	FALSE
SPC10213	clade_466	clade_85	hetero		FALSE	FALSE
SPC10233	clade_466	clade_262	hetero		FALSE	FALSE
SPC10243	clade_466	clade_408	hetero		FALSE	FALSE
SPC10298	clade_466	clade_172	hetero		FALSE	FALSE
SPC10301	clade_466	clade_172	hetero	+++	FALSE	FALSE
SPC10304	clade_466	clade_262	hetero		FALSE	FALSE
SPC10355	clade_466	clade_408	hetero		FALSE	FALSE
SPC10363	clade_466	clade_444	hetero	–	FALSE	FALSE
SPC10386	clade_466	clade_478	hetero	–	FALSE	FALSE
SPC10388	clade_466	clade_466	hetero	––	FALSE	FALSE
SPC10390	clade_466	clade_260	hetero	–	FALSE	FALSE
SPC10403	clade_466	clade_309	hetero	–	FALSE	FALSE
SPC10414	clade_466	clade_500	hetero		FALSE	FALSE
SPC10415	clade_466	clade_309	hetero	–	FALSE	FALSE
SPC00057	clade_360	clade_360	homo		FALSE	
SPC00061	clade_360	clade_444	hetero		FALSE	FALSE
SPC00080	clade_360	clade_393	hetero		FALSE	TRUE
SPC10001	clade_360	clade_479	hetero		FALSE	TRUE
SPC10019	clade_360	clade_110	hetero		FALSE	TRUE

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SPC10030	clade_360	clade_38	hetero		FALSE	FALSE
SPC10048	clade_360	clade_286	hetero		FALSE	FALSE
SPC10081	clade_360	clade_378	hetero		FALSE	FALSE
SPC10097	clade_360	clade_553	hetero	+++	FALSE	TRUE
SPC10110	clade_360	clade_92	hetero	++++	TRUE	TRUE
SPC10197	clade_360	clade_309	hetero		FALSE	FALSE
SPC10211	clade_360	clade_170	hetero		FALSE	FALSE
SPC10213	clade_360	clade_85	hetero		FALSE	FALSE
SPC10233	clade_360	clade_262	hetero		FALSE	FALSE
SPC10243	clade_360	clade_408	hetero	+	FALSE	FALSE
SPC10298	clade_360	clade_172	hetero	++++	FALSE	TRUE
SPC10301	clade_360	clade_172	hetero	+++	FALSE	FALSE
SPC10304	clade_360	clade_262	hetero		FALSE	FALSE
SPC10355	clade_360	clade_408	hetero	++++	TRUE	TRUE
SPC10363	clade_360	clade_444	hetero	++++	FALSE	TRUE
SPC10386	clade_360	clade_478	hetero	++	FALSE	TRUE
SPC10388	clade_360	clade_466	hetero	++++	FALSE	TRUE
SPC10390	clade_360	clade_260	hetero	+++	FALSE	TRUE
SPC10403	clade_360	clade_309	hetero	++	FALSE	TRUE
SPC10414	clade_360	clade_500	hetero	+++	FALSE	TRUE
SPC10415	clade_360	clade_309	hetero	++	FALSE	TRUE
SPC00061	clade_444	clade_444	homo		FALSE	
SPC00080	clade_444	clade_393	hetero		FALSE	TRUE
SPC10001	clade_444	clade_479	hetero		FALSE	FALSE
SPC10019	clade_444	clade_110	hetero		FALSE	FALSE
SPC10030	clade_444	clade_38	hetero		FALSE	FALSE
SPC10048	clade_444	clade_286	hetero		FALSE	FALSE
SPC10081	clade_444	clade_378	hetero		FALSE	FALSE
SPC10097	clade_444	clade_553	hetero	+	FALSE	TRUE
SPC10110	clade_444	clade_92	hetero	+	FALSE	TRUE
SPC10197	clade_444	clade_309	hetero		FALSE	FALSE
SPC10211	clade_444	clade_170	hetero		FALSE	FALSE
SPC10213	clade_444	clade_85	hetero		FALSE	FALSE
SPC10233	clade_444	clade_262	hetero	+	FALSE	FALSE
SPC10243	clade_444	clade_408	hetero	+	FALSE	FALSE
SPC10298	clade_444	clade_172	hetero	++	FALSE	FALSE
SPC10301	clade_444	clade_172	hetero	++	FALSE	FALSE
SPC10304	clade_444	clade_262	hetero		FALSE	FALSE
SPC10355	clade_444	clade_408	hetero	−	FALSE	FALSE
SPC10363	clade_444	clade_444	hetero	−−−	FALSE	FALSE
SPC10386	clade_444	clade_478	hetero		FALSE	FALSE
SPC10388	clade_444	clade_466	hetero		FALSE	FALSE
SPC10390	clade_444	clade_260	hetero		FALSE	FALSE
SPC10403	clade_444	clade_309	hetero	−	FALSE	FALSE
SPC10414	clade_444	clade_500	hetero		FALSE	FALSE
SPC10415	clade_444	clade_309	hetero	−	FALSE	FALSE
SPC00080	clade_393	clade_393	homo		FALSE	
SPC10001	clade_393	clade_479	hetero		FALSE	TRUE
SPC10019	clade_393	clade_110	hetero		FALSE	TRUE
SPC10030	clade_393	clade_38	hetero		FALSE	FALSE
SPC10048	clade_393	clade_286	hetero	++	FALSE	TRUE
SPC10081	clade_393	clade_378	hetero	+	FALSE	TRUE
SPC10097	clade_393	clade_553	hetero	+++	FALSE	TRUE
SPC10110	clade_393	clade_92	hetero	++++	FALSE	TRUE
SPC10197	clade_393	clade_309	hetero		FALSE	FALSE
SPC10211	clade_393	clade_170	hetero	+++	FALSE	TRUE
SPC10213	clade_393	clade_85	hetero		FALSE	FALSE
SPC10233	clade_393	clade_262	hetero	+	FALSE	FALSE
SPC10243	clade_393	clade_408	hetero	++	FALSE	FALSE
SPC10298	clade_393	clade_172	hetero		FALSE	FALSE
SPC10301	clade_393	clade_172	hetero		FALSE	FALSE
SPC10304	clade_393	clade_262	hetero	−	FALSE	FALSE
SPC10355	clade_393	clade_408	hetero		FALSE	FALSE
SPC10363	clade_393	clade_444	hetero	−−	FALSE	FALSE
SPC10386	clade_393	clade_478	hetero	−−	FALSE	FALSE
SPC10388	clade_393	clade_466	hetero	−	FALSE	FALSE
SPC10390	clade_393	clade_260	hetero		FALSE	FALSE
SPC10403	clade_393	clade_309	hetero	−−	FALSE	FALSE
SPC10414	clade_393	clade_500	hetero		FALSE	FALSE
SPC10415	clade_393	clade_309	hetero	−−−	FALSE	FALSE
SPC10001	clade_479	clade_479	homo		FALSE	
SPC10019	clade_479	clade_110	hetero		FALSE	TRUE
SPC10030	clade_479	clade_38	hetero		FALSE	FALSE
SPC10048	clade_479	clade_286	hetero		FALSE	FALSE
SPC10081	clade_479	clade_378	hetero		FALSE	TRUE
SPC10097	clade_479	clade_553	hetero	+++	FALSE	TRUE
SPC10110	clade_479	clade_92	hetero	++++	FALSE	TRUE
SPC10197	clade_479	clade_309	hetero		FALSE	FALSE
SPC10211	clade_479	clade_170	hetero	+	FALSE	FALSE
SPC10213	clade_479	clade_85	hetero		FALSE	FALSE

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SPC10233	clade_479	clade_262	hetero		FALSE	FALSE
SPC10243	clade_479	clade_408	hetero	+++	FALSE	FALSE
SPC10298	clade_479	clade_172	hetero		FALSE	FALSE
SPC10301	clade_479	clade_172	hetero	++++	TRUE	TRUE
SPC10304	clade_479	clade_262	hetero		FALSE	FALSE
SPC10355	clade_479	clade_408	hetero		FALSE	FALSE
SPC10363	clade_479	clade_444	hetero		FALSE	FALSE
SPC10386	clade_479	clade_478	hetero	–	FALSE	FALSE
SPC10388	clade_479	clade_466	hetero	–	FALSE	FALSE
SPC10390	clade_479	clade_260	hetero	––	FALSE	FALSE
SPC10403	clade_479	clade_309	hetero	–––	FALSE	FALSE
SPC10414	clade_479	clade_500	hetero	––	FALSE	FALSE
SPC10415	clade_479	clade_309	hetero	––	FALSE	FALSE
SPC10019	clade_110	clade_110	homo		FALSE	
SPC10030	clade_110	clade_38	hetero		FALSE	FALSE
SPC10048	clade_110	clade_286	hetero		FALSE	FALSE
SPC10081	clade_110	clade_378	hetero		FALSE	TRUE
SPC10097	clade_110	clade_553	hetero	++++	TRUE	TRUE
SPC10110	clade_110	clade_92	hetero	++++	TRUE	TRUE
SPC10197	clade_110	clade_309	hetero		FALSE	FALSE
SPC10211	clade_110	clade_170	hetero	++	FALSE	FALSE
SPC10213	clade_110	clade_85	hetero		FALSE	FALSE
SPC10233	clade_110	clade_262	hetero	+	FALSE	FALSE
SPC10243	clade_110	clade_408	hetero	+++	FALSE	TRUE
SPC10298	clade_110	clade_172	hetero	–	FALSE	FALSE
SPC10301	clade_110	clade_172	hetero	++++	TRUE	TRUE
SPC10304	clade_110	clade_262	hetero		FALSE	FALSE
SPC10355	clade_110	clade_408	hetero		FALSE	FALSE
SPC10363	clade_110	clade_444	hetero	–	FALSE	FALSE
SPC10386	clade_110	clade_478	hetero	–	FALSE	FALSE
SPC10388	clade_110	clade_466	hetero	––	FALSE	FALSE
SPC10390	clade_110	clade_260	hetero	–––	FALSE	FALSE
SPC10403	clade_110	clade_309	hetero	–	FALSE	FALSE
SPC10414	clade_110	clade_500	hetero		FALSE	FALSE
SPC10415	clade_110	clade_309	hetero		FALSE	FALSE
SPC10030	clade_38	clade_38	homo	+	FALSE	
SPC10048	clade_38	clade_286	hetero		FALSE	FALSE
SPC10081	clade_38	clade_378	hetero		FALSE	FALSE
SPC10097	clade_38	clade_553	hetero	++++	TRUE	TRUE
SPC10110	clade_38	clade_92	hetero	++++	TRUE	TRUE
SPC10197	clade_38	clade_309	hetero		FALSE	FALSE
SPC10211	clade_38	clade_170	hetero		FALSE	FALSE
SPC10213	clade_38	clade_85	hetero		FALSE	FALSE
SPC10233	clade_38	clade_262	hetero	++++	TRUE	FALSE
SPC10243	clade_38	clade_408	hetero	+++	FALSE	FALSE
SPC10298	clade_38	clade_172	hetero	++++	TRUE	TRUE
SPC10301	clade_38	clade_172	hetero	++++	TRUE	TRUE
SPC10304	clade_38	clade_262	hetero		FALSE	FALSE
SPC10355	clade_38	clade_408	hetero		FALSE	FALSE
SPC10363	clade_38	clade_444	hetero		FALSE	FALSE
SPC10386	clade_38	clade_478	hetero		FALSE	FALSE
SPC10388	clade_38	clade_466	hetero	–	FALSE	FALSE
SPC10390	clade_38	clade_260	hetero	–	FALSE	FALSE
SPC10403	clade_38	clade_309	hetero		FALSE	FALSE
SPC10414	clade_38	clade_500	hetero		FALSE	FALSE
SPC10415	clade_38	clade_309	hetero		FALSE	FALSE
SPC10048	clade_286	clade_286	homo	+	FALSE	
SPC10081	clade_286	clade_378	hetero		FALSE	FALSE
SPC10097	clade_286	clade_553	hetero	++++	FALSE	TRUE
SPC10110	clade_286	clade_92	hetero	++++	FALSE	TRUE
SPC10197	clade_286	clade_309	hetero		FALSE	FALSE
SPC10211	clade_286	clade_170	hetero		FALSE	FALSE
SPC10213	clade_286	clade_85	hetero		FALSE	FALSE
SPC10233	clade_286	clade_262	hetero	+	FALSE	FALSE
SPC10243	clade_286	clade_408	hetero	+	FALSE	FALSE
SPC10298	clade_286	clade_172	hetero	++++	TRUE	TRUE
SPC10301	clade_286	clade_172	hetero	++++	TRUE	FALSE
SPC10304	clade_286	clade_262	hetero		FALSE	FALSE
SPC10355	clade_386	clade_408	hetero		FALSE	FALSE
SPC10363	clade_286	clade_444	hetero		FALSE	FALSE
SPC10386	clade_286	clade_478	hetero	–	FALSE	FALSE
SPC10388	clade_286	clade_466	hetero		FALSE	FALSE
SPC10390	clade_386	clade_380	hetero		FALSE	FALSE
SPC10403	clade_286	clade_309	hetero		FALSE	FALSE
SPC10414	clade_386	clade_500	hetero		FALSE	FALSE
SPC10415	clade_386	clade_309	hetero	++	FALSE	FALSE
SPC10081	clade_378	clade_378	homo		FALSE	
SPC10097	clade_378	clade_553	hetero	+++	FALSE	TRUE
SPC10110	clade_378	clade_92	hetero	++++	TRUE	TRUE
SPC10197	clade_378	clade_309	hetero		FALSE	FALSE

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SPC10211	clade_378	clade_170	hetero		FALSE	FALSE
SPC10213	clade_378	clade_85	hetero		FALSE	FALSE
SPC10233	clade_378	clade_262	hetero	+++	FALSE	FALSE
SPC10243	clade_378	clade_408	hetero	++++	FALSE	TRUE
SPC10298	clade_378	clade_172	hetero	++++	FALSE	TRUE
SPC10301	clade_378	clade_172	hetero	++++	TRUE	TRUE
SPC10304	clade_378	clade_262	hetero		FALSE	FALSE
SPC10355	clade_378	clade_408	hetero		FALSE	FALSE
SPC10363	clade_378	clade_444	hetero		FALSE	FALSE
SPC10386	clade_378	clade_478	hetero	--	FALSE	FALSE
SPC10388	clade_378	clade_466	hetero		FALSE	FALSE
SPC10390	clade_378	clade_260	hetero	-	FALSE	FALSE
SPC10403	clade_378	clade_309	hetero		FALSE	FALSE
SPC10414	clade_378	clade_500	hetero		FALSE	TRUE
SPC10415	clade_378	clade_309	hetero	+++	FALSE	TRUE
SPC10097	clade_553	clade_553	homo	++	FALSE	
SPC10110	clade_553	clade_92	hetero	++++	TRUE	TRUE
SPC10197	clade_553	clade_309	hetero	++++	FALSE	FALSE
SPC10211	clade_553	clade_170	hetero	++++	FALSE	FALSE
SPC10213	clade_553	clade_85	hetero	++++	FALSE	TRUE
SPC10233	clade_553	clade_262	hetero	++++	TRUE	FALSE
SPC10243	clade_553	clade_408	hetero	++++	TRUE	TRUE
SPC10298	clade_553	clade_172	hetero	++++	TRUE	TRUE
SPC10301	clade_553	clade_172	hetero	++++	TRUE	TRUE
SPC10304	clade_553	clade_262	hetero	++++	FALSE	TRUE
SPC10355	clade_553	clade_408	hetero	++++	FALSE	TRUE
SPC10363	clade_553	clade_444	hetero	+++	FALSE	TRUE
SPC10386	clade_553	clade_478	hetero	++++	FALSE	TRUE
SPC10388	clade_553	clade_466	hetero	++++	FALSE	TRUE
SPC10390	clade_553	clade_260	hetero	++++	FALSE	TRUE
SPC10403	clade_553	clade_309	hetero	+++	FALSE	TRUE
SPC10414	clade_553	clade_500	hetero	++++	FALSE	TRUE
SPC10415	clade_553	clade_309	hetero	++++	TRUE	TRUE
SPC10110	clade_92	clade_92	homo	+++	FALSE	
SPC10197	clade_92	clade_309	hetero	+++	FALSE	FALSE
SPC10211	clade_92	clade_170	hetero	++++	TRUE	TRUE
SPC10213	clade_92	clade_85	hetero	++++	TRUE	TRUE
SPC10233	clade_92	clade_262	hetero	++++	FALSE	FALSE
SPC10243	clade_92	clade_408	hetero	++++	FALSE	FALSE
SPC10298	clade_92	clade_172	hetero	++++	TRUE	TRUE
SPC10301	clade_92	clade_172	hetero	++++	TRUE	FALSE
SPC10304	clade_92	clade_262	hetero	+++	FALSE	TRUE
SPC10355	clade_92	clade_408	hetero	++++	FALSE	TRUE
SPC10363	clade_92	clade_444	hetero	++++	FALSE	TRUE
SPC10386	clade_92	clade_478	hetero	+	FALSE	FALSE
SPC10388	clade_92	clade_466	hetero	+++	FALSE	TRUE
SPC10390	clade_92	clade_260	hetero	++++	FALSE	TRUE
SPC10403	clade_92	clade_309	hetero	+++	FALSE	TRUE
SPC10414	clade_92	clade_500	hetero	++++	TRUE	TRUE
SPC10415	clade_92	clade_309	hetero	++++	FALSE	FALSE
SPC10197	clade_309	clade_309	homo	++++	TRUE	
SPC10211	clade_309	clade_170	hetero	++++	TRUE	FALSE
SPC10213	clade_309	clade_85	hetero	++++	TRUE	FALSE
SPC10233	clade_309	clade_262	hetero	++++	TRUE	FALSE
SPC10243	clade_309	clade_408	hetero	++++	TRUE	TRUE
SPC10298	clade_309	clade_172	hetero	+++	FALSE	FALSE
SPC10301	clade_309	clade_172	hetero	++++	TRUE	FALSE
SPC10304	clade_309	clade_262	hetero	++++	FALSE	TRUE
SPC10355	clade_309	clade_408	hetero	+++	FALSE	FALSE
SPC10363	clade_309	clade_444	hetero	+	FALSE	FALSE
SPC10386	clade_309	clade_478	hetero		FALSE	FALSE
SPC10388	clade_309	clade_466	hetero		FALSE	FALSE
SPC10390	clade_309	clade_260	hetero		FALSE	FALSE
SPC10403	clade_309	clade_309	hetero		FALSE	FALSE
SPC10414	clade_309	clade_500	hetero		FALSE	FALSE
SPC10415	clade_309	clade_309	hetero		FALSE	FALSE
SPC10211	clade_170	clade_170	homo	++++	TRUE	
SPC10213	clade_170	clade_85	hetero	++++	TRUE	FALSE
SPC10233	clade_170	clade_262	hetero	++++	TRUE	FALSE
SPC10243	clade_170	clade_408	hetero	++++	FALSE	FALSE
SPC10298	clade_170	clade_172	hetero	++++	FALSE	FALSE
SPC10301	clade_170	clade_172	hetero	++++	TRUE	FALSE
SPC10304	clade_170	clade_262	hetero	+++	FALSE	FALSE
SPC10355	clade_170	clade_408	hetero	++	FALSE	FALSE
SPC10363	clade_170	clade_444	hetero		FALSE	FALSE
SPC10386	clade_170	clade_478	hetero		FALSE	FALSE
SPC10388	clade_170	clade_466	hetero		FALSE	FALSE
SPC10390	clade_170	clade_260	hetero		FALSE	FALSE
SPC10403	clade_170	clade_309	hetero		FALSE	FALSE
SPC10414	clade_170	clade_500	hetero		FALSE	FALSE

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SPC10415	clade_170	clade_309	hetero		FALSE	FALSE
SPC10213	clade_85	clade_85	homo	+++	FALSE	
SPC10233	clade_85	clade_262	hetero	++++	TRUE	FALSE
SPC10243	clade_85	clade_408	hetero	++++	TRUE	FALSE
SPC10298	clade_85	clade_172	hetero	++++	FALSE	FALSE
SPC10301	clade_85	clade_172	hetero	++++	TRUE	FALSE
SPC10304	clade_85	clade_262	hetero	++	FALSE	FALSE
SPC10355	clade_85	clade_408	hetero	+++	FALSE	FALSE
SPC10363	clade_85	clade_444	hetero		FALSE	FALSE
SPC10386	clade_85	clade_478	hetero		FALSE	FALSE
SPC10388	clade_85	clade_466	hetero		FALSE	FALSE
SPC10390	clade_38	clade_260	hetero		FALSE	FALSE
SPC10403	clade_38	clade_309	hetero	–	FALSE	FALSE
SPC10414	clade_85	clade_500	hetero	–	FALSE	FALSE
SPC10415	clade_38	clade_309	hetero		FALSE	FALSE
SPC10233	clade_262	clade_262	homo	++++	TRUE	
SPC10243	clade_262	clade_408	hetero	++++	TRUE	TRUE
SPC10298	clade_262	clade_172	hetero	++++	TRUE	FALSE
SPC10301	clade_262	clade_172	hetero	++++	TRUE	FALSE
SPC10304	clade_262	clade_262	hetero	++++	FALSE	FALSE
SPC10355	clade_262	clade_408	hetero	+	FALSE	FALSE
SPC10363	clade_262	clade_444	hetero		FALSE	FALSE
SPC10386	clade_262	clade_478	hetero		FALSE	FALSE
SPC10388	clade_262	clade_466	hetero		FALSE	FALSE
SPC10390	clade_262	clade_260	hetero		FALSE	FALSE
SPC10403	clade_262	clade_309	hetero		FALSE	FALSE
SPC10414	clade_262	clade_500	hetero		FALSE	FALSE
SPC10415	clade_262	clade_309	hetero		FALSE	FALSE
SPC10243	clade_408	clade_408	homo	++++	TRUE	
SPC10298	clade_408	clade_172	hetero	++++	TRUE	FALSE
SPC10301	clade_408	clade_172	hetero	++++	TRUE	TRUE
SPC10304	clade_408	clade_262	hetero	++++	FALSE	TRUE
SPC10355	clade_408	clade_408	hetero	+++	FALSE	FALSE
SPC10363	clade_408	clade_444	hetero	++++	FALSE	TRUE
SPC10386	clade_408	clade_478	hetero	++	FALSE	FALSE
SPC10388	clade_408	clade_466	hetero	+++	FALSE	FALSE
SPC10390	clade_408	clade_260	hetero	+++	FALSE	FALSE
SPC10403	clade_408	clade_309	hetero	+++	FALSE	FALSE
SPC10414	clade_408	clade_500	hetero	+++	FALSE	FALSE
SPC10415	clade_408	clade_309	hetero	++	FALSE	FALSE
SPC10298	clade_172	clade_172	homo	++++	TRUE	
SPC10301	clade_172	clade_172	hetero	++++	TRUE	TRUE
SPC10304	clade_172	clade_262	hetero	++++	FALSE	TRUE
SPC10355	clade_172	clade_408	hetero	+	FALSE	FALSE
SPC10363	clade_172	clade_444	hetero		FALSE	FALSE
SPC10386	clade_172	clade_478	hetero		FALSE	FALSE
SPC10388	clade_172	clade_466	hetero	++	FALSE	FALSE
SPC10390	clade_172	clade_260	hetero		FALSE	FALSE
SPC10403	clade_172	clade_309	hetero		FALSE	FALSE
SPC10414	clade_172	clade_500	hetero		FALSE	FALSE
SPC10415	clade_172	clade_309	hetero		FALSE	FALSE
SPC10301	clade_172	clade_172	homo	++++	TRUE	
SPC10304	clade_172	clade_262	hetero	++++	TRUE	TRUE
SPC10355	clade_172	clade_408	hetero	++++	FALSE	FALSE
SPC10363	clade_172	clade_444	hetero	++++	TRUE	TRUE
SPC10386	clade_172	clade_478	hetero		FALSE	FALSE
SPC10388	clade_172	clade_466	hetero	++++	TRUE	FALSE
SPC10390	clade_172	clade_260	hetero	++++	FALSE	FALSE
SPC10403	clade_172	clade_309	hetero	+	FALSE	FALSE
SPC10414	clade_172	clade_500	hetero	+++	FALSE	FALSE
SPC10415	clade_172	clade_309	hetero	+++	FALSE	FALSE
SPC10304	clade_262	clade_262	homo		FALSE	
SPC10355	clade_262	clade_408	hetero	++	FALSE	FALSE
SPC10363	clade_262	clade_444	hetero	+	FALSE	TRUE
SPC10386	clade_262	clade_478	hetero	+	FALSE	TRUE
SPC10388	clade_262	clade_466	hetero		FALSE	FALSE
SPC10390	clade_262	clade_260	hetero	++++	FALSE	TRUE
SPC10403	clade_262	clade_309	hetero		FALSE	TRUE
SPC10414	clade_262	clade_380	hetero		FALSE	FALSE
SPC10415	clade_262	clade_309	hetero	++++	TRUE	TRUE
SPC10355	clade_408	clade_408	homo	++	FALSE	
SPC10363	clade_408	clade_444	hetero	+	FALSE	FALSE
SPC10386	clade_408	clade_478	hetero	+++	FALSE	TRUE
SPC10388	clade_408	clade_466	hetero	+	FALSE	FALSE
SPC10390	clade_408	clade_260	hetero	+++	FALSE	FALSE
SPC10403	clade_408	clade_309	hetero		FALSE	FALSE
SPC10414	clade_408	clade_500	hetero		FALSE	FALSE
SPC10415	clade_408	clade_309	hetero	++++	TRUE	TRUE
SPC10363	clade_444	clade_444	homo	+	FALSE	
SPC10386	clade_444	clade_478	hetero	+	FALSE	TRUE

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SPC10388	clade_444	clade_466	hetero	+	FALSE	FALSE
SPC10390	clade_444	clade_260	hetero		FALSE	FALSE
SPC10403	clade_444	clade_309	hetero		FALSE	FALSE
SPC10414	clade_444	clade_500	hetero		FALSE	FALSE
SPC10415	clade_444	clade_309	hetero	+	FALSE	FALSE
SPC10386	clade_478	clade_478	homo		FALSE	
SPC10388	clade_478	clade_466	hetero	+	FALSE	TRUE
SPC10390	clade_478	clade_260	hetero	++	FALSE	TRUE
SPC10403	clade_478	clade_309	hetero		FALSE	TRUE
SPC10414	clade_478	clade_500	hetero		FALSE	TRUE
SPC10415	clade_478	clade_309	hetero	++++	TRUE	TRUE
SPC10388	clade_466	clade_466	homo		FALSE	
SPC10390	clade_466	clade_260	hetero		FALSE	FALSE
SPC10403	clade_466	clade_309	hetero		FALSE	FALSE
SPC10414	clade_466	clade_500	hetero		FALSE	TRUE
SPC10415	clade_466	clade_309	hetero		FALSE	FALSE
SPC10390	clade_260	clade_260	homo	++	FALSE	
SPC10403	clade_260	clade_309	hetero		FALSE	FALSE
SPC10414	clade_260	clade_500	hetero		FALSE	FALSE
SPC10415	clade_260	clade_309	hetero	++++	TRUE	TRUE
SPC10403	clade_309	clade_309	homo		FALSE	
SPC10414	clade_309	clade_500	hetero		FALSE	TRUE
SPC10415	clade_309	clade_309	hetero		FALSE	FALSE
SPC10414	clade_500	clade_500	homo		FALSE	
SPC10415	clade_500	clade_309	hetero		FALSE	FALSE
SPC10415	clade_309	clade_309	homo	++++	TRUE	
SPC10155	clade_553	clade_252	hetero		FALSE	FALSE
SPC10167	clade_553	clade_253	hetero	−	FALSE	FALSE
SPC10202	clade_553	clade_351	hetero	++	FALSE	FALSE
SPC10238	clade_553	clade_354	hetero	++++	TRUE	TRUE
SPC10256	clade_553	clade_252	hetero	++++	TRUE	TRUE
SPC10313	clade_553	clade_260	hetero	+++	FALSE	TRUE
SPC10325	clade_553	clade_408	hetero	++++	FALSE	TRUE
SPC10358	clade_553	clade_494	hetero	++++	TRUE	TRUE
SPC10468	clade_553	clade_360	hetero	++++	FALSE	TRUE
SPC10470	clade_553	clade_537	hetero	++++	FALSE	TRUE
SPC10567	clade_553	clade_444	hetero	++++	FALSE	TRUE
SPC10097	clade_252	clade_553	hetero		FALSE	FALSE
SPC10155	clade_252	clade_252	homo	++	FALSE	
SPC10167	clade_252	clade_253	hetero	++++	FALSE	TRUE
SPC10202	clade_252	clade_351	hetero	++++	TRUE	TRUE
SPC10238	clade_252	clade_354	hetero	++++	TRUE	TRUE
SPC10256	clade_252	clade_252	hetero	++++	TRUE	TRUE
SPC10304	clade_252	clade_262	hetero	++++	TRUE	TRUE
SPC10313	clade_252	clade_260	hetero	++++	TRUE	TRUE
SPC10325	clade_252	clade_408	hetero	++++	TRUE	TRUE
SPC10355	clade_252	clade_408	hetero	+	FALSE	FALSE
SPC10358	clade_252	clade_494	hetero	++++	TRUE	TRUE
SPC10386	clade_252	clade_478	hetero	++++	TRUE	TRUE
SPC10390	clade_252	clade_260	hetero	++++	TRUE	TRUE
SPC10415	clade_252	clade_309	hetero	++++	TRUE	TRUE
SPC10468	clade_252	clade_360	hetero	++++	TRUE	TRUE
SPC10470	clade_252	clade_537	hetero	++++	TRUE	TRUE
SPC10567	clade_252	clade_444	hetero	++++	TRUE	TRUE
SPC10155	clade_253	clade_252	hetero	++++	FALSE	TRUE
SPC10167	clade_253	clade_253	homo	+	FALSE	
SPC10202	clade_253	clade_351	hetero	++	FALSE	FALSE
SPC10238	clade_253	clade_354	hetero	++++	TRUE	TRUE
SPC10256	clade_253	clade_252	hetero	++++	TRUE	TRUE
SPC10304	clade_253	clade_262	hetero	++++	FALSE	TRUE
SPC10313	clade_253	clade_260	hetero		FALSE	FALSE
SPC10325	clade_253	clade_408	hetero	++++	FALSE	TRUE
SPC10355	clade_253	clade_408	hetero	+++	FALSE	TRUE
SPC10358	clade_253	clade_494	hetero	++++	TRUE	TRUE
SPC10386	clade_253	clade_478	hetero		FALSE	FALSE
SPC10390	clade_253	clade_260	hetero	++++	FALSE	TRUE
SPC10415	clade_253	clade_309	hetero	++++	TRUE	TRUE
SPC10468	clade_253	clade_360	hetero	++++	TRUE	TRUE
SPC10470	clade_253	clade_537	hetero		FALSE	FALSE
SPC10567	clade_253	clade_444	hetero	++++	TRUE	TRUE
SPC10167	clade_351	clade_253	hetero	++	FALSE	FALSE
SPC10202	clade_351	clade_351	homo	+++	FALSE	
SPC10238	clade_351	clade_354	hetero	++++	TRUE	TRUE
SPC10256	clade_351	clade_252	hetero	++++	TRUE	TRUE
SPC10304	clade_351	clade_262	hetero	++++	FALSE	TRUE
SPC10313	clade_351	clade_260	hetero	+++	FALSE	TRUE
SPC10325	clade_351	clade_408	hetero	++++	FALSE	TRUE
SPC10355	clade_351	clade_408	hetero	++++	FALSE	TRUE
SPC10358	clade_351	clade_494	hetero	++++	FALSE	TRUE
SPC10386	clade_351	clade_478	hetero	++++	TRUE	TRUE

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SPC10390	clade_351	clade_260	hetero	++++	TRUE	TRUE
SPC10415	clade_351	clade_309	hetero	++++	TRUE	TRUE
SPC10468	clade_351	clade_360	hetero	++++	TRUE	TRUE
SPC10470	clade_351	clade_537	hetero	++++	TRUE	TRUE
SPC10567	clade_351	clade_444	hetero	++++	TRUE	TRUE
SPC10202	clade_354	clade_351	hetero	++++	TRUE	TRUE
SPC10238	clade_354	clade_354	homo	++++	TRUE	
SPC10256	clade_354	clade_252	hetero	++++	TRUE	TRUE
SPC10304	clade_354	clade_262	hetero	++++	TRUE	TRUE
SPC10313	clade_354	clade_260	hetero	++++	TRUE	TRUE
SPC10325	clade_354	clade_408	hetero	++++	TRUE	TRUE
SPC10355	clade_354	clade_408	hetero	++++	TRUE	TRUE
SPC10358	clade_354	clade_494	hetero	++++	TRUE	TRUE
SPC10386	clade_354	clade_478	hetero	++++	TRUE	TRUE
SPC10390	clade_354	clade_260	hetero	++++	TRUE	TRUE
SPC10415	clade_354	clade_309	hetero	++++	TRUE	TRUE
SPC10468	clade_354	clade_360	hetero	++++	TRUE	TRUE
SPC10470	clade_354	clade_537	hetero	++++	TRUE	TRUE
SPC10567	clade_354	clade_444	hetero	++++	TRUE	TRUE
SPC10238	clade_252	clade_354	hetero	++++	TRUE	TRUE
SPC10256	clade_252	clade_252	homo	++++	TRUE	
SPC10304	clade_252	clade_262	hetero	++++	TRUE	TRUE
SPC10313	clade_252	clade_260	hetero	++++	TRUE	TRUE
SPC10325	clade_252	clade_408	hetero	++++	TRUE	TRUE
SPC10355	clade_252	clade_408	hetero	++++	TRUE	TRUE
SPC10358	clade_252	clade_494	hetero	++++	TRUE	TRUE
SPC10386	clade_252	clade_478	hetero	++++	TRUE	TRUE
SPC10390	clade_252	clade_260	hetero	++++	TRUE	TRUE
SPC10415	clade_252	clade_309	hetero	++++	TRUE	TRUE
SPC10468	clade_252	clade_360	hetero	++++	TRUE	TRUE
SPC10470	clade_252	clade_537	hetero	++++	TRUE	TRUE
SPC10567	clade_252	clade_444	hetero	++++	TRUE	TRUE
SPC10256	clade_262	clade_252	hetero	++++	TRUE	TRUE
SPC10313	clade_262	clade_260	hetero		FALSE	TRUE
SPC10325	clade_262	clade_408	hetero	+++	FALSE	TRUE
SPC10358	clade_262	clade_494	hetero	++++	FALSE	TRUE
SPC10468	clade_262	clade_360	hetero	++++	TRUE	TRUE
SPC10470	clade_262	clade_537	hetero	++++	FALSE	TRUE
SPC10567	clade_262	clade_444	hetero	++++	FALSE	TRUE
SPC10304	clade_260	clade_262	hetero		FALSE	TRUE
SPC10313	clade_260	clade_260	homo		FALSE	
SPC10325	clade_260	clade_408	hetero		FALSE	FALSE
SPC10355	clade_260	clade_408	hetero	++	FALSE	TRUE
SPC10358	clade_260	clade_494	hetero		FALSE	TRUE
SPC10386	clade_260	clade_478	hetero		FALSE	TRUE
SPC10390	clade_260	clade_260	hetero	++++	TRUE	TRUE
SPC10415	clade_260	clade_309	hetero	++++	TRUE	TRUE
SPC10468	clade_260	clade_360	hetero	++	FALSE	TRUE
SPC10470	clade_260	clade_537	hetero		FALSE	TRUE
SPC10567	clade_260	clade_444	hetero		FALSE	TRUE
SPC10313	clade_408	clade_260	hetero		FALSE	FALSE
SPC10325	clade_408	clade_408	homo	++	FALSE	
SPC10355	clade_408	clade_408	hetero	+++	FALSE	TRUE
SPC10358	clade_408	clade_494	hetero	+++	FALSE	TRUE
SPC10386	clade_408	clade_478	hetero	+	FALSE	FALSE
SPC10390	clade_408	clade_260	hetero	++++	TRUE	TRUE
SPC10415	clade_408	clade_309	hetero	++++	TRUE	TRUE
SPC10468	clade_408	clade_360	hetero	++++	FALSE	TRUE
SPC10470	clade_408	clade_537	hetero	++	FALSE	TRUE
SPC10567	clade_408	clade_444	hetero	+	FALSE	TRUE
SPC10325	clade_408	clade_408	hetero	+++	FALSE	TRUE
SPC10358	clade_408	clade_494	hetero	+++	FALSE	TRUE
SPC10468	clade_408	clade_360	hetero	++++	FALSE	TRUE
SPC10470	clade_408	clade_537	hetero	++++	FALSE	TRUE
SPC10567	clade_408	clade_444	hetero	+	FALSE	TRUE
SPC10355	clade_494	clade_408	hetero	+++	FALSE	TRUE
SPC10358	clade_494	clade_494	homo		FALSE	
SPC10386	clade_494	clade_478	hetero		FALSE	FALSE
SPC10390	clade_494	clade_260	hetero	+++	FALSE	TRUE
SPC10415	clade_494	clade_309	hetero	++++	TRUE	TRUE
SPC10468	clade_494	clade_360	hetero	++++	TRUE	TRUE
SPC10470	clade_494	clade_537	hetero		FALSE	FALSE
SPC10567	clade_494	clade_444	hetero		FALSE	TRUE
SPC10358	clade_478	clade_494	hetero		FALSE	FALSE
SPC10468	clade_478	clade_360	hetero	++++	FALSE	TRUE
SPC10470	clade_478	clade_537	hetero		FALSE	FALSE
SPC10567	clade_478	clade_444	hetero		FALSE	TRUE
SPC10468	clade_260	clade_360	hetero	++++	FALSE	TRUE
SPC10470	clade_260	clade_537	hetero	+++	FALSE	TRUE
SPC10567	clade_260	clade_444	hetero	++	FALSE	TRUE

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SPC10468		clade_309	clade_360		hetero	++++	TRUE	TRUE
SPC10470		clade_309	clade_537		hetero	++++	TRUE	TRUE
SPC10567		clade_309	clade_444		hetero	++++	TRUE	TRUE
SPC10415		clade_360	clade_309		hetero	++++	TRUE	TRUE
SPC10468		clade_360	clade_360		homo	+	FALSE	
SPC10470		clade_360	clade_537		hetero	+++	FALSE	TRUE
SPC10567		clade_360	clade_444		hetero	+++	FALSE	TRUE
SPC10468		clade_537	clade_360		hetero	+++	FALSE	TRUE
SPC10470		clade_537	clade_537		homo		FALSE	
SPC10567		clade_537	clade_444		hetero		FALSE	TRUE
SPC10470		clade_444	clade_537		hetero		FALSE	TRUE
SPC10567		clade_444	clade_444		homo		FALSE	
SPC10097	SPC10097	clade_553	clade_553	clade_553	homo	++	FALSE	
SPC10097	SPC10304	clade_553	clade_553	clade_262	semi	++	FALSE	FALSE
SPC10097	SPC10325	clade_553	clade_553	clade_408	semi	++++	FALSE	FALSE
SPC10097	SPC10355	clade_553	clade_553	clade_408	semi	+++	FALSE	FALSE
SPC10097	SPC10386	clade_553	clade_553	clade_478	semi	++++	FALSE	TRUE
SPC10097	SPC10390	clade_553	clade_553	clade_260	semi	+++	FALSE	FALSE
SPC10097	SPC10415	clade_553	clade_553	clade_309	semi	++++	FALSE	TRUE
SPC10097	SPC10567	clade_553	clade_553	clade_444	semi		FALSE	FALSE
SPC10304	SPC10304	clade_553	clade_262	clade_262	semi		FALSE	FALSE
SPC10304	SPC10325	clade_553	clade_262	clade_408	hetero	++++	FALSE	FALSE
SPC10304	SPC10355	clade_553	clade_262	clade_408	hetero	+++	FALSE	FALSE
SPC10304	SPC10386	clade_553	clade_262	clade_478	hetero	++++	FALSE	TRUE
SPC10304	SPC10390	clade_553	clade_262	clade_260	hetero	+++	FALSE	FALSE
SPC10304	SPC10415	clade_553	clade_262	clade_309	hetero	++++	FALSE	TRUE
SPC10304	SPC10567	clade_553	clade_262	clade_444	hetero	+++	FALSE	TRUE
SPC10325	SPC10325	clade_553	clade_408	clade_408	semi	+++	FALSE	FALSE
SPC10325	SPC10355	clade_553	clade_408	clade_408	hetero	++++	FALSE	FALSE
SPC10325	SPC10386	clade_553	clade_408	clade_478	hetero	++++	FALSE	TRUE
SPC10325	SPC10390	clade_553	clade_408	clade_260	hetero	++++	FALSE	FALSE
SPC10325	SPC10415	clade_553	clade_408	clade_309	hetero	++++	FALSE	TRUE
SPC10325	SPC10567	clade_553	clade_408	clade_444	hetero	++++	FALSE	FALSE
SPC10355	SPC10355	clade_553	clade_408	clade_408	semi		FALSE	FALSE
SPC10355	SPC10386	clade_553	clade_408	clade_478	hetero		FALSE	FALSE
SPC10355	SPC10390	clade_553	clade_408	clade_260	hetero	+	FALSE	FALSE
SPC10355	SPC10415	clade_553	clade_408	clade_309	hetero	++++	FALSE	TRUE
SPC10355	SPC10567	clade_553	clade_408	clade_444	hetero		FALSE	FALSE
SPC10386	SPC10386	clade_553	clade_478	clade_478	semi	++++	FALSE	TRUE
SPC10386	SPC10390	clade_553	clade_478	clade_260	hetero	+++	FALSE	FALSE
SPC10386	SPC10415	clade_553	clade_478	clade_309	hetero	++++	TRUE	TRUE
SPC10386	SPC10567	clade_553	clade_478	clade_444	hetero	+++	FALSE	TRUE
SPC10390	SPC10390	clade_553	clade_260	clade_260	semi	+++	FALSE	FALSE
SPC10390	SPC10415	clade_553	clade_260	clade_309	hetero	++++	FALSE	TRUE
SPC10390	SPC10567	clade_553	clade_260	clade_444	hetero	++++	FALSE	FALSE
SPC10415	SPC10415	clade_553	clade_309	clade_309	semi	++++	FALSE	FALSE
SPC10415	SPC10567	clade_553	clade_309	clade_444	hetero	++++	FALSE	TRUE
SPC10567	SPC10567	clade_553	clade_444	clade_444	semi	+	FALSE	TRUE
SPC10304	SPC10304	clade_262	clade_262	clade_262	homo		FALSE	
SPC10304	SPC10325	clade_262	clade_262	clade_408	semi		FALSE	FALSE
SPC10304	SPC10355	clade_262	clade_262	clade_408	semi		FALSE	FALSE
SPC10304	SPC10386	clade_262	clade_262	clade_478	semi		FALSE	FALSE
SPC10304	SPC10390	clade_262	clade_262	clade_260	semi		FALSE	FALSE
SPC10304	SPC10415	clade_262	clade_262	clade_309	semi	++++	FALSE	TRUE
SPC10304	SPC10567	clade_262	clade_262	clade_444	semi	−−	FALSE	FALSE
SPC10325	SPC10325	clade_262	clade_408	clade_408	semi	++	FALSE	FALSE
SPC10325	SPC10355	clade_262	clade_408	clade_408	hetero		FALSE	FALSE
SPC10325	SPC10386	clade_262	clade_408	clade_478	hetero	+++	FALSE	FALSE
SPC10325	SPC10390	clade_262	clade_408	clade_260	hetero	+++	FALSE	FALSE
SPC10325	SPC10415	clade_262	clade_408	clade_309	hetero	++++	FALSE	FALSE
SPC10325	SPC10567	clade_262	clade_408	clade_444	hetero	−−	FALSE	FALSE
SPC10355	SPC10355	clade_262	clade_408	clade_408	semi		FALSE	FALSE
SPC10355	SPC10386	clade_262	clade_408	clade_478	hetero		FALSE	FALSE
SPC10355	SPC10390	clade_262	clade_408	clade_260	hetero		FALSE	FALSE
SPC10355	SPC10415	clade_262	clade_408	clade_309	hetero	++++	FALSE	FALSE
SPC10355	SPC10567	clade_262	clade_408	clade_444	hetero	−−−	FALSE	FALSE
SPC10386	SPC10386	clade_262	clade_478	clade_478	semi	−−−	FALSE	FALSE
SPC10386	SPC10390	clade_262	clade_478	clade_260	hetero		FALSE	FALSE
SPC10386	SPC10415	clade_262	clade_478	clade_309	hetero	++++	FALSE	TRUE
SPC10386	SPC10567	clade_262	clade_478	clade_444	hetero	−	FALSE	FALSE
SPC10390	SPC10390	clade_262	clade_260	clade_260	semi	+	FALSE	FALSE
SPC10390	SPC10415	clade_262	clade_260	clade_309	hetero	++++	FALSE	TRUE
SPC10390	SPC10567	clade_262	clade_260	clade_444	hetero		FALSE	FALSE
SPC10415	SPC10415	clade_262	clade_309	clade_309	semi	++++	FALSE	FALSE
SPC10415	SPC10567	clade_262	clade_309	clade_444	hetero	++++	FALSE	TRUE
SPC10567	SPC10567	clade_262	clade_444	clade_444	semi	−	FALSE	FALSE
SPC10325	SPC10325	clade_408	clade_408	clade_408	homo		FALSE	
SPC10325	SPC10355	clade_408	clade_408	clade_408	semi		FALSE	FALSE
SPC10325	SPC10386	clade_408	clade_408	clade_478	semi	++	FALSE	FALSE
SPC10325	SPC10390	clade_408	clade_408	clade_260	semi	++	FALSE	FALSE

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SPC10325	SPC10415	clade_408	clade_408	clade_309	semi	++++	FALSE	FALSE
SPC10325	SPC10567	clade_408	clade_408	clade_444	semi		FALSE	FALSE
SPC10355	SPC10355	clade_408	clade_408	clade_408	semi	–	FALSE	FALSE
SPC10355	SPC10386	clade_408	clade_408	clade_478	hetero	–	FALSE	FALSE
SPC10355	SPC10390	clade_408	clade_408	clade_260	hetero		FALSE	FALSE
SPC10355	SPC10415	clade_408	clade_408	clade_309	hetero	++++	FALSE	FALSE
SPC10355	SPC10567	clade_408	clade_408	clade_444	hetero	–	FALSE	FALSE
SPC10386	SPC10386	clade_408	clade_478	clade_478	semi		FALSE	FALSE
SPC10386	SPC10390	clade_408	clade_478	clade_260	hetero	++++	FALSE	FALSE
SPC10386	SPC10415	clade_408	clade_478	clade_309	hetero	++++	FALSE	TRUE
SPC10386	SPC10567	clade_408	clade_478	clade_444	hetero		FALSE	FALSE
SPC10390	SPC10390	clade_408	clade_260	clade_260	semi	++	FALSE	FALSE
SPC10390	SPC10415	clade_408	clade_260	clade_309	hetero	++++	FALSE	FALSE
SPC10390	SPC10567	clade_408	clade_260	clade_444	hetero	+	FALSE	FALSE
SPC10415	SPC10415	clade_408	clade_309	clade_309	semi	++++	FALSE	FALSE
SPC10415	SPC10567	clade_408	clade_309	clade_444	hetero	++++	FALSE	TRUE
SPC10567	SPC10567	clade_408	clade_444	clade_444	semi	––	FALSE	FALSE
SPC10355	SPC10355	clade_408	clade_408	clade_408	homo		FALSE	
SPC10355	SPC10386	clade_408	clade_408	clade_478	semi		FALSE	FALSE
SPC10355	SPC10390	clade_408	clade_408	clade_260	semi	+	FALSE	FALSE
SPC10355	SPC10415	clade_408	clade_408	clade_309	semi	++++	FALSE	FALSE
SPC10355	SPC10567	clade_408	clade_408	clade_444	semi		FALSE	FALSE
SPC10386	SPC10386	clade_408	clade_478	clade_478	semi	–	FALSE	FALSE
SPC10386	SPC10390	clade_408	clade_478	clade_260	hetero	+	FALSE	FALSE
SPC10386	SPC10415	clade_408	clade_478	clade_309	hetero	++++	FALSE	TRUE
SPC10386	SPC10567	clade_408	clade_478	clade_444	hetero		FALSE	FALSE
SPC10390	SPC10390	clade_408	clade_260	clade_260	semi	+++	FALSE	FALSE
SPC10390	SPC10415	clade_408	clade_260	clade_309	hetero	++++	FALSE	TRUE
SPC10390	SPC10567	clade_408	clade_260	clade_444	hetero		FALSE	FALSE
SPC10415	SPC10415	clade_408	clade_309	clade_309	semi	++++	FALSE	FALSE
SPC10415	SPC10567	clade_408	clade_309	clade_444	hetero	++++	FALSE	TRUE
SPC10567	SPC10567	clade_408	clade_444	clade_444	semi		FALSE	FALSE
SPC10386	SPC10386	clade_478	clade_478	clade_478	homo	–	FALSE	
SPC10386	SPC10390	clade_478	clade_478	clade_260	semi		FALSE	FALSE
SPC10386	SPC10415	clade_478	clade_478	clade_309	semi	++++	FALSE	TRUE
SPC10386	SPC10567	clade_478	clade_478	clade_444	semi	–––	FALSE	FALSE
SPC10390	SPC10390	clade_478	clade_260	clade_260	semi	+++	FALSE	FALSE
SPC10390	SPC10415	clade_478	clade_260	clade_309	hetero	++++	FALSE	TRUE
SPC10390	SPC10567	clade_478	clade_260	clade_444	hetero		FALSE	FALSE
SPC10415	SPC10415	clade_478	clade_309	clade_309	semi	++++	FALSE	TRUE
SPC10415	SPC10567	clade_478	clade_309	clade_444	hetero	++++	FALSE	TRUE
SPC10567	SPC10567	clade_478	clade_444	clade_444	semi	–	FALSE	FALSE
SPC10390	SPC10390	clade_260	clade_260	clade_260	homo	+++	FALSE	
SPC10390	SPC10415	clade_260	clade_260	clade_309	semi	++++	FALSE	TRUE
SPC10390	SPC10567	clade_260	clade_260	clade_444	semi	+	FALSE	FALSE
SPC10415	SPC10415	clade_260	clade_309	clade_309	semi		FALSE	FALSE
SPC10415	SPC10567	clade_260	clade_309	clade_444	hetero	++++	FALSE	TRUE
SPC10567	SPC10567	clade_260	clade_444	clade_444	semi		FALSE	FALSE
SPC10415	SPC10415	clade_309	clade_309	clade_309	homo	++++	FALSE	
SPC10415	SPC10567	clade_309	clade_309	clade_444	semi	++++	FALSE	TRUE
SPC10567	SPC10567	clade_309	clade_444	clade_444	semi	++++	FALSE	TRUE
SPC10567	SPC10567	clade_444	clade_444	clade_444	homo		FALSE	
SPC10155	SPC10155	clade_553	clade_252	clade_252	semi	++++	FALSE	FALSE
SPC10155	SPC10167	clade_553	clade_382	clade_253	hetero	++++	FALSE	FALSE
SPC10155	SPC10202	clade_553	clade_252	clade_351	hetero	++++	FALSE	FALSE
SPC10155	SPC10238	clade_553	clade_252	clade_354	hetero	++++	FALSE	FALSE
SPC10155	SPC10256	clade_553	clade_252	clade_252	hetero	++++	TRUE	TRUE
SPC10155	SPC10313	clade_553	clade_252	clade_260	hetero	++++	FALSE	TRUE
SPC10155	SPC10358	clade_553	clade_252	clade_494	hetero	++++	FALSE	TRUE
SPC10155	SPC10468	clade_553	clade_252	clade_360	hetero	++++	FALSE	TRUE
SPC10155	SPC10470	clade_553	clade_252	clade_537	hetero	++++	FALSE	FALSE
SPC10155	SPC10613	clade_553	clade_382	clade_309	hetero	++++	FALSE	
SPC10167	SPC10167	clade_553	clade_253	clade_253	semi	+++	FALSE	FALSE
SPC10167	SPC10202	clade_553	clade_253	clade_351	hetero	++++	FALSE	FALSE
SPC10167	SPC10238	clade_553	clade_253	clade_354	hetero	++++	FALSE	TRUE
SPC10167	SPC10256	clade_553	clade_253	clade_252	hetero	++++	TRUE	TRUE
SPC10167	SPC10313	clade_553	clade_253	clade_260	hetero	++++	FALSE	FALSE
SPC10167	SPC10358	clade_553	clade_253	clade_494	hetero	++++	FALSE	TRUE
SPC10167	SPC10468	clade_553	clade_253	clade_360	hetero	++++	FALSE	FALSE
SPC10167	SPC10470	clade_553	clade_253	clade_537	hetero	++++	FALSE	FALSE
SPC10167	SPC10613	clade_553	clade_253	clade_309	hetero	++++	FALSE	
SPC10202	SPC10202	clade_553	clade_351	clade_351	semi	++++	FALSE	FALSE
SPC10202	SPC10238	clade_553	clade_351	clade_354	hetero	++++	FALSE	FALSE
SPC10202	SPC10256	clade_553	clade_351	clade_252	hetero	++++	TRUE	TRUE
SPC10202	SPC10313	clade_553	clade_351	clade_260	hetero	++++	FALSE	FALSE
SPC10202	SPC10358	clade_553	clade_351	clade_494	hetero	++++	FALSE	TRUE
SPC10202	SPC10468	clade_553	clade_351	clade_360	hetero	++++	FALSE	FALSE
SPC10202	SPC10470	clade_553	clade_351	clade_537	hetero	++++	FALSE	TRUE
SPC10202	SPC10613	clade_553	clade_351	clade_309	hetero	++++	FALSE	
SPC10238	SPC10238	clade_553	clade_354	clade_354	semi	++++	FALSE	FALSE

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SPC10238	SPC10256	clade_553	clade_354	clade_252	hetero	++++	TRUE	FALSE
SPC10238	SPC10313	clade_553	clade_354	clade_260	hetero	++++	FALSE	FALSE
SPC10238	SPC10358	clade_553	clade_354	clade_494	hetero	++++	FALSE	TRUE
SPC10238	SPC10468	clade_553	clade_354	clade_360	hetero	++++	FALSE	FALSE
SPC10238	SPC10470	clade_553	clade_354	clade_537	hetero	++++	FALSE	TRUE
SPC10238	SPC10613	clade_553	clade_354	clade_309	hetero	++++	FALSE	
SPC10256	SPC10256	clade_553	clade_252	clade_252	semi	++++	TRUE	FALSE
SPC10256	SPC10313	clade_553	clade_252	clade_260	hetero	++++	TRUE	TRUE
SPC10256	SPC10358	clade_553	clade_252	clade_494	hetero	++++	TRUE	TRUE
SPC10256	SPC10468	clade_553	clade_252	clade_360	hetero	++++	TRUE	TRUE
SPC10256	SPC10470	clade_553	clade_252	clade_537	hetero	++++	TRUE	TRUE
SPC10256	SPC10613	clade_553	clade_252	clade_309	hetero	++++	TRUE	
SPC10313	SPC10313	clade_553	clade_260	clade_260	semi	++	FALSE	TRUE
SPC10313	SPC10358	clade_553	clade_260	clade_494	hetero	++++	FALSE	TRUE
SPC10313	SPC10468	clade_553	clade_260	clade_360	hetero	++++	FALSE	TRUE
SPC10313	SPC10470	clade_553	clade_260	clade_537	hetero	++++	FALSE	TRUE
SPC10313	SPC10613	clade_553	clade_260	clade_309	hetero	++++	FALSE	
SPC10358	SPC10358	clade_553	clade_494	clade_494	semi	++++	FALSE	TRUE
SPC10358	SPC10468	clade_553	clade_494	clade_360	hetero	++++	FALSE	TRUE
SPC10358	SPC10470	clade_553	clade_494	clade_537	hetero	++++	FALSE	TRUE
SPC10358	SPC10613	clade_553	clade_494	clade_309	hetero	++++	FALSE	
SPC10468	SPC10468	clade_553	clade_360	clade_360	semi	++++	FALSE	TRUE
SPC10468	SPC10470	clade_553	clade_360	clade_537	hetero	++++	FALSE	TRUE
SPC10468	SPC10613	clade_553	clade_360	clade_309	hetero	++++	FALSE	
SPC10155	SPC10155	clade_262	clade_252	clade_252	semi	----	FALSE	FALSE
SPC10155	SPC10167	clade_262	clade_252	clade_253	hetero	----	FALSE	FALSE
SPC10155	SPC10202	clade_262	clade_252	clade_351	hetero		FALSE	FALSE
SPC10155	SPC10238	clade_262	clade_252	clade_354	hetero	++++	TRUE	TRUE
SPC10155	SPC10256	clade_262	clade_252	clade_252	hetero	++++	TRUE	TRUE
SPC10155	SPC10313	clade_262	clade_252	clade_260	hetero	++++	FALSE	FALSE
SPC10155	SPC10358	clade_262	clade_252	clade_494	hetero	++++	TRUE	TRUE
SPC10155	SPC10468	clade_262	clade_252	clade_360	hetero	++++	FALSE	TRUE
SPC10155	SPC10470	clade_262	clade_252	clade_537	hetero	++	FALSE	FALSE
SPC10155	SPC10613	clade_262	clade_252	clade_309	hetero	++++	FALSE	
SPC10167	SPC10167	clade_262	clade_253	clade_253	semi	----	FALSE	FALSE
SPC10167	SPC10202	clade_262	clade_553	clade_351	hetero	----	FALSE	FALSE
SPC10167	SPC10238	clade_262	clade_253	clade_354	hetero	++++	FALSE	TRUE
SPC10167	SPC10256	clade_262	clade_253	clade_252	hetero	++++	TRUE	TRUE
SPC10167	SPC10313	clade_262	clade_253	clade_260	hetero	----	FALSE	FALSE
SPC10167	SPC10358	clade_262	clade_253	clade_494	hetero	----	FALSE	FALSE
SPC10167	SPC10468	clade_262	clade_553	clade_360	hetero		FALSE	FALSE
SPC10167	SPC10470	clade_262	clade_253	clade_537	hetero	----	FALSE	FALSE
SPC10167	SPC10613	clade_262	clade_553	clade_309	hetero	++	FALSE	
SPC10202	SPC10202	clade_262	clade_351	clade_351	semi	----	FALSE	FALSE
SPC10202	SPC10238	clade_262	clade_351	clade_354	hetero	++++	FALSE	FALSE
SPC10202	SPC10256	clade_262	clade_351	clade_252	hetero	++++	TRUE	TRUE
SPC10202	SPC10313	clade_262	clade_351	clade_260	hetero	----	FALSE	FALSE
SPC10202	SPC10358	clade_262	clade_351	clade_494	hetero	--	FALSE	FALSE
SPC10202	SPC10468	clade_262	clade_351	clade_360	hetero	++++	FALSE	FALSE
SPC10202	SPC10470	clade_262	clade_351	clade_537	hetero	++++	FALSE	TRUE
SPC10202	SPC10613	clade_262	clade_351	clade_309	hetero	++++	FALSE	
SPC10238	SPC10238	clade_262	clade_354	clade_354	semi	++++	FALSE	FALSE
SPC10238	SPC10256	clade_262	clade_354	clade_252	hetero	++++	TRUE	TRUE
SPC10238	SPC10313	clade_262	clade_354	clade_260	hetero	++++	FALSE	TRUE
SPC10238	SPC10358	clade_262	clade_354	clade_494	hetero	++++	FALSE	TRUE
SPC10238	SPC10468	clade_262	clade_354	clade_360	hetero	++++	FALSE	FALSE
SPC10238	SPC10470	clade_262	clade_354	clade_537	hetero	++++	FALSE	TRUE
SPC10238	SPC10613	clade_262	clade_354	clade_309	hetero	++++	TRUE	
SPC10256	SPC10256	clade_262	clade_252	clade_252	semi	++++	TRUE	FALSE
SPC10256	SPC10313	clade_262	clade_252	clade_260	hetero	++++	TRUE	TRUE
SPC10256	SPC10358	clade_262	clade_252	clade_494	hetero	++++	TRUE	TRUE
SPC10256	SPC10468	clade_262	clade_252	clade_360	hetero	++++	TRUE	TRUE
SPC10256	SPC10470	clade_262	clade_252	clade_537	hetero	++++	TRUE	TRUE
SPC10256	SPC10613	clade_262	clade_252	clade_309	hetero	++++	TRUE	
SPC10313	SPC10313	clade_262	clade_260	clade_260	semi		FALSE	TRUE
SPC10313	SPC10358	clade_262	clade_260	clade_494	hetero		FALSE	FALSE
SPC10313	SPC10468	clade_262	clade_260	clade_360	hetero	++++	FALSE	TRUE
SPC10313	SPC10470	clade_262	clade_260	clade_537	hetero	+	FALSE	TRUE
SPC10313	SPC10613	clade_262	clade_260	clade_309	hetero	++++	FALSE	
SPC10358	SPC10358	clade_262	clade_494	clade_494	semi	----	FALSE	FALSE
SPC10358	SPC10468	clade_262	clade_494	clade_360	hetero	++++	FALSE	FALSE
SPC10358	SPC10470	clade_262	clade_494	clade_537	hetero	--	FALSE	FALSE
SPC10358	SPC10613	clade_262	clade_494	clade_309	hetero	++++	FALSE	
SPC10468	SPC10468	clade_262	clade_360	clade_360	semi	++++	FALSE	FALSE
SPC10468	SPC10470	clade_262	clade_360	clade_537	hetero	++++	FALSE	TRUE
SPC10468	SPC10613	clade_262	clade_360	clade_309	hetero	++++	FALSE	
SPC10155	SPC10155	clade_408	clade_252	clade_252	semi	++++	FALSE	FALSE
SPC10155	SPC10167	clade_408	clade_252	clade_253	hetero	++++	FALSE	FALSE
SPC10155	SPC10202	clade_408	clade_252	clade_351	hetero	++++	FALSE	FALSE
SPC10155	SPC10238	clade_408	clade_252	clade_354	hetero	++++	TRUE	TRUE

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SPC10155	SPC10256	clade_408	clade_252	clade_252	hetero	++++	TRUE	TRUE
SPC10155	SPC10313	clade_408	clade_252	clade_260	hetero	++++	FALSE	TRUE
SPC10155	SPC10358	clade_408	clade_252	clade_494	hetero	++++	FALSE	TRUE
SPC10155	SPC10468	clade_408	clade_252	clade_360	hetero	++++	FALSE	TRUE
SPC10155	SPC10470	clade_408	clade_252	clade_537	hetero	++++	FALSE	TRUE
SPC10155	SPC10613	clade_408	clade_252	clade_309	hetero	++++	FALSE	
SPC10167	SPC10167	clade_408	clade_253	clade_253	semi	+++	FALSE	FALSE
SPC10167	SPC10202	clade_408	clade_253	clade_351	hetero	–	FALSE	FALSE
SPC10167	SPC10238	clade_408	clade_253	clade_354	hetero	++++	FALSE	FALSE
SPC10167	SPC10256	clade_408	clade_253	clade_252	hetero	++++	TRUE	TRUE
SPC10167	SPC10313	clade_408	clade_253	clade_260	hetero		FALSE	FALSE
SPC10167	SPC10358	clade_408	clade_253	clade_494	hetero		FALSE	FALSE
SPC10167	SPC10468	clade_408	clade_253	clade_360	hetero	++++	FALSE	FALSE
SPC10167	SPC10470	clade_408	clade_253	clade_537	hetero	++++	FALSE	FALSE
SPC10167	SPC10613	clade_408	clade_253	clade_309	hetero	++++	FALSE	
SPC10202	SPC10202	clade_408	clade_351	clade_351	semi	––––	FALSE	FALSE
SPC10202	SPC10238	clade_408	clade_351	clade_354	hetero	++++	FALSE	FALSE
SPC10202	SPC10256	clade_408	clade_351	clade_252	hetero	++++	TRUE	TRUE
SPC10202	SPC10313	clade_408	clade_351	clade_260	hetero	––––	FALSE	FALSE
SPC10202	SPC10358	clade_408	clade_351	clade_494	hetero	––	FALSE	FALSE
SPC10202	SPC10468	clade_408	clade_381	clade_360	hetero	++++	FALSE	FALSE
SPC10202	SPC10470	clade_408	clade_351	clade_537	hetero	++++	FALSE	FALSE
SPC10202	SPC10613	clade_408	clade_351	clade_309	hetero	++++	FALSE	
SPC10238	SPC10238	clade_408	clade_354	clade_354	semi	++++	FALSE	FALSE
SPC10238	SPC10256	clade_408	clade_354	clade_252	hetero	++++	TRUE	FALSE
SPC10238	SPC10313	clade_408	clade_354	clade_260	hetero	++++	FALSE	FALSE
SPC10238	SPC10358	clade_408	clade_354	clade_494	hetero	++++	FALSE	FALSE
SPC10238	SPC10468	clade_408	clade_354	clade_360	hetero	++++	FALSE	FALSE
SPC10238	SPC10470	clade_408	clade_354	clade_537	hetero	++++	TRUE	TRUE
SPC10238	SPC10613	clade_408	clade_354	clade_309	hetero	++++	TRUE	
SPC10256	SPC10256	clade_408	clade_252	clade_252	semi	++++	TRUE	FALSE
SPC10256	SPC10313	clade_408	clade_252	clade_260	hetero	++++	TRUE	TRUE
SPC10256	SPC10358	clade_408	clade_252	clade_494	hetero	++++	TRUE	TRUE
SPC10256	SPC10468	clade_408	clade_252	clade_360	hetero	++++	TRUE	TRUE
SPC10256	SPC10470	clade_408	clade_252	clade_537	hetero	++++	TRUE	TRUE
SPC10256	SPC10613	clade_408	clade_252	clade_500	hetero	++++	TRUE	
SPC10313	SPC10313	clade_408	clade_260	clade_260	semi	––––	FALSE	FALSE
SPC10313	SPC10358	clade_408	clade_260	clade_494	hetero	––––	FALSE	FALSE
SPC10313	SPC10468	clade_408	clade_260	clade_360	hetero	++++	FALSE	FALSE
SPC10313	SPC10470	clade_408	clade_260	clade_537	hetero	+++	FALSE	TRUE
SPC10313	SPC10613	clade_408	clade_260	clade_309	hetero	++++	FALSE	
SPC10358	SPC10358	clade_408	clade_494	clade_494	semi	––––	FALSE	FALSE
SPC10358	SPC10468	clade_408	clade_494	clade_360	hetero	++++	FALSE	FALSE
SPC10358	SPC10470	clade_408	clade_494	clade_537	hetero		FALSE	FALSE
SPC10358	SPC10613	clade_408	clade_494	clade_309	hetero	++++	FALSE	
SPC10468	SPC10468	clade_408	clade_360	clade_360	semi		FALSE	FALSE
SPC10468	SPC10470	clade_408	clade_360	clade_537	hetero	++++	FALSE	TRUE
SPC10468	SPC10613	clade_408	clade_360	clade_309	hetero	++++	FALSE	
SPC10155	SPC10155	clade_408	clade_252	clade_252	semi	++++	FALSE	TRUE
SPC10155	SPC10167	clade_408	clade_252	clade_253	hetero	++++	FALSE	TRUE
SPC10155	SPC10202	clade_408	clade_252	clade_351	hetero	++++	FALSE	TRUE
SPC10155	SPC10238	clade_408	clade_252	clade_354	hetero	++++	TRUE	TRUE
SPC10155	SPC10256	clade_408	clade_252	clade_252	hetero	++++	TRUE	TRUE
SPC10155	SPC10313	clade_408	clade_252	clade_260	hetero	++++	FALSE	TRUE
SPC10155	SPC10358	clade_408	clade_252	clade_494	hetero	++++	FALSE	TRUE
SPC10155	SPC10468	clade_408	clade_252	clade_360	hetero	++++	FALSE	TRUE
SPC10155	SPC10470	clade_408	clade_252	clade_537	hetero	++++	FALSE	FALSE
SPC10155	SPC10613	clade_408	clade_382	clade_309	hetero	++++	FALSE	
SPC10167	SPC10167	clade_408	clade_553	clade_553	semi		FALSE	FALSE
SPC10167	SPC10202	clade_408	clade_253	clade_351	hetero	–––	FALSE	FALSE
SPC10167	SPC10238	clade_408	clade_253	clade_354	hetero	++++	FALSE	TRUE
SPC10167	SPC10256	clade_408	clade_553	clade_252	hetero	++++	TRUE	TRUE
SPC10167	SPC10313	clade_408	clade_253	clade_260	hetero	–	FALSE	FALSE
SPC10167	SPC10358	clade_408	clade_253	clade_494	hetero		FALSE	FALSE
SPC10167	SPC10468	clade_408	clade_253	clade_360	hetero	++	FALSE	FALSE
SPC10167	SPC10470	clade_408	clade_253	clade_537	hetero	––––	FALSE	FALSE
SPC10167	SPC10613	clade_408	clade_253	clade_309	hetero	–––	FALSE	
SPC10202	SPC10202	clade_408	clade_351	clade_351	semi	––	FALSE	FALSE
SPC10202	SPC10238	clade_408	clade_351	clade_354	hetero	++++	TRUE	TRUE
SPC10202	SPC10256	clade_408	clade_351	clade_252	hetero	++++	TRUE	TRUE
SPC10202	SPC10313	clade_408	clade_351	clade_260	hetero	––––	FALSE	FALSE
SPC10202	SPC10358	clade_408	clade_351	clade_494	hetero		FALSE	FALSE
SPC10202	SPC10468	clade_408	clade_351	clade_360	hetero	+++	FALSE	FALSE
SPC10202	SPC10470	clade_408	clade_351	clade_537	hetero	+++	FALSE	FALSE
SPC10202	SPC10613	clade_408	clade_351	clade_309	hetero	++++	FALSE	
SPC10238	SPC10238	clade_408	clade_354	clade_354	semi	++++	TRUE	FALSE
SPC10238	SPC10256	clade_408	clade_354	clade_252	hetero	++++	TRUE	FALSE
SPC10238	SPC10313	clade_408	clade_354	clade_260	hetero	++++	FALSE	TRUE
SPC10238	SPC10358	clade_408	clade_354	clade_494	hetero	++++	TRUE	TRUE
SPC10238	SPC10468	clade_408	clade_354	clade_360	hetero	++++	FALSE	TRUE

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SPC10238	SPC10470	clade_408	clade_354	clade_537	hetero	++++	TRUE	TRUE
SPC10238	SPC10613	clade_408	clade_354	clade_309	hetero	++++	TRUE	
SPC10256	SPC10256	clade_408	clade_252	clade_252	semi	++++	TRUE	FALSE
SPC10256	SPC10313	clade_408	clade_252	clade_260	hetero	++++	TRUE	TRUE
SPC10256	SPC10358	clade_408	clade_252	clade_494	hetero	++++	TRUE	TRUE
SPC10256	SPC10468	clade_408	clade_252	clade_360	hetero	++++	TRUE	TRUE
SPC10256	SPC10470	clade_408	clade_252	clade_537	hetero	++++	TRUE	TRUE
SPC10256	SPC10613	clade_408	clade_252	clade_309	hetero	++++	TRUE	
SPC10313	SPC10313	clade_408	clade_260	clade_260	semi	−−−	FALSE	FALSE
SPC10313	SPC10358	clade_408	clade_260	clade_494	hetero	++	FALSE	FALSE
SPC10313	SPC10468	clade_408	clade_260	clade_360	hetero	+++	FALSE	FALSE
SPC10313	SPC10470	clade_408	clade_260	clade_537	hetero	++++	FALSE	TRUE
SPC10313	SPC10613	clade_408	clade_260	clade_309	hetero	−−−	FALSE	
SPC10358	SPC10358	clade_408	clade_494	clade_494	semi	++++	FALSE	TRUE
SPC10358	SPC10468	clade_408	clade_494	clade_360	hetero	++++	FALSE	TRUE
SPC10358	SPC10470	clade_408	clade_494	clade_537	hetero	+++	FALSE	FALSE
SPC10358	SPC10613	clade_408	clade_494	clade_309	hetero	++++	FALSE	
SPC10468	SPC10468	clade_408	clade_360	clade_360	semi	++++	FALSE	FALSE
SPC10468	SPC10470	clade_408	clade_360	clade_537	hetero	++++	FALSE	TRUE
SPC10468	SPC10613	clade_408	clade_360	clade_500	hetero	++++	FALSE	
SPC10155	SPC10155	clade_478	clade_252	clade_252	semi		FALSE	FALSE
SPC10155	SPC10167	clade_478	clade_252	clade_253	hetero	++++	FALSE	FALSE
SPC10155	SPC10202	clade_478	clade_252	clade_351	hetero		FALSE	FALSE
SPC10155	SPC10238	clade_478	clade_252	clade_354	hetero	++++	FALSE	TRUE
SPC10155	SPC10256	clade_478	clade_252	clade_252	hetero	++++	TRUE	TRUE
SPC10155	SPC10313	clade_478	clade_252	clade_260	hetero	++++	FALSE	TRUE
SPC10155	SPC10358	clade_478	clade_252	clade_494	hetero	++++	FALSE	TRUE
SPC10155	SPC10468	clade_478	clade_252	clade_360	hetero	++++	FALSE	FALSE
SPC10155	SPC10470	clade_478	clade_252	clade_537	hetero	−−−−	FALSE	FALSE
SPC10155	SPC10613	clade_478	clade_252	clade_309	hetero	++++	FALSE	
SPC10167	SPC10167	clade_478	clade_253	clade_253	semi	−−−−	FALSE	FALSE
SPC10167	SPC10202	clade_478	clade_253	clade_351	hetero	−−−−	FALSE	FALSE
SPC10167	SPC10238	clade_478	clade_253	clade_354	hetero	++++	FALSE	TRUE
SPC10167	SPC10256	clade_478	clade_253	clade_252	hetero	++++	TRUE	TRUE
SPC10167	SPC10313	clade_478	clade_253	clade_260	hetero	−−−−	FALSE	FALSE
SPC10167	SPC10358	clade_478	clade_253	clade_494	hetero	−−−−	FALSE	FALSE
SPC10167	SPC10468	clade_478	clade_253	clade_360	hetero	−−−	FALSE	FALSE
SPC10167	SPC10470	clade_478	clade_253	clade_537	hetero	−−−−	FALSE	FALSE
SPC10167	SPC10613	clade_478	clade_253	clade_309	hetero	−−−	FALSE	
SPC10202	SPC10202	clade_478	clade_351	clade_351	semi	−−−−	FALSE	FALSE
SPC10202	SPC10238	clade_478	clade_351	clade_354	hetero	++++	FALSE	FALSE
SPC10202	SPC10256	clade_478	clade_351	clade_252	hetero	++++	TRUE	TRUE
SPC10202	SPC10313	clade_478	clade_351	clade_260	hetero	−−−−	FALSE	FALSE
SPC10202	SPC10358	clade_478	clade_351	clade_494	hetero	−−−−	FALSE	FALSE
SPC10202	SPC10468	clade_478	clade_351	clade_360	hetero	−	FALSE	FALSE
SPC10202	SPC10470	clade_478	clade_351	clade_537	hetero	−−−−	FALSE	FALSE
SPC10202	SPC10613	clade_478	clade_351	clade_309	hetero	++++	FALSE	
SPC10238	SPC10238	clade_478	clade_354	clade_354	semi	++++	FALSE	FALSE
SPC10238	SPC10256	clade_478	clade_354	clade_252	hetero	++++	TRUE	FALSE
SPC10238	SPC10313	clade_478	clade_354	clade_260	hetero	++++	FALSE	TRUE
SPC10238	SPC10358	clade_478	clade_354	clade_494	hetero	++++	TRUE	TRUE
SPC10238	SPC10468	clade_478	clade_354	clade_360	hetero	++++	FALSE	TRUE
SPC10238	SPC10470	clade_478	clade_354	clade_537	hetero	++++	FALSE	TRUE
SPC10238	SPC10613	clade_478	clade_354	clade_309	hetero	++++	TRUE	
SPC10256	SPC10256	clade_478	clade_252	clade_252	semi	++++	TRUE	FALSE
SPC10256	SPC10313	clade_478	clade_252	clade_260	hetero	++++	TRUE	TRUE
SPC10256	SPC10358	clade_478	clade_252	clade_494	hetero	++++	TRUE	TRUE
SPC10256	SPC10468	clade_478	clade_252	clade_360	hetero	++++	TRUE	TRUE
SPC10256	SPC10470	clade_478	clade_252	clade_537	hetero	++++	TRUE	TRUE
SPC10256	SPC10613	clade_478	clade_252	clade_309	hetero	++++	TRUE	
SPC10313	SPC10313	clade_478	clade_260	clade_260	semi	−−−−	FALSE	FALSE
SPC10313	SPC10358	clade_478	clade_260	clade_494	hetero	−−−−	FALSE	FALSE
SPC10313	SPC10468	clade_478	clade_260	clade_360	hetero	++++	FALSE	TRUE
SPC10313	SPC10470	clade_478	clade_260	clade_537	hetero	−−−	FALSE	FALSE
SPC10313	SPC10613	clade_478	clade_260	clade_309	hetero		FALSE	
SPC10358	SPC10358	clade_478	clade_494	clade_494	semi	−−−−	FALSE	FALSE
SPC10358	SPC10468	clade_478	clade_494	clade_360	hetero	++++	FALSE	FALSE
SPC10358	SPC10470	clade_478	clade_494	clade_537	hetero	−−	FALSE	FALSE
SPC10358	SPC10613	clade_478	clade_494	clade_309	hetero	++++	FALSE	
SPC10468	SPC10468	clade_478	clade_360	clade_360	semi	++++	FALSE	FALSE
SPC10468	SPC10470	clade_478	clade_360	clade_537	hetero	++++	FALSE	TRUE
SPC10468	SPC10613	clade_478	clade_360	clade_309	hetero	++++	FALSE	
SPC10155	SPC10155	clade_260	clade_252	clade_252	semi	++++	FALSE	FALSE
SPC10155	SPC10167	clade_260	clade_252	clade_253	hetero	++++	FALSE	TRUE
SPC10155	SPC10202	clade_260	clade_252	clade_351	hetero	++++	FALSE	FALSE
SPC10155	SPC10238	clade_260	clade_252	clade_354	hetero	++++	FALSE	FALSE
SPC10155	SPC10256	clade_260	clade_252	clade_252	hetero	++++	TRUE	TRUE
SPC10155	SPC10313	clade_260	clade_252	clade_260	hetero	++++	FALSE	FALSE
SPC10155	SPC10358	clade_260	clade_252	clade_494	hetero	++++	FALSE	TRUE
SPC10155	SPC10468	clade_260	clade_252	clade_360	hetero	++++	FALSE	TRUE

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SPC10155	SPC10470	clade_260	clade_252	clade_537	hetero	++++	FALSE	FALSE		
SPC10155	SPC10613	clade_260	clade_252	clade_309	hetero	++++	FALSE			
SPC10167	SPC10167	clade_260	clade_253	clade_253	semi	----	FALSE	FALSE		
SPC10167	SPC10202	clade_260	clade_253	clade_351	hetero	----	FALSE	FALSE		
SPC10167	SPC10238	clade_260	clade_253	clade_354	hetero	++++	FALSE	TRUE		
SPC10167	SPC10256	clade_260	clade_253	clade_252	hetero	++++	TRUE	TRUE		
SPC10167	SPC10313	clade_260	clade_253	clade_260	hetero	----	FALSE	FALSE		
SPC10167	SPC10358	clade_260	clade_253	clade_494	hetero	----	FALSE	FALSE		
SPC10167	SPC10468	clade_260	clade_253	clade_360	hetero	+++	FALSE	FALSE		
SPC10167	SPC10470	clade_260	clade_253	clade_537	hetero	----	FALSE	FALSE		
SPC10167	SPC10613	clade_260	clade_253	clade_309	hetero		FALSE			
SPC10202	SPC10202	clade_260	clade_351	clade_351	semi	----	FALSE	FALSE		
SPC10202	SPC10238	clade_260	clade_351	clade_354	hetero	++++	FALSE	FALSE		
SPC10202	SPC10256	clade_260	clade_351	clade_252	hetero	++++	TRUE	FALSE		
SPC10202	SPC10313	clade_260	clade_351	clade_260	hetero	----	FALSE	FALSE		
SPC10202	SPC10358	clade_260	clade_351	clade_494	hetero	---	FALSE	FALSE		
SPC10202	SPC10468	clade_260	clade_351	clade_360	hetero	++++	FALSE	FALSE		
SPC10202	SPC10470	clade_260	clade_351	clade_537	hetero	++++	FALSE	FALSE		
SPC10202	SPC10613	clade_260	clade_351	clade_309	hetero	++++	FALSE			
SPC10238	SPC10238	clade_260	clade_354	clade_354	semi	++++	FALSE	FALSE		
SPC10238	SPC10256	clade_260	clade_354	clade_252	hetero	++++	TRUE	FALSE		
SPC10238	SPC10313	clade_260	clade_354	clade_260	hetero	++++	FALSE	TRUE		
SPC10238	SPC10358	clade_260	clade_354	clade_494	hetero	++++	FALSE	TRUE		
SPC10238	SPC10468	clade_260	clade_354	clade_360	hetero	++++	TRUE	TRUE		
SPC10238	SPC10470	clade_260	clade_354	clade_537	hetero	++++	TRUE	TRUE		
SPC10238	SPC10613	clade_260	clade_354	clade_309	hetero	++++	TRUE			
SPC10256	SPC10256	clade_260	clade_252	clade_252	semi	++++	TRUE	FALSE		
SPC10256	SPC10313	clade_260	clade_252	clade_260	hetero	++++	TRUE	TRUE		
SPC10256	SPC10358	clade_260	clade_252	clade_494	hetero	++++	TRUE	TRUE		
SPC10256	SPC10468	clade_260	clade_252	clade_360	hetero	++++	TRUE	TRUE		
SPC10256	SPC10470	clade_260	clade_252	clade_537	hetero	++++	TRUE	TRUE		
SPC10256	SPC10613	clade_260	clade_252	clade_309	hetero	++++	TRUE			
SPC10313	SPC10313	clade_260	clade_260	clade_260	semi	----	FALSE	FALSE		
SPC10313	SPC10358	clade_260	clade_260	clade_494	hetero	----	FALSE	FALSE		
SPC10313	SPC10468	clade_260	clade_260	clade_360	hetero	++++	FALSE	FALSE		
SPC10313	SPC10470	clade_260	clade_260	clade_537	hetero	+++	FALSE	TRUE		
SPC10313	SPC10613	clade_260	clade_260	clade_309	hetero	++++	FALSE			
SPC10358	SPC10358	clade_260	clade_494	clade_494	semi		FALSE	FALSE		
SPC10358	SPC10468	clade_260	clade_494	clade_360	hetero	++++	FALSE	FALSE		
SPC10358	SPC10470	clade_260	clade_494	clade_537	hetero	+++	FALSE	FALSE		
SPC10358	SPC10613	clade_260	clade_494	clade_309	hetero	++++	FALSE			
SPC10468	SPC10468	clade_260	clade_360	clade_360	semi	++++	FALSE	FALSE		
SPC10468	SPC10470	clade_260	clade_360	clade_537	hetero	++++	FALSE	TRUE		
SPC10468	SPC10613	clade_260	clade_360	clade_309	hetero	++++	FALSE			
SPC10155	SPC10155	clade_309	clade_252	clade_252	semi	++++	TRUE	TRUE	++++	FALSE
SPC10155	SPC10167	clade_309	clade_252	clade_253	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10155	SPC10202	clade_309	clade_252	clade_351	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10155	SPC10238	clade_309	clade_252	clade_354	hetero	++++	TRUE	TRUE		FALSE
SPC10155	SPC10256	clade_309	clade_252	clade_252	hetero	++++	TRUE	TRUE	++++	TRUE
SPC10155	SPC10313	clade_309	clade_252	clade_260	hetero	++++	FALSE	TRUE	++++	FALSE
SPC10155	SPC10358	clade_309	clade_252	clade_494	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10155	SPC10468	clade_309	clade_252	clade_360	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10155	SPC10470	clade_309	clade_252	clade_537	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10155	SPC10613	clade_309	clade_252	clade_309	hetero	++++	TRUE		++++	FALSE
SPC10167	SPC10167	clade_309	clade_253	clade_253	semi	++++	FALSE	TRUE	++++	FALSE
SPC10167	SPC10202	clade_309	clade_253	clade_351	hetero	++++	FALSE	TRUE	++++	FALSE
SPC10167	SPC10238	clade_309	clade_253	clade_354	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10167	SPC10256	clade_309	clade_253	clade_252	hetero	++++	TRUE	TRUE	++++	TRUE
SPC10167	SPC10313	clade_309	clade_253	clade_260	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10167	SPC10358	clade_309	clade_253	clade_494	hetero	++++	TRUE	TRUE	+++	FALSE
SPC10167	SPC10468	clade_309	clade_253	clade_360	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10167	SPC10470	clade_309	clade_253	clade_537	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10167	SPC10613	clade_309	clade_253	clade_309	hetero	++++	TRUE		++++	FALSE
SPC10202	SPC10202	clade_309	clade_351	clade_351	semi	++++	TRUE	TRUE	++++	FALSE
SPC10202	SPC10238	clade_309	clade_351	clade_354	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10202	SPC10256	clade_309	clade_351	clade_252	hetero	++++	TRUE	TRUE	++++	TRUE
SPC10202	SPC10313	clade_309	clade_351	clade_260	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10202	SPC10358	clade_309	clade_351	clade_494	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10202	SPC10468	clade_309	clade_351	clade_360	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10202	SPC10470	clade_309	clade_351	clade_537	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10202	SPC10613	clade_309	clade_351	clade_309	hetero	++++	TRUE		++++	FALSE
SPC10238	SPC10238	clade_309	clade_354	clade_354	semi	++++	TRUE	TRUE	++++	TRUE
SPC10238	SPC10256	clade_309	clade_354	clade_252	hetero	++++	TRUE	FALSE	++++	TRUE
SPC10238	SPC10313	clade_309	clade_354	clade_260	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10238	SPC10358	clade_309	clade_354	clade_494	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10238	SPC10468	clade_309	clade_354	clade_360	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10238	SPC10470	clade_309	clade_354	clade_537	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10238	SPC10613	clade_309	clade_354	clade_309	hetero	++++	TRUE		++++	TRUE
SPC10256	SPC10256	clade_309	clade_252	clade_252	semi	++++	TRUE	FALSE	++++	TRUE
SPC10256	SPC10313	clade_309	clade_252	clade_260	hetero	++++	TRUE	TRUE	++++	TRUE

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SPC10256	SPC10358	clade_309	clade_252	clade_494	hetero	++++	TRUE	TRUE	++++	TRUE
SPC10256	SPC10468	clade_309	clade_252	clade_360	hetero	++++	TRUE	TRUE	++++	TRUE
SPC10256	SPC10470	clade_309	clade_252	clade_537	hetero	++++	TRUE	TRUE	++++	TRUE
SPC10256	SPC10613	clade_309	clade_252	clade_309	hetero	++++	TRUE		++++	TRUE
SPC10313	SPC10313	clade_309	clade_260	clade_260	semi	++++	TRUE	TRUE	++++	FALSE
SPC10313	SPC10358	clade_309	clade_260	clade_494	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10313	SPC10468	clade_309	clade_260	clade_360	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10313	SPC10470	clade_309	clade_260	clade_537	hetero	++++	TRUE	TRUE	+	FALSE
SPC10313	SPC10613	clade_309	clade_260	clade_309	hetero	++++	TRUE		+	FALSE
SPC10358	SPC10358	clade_309	clade_494	clade_494	semi	++++	TRUE	TRUE	++++	FALSE
SPC10358	SPC10468	clade_309	clade_494	clade_360	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10358	SPC10470	clade_309	clade_494	clade_537	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10358	SPC10613	clade_309	clade_494	clade_309	hetero	++++	TRUE		++++	FALSE
SPC10468	SPC10468	clade_309	clade_360	clade_360	semi	++++	TRUE	TRUE	++++	TRUE
SPC10468	SPC10470	clade_309	clade_360	clade_537	hetero	++++	TRUE	TRUE	++++	FALSE
SPC10468	SPC10613	clade_309	clade_360	clade_309	hetero	++++	TRUE		++++	FALSE
SPC10155	SPC10155	clade_444	clade_252	clade_252	semi	++++	FALSE	FALSE		
SPC10155	SPC10167	clade_444	clade_252	clade_253	hetero	+++	FALSE	FALSE		
SPC10155	SPC10202	clade_444	clade_252	clade_351	hetero	++++	FALSE	FALSE		
SPC10155	SPC10238	clade_444	clade_252	clade_354	hetero	++++	TRUE	TRUE		
SPC10155	SPC10256	clade_444	clade_252	clade_252	hetero	++++	TRUE	TRUE		
SPC10155	SPC10313	clade_444	clade_252	clade_260	hetero	++++	FALSE	TRUE		
SPC10155	SPC10358	clade_444	clade_252	clade_494	hetero	++++	FALSE	TRUE		
SPC10155	SPC10468	clade_444	clade_252	clade_360	hetero	++++	FALSE	TRUE		
SPC10155	SPC10470	clade_444	clade_252	clade_537	hetero	+	FALSE	FALSE		
SPC10155	SPC10613	clade_444	clade_252	clade_309	hetero	++++	FALSE			
SPC10167	SPC10167	clade_444	clade_253	clade_253	semi	−−−−	FALSE	FALSE		
SPC10167	SPC10202	clade_444	clade_253	clade_351	hetero	−−−−	FALSE	FALSE		
SPC10167	SPC10238	clade_444	clade_253	clade_354	hetero	++++	FALSE	TRUE		
SPC10167	SPC10256	clade_444	clade_253	clade_252	hetero	++++	TRUE	TRUE		
SPC10167	SPC10313	clade_444	clade_253	clade_260	hetero	−−−−	FALSE	FALSE		
SPC10167	SPC10358	clade_444	clade_253	clade_494	hetero	−−−−	FALSE	FALSE		
SPC10167	SPC10468	clade_444	clade_253	clade_360	hetero	++++	FALSE	FALSE		
SPC10167	SPC10470	clade_444	clade_253	clade_537	hetero	−−−−	FALSE	FALSE		
SPC10167	SPC10613	clade_444	clade_253	clade_309	hetero	−−−−	FALSE			
SPC10202	SPC10202	clade_444	clade_351	clade_351	semi		FALSE	FALSE		
SPC10202	SPC10238	clade_444	clade_351	clade_354	hetero	++++	FALSE	TRUE		
SPC10202	SPC10256	clade_444	clade_351	clade_252	hetero	++++	TRUE	TRUE		
SPC10202	SPC10313	clade_444	clade_351	clade_260	hetero	−−−−	FALSE	FALSE		
SPC10202	SPC10358	clade_444	clade_351	clade_494	hetero	−−−	FALSE	FALSE		
SPC10202	SPC10468	clade_444	clade_351	clade_360	hetero	++++	FALSE	FALSE		
SPC10202	SPC10470	clade_444	clade_351	clade 537	hetero		FALSE	FALSE		
SPC10202	SPC10613	clade_444	clade_351	clade_309	hetero	++++	FALSE			
SPC10238	SPC10238	clade_444	clade_354	clade_354	semi	++++	FALSE	FALSE		
SPC10238	SPC10256	clade_444	clade_354	clade_252	hetero	++++	TRUE	FALSE		
SPC10238	SPC10313	clade_444	clade_354	clade_260	hetero	++++	TRUE	TRUE		
SPC10238	SPC10358	clade_444	clade_354	clade_494	hetero	++++	TRUE	TRUE		
SPC10238	SPC10468	clade_444	clade_354	clade_360	hetero	++++	TRUE	TRUE		
SPC10238	SPC10470	clade_444	clade_354	clade_537	hetero	++++	TRUE	TRUE		
SPC10238	SPC10613	clade_444	clade_354	clade_309	hetero	++++	TRUE			
SPC10256	SPC10256	clade_444	clade_252	clade_252	semi	++++	TRUE	FALSE		
SPC10256	SPC10313	clade_444	clade_252	clade_260	hetero	++++	TRUE	TRUE		
SPC10256	SPC10358	clade_444	clade_252	clade_494	hetero	++++	TRUE	TRUE		
SPC10256	SPC10468	clade_444	clade_252	clade_360	hetero	++++	TRUE	TRUE		
SPC10256	SPC10470	clade_444	clade_252	clade_537	hetero	++++	TRUE	TRUE		
SPC10256	SPC10613	clade_444	clade_252	clade_309	hetero	++++	TRUE			
SPC10313	SPC10313	clade_444	clade_260	clade_260	semi	−−−−	FALSE	FALSE		
SPC10313	SPC10358	clade_444	clade_260	clade_494	hetero	−−−−	FALSE	FALSE		
SPC10313	SPC10468	clade_444	clade_260	clade_360	hetero	++	FALSE	TRUE		
SPC10313	SPC10470	clade_444	clade_260	clade_537	hetero	−	FALSE	TRUE		
SPC10313	SPC10613	clade_444	clade_260	clade_309	hetero	++++	FALSE			
SPC10358	SPC10358	clade_444	clade_494	clade_494	semi	−	FALSE	FALSE		
SPC10358	SPC10468	clade_444	clade_494	clade_360	hetero	++++	FALSE	FALSE		
SPC10358	SPC10470	clade_444	clade_494	clade_537	hetero	−	FALSE	FALSE		
SPC10358	SPC10613	clade_444	clade_494	clade_309	hetero	++++	FALSE			
SPC10468	SPC10468	clade_444	clade_360	clade_360	semi	++++	FALSE	FALSE		
SPC10468	SPC10470	clade_444	clade_360	clade_537	hetero	++++	FALSE	TRUE		
SPC10468	SPC10613	clade_444	clade_360	clade_309	hetero	++++	FALSE			
SPC10097	SPC10097	clade_252	clade_553	clade_553	semi		FALSE	FALSE		
SPC10097	SPC10304	clade_252	clade_553	clade_262	hetero	++++	FALSE	FALSE		
SPC10097	SPC10325	clade_252	clade_553	clade_408	hetero	++++	FALSE	FALSE		
SPC10097	SPC10355	clade_252	clade_553	clade_408	hetero	++++	FALSE	FALSE		
SPC10097	SPC10386	clade_252	clade_553	clade_478	hetero	++++	FALSE	FALSE		
SPC10097	SPC10390	clade_252	clade_553	clade_260	hetero	+++	FALSE	FALSE		
SPC10097	SPC10415	clade_252	clade_553	clade_309	hetero	++++	TRUE	TRUE		
SPC10097	SPC10567	clade_252	clade_553	clade_444	hetero	++++	FALSE	FALSE		
SPC10304	SPC10304	clade_252	clade_262	clade_262	semi	−−	FALSE	FALSE		
SPC10304	SPC10325	clade_252	clade_262	clade_408	hetero	++++	FALSE	FALSE		
SPC10304	SPC10355	clade_252	clade_262	clade_408	hetero		FALSE	FALSE		
SPC10304	SPC10386	clade_252	clade_262	clade_478	hetero	−−−−	FALSE	FALSE		

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SPC10304	SPC10390	clade_252	clade_262	clade_260	hetero	++++	FALSE	TRUE
SPC10304	SPC10415	clade_252	clade_262	clade_309	hetero	++++	FALSE	TRUE
SPC10304	SPC10567	clade_252	clade_262	clade_444	hetero		FALSE	FALSE
SPC10325	SPC10325	clade_252	clade_408	clade_408	semi	++++	FALSE	FALSE
SPC10325	SPC10355	clade_252	clade_408	clade_408	hetero	++++	FALSE	FALSE
SPC10325	SPC10386	clade_252	clade_408	clade_478	hetero	+++	FALSE	FALSE
SPC10325	SPC10390	clade_252	clade_408	clade_260	hetero	++++	FALSE	FALSE
SPC10325	SPC10415	clade_252	clade_408	clade_309	hetero	++++	FALSE	TRUE
SPC10325	SPC10567	clade_252	clade_408	clade_444	hetero	++++	FALSE	TRUE
SPC10355	SPC10355	clade_252	clade_408	clade_408	semi	−−−	FALSE	FALSE
SPC10355	SPC10386	clade_252	clade_408	clade_478	hetero	−−−−	FALSE	FALSE
SPC10355	SPC10390	clade_252	clade_408	clade_260	hetero	++++	FALSE	TRUE
SPC10355	SPC10415	clade_252	clade_408	clade_309	hetero	++++	FALSE	FALSE
SPC10355	SPC10567	clade_252	clade_408	clade_444	hetero	+++	FALSE	FALSE
SPC10386	SPC10386	clade_252	clade_478	clade_478	semi	−−	FALSE	FALSE
SPC10386	SPC10390	clade_252	clade_478	clade_260	hetero	++++	FALSE	FALSE
SPC10386	SPC10415	clade_252	clade_478	clade_309	hetero	++++	FALSE	TRUE
SPC10386	SPC10567	clade_252	clade_478	clade_444	hetero		FALSE	FALSE
SPC10390	SPC10390	clade_252	clade_260	clade_260	semi	++++	FALSE	FALSE
SPC10390	SPC10415	clade_252	clade_260	clade_309	hetero	++++	FALSE	TRUE
SPC10390	SPC10567	clade_252	clade_260	clade_444	hetero	++++	FALSE	TRUE
SPC10415	SPC10415	clade_252	clade_309	clade_309	semi	++++	FALSE	TRUE
SPC10415	SPC10567	clade_252	clade_309	clade_444	hetero	++++	TRUE	TRUE
SPC10567	SPC10567	clade_252	clade_444	clade_444	semi	+++	FALSE	TRUE
SPC10097	SPC10097	clade_253	clade_553	clade_553	semi	−	FALSE	FALSE
SPC10097	SPC10304	clade_253	clade_553	clade_262	hetero		FALSE	FALSE
SPC10097	SPC10325	clade_253	clade_553	clade_408	hetero	−−−	FALSE	FALSE
SPC10097	SPC10355	clade_253	clade_553	clade_408	hetero	−−−−	FALSE	FALSE
SPC10097	SPC10386	clade_253	clade_553	clade_478	hetero	−−−	FALSE	FALSE
SPC10097	SPC10390	clade_253	clade_553	clade_260	hetero	++	FALSE	FALSE
SPC10097	SPC10415	clade_253	clade_553	clade_309	hetero	++++	FALSE	TRUE
SPC10097	SPC10567	clade_253	clade_553	clade_444	hetero		FALSE	FALSE
SPC10304	SPC10304	clade_253	clade_262	clade_262	semi	−−−−	FALSE	FALSE
SPC10304	SPC10325	clade_253	clade_262	clade_408	hetero		FALSE	FALSE
SPC10304	SPC10355	clade_253	clade_262	clade_408	hetero	−−−−	FALSE	FALSE
SPC10304	SPC10386	clade_253	clade_262	clade_478	hetero	−−−−	FALSE	FALSE
SPC10304	SPC10390	clade_253	clade_262	clade_260	hetero	++++	FALSE	FALSE
SPC10304	SPC10415	clade_253	clade_262	clade_309	hetero	++++	FALSE	TRUE
SPC10304	SPC10567	clade_253	clade_262	clade_444	hetero	−−−−	FALSE	FALSE
SPC10325	SPC10325	clade_253	clade_408	clade_408	semi	−	FALSE	FALSE
SPC10325	SPC10355	clade_253	clade_408	clade_408	hetero	−−−−	FALSE	FALSE
SPC10325	SPC10386	clade_253	clade_408	clade_478	hetero	−−−−	FALSE	FALSE
SPC10325	SPC10390	clade_253	clade_408	clade_360	hetero	++	FALSE	FALSE
SPC10325	SPC10415	clade_253	clade_408	clade_309	hetero	++++	FALSE	TRUE
SPC10325	SPC10567	clade_253	clade_408	clade_444	hetero	++++	FALSE	FALSE
SPC10355	SPC10355	clade_253	clade_408	clade_408	semi	−−−−	FALSE	FALSE
SPC10355	SPC10386	clade_253	clade_408	clade_478	hetero	−−−−	FALSE	FALSE
SPC10355	SPC10390	clade_253	clade_408	clade_260	hetero	++++	FALSE	FALSE
SPC10355	SPC10415	clade_253	clade_408	clade_309	hetero	++++	FALSE	FALSE
SPC10355	SPC10567	clade_253	clade_408	clade_444	hetero	−−−−	FALSE	FALSE
SPC10386	SPC10386	clade_253	clade_478	clade_478	semi	−−−−	FALSE	FALSE
SPC10386	SPC10390	clade_253	clade_478	clade_260	hetero	−−−−	FALSE	FALSE
SPC10386	SPC10415	clade_253	clade_478	clade_309	hetero	++++	FALSE	TRUE
SPC10386	SPC10567	clade_253	clade_478	clade_444	hetero	−−−−	FALSE	FALSE
SPC10390	SPC10390	clade_253	clade_260	clade_260	semi	−−−	FALSE	FALSE
SPC10390	SPC10415	clade_253	clade_260	clade_309	hetero	++++	FALSE	TRUE
SPC10390	SPC10567	clade_253	clade_260	clade_444	hetero	+	FALSE	FALSE
SPC10415	SPC10415	clade_253	clade_309	clade_309	semi	++++	FALSE	TRUE
SPC10415	SPC10567	clade_253	clade_309	clade_444	hetero	++++	FALSE	TRUE
SPC10567	SPC10567	clade_253	clade_444	clade_444	semi	−−−−	FALSE	FALSE
SPC10097	SPC10097	clade_351	clade_553	clade_553	semi	+++	FALSE	FALSE
SPC10097	SPC10304	clade_351	clade_553	clade_262	hetero	++++	FALSE	FALSE
SPC10097	SPC10325	clade_351	clade_553	clade_408	hetero	+++	FALSE	FALSE
SPC10097	SPC10355	clade_351	clade_553	clade_408	hetero	++++	FALSE	FALSE
SPC10097	SPC10386	clade_351	clade_553	clade_478	hetero	++++	FALSE	FALSE
SPC10097	SPC10390	clade_351	clade_553	clade_260	hetero	++++	FALSE	TRUE
SPC10097	SPC10415	clade_351	clade_553	clade_309	hetero	++++	FALSE	TRUE
SPC10097	SPC10567	clade_351	clade_553	clade_444	hetero	++++	FALSE	TRUE
SPC10304	SPC10304	clade_351	clade_262	clade_262	semi	−−−	FALSE	FALSE
SPC10304	SPC10325	clade_351	clade_262	clade_408	hetero	++	FALSE	FALSE
SPC10304	SPC10355	clade_351	clade_262	clade_408	hetero	−−	FALSE	FALSE
SPC10304	SPC10386	clade_351	clade_262	clade_478	hetero	−−−−	FALSE	FALSE
SPC10304	SPC10390	clade_351	clade_262	clade_260	hetero	++++	FALSE	FALSE
SPC10304	SPC10415	clade_351	clade_262	clade_309	hetero	++++	FALSE	TRUE
SPC10304	SPC10567	clade_351	clade_262	clade_444	hetero	++++	FALSE	FALSE
SPC10325	SPC10325	clade_351	clade_408	clade_408	semi	++	FALSE	FALSE
SPC10325	SPC10355	clade_351	clade_408	clade_408	hetero	+++	FALSE	FALSE
SPC10325	SPC10386	clade_351	clade_408	clade_478	hetero		FALSE	FALSE
SPC10325	SPC10390	clade_351	clade_408	clade_260	hetero	++++	FALSE	FALSE
SPC10325	SPC10415	clade_351	clade_408	clade_309	hetero	++++	FALSE	TRUE

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SPC10325	SPC10567	clade_351	clade_408	clade_444	hetero	++++	FALSE	FALSE
SPC10355	SPC10355	clade_351	clade_408	clade_408	semi	–	FALSE	FALSE
SPC10355	SPC10386	clade_351	clade_408	clade_478	hetero	–	FALSE	FALSE
SPC10355	SPC10390	clade_351	clade_408	clade_260	hetero	+++	FALSE	FALSE
SPC10355	SPC10415	clade_351	clade_408	clade_309	hetero	++++	FALSE	FALSE
SPC10355	SPC10567	clade_351	clade_408	clade_444	hetero	+++	FALSE	FALSE
SPC10386	SPC10386	clade_351	clade_478	clade_478	semi	–––	FALSE	FALSE
SPC10386	SPC10390	clade_351	clade_478	clade_260	hetero	+++	FALSE	FALSE
SPC10386	SPC10415	clade_351	clade_478	clade_309	hetero	++++	FALSE	TRUE
SPC10386	SPC10567	clade_351	clade_478	clade_444	hetero		FALSE	FALSE
SPC10390	SPC10390	clade_351	clade_260	clade_260	semi	++++	FALSE	FALSE
SPC10390	SPC10415	clade_351	clade_260	clade_309	hetero	++++	FALSE	TRUE
SPC10390	SPC10567	clade_351	clade_260	clade_444	hetero	++++	FALSE	TRUE
SPC10415	SPC10415	clade_351	clade_309	clade_309	semi	++++	FALSE	TRUE
SPC10415	SPC10567	clade_351	clade_309	clade_444	hetero	++++	TRUE	TRUE
SPC10567	SPC10567	clade_351	clade_444	clade_444	semi	+++	FALSE	TRUE
SPC10097	SPC10097	clade_354	clade_553	clade_553	semi	++++	FALSE	FALSE
SPC10097	SPC10304	clade_354	clade_553	clade_262	hetero	++++	FALSE	FALSE
SPC10097	SPC10325	clade_354	clade_553	clade_408	hetero	++++	FALSE	FALSE
SPC10097	SPC10355	clade_354	clade_553	clade_408	hetero	++++	FALSE	TRUE
SPC10097	SPC10386	clade_354	clade_553	clade_478	hetero	++++	FALSE	TRUE
SPC10097	SPC10390	clade_354	clade_553	clade_260	hetero	++++	FALSE	TRUE
SPC10097	SPC10415	clade_354	clade_553	clade_309	hetero	++++	FALSE	FALSE
SPC10097	SPC10567	clade_354	clade_553	clade_444	hetero	++++	FALSE	TRUE
SPC10304	SPC10304	clade_354	clade_262	clade_262	semi	++++	FALSE	FALSE
SPC10304	SPC10325	clade_354	clade_262	clade_408	hetero	++++	FALSE	FALSE
SPC10304	SPC10355	clade_354	clade_262	clade_408	hetero	++++	FALSE	TRUE
SPC10304	SPC10386	clade_354	clade_262	clade_478	hetero	++++	FALSE	TRUE
SPC10304	SPC10390	clade_354	clade_262	clade_260	hetero	++++	FALSE	FALSE
SPC10304	SPC10415	clade_354	clade_262	clade_309	hetero	++++	FALSE	TRUE
SPC10304	SPC10567	clade_354	clade_262	clade_444	hetero	++++	TRUE	TRUE
SPC10325	SPC10325	clade_354	clade_408	clade_408	semi	++++	FALSE	FALSE
SPC10325	SPC10355	clade_354	clade_408	clade_408	hetero	++++	FALSE	TRUE
SPC10325	SPC10386	clade_354	clade_408	clade_478	hetero	++++	FALSE	TRUE
SPC10325	SPC10390	clade_354	clade_408	clade_260	hetero	++++	FALSE	TRUE
SPC10325	SPC10415	clade_354	clade_408	clade_309	hetero	++++	TRUE	TRUE
SPC10325	SPC10567	clade_354	clade_408	clade_444	hetero	++++	TRUE	TRUE
SPC10355	SPC10355	clade_354	clade_408	clade_408	semi	++++	FALSE	FALSE
SPC10355	SPC10386	clade_354	clade_408	clade_478	hetero	++++	FALSE	TRUE
SPC10355	SPC10390	clade_354	clade_408	clade_260	hetero	++++	FALSE	FALSE
SPC10355	SPC10415	clade_354	clade_408	clade_309	hetero	++++	FALSE	FALSE
SPC10355	SPC10567	clade_354	clade_408	clade_444	hetero	++++	TRUE	TRUE
SPC10386	SPC10386	clade_354	clade_478	clade_478	semi	++++	FALSE	TRUE
SPC10386	SPC10390	clade_354	clade_478	clade_260	hetero	++++	FALSE	TRUE
SPC10386	SPC10415	clade_354	clade_478	clade_309	hetero	++++	TRUE	TRUE
SPC10386	SPC10567	clade_354	clade_478	clade_444	hetero	++++	TRUE	TRUE
SPC10390	SPC10390	clade_354	clade_260	clade_260	semi	++++	FALSE	TRUE
SPC10390	SPC10415	clade_354	clade_260	clade_309	hetero	++++	TRUE	TRUE
SPC10390	SPC10567	clade_354	clade_260	clade_444	hetero	++++	TRUE	TRUE
SPC10415	SPC10415	clade_354	clade_309	clade_309	semi	++++	FALSE	FALSE
SPC10415	SPC10567	clade_354	clade_309	clade_444	hetero	++++	TRUE	TRUE
SPC10567	SPC10567	clade_354	clade_444	clade_444	semi	++++	TRUE	TRUE
SPC10097	SPC10097	clade_252	clade_553	clade_553	semi	++++	TRUE	TRUE
SPC10097	SPC10304	clade_252	clade_553	clade_262	hetero	++++	TRUE	TRUE
SPC10097	SPC10325	clade_252	clade_553	clade_408	hetero	++++	TRUE	FALSE
SPC10097	SPC10355	clade_252	clade_553	clade_408	hetero	++++	TRUE	TRUE
SPC10097	SPC10386	clade_252	clade_553	clade_478	hetero	++++	TRUE	TRUE
SPC10097	SPC10390	clade_252	clade_553	clade_260	hetero	++++	TRUE	TRUE
SPC10097	SPC10415	clade_252	clade_553	clade_309	hetero	++++	TRUE	TRUE
SPC10097	SPC10567	clade_252	clade_553	clade_444	hetero	++++	TRUE	TRUE
SPC10304	SPC10304	clade_252	clade_262	clade_262	semi	++++	TRUE	TRUE
SPC10304	SPC10325	clade_252	clade_262	clade_408	hetero	++++	TRUE	TRUE
SPC10304	SPC10355	clade_252	clade_262	clade_408	hetero	++++	TRUE	TRUE
SPC10304	SPC10386	clade_252	clade_262	clade_478	hetero	++++	TRUE	TRUE
SPC10304	SPC10390	clade_252	clade_262	clade_260	hetero	++++	TRUE	TRUE
SPC10304	SPC10415	clade_252	clade_262	clade_309	hetero	++++	TRUE	TRUE
SPC10304	SPC10567	clade_252	clade_262	clade_444	hetero	++++	TRUE	TRUE
SPC10325	SPC10325	clade_252	clade_408	clade_408	semi	++++	TRUE	TRUE
SPC10325	SPC10355	clade_252	clade_408	clade_408	hetero	++++	TRUE	FALSE
SPC10325	SPC10386	clade_252	clade_408	clade_478	hetero	++++	TRUE	TRUE
SPC10325	SPC10390	clade_252	clade_408	clade_260	hetero	++++	TRUE	TRUE
SPC10325	SPC10415	clade_252	clade_408	clade_309	hetero	++++	TRUE	TRUE
SPC10325	SPC10567	clade_252	clade_408	clade_444	hetero	++++	TRUE	TRUE
SPC10355	SPC10355	clade_252	clade_408	clade_408	semi	++++	TRUE	FALSE
SPC10355	SPC10386	clade_252	clade_408	clade_478	hetero	++++	TRUE	TRUE
SPC10355	SPC10390	clade_252	clade_408	clade_260	hetero	++++	TRUE	TRUE
SPC10355	SPC10415	clade_252	clade_408	clade_309	hetero	++++	TRUE	FALSE
SPC10355	SPC10567	clade_252	clade_408	clade_444	hetero	++++	TRUE	TRUE
SPC10386	SPC10386	clade_252	clade_478	clade_478	semi	++++	TRUE	TRUE
SPC10386	SPC10390	clade_252	clade_478	clade_260	hetero	++++	TRUE	TRUE

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SPC10386	SPC10415	clade_252	clade_478	clade_309	hetero	++++	TRUE	TRUE
SPC10386	SPC10567	clade_252	clade_478	clade_444	hetero	++++	TRUE	TRUE
SPC10390	SPC10390	clade_252	clade_260	clade_260	semi	++++	TRUE	TRUE
SPC10390	SPC10415	clade_252	clade_260	clade_309	hetero	++++	TRUE	TRUE
SPC10390	SPC10567	clade_252	clade_260	clade_444	hetero	++++	TRUE	TRUE
SPC10415	SPC10415	clade_252	clade_309	clade_309	semi	++++	TRUE	TRUE
SPC10415	SPC10567	clade_252	clade_309	clade_444	hetero	++++	TRUE	TRUE
SPC10567	SPC10567	clade_252	clade_444	clade_444	semi	++++	TRUE	TRUE
SPC10097	SPC10097	clade_260	clade_553	clade_553	semi	+	FALSE	FALSE
SPC10097	SPC10304	clade_260	clade_553	clade_262	hetero	++++	FALSE	TRUE
SPC10097	SPC10325	clade_260	clade_553	clade_408	hetero	++++	FALSE	FALSE
SPC10097	SPC10355	clade_260	clade_553	clade_408	hetero	++++	FALSE	FALSE
SPC10097	SPC10386	clade_260	clade_553	clade_478	hetero	++++	FALSE	TRUE
SPC10097	SPC10390	clade_260	clade_553	clade_260	hetero	++++	FALSE	TRUE
SPC10097	SPC10415	clade_260	clade_553	clade_309	hetero	++++	FALSE	TRUE
SPC10097	SPC10567	clade_260	clade_553	clade_444	hetero	++++	FALSE	TRUE
SPC10304	SPC10304	clade_260	clade_262	clade_262	semi		FALSE	FALSE
SPC10304	SPC10325	clade_260	clade_262	clade_408	hetero	++++	FALSE	TRUE
SPC10304	SPC10355	clade_260	clade_262	clade_408	hetero	++++	FALSE	TRUE
SPC10304	SPC10386	clade_260	clade_262	clade_478	hetero		FALSE	FALSE
SPC10304	SPC10390	clade_260	clade_262	clade_260	hetero	++++	FALSE	TRUE
SPC10304	SPC10415	clade_260	clade_262	clade_309	hetero	++++	FALSE	TRUE
SPC10304	SPC10567	clade_260	clade_262	clade_444	hetero	+	FALSE	TRUE
SPC10325	SPC10325	clade_260	clade_408	clade_408	semi	++++	FALSE	FALSE
SPC10325	SPC10355	clade_260	clade_408	clade_408	hetero		FALSE	FALSE
SPC10325	SPC10386	clade_260	clade_408	clade_478	hetero	++++	FALSE	TRUE
SPC10325	SPC10390	clade_260	clade_408	clade_260	hetero	++++	FALSE	TRUE
SPC10325	SPC10415	clade_260	clade_408	clade_309	hetero	++++	FALSE	TRUE
SPC10325	SPC10567	clade_260	clade_408	clade_444	hetero	++++	FALSE	TRUE
SPC10355	SPC10355	clade_260	clade_408	clade_408	semi	–	FALSE	FALSE
SPC10355	SPC10386	clade_260	clade_408	clade_478	hetero		FALSE	FALSE
SPC10355	SPC10390	clade_260	clade_408	clade_260	hetero	++++	FALSE	FALSE
SPC10355	SPC10415	clade_260	clade_408	clade_309	hetero	++++	FALSE	TRUE
SPC10355	SPC10567	clade_260	clade_408	clade_444	hetero	++++	FALSE	TRUE
SPC10386	SPC10386	clade_260	clade_478	clade_478	semi	––––	FALSE	FALSE
SPC10386	SPC10390	clade_260	clade_478	clade_260	hetero	++++	FALSE	TRUE
SPC10386	SPC10415	clade_260	clade_478	clade_309	hetero	++++	FALSE	TRUE
SPC10386	SPC10567	clade_260	clade_478	clade_444	hetero	+++	FALSE	TRUE
SPC10390	SPC10390	clade_260	clade_260	clade_260	semi	++++	FALSE	TRUE
SPC10390	SPC10415	clade_260	clade_260	clade_309	hetero	++++	TRUE	TRUE
SPC10390	SPC10567	clade_260	clade_260	clade_444	hetero	++++	FALSE	TRUE
SPC10415	SPC10415	clade_260	clade_309	clade_309	semi	++++	TRUE	TRUE
SPC10415	SPC10567	clade_260	clade_309	clade_444	hetero	++++	TRUE	TRUE
SPC10567	SPC10567	clade_260	clade_444	clade_444	semi	++++	FALSE	TRUE
SPC10097	SPC10097	clade_494	clade_553	clade_553	semi	++++	FALSE	TRUE
SPC10097	SPC10304	clade_494	clade_553	clade_262	hetero	++++	FALSE	TRUE
SPC10097	SPC10325	clade_494	clade_553	clade_408	hetero	++++	FALSE	TRUE
SPC10097	SPC10355	clade_494	clade_553	clade_408	hetero	++++	FALSE	TRUE
SPC10097	SPC10386	clade_494	clade_553	clade_478	hetero	++++	FALSE	TRUE
SPC10097	SPC10390	clade_494	clade_553	clade_260	hetero	++++	FALSE	TRUE
SPC10097	SPC10415	clade_494	clade_553	clade_309	hetero	++++	TRUE	TRUE
SPC10097	SPC10567	clade_494	clade_553	clade_444	hetero	++++	FALSE	TRUE
SPC10304	SPC10304	clade_494	clade_262	clade_262	semi	––––	FALSE	FALSE
SPC10304	SPC10325	clade_494	clade_262	clade_408	hetero	+	FALSE	FALSE
SPC10304	SPC10355	clade_494	clade_262	clade_408	hetero	++	FALSE	FALSE
SPC10304	SPC10386	clade_494	clade_262	clade_478	hetero	––––	FALSE	FALSE
SPC10304	SPC10390	clade_494	clade_262	clade_260	hetero	++++	FALSE	TRUE
SPC10304	SPC10415	clade_494	clade_262	clade_309	hetero	++++	TRUE	TRUE
SPC10304	SPC10567	clade_494	clade_262	clade_444	hetero	––––	FALSE	FALSE
SPC10325	SPC10325	clade_494	clade_408	clade_408	semi	++++	FALSE	FALSE
SPC10325	SPC10355	clade_494	clade_408	clade_408	hetero	++	FALSE	FALSE
SPC10325	SPC10386	clade_494	clade_408	clade_478	hetero	++++	FALSE	FALSE
SPC10325	SPC10390	clade_494	clade_408	clade_260	hetero	++++	FALSE	FALSE
SPC10325	SPC10415	clade_494	clade_408	clade_309	hetero	++++	TRUE	TRUE
SPC10325	SPC10567	clade_494	clade_408	clade_444	hetero	++++	FALSE	TRUE
SPC10355	SPC10355	clade_494	clade_408	clade_408	semi	––	FALSE	FALSE
SPC10355	SPC10386	clade_494	clade_408	clade_478	hetero	–––	FALSE	FALSE
SPC10355	SPC10390	clade_494	clade_408	clade_260	hetero	++++	FALSE	FALSE
SPC10355	SPC10415	clade_494	clade_408	clade_309	hetero	++++	FALSE	TRUE
SPC10355	SPC10567	clade_494	clade_408	clade_444	hetero	++	FALSE	FALSE
SPC10386	SPC10386	clade_494	clade_478	clade_478	semi	–––	FALSE	FALSE
SPC10386	SPC10390	clade_494	clade_478	clade_260	hetero		FALSE	FALSE
SPC10386	SPC10415	clade_494	clade_478	clade_309	hetero	++++	TRUE	TRUE
SPC10386	SPC10567	clade_494	clade_478	clade_444	hetero	––––	FALSE	FALSE
SPC10390	SPC10390	clade_494	clade_260	clade_260	semi		FALSE	FALSE
SPC10390	SPC10415	clade_494	clade_260	clade_309	hetero	++++	TRUE	TRUE
SPC10390	SPC10567	clade_494	clade_260	clade_444	hetero	+++	FALSE	FALSE
SPC10415	SPC10415	clade_494	clade_309	clade_309	semi	++++	TRUE	TRUE
SPC10415	SPC10567	clade_494	clade_309	clade_444	hetero	++++	TRUE	TRUE
SPC10567	SPC10567	clade_494	clade_444	clade_444	semi	–	FALSE	FALSE

TABLE 4a-continued

SPC10097	SPC10097	clade_360	clade_553	clade_553	semi		FALSE	FALSE
SPC10097	SPC10304	clade_360	clade_553	clade_262	hetero		FALSE	FALSE
SPC10097	SPC10325	clade_360	clade_553	clade_408	hetero	+	FALSE	FALSE
SPC10097	SPC10355	clade_360	clade_553	clade_408	hetero	++++	FALSE	FALSE
SPC10097	SPC10386	clade_360	clade_553	clade_478	hetero	++++	FALSE	FALSE
SPC10097	SPC10390	clade_360	clade_553	clade_260	hetero	++++	FALSE	TRUE
SPC10097	SPC10415	clade_360	clade_553	clade_309	hetero	++++	FALSE	TRUE
SPC10097	SPC10567	clade_360	clade_553	clade_444	hetero	++++	FALSE	TRUE
SPC10304	SPC10304	clade_360	clade_262	clade_262	semi	–	FALSE	FALSE
SPC10304	SPC10325	clade_360	clade_262	clade_408	hetero		FALSE	FALSE
SPC10304	SPC10355	clade_360	clade_262	clade_408	hetero		FALSE	FALSE
SPC10304	SPC10386	clade_360	clade_262	clade_478	hetero	++++	FALSE	FALSE
SPC10304	SPC10390	clade_360	clade_262	clade_260	hetero	++++	FALSE	TRUE
SPC10304	SPC10415	clade_360	clade_262	clade_309	hetero	++++	FALSE	TRUE
SPC10304	SPC10567	clade_360	clade_262	clade_444	hetero	++++	FALSE	TRUE
SPC10325	SPC10325	clade_360	clade_408	clade_408	semi	++	FALSE	FALSE
SPC10325	SPC10355	clade_360	clade_408	clade_408	hetero	++++	FALSE	FALSE
SPC10325	SPC10386	clade_360	clade_408	clade_478	hetero	+++	FALSE	FALSE
SPC10325	SPC10390	clade_360	clade_408	clade_260	hetero	++++	FALSE	FALSE
SPC10325	SPC10415	clade_360	clade_408	clade_309	hetero	++++	TRUE	TRUE
SPC10325	SPC10567	clade_360	clade_408	clade_444	hetero	++++	FALSE	TRUE
SPC10355	SPC10355	clade_360	clade_408	clade_408	semi		FALSE	FALSE
SPC10355	SPC10386	clade_360	clade_408	clade_478	hetero		FALSE	FALSE
SPC10355	SPC10390	clade_360	clade_408	clade_260	hetero	++++	FALSE	TRUE
SPC10355	SPC10415	clade_360	clade_408	clade_309	hetero	++++	FALSE	TRUE
SPC10355	SPC10567	clade_360	clade_408	clade_444	hetero	++++	FALSE	FALSE
SPC10386	SPC10386	clade_360	clade_478	clade_478	semi		FALSE	FALSE
SPC10386	SPC10390	clade_360	clade_478	clade_260	hetero	++++	FALSE	TRUE
SPC10386	SPC10415	clade_360	clade_478	clade_309	hetero	++++	FALSE	TRUE
SPC10386	SPC10567	clade_360	clade_478	clade_444	hetero	++++	FALSE	TRUE
SPC10390	SPC10390	clade_360	clade_260	clade_260	semi	++++	FALSE	TRUE
SPC10390	SPC10415	clade_360	clade_260	clade_309	hetero	++++	TRUE	TRUE
SPC10390	SPC10567	clade_360	clade_260	clade_444	hetero	++++	FALSE	TRUE
SPC10415	SPC10415	clade_360	clade_309	clade_309	semi	++++	TRUE	TRUE
SPC10415	SPC10567	clade_360	clade_309	clade_444	hetero	++++	TRUE	TRUE
SPC10567	SPC10567	clade_360	clade_444	clade_444	semi	++++	FALSE	TRUE

TABLE 4b

OTU1	ID1	OTU2	ID2	OTU3	ID3	Clade1	Clade2	Clade3
Bacteroides sp. 1_1_6	295	*Clostridium* sp. HGF2	628			clade_65	clade_351	
Bacteroides sp. 1_1_6	295	*Bifidobacterium pseudocatenulatum*	357			clade_65	clade_172	
Bacteroides sp. 1_1_6	295	*Clostridium symbiosum*	652			clade_65	clade_408	
Bacteroides sp. 3_1_23	308	*Clostridium nexile*	607			clade_38	clade_262	
Bacteroides sp. 3_1_23	308	*Bifidobacterium pseudocatenulatum*	357			clade_38	clade_172	
Bacteroides sp. 3_1_23	308	*Clostridium symbiosum*	652			clade_38	clade_408	
Streptococcus thermophilus	1883	*Bifidobacterium pseudocatenulatum*	357			clade_98	clade_172	
Clostridium nexile	607	*Bifidobacterium pseudocatenulatum*	357			clade_262	clade_172	
Parabacteroides merdae	1420	*Bifidobacterium pseudocatenulatum*	357			clade_286	clade_172	
Clostridium tertium	653	*Clostridium mayombei*	605			clade_252	clade_354	
Clostridium tertium	653	*Clostridium butyricum*	561			clade_252	clade_252	
Clostridium tertium	653	*Coprococcus comes*	674			clade_252	clade_262	
Clostridium tertium	653	*Clostridium hylemonae*	593			clade_252	clade_260	
Clostridium tertium	653	*Clostridium orbiscindens*	609			clade_252	clade_494	
Clostridium tertium	653	Lachnospiraceae *bacterium* 5_1_57FAA	1054			clade_252	clade_260	
Clostridium tertium	653	*Ruminococcus gnavus*	1661			clade_252	clade_360	
Clostridium tertium	653	*Ruminococcus bromii*	1657			clade_252	clade_537	
Clostridium disporicum	579	*Clostridium mayombei*	605			clade_253	clade_354	
Clostridium disporicum	579	*Clostridium butyricum*	561			clade_253	clade_252	
Clostridium disporicum	579	*Clostridium orbiscindens*	609			clade_253	clade_494	
Clostridium disporicum	579	*Ruminococcus gnavus*	1661			clade_253	clade_360	
Clostridium mayombei	605	*Clostridium butyricum*	561			clade_354	clade_252	
Clostridium mayombei	605	*Coprococcus comes*	674			clade_354	clade_262	
Clostridium mayombei	605	*Clostridium hylemonae*	593			clade_354	clade_260	
Clostridium mayombei	605	*Clostridium symbiosum*	652			clade_354	clade_408	
Clostridium mayombei	605	*Clostridium orbiscindens*	609			clade_354	clade_494	
Clostridium mayombei	605	Lachnospiraceae *bacterium* 5_1_57FAA	1054			clade_354	clade_260	
Clostridium mayombei	605	*Ruminococcus gnavus*	1661			clade_354	clade_360	
Clostridium mayombei	605	*Ruminococcus bromii*	1657			clade_354	clade_537	

TABLE 4b-continued

OTU1	ID1	OTU2	ID2	OTU3	ID3	Clade1	Clade2	Clade3
Clostridium butyricum	561	*Clostridium mayombei*	605			clade_252	clade_354	
Clostridium butyricum	561	*Coprococcus comes*	674			clade_252	clade_262	
Clostridium butyricum	561	*Clostridium hylemonae*	593			clade_252	clade_260	
Clostridium butyricum	561	*Clostridium symbiosum*	652			clade_252	clade_408	
Clostridium butyricum	561	*Clostridium orbiscindens*	609			clade_252	clade_494	
Clostridium butyricum	561	Lachnospiraceae *bacterium* 5_1_57FAA	1054			clade_252	clade_260	
Clostridium butyricum	561	*Ruminococcus gnavus*	1661			clade_252	clade_360	
Clostridium butyricum	561	*Ruminococcus bromii*	1657			clade_252	clade_537	
Coprococcus comes	674	*Clostridium butyricum*	561			clade_262	clade_252	
Coprococcus comes	674	*Ruminococcus gnavus*	1661			clade_262	clade_360	
Clostridium hylemonae	593	Lachnospiraceae *bacterium* 5_1_57FAA	1054			clade_260	clade_260	
Clostridium orbiscindens	609	*Ruminococcus gnavus*	1661			clade_494	clade_360	
Coprococcus comes	674	*Clostridium tertium*	653	*Clostridium mayombei*	605	clade_262	clade_252	clade_354
Coprococcus comes	674	*Clostridium tertium*	653	*Clostridium butyricum*	561	clade_262	clade_252	clade_252
Coprococcus comes	674	*Clostridium tertium*	653	*Clostridium orbiscindens*	609	clade_262	clade_252	clade_494
Coprococcus comes	674	*Clostridium disporicum*	579	*Clostridium butyricum*	561	clade_262	clade_253	clade_252
Coprococcus comes	674	*Clostridium mayombei*	605	*Clostridium butyricum*	561	clade_262	clade_354	clade_252
Coprococcus comes	674	*Clostridium butyricum*	561	*Clostridium hylemonae*	593	clade_262	clade_252	clade_260
Coprococcus comes	674	*Clostridium butyricum*	561	*Clostridium orbiscindens*	609	clade_262	clade_252	clade_494
Coprococcus comes	674	*Clostridium butyricum*	561	*Ruminococcus gnavus*	1661	clade_262	clade_252	clade_360
Coprococcus comes	674	*Clostridium butyricum*	561	*Ruminococcus bromii*	1657	clade_262	clade_252	clade_537
Clostridium symbiosum	652	*Clostridium tertium*	653	*Clostridium mayombei*	605	clade_408	clade_252	clade_354
Clostridium symbiosum	652	*Clostridium tertium*	653	*Clostridium butyricum*	561	clade_408	clade_252	clade_252
Clostridium symbiosum	652	*Clostridium disporicum*	579	*Clostridium butyricum*	561	clade_408	clade_253	clade_252
Clostridium symbiosum	652	*Clostridium mayombei*	605	*Clostridium orbiscindens*	609	clade_408	clade_354	clade_494
Clostridium symbiosum	652	*Clostridium mayombei*	605	*Ruminococcus bromii*	1657	clade_408	clade_354	clade_537
Clostridium symbiosum	652	*Clostridium butyricum*	561	*Clostridium hylemonae*	593	clade_408	clade_252	clade_260
Clostridium symbiosum	652	*Clostridium butyricum*	561	*Clostridium orbiscindens*	609	clade_408	clade_252	clade_494
Clostridium symbiosum	652	*Clostridium butyricum*	561	*Ruminococcus gnavus*	1661	clade_408	clade_252	clade_360
Clostridium symbiosum	652	*Clostridium butyricum*	561	*Ruminococcus bromii*	1657	clade_408	clade_252	clade_537
Lachnospiraceae *bacterium* 5_1_57FAA	1054	*Clostridium tertium*	653	*Clostridium butyricum*	561	clade_260	clade_252	clade_252
Lachnospiraceae *bacterium* 5_1_57FAA	1054	*Clostridium disporicum*	579	*Clostridium butyricum*	561	clade_260	clade_253	clade_252
Lachnospiraceae *bacterium* 5_1_57FAA	1054	*Clostridium mayombei*	605	*Ruminococcus gnavus*	1661	clade_260	clade_354	clade_360
Lachnospiraceae *bacterium* 5_1_57FAA	1054	*Clostridium mayombei*	605	*Ruminococcus bromii*	1657	clade_260	clade_354	clade_537
Lachnospiraceae *bacterium* 5_1_57FAA	1054	*Clostridium butyricum*	561	*Clostridium hylemonae*	593	clade_260	clade_252	clade_260
Lachnospiraceae *bacterium* 5_1_57FAA	1054	*Clostridium butyricum*	561	*Clostridium orbiscindens*	609	clade_260	clade_252	clade_494
Lachnospiraceae *bacterium* 5_1_57FAA	1054	*Clostridium butyricum*	561	*Ruminococcus gnavus*	1661	clade_260	clade_252	clade_360
Lachnospiraceae *bacterium* 5_1_57FAA	1054	*Clostridium butyricum*	561	*Ruminococcus bromii*	1657	clade_260	clade_252	clade_537
Clostridium butyricum	561	*Coprococcus comes*	674	*Clostridium symbiosum*	652	clade_252	clade_262	clade_408
Clostridium butyricum	561	*Coprococcus comes*	674	Lachnospiraceae *bacterium* 5_1_57FAA	1054	clade_252	clade_262	clade_260
Clostridium butyricum	561	*Clostridium symbiosum*	652	Lachnospiraceae *bacterium* 5_1_57FAA	1054	clade_252	clade_408	clade_260

TABLE 5

Expt.	SP Arm	Key	Seres Arm	Treatment	OTU1	OTU2	OTU3	CivSim Inhibition (log10 CFU/mL)	CivSim Inhibition (Confidence)	Mean Min. Rel. Weight	Mean Max. Clin. Score	Cum.			
Mort.	Clade1	Clade2	Clade3												
SP427	1	PBS	1	Vehicle				n/a	n/a	0.82	2.6	3			
SP427	14	V5F	2	FSV33_10pct (EMT)				n/a	n/a	0.99	0	0			
SP427	6	79F	3	DE277512.1	*Collinsella aerofaciens*	*Faecalibacterium prausnitzii*	*Blautia producta*	1.84	++++	0.85	1.4	1	clade_553	clade_478	clade_309
SP427	12	V7B	4	DE643314.1	*Faecalibacterium prausnitzii*	*Blautia producta*	*Eubacterium rectale*	1.27	++++	0.81	2	3	clade_478	clade_309	clade_444
SP427	5	699	5	DE022136.1	*Clostridium bolteae*	Lachnospiraceae *bacterium* 5_1_57FAA	*Blautia producta*	0.81	++++	0.93	1	0	clade_408	clade_260	clade_309
SP427	10	E4W	6	DE061176.1	*Collinsella aerofaciens*	*Clostridium butyricum*	*Ruminococcus gnavus*	2.87	++++	0.96	0.2	0	clade_553	clade_252	clade_360
SP427	7	852	7	DE554703.1	*Collinsella aerofaciens*	*Clostridium butyricum*	*Clostridium hylemonae*	3.09	++++	0.78	1.7	2	clade_553	clade_252	clade_260
SP427	3	366	8	DE705158.1	*Coprococcus comes*	*Clostridium innocuum*	*Clostridium butyricum*	3	++++	0.85	1.7	2	clade_262	clade_351	clade_252
SP427	9	CE3	9	DE266960.1	*Clostridium bolteae*	*Clostridium butyricum*	*Ruminococcus gnavus*	3.06	++++	0.91	1	0	clade_408	clade_252	clade_360
SP427	8	BCY	10	DE897971.1	*Clostridium mayombei*	*Collinsella aerofaciens*	*Clostridium symbiosum*	1.46	++++	0.83	1.7	1	clade_354	clade_553	clade_408
SP427	13	Y4K	11	DE001210.1	*Clostridium tertium*	*Collinsella aerofaciens*	*Blautia producta*	2	++++	0.88	1.1	0	clade_252	clade_553	clade_309
SP427	11	FBK	12	DE586246.1	*Clostridium mayombei*	Lachnospiraceae *bacterium* 5_1_57FAA	*Blautia producta*	1.88	++++	0.94	1	0	clade_354	clade_260	clade_309
SP427	4	3R1	13	DE844277.1	*Coprococcus comes*	*Clostridium mayombei*	*Blautia* sp. M25	2.1	++++	0.88	1	0	clade_262	clade_354	clade_309
SP427	2	1HR	14	DE208485.1	*Coprococcus comes*	*Clostridium tertium*	*Clostridium orbiscindens*	1.8	++++	0.89	1.3	1	clade_262	clade_252	clade_494
SP427	31	PBS NoCdiff	31	Naive				n/a	n/a	1	0	0			

TABLE 6

Initial Concentration of VRE (log10 CFU/mL)	Initial Concentration of Ecobiotic ™ Composition (log10 CFU/mL/Strain)	Incu-bation Time (h)	Inhibition (log10 CFU/mL)
3	6	15	3.1
3	4	15	1.3
2	6	15	5.2
2	4	15	1.6
3	6	24	2.7
3	4	24	0.7
2	6	24	4.6

TABLE 7

Initial Concentration of *K. pneumoniae* (log10 CFU/mL)	Initial Concentration of Ecobiotic ™ Composition (log10 CFU/mL/Strain)	Incu-bation Time (h)	Inhibition (log10 CFU/mL)
3	6	15	2.5
3	4	15	0.4
2	6	15	4.2
2	4	15	1.2
3	6	24	1.7

TABLE 7-continued

Initial Concentration of *K. pneumoniae* (log10 CFU/mL)	Initial Concentration of Ecobiotic ™ Composition (log10 CFU/mL/Strain)	Incu-bation Time (h)	Inhibition (log10 CFU/mL)
3	4	24	0.2
2	6	24	3.1
2	4	24	0.1

TABLE 8

Initial Concentration of *M. morganii* (log10 CFU/mL)	Initial Concentration of Ecobiotic ™ Composition (log10 CFU/mL/Strain)	Incu-bation Time (h)	Inhibition (log10 CFU/mL)
3	6	15	4.3
3	4	15	2.1
2	6	15	5.8
2	4	15	3.3
3	6	24	3.9
3	4	24	1.4
2	6	24	5.1
2	4	24	2.5

TABLE 9

SP Expt.	SP Arm	Test Article	Target Dose (CFU/OTU/mouse)	Cumulative Mortality (%)	Mean Min. Rel. Weight	Mean Max. Clin. Score (Death = 4)
SP-327	3	Vehicle Control		30	0.89	2.2
SP-327	4	Vanco. Positive Control		0	0.99	1
SP-327	12	N1957	2.0E+07	0	0.87	0
SP-327	13	N1957	2.0E+06	40	0.86	2.2
SP-327	14	N1957	2.0E+05	50	0.80	2.8
SP-338	1	Vehicle Control		60	0.81	3.2
SP-338	2	Vanco. Positive Control		0	1.00	0
SP-338	3	10% fecal suspension		0	0.95	1
SP-338	5	N1957	2.0E+07	10	0.80	2
SP-338	6	N1957	2.0E+06	0	0.97	1
SP-338	7	N1957	2.0E+05	20	0.85	1.7
SP-338	11	N1957	2.0E+07	20	0.86	2
SP-338	12	N1957	2.0E+06	30	0.83	2.5
SP-338	13	N1961	2.0E+07	10	0.93	1.3
SP-338	14	N1955	2.0E+07	0	0.91	1.2
SP-338	15	N1955	2.0E+06	10	0.90	1.5
SP-338	16	N1955	2.0E+05	10	0.89	2.7
SP-338	17	N1967	2.0E+07	10	0.94	1.4
SP-338	18	N1983	2.0E+07	0	0.92	1
SP-338	19	N1989	2.0E+07	10	0.91	1.3
SP-338	20	N1996	2.0E+07	10	0.93	1.3
SP-338	21	Naïve		0	1.00	0
SP-339	1	Vehicle Control		20	0.88	2.2
SP-339	2	Vanco. Positive Control		0	0.99	0
SP-339	3	10% fecal suspension		0	0.97	0
SP-339	4	N1995	2.0E+07	20	0.83	2.1
SP-339	5	N1995	2.0E+06	10	0.91	1.5
SP-339	6	N1995	2.0E+05	0	0.96	1.2
SP-339	7	N1950	2.0E+07	0	0.94	1
SP-339	8	N1994	2.0E+07	20	0.87	1.8
SP-339	9	N1997	2.0E+07	0	0.95	1.2
SP-339	10	N1967	2.0E+07	0	0.93	1.2
SP-339	11	N1983	2.0E+07	10	0.83	2.2
SP-339	12	N1989	2.0E+07	0	0.88	1.5
SP-339	13	N1996	2.0E+07	0	0.97	1
SP-339	14	N2002	2.0E+07	20	0.92	2
SP-339	15	N2000	2.0E+07	0	0.98	1.2
SP-339	21	Naïve		0	0.98	0
SP-342	1	Vehicle Control		40	0.85	2.5
SP-342	2	Vanco. Positive Control		0	1.00	0
SP-342	5	N1957	2.0E+08	0	0.94	0.2

TABLE 9-continued

SP Expt.	SP Arm	Test Article	Target Dose (CFU/OTU/ mouse)	Cumulative Mortality (%)	Mean Min. Rel. Weight	Mean Max. Clin. Score (Death = 4)
SP-342	6	N1957	2.0E+07	0	0.96	0
SP-342	7	N1957	2.0E+06	10	0.88	1.3
SP-342	8	N1980	2.0E+08	10	0.92	1.8
SP-342	9	N1998	2.0E+08	20	0.83	2.8
SP-342	10	N1976	2.0E+08	10	0.92	1.4
SP-342	11	N1987	2.0E+08	10	0.93	1.6
SP-342	12	N2005	2.0E+08	20	0.86	2.4
SP-342	13	N1958	2.0E+08	0	0.94	1.5
SP-342	14	N2004	2.0E+08	10	0.93	1.4
SP-342	15	N1949	2.0E+08	10	0.87	1.5
SP-342	18	N1970	2.0E+08	50	0.81	3
SP-342	21	Naïve		0	0.99	0
SP-361	1	Vehicle Control		30	0.88	2.6
SP-361	2	10% fecal suspension		0	0.99	0
SP-361	3	N435	1.0E+07	80	0.83	3.6
SP-361	4	N1979	1.0E+07	0	0.97	0
SP-361	5	N414	1.0E+07	0	0.97	0
SP-361	6	N512	1.0E+07	20	0.94	1.6
SP-361	7	N582	1.0E+07	10	0.93	0.9
SP-361	8	N571	1.0E+07	30	0.88	2.1
SP-361	9	N510	1.0E+07	0	0.93	0.3
SP-361	10	N1981	1.0E+07	40	0.83	2.8
SP-361	11	N1969	1.0E+07	80	0.82	3.6
SP-361	12	N461	1.0E+07	10	0.89	1.2
SP-361	13	N460	1.0E+07	0	0.93	1.1
SP-361	14	N1959	1.0E+07	30	0.89	1.9
SP-361	15	N2006	1.0E+07	30	0.89	1.9
SP-361	16	N1953	1.0E+07	10	0.83	2.3
SP-361	17	N1960	1.0E+07	0	0.92	1
SP-361	18	N2007	1.0E+07	10	0.91	0.9
SP-361	19	N1978	1.0E+07	10	0.91	1.3
SP-361	20	N1972	1.0E+07	30	0.83	2.6
SP-361	21	Naïve		0	1.00	0
SP-363	1	Vehicle Control		30	0.85	2.6
SP-363	2	10% fecal suspension		0	0.95	0
SP-363	8	N1974	1.0E+07	60	0.81	3.2
SP-363	9	N582	1.0E+07	60	0.81	3.2
SP-363	10	N435	1.0E+07	30	0.86	2.1
SP-363	11	N414	1.0E+07	40	0.83	2.5
SP-363	12	N457	1.0E+07	30	0.83	2.2
SP-363	13	N511	1.0E+07	20	0.87	2
SP-363	14	N513	1.0E+07	0	0.88	0.2
SP-363	15	N682	1.0E+07	30	0.82	2.6
SP-363	16	N736	1.0E+07	40	0.82	2.8
SP-363	17	N732	1.0E+07	10	0.86	1.3
SP-363	18	N1948	1.0E+07	60	0.85	3.2
SP-363	19	N853	1.0E+07	10	0.85	2.2
SP-363	20	N1979	1.0E+07	60	0.78	3.2
SP-363	21	N879	1.0E+07	40	0.83	2.8
SP-363	22	N999	1.0E+07	20	0.88	2.4
SP-363	23	N975	1.0E+07	30	0.80	2.6
SP-363	24	N861	1.0E+07	50	0.85	3
SP-363	25	N1095	1.0E+07	80	0.83	3.6
SP-363	26	Naïve		0	1.00	0
SP-364	1	Vehicle Control		40	0.83	2.8
SP-364	4	N582	1.0E+07	0	0.81	0.9
SP-364	5	N582	1.0E+06	0	0.84	0.9
SP-364	6	N582	1.0E+05	40	0.76	2.5
SP-364	13	N414	1.0E+07	0	0.84	0
SP-364	14	N414	1.0E+06	30	0.79	2.4
SP-364	15	N414	1.0E+05	10	0.76	2
SP-364	22	10% fecal suspension		0	0.97	0
SP-364	23	Nave		0	0.99	0
SP-365	1	Vehicle Control		40	0.83	2.8
SP-365	4	10% fecal suspension		0	0.98	0
SP-365	13	N582	1.0E+07	60	0.80	3.2
SP-365	14	N582	1.0E+06	10	0.89	1.5
SP-365	15	N414	1.0E+07	20	0.86	1.7
SP-365	16	N414	1.0E+06	80	0.83	3.5
SP-365	21	Naïve		0	1.00	0
SP-366	1	Vehicle Control		20	0.82	2.4
SP-366	4	10% fecal suspension		0	0.93	1
SP-366	7	N582	1.0E+07	0	0.86	1
SP-366	10	N414	1.0E+07	20	0.83	2.4
SP-366	13	N402	1.0E+07	30	0.81	2.1

TABLE 9-continued

SP Expt.	SP Arm	Test Article	Target Dose (CFU/OTU/ mouse)	Cumulative Mortality (%)	Mean Min. Rel. Weight	Mean Max. Clin. Score (Death = 4)
SP-366	16	N1982	1.0E+07	0	0.90	1.1
SP-366	19	N460	1.0E+07	10	0.83	2.2
SP-366	22	N513	6.7E+06	40	0.82	2.8
SP-366	23	N1966	1.0E+07	0	0.90	0.5
SP-366	24	N1977	1.0E+07	20	0.83	1.9
SP-366	25	N1979	1.0E+07	20	0.83	2.4
SP-366	26	N682	1.0E+07	20	0.83	2.3
SP-366	27	N1947	1.0E+07	10	0.82	1.3
SP-366	28	N582	1.0E+07	20	0.82	1.8
SP-366	29	N414	1.0E+07	0	0.85	1.5
SP-366	30	N603	1.0E+07	30	0.82	2.2
SP-366	31	Naïve		0	0.99	0
SP-368	1	Vehicle Control		50	0.85	2.8
SP-368	2	10% fecal suspension		0	0.97	0
SP-368	7	N1966	1.0E+07	0	0.89	1
SP-368	8	N1966	1.0E+06	10	0.91	1.5
SP-368	9	N1966	1.0E+05	50	0.82	3.1
SP-368	21	Naïve		0	1.00	0
SP-374	1	Vehicle Control		100	0.83	4
SP-374	4	10% fecal suspension		10	0.89	0.5
SP-374	11	N1966	1.0E+08	0	0.87	1
SP-374	12	N1966	1.0E+08	0	0.91	0.5
SP-374	13	N1966	1.0E+07	10	0.88	1.3
SP-374	14	N1966	1.0E+06	50	0.79	3
SP-374	15	N584	1.0E+08	0	0.89	1
SP-374	16	N584	1.0E+07	30	0.84	2.4
SP-374	17	N1962	1.0E+07	0	0.93	0
SP-374	18	N382	1.0E+07	10	0.85	1.5
SP-374	19	N1964	1.0E+07	20	0.89	1.8
SP-374	20	N1965	1.0E+07	30	0.85	2.1
SP-374	21	N306	1.0E+07	10	0.90	0.4
SP-374	22	N1988	1.0E+07	0	0.89	1
SP-374	23	N2003	1.0E+07	0	0.92	1.2
SP-374	24	N1993	1.0E+07	20	0.77	2.4
SP-374	25	Naïve		0	0.99	0
SP-376	1	Vehicle Control		60	0.83	3.2
SP-376	2	10% fecal suspension		0	0.98	0
SP-376	3	N1966	1.0E+08	30	0.79	2.4
SP-376	4	N1966	1.0E+07	0	0.95	0
SP-376	5	N1966	1.0E+08	30	0.79	2.6
SP-376	6	N1966	1.0E+07	10	0.88	2.2
SP-376	7	N1986	1.0E+07	40	0.80	2.8
SP-376	8	N1962	1.0E+08	0	0.98	0
SP-376	9	N1962	1.0E+07	0	0.95	0
SP-376	10	N1963	1.0E+07	40	0.81	2.6
SP-376	11	N1984	1.0E+08	0	0.97	0
SP-376	12	N1984	1.0E+07	0	0.90	1.1
SP-376	13	N1990	1.0E+08	0	0.92	1
SP-376	14	N1990	1.0E+07	0	0.92	1
SP-376	15	N1999	1.0E+08	10	0.87	1.4
SP-376	16	N1999	1.0E+07	0	0.93	0
SP-376	17	N1968	1.0E+07	50	0.78	3
SP-376	18	N1951	1.0E+07	0	0.93	1
SP-376	19	N1991	1.0E+07	0	0.93	1.1
SP-376	20	N1975	1.0E+07	50	0.78	3
SP-376	21	Naïve		0	0.99	0
SP-383	1	Vehicle Control		100	0.83	4
SP-383	2	10% fecal suspension		0	0.92	0.1
SP-383	9	N1962	1.0E+09	10	0.95	1.3
SP-383	10	N1962	1.0E+08	10	0.93	1.3
SP-383	11	N1962	1.0E+07	0	0.92	1
SP-383	12	N1984	1.0E+09	0	0.89	1
SP-383	13	N1984	1.0E+08	10	0.94	1.3
SP-383	14	N1984	1.0E+07	10	0.90	1.3
SP-383	21	Naïve		0	1.00	0
SP-390	1	Vehicle Control		80	0.82	3.6
SP-390	2	10% fecal suspension		0	0.98	0.1
SP-390	3	N1962	2.0E+07	0	0.97	0
SP-390	4	N1962	2.0E+06	0	0.98	0
SP-390	5	N1984	2.0E+07	0	0.95	1
SP-390	6	N1984	2.0E+06	0	0.95	0.1
SP-390	9	N1962	2.0E+07	0	0.93	1
SP-390	10	N1962	2.0E+06	10	0.93	1.3
SP-390	11	N1984	2.0E+07	20	0.86	2.2
SP-390	12	N1984	2.0E+09	30	0.88	2.1

TABLE 9-continued

SP Expt.	SP Arm	Test Article	Target Dose (CFU/OTU/ mouse)	Cumulative Mortality (%)	Mean Min. Rel. Weight	Mean Max. Clin. Score (Death = 4)
SP-390	13	N1952	2.0E+07	0	0.89	1
SP-390	14	N2001	2.0E+07	0	0.95	0.2
SP-390	15	N1973	2.0E+07	10	0.90	0.7
SP-390	16	N1954	2.0E+07	0	0.94	1.1
SP-390	17	N1985	2.0E+07	10	0.86	1.8
SP-390	18	N1971	2.0E+07	0	0.89	0.9
SP-390	19	N1956	2.0E+07	0	0.95	0
SP-390	20	N1992	2.0E+07	0	0.95	0
SP-390	31	Naïve		0	0.98	0

TABLE 10

Network Ecology ID	Exemplary Network Clades	Exemplary Network OTUs
N306	clade_252, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_351 or clade_351e), (clade_38 or clade_38e or clade_38i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), (clade_478 or clade_478i), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i), clade_494, (clade_553 or clade_553i)	*Blautia producta*, *Clostridium hylemonae*, *Clostridium innocuum*, *Clostridium orbiscindens*, *Clostridium symbiosum*, *Clostridium tertium*, *Collinsella aerofaciens*, *Coprobacillus* sp. D7, *Coprococcus comes*, *Eubacterium rectale*, *Eubacterium* sp. WAL 14571, *Faecalibacterium prausnitzii*, Lachnospiraceae bacterium 5_1_57FAA, *Roseburia faecalis*, *Ruminococcus obeum*, *Ruminococcus torques*
N382	clade_252, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_351 or clade_351e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), clade_494, clade_537, (clade_553 or clade_553i)	*Blautia producta*, *Clostridium hylemonae*, *Clostridium innocuum*, *Clostridium orbiscindens*, *Clostridium symbiosum*, *Clostridium tertium*, *Collinsella aerofaciens*, *Coprococcus comes*, Lachnospiraceae bacterium 5_1_57FAA, *Ruminococcus bromii*, *Ruminococcus gnavus*
N402	(clade_262 or clade_262i), clade_286, (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_444 or clade_444i), clade_466, (clade_478 or clade_478i), clade_500	*Alistipes shahii*, *Coprococcus comes*, *Dorea formicigenerans*, *Dorea longicatena*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Odoribacter splanchnicus*, *Parabacteroides merdae*, *Ruminococcus obeum*, *Ruminococcus torques*
N414	(clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_444 or clade_444i), clade_466, (clade_478 or clade_478i), (clade_522 or clade_522i)	*Coprococcus comes*, *Dorea formicigenerans*, *Dorea longicatena*, *Eubacterium eligens*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Odoribacter splanchnicus*, *Ruminococcus obeum*, *Ruminococcus torques*
N435	(clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_444 or clade_444i), clade_466, (clade_478 or clade_478i)	*Coprococcus comes*, *Dorea formicigenerans*, *Dorea longicatena*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Odoribacter splanchnicus*, *Ruminococcus obeum*, *Ruminococcus torques*
N457	(clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_444 or clade_444i), (clade_478 or clade_478i), (clade_522 or clade_522i)	*Dorea longicatena*, *Eubacterium eligens*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Roseburia intestinalis*, *Ruminococcus obeum*, *Ruminococcus torques*
N460	(clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_444 or clade_444i), clade_466, (clade_478 or clade_478i)	*Coprococcus comes*, *Dorea formicigenerans*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Odoribacter splanchnicus*, *Ruminococcus obeum*, *Ruminococcus torques*

TABLE 10-continued

Network Ecology ID	Exemplary Network Clades	Exemplary Network OTUs
N461	(clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_444 or clade_444i), (clade_478 or clade_478i)	*Coprococcus comes*, *Dorea formicigenerans*, *Dorea longicatena*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Ruminococcus obeum*, *Ruminococcus torques*
N510	(clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_444 or clade_444i), (clade_478 or clade_478i), (clade_522 or clade_522i)	*Dorea longicatena*, *Eubacterium eligens*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Ruminococcus obeum*, *Ruminococcus torques*
N511	(clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_444 or clade_444i), (clade_478 or clade_478i)	*Coprococcus comes*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Roseburia intestinalis*, *Ruminococcus obeum*, *Ruminococcus torques*
N512	(clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_444 or clade_444i)	*Coprococcus comes*, *Dorea formicigenerans*, *Dorea longicatena*, *Eubacterium rectale*, *Ruminococcus obeum*, *Ruminococcus torques*
N513	(clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_444 or clade_444i), (clade_478 or clade_478i)	*Coprococcus comes*, *Dorea formicigenerans*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Ruminococcus obeum*, *Ruminococcus torques*
N571	(clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_444 or clade_444i), (clade_478 or clade_478i)	*Dorea longicatena*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Ruminococcus obeum*, *Ruminococcus torques*
N582	(clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), (clade_478 or clade_478i)	*Clostridium symbiosum*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Ruminococcus obeum*, *Ruminococcus torques*
N584	(clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_553 or clade_553i)	*Blautia producta*, *Clostridium symbiosum*, *Collinsella aerofaciens*, *Coprococcus comes*, Lachnospiraceae bacterium 5_1_57FAA
N603	(clade_172 or clade_172i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_444 or clade_444i), (clade_478 or clade_478i)	*Bifidobacterium adolescentis*, *Dorea longicatena*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Ruminococcus obeum*
N682	clade_170, (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_444 or clade_444i), (clade_478 or clade_478i)	*Bacteroides caccae*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Ruminococcus obeum*
N732	(clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_478 or clade_478i)	*Coprococcus comes*, *Faecalibacterium prausnitzii*, *Ruminococcus obeum*, *Ruminococcus torques*
N736	(clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_478 or clade_478i)	*Dorea longicatena*, *Faecalibacterium prausnitzii*, *Ruminococcus obeum*, *Ruminococcus torques*
N853	(clade_262 or clade_262i), (clade_444 or clade_444i), (clade_478 or clade_478i)	*Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Ruminococcus torques*
N861	(clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h)	*Clostridium hathewayi*, *Ruminococcus obeum*, *Ruminococcus torques*
N879	(clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_444 or clade_444i), (clade_478 or clade_478i)	*Faecalibacterium prausnitzii*, *Roseburia intestinalis*, *Ruminococcus obeum*

TABLE 10-continued

Network Ecology ID	Exemplary Network Clades	Exemplary Network OTUs
N975	clade_170, (clade_262 or clade_262i), (clade_360 or clade_360c or clade_360g or clade_360h clade_360i)	*Bacteroides caccae*, *Coprococcus comes*, *Dorea longicatena*
N999	(clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_478 or clade_478i)	*Dorea formicigenerans*, *Faecalibacterium prausnitzii*, *Ruminococcus obeum*
N1095	(clade_444 or clade_444i), (clade_522 or clade_522i)	*Eubacterium eligens*, *Eubacterium rectale*
N1947	(clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_38 or clade_38e or clade_38i), (clade_444 or clade_444i), (clade_478 or clade_478i), (clade_553 or clade_553i), (clade_92 or clade_92e or clade_92i)	*Bacteroides* sp. 3_1_23, *Collinsella aerofaciens*, *Dorea longicatena*, *Escherichia coli*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Roseburia intestinalis*, *Ruminococcus obeum*, *Ruminococcus torques*
N1948	(clade_262 or clade_262i), (clade_38 or clade_38e or clade_38i), (clade_478 or clade_478i), (clade_65 or clade_65e)	*Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, *Faecalibacterium prausnitzii*, *Ruminococcus torques*
N1949	(clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_378 or clade_378e), (clade_38 or clade_38e or clade_38i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i)	*Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, *Bacteroides vulgatus*, *Blautia producta*, *Enterococcus faecalis*, Erysipelotrichaceae bacterium 3_1_53, *Escherichia coli*
N1950	clade_253, (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_378 or clade_378e), (clade_38 or clade_38e or clade_38i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_65 or clade_65e)	*Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, *Bacteroides vulgatus*, *Blautia producta*, *Clostridium disporicum*, Erysipelotrichaceae bacterium 3_1_53
N1951	(clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i)	*Blautia producta*, *Clostridium bolteae*, *Clostridium hylemonae*, *Clostridium symbiosum*, *Coprococcus comes*, *Eubacterium rectale*, Lachnospiraceae bacterium 5_1_57FAA, *Ruminococcus gnavus*
N1952	clade_252, clade_253, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_351 or clade_351e), (clade_354 or clade_354e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), clade_494, clade_537, (clade_553 or clade_553i)	*Blautia producta*, *Clostridium bolteae*, *Clostridium butyricum*, *Clostridium disporicum*, *Clostridium hylemonae*, *Clostridium innocuum*, *Clostridium mayombei*, *Clostridium orbiscindens*, *Clostridium symbiosum*, *Clostridium tertium*, *Collinsella aerofaciens*, *Coprococcus comes*, Lachnospiraceae bacterium 5_1_57FAA, *Ruminococcus bromii*, *Ruminococcus gnavus*
N1953	clade_253, (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_354 or clade_354e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_378 or clade_378e), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), clade_466, (clade_478 or clade_478i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i)	*Bacteroides* sp. 1_1_6, *Bacteroides vulgatus*, *Clostridium disporicum*, *Clostridium mayombei*, *Clostridium symbiosum*, *Coprobacillus* sp. D7, *Coprococcus comes*, *Dorea formicigenerans*, *Enterococcus faecalis*, Erysipelotrichaceae bacterium 3_1_53, *Escherichia coli*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Odoribacter splanchnicus*, *Ruminococcus obeum*
N1954	clade_252, clade_253, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_351 or	*Blautia* sp. M25, *Clostridium bolteae*, *Clostridium butyricum*, *Clostridium disporicum*, *Clostridium hylemonae*, *Clostridium innocuum*, *Clostridium mayombei*, *Clostridium orbiscindens*, *Clostridium symbiosum*, *Clostridium tertium*, *Collinsella*

TABLE 10-continued

Network Ecology ID	Exemplary Network Clades	Exemplary Network OTUs
	clade_351e), (clade_354 or clade_354e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), clade_494, clade_537, (clade_553 or clade_553i)	*aerofaciens*, *Coprococcus comes*, Lachnospiraceae bacterium 5_1_57FAA, *Ruminococcus bromii*, *Ruminococcus gnavus*
N1955	(clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_354 or clade_354e), (clade_378 or clade_378e), (clade_38 or clade_38e or clade_38i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i)	*Bacteroides* sp. 1_1_6, *Bacteroides* sp. 2_1_22, *Bacteroides* sp. 3_1_23, *Bacteroides vulgatus*, *Blautia producta*, *Clostridium sordellii*, *Coprobacillus* sp. D7, *Enterococcus faecalis*, *Enterococcus faecium*, Erysipelotrichaceae bacterium 3_1_53, *Escherichia coli*
N1956	clade_252, clade_253, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_351 or clade_351e), (clade_354 or clade_354e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_380i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), clade_494, clade_537, (clade_553 or clade_553i)	*Blautia producta*, *Clostridium bolteae*, *Clostridium butyricum*, *Clostridium disporicum*, *Clostridium hylemonae*, *Clostridium innocuum*, *Clostridium mayombei*, *Clostridium nexile*, *Clostridium orbiscindens*, *Clostridium symbiosum*, *Clostridium tertium*, *Collinsella aerofaciens*, Lachnospiraceae bacterium 5_1_57FAA, *Ruminococcus bromii*, *Ruminococcus gnavus*
N1957	(clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_351 or clade_351e), (clade_354 or clade_354e), (clade_378 or clade_378e), (clade_38 or clade_38e or clade_38i), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i)	*Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, *Bacteroides vulgatus*, *Blautia producta*, *Clostridium innocuum*, *Clostridium sordellii*, *Coprobacillus* sp. D7, *Enterococcus faecalis*, *Escherichia coli*
N1958	(clade_351 or clade_351e), (clade_354 or clade_354e), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i)	*Clostridium innocuum*, *Clostridium sordellii*, *Coprobacillus* sp. D7
N1959	clade_253, (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_378 or clade_378e), (clade_38 or clade_38e or clade_38i), (clade_444 or clade_444i), (clade_478 or clade_478i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_497 or clade_497e or clade_497f), (clade_522 or clade_522i), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i)	*Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, *Bacteroides vulgatus*, *Blautia producta*, *Clostridium disporicum*, *Coprococcus comes*, *Dorea formicigenerans*, *Dorea longicatena*, *Enterococcus faecalis*, Erysipelotrichaceae bacterium 3_1_53, *Escherichia coli*, *Eubacterium eligens*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Ruminococcus obeum*, *Ruminococcus torques*
N1960	clade_253, (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_354 or clade_354e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), (clade_478 or clade_478i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i), (clade_497 or clade_497e or clade_497f), (clade_92 or clade_92e or clade_92i)	*Clostridium disporicum*, *Clostridium mayombei*, *Clostridium symbiosum*, *Coprobacillus* sp. D7, *Coprococcus comes*, *Dorea formicigenerans*, *Enterococcus faecalis*, Erysipelotrichaceae bacterium 3_1_53, *Escherichia coli*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Ruminococcus obeum*
N1961	(clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_351 or clade_351e), (clade_378 or clade_378e), (clade_38 or clade_38e or clade_38i), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i)	*Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, *Bacteroides vulgatus*, *Blautia producta*, *Clostridium innocuum*, *Enterococcus faecalis*, *Escherichia coli*

TABLE 10-continued

Network Ecology ID	Exemplary Network Clades	Exemplary Network OTUs
N1962	clade_252, clade_253, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_351 or clade_351e), (clade_354 or clade_354e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), clade_494, clade_537, (clade_553 or clade_553i)	*Blautia producta*, *Clostridium bolteae*, *Clostridium butyricum*, *Clostridium disporicum*, *Clostridium hylemonae*, *Clostridium innocuum*, *Clostridium mayombei*, *Clostridium orbiscindens*, *Clostridium symbiosum*, *Clostridium tertium*, *Collinsella aerofaciens*, *Coprococcus comes*, Lachnospiraceae bacterium 5_1_S7FAA, *Ruminococcus bromii*, *Ruminococcus gnavus*
N1963	clade_252, clade_253, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_351 or clade_351e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), clade_494, clade_537, (clade_553 or clade_553i)	*Blautia producta*, *Clostridium disporicum*, *Clostridium hylemonae*, *Clostridium innocuum*, *Clostridium orbiscindens*,*Clostridium symbiosum*, *Clostridium tertium*, *Collinsella aerofaciens*, *Coprococcus comes*, Lachnospiraceae bacterium 5_1_57FAA, *Ruminococcus bromii*, *Ruminococcus gnavus*
N1964	clade_170, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), clade_286, (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade 360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), clade_485, clade_500, clade_537, (clade_553or clade_553i), clade_85	*Alistipes shahii*, *Bacteroides caccae*, *Bacteroides stercoris*, *Blautia producta*, *Clostridium hathewayi*, *Clostridium symbiosum*, *Collinsella aerofaciens*, *Coprococcus comes*, *Dorea formicigenerans*, *Eubacterium rectale*, *Holdemania filiformis*, Lachnospiraceae bacterium 5_1_57FAA, *Parabacteroides merdae*, *Ruminococcus bromii*, *Ruminococcus obeum*, *Ruminococcus torques*
N1965	clade_252, clade_253, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_354 or clade_354e), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), (clade_478 or clade_478i), clade_537, (clade_553 or clade_553i)	*Blautia products*, *Clostridium bolteae*, *Clostridium butyricum*, *Clostridium disporicum*, *Clostridium mayombel*, *Clostridium symbiosum*, *Collinsella aerofaciens*, *Coprococcus comes*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, Lachnospiraceae bacterium 5_1_57FAA, *Roseburia intestinalis*, *Ruminococcus bromii*, *Ruminococcus obeum*
N1966	clade_252, clade_253, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_354 or clade_354e), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_553 or clade_553i)	*Blautia producta*, *Clostridium bolteae*, *Clostridium butyricum*, *Clostridium disporicum*, *Clostridium mayombei*, *Clostridium symbiosum*, *Collinsella aerofaciens*, *Coprococcus comes*, Lachnospiraceae bacterium 5_1_57FAA
N1967	(clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_378 or clade_378e), (clade_38 or clade_38e or clade_38i), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i)	*Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, *Bacteroides vulgatus*, *Blautia producta*, *Enterococcus faecium*, *Escherichia coli*
N1968	clade_252, clade_253, (clade_351 or clade_351e), (clade_354 or clade_354e), (clade_478 or clade_478i), clade_494, clade_537, (clade_553 or clade_553i)	*Clostridium butyricum*, *Clostridium disporicum*, *Clostridium innocuum*, *Clostridium mayombei*, *Clostridium orbiscindens*, *Clostridium tertium*, *Collinsella aerofaciens*, *Faecalibacterium prausnitzii*, *Ruminococcus bromii*
N1969	clade_252, clade_253, (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_354 or clade_354e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_479 or clade_479c or clade_479g or clade_479h)	*Blautia producta*, *Clostridium butyricum*, *Clostridium disporicum*, *Clostridium mayombei*, *Dorea formicigenerans*, Erysipelotrichaceae bacterium 3_1_53, *Ruminococcus torques*

TABLE 10-continued

Network Ecology ID	Exemplary Network Clades	Exemplary Network OTUs
N1970	(clade_351 or clade_351e), (clade_354 or clade_354e), (clade_378 or clade_378e), (clade_38 or clade_38e or clade_38i), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i), (clade_497 or clade_497e or clade_497f), (cladc_65 or clade_65e), (clade_92 or clade_92e or clade_92i)	*Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, *Bacteroides vulgatus*, *Clostridium innocuum*, *Clostridium sordellii*, *Coprobacillus* sp. D7, *Enterococcus faecalis*, *Escherichia coli*
N1971	clade_252, clade_253, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_351 or clade_351e), (clade_354 or clade_354e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), clade_494, clade_537, (clade_553 or clade_553i)	*Clostridium bolteae*, *Clostridium butyricum*, *Clostridium disporicum*, *Clostridium hylemonae*, *Clostridium innocuum*, *Clostridium mayombei*, *Clostridium orbiscindens*, *Clostridium symbiosum*, *Clostridium tertium*, *Collinsessa aerofaciens*, *Coprococcus comes*, Lachnospiraceae bacterium 5_1_57FAA, *Ruminococcus bromii*, *Ruminococcus gnavus*
N1972	clade_253, (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_354 or clade_354e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), (clade_478 or clade_478i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i)	*Clostridium disporicum*, *Clostridium mayombei*, *Clostridium symbiosum*, *Coprobacillus* sp. D7, *Coprococcus comes*, *Dorea formicigenerans*, Erysipelotrichaceae bacterium 3_1_53, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Ruminococcus obeum*
N1973	clade_252, clade_253, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_351 or clade_351e), (clade_354 or clade_354e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), clade_494, clade_537	*Blautia producta*, *Clostridium bolteae*, *Clostridium butyricum*, *Clostridium disporicum*, *Clostridium hylemonae*, *Clostridium innocuum*, *Clostridium mayombei*, *Clostridium orbiscindens*, *Clostridium symbiosum*, *Clostridium tertium*, *Coprococcus comes*, Lachnospiraceae bacterium 5_1_57FAA, *Ruminococcus bromii*, *Ruminococcus gnavus*
N1974	(clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309f), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_444 or clade_444i), clade_466, (clade_478 or clade_478i)	*Coprococcus comes*, *Dorea formicigenerans*, *Dorea longicatena*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Odoribacter splanchnicus*, *Roseburia intestinalis*, *Ruminococcus obeum*, *Ruminococcus torques*
N1975	(clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_351 or clade_351e), (clade_354 or clade_354e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), (clade_478 or clade_478i), clade_494, clade_537, (clade_553 or clade_553i)	*Blautia producta*, *Clostridium bolteae*, *Clostridium hylemonae*, *Clostridium innocuum*, *Clostridium mayombei*, *Clostridium orbiscindens*, *Clostridium symbiosum*, *Collinsella aerofaciens*, *Coprococcus comes*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, Lachnospiraceae bacterium 5_1_57FAA, *Ruminococcus bromii*, *Ruminococcus gnavus*
N1976	(clade_351 or clade_351e), (clade_378 or clade_378e), (clade_38 or clade_38e or clade_38i), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e)	*Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, *Bacteroides vulgatus*, *Clostridium innocuum*, *Coprobacillus* sp. D7, *Enterococcus faecalis*
N1977	clade_252, clade_253, (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_354 or clade_354e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_479 or clade_479c or clade_479g or clade_479h)	*Blautia producta*, *Clostridium butyricum*, *Clostridium disporicum*, *Clostridium mayombei*, *Dorea formicigenerans*, Erysipelotrichaceae bacterium 3_1_53, *Eubacterium tenue*, *Ruminococcus torques*
N1978	clade_253, (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_354 or clade_354e), (clade_360 or	*Bacteroides* sp. 1_1_6, *Bacteroides vulgatus*, *Clostridium disporicum*, *Clostridium mayombei*, *Clostridium symbiosum*, *Coprobacillus* sp. D7, *Coprococcus comes*, *Dorea formicigenerans*,

TABLE 10-continued

Network Ecology ID	Exemplary Network Clades	Exemplary Network OTUs
	clade_360c or clade_360g or clade_360h or clade_360i), (clade_378 or clade_378e), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), clade_466, (clade_478 or clade_478i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i)	Erysipelotrichaceae bacterium 3_1_53, *Escherichia coli*, *Eubacterium rectale*, *Faecalibacterium prausnttzii*, *Odoribacter splanchnicus*, *Ruminococcus obeum*
N1979	(clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_444 or clade_444i), (clade_478 or clade_478i), (clade_65 or clade_65e)	*Bacteroides* sp. 1_1_6, *Coprococcus comes*, *Dorea formicigenerans*, *Dorea longicatena*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Ruminococcus obeum*, *Ruminococcus torques*
N1980	(clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_351 or clade_351e), (clade_354 or clade_354e), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i), (clade_497 or clade_497e or clade_497f), (clade_92 or clade_92e or clade_92i)	*Blautia producta*, *Clostridium innocuum*, *Clostridium sordellii*, *Coprobacillus* sp. D7, *Enterococcus faecalis*, *Escherichia coli*
N1981	(clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_38 or clade_38e or clade_38i), (clade_444 or clade_444i), (clade_478 or clade_478i), (clade_522 or clade_522i)	*Bacteroides* sp. 3_1_23, *Dorea longicatena*, *Eubacterium eligens*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Ruminococcus obeum*, *Ruminococcus torques*
N1982	clade_252, clade_253, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_354 or clade_354e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_553 or clade_553i)	*Blautia producta*, *Clostridium bolteae*, *Clostridium butyricum*, *Clostridium disporicum*, *Clostridium mayombei*, *Clostridium symbiosum*, *Collinsella aerofaciens*, *Coprococcus comes*, *Dorea formicigenerans*, Erysipelotrichaceae bacterium 3_1_53, *Eubacterium rectale*, Lachnospiraceae bacterium 5_1_57FAA, *Ruminococcus obeum*, *Ruminococcus torques*
N1983	clade_252, clade_253, (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_354 or clade_354e), (clade_378 or clade_378e), (clade_38 or clade_38e or clade_38i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i)	*Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, *Bacteroides vulgatus*, *Blautia producta*, *Clostridium butyricum*, *Clostridium disporicum*, *Clostridium mayombei*, *Enterococcus faecium*, Erysipelotrichaceae bacterium 3_1_53, *Escherichia coli*
N1984	clade_253, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_351 or clade_351e), (clade_354 or clade_354e), (clade_403 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), clade_494, (clade_553 or clade_553i)	*Blautia producta*, *Clostridium disporicum*, *Clostridium innocuum*, *Clostridium mayombei*, *Clostridium orbiscindens*, *Clostridium symbiosum*, *Collinsella aerofaciens*, *Eubacterium rectale*, Lachnospiraceae bacterium 5_1_57FAA
N1985	clade_252, clade_253, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_351 or clade_351e), (clade_354 or clade_354e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), clade_494, clade_537, (clade_553 or clade_553i)	*Blautia glucerasei*, *Clostridium bolteae*, *Clostridium butyricum*, *Clostridium disporicum*, *Clostridium hylemonae*, *Clostridium innocuum*, *Clostridium mayombei*, *Clostridium orbiscindens*, *Clostridium symbiosum*, *Clostridium tertiurm*, *Collinsella aerofaciens*, *Coprococcus comes*, Lachnospiraceae bacterium 5_1_57FAA, *Ruminococcus bromii*, *Ruminococcus gnavus*
N1986	clade_253, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or	*Blautia producta*, *Clostridium disporicum*, *Clostridium symbiosum*, *Collinsella aerofaciens*,

TABLE 10-continued

Network Ecology ID	Exemplary Network Clades	Exemplary Network OTUs
	clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_553 or clade_553i)	*Coprococcus comes*, Lachnospiraceae bacterium 5_1_57FAA
N1987	(clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_354 or clade_354e), (clade_92 or clade_92e or clade_92i)	*Blautia producta*, *Clostridium sordellii*, *Escherichia coli*
N1988	clade_252, clade_253, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_354 or clade_354e), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), (clade_478 or clade_478i), clade_485, clade_500, (clade_553 or clade_553i)	*Alistipes shahii*, *Blautia producta*, *Clostridium bolteae*, *Clostridium butyricum*, *Clostridium disporicum*, *Clostridium mayombei*, *Clostridium symbiosum*, *Collinsella aerofaciens*, *Coprococcus comes*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Holdemania filiformis*, Lachnospiraceae bacterium 5_1_57FAA, *Roseburia intestinalis*, *Ruminococcus obeum*, *Ruminococcus torques*
N1989	clade_252, clade_253, (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_354 or clade_354e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_378 or clade_378e), (clade_38 or clade_38e or clade_38i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i)	*Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, *Bacteroides vulgatus*, *Blautia producta*, *Clostridium butyricum*, *Clostridium disporicum*, *Clostridium mayombei*, *Dorea formicigenerans*, *Enterococcus faecium*, Erysipelotrichaceae bacterium 3_1_53, *Escherichia coli*, *Eubacterium tenue*, *Ruminococcus torques*
N1990	clade_252, clade_253, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_351 or clade_351e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), (clade_478 or clade_478i), clade_494, clade_537, (clade_553 or clade_553i)	*Blautia producta*, *Clostridium disporicum*, *Clostridium hylemonae*, *Clostridium innocuum*, *Clostridium orbiscindens*, *Clostridium symbiosum*, *Clostridium tertium*, *Collinsella aerofaciens*, *Coprococcus comes*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, Lachnospiraceae bacterium 5_1_57FAA, *Ruminococcus bromii*, *Ruminococcus gnavus*
N1991	clade_252, clade_253, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_351 or clade_351e), (clade_354 or clade_354e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), (clade_553 or clade_553i)	*Blautia producta*, *Clostridium bolteae*, *Clostridium butyricum*, *Clostrtdium disporicum*, *Clostridium hylemonae*, *Clostridium innocuum*, *Clostridium mayombei*, *Clostridium symbiosum*, *Clostridium tertium*, *Collinsella aerofaciens*, *Coprococcus comes*, *Eubacterium rectale*, Lachnospiraceae bacterium 5_1_57FAA, *Ruminococcus gnavus*
N1992	clade_252, clade_253, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_351 or clade_351e), (clade_354 or clade_354e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), clade_494, clade_537, (clade_553 or clade_553i)	*Blautia producta*, *Clostridium bolteae*, *Clostridium butyricum*, *Clostridium disporicum*, *Clostridium hylemonae*, *Clostridium innocuum*, *Clostridium mayombei*, *Clostridium orbiscindens*, *Clostridium symbiosum*, *Clostridium tertium*, *Collinsella aerofaciens*, Lachnospiraceae bacterium 1_4_56FAA, Lachnospiraceae bacterium 5_1_57FAA, *Ruminococcus bromii*, *Ruminococcus gnavus*
N1993	clade_110, clade_170, (clade_378 or clade_378e), (clade_38 or clade_38e or clade_38i), (clade_65 or clade_65e), clade_85	*Bacteroides caccae*, *Bacteroides eggerthii*, *Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, *Bacteroides stercoris*, *Bacteroides uniformis*, *Bacteroides vulgatus*
N1994	clade_253, (clade_378 or clade_378e), (clade_38 or clade_38e or clade_38i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i)	*Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, *Bacteroides vulgatus*, *Clostridium disporicum*, *Enterococcus faecium*, Erysipelotrichaceae bacterium 3_1_53, *Escherichia coli*
N1995	clade_253, (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or	*Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, *Bacteroides vulgatus*, *Blautia producta*, *Clostridium*

TABLE 10-continued

Network Ecology ID	Exemplary Network Clades	Exemplary Network OTUs
	clade_309i), (clade_378 or clade_378e), (clade_38 or clade_38e or clade_38i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i)	*disporicum*, *Enterococcus faecium*, Erysipelotrichaceae bacterium 3_1_53, *Escherichia coli*
N1996	clade_253, (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_479 or clade_479c or clade_479g or clade_479h)	*Blautia producta*, *Clostridium disporicum*, Erysipelotrichaceae bacterium 3_1_53
N1997	clade_253, (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_497 or clade_497e or clade_497f), (clade_92 or clade_92e or clade_92i)	*Blautia producta*, Clostridium disporicum, *Enterococcus faecium*, Erysipelotrichaceae bacterium 3_1_53, *Escherichia coli*
N1998	(clade_378 or clade_378e), (clade_38 or clade_38e or clade_38i), (clade_65 or clade_65e)	*Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, *Bacteroides vulgatus*
N1999	clade_252, clade_253, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_354 or clade_354e), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), (clade_478 or clade_478i), (clade_553 or clade_553i)	*Blautia producta*, *Clostridium bolteae*, *Clostridium butyricum*, *Clostridium disporicum*, *Clostridium mayombei*, *Clostridium symbiosum*, *Collinsella aerofaciens*, *Coprococcus comes*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, Lachnospiraceae bacterium 5_1_57FAA
N2000	clade_252, clade_253, (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_354 or clade_354e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_479 or clade_479c or clade_479g or clade_479h)	*Blautia producta*, *Clostridium butyricum*, *Clostridium disporicum*, *Clostridium mayombei*, *Dorea formicigenerans*, Erysipelotrichaceae bacterium 3_1_53, *Eubacterium tenue*, *Ruminococcus torques*
N2001	clade_252, clade_253, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_351 or clade_351e), (clade_354 or clade_354e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), clade_494, clade_537, (clade_553 or clade_553i)	*Blautia producta*, *Clostridium bolteae*, *Clostridium disporicum*, *Clostridium hylemonae*, *Clostridium innocuum*, *Clostridium mayombei*, *Clostridium orbiscindens*, *Clostridium symbiosum*, *Clostridium tertium*, *Collinsella aerofaciens*, *Coprococcus comes*, Lachnospiraceae bacterium 5_1_57FAA, *Ruminococcus bromii*, *Ruminococcus gnavus*
N2002	clade_252, clade_253, (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_354 or clade_354e), (clade_479 or clade_479c or clade_479g or clade_479h)	*Blautia producta*, *Clostridium butyricum*, *Clostridium disporicum*, *Clostridium mayombei*, Erysipelotrichaceae bacterium 3_1_53
N2003	clade_252, clade_253, (clade_260 or clade_260c or clade_260g or clade_260h), (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_354 or clade_354e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_38 or clade_38e or clade_38i), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i), clade_537, (clade_553 or clade_553i)	*Blautia producta*, *Clostridium bolteae*, *Clostridium butyricum*, *Clostridium disporicum*, *Clostridium hylemonae*, *Clostridium mayombei*, *Clostridium sordellii*, *Clostridium symbiosum*, *Clostridium tertium*, *Collinsella aerofaciens*, *Coprobacillus* sp. D7, *Coprococcus comes*, *Eubacterium* sp. WAL 14571, Lachnospiraceae bacterium 5_1_57FAA, *Rumtinococcus bromii*, *Ruminococcus gnavus*
N2004	(clade_351 or clade_351e), (clade_378 or clade_378e), (clade_38 or clade_38e or clade_38i), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i)	*Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, *Bacteroides vulgatus*, *Clostridium innocuum*, *Enterococcus faecalis*, *Escherichia coli*

TABLE 10-continued

Network Ecology ID	Exemplary Network Clades	Exemplary Network OTUs
N2005	(clade_309 or clade_309c or dade_309e or clade_309g or clade_309h or clade_309i) (clade_378 or clade_378e), (clade_38 or clade_38e or clade_38i), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e), (clade_92 or clade_92e or clade_92i)	*Bacteroides* sp. 1_1_6, *Bacteroides* sp. 3_1_23, *Bacteroides vulgatus*, *Blautia producta*, *Enterococcus faecalis*, *Escherichia coli*
N2006	clade_170, (clade_262 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_444 or clade_444i), (clade_478 or clade_478i), (clade_65 or dade_65e)	*Bacteroides caccae*, *Bacteroides* sp. 1_1_6, *Coprococcus comes*, *Dorea formicigenerans*, *Dorea longicatena*, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Ruminococcus obeum*, *Ruminococcus torques*
N2007	clade_253, (clade_202 or clade_262i), (clade_309 or clade_309c or clade_309e or clade_309g or clade_309h or clade_309i), (clade_354 or clade_354e), (clade_360 or clade_360c or clade_360g or clade_360h or clade_360i), (clade_378 or clade_378e), (clade_408 or clade_408b or clade_408d or clade_408f or clade_408g or clade_408h), (clade_444 or clade_444i), clade_466, (clade_478 or clade_478i), (clade_479 or clade_479c or clade_479g or clade_479h), (clade_481 or clade_481a or clade_481b or clade_481e or clade_481g or clade_481h or clade_481i), (clade_497 or clade_497e or clade_497f), (clade_65 or clade_65e)	*Bacteroides* sp. 1_1_6, *Bacteroides vulgatus*, *Clostridium disporicurn*, *Clostridium mayombei*, *Clostridium symbiosum*, *Coprobacillus* sp. D7, *Coprococcus comes*, *Dorea formicigenerans*, *Enterococcus faecalis*, Erysipelotrichaceae bacterium 3_1_53, *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Odoribacter splanchnicus*, *Ruminococcus obeum*

TABLE 11

OTU	Clade	Genus	Family	Order	Spore Former	Percent of Dose Ecologies Occurs	Percent of Engrafted Ecologies Occurs	Percent of Augemented Ecologies Occurs	Keystone OTU	CES
Blautia_luti	clade_309	Blautia	Lachnospiraceae	Clostridiales	yes	100	100	0	0	4.0
Blautia_schinkii	clade_309	Blautia	Lachnospiraceae	Clostridiales	yes	100	93	7	0	4.0
Blautia_sp_M25	clade_309	Blautia	Lachnospiraceae	Clostridiales	yes	100	93	7	1	5.0
Subdoligranulum_variabile	clade_478	Subdoligranulum	Ruminococcaceae	Clostridiales	yes	100	93	0	1	5.0
Eubacterium_rectale	clade_444	Eubacterium	Eubacteriaceae	Clostridiales	yes	100	87	13	1	5.0
Lachnospiraceae_bacterium_2_1_58FAA	clade_360	unassigned	Lachnospiraceae	Clostridiales	yes	100	80	7	0	4.0
Clostridium_leptum	clade_537	Clostridium	Clostridiaceae	Clostridiales	yes	100	73	13	1	4.0
Faecalibacterium_prausnitzii	clade_478	Faecalibacterium	Ruminococcaceae	Clostridiales	yes	100	73	0	1	4.0
Ruminococcus_bromii	clade_537	Ruminococcus	Ruminococcaceae	Clostridiales	yes	100	67	7	1	4.0
Clostridium_citroniae	clade_408	Clostridium	Clostridiaceae	Clostridiales	yes	100	53	27	0	3.0
Christensenella_minuta	clade_558	Christensenella	Christensenellaceae	Clostridiales	yes	100	0	27	0	2.0
Ruminococcus_torques	clade_262	Blautia	Lachnospiraceae	Clostridiales	yes	93	87	7	1	5.0
Dorea_longicatena	clade_360	Dorea	Lachnospiraceae	Clostridiales	yes	93	80	20	1	5.0
Eubacterium_hadrum	clade_408	Anaerostipes	Lachnospiraceae	Clostridiales	yes	93	80	7	1	5.0
Blautia_hansenii	clade_309	Blautia	Lachnospiraceae	Clostridiales	yes	93	73	13	0	3.0
Clostridium_ramosum	clade_481	unassigned	Erysipelotrichaceae	Erysipelotrichales	yes	93	73	13	0	3.0
Ruminococcus_lactaris	clade_262	Ruminococcus	Ruminococcaceae	Clostridiales	yes	93	67	33	1	4.0
Clostridiales_sp_SS3_4	clade_246	Clostridiales	unclassified	Clostridiales	yes	86	73	20	0	3.0
Dorea_formicigenerans	clade_360	Dorea	Lachnospiraceae	Clostridiales	yes	86	53	7	1	4.0
Coprococcus_comes	clade_262	Coprococcus	Lachnospiraceae	Clostridiales	yes	86	47	0	1	4.0
Lachnospiraceae_bacterium_A4	clade_408	unassigned	Lachnospiraceae	Clostridiales	yes	86	40	20	0	2.4
Eubacterium_hallii	clade_396	Eubacterium	Eubacteriaceae	Clostridiales	yes	86	40	7	1	3.4
Eubacterium_brachy	clade_533	Eubacterium	Eubacteriaceae	Clostridiales	yes	86	13	0	0	2.4
Ruminococcus_callidus	clade_406	Ruminococcus	Ruminococcaceae	Clostridiales	yes	79	67	13	0	2.3
Clostridium_bartlettii	clade_354	unassigned	Peptostreptococcaceae	Clostridiales	yes	79	67	7	0	2.3
Clostridium_sporosphaeroides	clade_537	Clostridium	Clostridiaceae	Clostridiales	yes	79	60	20	0	2.3
Clostridium_bifermentans	clade_354	unassigned	Peptostreptococcaceae	Clostridiales	yes	79	53	13	0	2.3
Turicibacter_sanguinis	clade_555	Turicibacter	Erysipelotrichaceae	Erysipelotrichales	yes	79	40	13	0	1.7
Ruminococcus_albus	clade_516	Ruminococcus	Ruminococcaceae	Clostridiales	yes	71	60	27	0	2.3
Eubacterium_ramulus	clade_482	Eubacterium	Eubacteriaceae	Clostridiales	yes	71	47	33	0	2.3
Eubacterium_desmolans	clade_572	Eubacterium	Eubacteriaceae	Clostridiales	yes	71	40	27	0	1.7
Coprococcus_catus	clade_393	Coprococcus	Lachnospiraceae	Clostridiales	yes	71	33	7	1	2.7
Clostridium_oroticum	clade_96	Clostridium	Clostridiaceae	Clostridiales	yes	71	27	7	0	1.7
Blautia_glucerasea	clade_309	Blautia	Lachnospiraceae	Clostridiales	yes	64	33	33	0	1.7
Lachnospiraceae_bacterium_3_1_57FAA	clade_408	unassigned	Lachnospiraceae	Clostridiales	yes	64	33	13	1	2.7
Clostridium_viride	clade_540	Clostridium	Clostridiaceae	Clostridiales	yes	64	27	13	0	1.7
Ruminococcus_obeum	clade_309	Blautia	Lachnospiraceae	Clostridiales	yes	64	27	7	1	2.7
Eubacterium_ruminantium	clade_543	Eubacterium	Eubacteriaceae	Clostridiales	yes	57	20	27	0	1.7
Clostridium_thermocellum	clade_495	Clostridium	Clostridiaceae	Clostridiales	yes	50	47	20	0	2.3
Oscillibacter_valericigenes	clade_540	Oscillibacter	Oscillospiraceae	Clostridiales	yes	50	40	60	1	2.7
Eubacterium_coprostanoligenes	clade_537	Eubacterium	Eubacteriaceae	Clostridiales	yes	50	27	13	1	2.7
Clostridium_disporicum	clade_253	Clostridium	Clostridiaceae	Clostridiales	yes	50	20	27	0	1.7
Clostridium_mayombei	clade_354	Clostridium	Clostridiaceae	Clostridiales	yes	50	20	0	0	1.7
Roseburia_faecalis	clade_444	Roseburia	Lachnospiraceae	Clostridiales	yes	43	27	60	0	1.7
Lachnospiraceae_bacterium_1_4_56FAA	clade_262	unassigned	Lachnospiraceae	Clostridiales	yes	43	13	20	1	2.7
Clostridium_spiroforme	clade_481	unassigned	Erysipelotrichaceae	Erysipelotrichales	yes	36	33	33	0	1.7

TABLE 11-continued

OTU	Clade	Genus	Family	Order	Spore Former	Percent of Dose Ecologies Occurs	Percent of Engrafted Ecologies Occurs	Percent of Augemented Ecologies Occurs	Keystone OTU	CES
Eubacterium_siraeum	clade_538	*Eubacterium*	Eubacteriaceae	Clostridiales	yes	36	27	60	1	2.7
Lachnospira_pectinoschiza	clade_522	*Lachnospira*	Lachnospiraceae	Clostridiales	yes	36	27	53	0	1.7
Papillibacter_cinnamivorans	clade_572	*Papillibacter*	Ruminococcaceae	Clostridiales	yes	36	27	53	0	1.7
Clostridium_tyrobutyricum	clade_430	*Clostridium*	Clostridiaceae	Clostridiales	yes	36	27	13	0	1.7
Roseburia_inulinivorans	clade_444	*Roseburia*	Lachnospiraceae	Clostridiales	yes	36	20	60	1	2.7
Ethanoligenens_harbinense	clade_439	*Ethanoligenens*	Ruminococcaceae	Clostridiales	yes	36	20	27	0	1.7
Eggerthella_lenta	clade_566	*Eggerthella*	Coriobacteriaceae	Coriobacteriales	yes	36	13	40	0	1.7
Clostridium_orbiscindens	clade_494	unassigned	unassigned	Clostridiales	yes	29	20	60	0	1.1
Eubacterium_ventriosum	clade_519	*Eubacterium*	Eubacteriaceae	Clostridiales	yes	29	20	40	0	1.1
Clostridium_paraputrificum	clade_223	*Clostridium*	Clostridiaceae	Clostridiales	yes	29	13	33	0	1.1
*Clostridium*_sp_YIT_12069	clade_537	*Clostridium*	Clostridiaceae	Clostridiales	yes	29	7	53	0	1.1
Eubacterium_barkeri	clade_512	*Eubacterium*	Eubacteriaceae	Clostridiales	yes	29	0	13	0	0.7
Eubacterium_biforme	clade_385	unassigned	Erysipelotrichaceae	Erysipelotrichales	yes	29	0	13	0	0.7
Alkaliphilus_oremlandii	clade_554	*Alkaliphilus*	Clostridiaceae	Clostridiales	yes	29	0	7	0	0.7
Lachnospiraceae_*bacterium*_5_1_57FAA	clade_260	unassigned	Lachnospiraceae	Clostridiales	yes	21	20	60	0	1.1
Eubacterium_eligens	clade_522	*Eubacterium*	Eubacteriaceae	Clostridiales	yes	21	20	33	0	1.1
*Bacillus*_sp_9_3AIA	clade_527	*Bacillus*	Bacillaceae	Bacillales	yes	21	13	73	0	1.1
*Eubacterium*_sp_WAL_14571	clade_384	*Eubacterium*	Eubacteriaceae	Clostridiales	yes	21	7	60	0	1.1
Anaerosporobacter_mobilis	clade_396	*Anaerosporobacter*	Clostridiaceae	Clostridiales	yes	21	7	47	0	1.1
Coprococcus_eutactus	clade_543	*Coprococcus*	Lachnospiraceae	Clostridiales	yes	21	7	20	0	1.1
*Eubacterium*_sp_oral_clone_JH012	clade_476	*Eubacterium*	Eubacteriaceae	Clostridiales	yes	21	7	13	0	1.1
Lachnospira_multipara	clade_522	*Lachnospira*	Lachnospiraceae	Clostridiales	yes	21	7	13	0	1.1
Clostridium_carnis	clade_253	*Clostridium*	Clostridiaceae	Clostridiales	yes	21	7	0	0	1.1
Clostridium_colinum	clade_576	*Clostridium*	Clostridiaceae	Clostridiales	yes	21	7	0	0	1.1
Clostridium_hylemonae	clade_260	*Clostridium*	Clostridiaceae	Clostridiales	yes	21	0	40	0	0.7
Gloeobacter_violaceus	clade_596	*Gloeobacter*	unassigned	Gloeobacterales	yes	21	0	7	0	0.7
Clostridium_algidicarnis	clade_430	*Clostridium*	Clostridiaceae	Clostridiales	yes	14	20	67	0	1.1
Holdemania_filiformis	clade_485	*Holdemania*	Erysipelotrichaceae	Erysipelotrichales	yes	14	13	73	1	2.1
Clostridium_aldenense	clade_408	*Clostridium*	Clostridiaceae	Clostridiales	yes	14	13	67	0	1.1
Sporobacter_termitidis	clade_572	*Sporobacter*	Ruminococcaceae	Clostridiales	yes	14	13	67	1	2.1
*Ruminococcus*_sp_ID8	clade_360	*Ruminococcus*	Ruminococcaceae	Clostridiales	yes	14	13	47	0	1.1
Lachnospiraceae_*bacterium*_4_1_37FAA	clade_360	unassigned	Lachnospiraceae	Clostridiales	yes	14	0	60	0	0.7
*Ruminococcus*_sp_18P13	clade_406	*Ruminococcus*	Ruminococcaceae	Clostridiales	yes	14	0	7	0	0.7
Blautia_hydrogenotrophica	clade_368	*Blautia*	Lachnospiraceae	Clostridiales	yes	14	0	0	0	0.7
Anaerotruncus_colihominis	clade_516	*Anaerotruncus*	Ruminococcaceae	Clostridiales	yes	7	7	93	0	1.1
Clostridium_symbiosum	clade_408	*Clostridium*	Clostridiaceae	Clostridiales	yes	7	7	87	0	1.1
Clostridium_lactatifermentans	clade_576	*Clostridium*	Clostridiaceae	Clostridiales	yes	7	7	80	1	2.1
Lactobacillus_rogosae	clade_522	*Lactobacillus*	Lactobacillaceae	Lactobacillases	yes	7	7	80	0	1.1
*Clostridium*_sp_HGF2	clade_351	*Clostridium*	Clostridiaceae	Clostridiales	yes	7	7	47	0	1.1
*Clostridium*_sp_SY8519	clade_482	*Clostridium*	Clostridiaceae	Clostridiales	yes	7	7	47	0	1.1
Desulfotomaculum_nigrificans	clade_560	*Desulfotomaculum*	Peptococcaceae	Clostridiales	yes	7	7	27	0	1.1
Eubacterium_cylindroides	clade_385	unassigned	Erysipelotrichaceae	Erysipelotrichales	yes	7	7	7	0	1.1
*Ruminococcus*_sp_K_1	clade_309	*Ruminococcus*	Ruminococcaceae	Clostridiales	yes	7	7	0	0	1.1
*Lachnospiraceae_bacterium*_oral_taxon_F15	clade_393	unassigned	Lachnospiraceae	Clostridiales	yes	7	0	53	0	0.7
Clostridium_nexile	clade_262	*Clostridium*	Clostridiaceae	Clostridiales	yes	7	0	40	1	1.7
Acetanaerobacterium_elongatum	clade_439	*Acetanaerobacterium*	Ruminococcaceae	Clostridiales	yes	7	0	7	0	0.7
Butyricicoccus_pullicaecorum	clade_572	*Butyricicoccus*	Clostridiaceae	Clostridiales	yes	7	0	0	1	1.7
Clostridium_butyricum	clade_252	*Clostridium*	Clostridiaceae	Clostridiales	yes	7	0	0	0	0.7

TABLE 11-continued

OTU	Clade	Genus	Family	Order	Spore Former	Percent of Dose Ecologies Occurs	Percent of Engrafted Ecologies Occurs	Percent of Augemented Ecologies Occurs	Keystone OTU	CES
Solobacterium_*moorei*	clade_388	*Solobacterium*	Erysipelotrichaceae	Erysipelotrichales	yes	7	0	0	0	0.7
Bacteroides_uniformis	clade_110	*Bacteroides*	Bacteroidaceae	Bacteroidales	no		87	13	0	4.0
*Alloscardovia*_sp_OB7196	clade_475	*Alloscardovia*	Bifidobacteriaceae	Bifidobacteriales	no		53	33	0	2.3
*Clostridiales_bacterium*_oral_clone_P4PA	clade_558	unassigned	unassigned	Clostridiales	no		47	27	0	2.3
Enterococcus_faecium	clade_497	*Enterococcus*	Enterococcaceae	Lactobacillales	no		47	7	0	3.0
*Clostridiales_bacterium*_oral_taxon_F32	clade_584	unassigned	unassigned	Clostridiales	no		40	0	0	1.7
Bifidobacterium_breve	clade_172	*Bifidobacterium*	Bifidobacteriaceae	Bifidobacteriales	no		33	7	0	2.4
Bifidobacterium_longum	clade_172	*Bifidobacterium*	Bifidobacteriaceae	Bifidobacteriales	no		33	0	1	2.7
Bacteroides_oleiciplenus	clade_85	*Bacteroides*	Bacteroidaceae	Bacteroidales	no		27	47	0	1.7
Dialister_invisus	clade_506	*Dialister*	Veillonellaceae	Selenomonadales	no		27	7	0	1.7
Anaerobaculum_hydrogeniformans	clade_591	*Anaerobaculum*	Synergistaceae	Synergistales	no		20	47	0	1.7
*Streptococcus*_sp_oral_taxon_G63	clade_98	*Streptococcus*	Streptococcaceae	Lactobacillales	no		20	27	0	1.7
Streptococcus_thermophilus	clade_98	*Streptococcus*	Streptococcaceae	Lactobacillales	no		20	13	0	2.4
Dialister_micraerophilus	clade_506	*Dialister*	Veillonellaceae	Selenomonadales	no		20	13	0	1.7
Bifidobacterium_animalis	clade_172	*Bifidobacterium*	Bifidobacteriaceae	Bifidobacteriales	no		20	7	0	1.7
Lactobacillus_iners	clade_398	*Lactobacillus*	Lactobacillaceae	Lactobacillales	no		20	0	0	1.7
Butyrivibrio_fibrisolvens	clade_444	*Butyrivibrio*	Lachnospiraceae	Clostridiales	no		13	67	0	1.1
*Streptococcus*_sp_ACS2	clade_98	*Streptococcus*	Streptococcaceae	Lactobacillales	no		13	33	0	1.7
Lactococcus_lactis	clade_401	*Lactococcus*	Streptococcaceae	Lactobacillales	no		13	13	0	1.7
Lactobacillus_delbrueckii	clade_72	*Lactobacillus*	Lactobacillaceae	Lactobacillales	no		13	13	0	1.7
Cytophaga_xylanolytica	clade_561	*Cytophaga*	Cytophagaceae	Cytophagales	no		7	53	0	1.1
Streptococcus_gallolyticus	clade_98	*Streptococcus*	Streptococcaceae	Lactobacillales	no		7	33	0	1.1
Marvinbryantia_formatexigens	clade_309	unassigned	Lachnospiraceae	Clostridiales	no		7	33	0	1.1
Akkermansia_muciniphila	clade_583	*Akkermansia*	Verrucomicrobiaceae	Verrucomicrobiales	no		7	20	1	2.7
Bacteroides_dorei	clade_378	*Bacteroides*	Bacteroidaceae	Bacteroidales	no		7	20	1	2.7
*Megasphaera*_genomosp_type_1_28L	clade_506	*Megasphaera*	Veillonellaceae	Selenomonadales	no		7	20	0	1.1
Lactobacillus_hominis	clade_398	*Lactobacillus*	Lactobacillaceae	Lactobacillales	no		7	13	0	1.7
Actinomyces_oricola	clade_54	*Actinomyces*	Actinomycetaceae	Actinomycetales	no		7	13	0	1.1
Streptobacillus_moniliformis	clade_532	*Streptobacillus*	Leptotrichiaceae	Fusobacteriales	no		7	13	0	1.1
*Streptococcus*_sp_oral_clone_ASCB06	clade_98	*Streptococcus*	Streptococcaceae	Lactobacillales	no		7	7	0	1.1
*Bacteroides*_sp_D20	clade_110	*Bacteroides*	Bacteroidaceae	Bacteroidales	no		7	7	1	2.1
Collinsella_intestinalis	clade_553	*Collinsella*	Coriobacteriaceae	Coriobacteriales	no		7	0	0	1.7
Methanosphaera_stadtmanae	clade_595	*Methanosphaera*	Methanobacteriaceae	Methanobacteriales	no		7	0	0	1.7
Streptococcus_vestibularis	clade_98	*Streptococcus*	Streptococcaceae	Lactobacillales	no		7	0	0	1.1
*Clostridiaceae_bacterium*_END_2	clade_368	unassigned	Clostridiaceae	Clostridiales	no		0	53	0	0.7
Parabacteroides_distasonis	clade_335	*Parabacteroides*	Porphyromonadaceae	Bacteroidales	no		0	27	0	1.3
Veillonella_dispar	clade_358	*Veillonella*	Veillonellaceae	Selenomonadales	no		0	27	0	0.7
Bacteroides_caccae	clade_170	*Bacteroides*	Bacteroidaceae	Bacteroidales	no		0	20	1	1.7
*Veillonella*_sp_3_1_44	clade_358	*Veillonella*	Veillonellaceae	Selenomonadales	no		0	20	0	0.7
Megasphaera_micronuciformis	clade_493	*Megasphaera*	Veillonellaceae	Selenomonadales	no		0	20	0	0.7
Oxalobacter_formigenes	clade_357	*Oxalobacter*	Oxalobacteraceae	Burkholderiales	no		0	20	0	0.7
Streptococcus_parasanguinis	clade_98	*Streptococcus*	Streptococcaceae	Lactobacillales	no		0	13	1	2.3
Bacteroides_fragilis	clade_65	*Bacteroides*	Bacteroidaceae	Bacteroidales	no		0	13	0	0.7
*Bacteroides*_sp_4_1_36	clade_110	*Bacteroides*	Bacteroidaceae	Bacteroidales	no		0	13	0	0.7
*Lactobacillus*_sp_BT6	clade_373	*Lactobacillus*	Lactobacillaceae	Lactobacillales	no		0	13	0	0.7
*Bacteroides*_sp_1_1_14	clade_65	*Bacteroides*	Bacteroidaceae	Bacteroidales	no		0	13	0	0.7
Escherichia_hermannii	clade_92	*Escherichia*	Enterobacteriaceae	Enterobacteriales	no		0	13	0	0.7
*Escherichia*_sp_B4	clade_92	*Escherichia*	Enterobacteriaceae	Enterobacteriales	no		0	13	0	0.7

TABLE 11-continued

OTU	Clade	Genus	Family	Order	Spore Former	Percent of Dose Ecologies Occurs	Percent of Engrafted Ecologies Occurs	Percent of Augemented Ecologies Occurs	Keystone OTU	CES
Gemella_moribillorum	clade_450	*Gemella*	unassigned	Bacillales	no		0	13	0	0.7
Klebsiella_variicola	clade_92	*Klebsiella*	Enterobacteriaceae	Enterobacteriales	no		0	13	0	0.7
Phascolarctobacterium_succinatutens	clade_556	*Phascolarctobacterium*	Acidaminococcaceae	Selenomonadales	no		0	13	0	0.7
*Streptococcus*_sp_CM7	clade_60	*Streptococcus*	Streptococcaceae	Lactobacillales	no		0	13	0	0.7
Bilophila_wadsworthia	clade_521	*Bilophila*	Desulfovibrionaceae	Desulfovibrionales	no		0	7	1	2.3
*Streptococcus*_sp_oral_clone_GM006	clade_98	*Streptococcus*	Streptococcaceae	Lactobacillales	no		0	7	0	1.3
Adlercreutzia_equolifaciens	clade_566	*Adlercreutzia*	Coriobacteriaceae	Coriobacteriales	no		0	7	0	0.7
Lactobacillus_murinus	clade_449	*Lactobacillus*	Lactobacillaceae	Lactobacillales	no		0	7	0	0.7
Helicobacter_pullorum	clade_489	*Helicobacter*	Helicobacteraceae	Campylobacterales	no		0	7	0	0.7
Alistipes_finegoldii	clade_500	*Alistipes*	Rikenellaceae	Bacteroidales	no		0	7	0	0.7
Averyella_dalhousiensis	clade_92	*Averyella*	Enterobacteriaceae	Enterobacteriales	no		0	7	0	0.7
Desulfovibrio_desulfuricans	clade_445	*Desulfovibrio*	Desulfovibrionaceae	Desulfovibrionales	no		0	7	0	0.7
Plesiomonas_shigelloides	clade_92	*Plesiomonas*	Enterobacteriaceae	Enterobacteriales	no		0	7	0	0.7
Actinomyces_israelii	clade_212	*Actinomyces*	Actinomycetaceae	Actinomycetales	no		0	7	0	0.7
*Bacteroidales*_genomosp_P1	clade_529	unassigned	unassigned	Bacteroidales	no		0	7	0	0.7
Bifidobacterium_bifidum	clade_293	*Bifidobacterium*	Bifidobacteriaceae	Bifidobacteriales	no		0	7	0	0.7
Cedecea_davisae	clade_92	*Cedecea*	Enterobacteriaceae	Enterobacteriales	no		0	7	0	0.7
Gardnerella_*vaginalis*	clade_344	*Gardnerella*	Bifidobacteriaceae	Bifidobacteriales	no		0	7	0	0.7
Lactobacillus_fermentum	clade_313	*Lactobacillus*	Lactobacillaceae	Lactobacillales	no		0	7	0	0.7
Lactobacillus_reuteri	clade_313	*Lactobacillus*	Lactobacillaceae	Lactobacillales	no		0	8	0	0.7
Lactococcus_raffinolactis	clade_524	*Lactococcus*	Streptococcaceae	Lactobacillales	no		0	7	0	0.7
Pediococcus_pentosaceus	clade_372	*Pediococcus*	Lactobacillaceae	Lactobacillales	no		0	7	0	0.7
Prevotella_denticola	clade_83	*Prevotella*	Prevotellaceae	Bacteroidales	no		0	7	0	0.7
Rothia_mucilaginosa	clade_271	*Rothia*	Micrococcaceae	Actinomycetales	no		0	7	0	0.7
Sutterella_stercoricanis	clade_432	*Sutterella*	Sutterellaceae	Burkholderiales	no		0	7	0	0.7
*Eggerthelia*_sp_1_3_56FAA	clade_566	*Eggerthella*	Coriobacteriaceae	Coriobacteriales	no		0	0	0	1.3
*Coriobacteriaceae_bacterium*_JC110	clade_566	unassigned	Coriobacteriaceae	Coriobacteriales	no		0	0	0	1.3
Megamonas_funiformis	clade_542	*Megamonas*	Veillonellaceae	Selenomonadales	no		0	0	0	1.3
Gordonibacter_pamelaeae	clade_566	*Gordonibacter*	Coriobacteriaceae	Coriobacteriales	no		0	0	0	1.3
*Bifidobacterium*_sp_HM2	clade_172	*Bifidobacterium*	Bifidobacteriaceae	Bifidobacteriales	no		0	0	0	0.7
Bacteroides_stercoris	clade_85	*Bacteroides*	Bacteroidaceae	Bacteroidales	no		0	0	1	1.7
Bifidobacterium_angulatum	clade_172	*Bifidobacterium*	Bifidobacteriaceae	Bifidobacteriales	no		0	0	0	0.7
Parasutterella_excrementihominis	clade_432	*Parasutterella*	Sutterellaceae	Burkholderiales	no		0	0	0	0.7
Phascolarctobacterium_faecium	clade_556	*Phascolarctobacterium*	Acidaminococcaceae	Selenomonadales	no		0	0	0	0.7
Cryptobacterium_curtum	clade_566	*Cryptobacterium*	Coriobacteriaceae	Coriobacteriales	no		0	0	0	0.7
*Prevotella*_sp_BI_42	clade_168	*Prevotella*	Prevotellaceae	Bacteroidales	no		0	0	0	0.7
Slackia_isoflavoniconvertens	clade_566	*Slackia*	Coriobacteriaceae	Coriobacteriales	no		0	0	0	0.7
*Acidaminococcus*_sp_D21	clade_556	*Acidaminococcus*	Acidaminococcaceae	Selenomonadales	no		0	0	0	0.7
Atopobium_vaginae	clade_539	*Atopobium*	Coriobacteriaceae	Coriobacteriales	no		0	0	0	0.7
Catabacter_hongkongensis	clade_558	*Catabacter*	Catabacteriaceae	Clostridiales	no		0	0	0	0.7
Lactobacillus_ruminis	clade_449	*Lactobacillus*	Lactobacillaceae	Lactobacillales	no		0	0	0	0.7
Lactobacillus_senioris	clade_398	*Lactobacillus*	Lactobacillaceae	Lactobacillales	no		0	0	0	0.7
Morganella_morganii	clade_89	*Morganella*	Enterobacteriaceae	Enterobacteriales	no		0	0	0	0.7
Parabacteroides_merdae	clade_286	*Parabacteroides*	Porphyromonadaceae	Bacteroidales	no		0	0	1	1.7
Peptoniphilus_harei	clade_389	*Peptoniphilus*	Clostridiales Family XI	Clostridiales	no		0	0	0	0.7
Streptococcus_downei	clade_441	*Streptococcus*	Streptococcaceae	Lactobacillales	no		0	0	0	0.7

TABLE 12

OTC1 of Composition	OTC2 of Composition	OTC3 of Composition	Clade of OTU1	Clade of OTU2	Clade of OTU3 (if applicable)	*C. diff* Inhibition Score	Percent of Dose Ecologies Occurs	Percent of Post-treatment Ecologies
Dorea longicatena	*Eubacterium rectale*		clade_360	clade_444		++++	92.9	100.0
Ruminococcus torques	*Ruminococcus torques*		clade_262	clade_262		++++	92.9	93.3
Coprococcus comes	*Eubacterium rectale*		clade_262	clade_444		++++	85.7	46.7
Coprococcus comes	*Ruminococcus bromii*		clade_262	clade_537		++++	85.7	26.7
Ruminococcus torques	*Coprococcus comes*		clade_262	clade_262		++++	78.6	40.0
Ruminococcus obeum	*Ruminococcus obeum*		clade_309	clade_309		++++	64.3	33.3
Ruminococcus obeum	*Coprococcus comes*		clade_309	clade_262		++++	64.3	20.0
Ruminococcus obeum	*Ruminococcus torques*		clade_309	clade_262		++++	57.1	33.3
Clostridium disporicum	*Eubacterium rectale*		clade_253	clade_444		++++	50.0	46.7
Ciostridium mayombei	*Eubacterium rectale*	*Eubacterium rectale*	clade_354	clade_444	clade_444	++++	50.0	20.0
Ciostridium mayombei	*Faecalibacterium prausnitzii*	*Eubacterium rectale*	clade_354	clade_478	clade_444	++++	50.0	20.0
Ciostridium mayombei	*Faecalibacterium prausnitzii*	*Faecalibacterium prausnitzii*	clade_354	clade_478	clade_478	++++	50.0	20.0
Eubacterium rectale	*Clostridium mayombei*	*Blautia* sp. M25	clade_444	clade_354	clade-309	++++	50.0	20.0
Eubacterium rectale	*Clostridium mayombei*	*Clostridium mayombei*	clade_444	clade_354	clade_354	++++	50.0	20.0
Eubacterium rectale	*Clostridium mayombei*	*Ruminococcus bromii*	clade_444	clade_354	clade_537	++++	50.0	20.0
Faecalibacterium prausnitzii	*Clostridium mayombei*	*Blautia* sp. M25	clade_478	clade_354	clade_309	++++	50.0	20.0
Faecalibacterium prausnitzii	*Clostridium mayombei*	*Clostridium mayombei*	clade_478	clade_354	clade_354	++++	50.0	20.0
Faecalibacterium prausnitzii	*Clostridium mayombei*	*Ruminococcus bromii*	clade_478	clade_354	clade_537	++++	50.0	20.0
Clostridium mayombei	*Clostridium mayombei*		clade_354	clade_354		++++	50.0	20.0
Clostridium mayombei	*Faecalibacterium prausnitzii*		clade_354	clade_478		++++	50.0	20.0
Clostridium mayombei	*Ruminococcus bromii*		clade_354	clade_537		++++	50.0	20.0
Clostridium mayombei	*Eubacterium rectale*		clade_354	clade_444		++++	50.0	20.0
Eubacterium rectale	*Clostridium disporicum*	*Clostridium mayombei*	clade_444	clade_253	clade_354	++++	42.9	13.3
Faecalibacterium prausnitzii	*Clostridium disporicum*	*Clostridium mayombei*	clade_478	clade_253	clade_354	++++	42.9	13.3
Clostridium disporicum	*Clostridium mayombei*		clade_253	clade_354		++++	42.9	13.3
Clostridium disporicum	*Coprococcus comes*		clade_253	clade_262		++++	35.7	13.3
Eubacterium rectale	*Clostridium orbiscindens*	*Blautia* sp. M25	clade_444	clade_494	clade_309	++++	28.6	80.0
Faecalibacterium prausnitzii	*Clostridium orbiscindens*	*Blautia* sp. M25	clade_478	clade_494	clade_309	++++	28.6	73.3
Clostridium disporicum	*Clostridium orbiscindens*		clade_253	clade_494		++++	28.6	46.7
Clostridium hylemonae	*Eubacterium rectale*	*Eubacterium rectale*	clade_260	clade_444	clade_444	++++	21.4	40.0
Eubacterium rectale	*Clostridium hylemonae*	*Blautia* sp. M25	clade_444	clade_260	clade_309	++++	21.4	40.0
Coprococcus comes	*Clostridium orbiscindens*	*Blautia* sp. M25	clade_262	clade_494	clade_309	++++	21.4	33.3
Coprococcus comes	*Clostridium orbiscindens*		clade_262	clade_494		++++	21.4	33.3
Eubacterium rectale	*Clostridium mayombei*	*Clostridium orbiscindens*	clade_444	clade_354	clade_494	++++	21.4	20.0
Faecalibacterium prausnitzii	*Clostridium mayombei*	*Clostridium orbiscindens*	clade_478	clade_354	clade_494	++++	21.4	20.0
Clostridium mayombei	*Clostridium orbiscindens*		clade_354	clade_494		++++	21.4	20.0
Clostridium disporicum	Lachnospiraceae *bacterium* 5_1_57FAA		clade_253	clade_260		++++	14.3	40.0
Clostridium hylemonae	Lachnospiraceae *bacterium* 5_1_57FAA	Lachnospiraceae *bacterium* 5_1_57FAA	clade_260	clade_260	clade_260	++++	14.3	26.7
Clostridium hylemonae	Lachnospiraceae *bacterium* 5_1_57FAA		clade_260	clade_260		++++	14.3	26.7
Clostridium hylemonae	Lachnospiraceae *bacterium* 5_1_57FAA	*Eubacterium rectale*	clade_260	clade_260	clade_444	++++	14.3	26.7
Lachnospiraceae *bacterium* 5_1_57FAA	*Clostridium hylemonae*	*Blautia* sp. M25	clade_260	clade_260	clade_309	++++	14.3	26.7
Bacteroides caccae	*Bacteroides caccae*		clade_170	clade_170		++++	14.3	20.0

TABLE 12-continued

OTC1 of Composition	OTC2 of Composition	OTC3 of Composition	Clade of OTU1	Clade of OTU2	Clade of OTU3 (if applicable)	*C. diff* Inhibition Score	Percent of Dose Ecologies Occurs	Percent of Post-treatment Ecologies
Clostridium hylemonae	*Faecalibacterium prausnitzii*	Lachnospiraceae *bacterium* 5_1_57FAA	clade_260	clade_478	clade_260	++++	14.3	20.0
Bacteroides caccae	*Ruminococcus torques*		clade_170	clade_262		++++	14.3	20.0
Clostridium mayombei	*Faecalibacterium prausnitzii*	Lachnospiraceae *bacterium* 5_1_57FAA	clade_354	clade_478	clade_260	++++	14.3	13.3
Clostridium mayombei	Lachnospiraceae *bacterium* 5_1_57FAA	*Eubacterium rectale*	clade_354	clade_260	clade_444	++++	14.3	13.3
Clostridium mayombei	Lachnospiraceae *bacterium* 5_1_57FAA	Lachnospiraceae *bacterium* 5_1_57FAA	clade_354	clade_260	clade_260	++++	14.3	13.3
Lachnospiraceae *bacterium* 5_1_57FAA	*Clostridium mayombei*	*Blautia* sp. M25	clade_260	clade_354	clade_309	++++	14.3	13.3
Lachnospiraceae *bacterium* 5_1_57FAA	*Clostridium mayombei*	*Clostridium mayombei*	clade_260	clade_354	clade_354	++++	14.3	13.3
Lachnospiraceae *bacterium* 5_1_57FAA	*Clostridium mayombei*	*Ruminococcus bromii*	clade_260	clade_354	clade_537	++++	14.3	13.3
Clostridium mayombei	Lachnospiraceae *bacterium* 5_1_57FAA		clade_354	clade_260		++++	14.3	13.3
Coprococcus comes	*Clostridium hylemonae*	*Blautia* sp. M25	clade_262	clade_260	clade_309	++++	14.3	13.3
Lachnospiraceae *bacterium* 5_1_57FAA	*Clostridium disporicum*	*Clostridium mayombei*	clade_260	clade_253	clade_354	++++	14.3	6.7
Dorea longicatena	*Clostridium symbiosum*		clade_360	clade_408		++++	7.1	93.3
Clostridium symbiosum	*Clostridium orbiscindens*	*Clostridium orbiscindens*	clade_408	clade_494	clade_494	++++	7.1	80.0
Clostridium symbiosum	*Clostridium orbiscindens*	*Blautia* sp. M25	clade_408	clade_494	clade_309	++++	7.1	80.0
Clostridium orbiscindens	*Clostridium symbiosum*	Lachnospiraceae *bacterium* 5_1_57FAA	clade_494	clade_408	clade_260	++++	7.1	66.7
Lachnospiraceae *bacterium* 5_1_57FAA	*Clostridium orbiscindens*	*Blautia* sp. M25	clade_260	clade_494	clade_309	++++	7.1	66.7
Clostridium symbiosum	*Ruminococcus bromii*		clade_408	clade_537		++++	7.1	66.7
Clostridium disporicum	*Clostridium symbiosum*	Lachnospiraceae *bacterium* 5_1_57FAA	clade_253	clade_408	clade_260	++++	7.1	40.0
Clostridium hylemonae	*Clostridium symbiosum*	*Eubacterium rectale*	clade_260	clade_408	clade_444	++++	7.1	40.0
Coprococcus comes	Lachnospiraceae *bacterium* 5_1_57FAA		clade_262	clade_260		++++	7.1	33.3
Clostridium symbiosum	*Clostridium hylemonae*	*Ruminococcus bromii*	clade_408	clade_260	clade_537	++++	7.1	33.3
Clostridium hylemonae	*Clostridium symbiosum*	Lachnospiraceae *bacterium* 5_1_57FAA	clade_260	clade_408	clade_260	++++	7.1	26.7
Clostridium mayombei	*Clostridium symbiosum*	*Clostridium symbiosum*	clade_354	clade_408	clade_408	++++	7.1	20.0
Clostridium mayombei	*Clostridium symbiosum*	*Eubacterium rectale*	clade_354	clade_408	clade_444	++++	7.1	20.0
Clostridium mayombei	*Clostridium symbiosum*	*Faecalibacterium prausnitzii*	clade_354	clade_408	clade_478	++++	7.1	20.0
Clostridium symbiosum	*Clostridium mayombei*	*Blautia* sp. M25	clade_408	clade_354	clade_309	++++	7.1	20.0
Clostridium symbiosum	*Clostridium mayombei*	*Clostridium mayombei*	clade_408	clade_354	clade_354	++++	7.1	20.0
Clostridium symbiosum	*Clostridium mayombei*	*Clostridium orbiscindens*	clade_408	clade_354	clade_494	++++	7.1	20.0
Clostridium symbiosum	*Clostridium mayombei*	*Ruminococcus bromii*	clade_408	clade_354	clade_537	++++	7.1	20.0
Clostridium mayombei	*Clostridium symbiosum*		clade_354	clade_408		++++	7.1	20.0
Clostridium mayombei	*Clostridium symbiosum*	Lachnospiraceae *bacterium* 5_1_57FAA	clade_354	clade_408	clade_260	++++	7.1	13.3
Clostridium symbiosum	*Clostridium disporicum*	*Clostridium mayombei*	clade_408	clade_253	clade_354	++++	7.1	13.3
Clostridium symbiosum	*Clostridium mayombei*	*Clostridium hylemonae*	clade_408	clade_354	clade_260	++++	7.1	13.3
Eubacterium rectale	*Clostridium mayombei*	*Clostridium hylemonae*	clade_444	clade_354	clade_260	++++	7.1	13.3

TABLE 12-continued

OTC1 of Composition	OTC2 of Composition	OTC3 of Composition	Clade of OTU1	Clade of OTU2	Clade of OTU3 (if applicable)	*C. diff* Inhibition Score	Percent of Dose Ecologies Occurs	Percent of Post-treatment Ecologies
Faecalibacterium prausnitzii	*Clostridium mayombei*	*Clostridium hylemonae*	clade_478	clade_354	clade_260	++++	7.1	13.3
Lachnospiraceae bacterium 5_1_57FAA	*Clostridium mayombei*	*Clostridium orbiscindens*	clade_260	clade_354	clade_494	++++	7.1	13.3
Clostridium mayombei	*Clostridium hylemonae*		clade_354	clade_260		++++	7.1	13.3
Clostridium hylemonae	*Coprococcus comes*	Lachnospiraceae *bacterium* 5_1_57FAA	clade_260	clade_262	clade_260	++++	7.1	6.7
Lachnospiraceae *bacterium* 5_1_57FAA	*Clostridium mayombei*	*Clostridium hylemonae*	clade_260	clade_354	clade_260	++++	7.1	6.7

TABLE 13

Num. of OTUs	OTU Clade									
	clade_309 *Blautia luti*	clade_309 *Blautia schinkii*	clade_309 *Blautia* sp. M25	clade_408 *Clostridium citroniae*	clade_537 *Clostridium leptum*	clade_444 *Eubacterium rectale*	clade_478 *Faecalibacterium prausnitzii*	clade_360 Lachnospiraceae *bacterium* 2_1_58FAA	clade_537 *Ruminococcus bromii*	clade_478 *Subdoligranulum variabile*
10	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
9	Y	Y	Y	Y	Y	Y	Y	Y	Y	
9	Y	Y	Y	Y	Y	Y	Y	Y		Y
9	Y	Y	Y	Y	Y	Y	Y		Y	Y
9	Y	Y	Y	Y	Y	Y		Y	Y	Y
9	Y	Y	Y	Y	Y		Y	Y	Y	Y
9	Y	Y	Y	Y		Y	Y	Y	Y	Y
9	Y	Y	Y		Y	Y	Y	Y	Y	Y
9	Y	Y		Y	Y	Y	Y	Y	Y	Y
9	Y		Y	Y	Y	Y	Y	Y	Y	Y
9		Y	Y	Y	Y	Y	Y	Y	Y	Y
8	Y	Y	Y	Y	Y	Y	Y	Y		
8	Y	Y	Y	Y	Y	Y	Y		Y	
8	Y	Y	Y	Y	Y	Y	Y			Y
8	Y	Y	Y	Y	Y	Y		Y	Y	
8	Y	Y	Y	Y	Y	Y		Y		Y
8	Y	Y	Y	Y	Y	Y			Y	Y
8	Y	Y	Y	Y	Y		Y	Y	Y	
8	Y	Y	Y	Y	Y		Y	Y		Y
8	Y	Y	Y	Y	Y		Y		Y	Y
8	Y	Y	Y	Y	Y			Y	Y	Y
8	Y	Y	Y	Y		Y	Y	Y	Y	
8	Y	Y	Y	Y		Y	Y	Y		Y
8	Y	Y	Y	Y		Y	Y		Y	Y
8	Y	Y	Y	Y		Y		Y	Y	Y
8	Y	Y	Y	Y			Y	Y	Y	Y
8	Y	Y	Y		Y	Y	Y	Y	Y	
8	Y	Y	Y		Y	Y	Y	Y		Y
8	Y	Y	Y		Y	Y	Y		Y	Y
8	Y	Y	Y		Y	Y		Y	Y	Y
8	Y	Y	Y		Y		Y	Y	Y	Y
8	Y	Y	Y			Y	Y	Y	Y	Y
8	Y	Y		Y	Y	Y	Y	Y	Y	
8	Y	Y		Y	Y	Y	Y	Y		Y
8	Y	Y		Y	Y	Y	Y		Y	Y
8	Y	Y		Y	Y	Y		Y	Y	Y
8	Y	Y		Y	Y		Y	Y	Y	Y
8	Y	Y		Y		Y	Y	Y	Y	Y
8	Y	Y			Y	Y	Y	Y	Y	Y
8	Y		Y	Y	Y	Y	Y	Y	Y	
8	Y		Y	Y	Y	Y	Y	Y		Y
8	Y		Y	Y	Y	Y	Y		Y	Y
8	Y		Y	Y	Y	Y		Y	Y	Y
8	Y		Y	Y	Y		Y	Y	Y	Y
8	Y		Y	Y		Y	Y	Y	Y	Y

TABLE 13-continued

Num. of OTUs	OTU Clade									
	clade_309 *Blautia luti*	clade_309 *Blautia schinkii*	clade_309 *Blautia* sp. M25	clade_408 *Clostridium citroniae*	clade_537 *Clostridium leptum*	clade_444 *Eubacterium rectale*	clade_478 *Faecalibacterium prausnitzii*	clade_360 Lachnospiraceae *bacterium* 2_1_58FAA	clade_537 *Ruminococcus bromii*	clade_478 *Subdoligranulum variabile*
8	Y		Y		Y	Y	Y	Y	Y	Y
8	Y			Y	Y	Y	Y	Y	Y	Y
8		Y	Y	Y	Y	Y	Y	Y	Y	
8		Y	Y	Y	Y	Y	Y	Y		Y
8		Y	Y	Y	Y	Y	Y		Y	Y
8		Y	Y	Y	Y	Y		Y	Y	Y
8		Y	Y	Y	Y		Y	Y	Y	Y
8		Y	Y	Y		Y	Y	Y	Y	Y
8		Y	Y		Y	Y	Y	Y	Y	Y
8		Y		Y	Y	Y	Y	Y	Y	Y
8			Y	Y	Y	Y	Y	Y	Y	Y
7	Y	Y	Y	Y	Y	Y	Y			
7	Y	Y	Y	Y	Y	Y		Y		
7	Y	Y	Y	Y	Y	Y			Y	
7	Y	Y	Y	Y	Y	Y				Y
7	Y	Y	Y	Y	Y		Y	Y		
7	Y	Y	Y	Y	Y		Y		Y	
7	Y	Y	Y	Y	Y		Y			Y
7	Y	Y	Y	Y	Y			Y	Y	
7	Y	Y	Y	Y	Y			Y		Y
7	Y	Y	Y	Y	Y				Y	Y
7	Y	Y	Y	Y		Y	Y	Y		
7	Y	Y	Y	Y		Y	Y		Y	
7	Y	Y	Y	Y		Y	Y			Y
7	Y	Y	Y	Y		Y		Y	Y	
7	Y	Y	Y	Y		Y		Y		Y
7	Y	Y	Y	Y		Y			Y	Y
7	Y	Y	Y	Y			Y	Y	Y	
7	Y	Y	Y	Y			Y	Y		Y
7	Y	Y	Y	Y			Y		Y	Y
7	Y	Y	Y	Y				Y	Y	Y
7	Y	Y	Y		Y	Y	Y	Y		
7	Y	Y	Y		Y	Y	Y		Y	
7	Y	Y	Y		Y	Y	Y			Y
7	Y	Y	Y		Y	Y		Y	Y	
7	Y	Y	Y		Y	Y		Y		Y
7	Y	Y	Y		Y	Y			Y	Y
7	Y	Y	Y		Y		Y	Y	Y	
7	Y	Y	Y		Y		Y	Y		Y
7	Y	Y	Y		Y		Y		Y	Y
7	Y	Y	Y		Y			Y	Y	Y
7	Y	Y	Y			Y	Y	Y	Y	
7	Y	Y	Y			Y	Y	Y		Y
7	Y	Y	Y			Y	Y		Y	Y
7	Y	Y	Y			Y		Y	Y	Y
7	Y	Y	Y				Y	Y	Y	Y
7	Y	Y		Y	Y	Y	Y	Y		

TABLE 13-continued

Num. of OTUs	OTU Clade clade_309 *Blautia luti*	clade_309 *Blautia schinkii*	clade_309 *Blautia* sp. M25	clade_408 *Clostridium citroniae*	clade_537 *Clostridium leptum*	clade_444 *Eubacterium rectale*	clade_478 *Faecalibacterium prausnitzii*	clade_360 Lachnospiraceae *bacterium* 2_1_58FAA	clade_537 *Ruminococcus bromii*	clade_478 *Subdoligranulum variabile*
7	Y	Y		Y	Y	Y	Y		Y	
7	Y	Y		Y	Y	Y	Y			Y
7	Y	Y		Y	Y	Y		Y	Y	
7	Y	Y		Y	Y	Y		Y		Y
7	Y	Y		Y	Y	Y			Y	Y
7	Y	Y		Y	Y		Y	Y	Y	
7	Y	Y		Y	Y		Y	Y		Y
7	Y	Y		Y	Y		Y		Y	Y
7	Y	Y		Y	Y			Y	Y	Y
7	Y	Y		Y		Y	Y	Y	Y	
7	Y	Y		Y		Y	Y	Y		Y
7	Y	Y		Y		Y	Y		Y	Y
7	Y	Y		Y		Y		Y	Y	Y
7	Y	Y		Y			Y	Y	Y	Y
7	Y	Y			Y	Y	Y	Y	Y	
7	Y	Y			Y	Y	Y	Y		Y
7	Y	Y			Y	Y	Y		Y	Y
7	Y	Y			Y	Y		Y	Y	Y
7	Y	Y			Y		Y	Y	Y	Y
7	Y	Y				Y	Y	Y	Y	Y
7	Y		Y	Y	Y	Y	Y	Y		
7	Y		Y	Y	Y	Y	Y		Y	
7	Y		Y	Y	Y	Y	Y			Y
7	Y		Y	Y	Y	Y		Y	Y	
7	Y		Y	Y	Y	Y		Y		Y
7	Y		Y	Y	Y	Y			Y	Y
7	Y		Y	Y	Y		Y	Y	Y	
7	Y		Y	Y	Y		Y	Y		Y
7	Y		Y	Y	Y		Y		Y	Y
7	Y		Y	Y	Y			Y	Y	Y
7	Y		Y	Y		Y	Y	Y	Y	
7	Y		Y	Y		Y	Y	Y		Y
7	Y		Y	Y		Y	Y		Y	Y
7	Y		Y	Y		Y		Y	Y	Y
7	Y		Y	Y			Y	Y	Y	Y
7	Y		Y		Y	Y	Y	Y	Y	
7	Y		Y		Y	Y	Y	Y		Y
7	Y		Y		Y	Y	Y		Y	Y
7	Y		Y		Y	Y		Y	Y	Y
7	Y		Y		Y		Y	Y	Y	Y
7	Y		Y			Y	Y	Y	Y	Y
7	Y			Y	Y	Y	Y	Y	Y	
7	Y			Y	Y	Y	Y	Y		Y
7	Y			Y	Y	Y	Y		Y	Y
7	Y			Y	Y	Y		Y	Y	Y
7	Y			Y	Y		Y	Y	Y	Y
7	Y			Y		Y	Y	Y	Y	Y

TABLE 13-continued

Num. of OTUs	OTU Clade									
	clade_309 *Blautia luti*	clade_309 *Blautia schinkii*	clade_309 *Blautia* sp. M25	clade_408 *Clostridium citroniae*	clade_537 *Clostridium leptum*	clade_444 *Eubacterium rectale*	clade_478 *Faecalibacterium prausnitzii*	clade_360 Lachnospiraceae *bacterium* 2_1_58FAA	clade_537 *Ruminococcus bromii*	clade_478 *Subdoligranulum variabile*
7	Y				Y	Y	Y	Y	Y	Y
7		Y	Y	Y	Y	Y	Y	Y		
7		Y	Y	Y	Y	Y	Y		Y	
7		Y	Y	Y	Y	Y	Y			Y
7		Y	Y	Y	Y	Y		Y	Y	
7		Y	Y	Y	Y	Y		Y		Y
7		Y	Y	Y	Y	Y			Y	Y
7		Y	Y	Y	Y		Y	Y	Y	
7		Y	Y	Y	Y		Y	Y		Y
7		Y	Y	Y	Y		Y		Y	Y
7		Y	Y	Y	Y			Y	Y	Y
7		Y	Y	Y		Y	Y	Y	Y	
7		Y	Y	Y		Y	Y	Y		Y
7		Y	Y	Y		Y	Y		Y	Y
7		Y	Y	Y		Y		Y	Y	Y
7		Y	Y	Y			Y	Y	Y	Y
7		Y	Y		Y	Y	Y	Y	Y	
7		Y	Y		Y	Y	Y	Y		Y
7		Y	Y		Y	Y	Y		Y	Y
7		Y	Y		Y	Y		Y	Y	Y
7		Y	Y		Y		Y	Y	Y	Y
7		Y	Y			Y	Y	Y	Y	Y
7		Y		Y	Y	Y	Y	Y	Y	
7		Y		Y	Y	Y	Y	Y		Y
7		Y		Y	Y	Y	Y		Y	Y
7		Y		Y	Y	Y		Y	Y	Y
7		Y		Y	Y		Y	Y	Y	Y
7		Y		Y		Y	Y	Y	Y	Y
7		Y			Y	Y	Y	Y	Y	Y
7			Y	Y	Y	Y	Y	Y	Y	
7			Y	Y	Y	Y	Y	Y		Y
7			Y	Y	Y	Y	Y		Y	Y
7			Y	Y	Y	Y		Y	Y	Y
7			Y	Y	Y		Y	Y	Y	Y
7			Y	Y		Y	Y	Y	Y	Y
7			Y		Y	Y	Y	Y	Y	Y
7				Y	Y	Y	Y	Y	Y	Y
6	Y	Y	Y	Y	Y	Y				
6	Y	Y	Y	Y	Y		Y			
6	Y	Y	Y	Y	Y			Y		
6	Y	Y	Y	Y	Y				Y	
6	Y	Y	Y	Y	Y					Y
6	Y	Y	Y	Y		Y	Y			
6	Y	Y	Y	Y		Y		Y		
6	Y	Y	Y	Y		Y			Y	
6	Y	Y	Y	Y		Y				Y
6	Y	Y	Y	Y			Y	Y		

TABLE 13-continued

Num. of OTUs	OTU Clade									
	clade_309 *Blautia luti*	clade_309 *Blautia schinkii*	clade_309 *Blautia* sp. M25	clade_408 *Clostridium citroniae*	clade_537 *Clostridium leptum*	clade_444 *Eubacterium rectale*	clade_478 *Faecalibacterium prausnitzii*	clade_360 Lachnospiraceae *bacterium* 2_1_58FAA	clade_537 *Ruminococcus bromii*	clade_478 *Subdoligranulum variabile*
6	Y	Y	Y	Y			Y		Y	
6	Y	Y	Y	Y			Y			Y
6	Y	Y	Y	Y				Y	Y	
6	Y	Y	Y	Y				Y		Y
6	Y	Y	Y	Y					Y	Y
6	Y	Y	Y		Y	Y	Y			
6	Y	Y	Y		Y	Y		Y		
6	Y	Y	Y		Y	Y			Y	
6	Y	Y	Y		Y	Y				Y
6	Y	Y	Y		Y		Y	Y		
6	Y	Y	Y		Y		Y		Y	
6	Y	Y	Y		Y		Y			Y
6	Y	Y	Y		Y			Y	Y	
6	Y	Y	Y		Y			Y		Y
6	Y	Y	Y		Y				Y	Y
6	Y	Y	Y			Y	Y	Y		
6	Y	Y	Y			Y	Y		Y	
6	Y	Y	Y			Y	Y			Y
6	Y	Y	Y			Y		Y	Y	
6	Y	Y	Y			Y		Y		Y
6	Y	Y	Y			Y			Y	Y
6	Y	Y	Y				Y	Y	Y	
6	Y	Y	Y				Y	Y		Y
6	Y	Y	Y				Y		Y	Y
6	Y	Y	Y					Y	Y	Y
6	Y	Y		Y	Y	Y	Y			
6	Y	Y		Y	Y	Y		Y		
6	Y	Y		Y	Y	Y			Y	
6	Y	Y		Y	Y	Y				Y
6	Y	Y		Y	Y		Y	Y		
6	Y	Y		Y	Y		Y		Y	
6	Y	Y		Y	Y		Y			Y
6	Y	Y		Y	Y			Y	Y	
6	Y	Y		Y	Y			Y		Y
6	Y	Y		Y	Y				Y	Y
6	Y	Y		Y		Y	Y	Y		
6	Y	Y		Y		Y	Y		Y	
6	Y	Y		Y		Y	Y			Y
6	Y	Y		Y		Y		Y	Y	
6	Y	Y		Y		Y		Y		Y
6	Y	Y		Y		Y			Y	Y
6	Y	Y		Y			Y	Y	Y	
6	Y	Y		Y			Y	Y		Y
6	Y	Y		Y			Y		Y	Y
6	Y	Y		Y				Y	Y	Y
6	Y	Y			Y	Y	Y	Y		
6	Y	Y			Y	Y	Y		Y	

TABLE 13-continued

Num. of OTUs	OTU Clade									
	clade_309 *Blautia luti*	clade_309 *Blautia schinkii*	clade_309 *Blautia* sp. M25	clade_408 *Clostridium citroniae*	clade_537 *Clostridium leptum*	clade_444 *Eubacterium rectale*	clade_478 *Faecalibacterium prausnitzii*	clade_360 Lachnospiraceae *bacterium* 2_1_58FAA	clade_537 *Ruminococcus bromii*	clade_478 *Subdoligranulum variabile*
6	Y	Y			Y	Y	Y			Y
6	Y	Y			Y	Y		Y	Y	
6	Y	Y			Y	Y		Y		Y
6	Y	Y			Y	Y			Y	Y
6	Y	Y			Y		Y	Y	Y	
6	Y	Y			Y		Y	Y		Y
6	Y	Y			Y		Y		Y	Y
6	Y	Y			Y			Y	Y	Y
6	Y	Y				Y	Y	Y	Y	
6	Y	Y				Y	Y	Y		Y
6	Y	Y				Y	Y		Y	Y
6	Y	Y				Y		Y	Y	Y
6	Y	Y					Y	Y	Y	Y
6	Y		Y	Y	Y	Y	Y			
6	Y		Y	Y	Y	Y		Y		
6	Y		Y	Y	Y	Y			Y	
6	Y		Y	Y	Y	Y				Y
6	Y		Y	Y	Y		Y	Y		
6	Y		Y	Y	Y		Y		Y	
6	Y		Y	Y	Y		Y			Y
6	Y		Y	Y	Y			Y	Y	
6	Y		Y	Y	Y			Y		Y
6	Y		Y	Y	Y				Y	Y
6	Y		Y	Y		Y	Y	Y		
6	Y		Y	Y		Y	Y		Y	
6	Y		Y	Y		Y	Y			Y
6	Y		Y	Y		Y		Y	Y	
6	Y		Y	Y		Y		Y		Y
6	Y		Y	Y		Y			Y	Y
6	Y		Y	Y			Y	Y	Y	
6	Y		Y	Y			Y	Y		Y
6	Y		Y	Y			Y		Y	Y
6	Y		Y	Y				Y	Y	Y
6	Y		Y		Y	Y	Y	Y		
6	Y		Y		Y	Y	Y		Y	
6	Y		Y		Y	Y	Y			Y
6	Y		Y		Y	Y		Y	Y	
6	Y		Y		Y	Y		Y		Y
6	Y		Y		Y	Y			Y	Y
6	Y		Y		Y		Y	Y	Y	
6	Y		Y		Y		Y	Y		Y
6	Y		Y		Y		Y		Y	Y
6	Y		Y		Y			Y	Y	Y
6	Y		Y			Y	Y	Y	Y	
6	Y		Y			Y	Y	Y		Y
6	Y		Y			Y	Y		Y	Y
6	Y		Y			Y		Y	Y	Y

TABLE 13-continued

Num. of OTUs	OTU Clade clade_309 *Blautia luti*	clade_309 *Blautia schinkii*	clade_309 *Blautia* sp. M25	clade_408 *Clostridium citroniae*	clade_537 *Clostridium leptum*	clade_444 *Eubacterium rectale*	clade_478 *Faecalibacterium prausnitzii*	clade_360 Lachnospiraceae *bacterium* 2_1_58FAA	clade_537 *Ruminococcus bromii*	clade_478 *Subdoligranulum variabile*
6	Y		Y				Y	Y	Y	Y
6	Y			Y	Y	Y	Y	Y		
6	Y			Y	Y	Y	Y		Y	
6	Y			Y	Y	Y	Y			Y
6	Y			Y	Y	Y		Y	Y	
6	Y			Y	Y	Y		Y		Y
6	Y			Y	Y	Y			Y	Y
6	Y			Y	Y		Y	Y	Y	
6	Y			Y	Y		Y	Y		Y
6	Y			Y	Y		Y		Y	Y
6	Y			Y	Y			Y	Y	Y
6	Y			Y		Y	Y	Y	Y	
6	Y			Y		Y	Y	Y		Y
6	Y			Y		Y	Y		Y	Y
6	Y			Y		Y		Y	Y	Y
6	Y			Y			Y	Y	Y	Y
6	Y				Y	Y	Y	Y	Y	
6	Y				Y	Y	Y	Y		Y
6	Y				Y	Y	Y		Y	Y
6	Y				Y	Y		Y	Y	Y
6	Y				Y		Y	Y	Y	Y
6	Y					Y	Y	Y	Y	Y
6		Y	Y	Y	Y	Y	Y			
6		Y	Y	Y	Y	Y		Y		
6		Y	Y	Y	Y	Y			Y	
6		Y	Y	Y	Y	Y				Y
6		Y	Y	Y	Y		Y	Y		
6		Y	Y	Y	Y		Y		Y	
6		Y	Y	Y	Y		Y			Y
6		Y	Y	Y	Y			Y	Y	
6		Y	Y	Y	Y			Y		Y
6		Y	Y	Y	Y				Y	Y
6		Y	Y	Y		Y	Y	Y		
6		Y	Y	Y		Y	Y		Y	
6		Y	Y	Y		Y	Y			Y
6		Y	Y	Y		Y		Y	Y	
6		Y	Y	Y		Y		Y		Y
6		Y	Y	Y		Y			Y	Y
6		Y	Y	Y			Y	Y	Y	
6		Y	Y	Y			Y	Y		Y
6		Y	Y	Y			Y		Y	Y
6		Y	Y	Y				Y	Y	Y
6		Y	Y		Y	Y	Y	Y		
6		Y	Y		Y	Y	Y		Y	
6		Y	Y		Y	Y	Y			Y
6		Y	Y		Y	Y		Y	Y	
6		Y	Y		Y	Y		Y		Y

TABLE 13-continued

Num. of OTUs	OTU Clade clade_309 *Blautia luti*	clade_309 *Blautia schinkii*	clade_309 *Blautia* sp. M25	clade_408 *Clostridium citroniae*	clade_537 *Clostridium leptum*	clade_444 *Eubacterium rectale*	clade_478 *Faecalibacterium prausnitzii*	clade_360 Lachnospiraceae *bacterium* 2_1_58FAA	clade_537 *Ruminococcus bromii*	clade_478 *Subdoligranulum variabile*
6		Y	Y		Y	Y			Y	Y
6		Y	Y		Y		Y	Y	Y	
6		Y	Y		Y		Y	Y		Y
6		Y	Y		Y		Y		Y	Y
6		Y	Y		Y			Y	Y	Y
6		Y	Y			Y	Y	Y	Y	
6		Y	Y			Y	Y	Y		Y
6		Y	Y			Y	Y		Y	Y
6		Y	Y			Y		Y	Y	Y
6		Y	Y				Y	Y	Y	Y
6		Y		Y	Y	Y	Y	Y		
6		Y		Y	Y	Y	Y		Y	
6		Y		Y	Y	Y	Y			Y
6		Y		Y	Y	Y		Y	Y	
6		Y		Y	Y	Y		Y		Y
6		Y		Y	Y	Y			Y	Y
6		Y		Y	Y		Y	Y	Y	
6		Y		Y	Y		Y	Y		Y
6		Y		Y	Y		Y		Y	Y
6		Y		Y	Y			Y	Y	Y
6		Y		Y		Y	Y	Y	Y	
6		Y		Y		Y	Y	Y		Y
6		Y		Y		Y	Y		Y	Y
6		Y		Y		Y		Y	Y	Y
6		Y		Y			Y	Y	Y	Y
6		Y			Y	Y	Y	Y	Y	
6		Y			Y	Y	Y	Y		Y
6		Y			Y	Y	Y		Y	Y
6		Y			Y	Y		Y	Y	Y
6		Y			Y		Y	Y	Y	Y
6		Y				Y	Y	Y	Y	Y
6			Y	Y	Y	Y	Y	Y		
6			Y	Y	Y	Y	Y		Y	
6			Y	Y	Y	Y	Y			Y
6			Y	Y	Y	Y		Y	Y	
6			Y	Y	Y	Y		Y		Y
6			Y	Y	Y	Y			Y	Y
6			Y	Y	Y		Y	Y	Y	
6			Y	Y	Y		Y	Y		Y
6			Y	Y	Y		Y		Y	Y
6			Y	Y	Y			Y	Y	Y
6			Y	Y		Y	Y	Y	Y	
6			Y	Y		Y	Y	Y		Y
6			Y	Y		Y	Y		Y	Y
6			Y	Y		Y		Y	Y	Y
6			Y	Y			Y	Y	Y	Y
6			Y		Y	Y	Y	Y	Y	

TABLE 13-continued

Num. of OTUs	OTU Clade clade_309 *Blautia luti*	clade_309 *Blautia schinkii*	clade_309 *Blautia* sp. M25	clade_408 *Clostridium citroniae*	clade_537 *Clostridium leptum*	clade_444 *Eubacterium rectale*	clade_478 *Faecalibacterium prausnitzii*	clade_360 Lachnospiraceae *bacterium* 2_1_58FAA	clade_537 *Ruminococcus bromii*	clade_478 *Subdoligranulum variabile*
6			Y		Y	Y	Y	Y		Y
6			Y		Y	Y	Y		Y	Y
6			Y		Y	Y		Y	Y	Y
6			Y		Y		Y	Y	Y	Y
6			Y			Y	Y	Y	Y	Y
6				Y	Y	Y	Y	Y	Y	
6				Y	Y	Y	Y	Y		Y
6				Y	Y	Y	Y		Y	Y
6				Y	Y	Y		Y	Y	Y
6				Y	Y		Y	Y	Y	Y
6				Y		Y	Y	Y	Y	Y
6					Y	Y	Y	Y	Y	Y
5	Y	Y	Y	Y	Y					
5	Y	Y	Y	Y		Y				
5	Y	Y	Y	Y			Y			
5	Y	Y	Y	Y				Y		
5	Y	Y	Y	Y					Y	
5	Y	Y	Y	Y						Y
5	Y	Y	Y		Y	Y				
5	Y	Y	Y		Y		Y			
5	Y	Y	Y		Y			Y		
5	Y	Y	Y		Y				Y	
5	Y	Y	Y		Y					Y
5	Y	Y	Y			Y	Y			
5	Y	Y	Y			Y		Y		
5	Y	Y	Y			Y			Y	
5	Y	Y	Y			Y				Y
5	Y	Y	Y				Y	Y		
5	Y	Y	Y				Y		Y	
5	Y	Y	Y				Y			Y
5	Y	Y	Y					Y	Y	
5	Y	Y	Y					Y		Y
5	Y	Y	Y						Y	Y
5	Y	Y		Y	Y	Y				
5	Y	Y		Y	Y		Y			
5	Y	Y		Y	Y			Y		
5	Y	Y		Y	Y				Y	
5	Y	Y		Y	Y					Y
5	Y	Y		Y		Y	Y			
5	Y	Y		Y		Y		Y		
5	Y	Y		Y		Y			Y	
5	Y	Y		Y		Y				Y
5	Y	Y		Y			Y	Y		
5	Y	Y		Y			Y		Y	
5	Y	Y		Y			Y			Y
5	Y	Y		Y				Y	Y	
5	Y	Y		Y				Y		Y

TABLE 13-continued

Num. of OTUs	OTU Clade									
	clade_309 *Blautia luti*	clade_309 *Blautia schinkii*	clade_309 *Blautia* sp. M25	clade_408 *Clostridium citroniae*	clade_537 *Clostridium leptum*	clade_444 *Eubacterium rectale*	clade_478 *Faecalibacterium prausnitzii*	clade_360 Lachnospiraceae *bacterium* 2_1_58FAA	clade_537 *Ruminococcus bromii*	clade_478 *Subdoligranulum variabile*
5	Y	Y		Y					Y	Y
5	Y	Y			Y	Y	Y			
5	Y	Y			Y	Y		Y		
5	Y	Y			Y	Y			Y	
5	Y	Y			Y	Y				Y
5	Y	Y			Y		Y	Y		
5	Y	Y			Y		Y		Y	
5	Y	Y			Y		Y			Y
5	Y	Y			Y			Y	Y	
5	Y	Y			Y			Y		Y
5	Y	Y			Y				Y	Y
5	Y	Y				Y	Y	Y		
5	Y	Y				Y	Y		Y	
5	Y	Y				Y	Y			Y
5	Y	Y				Y		Y	Y	
5	Y	Y				Y		Y		Y
5	Y	Y				Y			Y	Y
5	Y	Y					Y	Y	Y	
5	Y	Y					Y	Y		Y
5	Y	Y					Y		Y	Y
5	Y	Y						Y	Y	Y
5	Y		Y	Y	Y	Y				
5	Y		Y	Y	Y		Y			
5	Y		Y	Y	Y			Y		
5	Y		Y	Y	Y				Y	
5	Y		Y	Y	Y					Y
5	Y		Y	Y		Y	Y			
5	Y		Y	Y		Y		Y		
5	Y		Y	Y		Y			Y	
5	Y		Y	Y		Y				Y
5	Y		Y	Y			Y	Y		
5	Y		Y	Y			Y		Y	
5	Y		Y	Y			Y			Y
5	Y		Y	Y				Y	Y	
5	Y		Y	Y				Y		Y
5	Y		Y	Y					Y	Y
5	Y		Y		Y	Y	Y			
5	Y		Y		Y	Y		Y		
5	Y		Y		Y	Y			Y	
5	Y		Y		Y	Y				Y
5	Y		Y		Y		Y	Y		
5	Y		Y		Y		Y		Y	
5	Y		Y		Y		Y			Y
5	Y		Y		Y			Y	Y	
5	Y		Y		Y			Y		Y
5	Y		Y		Y				Y	Y
5	Y		Y			Y	Y	Y		

TABLE 13-continued

Num. of OTUs	OTU Clade									
	clade_309 *Blautia luti*	clade_309 *Blautia schinkii*	clade_309 *Blautia* sp. M25	clade_408 *Clostridium citroniae*	clade_537 *Clostridium leptum*	clade_444 *Eubacterium rectale*	clade_478 *Faecalibacterium prausnitzii*	clade_360 Lachnospiraceae *bacterium* 2_1_58FAA	clade_537 *Ruminococcus bromii*	clade_478 *Subdoligranulum variabile*
5	Y		Y			Y	Y		Y	
5	Y		Y			Y	Y			Y
5	Y		Y			Y		Y	Y	
5	Y		Y			Y		Y		Y
5	Y		Y			Y			Y	Y
5	Y		Y				Y	Y	Y	
5	Y		Y				Y	Y		Y
5	Y		Y				Y		Y	Y
5	Y		Y					Y	Y	Y
5	Y			Y	Y	Y	Y			
5	Y			Y	Y	Y		Y		
5	Y			Y	Y	Y			Y	
5	Y			Y	Y	Y				Y
5	Y			Y	Y		Y	Y		
5	Y			Y	Y		Y		Y	
5	Y			Y	Y		Y			Y
5	Y			Y	Y			Y	Y	
5	Y			Y	Y			Y		Y
5	Y			Y	Y				Y	Y
5	Y			Y		Y	Y	Y		
5	Y			Y		Y	Y		Y	
5	Y			Y		Y	Y			Y
5	Y			Y		Y		Y	Y	
5	Y			Y		Y		Y		Y
5	Y			Y		Y			Y	Y
5	Y			Y			Y	Y	Y	
5	Y			Y			Y	Y		Y
5	Y			Y			Y		Y	Y
5	Y			Y				Y	Y	Y
5	Y				Y	Y	Y	Y		
5	Y				Y	Y	Y		Y	
5	Y				Y	Y	Y			Y
5	Y				Y	Y		Y	Y	
5	Y				Y	Y		Y		Y
5	Y				Y	Y			Y	Y
5	Y				Y		Y	Y	Y	
5	Y				Y		Y	Y		Y
5	Y				Y		Y		Y	Y
5	Y				Y			Y	Y	Y
5	Y					Y	Y	Y	Y	
5	Y					Y	Y	Y		Y
5	Y					Y	Y		Y	Y
5	Y					Y		Y	Y	Y
5	Y						Y	Y	Y	Y
5		Y	Y	Y	Y	Y				
5		Y	Y	Y	Y		Y			
5		Y	Y	Y	Y			Y		

TABLE 13-continued

Num. of OTUs	OTU Clade clade_309 *Blautia luti*	clade_309 *Blautia schinkii*	clade_309 *Blautia* sp. M25	clade_408 *Clostridium citroniae*	clade_537 *Clostridium leptum*	clade_444 *Eubacterium rectale*	clade_478 *Faecalibacterium prausnitzii*	clade_360 Lachnospiraceae *bacterium* 2_1_58FAA	clade_537 *Ruminococcus bromii*	clade_478 *Subdoligranulum variabile*
5		Y	Y	Y	Y				Y	
5		Y	Y	Y	Y					Y
5		Y	Y	Y		Y	Y			
5		Y	Y	Y		Y		Y		
5		Y	Y	Y		Y			Y	
5		Y	Y	Y		Y				Y
5		Y	Y	Y			Y	Y		
5		Y	Y	Y			Y		Y	
5		Y	Y	Y			Y			Y
5		Y	Y	Y				Y	Y	
5		Y	Y	Y				Y		Y
5		Y	Y	Y					Y	Y
5		Y	Y		Y	Y	Y			
5		Y	Y		Y	Y		Y		
5		Y	Y		Y	Y			Y	
5		Y	Y		Y	Y				Y
5		Y	Y		Y		Y	Y		
5		Y	Y		Y		Y		Y	
5		Y	Y		Y		Y			Y
5		Y	Y		Y			Y	Y	
5		Y	Y		Y			Y		Y
5		Y	Y		Y				Y	Y
5		Y	Y			Y	Y	Y		
5		Y	Y			Y	Y		Y	
5		Y	Y			Y	Y			Y
5		Y	Y			Y		Y	Y	
5		Y	Y			Y		Y		Y
5		Y	Y			Y			Y	Y
5		Y	Y				Y	Y	Y	
5		Y	Y				Y	Y		Y
5		Y	Y				Y		Y	Y
5		Y	Y					Y	Y	Y
5		Y		Y	Y	Y	Y			
5		Y		Y	Y	Y		Y		
5		Y		Y	Y	Y			Y	
5		Y		Y	Y	Y				Y
5		Y		Y	Y		Y	Y		
5		Y		Y	Y		Y		Y	
5		Y		Y	Y		Y			Y
5		Y		Y	Y			Y	Y	
5		Y		Y	Y			Y		Y
5		Y		Y	Y				Y	Y
5		Y		Y		Y	Y	Y		
5		Y		Y		Y	Y		Y	
5		Y		Y		Y	Y			Y
5		Y		Y		Y		Y	Y	
5		Y		Y		Y		Y		Y

TABLE 13-continued

	OTU Clade									
Num. of OTUs	clade_309 *Blautia luti*	clade_309 *Blautia schinkii*	clade_309 *Blautia* sp. M25	clade_408 *Clostridium citroniae*	clade_537 *Clostridium leptum*	clade_444 *Eubacterium rectale*	clade_478 *Faecalibacterium prausnitzii*	clade_360 Lachnospiraceae *bacterium* 2_1_58FAA	clade_537 *Ruminococcus bromii*	clade_478 *Subdoligranulum variabile*
5		Y		Y		Y			Y	Y
5		Y		Y			Y	Y	Y	
5		Y		Y			Y	Y		Y
5		Y		Y			Y		Y	Y
5		Y		Y				Y	Y	Y
5		Y			Y	Y	Y	Y		
5		Y			Y	Y	Y		Y	
5		Y			Y	Y	Y			Y
5		Y			Y	Y		Y	Y	
5		Y			Y	Y		Y		Y
5		Y			Y	Y			Y	Y
5		Y			Y		Y	Y	Y	
5		Y			Y		Y	Y		Y
5		Y			Y		Y		Y	Y
5		Y			Y			Y	Y	Y
5		Y				Y	Y	Y	Y	
5		Y				Y	Y	Y		Y
5		Y				Y	Y		Y	Y
5		Y				Y		Y	Y	Y
5		Y					Y	Y	Y	Y
5			Y	Y	Y	Y	Y			
5			Y	Y	Y	Y		Y		
5			Y	Y	Y	Y			Y	
5			Y	Y	Y	Y				Y
5			Y	Y	Y		Y	Y		
5			Y	Y	Y		Y		Y	
5			Y	Y	Y		Y			Y
5			Y	Y	Y			Y	Y	
5			Y	Y	Y			Y		Y
5			Y	Y	Y				Y	Y
5			Y	Y		Y	Y	Y		
5			Y	Y		Y	Y		Y	
5			Y	Y		Y	Y			Y
5			Y	Y		Y		Y	Y	
5			Y	Y		Y		Y		Y
5			Y	Y		Y			Y	Y
5			Y	Y			Y	Y	Y	
5			Y	Y			Y	Y		Y
5			Y	Y			Y		Y	Y
5			Y	Y				Y	Y	Y
5			Y		Y	Y	Y	Y		
5			Y		Y	Y	Y		Y	
5			Y		Y	Y	Y			Y
5			Y		Y	Y		Y	Y	
5			Y		Y	Y		Y		Y
5			Y		Y	Y			Y	Y
5			Y		Y		Y	Y	Y	

TABLE 13-continued

Num. of OTUs	OTU Clade clade_309 *Blautia luti*	clade_309 *Blautia schinkii*	clade_309 *Blautia* sp. M25	clade_408 *Clostridium citroniae*	clade_537 *Clostridium leptum*	clade_444 *Eubacterium rectale*	clade_478 *Faecalibacterium prausnitzii*	clade_360 Lachnospiraceae *bacterium* 2_1_58FAA	clade_537 *Ruminococcus bromii*	clade_478 *Subdoligranulum variabile*
5			Y		Y		Y	Y		Y
5			Y		Y		Y		Y	Y
5			Y		Y			Y	Y	Y
5			Y			Y	Y	Y	Y	
5			Y			Y	Y	Y		Y
5			Y			Y	Y		Y	Y
5			Y			Y		Y	Y	Y
5			Y				Y	Y	Y	Y
5				Y	Y	Y	Y	Y		
5				Y	Y	Y	Y		Y	
5				Y	Y	Y	Y			Y
5				Y	Y	Y		Y	Y	
5				Y	Y	Y		Y		Y
5				Y	Y	Y			Y	Y
5				Y	Y		Y	Y	Y	
5				Y	Y		Y	Y		Y
5				Y	Y		Y		Y	Y
5				Y	Y			Y	Y	Y
5				Y		Y	Y	Y	Y	
5				Y		Y	Y	Y		Y
5				Y		Y	Y		Y	Y
5				Y		Y		Y	Y	Y
5				Y			Y	Y	Y	Y
5					Y	Y	Y	Y	Y	
5					Y	Y	Y	Y		Y
5					Y	Y	Y		Y	Y
5					Y	Y		Y	Y	Y
5					Y		Y	Y	Y	Y
5						Y	Y	Y	Y	Y
4	Y	Y	Y	Y						
4	Y	Y	Y		Y					
4	Y	Y	Y			Y				
4	Y	Y	Y				Y			
4	Y	Y	Y					Y		
4	Y	Y	Y						Y	
4	Y	Y	Y							Y
4	Y	Y		Y	Y					
4	Y	Y		Y		Y				
4	Y	Y		Y			Y			
4	Y	Y		Y				Y		
4	Y	Y		Y					Y	
4	Y	Y		Y						Y
4	Y	Y			Y	Y				
4	Y	Y			Y		Y			
4	Y	Y			Y			Y		
4	Y	Y			Y				Y	
4	Y	Y			Y					Y

TABLE 13-continued

Num. of OTUs	OTU Clade									
	clade_309 *Blautia luti*	clade_309 *Blautia schinkii*	clade_309 *Blautia* sp. M25	clade_408 *Clostridium citroniae*	clade_537 *Clostridium leptum*	clade_444 *Eubacterium rectale*	clade_478 *Faecalibacterium prausnitzii*	clade_360 Lachnospiraceae *bacterium* 2_1_58FAA	clade_537 *Ruminococcus bromii*	clade_478 *Subdoligranulum variabile*
4	Y	Y				Y	Y			
4	Y	Y				Y		Y		
4	Y	Y				Y			Y	
4	Y	Y				Y				Y
4	Y	Y					Y	Y		
4	Y	Y					Y		Y	
4	Y	Y					Y			Y
4	Y	Y						Y	Y	
4	Y	Y						Y		Y
4	Y	Y							Y	Y
4	Y		Y	Y	Y					
4	Y		Y	Y		Y				
4	Y		Y	Y			Y			
4	Y		Y	Y				Y		
4	Y		Y	Y					Y	
4	Y		Y	Y						Y
4	Y		Y		Y	Y				
4	Y		Y		Y		Y			
4	Y		Y		Y			Y		
4	Y		Y		Y				Y	
4	Y		Y		Y					Y
4	Y		Y			Y	Y			
4	Y		Y			Y		Y		
4	Y		Y			Y			Y	
4	Y		Y			Y				Y
4	Y		Y				Y	Y		
4	Y		Y				Y		Y	
4	Y		Y				Y			Y
4	Y		Y					Y	Y	
4	Y		Y					Y		Y
4	Y		Y						Y	Y
4	Y			Y	Y	Y				
4	Y			Y	Y		Y			
4	Y			Y	Y			Y		
4	Y			Y	Y				Y	
4	Y			Y	Y					Y
4	Y			Y		Y	Y			
4	Y			Y		Y		Y		
4	Y			Y		Y			Y	
4	Y			Y		Y				Y
4	Y			Y			Y	Y		
4	Y			Y			Y		Y	
4	Y			Y			Y			Y
4	Y			Y				Y	Y	
4	Y			Y				Y		Y
4	Y			Y					Y	Y
4	Y				Y	Y	Y			

TABLE 13-continued

Num. of OTUs	OTU Clade									
	clade_309 *Blautia luti*	clade_309 *Blautia schinkii*	clade_309 *Blautia* sp. M25	clade_408 *Clostridium citroniae*	clade_537 *Clostridium leptum*	clade_444 *Eubacterium rectale*	clade_478 *Faecalibacterium prausnitzii*	clade_360 Lachnospiraceae *bacterium* 2_1_58FAA	clade_537 *Ruminococcus bromii*	clade_478 *Subdoligranulum variabile*
4	Y				Y	Y		Y		
4	Y				Y	Y			Y	
4	Y				Y	Y				Y
4	Y				Y		Y	Y		
4	Y				Y		Y		Y	
4	Y				Y		Y			Y
4	Y				Y			Y	Y	
4	Y				Y			Y		Y
4	Y				Y				Y	Y
4	Y					Y	Y	Y		
4	Y					Y	Y		Y	
4	Y					Y	Y			Y
4	Y					Y		Y	Y	
4	Y					Y		Y		Y
4	Y					Y			Y	Y
4	Y						Y	Y	Y	
4	Y						Y	Y		Y
4	Y						Y		Y	Y
4	Y							Y	Y	Y
4		Y	Y	Y	Y					
4		Y	Y	Y		Y				
4		Y	Y	Y			Y			
4		Y	Y	Y				Y		
4		Y	Y	Y					Y	
4		Y	Y	Y						Y
4		Y	Y		Y	Y				
4		Y	Y		Y		Y			
4		Y	Y		Y			Y		
4		Y	Y		Y				Y	
4		Y	Y		Y					Y
4		Y	Y			Y	Y			
4		Y	Y			Y		Y		
4		Y	Y			Y			Y	
4		Y	Y			Y				Y
4		Y	Y				Y	Y		
4		Y	Y				Y		Y	
4		Y	Y				Y			Y
4		Y	Y					Y	Y	
4		Y	Y					Y		Y
4		Y	Y						Y	Y
4		Y		Y	Y	Y				
4		Y		Y	Y		Y			
4		Y		Y	Y			Y		
4		Y		Y	Y				Y	
4		Y		Y	Y					Y
4		Y		Y		Y	Y			
4		Y		Y		Y		Y		

TABLE 13-continued

Num. of OTUs	OTU Clade clade_309 *Blautia luti*	clade_309 *Blautia schinkii*	clade_309 *Blautia* sp. M25	clade_408 *Clostridium citroniae*	clade_537 *Clostridium leptum*	clade_444 *Eubacterium rectale*	clade_478 *Faecalibacterium prausnitzii*	clade_360 Lachnospiraceae *bacterium* 2_1_58FAA	clade_537 *Ruminococcus bromii*	clade_478 *Subdoligranulum variabile*
4		Y		Y		Y			Y	
4		Y		Y		Y				Y
4		Y		Y			Y	Y		
4		Y		Y			Y		Y	
4		Y		Y			Y			Y
4		Y		Y				Y	Y	
4		Y		Y				Y		Y
4		Y		Y					Y	Y
4		Y			Y	Y	Y			
4		Y			Y	Y		Y		
4		Y			Y	Y			Y	
4		Y			Y	Y				Y
4		Y			Y		Y	Y		
4		Y			Y		Y		Y	
4		Y			Y		Y			Y
4		Y			Y			Y	Y	
4		Y			Y			Y		Y
4		Y			Y				Y	Y
4		Y				Y	Y	Y		
4		Y				Y	Y		Y	
4		Y				Y	Y			Y
4		Y				Y		Y	Y	
4		Y				Y		Y		Y
4		Y				Y			Y	Y
4		Y					Y	Y	Y	
4		Y					Y	Y		Y
4		Y					Y		Y	Y
4		Y						Y	Y	Y
4			Y	Y	Y	Y				
4			Y	Y	Y		Y			
4			Y	Y	Y			Y		
4			Y	Y	Y				Y	
4			Y	Y	Y					Y
4			Y	Y		Y	Y			
4			Y	Y		Y		Y		
4			Y	Y		Y			Y	
4			Y	Y		Y				Y
4			Y	Y			Y	Y		
4			Y	Y			Y		Y	
4			Y	Y			Y			Y
4			Y	Y				Y	Y	
4			Y	Y				Y		Y
4			Y	Y					Y	Y
4			Y		Y	Y	Y			
4			Y		Y	Y		Y		
4			Y		Y	Y			Y	
4			Y		Y	Y				Y

TABLE 13-continued

Num. of OTUs	OTU Clade clade_309 *Blautia luti*	clade_309 *Blautia schinkii*	clade_309 *Blautia* sp. M25	clade_408 *Clostridium citroniae*	clade_537 *Clostridium leptum*	clade_444 *Eubacterium rectale*	clade_478 *Faecalibacterium prausnitzii*	clade_360 Lachnospiraceae *bacterium* 2_1_58FAA	clade_537 *Ruminococcus bromii*	clade_478 *Subdoligranulum variabile*
4			Y		Y		Y	Y		
4			Y		Y		Y		Y	
4			Y		Y		Y			Y
4			Y		Y			Y	Y	
4			Y		Y			Y		Y
4			Y		Y				Y	Y
4			Y			Y	Y	Y		
4			Y			Y	Y		Y	
4			Y			Y	Y			Y
4			Y			Y		Y	Y	
4			Y			Y		Y		Y
4			Y			Y			Y	Y
4			Y				Y	Y	Y	
4			Y				Y	Y		Y
4			Y				Y		Y	Y
4			Y					Y	Y	Y
4				Y	Y	Y	Y			
4				Y	Y	Y		Y		
4				Y	Y	Y			Y	
4				Y	Y	Y				Y
4				Y	Y		Y	Y		
4				Y	Y		Y		Y	
4				Y	Y		Y			Y
4				Y	Y			Y	Y	
4				Y	Y			Y		Y
4				Y	Y				Y	Y
4				Y		Y	Y	Y		
4				Y		Y	Y		Y	
4				Y		Y	Y			Y
4				Y		Y		Y	Y	
4				Y		Y		Y		Y
4				Y		Y			Y	Y
4				Y			Y	Y	Y	
4				Y			Y	Y		Y
4				Y			Y		Y	Y
4				Y				Y	Y	Y
4					Y	Y	Y	Y		
4					Y	Y	Y		Y	
4					Y	Y	Y			Y
4					Y	Y		Y	Y	
4					Y	Y		Y		Y
4					Y	Y			Y	Y
4					Y		Y	Y	Y	
4					Y		Y	Y		Y
4					Y		Y		Y	Y
4					Y			Y	Y	Y
4						Y	Y	Y	Y	

TABLE 13-continued

Num. of OTUs	OTU Clade									
	clade_309 *Blautia luti*	clade_309 *Blautia schinkii*	clade_309 *Blautia* sp. M25	clade_408 *Clostridium citroniae*	clade_537 *Clostridium leptum*	clade_444 *Eubacterium rectale*	clade_478 *Faecalibacterium prausnitzii*	clade_360 Lachnospiraceae *bacterium* 2_1_58FAA	clade_537 *Ruminococcus bromii*	clade_478 *Subdoligranulum variabile*
4						Y	Y	Y		Y
4						Y	Y		Y	Y
4						Y		Y	Y	Y
4							Y	Y	Y	Y

TABLE 14

Post-Treatment Ecologies				
OTU 1	OTU 2	OTU 3	Percent of doses in which the ternary is present	Percent of post-treatment patients in which the ternary is present
Blautia sp. M25	*Faecalibacterium prausnitzii*	*Ruminococcus bromii*	100	75
Blautia sp. M25	*Faecalibacterium prausnitzii*	*Ruminococcus obeum*	100	89
Blautia sp. M25	*Faecalibacterium prausnitzii*	*Ruminococcus* sp. 5_1_39BFAA	100	89
Blautia sp. M25	*Faecalibacterium prausnitzii*	*Ruminococcus torques*	100	93
Blautia sp. M25	*Ruminococcus bromii*	*Ruminococcus obeum*	100	71
Blautia sp. M25	*Ruminococcus bromii*	*Ruminococcus* sp. 5_1_39BFAA	100	71
Blautia sp. M25	*Ruminococcus bromii*	*Ruminococcus torques*	100	75
Blautia sp. M25	*Ruminococcus obeum*	*Ruminococcus* sp. 5_1_39BFAA	100	86
Blautia sp. M25	*Ruminococcus obeum*	*Ruminococcus torques*	100	89
Blautia sp. M25	*Ruminococcus* sp. 5_1_39BFAA	*Ruminococcus torques*	100	89
Clostridium hathewayi	*Clostridium orbiscindens*	*Escherichia coli*	0	100
Clostridium hathewayi	*Clostridium orbiscindens*	Lachnospiraceae bacterium 3_1_57FAA_CT1	0	100
Clostridium hathewayi	*Clostridium orbiscindens*	*Ruminococcus torques*	0	100
Clostridium hathewayi	*Escherichia coli*	Lachnospiraceae bacterium 3_1_57FAA_CT1	0	100
Clostridium hathewayi	*Escherichia coli*	*Ruminococcus torques*	0	100
Clostridium orbiscindens	*Escherichia coli*	Lachnospiraceae bacterium 3_1_57FAA_CT1	0	100
Clostridium orbiscindeas	*Escherichia coli*	*Ruminococcus torques*	0	100
Clostridium orbiscindens	*Ruminococcus torques*	*Streptococcus salivarius*	0	100
Faecalibacterium prausnitzii	*Ruminococcus bromii*	*Ruminococcus obeum*	100	75
Faecalibacterium prausnitzii	*Ruminococcus bromii*	*Ruminococcus* sp. 5_1_39BFAA	100	71
Faecalibacterium prausnitzii	*Ruminococcus bromii*	*Ruminococcus torques*	100	75
Faecalibacterium prausnitzii	*Ruminococcus obeum*	*Ruminococcus* sp. 5_1_39BFAA	100	86
Faecalibacterium *prausnitzii*	*Ruminococcus obeum*	*Ruminococcus torques*	100	93
Faecalibacterium prausnitzii	*Ruminococcus* sp. 5_1_39BFAA	*Ruminococcus torques*	100	89
Ruminococcus bromii	*Ruminococcus obeum*	*Ruminococcus* sp. 5_1_9BFAA	100	68
Ruminococcus bromii	*Ruminococcus obeum*	*Ruminococcus torques*	100	79
Ruminococcus bromii	*Ruminococcus* sp. 5_1_39BFAA	*Ruminococcus torques*	100	71
Ruminococcus obeum	*Ruminococcus* sp. 5_1_39BFAA	*Ruminococcus torques*	100	89

TABLE 15

Engrafting OTUs		
OTU 1	OTU 2	OTU 3
Blautia sp. M25	*Clostridiales* sp. SSC/2	*Coprococcus catus*
Blautia sp. M25	*Clostridiales* sp. SSC/2	*Faecalibacterium prausnitzii*
Blautia sp. M25	*Clostridiales* sp. SSC/2	*Ruminococcus bromii*
Blautia sp. M25	*Clostridiales* sp. SSC/2	*Ruminococcus obeum*
Blautia sp. M25	*Clostridiales* sp. SSC/2	*Ruminococcus* sp. 5_1_39BFAA
Blautia sp. M25	*Clostridiales* sp. SSC/2	*Ruminococcus torques*
Blautia sp. M25	*Coproeoceus catus*	*Faecalibacterium prausnitzii*
Blautia sp. M25	*Coprococcus catus*	*Ruminococcus bromii*
Blautia sp. M25	*Coproeoceus catus*	*Ruminococcus obeum*
Blautia sp. M25	*Coprococcus catus*	*Ruminococcus* sp. 5_1_39BFAA

TABLE 15-continued

Engrafting OTUs		
OTU 1	OTU 2	OTU 3
Blautia sp. M25	*Coprococcus catus*	*Ruminococcus torques*
Blautia sp. M25	*Eubacterium hallii*	*Faecalibacterium prausnitzii*
Blautia sp. M25	*Eubacterium hallii*	*Ruminococcus bromii*
Blautia sp. M25	*Eubacterium hallii*	*Ruminococcus obeum*
Blautia sp. M25	*Eubacterium hallii*	*Ruminococcus* sp. 5_1_39BFAA
Blautia sp. M25	*Eubacterium hallii*	*Ruminococcus torques*
Blautia sp. M25	*Faecalibacterium prausnitzii*	*Gemmiger formicilis*
Blautia sp. M25	*Gemmiger formicilis*	*Ruminococcus bromii*
Blautia sp. M25	*Gemmiger formicilis*	*Ruminococcus obeum*
Blautia sp. M25	*Gemmiger formicilis*	*Ruminococcus* sp. 5_1_39BFAA
Blautia sp. M25	*Gemmiger formicilis*	*Ruminococcus torques*
Clostridiales sp. SSC/2	*Coprococcus catus*	*Faecalibacterium prausnitzii*
Clostridiales sp. SSC/2	*Coprococcus catus*	*Ruminococcus bromii*
Clostridiales sp. SSC/2	*Coprococcus catus*	*Ruminococcus obeum*
Clostridiales sp. SSC/2	*Coprococcus catus*	*Ruminococcus* sp. 5_1_39BFAA
Clostridiales sp. SSC/2	*Coprococcus catus*	*Ruminococcus torques*
Clostridiales sp. SSC/2	*Faecalibacterium prausnitzii*	*Ruminococcus bromii*
Clostridiales sp. SSC/2	*Faecalibacterium prausnitzii*	*Ruminococcus obeum*
Clostridiales sp. SSC/2	*Faecalibacterium prausnitzii*	*Ruminococcus* sp. 5_1_39BFAA
Clostridiales sp. SSC/2	*Faecalibacterium prausnitzii*	*Ruminococcus torques*
Clostridiales sp. SSC/2	*Ruminococcus bromii*	*Ruminococcus obeum*
Clostridiales sp. SSC/2	*Ruminococcus bromii*	*Ruminococcus* sp. 5_1_39BFAA
Clostridiales sp. SSC/2	*Ruminococcus bromii*	*Ruminococcus torques*
Clostridiales sp. SSC/2	*Ruminococcus obeum*	*Ruminococcus* sp. 5_1_39BFAA
Clostridiales sp. SSC/2	*Ruminococcus obeum*	*Ruminococcus torques*
Clostridiales sp. SSC/2	*Ruminococcus* sp. 5_1_39BFAA	*Ruminococcus torques*
Coprococcus catus	*Faecalibacterium prausnitzii*	*Ruminococcus bromii*
Coprococcus catus	*Faecalibacterium prausnitzii*	*Ruminococcus obeum*
Coprococcus catus	*Faecalibacterium prausnitzii*	*Ruminococcus* sp. 5_1_39BFAA
Coprococcus catus	*Faecalibacterium prausnitzii*	*Ruminococcus torques*
Coprococcus catus	*Ruminococcus bromii*	*Ruminococcus obeum*
Coprococcus catus	*Ruminococcus bromii*	*Ruminococcus* sp. 5_1_39BFAA
Coprococcus catus	*Ruminococcus bromii*	*Ruminococcus torques*
Coprococcus catus	*Ruminococcus obeum*	*Ruminococcus* sp. 5_1_39BFAA
Coprococcus catus	*Ruminococcus obeum*	*Ruminococcus torques*
Coprococcus catus	*Ruminococcus* sp. 5_1_39BFAA	*Ruminococcus torques*
Eubacterium hallii	*Faecalibacterium prausnitzii*	*Ruminococcus bromii*
Eubacterium hallii	*Faecalibacterium prausnitzii*	*Ruminococcus obeum*
Eubacterium hallii	*Faecalibacterium prausnitzii*	*Ruminococcus* sp. 5_1_39BFAA
Eubacterium hallii	*Faecalibacterium prausnitzii*	*Ruminococcus torques*
Eubacterium hallii	*Ruminococcus bromii*	*Ruminococcus obeum*
Eubacterium hallii	*Ruminococcus bromii*	*Ruminococcus* sp. 5_1_39BFAA
Eubacterium hallii	*Ruminococcus bromii*	*Ruminococcus torques*
Eubacterium hallii	*Ruminococcus obeum*	*Ruminococcus* sp. 5_1_39BFAA
Eubacterium hallii	*Ruminococcus obeum*	*Ruminococcus torques*
Eubacterium hallii	*Ruminococcus* sp. 5_1_39BFAA	*Ruminococcus torques*
Faecalibacterium prausnitzii	*Gemmiger formicilis*	*Ruminococcus bromii*
Faecalibacterium prausnitzii	*Gemmiger formicilis*	*Ruminococcus obeum*
Faecalibacterium prausnitzii	*Gemmiger formicilis*	*Ruminococcus* sp. 5_1_39BFAA
Faecalibacterium prausnitzii	*Gemmiger formicilis*	*Ruminococcus torques*
Gemmiger formicilis	*Ruminococcus bromii*	*Ruminococcus obeum*
Gemmiger formicilis	*Ruminococcus bromii*	*Ruminococcus* sp. 5_1_39BFAA
Gemmiger formicilis	*Ruminococcus bromii*	*Ruminococcus torques*
Gemmiger formicilis	*Ruminococcus obeum*	*Ruminococcus* sp. 5_1_39BFAA

TABLE 15-continued

Engrafting OTUs		
OTU 1	OTU 2	OTU 3
Gemmiger formicilis	*Ruminococcus obeum*	*Ruminococcus torques*
Gemmiger formicilis	*Ruminococcus* sp. 5_1_39BFAA	*Ruminococcus torques*

TABLE 16

Augmenting OTUs			
OTU 1	OTU 2	OTU 3	Percent of post-treatment patients in which the ternary is present
Anaerotruncus colihominis	*Clostridium orbiscindens*	*Escherichia coli*	75
Clostridium lactatifermentans	*Clostridium orbiscindens*	*Escherichia coli*	79
Clostridium lactatifermentans	*Clostridium orbiscindens*	*Streptococcus salivarius*	79
Clostridium lactatifermentans	*Escherichia coli*	*Streptococcus salivarius*	75
Clostridium orbiscindens	*Clostridium* sp. NML 04A032	*Escherichia coli*	89
Clostridium orbiscindens	*Clostridium* sp. NML 04A032	*Oscillibacter* sp. G2	89
Clostridium orbiscindens	*Clostridium* sp. NML 04A032	*Streptococcus salivarius*	93
Clostridium orbiscindens	*Escherichia coli*	*Klebsiella* sp. SRC_DSD2	75
Clostridium orbiscindens	*Escherichia coli*	*Oscillibacter* sp. G2	89
Clostridium orbiscindens	*Escherichia coli*	*Streptococcus salivarius*	96
Clostridium orbiscindens	*Oscillibacter* sp. G2	*Streptococcus salivarius*	89
Clostridium sp. NML 04A032	*Escherichia coli*	*Oscillibacter* sp. G2	82
Clostridium sp. NML 04A032	*Escherichia coli*	*Streptococcus salivarius*	86
Clostridium sp. NML 04A032	*Oscillibacter* sp. G2	*Streptococcus salivarius*	86
Escherichia coli	*Oscillibacter* sp. G2	*Streptococcus salivarius*	82

TABLE 17

Ternary OTU combinations in administered spore ecology doses resulting in augmentation or engraftment of the OTU *Clostridiales* sp. SM4/1			
OTU 1	OTU 2	OTU 3	Percent of doses in which the ternary is present
Clostridium saccharogumia	*Eubacterium rectale*	*Faecalibacterium prausnitzii*	85
Clostridium saccharogumia	*Eubacterium rectale*	*Ruminococcus torques*	85
Clostridium saccharogumia	*Faecalibacterium prausnitzii*	*Ruminococcus torques*	85
Blautia sp. M25	*Clostridium saccharogumia*	*Eubacterium rectale*	85
Blautia sp. M25	*Clostridium saccharogumia*	*Faecalibacterium prausnitzii*	85
Blautia sp. M25	*Clostridium saccharogumia*	*Ruminococcus torques*	85
Clostridium saccharogumia	*Eubacterium rectale*	*Ruminococcus obeum*	85
Clostridium saccharogumia	*Eubacterium rectale*	*Ruminococcus* sp. 5_1_39BFAA	85
Clostridium saccharogumia	*Faecalibacterium prausnitzii*	*Ruminococcus obeum*	85
Clostridium saccharogumia	*Faecalibacterium prausnitzii*	*Ruminococcus* sp. 5_1_39BFAA	85
Clostridium saccharogumia	*Ruminococcus obeum*	*Ruminococcus torques*	85
Clostridium saccharogumia	*Ruminococcus* sp. 5_1_39BFAA	*Ruminococcus torques*	85
Blautia sp. M25	*Clostridium saccharogumia*	*Ruminococcus* sp. 5_1_39BFAA	85

TABLE 17-continued

Ternary OTU combinations in administered spore ecology doses resulting in augmentation or engraftment of the OTU *Clostridiales* sp. SM4/1

OTU 1	OTU 2	OTU 3	Percent of doses in which the ternary is present
Blautia sp. M25	*Clostridium saccharogumia*	*Ruminococcus obeum*	85
Clostridium saccharogumia	*Ruminococcus obeum*	*Ruminococcus* sp. 5_1_39BFAA	85
Clostridium saccharogumia	*Faecalibacterium prausnitzii*	*Ruminococcus bromii*	85
Blautia sp. M25	*Clostridium saccharogumia*	*Ruminococcus bromii*	85
Clostridium saccharogumia	*Eubacterium rectale*	*Ruminococcus bromii*	85
Clostridium saccharogumia	*Ruminococcus bromii*	*Ruminococcus obeum*	85
Clostridium saccharogumia	*Ruminococcus bromii*	*Ruminococcus* sp. 5_1_39BFAA	85
Clostridium saccharogumia	*Ruminococcus bromii*	*Ruminococcus torques*	85
Clostridiales sp. SSC/2	*Clostridium saccharogumia*	*Eubacterium rectale*	80
Clostridiales sp. SSC/2	*Clostridium saccharogumia*	*Faecalibacterium prausnitzii*	80
Clostridiales sp. SSC/2	*Clostridium saccharogumia*	*Ruminococcus torques*	80
Blautia sp. M25	*Clostridiales* sp. *SSC/2*	*Clostridium saccharogumia*	80
Blautia sp. M25	*Clostridium saccharogumia*	*Ruminococcus lactaris*	80
Clostridiales sp. *SSC/2*	*Clostridium saccharogumia*	*Ruminococcus obeum*	80
Clostridiales sp. SSC/2	*Clostridium saccharogumia*	*Ruminococcus* sp. 5_1_39BFAA	80
Clostridium lavalense	*Clostridium saccharogumia*	*Eubacterium rectale*	80
Clostridium lavalense	*Clostridium saccharogumia*	*Faecalibacterium prausnitzii*	80
Clostridium lavalense	*Clostridium saccharogumia*	*Ruminococcus torques*	80
Clostridium saccharogumia	*Eubacterium rectale*	*Ruminococcus lactaris*	80
Clostridium saccharogumia	*Ruminococcus lactaris*	*Ruminococcus* sp. 5_1_39BFAA	80
Clostridium saccharogumia	*Ruminococcus lactaris*	*Ruminococcus torques*	80
Blautia sp. M25	*Clostridium lavalense*	*Clostridium saccharogumia*	80
Clostridium asparagiforme	*Clostridium saccharogumia*	*Eubacterium rectale*	80
Clostridium asparagiforme	*Clostridium saccharogumia*	*Faecalibacterium prausnitzii*	80
Clostridium asparagiforme	*Clostridium saccharogumia*	*Ruminococcus torques*	80
Clostridium lavalense	*Clostridium saccharogumia*	*Ruminococcus obeum*	80
Clostridium lavalense	*Clostridium saccharogumia*	*Ruminococcus* sp. 5_1_39BFAA	80
Clostridium saccharogumia	*Eubacterium rectale*	*Gemmiger formicilis*	80
Clostridium saccharogumia	*Faecalibacterium prausnitzii*	*Ruminococcus lactaris*	80
Clostridium saccharogumia	*Gemmiger formicilis*	*Ruminococcus torques*	80
Clostridium saccharogumia	*Ruminococcus lactaris*	*Ruminococcus obeum*	80
Clostridium saccharogumia	*Faecalibacterium prausnitzii*	*Gemmiger formicilis*	80
Blautia sp. M25	*Clostridium asparagiforme*	*Clostridium saccharogumia*	80
Blautia sp. M25	*Clostridium saccharogumia*	*Gemmiger formicilis*	80

TABLE 17-continued

Ternary OTU combinations in administered spore ecology doses resulting in augmentation or engraftment of the OTU *Clostridiales* sp. SM4/1

OTU 1	OTU 2	OTU 3	Percent of doses in which the ternary is present
Clostridium asparagiforme	*Clostridium saccharogumia*	*Ruminococcus obeum*	80
Clostridium asparagiforme	*Clostridium saccharogumia*	*Ruminococcus* sp. 5_1_39BFAA	80
Clostridium saccharogumia	*Coprococcus comes*	*Eubacterium rectale*	80
Clostridium saccharogumia	*Coprococcus comes*	*Faecalibacterium prausnitzii*	80
Clostridium saccharogumia	*Coprococcus comes*	*Ruminococcus obeum*	80
Clostridium saccharogumia	*Coprococcus comes*	*Ruminococcus torques*	80
Clostridium saccharogumia	*Dorea formicigenerans*	*Eubacterium rectale*	80
Clostridium saccharogumia	*Dorea formicigenerans*	*Faecalibacterium prausnitzii*	80
Clostridium saccharogumia	*Dorea formicigenerans*	*Ruminococcus obeum*	80
Clostridium saccharogumia	*Dorea formicigenerans*	*Ruminococcus torques*	80
Clostridium saccharogumia	*Gemmiger formicilis*	*Ruminococcus obeum*	80
Clostridium saccharogumia	*Gemmiger formicilis*	*Ruminococcus* sp. 5_1_39BFAA	80
Clostridium asparagiforme	*Clostridium lavalense*	*Clostridium saccharogumia*	80
Blautia sp. M25	*Clostridium saccharogumia*	*Coprococcus comes*	80
Blautia sp. M25	*Clostridium saccharogumia*	*Dorea formicigenerans*	80
Blautia sp. M25	*Clostridium saccharogumia*	*Eubacterium hallii*	80
Clostridiales sp. SSC/2	*Clostridium saccharogumia*	*Ruminococcus bromii*	80
Clostridium asparagiforme	*Clostridium saccharogumia*	*Ruminococcus bromii*	80
Clostridium asparagiforme	*Clostridium saccharogumia*	*Ruminococcus lactaris*	80
Clostridium lavalense	*Clostridium saccharogumia*	*Dorea formicigenerans*	80
Clostridium lavalense	*Clostridium saccharogumia*	*Ruminococcus bromii*	80
Clostridium lavalense	*Clostridium saccharogumia*	*Ruminococcus lactaris*	80
Clostridium saccharogumia	*Coprococcus comes*	*Ruminococcus* sp. 5_1_39BFAA	80
Clostridium saccharogumia	*Dorea formicigenerans*	*Ruminococcus lactaris*	80
Clostridium saccharogumia	*Dorea formicigenerans*	*Ruminococcus* sp. 5_1_39BFAA	80
Clostridium saccharogumia	*Eubacterium hallii*	*Eubacterium rectale*	80
Clostridium saccharogumia	*Eubacterium hallii*	*Faecalibacterium prausnitzii*	80
Clostridium saccharogumia	*Eubacterium hallii*	*Ruminococcus lactaris*	80
Clostridium saccharogumia	*Eubacterium hallii*	*Ruminococcus obeum*	80
Clostridium saccharogumia	*Eubacterium hallii*	*Ruminococcus* sp. 5_1_39BFAA	80
Clostridium saccharogumia	*Eubacterium hallii*	*Ruminococcus torques*	80
Clostridium saccharogumia	*Ruminococcus bromii*	*Ruminococcus lactaris*	80
Clostridium asparagiforme	*Clostridium saccharogumia*	*Coprococcus comes*	80
Clostridium asparagiforme	*Clostridium saccharogumia*	*Eubacterium hallii*	80

TABLE 17-continued

Ternary OTU combinations in administered spore ecology doses resulting in augmentation or engraftment of the OTU *Clostridiales* sp. SM4/1

OTU 1	OTU 2	OTU 3	Percent of doses in which the ternary is present
Clostridium lavalense	*Clostridium saccharogumia*	*Coprococcus comes*	80
Clostridium saccharogumia	*Coprococcus comes*	*Ruminococcus bromii*	80
Clostridium saccharogumia	*Coprococcus comes*	*Ruminococcus lactaris*	80
Clostridium saccharogumia	*Gemmiger formicilis*	*Ruminococcus bromii*	80
Blautia sp. M25	*Clostridium saccharogumia*	*Coprococcus catus*	80
Blautia sp. M25	*Clostridium saccharogumia*	*Dorea longicatena*	80
Clostridium asparagiforme	*Clostridium saccharogumia*	*Dorea formicigenerans*	80
Clostridium asparagiforme	*Clostridium saccharogumia*	*Dorea longicatena*	80
Clostridium lavalense	*Clostridium saccharogumia*	*Dorea longicatena*	80
Clostridium lavalense	*Clostridium saccharogumia*	*Eubacterium hallii*	80
Clostridium saccharogumia	*Coprococcus catus*	*Eubacterium rectale*	80
Clostridium saccharogumia	*Coprococcus catus*	*Faecalibacterium prausnitzii*	80
Clostridium saccharogumia	*Coprococcus catus*	*Ruminococcus* sp. 5_1_39BFAA	80
Clostridium saccharogumia	*Coprococcus catus*	*Ruminococcus torques*	80
Clostridium saccharogumia	*Coprococcus comes*	*Dorea formicigenerans*	80
Clostridium saccharogumia	*Coprococcus comes*	*Eubacterium hallii*	80
Clostridium saccharogumia	*Dorea formicigenerans*	*Eubacterium hallii*	80
Clostridium saccharogumia	*Dorea formicigenerans*	*Ruminococcus bromii*	80
Clostridium saccharogumia	*Dorea longicatena*	*Eubacterium rectale*	80
Clostridium saccharogumia	*Dorea longicatena*	*Faecalibacterium prausnitzii*	80
Clostridium saccharogumia	*Dorea longicatena*	*Ruminococcus obeum*	80
Clostridium saccharogumia	*Dorea longicatena*	*Ruminococcus* sp. 5_1_39BFAA	80
Clostridium saccharogumia	*Dorea longicatena*	*Ruminococcus torques*	80
Clostridium saccharogumia	*Eubacterium hallii*	*Ruminococcus bromii*	80
Clostridiales sp. SSC/2	*Clostridium saccharogumia*	*Coprococcus catus*	80
Clostridium saccharogumia	*Coprococcus catus*	*Ruminococcus obeum*	80
Clostridium saccharogumia	*Dorea longicatena*	*Eubacterium hallii*	80
Clostridium saccharogumia	*Dorea longicatena*	*Ruminococcus bromii*	80
Clostridium saccharogumia	*Dorea longicatena*	*Ruminococcus lactaris*	80
Clostridium saccharogumia	*Dorea formicigenerans*	*Dorea longicatena*	80
Clostridium saccharogumia	*Coprococcus catus*	*Ruminococcus bromii*	80
Clostridium saccharogumia	*Coprococcus comes*	*Dorea longicatena*	80
Blautia sp. M25	*Clostridium algidixylanolyticum*	*Faecalibacterium prausnitzii*	75
Clostridium algidixylanolyticum	*Faecalibacterium prausnitzii*	*Ruminococcus obeum*	75

TABLE 17-continued

Ternary OTU combinations in administered spore ecology doses resulting in augmentation or engraftment of the OTU *Clostridiales* sp. SM4/1

OTU 1	OTU 2	OTU 3	Percent of doses in which the ternary is present
Clostridium algidixylanolyticum	*Faecalibacterium prausnitzii*	*Ruminococcus torques*	75
Blautia sp. M25	*Clostridium algidixylanolyticum*	*Ruminococcus obeum*	75
Blautia sp. M25	*Clostridium algidixylanolyticum*	*Ruminococcus torques*	75
Clostridium algidixylanolyticum	*Faecalibacterium prausnitzii*	*Ruminococcus* sp. 5_1_39BFAA	75
Clostridium algidixylanolyticum	*Ruminococcus obeum*	*Ruminococcus torques*	75
Blautia sp. M25	*Clostridium algidixylanolyticum*	*Ruminococcus* sp. 5_1_39BFAA	75
Clostridium algidixylanolyticum	*Ruminococcus* sp. 5_1_39BFAA	*Ruminococcus torques*	75
Clostridium algidixylanolyticum	*Ruminococcus obeum*	*Ruminococcus* sp. 5_1_39BFAA	75
Clostridiales sp. SSC/2	*Clostridium algidixylanolyticum*	*Ruminococcus torques*	75
Clostridiales sp. SSC/2	*Clostridium algidixylanolyticum*	*Faecalibacterium prausnitzii*	75
Clostridiales sp. SSC/2	*Clostridium algidixylanolyticum*	*Ruminococcus obeum*	75
Clostridiales sp. SSC/2	*Clostridium algidixylanolyticum*	*Ruminococcus* sp. 5_1_39BFAA	75
Clostridium algidixylanolyticum	*Ruminococcus bromii*	*Ruminococcus torques*	75
Blautia sp. M25	*Clostridiales* sp. SSC/2	*Clostridium algidixylanolyticum*	75
Blautia sp. M25	*Clostridium algidixylanolyticum*	*Ruminococcus bromii*	75
Clostridium algidixylanolyticum	*Faecalibacterium prausnitzii*	*Ruminococcus bromii*	75
Clostridium algidixylanolyticum	*Ruminococcus bromii*	*Ruminococcus obeum*	75
Clostridium algidixylanolyticum	*Ruminococcus bromii*	*Ruminococcus* sp. 5_1_39BFAA	75
Clostridiales sp. SSC/2	*Clostridium algidixylanolyticum*	*Ruminococcus bromii*	75
Clostridium algidixylanolyticum	*Coprococcus catus*	*Faecalibacterium prausnitzii*	75
Clostridium algidixylanolyticum	*Coprococcus catus*	*Ruminococcus* sp. 5_1_39BFAA	75
Blautia sp. M25	*Clostridium algidixylanolyticum*	*Coprococcus catus*	75
Clostridium algidixylanolyticum	*Coprococcus catus*	*Ruminococcus obeum*	75
Clostridium algidixylanolyticum	*Coprococcus catus*	*Ruminococcus torques*	75
Clostridiales sp. SSC/2	*Clostridium saccharogumia*	*Ruminococcus lactaris*	75
Clostridiales sp. SSC/2	*Clostridium algidixylanolyticum*	*Coprococcus catus*	75
Clostridiales sp. SSC/2	*Clostridium saccharogumia*	*Gemmiger formicilis*	75
Clostridiales sp. SSC/2	*Clostridium lavalense*	*Clostridium saccharogumia*	75
Clostridiales sp. SSC/2	*Clostridium saccharogumia*	*Dorea formicigenerans*	75
Clostridium algidixylanolyticum	*Coprococcus catus*	*Ruminococcus bromii*	75
Clostridium lavalense	*Clostridium saccharogumia*	*Gemmiger formicilis*	75
Clostridium saccharogumia	*Gemmiger formicilis*	*Ruminococcus lactaris*	75
Clostridiales sp. SSC/2	*Clostridium asparagiforme*	*Clostridium saccharogumia*	75
Clostridiales sp. SSC/2	*Clostridium saccharogumia*	*Coprococcus comes*	75

TABLE 17-continued

Ternary OTU combinations in administered spore ecology doses resulting in augmentation or engraftment of the OTU *Clostridiales* sp. SM4/1

OTU 1	OTU 2	OTU 3	Percent of doses in which the ternary is present
Clostridiales sp. SSC/2	*Clostridium saccharogumia*	*Eubacterium hallii*	75
Clostridium saccharogumia	*Dorea formicigenerans*	*Gemmiger formicilis*	75
Clostridium asparagiforme	*Clostridium saccharogumia*	*Gemmiger formicilis*	75
Clostridium saccharogumia	*Coprococcus comes*	*Gemmiger formicilis*	75
Clostridium asparagiforme	*Clostridium saccharogumia*	*Coprococcus catus*	75
Clostridium lavalense	*Clostridium saccharogumia*	*Coprococcus catus*	75
Clostridium saccharogumia	*Eubacterium hallii*	*Gemmiger formicilis*	75
Blautia sp. M25	*Clostridium saccharogumia*	*Eubacterium ramulus*	75
Clostridiales sp. SSC/2	*Clostridium saccharogumia*	*Dorea longicatena*	75
Clostridiales sp. SSC/2	*Clostridium saccharogumia*	*Eubacterium ramulus*	75
Clostridium saccharogumia	*Coprococcus catus*	*Ruminococcus lactaris*	75
Clostridium saccharogumia	*Eubacterium ramulus*	*Eubacterium rectale*	75
Clostridium saccharogumia	*Eubacterium ramulus*	*Faecalibacterium prausnitzii*	75
Clostridium saccharogumia	*Eubacterium ramulus*	*Ruminococcus obeum*	75
Clostridium saccharogumia	*Eubacterium ramulus*	*Ruminococcus* sp. 5_1_39BFAA	75
Clostridium saccharogumia	*Eubacterium ramulus*	*Ruminococcus torques*	75
Clostridium asparagiforme	*Clostridium saccharogumia*	*Eubacterium ramulus*	75
Clostridium saccharogumia	*Coprococcus catus*	*Dorea longicatena*	75
Clostridium saccharogumia	*Coprococcus catus*	*Eubacterium hallii*	75
Clostridium saccharogumia	*Coprococcus catus*	*Gemmiger formicilis*	75
Clostridium saccharogumia	*Coprococcus comes*	*Eubacterium ramulus*	75
Clostridium saccharogumia	*Dorea longicatena*	*Gemmiger formicilis*	75
Clostridium saccharogumia	*Eubacterium hallii*	*Eubacterium ramulus*	75
Clostridium saccharogumia	*Eubacterium ramulus*	*Ruminococcus lactaris*	75
Clostridium lavalense	*Clostridium saccharogumia*	*Eubacterium ramulus*	75
Clostridium saccharogumia	*Coprococcus catus*	*Coprococcus comes*	75
Clostridium saccharogumia	*Coprococcus catus*	*Dorea formicigenerans*	75
Clostridium saccharogumia	*Dorea formicigenerans*	*Eubacterium ramulus*	75
Clostridium saccharogumia	*Eubacterium ramulus*	*Ruminococcus bromii*	75
Clostridium saccharogumia	*Coprococcus catus*	*Eubacterium ramulus*	75
Clostridium saccharogumia	*Dorea longicatena*	*Eubacterium ramulus*	75

TABLE 18

Ternary OTU combinations in administered spore ecology doses that result in augmentation or engraftment of the OTU *Clostridiales* sp. SSC/2.

OTU 1	OTU 2	OTU 3	Percent of doses in which the ternary is present
Faecalibacterium prausnitzii	*Ruminococcus obeum*	*Turicibacter sanguinis*	85
Faecalibacterium prausnitzii	*Ruminococcus torques*	*Turicibacter sanguinis*	85
Blautia sp. M25	*Faecalibacterium prausnitzii*	*Turicibacter sanguinis*	85
Blautia sp. M25	*Ruminococcus torques*	*Turicibacter sanguinis*	85
Faecalibacterium prausnitzii	*Ruminococcus* sp. 5_1_39BFAA	*Turicibacter sanguinis*	85
Ruminococcus obeum	*Ruminococcus torques*	*Turicibacter sanguinis*	85
Ruminococcus obeum	*Ruminococcus* sp. 5_1_39BFAA	*Turicibacter sanguinis*	85
Ruminococcus sp. 5_1_39BFAA	*Ruminococcus torques*	*Turicibacter sanguinis*	85
Blautia sp. M25	*Ruminococcus obeum*	*Turicibacter sanguinis*	85
Faecalibacterium prausnitzii	*Ruminococcus bromii*	*Turicibacter sanguinis*	85
Ruminococcus bromii	*Ruminococcus obeum*	*Turicibacter sanguinis*	85
Ruminococcus bromii	*Ruminococcus torques*	*Turicibacter sanguinis*	85
Blautia sp. M25	*Ruminococcus* sp. 5_1_39BFAA	*Turicibacter sanguinis*	85
Ruminococcus bromii	*Ruminococcus* sp. 5_1_39BFAA	*Turicibacter sanguinis*	85
Blautia sp. M25	*Ruminococcus bromii*	*Turicibacter sanguinis*	85
Clostridiales sp. SSC/2	*Faecalibacterium prausnitzii*	*Turicibacter sanguinis*	80
Clostridiales sp. SSC/2	*Ruminococcus obeum*	*Turicibacter sanguinis*	80
Clostridiales sp. SSC/2	*Ruminococcus* sp. 5_1_39BFAA	*Turicibacter sanguinis*	80
Clostridiales sp. SSC/2	*Ruminococcus torques*	*Turicibacter sanguinis*	80
Blautia sp. M25	*Clostridiales* sp. SSC/2	*Turicibacter sanguinis*	80
Blautia sp. M25	*Gemmiger formicilis*	*Turicibacter sanguinis*	80
Gemmiger formicilis	*Ruminococcus obeum*	*Turicibacter sanguinis*	80
Gemmiger formicilis	*Ruminococcus* sp. 5_1_39BFAA	*Turicibacter sanguinis*	80
Gemmiger formicilis	*Ruminococcus torques*	*Turicibacter sanguinis*	80
Faecalibacterium prausnitzii	*Gemmiger formicilis*	*Turicibacter sanguinis*	80
Clostridiales sp. SSC/2	*Ruminococcus bromii*	*Turicibacter sanguinis*	80
Eubacterium hallii	*Faecalibacterium prausnitzii*	*Turicibacter sanguinis*	80
Eubacterium hallii	*Ruminococcus obeum*	*Turicibacter sanguinis*	80
Eubacterium hallii	*Ruminococcus* sp. 5_1_39BFAA	*Turicibacter sanguinis*	80
Eubacterium hallii	*Ruminococcus torques*	*Turicibacter sanguinis*	80
Gemmiger formicilis	*Ruminococcus bromii*	*Turicibacter sanguinis*	80
Blautia sp. M25	*Eubacterium hallii*	*Turicibacter sanguinis*	80
Eubacterium hallii	*Ruminococcus bromii*	*Turicibacter sanguinis*	80
Blautia sp. M25	*Coprococcus catus*	*Turicibacter sanguinis*	80
Coprococcus catus	*Faecalibacterium prausnitzii*	*Turicibacter sanguinis*	80

TABLE 18-continued

Ternary OTU combinations in administered spore ecology doses that result in augmentation or engraftment of the OTU *Clostridiales* sp. SSC/2.

OTU 1	OTU 2	OTU 3	Percent of doses in which the ternary is present
Coprococcus catus	*Ruminococcus obeum*	*Turicibacter sanguinis*	80
Coprococcus catus	*Ruminococcus* sp. 5_1_39BFAA	*Turicibacter sanguinis*	80
Coprococcus catus	*Ruminococcus torques*	*Turicibacter sanguinis*	80
Clostridiales sp. SSC/2	*Coprococcus catus*	*Turicibacter sanguinis*	80
Coprococcus catus	*Ruminococcus bromii*	*Turicibacter sanguinis*	80
Clostridium bartlettii	*Faecalibacterium prausnitzii*	*Ruminococcus obeum*	75
Clostridium bartlettii	*Faecalibacterium prausnitzii*	*Ruminococcus torques*	75
Blautia sp. M25	*Clostridium bartlettii*	*Faecalibacterium prausnitzii*	75
Clostridium bartlettii	*Ruminococcus obeum*	*Ruminococcus torques*	75
Clostridium bartlettii	*Ruminococcus* sp. 5_1_39BFAA	*Ruminococcus torques*	75
Blautia sp. M25	*Clostridium bartlettii*	*Ruminococcus torques*	75
Clostridium bartlettii	*Faecalibacterium prausnitzii*	*Ruminococcus* sp. 5_1_39BFAA	75
Blautia sp. M25	*Clostridium bartlettii*	*Ruminococcus obeum*	75
Clostridium bartlettii	*Ruminococcus obeum*	*Ruminococcus* sp. 5_1_39BFAA	75
Blautia sp. M25	*Clostridium bartlettii*	*Ruminococcus* sp. 5_1_39BFAA	75
Clostridium bartlettii	*Ruminococcus bromii*	*Ruminococcus torques*	75
Clostridium bartlettii	*Faecalibacterium prausnitzii*	*Ruminococcus bromii*	75
Clostridium bartlettii	*Ruminococcus bromii*	*Ruminococcus obeum*	75
Blautia sp. M25	*Clostridium bartlettii*	*Ruminococcus bromii*	75
Clostridium bartlettii	*Ruminococcus bromii*	*Ruminococcus* sp. 5_1_39BFAA	75
Eubacterium rectale	*Faecalibacterium prausnitzii*	*Turicibacter sanguinis*	75
Eubacterium rectale	*Ruminococcus obeum*	*Turicibacter sanguinis*	75
Eubacterium rectale	*Ruminococcus* sp. 5_1_39BFAA	*Turicibacter sanguinis*	75
Eubacterium rectale	*Ruminococcus torques*	*Turicibacter sanguinis*	75
Blautia sp. M25	*Eubacterium rectale*	*Turicibacter sanguinis*	75
Clostridium asparagiforme	*Faecalibacterium prausnitzii*	*Turicibacter sanguinis*	75
Clostridium asparagiforme	*Ruminococcus obeum*	*Turicibacter sanguinis*	75
Clostridium asparagiforme	*Ruminococcus* sp. 5_1_39BFAA	*Turicibacter sanguinis*	75
Eubacterium rectale	*Ruminococcus bromii*	*Turicibacter sanguinis*	75
Clostridium asparagiforme	*Ruminococcus torques*	*Turicibacter sanguinis*	75
Blautia sp. M25	*Clostridium asparagiforme*	*Turicibacter sanguinis*	75
Clostridiales sp. SSC/2	*Eubacterium hallii*	*Turicibacter sanguinis*	75
Clostridiales sp. SSC/2	*Gemmiger formicilis*	*Turicibacter sanguinis*	75
Clostridium asparagiforme	*Ruminococcus bromii*	*Turicibacter sanguinis*	75
Clostridium sp. SS2/1	*Ruminococcus torques*	*Turicibacter sanguinis*	75

TABLE 18-continued

Ternary OTU combinations in administered spore ecology doses that result in augmentation or engraftment of the OTU *Clostridiales* sp. SSC/2.

OTU 1	OTU 2	OTU 3	Percent of doses in which the ternary is present
Clostridium asparagiforme	*Eubacterium hallii*	*Turicibacter sanguinis*	75
Clostridium sp. SS2/1	*Faecalibacterium prausnitzii*	*Turicibacter sanguinis*	75
Clostridium sp. SS2/1	*Ruminococcus obeum*	*Turicibacter sanguinis*	75
Clostridium sp. SS2/1	*Ruminococcus* sp. 5_1_39BFAA	*Turicibacter sanguinis*	75
Eubacterium hallii	*Gemmiger formicilis*	*Turicibacter sanguinis*	75
Clostridiales sp. SSC/2	*Clostridium* sp. SS2/1	*Turicibacter sanguinis*	75
Blautia sp. M25	*Clostridium* sp. SS2/1	*Turicibacter sanguinis*	75
Clostridium sp. SS2/1	*Ruminococcus bromii*	*Turicibacter sanguinis*	75
Clostridium sp. SS2/1	*Coprococcus catus*	*Turicibacter sanguinis*	75
Collinsella aerofaciens	*Faecalibacterium prausnitzii*	*Turicibacter sanguinis*	75
Collinsella aerofaciens	*Ruminococcus bromii*	*Turicibacter sanguinis*	75
Collinsella aerofaciens	*Ruminococcus obeum*	*Turicibacter sanguinis*	75
Collinsella aerofaciens	*Ruminococcus torques*	*Turicibacter sanguinis*	75
Coprococcus catus	*Eubacterium hallii*	*Turicibacter sanguinis*	75
Coprococcus catus	*Gemmiger formicilis*	*Turicibacter sanguinis*	75
Blautia sp. M25	*Collinsella aerofaciens*	*Turicibacter sanguinis*	75
Collinsella aerofaciens	*Eubacterium hallii*	*Turicibacter sanguinis*	75
Collinsella aerofaciens	*Ruminococcus* sp. 5_1_39BFAA	*Turicibacter sanguinis*	75

TABLE 19

Ternary OTU combinations in administered spore ecology doses that result in augmentation or engraftment of the OTU *Clostridium* sp. NML 04A032.

OTU 1	OTU 2	OTU 3	Percent of doses in which the ternary is present
Clostridium asparagiforme	*Faecalibacterium prausnitzii*	*Ruminococcus champanellensis*	80
Clostridium asparagiforme	*Ruminococcus champanellensis*	*Ruminococcus torques*	80
Clostridium asparagiforme	*Ruminococcus champanellensis*	*Ruminococcus obeum*	80
Blautia sp. M25	*Clostridium asparagiforme*	*Ruminococcus champanellensis*	80
Clostridium asparagiforme	*Ruminococcus champanellensis*	*Ruminococcus* sp. 5_1_39BFAA	80
Clostridium asparagiforme	*Ruminococcus bromii*	*Ruminococcus champanellensis*	80
Clostridium asparagiforme	*Eubacterium hallii*	*Ruminococcus champanellensis*	80
Clostridiales bacterium 1_7_47FAA	*Eubacterium rectale*	*Faecalibacterium prausnitzii*	75

TABLE 19-continued

Ternary OTU combinations in administered spore ecology doses that result in augmentation or engraftment of the OTU *Clostridium* sp. NML 04A032.

OTU 1	OTU 2	OTU 3	Percent of doses in which the ternary is present
Clostridiales bacterium 1_7_47FAA	*Eubacterium rectale*	*Ruminococcus obeum*	75
Clostridiales bacterium 1_7_47FAA	*Eubacterium rectale*	*Ruminococcus torques*	75
Blautia sp. M25	*Clostridiales bacterium* 1_7_47FAA	*Eubacterium rectale*	75
Clostridiales bacterium 1_7_47FAA	*Eubacterium rectale*	*Ruminococcus* sp. 5_1_39BFAA	75
Eubacterium rectale	*Faecalibacterium prausnitzii*	*Ruminococcus champanellensis*	75
Faecalibacterium prausnitzii	*Ruminococcus champanellensis*	*Ruminococcus lactaris*	75
Clostridiales bacterium 1_7_47FAA	*Clostridium asparagiforme*	*Faecalibacterium prausnitzii*	75
Clostridiales bacterium 1_7_47FAA	*Clostridium asparagiforme*	*Ruminococcus torques*	75
Blautia sp. M25	*Ruminococcus champanellensis*	*Ruminococcus lactaris*	75
Eubacterium rectale	*Ruminococcus champanellensis*	*Ruminococcus obeum*	75
Eubacterium rectale	*Ruminococcus champanellensis*	*Ruminococcus torques*	75
Ruminococcus champanellensis	*Ruminococcus lactaris*	*Ruminococcus obeum*	75
Clostridiales bacterium 1_7_47FAA	*Clostridium asparagiforme*	*Ruminococcus obeum*	75
Ruminococcus champanellensis	*Ruminococcus lactaris*	*Ruminococcus torques*	75
Blautia sp. M25	*Eubacterium rectale*	*Ruminococcus champanellensis*	75
Clostridium lavalense	*Faecalibacterium prausnitzii*	*Ruminococcus champanellensis*	75
Eubacterium rectale	*Ruminococcus champanellensis*	*Ruminococcus lactaris*	75
Clostridiales bacterium 1_7_47FAA	*Clostridium asparagiforme*	*Ruminococcus* sp. 5_1_39BFAA	75
Clostridiales bacterium 1_7_47FAA	*Eubacterium rectale*	*Ruminococcus bromii*	75
Clostridium lavalense	*Ruminococcus champanellensis*	*Ruminococcus obeum*	75
Clostridium lavalense	*Ruminococcus champanellensis*	*Ruminococcus torques*	75
Eubacterium rectale	*Ruminococcus champanellensis*	*Ruminococcus* sp. 5_1_39BFAA	75
Ruminococcus bromii	*Ruminococcus champanellensis*	*Ruminococcus lactaris*	75
Ruminococcus champanellensis	*Ruminococcus lactaris*	*Ruminococcus* sp. 5_1_39BFAA	75
Blautia sp. M25	*Clostridiales bacterium* 1_7_47FAA	*Clostridium asparagiforme*	75
Clostridium lavalense	*Eubacterium rectale*	*Ruminococcus champanellensis*	75
Clostridium lavalense	*Ruminococcus champanellensis*	*Ruminococcus lactaris*	75
Clostridium lavalense	*Ruminococcus champanellensis*	*Ruminococcus* sp. 5_1_39BFAA	75
Blautia sp. M25	*Clostridium lavalense*	*Ruminococcus champanellensis*	75
Clostridium asparagiforme	*Eubacterium rectale*	*Ruminococcus champanellensis*	75

TABLE 19-continued

Ternary OTU combinations in administered spore ecology doses that result in augmentation or engraftment of the OTU *Clostridium* sp. NML 04A032.

OTU 1	OTU 2	OTU 3	Percent of doses in which the ternary is present
Clostridium lavalense	*Ruminococcus bromii*	*Ruminococcus champanellensis*	75
Eubacterium rectale	*Ruminococcus bromii*	*Ruminococcus champanellensis*	75
Clostridiales bacterium 1_7_47FAA	*Clostridium asparagiforme*	*Ruminococcus bromii*	75
Clostridium asparagiforme	*Ruminococcus champanellensis*	*Ruminococcus lactaris*	75
Clostridiales bacterium 1_7_47FAA	*Clostridium asparagiforme*	*Eubacterium hallii*	75
Clostridium asparagiforme	*Clostridium lavalense*	*Ruminococcus champanellensis*	75
Coprococcus comes	*Eubacterium rectale*	*Ruminococcus champanellensis*	75
Coprococcus comes	*Faecalibacterium prausnitzii*	*Ruminococcus champanellensis*	75
Coprococcus comes	*Ruminococcus champanellensis*	*Ruminococcus obeum*	75
Coprococcus comes	*Ruminococcus champanellensis*	*Ruminococcus torques*	75
Dorea formicigenerans	*Eubacterium rectale*	*Ruminococcus champanellensis*	75
Dorea formicigenerans	*Faecalibacterium prausnitzii*	*Ruminococcus champanellensis*	75
Dorea formicigenerans	*Ruminococcus champanellensis*	*Ruminococcus obeum*	75
Dorea formicigenerans	*Ruminococcus champanellensis*	*Ruminococcus torques*	75
Dorea longicatena	*Ruminococcus champanellensis*	*Ruminococcus obeum*	75
Blautia sp. M25	*Coprococcus comes*	*Ruminococcus champanellensis*	75
Blautia sp. M25	*Dorea longicatena*	*Ruminococcus champanellensis*	75
Dorea formicigenerans	*Ruminococcus champanellensis*	*Ruminococcus lactaris*	75
Dorea longicatena	*Faecalibacterium prausnitzii*	*Ruminococcus champanellensis*	75
Dorea longicatena	*Ruminococcus champanellensis*	*Ruminococcus torques*	75
Eubacterium hallii	*Eubacterium rectale*	*Ruminococcus champanellensis*	75
Eubacterium hallii	*Ruminococcus champanellensis*	*Ruminococcus lactaris*	75
Blautia sp. M25	*Dorea formicigenerans*	*Ruminococcus champanellensis*	75
Clostridium asparagiforme	*Dorea longicatena*	*Ruminococcus champanellensis*	75
Clostridium asparagiforme	*Gemmiger formicilis*	*Ruminococcus champanellensis*	75
Clostridium lavalense	*Dorea formicigenerans*	*Ruminococcus champanellensis*	75
Dorea formicigenerans	*Ruminococcus champanellensis*	*Ruminococcus* sp. 5_1_39BFAA	75
Dorea longicatena	*Eubacterium rectale*	*Ruminococcus champanellensis*	75
Dorea longicatena	*Ruminococcus champanellensis*	*Ruminococcus lactaris*	75
Clostridiales sp. SSC/2	*Clostridium asparagiforme*	*Ruminococcus champanellensis*	75
Clostridium asparagiforme	*Coprococcus comes*	*Ruminococcus champanellensis*	75
Clostridium asparagiforme	*Dorea formicigenerans*	*Ruminococcus champanellensis*	75
Clostridium lavalense	*Dorea longicatena*	*Ruminococcus champanellensis*	75
Coprococcus comes	*Ruminococcus bromii*	*Ruminococcus champanellensis*	75
Coprococcus comes	*Ruminococcus champanellensis*	*Ruminococcus lactaris*	75
Coprococcus comes	*Ruminococcus champanellensis*	*Ruminococcus* sp. 5_1_39BFAA	75
Dorea longicatena	*Ruminococcus bromii*	*Ruminococcus champanellensis*	75
Dorea longicatena	*Ruminococcus champanellensis*	*Ruminococcus* sp. 5_1_39BFAA	75
Clostridium asparagiforme	*Coprococcus catus*	*Ruminococcus champanellensis*	75
Clostridium lavalense	*Eubacterium hallii*	*Ruminococcus champanellensis*	75
Dorea formicigenerans	*Eubacterium hallii*	*Ruminococcus champanellensis*	75
Dorea formicigenerans	*Dorea longicatena*	*Ruminococcus champanellensis*	75
Clostridium lavalense	*Coprococcus comes*	*Ruminococcus champanellensis*	75
Coprococcus comes	*Dorea formicigenerans*	*Ruminococcus champanellensis*	75
Coprococcus comes	*Eubacterium hallii*	*Ruminococcus champanellensis*	75
Dorea formicigenerans	*Ruminococcus bromii*	*Ruminococcus champanellensis*	75
Coprococcus comes	*Dorea longicatena*	*Ruminococcus champanellensis*	75
Dorea longicatena	*Eubacterium hallii*	*Ruminococcus champanellensis*	75
Clostridium asparagiforme	*Collinsella aerofaciens*	*Ruminococcus champanellensis*	75

TABLE 20

Ternary OTU combinations in administered spore ecology doses that result in augmentation or engraftment of the OTUs *Clostridium* sp. NML 04A032, *Ruminococcus lactaris*, and *Ruminococcus torques*.

OTU 1	OTU 2	OTU 3	Percent of doses in which the ternary is present
Clostridiales sp. SM4/1	*Faecalibacterium prausnitzii*	*Ruminococcus obeum*	75
Clostridiales sp. SM4/1	*Faecalibacterium prausnitzii*	*Ruminococcus torques*	75
Clostridiales sp. SM4/1	*Ruminococcus obeum*	*Ruminococcus torques*	75
Blautia sp. M25	*Clostridiales* sp. SM4/1	*Faecalibacterium prausnitzii*	75
Clostridiales sp. SM4/1	*Ruminococcus* sp. 5_1_39BFAA	*Ruminococcus torques*	75
Blautia sp. M25	*Clostridiales* sp. SM4/1	*Ruminococcus torques*	75
Clostridiales sp. SM4/1	*Faecalibacterium prausnitzii*	*Ruminococcus* sp. 5_1_39BFAA	75
Blautia sp. M25	*Clostridiales* sp. SM4/1	*Ruminococcus obeum*	75
Blautia sp. M25	*Clostridiales* sp. SM4/1	*Ruminococcus* sp. 5_1_39BFAA	75
Clostridiales sp. SM4/1	*Ruminococcus obeum*	*Ruminococcus* sp. 5_1_39BFAA	75
Clostridiales sp. SM4/1	*Clostridium asparagiforme*	*Faecalibacterium prausnitzii*	75
Clostridiales sp. SM4/1	*Clostridium asparagiforme*	*Ruminococcus torques*	75

TABLE 20-continued

Ternary OTU combinations in administered spore ecology doses that result in augmentation or engraftment of the OTUs *Clostridium* sp. NML 04A032, *Ruminococcus lactaris*, and *Ruminococcus torques*.

OTU 1	OTU 2	OTU 3	Percent of doses in which the ternary is present
Clostridiales sp. SM4/1	*Clostridium asparagiforme*	*Ruminococcus obeum*	75
Blautia sp. M25	*Clostridiales* sp. SM4/1	*Ruminococcus bromii*	75
Clostridiales sp. SM4/1	*Faecalibacterium prausnitzii*	*Ruminococcus bromii*	75
Clostridiales sp. SM4/1	*Ruminococcus bromii*	*Ruminococcus obeum*	75
Clostridiales sp. SM4/1	*Ruminococcus bromii*	*Ruminococcus* sp. 5_1_39BFAA	75
Clostridiales sp. SM4/1	*Ruminococcus bromii*	*Ruminococcus torques*	75
Blautia sp. M25	*Clostridiales* sp. SM4/1	*Clostridium asparagiforme*	75
Clostridiales sp. SM4/1	*Clostridium asparagiforme*	*Ruminococcus* sp. 5_1_39BFAA	75
Clostridiales sp. SM4/1	*Eubacterium hallii*	*Faecalibacterium prausnitzii*	75
Clostridiales sp. SM4/1	*Eubacterium hallii*	*Ruminococcus obeum*	75
Clostridiales sp. SM4/1	*Eubacterium hallii*	*Ruminococcus torques*	75
Blautia sp. M25	*Clostridiales* sp. SM4/1	*Eubacterium hallii*	75
Clostridiales sp. SM4/1	*Eubacterium hallii*	*Ruminococcus* sp. 5_1_39BFAA	75
Clostridiales sp. SM4/1	*Clostridium asparagiforme*	*Ruminococcus bromii*	75
Clostridiales sp. SM4/1	*Clostridium asparagiforme*	*Eubacterium hallii*	75
Clostridiales sp. SM4/1	*Eubacterium hallii*	*Ruminococcus bromii*	75

TABLE 21

Ternary OTU combinations in administered spore ecology doses that result in augmentation or engraftment of the OTUs *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Oscillibacter* sp. G2, *Ruminococcus lactaris*, and *Ruminococcus torques*.

OTU 1	OTU 2	OTU 3	Percent of doses in which the ternary is present
Clostridium bifermentans	*Faecalibacterium prausnitzii*	*Ruminococcus obeum*	75
Clostridium bifermentans	*Ruminococcus obeum*	*Ruminococcus torques*	75
Clostridium bifermentans	*Ruminococcus obeum*	*Ruminococcus* sp. 5_1_39BFAA	75
Blautia sp. M25	*Clostridium bifermentans*	*Faecalibacterium prausnitzii*	75
Blautia sp. M25	*Clostridium bifermentans*	*Ruminococcus torques*	75
Clostridium bifermentans	*Faecalibacterium prausnitzii*	*Ruminococcus* sp. 5_1_39BFAA	75
Clostridium bifermentans	*Faecalibacterium prausnitzii*	*Ruminococcus torques*	75
Clostridium bifermentans	*Ruminococcus* sp. 5_1_39BFAA	*Ruminococcus torques*	75
Blautia sp. M25	*Clostridium bifermentans*	*Ruminococcus obeum*	75
Blautia sp. M25	*Clostridium bifermentans*	*Ruminococcus* sp. 5_1_39BFAA	75

TABLE 21-continued

Ternary OTU combinations in administered spore ecology doses that result in augmentation or engraftment of the OTUs *Eubacterium rectale*, *Faecalibacterium prausnitzii*, *Oscillibacter* sp. G2, *Ruminococcus lactaris*, and *Ruminococcus torques*.

OTU 1	OTU 2	OTU 3	Percent of doses in which the ternary is present
Clostridium bifermentans	*Ruminococcus bromii*	*Ruminococcus obeum*	75
Blautia sp. M25	*Clostridium bifermentans*	*Gemmiger formicilis*	75
Clostridium bifermentans	*Faecalibacterium prausnitzii*	*Ruminococcus bromii*	75
Clostridium bifermentans	*Gemmiger formicilis*	*Ruminococcus obeum*	75
Clostridium bifermentans	*Gemmiger formicilis*	*Ruminococcus* sp. 5_1_39BFAA	75
Clostridium bifermentans	*Ruminococcus bromii*	*Ruminococcus torques*	75
Clostridium bifermentans	*Faecalibacterium prausnitzii*	*Gemmiger formicilis*	75
Blautia sp. M25	*Clostridium bifermentans*	*Ruminococcus bromii*	75
Clostridium bifermentans	*Gemmiger formicilis*	*Ruminococcus torques*	75
Clostridium bifermentans	*Ruminococcus bromii*	*Ruminococcus* sp. 5_1_39BFAA	75
Clostridium bifermentans	*Gemmiger formicilis*	*Ruminococcus bromii*	75

TABLE 22

Selected OTUs that may be present in the tables, specification, or in the art with their alternate names, e.g., the current name used per NCBI. Reference 1 is Kaur et al., "*Hungatella effluvii* gen. nov., sp. nov., an obligately anaerobic bacterium isolated from an effluent treatment plant, and reclassification of *Clostridium hathewayi* as *Hungatella hathewayi* gen. nov., comb. nov.", Int J Sys Evol Microbiol, March 2014, vol. 64, pp. 710-718. Reference 2 is Gerritsen et al., "Characterization of *Romboutsia ilealis* gen. nov., sp. nov., isolated from the gastro-intestinal tract of a rat, and proposal for the reclassification of five closely related members of the genus *Clostridium* into the genera *Romboutsia* gen. nov., *Intestinibacter* gen, nov., *Terrisporobacter* gen. nov. and *Asaccharospora* gen. nov.", Int J Sys Evol Microbiol, May 2014, vol. 64, pp. 1600-1616.

OTU name	Alternate OTU name	Reference
Clostridium hathewayi	*Hungatella hathewayi*	1
Clostridium lituseburense	*Romboutsia lituseburense*	2
Clostridium bartlettii	*Intestinibacter bartlettii*	2
Clostridium glycolicum	*Terrisporobacter glycolicus*	2
Clostridium mayombei	*Terrisporobacter mayombei*	2
Clostridium irregulare	*Asaccharospora irregularis*	2

SEQUENCE LISTING

The patent contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US12409197B2). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

What is claimed is:

1. A method of functionally populating the gastrointestinal tract of a human subject in need thereof, comprising administering to the subject a composition comprising: (a) a first species of an isolated bacterium capable of forming a spore, (b) a second species of isolated bacterium capable of forming a spore, and (c) a third species of isolated bacterium capable of forming a spore, and (d) a capsule;

wherein the capsule encapsulates the first species, the second species, and third species; wherein the first species, the second species, and the third species are not identical;

wherein the first species is *Clostridium orbiscindens*, the second species is *Clostridium bolteae*, and the third species is *Lachnospiraceae bacterium* 5_1_57FAA,

wherein the *Clostridium orbiscindens* comprises a 16S rDNA sequence that is at least 97% identical to a 16S rDNA the sequence set forth in SEQ ID NO: 609; and

wherein a combination of the first species, the second species, and the third species is capable of decreasing and/or inhibiting the growth and/or colonization of at least one type of pathogenic bacteria.

2. The method of claim **1**, wherein the combination of the first species, the second species, and the third species is capable of synergistically inhibiting the growth and/or colonization of the at least one type of pathogenic bacteria in a CivSim assay with a confidence interval of >99% (++++).

3. The method of claim **1**, wherein one or more of the first species, the second species, and the third species are in the form of spores.

4. The method of claim **3**, wherein the first species, the second species, and the third species are in the form of spores.

5. The method of claim **1**, wherein the composition further comprises a capsule, which encapsulates one or more of the first species, the second species, and the third species.

6. The method of claim **5**, wherein the capsule is enterically coated.

7. The method of claim **5**, wherein the capsule encapsulates the first species, the second species, and the third species.

8. The method of claim **1**, wherein the composition further comprises glycerol.

9. The method of claim **7**, wherein the composition further comprises glycerol.

10. The method of claim **1**, wherein the *Clostridium orbiscindens* comprises a 16S rDNA sequence that is at least 98% identical to the sequence set forth in SEQ ID NO: 609.

11. The method of claim **1**, wherein the *Clostridium bolteae* comprises a 16S rDNA sequence that is at least 97% identical to the sequence set forth in SEQ ID NO: 559.

12. The method of claim **1**, wherein the *Lachnospiraceae bacterium* 5_1_57FAA comprises a 16S rDNA sequence that is at least 97% identical to the sequence set forth in SEQ ID NO: 1054.

13. The method of claim **1**, wherein the *Clostridium orbiscindens* comprises a 16S rDNA sequence that is at least 99% identical to the sequence set forth in SEQ ID NO: 609.

14. The method of claim **1**, wherein the *Clostridium orbiscindens* comprises the 16S rDNA sequence set forth in SEQ ID NO: 609.

15. The method of claim **1**, wherein the *Clostridium bolteae* comprises a 16S rDNA sequence that is at least 98% identical to the sequence set forth in SEQ ID NO: 559.

16. The method of claim **1**, wherein the *Clostridium bolteae* comprises a 16S rDNA sequence that is at least 99% identical to the sequence set forth in SEQ ID NO: 559.

17. The method of claim **1**, wherein the *Clostridium bolteae* comprises the 16S rDNA sequence set forth in SEQ ID NO: 559.

18. The method of claim **1**, wherein the *Lachnospiraceae bacterium* 5_1_57FAA comprises a 16S rDNA sequence that is at least 98% identical to the sequence set forth in SEQ ID NO: 1054.

19. The method of claim **1**, wherein the *Lachnospiraceae bacterium* 5_1_57FAA comprises a 16S rDNA sequence that is at least 99% identical to the sequence set forth in SEQ ID NO: 1054.

20. The method of claim **1**, wherein the *Lachnospiraceae bacterium* 5_1_57FAA comprises the 16S rDNA sequence set forth in SEQ ID NO: 1054.

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